

Construction Theory for C_8 -Equivariant, Parity-Compatible Finite Quotients of the Bolza Surface Group

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Abstract

This manuscript builds finite quotients of the Bolza surface group from exact hyperbolic and group-theoretic data. Word, based-geometric, and global-systolic injectivity are kept distinct; all descent maps are proved well defined; and finite certificates are separated from numerical tractability. The algebraic construction proceeds through lower central and p -central series, graded Lie layers, C_8 -modules, two-step commutator pairings, Hall coordinates, and cocycle-certified finite nilpotent extensions.

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Conventions

The curvature scale is $R > 0$, so $\mathbb{H}_R^2$ has curvature $-R^{-2}$. Composition is functional: $(fg)(x) = f(g(x))$. Matrices in $SU(1,1)$ act on the left, and matrices differing by $-I$ define the same projective isometry. Conditional statements record all family-specific hypotheses. Invisible subsection labels provide internal cross-references.

Part I

Hyperbolic geometry and the Bolza surface

1 The hyperbolic plane

1.1 Intrinsic definition and normalization

Definition 1.1.1 (Hyperbolic plane at scale R). A hyperbolic plane of scale $R > 0$ is a connected, oriented, complete and simply connected two-dimensional Riemannian manifold whose Gaussian curvature is the constant $-R^{-2}$. An isomorphism of such objects is an orientation-preserving Riemannian isometry.

Constant negative curvature alone does not determine the global surface: a quotient by a freely acting discrete isometry group has the same local curvature but need not be simply connected. Simple connectedness without completeness also does not suffice, because a proper open metric ball has the same curvature as the ambient plane. The scale R is dimensional: multiplying a curvature- -1 metric by R^2 multiplies every distance by R and changes curvature to $-R^{-2}$.

Theorem 1.1.2 (Existence and uniqueness). *For each $R > 0$ a hyperbolic plane of scale R exists. Any two such planes are related by an orientation-preserving isometry.*

Proof. Existence is established in the next subsection by the disk metric. For uniqueness, let (M, g) and (N, h) satisfy the definition. Choose positively oriented orthonormal frames (p, e_1, e_2) and (q, f_1, f_2) . Along every unit-speed geodesic the normal Jacobi equation is

$$J'' - \frac{1}{R^2}J = 0.$$

The solution with $J(0) = 0$ and $J'(0) = v$ is $J(t) = R \sinh(t/R) P_t v$. Hence geodesic polar coordinates about either base point have metric

$$ds^2 = d\rho^2 + R^2 \sinh^2(\rho/R) d\theta^2. \quad (1.1)$$

Completeness makes both exponential maps globally defined. Simple connectedness and nonpositive curvature satisfy the hypotheses of the Cartan–Hadamard theorem, so those exponential maps are diffeomorphisms. If $L : T_p M \rightarrow T_q N$ is the oriented linear isometry taking e_i to f_i , define

$$F = \exp_q \circ L \circ \exp_p^{-1}.$$

Formula (1.1) gives $F^*h = g$, and the frame choice makes F orientation preserving. The standard space-form and Cartan–Hadamard results used here are stated with these same completeness, simple connectedness, and curvature hypotheses in [2, 1]. $\square$

1.2 The Poincaré disk metric

Definition 1.2.1 (Disk model). Let

$$\mathbb{D} = \{z = x + iy \in \mathbb{C} : |z| < 1\}, \quad g_R = \lambda(z)^2 (dx^2 + dy^2), \quad \lambda(z) = \frac{2R}{1 - |z|^2}. \quad (1.2)$$

The Euclidean unit circle is the ideal boundary and is not part of the Riemannian manifold.

Proposition 1.2.2 (Curvature, area, and completeness). *The metric g_R has Gaussian curvature $-R^{-2}$, area form*

$$dA_R = \frac{4R^2}{(1 - |z|^2)^2} dx dy, \quad (1.3)$$

and is complete.

Proof. For a conformal metric $e^{2u}(dx^2 + dy^2)$, evaluation of the Christoffel symbols gives

$$K = -e^{-2u}(u_{xx} + u_{yy}). \quad (1.4)$$

Here $u = \log(2R) - \log(1 - x^2 - y^2)$. With $r^2 = x^2 + y^2$,

$$u_x = \frac{2x}{1 - r^2}, \quad u_{xx} = \frac{2}{1 - r^2} + \frac{4x^2}{(1 - r^2)^2},$$

and the analogous y -formula yields

$$\Delta u = \frac{4}{1 - r^2} + \frac{4r^2}{(1 - r^2)^2} = \frac{4}{(1 - r^2)^2}.$$

Since $e^{-2u} = (1 - r^2)^2/(4R^2)$, (1.4) gives $K = -R^{-2}$. The square root of the metric determinant is λ^2 , proving (1.3).

If a piecewise C^1 curve $\gamma(t) = r(t)e^{i\theta(t)}$ approaches $|z| = 1$, then

$$L_R(\gamma) = \int \frac{2R\sqrt{\dot{r}^2 + r^2\dot{\theta}^2}}{1 - r^2} dt \geq \left| \int \frac{2R\dot{r}}{1 - r^2} dt \right| = 2R |\operatorname{artanh} r|_{\text{endpoints}},$$

which diverges. A finite-length Cauchy sequence therefore cannot escape through the ideal boundary. On each Euclidean disk $|z| \leq r_0 < 1$, the Euclidean and hyperbolic norms are uniformly equivalent, so every remaining Cauchy sequence converges. This proves completeness. The disk is connected and simply connected, completing the existence proof in the preceding subsection. $\square$

Setting $r = \tanh(\rho/(2R))$ gives

$$\frac{2R dr}{1 - r^2} = d\rho, \quad \frac{2Rr}{1 - r^2} = R \sinh(\rho/R),$$

and therefore

$$g_R = d\rho^2 + R^2 \sinh^2(\rho/R) d\theta^2. \quad (1.5)$$

1.3 Distance and geodesics

For $a \in \mathbb{D}$, define $m_a(z) = (z - a)/(1 - \bar{a}z)$. The denominator is nonzero in $\mathbb{D}$, and direct expansion gives

$$1 - |m_a(z)|^2 = \frac{(1 - |a|^2)(1 - |z|^2)}{|1 - \bar{a}z|^2}, \quad |m'_a(z)| = \frac{1 - |a|^2}{|1 - \bar{a}z|^2}. \quad (1.6)$$

Thus m_a preserves the metric and sends a to 0.

Theorem 1.3.1 (Exact distance formula). *For $z, w \in \mathbb{D}$,*

$$d_R(z, w) = 2R \operatorname{artanh} \left| \frac{z - w}{1 - \bar{z}w} \right|, \quad (1.7)$$

$$\cosh \frac{d_R(z, w)}{R} = 1 + \frac{2|z - w|^2}{(1 - |z|^2)(1 - |w|^2)}. \quad (1.8)$$

Proof. Metric invariance gives $d_R(z, w) = d_R(0, m_z(w))$. A radial segment from 0 to u has length

$$\int_0^{|u|} \frac{2R \, dt}{1 - t^2} = 2R \operatorname{artanh} |u|. \quad (1.9)$$

Every competing curve has length at least its absolute radial integral, so this segment minimizes and proves (1.7). If $t = |m_z(w)|$, then

$$\cosh(2 \operatorname{artanh} t) = \frac{1 + t^2}{1 - t^2},$$

while

$$|1 - \bar{z}w|^2 - |z - w|^2 = (1 - |z|^2)(1 - |w|^2).$$

Substitution gives (1.8). The formula is finite for interior points, vanishes precisely at $z = w$, and diverges when one point approaches the ideal boundary while the other stays in a compact subset. $\square$

Proposition 1.3.2 (Geodesic classification). *The complete unoriented geodesics in $\mathbb{D}$ are exactly the Euclidean diameters and the arcs of Euclidean circles orthogonal to $\partial\mathbb{D}$.*

Proof. Formula (1.9) proves that diameter segments minimize. Disk automorphisms are conformal Möbius maps, hence carry generalized circles to generalized circles and preserve orthogonality to $\partial\mathbb{D}$. They therefore carry diameters to orthogonal arcs. Conversely, an orthogonal circle has two distinct ideal endpoints. A disk automorphism sends those endpoints to -1 and 1 , making its arc the real diameter. Applying the inverse isometry proves that the original arc is geodesic. Uniqueness follows from the unique orthogonal circle through two interior points, equivalently from strict negative curvature. $\square$

1.4 All disk isometries

Theorem 1.4.1 (Orientation-preserving isometries). *Every orientation-preserving isometry of $(\mathbb{D}, g_R)$ has a unique expression*

$$f(z) = e^{i\vartheta} \frac{z - a}{1 - \bar{a}z}, \quad a \in \mathbb{D}, \quad e^{i\vartheta} \in S^1. \quad (1.10)$$

Every orientation-reversing isometry is $z \mapsto f(\bar{z})$ for a unique orientation-preserving f .

Proof. Equation (1.6) and rotational invariance of (1.2) show that every displayed map is an orientation-preserving isometry. Conversely, let F be one and put $a = F^{-1}(0)$. Then $G = F \circ m_a^{-1}$ fixes 0. The differential dG_0 preserves the Euclidean scalar product because the hyperbolic metric at 0 is $4R^2$ times that product. Orientation forces dG_0 to be rotation through an angle ϑ . Isometries take geodesics to geodesics and preserve arclength parameters. Every point lies on one radial geodesic from 0, hence

$$G(re^{i\alpha}) = re^{i(\alpha + \vartheta)}.$$

Thus $G(z) = e^{i\vartheta} z$, proving (1.10). The zero a and the derivative phase determine the parameters, so the expression is unique. Complex conjugation preserves (1.2) and reverses orientation; composition with it proves the last assertion. $\square$

A nonidentity orientation-preserving isometry extends to the closed disk. It is elliptic if it has an interior fixed point, parabolic if it has one ideal fixed point and no interior fixed point, and hyperbolic if it has two ideal fixed points. The identity is a separate case. Conjugation transports fixed sets, so this classification is conjugacy invariant.

1.5 The $SU(1, 1)$ and projective actions

Put $J = \text{diag}(1, -1)$. The equations $A^*JA = J$ and $\det A = 1$ are equivalent to

$$A = A(a, b) = \begin{pmatrix} a & b \\ \bar{b} & \bar{a} \end{pmatrix}, \quad |a|^2 - |b|^2 = 1. \quad (1.11)$$

The first column has J -norm 1; its J -orthogonal complement is spanned by $(b, \bar{a})^T$; determinant one fixes the remaining phase. Multiplication and inversion are

$$A(a, b)A(c, d) = A(ac + b\bar{d}, ad + b\bar{c}), \quad (1.12)$$

$$A(a, b)^{-1} = A(\bar{a}, -b). \quad (1.13)$$

Theorem 1.5.1 (Projective realization). *The formula*

$$A(a, b) \cdot z = \frac{az + b}{\bar{b}z + \bar{a}} \quad (1.14)$$

is a well-defined left action by all orientation-preserving disk isometries. Its kernel is $\{\pm I\}$, and therefore

$$\text{PSU}(1, 1) = \text{SU}(1, 1)/\{\pm I\} \cong \text{Isom}^+(\mathbb{H}_R^2). \quad (1.15)$$

Proof. Because $|a| > |b|$, the denominator in (1.14) cannot vanish for $|z| < 1$. Expanding with $|a|^2 - |b|^2 = 1$ gives

$$1 - |A \cdot z|^2 = \frac{1 - |z|^2}{|\bar{b}z + \bar{a}|^2}, \quad (A \cdot z)' = \frac{1}{(\bar{b}z + \bar{a})^2}. \quad (1.16)$$

Thus the metric density is invariant. Equation (1.12) proves $(AB) \cdot z = A \cdot (B \cdot z)$, and I acts trivially, establishing a left action and its well-definedness.

If A acts trivially, evaluation at 0 gives $b = 0$. Then $|a| = 1$ and (1.14) becomes $z \mapsto a^2 z$; hence $a^2 = 1$ and $A = \pm I$. Conversely, (1.10) is represented by

$$a_0 = e^{i\vartheta/2}(1 - |a|^2)^{-1/2}, \quad b_0 = -e^{i\vartheta/2}a(1 - |a|^2)^{-1/2}.$$

These satisfy $|a_0|^2 - |b_0|^2 = 1$, and substitution recovers (1.10). The action is surjective; quotienting by the computed kernel proves (1.15). $\square$

1.6 Translation length, axes, and conjugacy

Definition 1.6.1. For $g \in \text{Isom}^+(\mathbb{H}_R^2)$, set

$$\ell_{\mathbb{H}}(g) = \inf_{x \in \mathbb{H}_R^2} d_R(x, gx), \quad \text{Min}(g) = \{x : d_R(x, gx) = \ell_{\mathbb{H}}(g)\}. \quad (1.17)$$

Theorem 1.6.2 (Axis theorem and boundary cases). *If g is hyperbolic, its two ideal fixed points determine a unique invariant geodesic $\mathcal{A}_g$, and $\text{Min}(g) = \mathcal{A}_g$. The map g translates that axis by $\ell_{\mathbb{H}}(g) > 0$. If g is elliptic, $\ell_{\mathbb{H}}(g) = 0$ and the minimum is attained. If g is parabolic, $\ell_{\mathbb{H}}(g) = 0$ but the infimum is not attained.*

Proof. Conjugate a hyperbolic element so its ideal fixed points are $-1, 1$. In the arclength coordinate $u = 2R \operatorname{artanh} x$ on the real diameter, it acts as $u \mapsto u + L$, with $L > 0$ after orienting the axis. The hyperbolic quadrilateral identity in Fermi coordinates gives

$$\sinh \frac{d_R(x, gx)}{2R} = \cosh \frac{d_R(x, \mathcal{A}_g)}{R} \sinh \frac{L}{2R}. \quad (1.18)$$

The middle factor is at least one and equals one exactly on the axis. Thus the minimum is L , attained precisely there.

An elliptic element is conjugate to a rotation about 0, whose center realizes zero displacement. A parabolic element is conjugate in the upper-half-plane model to $z \mapsto z + c$, $c \neq 0$. At height y ,

$$\cosh \frac{d_R(iy, c + iy)}{R} = 1 + \frac{c^2}{2y^2} \longrightarrow 1.$$

The infimum is zero. Attainment would supply an interior fixed point, contradicting parabolicity. $\square$

For every isometry h , changing variables from x to hy gives

$$\ell_{\mathbb{H}}(hgh^{-1}) = \inf_x d_R(x, hgh^{-1}x) = \inf_y d_R(hy, hgy) = \inf_y d_R(y, gy) = \ell_{\mathbb{H}}(g). \quad (1.19)$$

Translation length is therefore conjugacy invariant. Based displacement is not; (1.18) records its dependence on distance from the axis.

1.7 Trace classification and the length formula

For $A = A(a, b)$, $\operatorname{tr} A = a + \bar{a} = 2 \operatorname{Re} a$ is real. Replacing the lift by $-A$ changes the sign but not $|\operatorname{tr} A|$, so absolute trace is well-defined projectively.

Theorem 1.7.1 (Trace trichotomy). *For $g = [A] \in \operatorname{PSU}(1, 1)$:*

- (i) $|\operatorname{tr} A| < 2$ if and only if g is elliptic;
- (ii) $|\operatorname{tr} A| = 2$ if and only if g is the identity ($A = \pm I$) or is parabolic;
- (iii) $|\operatorname{tr} A| > 2$ if and only if g is hyperbolic, and then

$$\ell_{\mathbb{H}}(g) = 2R \operatorname{arcosh} \frac{|\operatorname{tr} A|}{2}. \quad (1.20)$$

Proof. The characteristic polynomial is

$$p_A(\lambda) = \lambda^2 - (\operatorname{tr} A)\lambda + 1, \quad \Delta = (\operatorname{tr} A)^2 - 4.$$

For $\Delta < 0$, the conjugate eigenvalues lie on the unit circle and the projective map has an interior fixed point. For $\Delta = 0$, the repeated eigenvalue is 1 or -1 ; the diagonalizable case is $\pm I$, while the nondiagonalizable case has one ideal fixed point. For $\Delta > 0$, the two real reciprocal eigenvalues have distinct moduli and determine two ideal fixed points. Indeed, if $Av = \lambda v$, preservation of the J -Hermitian form gives

$$(|\lambda|^2 - 1)\langle v, v \rangle_J = 0.$$

Every eigenline with $|\lambda| \neq 1$ is therefore J -null and represents an ideal boundary point; in the nondiagonalizable repeated-root case the unique eigenline is likewise null. These are exactly the elliptic, identity/parabolic, and hyperbolic alternatives, so both directions have been proved.

A hyperbolic element is conjugate, after changing the lift sign if needed, to

$$H_t = \begin{pmatrix} \cosh(t/2) & \sinh(t/2) \\ \sinh(t/2) & \cosh(t/2) \end{pmatrix}, \quad t > 0. \quad (1.21)$$

It maps 0 to $\tanh(t/2)$ on its axis. Hence

$$\ell_{\mathbb{H}}([H_t]) = 2R \operatorname{artanh}(\tanh(t/2)) = Rt. \quad (1.22)$$

Since $\operatorname{tr} H_t = 2 \cosh(t/2)$, solving for t proves (1.20). Conjugacy invariance transports the formula to A . $\square$

Remark 1.7.2. The equality $|\operatorname{tr} A| = 2$ does not by itself mean parabolic: $\pm I$ is the identity. Elliptic and parabolic elements both have translation length zero, but only the elliptic case attains the minimum. These exceptional cases remain explicit in every later trace-cutoff argument.

Remark 1.7.3. Later quotients have the same local curvature but are not called the hyperbolic plane. Every curvature- -1 identity is rescaled by inserting x/R in dimensionless hyperbolic functions.

2 The regular octagon and the Bolza surface

2.1 Existence and uniqueness of the regular octagon

Definition 2.1.1. A centered regular hyperbolic octagon is a compact convex geodesic polygon $P \subset \mathbb{H}_R^2$ with center o , eight equal sides, eight equal interior angles, and rotational symmetry through $\pi/4$. For the Bolza octagon the common interior angle is fixed to $\pi/4$.

Join o to every vertex and every side midpoint. Reflection symmetry in the radius through a side midpoint bisects that side orthogonally, and rotation symmetry divides the full angle at o into eight angles $\pi/4$. Each of the resulting sixteen triangles therefore has angles

$$\angle O = \frac{\pi}{8}, \quad \angle V = \frac{\pi}{8}, \quad \angle M = \frac{\pi}{2}. \quad (2.1)$$

The angle sum is $3\pi/4 < \pi$, so a hyperbolic triangle with these angles exists and is unique up to isometry. Reflecting it across OM , then rotating the doubled sector successively by multiples of $\pi/4$, produces a compact regular octagon.

Proposition 2.1.2. *The centered regular octagon with interior angle $\pi/4$ exists and is unique up to an orientation-preserving isometry. Its rotations and reflections generate the local dihedral symmetry D_8 . The copies of the right triangle in (2.1) have disjoint interiors and reconstruct the entire octagon.*

Proof. The hyperbolic angle-angle-angle theorem determines a triangle up to isometry because negative curvature removes Euclidean scaling freedom. Gluing two copies along the leg OM makes an isosceles sector of angle $\pi/4$ at O and interior half-angles $\pi/8$ at its two outer vertices. Eight rotated sectors close because $8(\pi/4) = 2\pi$. Convexity follows because each interior angle is less than π . Conversely, bisecting any regular octagon with the stated angle produces the triangle (2.1); triangle uniqueness yields an isometry of sectors, and the rotational gluing extends it over the polygon. Orientation can be chosen after the extension. The reflected and rotated copies give the claimed symmetries. No side pairing or quotient assertion is used here. $\square$

2.2 Side length, inradius, and circumradius

Let $r_{\text{in}} = d_R(O, M)$, $r_{\text{out}} = d_R(O, V)$, and let $a_B = 2d_R(V, M)$ be the full side length. In a curvature- $-R^{-2}$ right triangle, with hypotenuse c , legs a, b , and opposite angles A, B , the right-triangle identities are

$$\cosh \frac{a}{R} = \frac{\cos A}{\sin B}, \quad \cosh \frac{b}{R} = \frac{\cos B}{\sin A}, \quad \cosh \frac{c}{R} = \cot A \cot B. \quad (2.2)$$

They follow from the hyperbolic cosine rules after setting the third angle to $\pi/2$.

Proposition 2.2.1 (Exact radii). *For the triangle (2.1),*

$$\cosh \frac{r_{\text{in}}}{R} = \cot \frac{\pi}{8} = 1 + \sqrt{2}, \quad (2.3)$$

$$a_B = 2r_{\text{in}} = 2R \operatorname{arccosh}(1 + \sqrt{2}), \quad (2.4)$$

$$\cosh \frac{r_{\text{out}}}{R} = \cot^2 \frac{\pi}{8} = (1 + \sqrt{2})^2 = 3 + 2\sqrt{2}. \quad (2.5)$$

In disk coordinates centered at $o = 0$, the Euclidean vertex radius is

$$\rho = \tanh \frac{r_{\text{out}}}{2R} = 2^{-1/4}. \quad (2.6)$$

Proof. In (2.2), both acute angles are $\pi/8$. The two legs opposite equal angles are equal, so $OM = VM$, giving (2.4). Moreover

$$\frac{\cos(\pi/8)}{\sin(\pi/8)} = \cot(\pi/8) = \frac{1 + \cos(\pi/4)}{\sin(\pi/4)} = 1 + \sqrt{2},$$

which proves (2.3). The hypotenuse identity gives (2.5). Finally,

$$\tanh^2 \frac{x}{2} = \frac{\cosh x - 1}{\cosh x + 1}.$$

Taking $x = r_{\text{out}}/R$ and using (2.5),

$$\rho^2 = \frac{(3 + 2\sqrt{2}) - 1}{(3 + 2\sqrt{2}) + 1} = \frac{2 + 2\sqrt{2}}{4 + 2\sqrt{2}} = \frac{1}{\sqrt{2}}.$$

The positive square root is $\rho = 2^{-1/4}$. All values are positive and finite, so no sign or ideal-vertex exception occurs. $\square$

2.3 Exact opposite-side pairings

Index vertices and oriented sides modulo eight by

$$v_k = \rho e^{(2k+1)\pi i/8}, \quad s_j = [v_{j-1}, v_j].$$

Put

$$\zeta = e^{\pi i/4}, \quad A = 1 + \sqrt{2}, \quad B = \sqrt{2 + 2\sqrt{2}}.$$

Since

$$A^2 = 3 + 2\sqrt{2}, \quad B^2 = 2 + 2\sqrt{2}, \quad A^2 - B^2 = 1, \quad (2.7)$$

the matrices

$$G_j = \begin{pmatrix} A & B\zeta^j \\ B\zeta^{-j} & A \end{pmatrix} \in \operatorname{SU}(1, 1), \quad g_j = [G_j] \in \operatorname{PSU}(1, 1) \quad (2.8)$$

are well defined. Because $\zeta^4 = -1$,

$$G_{j+4} = \begin{pmatrix} A & -B\zeta^j \\ -B\zeta^{-j} & A \end{pmatrix} = G_j^{-1}, \quad g_{j+4} = g_j^{-1}. \quad (2.9)$$

Proposition 2.3.1 (Endpoint calculation). *For every $j \in \mathbb{Z}/8\mathbb{Z}$,*

$$g_j(v_{j+3}) = v_j, \quad g_j(v_{j+4}) = v_{j-1}. \quad (2.10)$$

Thus g_j maps s_{j+4} to s_j and reverses its boundary orientation.

Proof. Let

$$C = \begin{pmatrix} e^{\pi i/8} & 0 \\ 0 & e^{-\pi i/8} \end{pmatrix}.$$

Its projective action is rotation by $\pi/4$, and matrix multiplication gives

$$G_j = C^j G_0 C^{-j}. \quad (2.11)$$

It is enough to verify $j = 0$. Cross multiplication shows that $g_0(v_3) = v_0$ is equivalent to

$$A(v_3 - v_0) + B(1 - v_0 v_3) = 0. \quad (2.12)$$

Now $v_0 v_3 = -\rho^2 = -1/\sqrt{2}$, and

$$v_3 - v_0 = \rho(e^{7\pi i/8} - e^{\pi i/8}) = -2\rho \sin(3\pi/8).$$

The positive quantities in (2.12) satisfy

$$\frac{B}{A} = \sqrt{\frac{A^2 - 1}{A^2}} = \tanh(\operatorname{arcosh} A) = \frac{2\rho \sin(3\pi/8)}{1 + \rho^2},$$

where the last equality follows after squaring both positive sides and using $\rho^2 = 1/\sqrt{2}$ and $4\sin^2(3\pi/8) = 2 + \sqrt{2}$. This proves (2.12). The calculation for $v_4 \mapsto v_7$ has the same real difference: $v_4 v_7 = -\rho^2$ and $v_4 - v_7 = -2\rho \sin(3\pi/8)$. Hence $g_0(v_4) = v_7$. Rotating both identities by (2.11) gives (2.10). A Möbius map takes the geodesic through the first endpoint pair to the geodesic through the second; the reversed endpoint order proves reversed boundary orientation. $\square$

2.4 The Poincaré polygon theorem and the vertex cycle

We use the following two-dimensional form of the Poincaré polygon theorem. It is the compact, finite-vertex specialization of the fundamental polyhedron theorem in [2]; compatible accounts for Fuchsian groups and fundamental polygons appear in [1, 3].

Theorem 2.4.1 (Poincaré polygon theorem, compact surface form). *Let P be a compact convex finite-sided polygon in $\mathbb{H}_R^2$. Suppose its sides are paired by orientation-preserving isometries so that:*

- (a) *every side is paired with exactly one distinct side, and the reverse pairing is the inverse isometry;*
- (b) *a pairing maps the paired side onto its mate with reversed boundary orientation and maps the exterior half-plane onto the interior half-plane adjacent to the mate;*
- (c) *every vertex equivalence class is a finite cycle, its cycle transformation is the identity, and the sum of the incident polygon angles in that cycle is 2π .*

Then the pairing group is discrete, the translates of P cover $\mathbb{H}_R^2$ with disjoint interiors, and P is a fundamental polygon. The quotient has no cone singularity at the vertex cycles. Its group presentation has the side inverse relations and the vertex-cycle relations.

The half-plane clause in (b) is not an extra numerical assumption here. Both paired sides are supporting geodesics of the convex polygon. An orientation-preserving isometry that reverses their boundary orientations takes the inward normal of the source to the outward normal at the mate; hence it takes the source exterior to the mate interior. We now verify the remaining hypotheses from the eight physical vertices, without double counting them.

Lemma 2.4.2 (Actual vertex cycle). *The four opposite-side pairings put all and only the eight vertices $v_0, \dots, v_7$ into one equivalence class. One traversal is*

$$\begin{array}{c|cccccccc} \text{vertex} & v_0 & v_3 & v_6 & v_1 & v_4 & v_7 & v_2 & v_5 & v_0 \\ \text{map} & g_4 & g_7 & g_2 & g_5 & g_0 & g_3 & g_6 & g_1 & - \end{array} \quad (2.13)$$

and the corresponding cycle transformation is

$$W = g_1 g_6 g_3 g_0 g_5 g_2 g_7 g_4 = 1 \quad \text{in } \text{PSU}(1, 1). \quad (2.14)$$

Proof. Each arrow in (2.13) is an instance of (2.10) or its inverse:

$$\begin{aligned} g_4(v_0) &= v_3, & g_7(v_3) &= v_6, & g_2(v_6) &= v_1, & g_5(v_1) &= v_4, \\ g_0(v_4) &= v_7, & g_3(v_7) &= v_2, & g_6(v_2) &= v_5, & g_1(v_5) &= v_0. \end{aligned}$$

The intermediate vertices are 0, 3, 6, 1, 4, 7, 2, 5 modulo eight, hence are pairwise distinct; the cycle contains exactly the eight physical vertices.

For completeness, the cycle word is checked at the matrix level rather than inferred from the desired quotient. Multiplication of two matrices (2.8) uses

$$G_j G_k = \begin{pmatrix} A^2 + B^2 \zeta^{j-k} & AB(\zeta^j + \zeta^k) \\ AB(\zeta^{-j} + \zeta^{-k}) & A^2 + B^2 \zeta^{k-j} \end{pmatrix}. \quad (2.15)$$

Applying (2.15) successively in the order 4, 7, 2, 5, 0, 3, 6, 1, and reducing after each multiplication with

$$\zeta^2 = i, \quad \zeta^4 = -1, \quad A = 1 + \sqrt{2}, \quad B^2 = 2A, \quad A^2 - B^2 = 1,$$

gives

$$G_1 G_6 G_3 G_0 G_5 G_2 G_7 G_4 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (2.16)$$

Equation (2.16) proves (2.14) independently of the polygon theorem. $\square$

Theorem 2.4.3 (Correct polygon-theorem application). *The group*

$$\Gamma_B = \langle g_0, g_1, g_2, g_3 \rangle \leq \text{PSU}(1, 1), \quad g_{j+4} = g_j^{-1}, \quad (2.17)$$

is discrete, cocompact, torsion-free, and acts freely; P is its fundamental polygon. The quotient $\Gamma_B \backslash \mathbb{H}_R^2$ is a smooth hyperbolic surface.

Proof. The polygon is compact and convex by the first subsection. Equations (2.9)–(2.10) prove the side-pairing, inverse, endpoint, orientation, and half-plane requirements. Lemma (2.13)–(2.16) proves that there is one vertex cycle and that its transformation is the identity. The angle sum is the sum over the eight physical corners:

$$\underbrace{\frac{\pi}{4} + \dots + \frac{\pi}{4}}_{\text{eight terms}} = 8 \frac{\pi}{4} = 2\pi. \quad (2.18)$$

This is the required cycle-angle equality. In particular, neither sixteen corners nor any division by four occurs. Every hypothesis of the stated Poincaré polygon theorem has now been checked, so discreteness, tiling, and the fundamental-domain conclusion follow.

The quotient is compact because its fundamental polygon is compact. Interior points have trivial stabilizer because distinct fundamental-domain interiors are disjoint. Let x lie in the relative interior of s_j and suppose $hx = x$. The isometry h permutes the exactly two tiles incident to x . If $hP = P$, equality of fundamental tiles and disjoint interiors forces $h = 1$. Otherwise tile-label uniqueness gives $hP = g_j P$, hence $h = g_j$. But $x = g_j x$ and $x \in s_j$ would imply

$x = g_j^{-1}x \in g_j^{-1}s_j = s_{j+4}$, impossible because opposite sides of the compact convex octagon are disjoint. Thus side-interior stabilizers are also trivial. The only possible vertex stabilizer has order $2\pi/(2\pi) = 1$ by the verified cycle angle and identity cycle transformation. Every finite-order nonidentity element of a discrete orientation-preserving hyperbolic group is elliptic and fixes an interior point. Since all point stabilizers are trivial, Γ_B is torsion-free, and the quotient is a surface rather than an orbifold. $\square$

2.5 The genus-two Bolza surface

Definition 2.5.1. The Bolza surface in this normalization is

$$\Sigma_B = \Gamma_B \backslash \mathbb{H}_R^2$$

with Γ_B and P constructed above.

The quotient metric is well defined: if $x' = \gamma x$ and $v' = d\gamma_x v$, then $g_R(v', v') = g_R(v, v)$ because every $\gamma \in \Gamma_B$ is an isometry. Freeness gives manifold charts and proper discontinuity makes the quotient Hausdorff. The orientation also descends because every generator lies in $\text{PSU}(1, 1)$. Thus the definition constructs a closed oriented Riemannian surface, not merely a topological edge quotient. The name *Bolza surface* refers here to this marked regular-octagon construction. No claim that an arbitrary genus-two hyperbolic surface is isometric to it is made.

Proposition 2.5.2 (Topology and area). *The surface Σ_B is closed, oriented, and has genus 2. Its area is*

$$\text{area}(\Sigma_B) = 4\pi R^2. \quad (2.19)$$

Proof. There is one face. The eight sides form four paired edges. Lemma (2.13) shows that all eight vertices yield one quotient vertex. Hence

$$\chi(\Sigma_B) = V - E + F = 1 - 4 + 1 = -2.$$

For a closed orientable surface, $\chi = 2 - 2g$, so $-2 = 2 - 2g$ and $g = 2$. Smoothness and compactness were proved above.

The hyperbolic polygon area formula gives

$$\text{area}(P) = R^2 \left((8 - 2)\pi - \sum_{k=0}^7 \frac{\pi}{4} \right) = R^2(6\pi - 2\pi) = 4\pi R^2.$$

Because boundary identifications affect a measure-zero set, this equals the quotient area. Independently, Gauss–Bonnet gives

$$-R^{-2} \text{area}(\Sigma_B) = 2\pi\chi(\Sigma_B) = -4\pi,$$

which yields the same value and supplies a consistency check. $\square$

Corollary 2.5.3. *The marked surface is unique up to orientation-preserving isometry once the centered regular octagon and the side pairings (2.8) are fixed up to global conjugacy.*

Proof. An oriented isometry between two marked octagons conjugates each prescribed side pairing to its mate by the endpoint uniqueness in (2.10). It therefore conjugates the two pairing groups and descends to an isometry of quotients. This is uniqueness of the construction data; it is neither a moduli-space minimality nor an optimality assertion. $\square$

2.6 Tessellation and the orbit of centers

The polygon theorem supplies a tessellation

$$\mathcal{T} = \{\gamma P : \gamma \in \Gamma_B\}.$$

Define the center orbit

$$\Lambda_B = \Gamma_B o = \{\gamma o : \gamma \in \Gamma_B\}.$$

Proposition 2.6.1. *The map $\Gamma_B \rightarrow \Lambda_B$, $\gamma \mapsto \gamma o$, is a bijection. Moreover γP and $\gamma g_j P$ share the side γs_j , and every side-adjacent pair arises uniquely in this way.*

Proof. Surjectivity is the definition of the orbit. If $\gamma o = \eta o$, then $\eta^{-1}\gamma$ fixes an interior point. A nonidentity orientation-preserving hyperbolic isometry fixing an interior point is elliptic, while Γ_B is torsion-free by the polygon-cycle analysis. Thus $\eta^{-1}\gamma = 1$ and $\gamma = \eta$, proving injectivity.

At the base tile, g_j maps s_{j+4} to s_j , so P and $g_j P$ lie on opposite sides of their common edge s_j . Applying γ shows that γP and $\gamma g_j P$ share γs_j . Conversely, the fundamental tessellation has exactly one adjacent tile across each side, and the side-pairing identifies it as $g_j P$. Distinct j give distinct sides, proving uniqueness. This also fixes the convention: the group acts on points on the left, while tile labels change by right multiplication. $\square$

2.7 The eight physical nearest-neighbor directions

Definition 2.7.1. The ordered physical shell is

$$S_8^{\text{geo}} = (g_0, g_1, \dots, g_7), \quad g_{j+4} = g_j^{-1}. \quad (2.20)$$

It is the geometric shell; the standard surface-presentation alphabet.

Proposition 2.7.2. *The eight elements in S_8^{geo} are distinct, inversion closed, and all move the center through the same distance a_B . More precisely,*

$$g_j o = \frac{B}{A} \zeta^j, \quad d_R(o, g_j o) = 2R \operatorname{artanh} \frac{B}{A} = 2R \operatorname{arcosh} A = a_B. \quad (2.21)$$

Consequently the center-orbit adjacency graph is simple and 8-regular.

Proof. Putting $z = 0$ in (1.14) gives the first identity. Since $B^2 = A^2 - 1$,

$$\frac{B}{A} = \sqrt{1 - A^{-2}} = \tanh(\operatorname{arcosh} A).$$

Equation (1.7) gives (2.21), and (2.3)–(2.4) identify the result with the side length. The arguments of the eight nonzero disk points are $j\pi/4$, so the images $g_j o$, and therefore the group elements g_j , are pairwise distinct. Equation (2.9) proves inversion closure. The tessellation proposition supplies exactly one neighbor for each side, giving degree eight and excluding loops or multiple edges. $\square$

2.8 Fundamental domains, covers, and quotient surfaces

Let $K \leq \Gamma_B$. Since Γ_B acts freely and properly discontinuously, so does K , and the orbit map induces

$$p_K : K \backslash \mathbb{H}_R^2 \longrightarrow \Gamma_B \backslash \mathbb{H}_R^2, \quad Kx \longmapsto \Gamma_B x. \quad (2.22)$$

Theorem 2.8.1 (Covering specification). *The map (2.22) is well defined and is a covering of degree $[\Gamma_B : K]$, with infinite degree allowed. It is a regular covering if and only if $K \triangleleft \Gamma_B$. In the normal finite-index case its deck group is canonically Γ_B/K , and*

$$\operatorname{area}(K \backslash \mathbb{H}_R^2) = 4\pi R^2 [\Gamma_B : K], \quad (2.23)$$

$$g(K \backslash \mathbb{H}_R^2) = 1 + [\Gamma_B : K]. \quad (2.24)$$

Proof. If $Kx = Ky$, then $y = kx$ for some $k \in K \subset \Gamma_B$, so $\Gamma_B y = \Gamma_B x$; this is the required coset independence. A sufficiently small metric ball U has disjoint translates under every nonidentity element of Γ_B . Its images in both quotients are evenly covered, proving the covering property. Fibers correspond to left cosets of K in Γ_B , so their cardinality is the index.

If K is normal, $\gamma Kx = K\gamma x$ is independent of the representative of γK , and it defines a deck transformation. Its kernel is K , giving Γ_B/K . Conversely, regularity makes deck transformations act transitively on a fiber; the standard lifting correspondence then forces every conjugate $\gamma K\gamma^{-1}$ to equal K . Thus normality is necessary and sufficient.

For finite index n , a fundamental region consists of n translates of P , proving (2.23). Euler characteristic multiplies under an n -sheeted finite cover:

$$2 - 2g(K \backslash \mathbb{H}_R^2) = n(2 - 2 \cdot 2) = -2n,$$

which rearranges to (2.24). Since every subgroup of the torsion-free group Γ_B is torsion-free, no orbifold exception occurs. $\square$

3 Finite regions in the group and in the tessellation

Fix the ordered symmetric geometric generating set S_8^{geo} from (2.20). All inequalities below use the closed cutoff convention.

3.1 Word balls

Definition 3.1.1. For $\gamma \in \Gamma_B$, its word length and radius- n word ball are

$$|\gamma|_{S_8^{\text{geo}}} = \min\{m : \gamma = s_1 \cdots s_m, s_i \in S_8^{\text{geo}}\}, \quad B_{\text{word}}(n) = \{\gamma : |\gamma|_{S_8^{\text{geo}}} \leq n\}. \quad (3.1)$$

The empty product represents 1, so $|1|_{S_8^{\text{geo}}} = 0$.

Proposition 3.1.2 (Finiteness and boundary values). *For every integer $n \geq 0$, $B_{\text{word}}(n)$ is finite and*

$$|B_{\text{word}}(n)| \leq 1 + 8 \sum_{k=0}^{n-1} 7^k = 1 + \frac{8(7^n - 1)}{6}. \quad (3.2)$$

For $n < 0$ the ball is empty. Equality in (3.2) is not asserted because surface relations can identify different reduced words.

Proof. A nonempty freely reduced word has eight choices for its first letter. Each later letter has at most seven choices because the inverse of the preceding letter would cancel. Thus there are at most $8 \cdot 7^{m-1}$ reduced words of length m . Every group element of length at most n is represented by one of these words, although more than one word may represent it. Summing over $m = 1, \dots, n$ and adjoining the empty word gives (3.2). The minimum in (3.1) exists because word lengths are nonnegative integers and S_8^{geo} generates the side-pairing group. $\square$

3.2 Geometric orbit balls

Definition 3.2.1. The centered geometric orbit ball at radius $D \geq 0$ is

$$B_{\text{geo}}(D) = \{\gamma \in \Gamma_B : d_R(o, \gamma o) \leq D\}. \quad (3.3)$$

It is a subset of the group, not the continuum metric ball $\overline{B}_{\mathbb{H}_R^2}(o, D) \subset \mathbb{H}_R^2$.

Theorem 3.2.2. *Every $B_{\text{geo}}(D)$ is finite. The closed inequality in (3.3) includes every orbit point exactly on the boundary.*

Proof. The closed hyperbolic ball $\overline{B}_{\mathbb{H}_R^2}(o, D)$ is compact because $\mathbb{H}_R^2$ is complete and finite dimensional. The orbit $\Gamma_B o$ is discrete: proper discontinuity supplies a neighborhood U_x at every x meeting only finitely many translates of a compact neighborhood of o . If infinitely many distinct orbit points lay in the compact ball, they would have an accumulation point there. A sufficiently small neighborhood of that point would then meet infinitely many translates, contradicting proper discontinuity. The orbit map is injective by the torsion-free argument in Statement 2.6, so finiteness of orbit points is finiteness of group elements. $\square$

An equivalent tessellation description is useful later:

$$d_R(o, \gamma o) \leq D \iff \text{the center of the tile } \gamma P \text{ lies in } \overline{B}_{\mathbb{H}_R^2}(o, D). \quad (3.4)$$

3.3 Based displacement balls

For an arbitrary base point $x \in \mathbb{H}_R^2$, define

$$B_x(D) = \{\gamma \in \Gamma_B : d_R(x, \gamma x) \leq D\}. \quad (3.5)$$

The proof for (3.3), with x in place of o , proves that (3.5) is finite.

Lemma 3.3.1 (Change of base point). *For all $x, y \in \mathbb{H}_R^2$ and $\gamma \in \Gamma_B$,*

$$|d_R(x, \gamma x) - d_R(y, \gamma y)| \leq 2d_R(x, y). \quad (3.6)$$

Consequently

$$B_x(D) \subseteq B_y(D + 2d_R(x, y)) \quad (3.7)$$

and the reverse inclusion holds after exchanging x, y .

Proof. The triangle inequality and isometry invariance give

$$d_R(x, \gamma x) \leq d_R(x, y) + d_R(y, \gamma y) + d_R(\gamma y, \gamma x) = d_R(y, \gamma y) + 2d_R(x, y).$$

Exchanging x and y yields the opposite inequality and hence (3.6). If the left displacement is at most D , (3.6) gives (3.7). $\square$

Based displacement is not conjugacy invariant. Indeed,

$$d_R(x, h\gamma h^{-1}x) = d_R(h^{-1}x, \gamma h^{-1}x),$$

which changes the base point from x to $h^{-1}x$. It is therefore incorrect to use a finite based ball as if it were already a finite set of short conjugacy classes. Its valid role is to certify geometry near one chosen lifted point.

3.4 Translation-length balls and finite conjugacy representatives

Define the raw translation-length set

$$T(L) = \{\gamma \in \Gamma_B : \ell_{\mathbb{H}}(\gamma) \leq L\}. \quad (3.8)$$

Translation length is conjugacy invariant by (1.19). Unless $\gamma = 1$, the conjugacy class of a surface-group element is infinite, so $T(L)$ is generally infinite as soon as it contains a nontrivial element. The finite object needed for a global systole test is the quotient by conjugacy.

Theorem 3.4.1 (Finitely many short conjugacy classes). *For every $L \geq 0$, the set of conjugacy classes*

$$\mathcal{C}(L) = \{[\gamma]_{\text{conj}} : \ell_{\mathbb{H}}(\gamma) \leq L\} \quad (3.9)$$

is finite. More precisely, let

$$D_P = \max_{x \in P} d_R(o, x) = r_{\text{out}}.$$

Every class in $\mathcal{C}(L)$ has a representative γ' satisfying

$$d_R(o, \gamma'o) \leq L + 2D_P. \quad (3.10)$$

Proof. The identity class is harmless. Let $\gamma \neq 1$. Since Γ_B is torsion-free, γ is not elliptic; since the quotient is compact, it has a positive uniform injectivity radius. To see this without a cusp classification, cover the compact quotient by finitely many evenly covered metric balls and take the positive minimum of their admissible radii. No nonidentity deck transformation can move any lifted center by less than twice that minimum. A parabolic element has translation-length infimum zero by Statement 1.6, contradicting this uniform lower bound. Thus γ is hyperbolic and has an axis $\mathcal{A}_\gamma$. Choose $x \in \mathcal{A}_\gamma$. The tessellation covers $\mathbb{H}_R^2$, so $x \in \eta P$ for some $\eta \in \Gamma_B$. Set $\gamma' = \eta^{-1}\gamma\eta$ and $y = \eta^{-1}x \in P$. Then y lies on the axis of γ' , and

$$\begin{aligned} d_R(o, \gamma'o) &\leq d_R(o, y) + d_R(y, \gamma'y) + d_R(\gamma'y, \gamma'o) \\ &= 2d_R(o, y) + \ell_{\mathbb{H}}(\gamma') \leq 2D_P + L. \end{aligned}$$

This proves (3.10). The map from the finite based ball $B_{\text{geo}}(L + 2D_P)$ to conjugacy classes is therefore surjective onto $\mathcal{C}(L)$. A surjective image of a finite set is finite, so (3.9) follows without an arbitrary representative choice. $\square$

This proof also explains the necessary distinction: the raw set (3.8) can be infinite, but a bounded based ball contains a complete finite choice of short conjugacy representatives after the axis has been moved to meet the fixed fundamental polygon.

3.5 Comparison of word and geometric metrics

Lemma 3.5.1 (One-sided Lipschitz estimate). *For every $\gamma \in \Gamma_B$,*

$$d_R(o, \gamma o) \leq a_B |\gamma|_{S_8^{\text{geo}}}. \quad (3.11)$$

Proof. Choose a shortest word $\gamma = s_1 \cdots s_m$, where $m = |\gamma|_{S_8^{\text{geo}}}$. Insert the orbit points of its prefixes and apply the triangle inequality:

$$\begin{aligned} d_R(o, \gamma o) &\leq \sum_{k=1}^m d_R(s_1 \cdots s_{k-1} o, s_1 \cdots s_k o) \\ &= \sum_{k=1}^m d_R(o, s_k o) = m a_B. \end{aligned}$$

The second line uses isometry invariance, and the last uses (2.21). $\square$

The constant in this one-sided estimate is geometrically sharp for the chosen shell: if $\gamma = g_j$, then $|g_j|_{S_8^{\text{geo}}} = 1$ and $d_R(o, g_j o) = a_B$, so equality holds. This does not make the whole inequality an equality. For a product, distinct steps can bend toward one another, and a defining relation can return a nonempty written path to the identity. In particular, written length and minimal word length must also be distinguished before (3.11) is applied.

Corollary 3.5.2. *For every $n \geq 0$,*

$$B_{\text{word}}(n) \subseteq B_{\text{geo}}(a_B n).$$

The converse inclusion requires an additive as well as multiplicative constant and is supplied by the next theorem.

Proof. If $|\gamma|_{S_8^{\text{geo}}} \leq n$, (3.11) gives $d_R(o, \gamma o) \leq a_B n$, which is precisely membership in the displayed geometric ball. No reverse inference has been used. $\square$

The converse cannot be obtained by reversing a triangle inequality. Cancellation and the surface relator can make a long written word represent an element of short minimal word length, while a geodesic can pass close to tile vertices and cross many sides over a short interval. A valid converse uses cocompactness and a finite auxiliary set, as proved next.

3.6 Effective quasi-isometry constants

Let $D_P = r_{\text{out}}$ and define the finite set

$$T = \{\eta \in \Gamma_B : d_R(o, \eta o) \leq 2D_P + R\}. \quad (3.12)$$

Because T is finite and S_8^{geo} generates Γ_B , the integer

$$M = \max_{\eta \in T} |\eta|_{S_8^{\text{geo}}} \quad (3.13)$$

exists. It is positive: $d_R(o, g_0 o) = a_B = 2r_{\text{in}}$, while $r_{\text{in}} < r_{\text{out}} = D_P$, so $a_B < 2D_P + R$ and $g_0 \in T$. Hence $M \geq 1$, justifying every division by M below. The finite tile-flood of the final subsection, followed by a Cayley breadth-first search, computes T and the maximum (3.13); no unproved optimality claim is attached to the resulting constant.

Theorem 3.6.1 (Orbit-map quasi-isometry). *For every $\gamma \in \Gamma_B$,*

$$\frac{R}{M} |\gamma|_{S_8^{\text{geo}}} - R \leq d_R(o, \gamma o) \leq a_B |\gamma|_{S_8^{\text{geo}}}. \quad (3.14)$$

Thus the orbit map from the Cayley graph to $\mathbb{H}_R^2$ is a quasi-isometric embedding with displayed constants; cocompactness makes it coarsely surjective with constant D_P .

Proof. Only the first inequality remains. Parameterize the geodesic from o to γo by arclength and choose points $y_0, \dots, y_m$ in order, with $y_0 = o$, $y_m = \gamma o$, successive distances at most R , and

$$m \leq \frac{d_R(o, \gamma o)}{R} + 1. \quad (3.15)$$

Choose h_k so that $y_k \in h_k P$, taking $h_0 = 1$ and $h_m = \gamma$. Because each tile center is at distance at most D_P from any point of its tile,

$$\begin{aligned} d_R(o, h_k^{-1} h_{k+1} o) &= d_R(h_k o, h_{k+1} o) \\ &\leq d_R(h_k o, y_k) + d_R(y_k, y_{k+1}) + d_R(y_{k+1}, h_{k+1} o) \\ &\leq 2D_P + R. \end{aligned}$$

Hence $t_k = h_k^{-1} h_{k+1} \in T$, and the telescoping product gives

$$\gamma = t_0 t_1 \cdots t_{m-1}.$$

By (3.13),

$$|\gamma|_{S_8^{\text{geo}}} \leq M m \leq M \left(\frac{d_R(o, \gamma o)}{R} + 1 \right).$$

Rearranging proves the lower inequality in (3.14). Every $x \in \mathbb{H}_R^2$ lies in some γP , so $d_R(x, \gamma o) \leq D_P$; this proves coarse surjectivity. $\square$

The constants are explicit once the finite set (3.12) is enumerated. They are not claimed optimal. Existence, explicit computability, and optimality are separate properties.

3.7 Exact enumeration of a geometric ball

For a cutoff $D \geq 0$, (3.14) implies

$$\gamma \in B_{\text{geo}}(D) \implies |\gamma|_{S_8^{\text{geo}}} \leq N(D) := \left\lfloor M \left(\frac{D}{R} + 1 \right) \right\rfloor. \quad (3.16)$$

Theorem 3.7.1 (Finite exact enumeration with a decidable cutoff). *Assume that, for every candidate in the finite word ball (3.16), the predicate $d_R(o, \gamma o) \leq D$ is decidable. This holds, for example, when the threshold $C_D = \cosh(D/R)$, rather than merely D , is supplied as a real algebraic number in an effectively ordered extension of the matrix coefficient field, or when the input supplies a certified comparison oracle. Enumerate every word of length at most $N(D)$, multiply its exact $\text{SU}(1, 1)$ matrices, identify G with $-G$, remove duplicate projective matrices, and retain precisely those satisfying $d_R(o, \gamma o) \leq D$. The output is exactly $B_{\text{geo}}(D)$.*

Proof. Every enumerated word represents an element of Γ_B ; the final distance predicate proves soundness. Conversely, (3.16) gives every member of the target ball a word of enumerated length, so its exact projective matrix occurs before duplicate removal. The equality criterion in $\text{PSU}(1, 1)$ is $G = H$ or $G = -H$, so duplicate removal neither loses nor invents group elements. The closed comparison retains boundary elements. These observations prove completeness.

Equation (1.8) makes the test a comparison between the algebraic matrix quantity and C_D . If C_D is real algebraic, the comparison is an exact sign test in the ordered compositum of its field with the matrix coefficient field. The weaker condition that D itself is algebraic is insufficient: for example, $D/R = 1$ is algebraic while $\cosh 1$ is transcendental. Without the stated decidability hypothesis, a noncomputable real cutoff cannot support an exact terminating algorithm. Certified interval arithmetic may instead produce three sets: included, excluded, and undecided. An interval containing a boundary value must be refined or remain undecided, never rounded silently. $\square$

As exact tests, $B_{\text{geo}}(0) = \{1\}$ because the action is free, and

$$\{1, g_0, \dots, g_7\} \subseteq B_{\text{geo}}(a_B).$$

For any $0 \leq D < a_B$, no tile center adjacent to o is included. The statement does not assert that no more distant word has smaller displacement; the enumeration, rather than that intuition, decides the full ball.

3.8 Completeness of tile flooding

Define

$$\mathcal{F}(D) = \{\gamma \in \Gamma_B : \gamma P \cap \overline{B}_{\mathbb{H}_R^2}(o, D) \neq \emptyset\}. \quad (3.17)$$

A tile-flood starts with P , maintains a visited set of tile labels, tests each unvisited neighbor across all eight sides, accepts exactly those satisfying (3.17), and repeats from every newly accepted tile. Thus each tile is enqueued at most once. If the metric ball meets a tessellation vertex, all tiles incident to that vertex are included by the closed intersection test.

Theorem 3.8.1 (Tile-flood termination and completeness). *With an exact predicate for (3.17), the flood terminates after finitely many tiles and returns exactly $\mathcal{F}(D)$. It contains every tile whose center belongs to $B_{\text{geo}}(D)$, and every returned tile center lies in $B_{\text{geo}}(D + D_P)$.*

Proof. If $x \in \gamma P \cap \overline{B}(o, D)$, then

$$d_R(o, \gamma o) \leq d_R(o, x) + d_R(x, \gamma o) \leq D + D_P.$$

Thus $\mathcal{F}(D)$ injects into the finite based ball $B_{\text{geo}}(D + D_P)$, proving finiteness and eventual termination.

To prove reachability, take $x \in \gamma P \cap \overline{B}(o, D)$ and the geodesic segment $[o, x]$. Convexity of the metric ball keeps the entire segment inside $\overline{B}(o, D)$. The closed tiles meeting the segment form a finite side-connected subcomplex from P to γP . If the segment runs along an edge, include the tiles on both sides. If it passes through a vertex, order all incident tiles around that vertex; consecutive tiles share a side, and every tile contains the same point of the segment. Thus every inserted tile passes the closed test, and the flood reaches γP . Soundness is immediate because it accepts only tiles passing (3.17), so its output is exactly $\mathcal{F}(D)$.

If $d_R(o, \gamma o) \leq D$, then the center γo itself lies in both γP and the metric ball, proving the first containment. The second is the displayed estimate at the start of the proof. $\square$

Tile flooding and word enumeration are therefore two independently complete finite procedures. Their output sets are not identical: one selects tiles intersecting a continuum ball, while the other selects centers inside a radius. Equations (3.17) and the two radius inclusions specify their exact interface.

Part II

Exact algebra and eightfold symmetry

4 The Bolza surface group

4.1 The standard genus-two presentation

Choose the quotient vertex as basepoint and choose a symplectic free basis a_1, b_1, a_2, b_2 of the fundamental group of the one-skeleton. These basis loops may be composites of the four paired-edge loops; they are not being identified term by term with those four loops. With the handle-adapted orientations, the single two-cell is attached along the word

$$\mathcal{R} = [a_1, b_1][a_2, b_2], \quad [x, y] = xyx^{-1}y^{-1}. \quad (4.1)$$

Theorem 4.1.1 (Presentation from the polygonal cell structure). *The fundamental group of the Bolza surface has presentation*

$$\pi_1(\Sigma_B) \cong \langle a_1, b_1, a_2, b_2 \mid [a_1, b_1][a_2, b_2] = 1 \rangle. \quad (4.2)$$

Proof. The quotient cell decomposition has one vertex, four oriented one-cells, and one two-cell by Statement 2.5. Its one-skeleton is therefore a wedge of four circles and has free fundamental group of rank four. Since the quotient is the closed oriented genus-two surface, choose the usual two-handle cut system based at the vertex. Its four oriented loops form a free basis of the one-skeleton; cutting the two handles along them reads successively $a_1, b_1, a_1^{-1}, b_1^{-1}$ and $a_2, b_2, a_2^{-1}, b_2^{-1}$. Hence the attaching word is (4.1). This is a free-basis change from the individual paired-edge loops, not merely a permutation or reversal of them. Van Kampen's theorem states that attaching this disk quotients the free group by the normal closure of its boundary word:

$$\pi_1(\Sigma_B) = F(a_1, b_1, a_2, b_2) / \langle\langle \mathcal{R} \rangle\rangle.$$

This is exactly (4.2). A different base edge or cyclic starting point conjugates or cyclically permutes the relator and gives an isomorphic presentation; it does not alter the fixed convention used below. $\square$

From now on Γ_B denotes both the side-pairing deck group and the abstract group (4.2), once the explicit isomorphism is proved in Statement 4.7.

4.2 Oriented geometric side-pairing generators

The unsigned tuple (g_0, g_1, g_2, g_3) records four opposite-side pairings, but the oriented boundary traversal alternates two of their directions. Define

$$h_0 = g_0, \quad h_1 = g_1^{-1}, \quad h_2 = g_2, \quad h_3 = g_3^{-1}. \quad (4.3)$$

Proposition 4.2.1 (Boundary word). *The oriented octagon boundary relation is*

$$\mathcal{B} = h_0 h_1 h_2 h_3 h_0^{-1} h_1^{-1} h_2^{-1} h_3^{-1} = 1. \quad (4.4)$$

Proof. Start at v_7 , traverse s_0 toward v_0 , and continue around the polygon boundary counter-clockwise. Each side label occurs once, and its paired opposite side is traversed in the reverse induced orientation. With the first four labels chosen as (4.3), the eight directed sides read

$$h_0, h_1, h_2, h_3, h_0^{-1}, h_1^{-1}, h_2^{-1}, h_3^{-1}.$$

The Poincaré polygon theorem gives one relation for the unique vertex cycle. To connect this boundary word to the cycle word actually multiplied in (2.16), put

$$X = G_1 G_6 G_3, \quad W = G_1 G_6 G_3 G_0 G_5 G_2 G_7 G_4.$$

Using $H_0 = G_0, H_1 = G_5, H_2 = G_2, H_3 = G_7$, the coherent lift of $\mathcal{B}$ is

$$\begin{aligned} M(\mathcal{B}) &= G_0 G_5 G_2 G_7 G_4 G_1 G_6 G_3 \\ &= X^{-1} W X = I, \end{aligned}$$

where $W = I$ by (2.16). Thus (4.4) holds already for the chosen $SU(1, 1)$ lifts. Starting instead at v_0 would read the cyclic conjugate $h_0^{-1} \mathcal{B} h_0$, not the displayed word itself. $\square$

The physical shell remains $(g_0, \dots, g_7)$. The tuple $(h_0, \dots, h_3)$ is a coordinate basis for the polygon relation, not a replacement for all eight directed neighbors.

4.3 Standard versus geometric generators

Definition 4.3.1 (Two alphabets). Let

$$\mathcal{A}_{\text{geo}} = \{h_0^{\pm 1}, h_1^{\pm 1}, h_2^{\pm 1}, h_3^{\pm 1}\}, \quad \mathcal{A}_{\text{pres}} = \{a_1^{\pm 1}, b_1^{\pm 1}, a_2^{\pm 1}, b_2^{\pm 1}\}. \quad (4.5)$$

A word carries its alphabet as part of its data.

The change of coordinates is

$$a_1 = h_0, \quad b_1 = h_1, \quad a_2 = h_1 h_0 h_2, \quad b_2 = h_3 h_0^{-1} h_1^{-1}. \quad (4.6)$$

Consequently a_2 and b_2 have displayed geometric words of length three. Their word-metric minimality is independent of the construction below.

Proposition 4.3.2 (Nonidentity of the shells). *The presentation alphabet and physical shell are not interchangeable. Equation (4.6) is a group-coordinate dictionary, whereas physical nearest-neighbor adjacency is defined only by $S_8^{\text{geo}} = (g_0, \dots, g_7)$.*

Proof. The physical shell contains the eight transformations taking the center to the eight centers of side-adjacent tiles, all at distance a_B . The lifts of all g_j have real upper-left entry A , and those of $g_j^{-1} = g_{j+4}$ have the same upper-left entry. Direct multiplication of (2.8) according to (4.6), using $A^2 - B^2 = 1$, gives

$$(A_2)_{11} = -A - 2iA^2, \quad (B_2)_{11} = -A + 2iA.$$

The upper-left entries of their inverses are the complex conjugates of these two numbers. Since $A > 0$, none is real, and therefore none of $a_2^{\pm 1}, b_2^{\pm 1}$ equals any g_j projectively: equality could only mean equality with G_j or $-G_j$, both having real upper-left entry. On the other hand,

$$\{a_1^{\pm 1}, b_1^{\pm 1}\} = \{g_0, g_4, g_5, g_1\}.$$

Thus the signed presentation shell and physical shell are genuinely different sets. More importantly, their ordered roles differ: a relation calculated in presentation coordinates may be transported through (4.6), but an adjacency operator must continue to use S_8^{geo} . $\square$

4.4 Equivalence of the boundary and surface relations

Theorem 4.4.1. *Under the substitutions (4.6),*

$$[a_1, b_1][a_2, b_2] = h_0 h_1 h_2 h_3 h_0^{-1} h_1^{-1} h_2^{-1} h_3^{-1}. \quad (4.7)$$

Hence the standard surface relator holds in the exact side-pairing group.

Proof. The first commutator is

$$[a_1, b_1] = h_0 h_1 h_0^{-1} h_1^{-1}. \quad (4.8)$$

For $c = h_1 h_0 h_2$ and $d = h_3 h_0^{-1} h_1^{-1}$,

$$c^{-1} = h_2^{-1} h_0^{-1} h_1^{-1}, \quad d^{-1} = h_1 h_0 h_3^{-1}.$$

Therefore

$$\begin{aligned} [c, d] &= h_1 h_0 h_2 h_3 h_0^{-1} h_1^{-1} h_2^{-1} h_0^{-1} h_1^{-1} h_1 h_0 h_3^{-1} \\ &\longrightarrow h_1 h_0 h_2 h_3 h_0^{-1} h_1^{-1} h_2^{-1} h_3^{-1}, \end{aligned} \quad (4.9)$$

where the only reductions are the adjacent pairs $h_1^{-1} h_1$ and $h_0^{-1} h_0$. Multiplying (4.8) and (4.9) gives

$$\begin{aligned} [a_1, b_1][a_2, b_2] &= h_0 h_1 h_0^{-1} h_1^{-1} h_1 h_0 h_2 h_3 h_0^{-1} h_1^{-1} h_2^{-1} h_3^{-1} \\ &\longrightarrow h_0 h_1 h_2 h_3 h_0^{-1} h_1^{-1} h_2^{-1} h_3^{-1}, \end{aligned}$$

after cancelling $h_1^{-1} h_1$ and $h_0^{-1} h_0$. This is (4.7), and (4.4) makes it the identity. The converse substitution is proved separately in Statement 6.5–Statement 6.6. $\square$

4.5 Reduced words and exact normal forms

A word is freely reduced if it contains no adjacent ss^{-1} . Free reduction terminates because every cancellation lowers length by two, and its result in a free group is unique. It is not a normal-form algorithm for the surface group: the freely reduced relator $\mathcal{R}$ represents the identity in Γ_B .

Fix a total order on the eight signed presentation letters and order all finite words first by length and then lexicographically.

Definition 4.5.1 (Shortlex group normal form). For $\gamma \in \Gamma_B$, define $\text{nf}(\gamma)$ to be the least word in shortlex order representing γ .

Equivalently, with $\bar{w}$ denoting the represented group element,

$$\text{nf}(\gamma) = \min_{\text{shortlex}} \{w \in \mathcal{A}_{\text{pres}}^* : \bar{w} = \gamma\}. \quad (4.11)$$

The candidate set is nonempty, but it is usually infinite; shortlex well-ordering, not finiteness of the entire set, is the existence argument.

Proposition 4.5.2. *The shortlex normal form exists, is unique, and is computable by exhaustive breadth-first enumeration together with exact projective matrix equality. This statement asserts decidability, not efficiency.*

Proof. Every element has at least one representing word. Lengths are nonnegative integers, so there is a least representing length; finitely many words have that length, so exactly one is lexicographically least. For computation, enumerate words in shortlex order, freely reducing before evaluation, and compare their side-pairing matrices to the matrix of the input word. The faithfulness of this *specific* matrix map follows without identifying two groups merely by their abstract isomorphism type. Indeed, (4.6) has the explicit inverse free-group substitution

$$h_0 = a_1, \quad h_1 = b_1, \quad h_2 = a_1^{-1}b_1^{-1}a_2, \quad h_3 = b_2b_1a_1. \quad (4.12)$$

Substitution in both directions returns each free generator, so this is a free-group isomorphism. Equation (4.7) sends $\mathcal{R}$ to $\mathcal{B}$; applying the inverse therefore sends $\langle\langle\mathcal{B}\rangle\rangle$ to $\langle\langle\mathcal{R}\rangle\rangle$. The Poincaré polygon theorem identifies $\langle h_0, h_1, h_2, h_3 \mid \mathcal{B} \rangle$ with the concrete side-pairing matrix group. Consequently (4.6) induces an isomorphism from (4.2) onto that concrete group, proving that the evaluated matrix map used in the search is faithful.

The input word itself eventually appears, so the search terminates. Faithfulness makes matrix equality equivalent to group equality, and enumeration order makes the first match the defined normal form. $\square$

A faster Dehn or automatic-group procedure may replace the exhaustive search only after its own hypotheses and rewrite completeness are verified.

4.6 Geometric meaning of elements and conjugacy

Proposition 4.6.1. *Each $\gamma \in \Gamma_B$ is a deck transformation of $\mathbb{H}_R^2 \rightarrow \Sigma_B$. Its orbit point γo is the center of γP , and every S_8^{geo} -word supplies a side-adjacent tile path. Primitive nontrivial conjugacy classes, where primitive means not a proper positive power in Γ_B , correspond bijectively to oriented primitive closed geodesics. Powers describe iterations, and inversion reverses orientation.*

Proof. The first two assertions are Statement 2.6 with the left-action/right-label convention. If $\gamma = s_1 \cdots s_m$, the prefix tiles

$$P, s_1P, s_1s_2P, \dots, s_1 \cdots s_mP$$

are consecutive after adopting the matching multiplication convention; equivalently one may reverse the written convention and use right label multiplication. What matters is that each step is one side pairing.

For $\gamma \neq 1$, cocompact torsion-freeness makes γ hyperbolic. Its axis projects to a closed geodesic of length $\ell_{\mathbb{H}}(\gamma)$. Replacing γ by $\eta\gamma\eta^{-1}$ moves the axis by η and gives the same projected oriented geodesic. Conversely, lifting an oriented closed geodesic and comparing its endpoints gives a unique deck transformation up to conjugacy. If $\gamma = \delta^m$, the projected path traverses the primitive geodesic m times; primitivity excludes this. The axis of γ^{-1} has the opposite translation orientation, so inversion reverses orientation. $\square$

4.7 Faithfulness of the geometric action

Let

$$\rho_{\text{geom}} : \langle a_1, b_1, a_2, b_2 \mid \mathcal{R} \rangle \longrightarrow \text{PSU}(1, 1)$$

send the generators by (4.6) and (4.3).

Theorem 4.7.1. *The homomorphism ρ_{geom} is well defined and injective. Its image is the side-pairing group Γ_B .*

Proof. Well-definedness requires the relator to map to the identity; this is exactly (4.7) followed by (4.4). The images generate the h_j , because the inverse formulas in Statement 6.5 express every h_j in the a_i, b_i . Thus the image is the side-pairing group.

The polygon theorem presents the side-pairing group with the four side generators and the single vertex-cycle relation (4.4). Equations (4.6)–(4.7) are an invertible Nielsen change of the free generators and carry that relation to $\mathcal{R}$. Hence the induced presentation map has an inverse and is injective. Equivalently, both groups identify with the deck transformation group of the universal cover of Σ_B . $\square$

This proof separates faithfulness from torsion-freeness. Torsion-freeness was proved geometrically in Statement 2.4; faithfulness identifies the abstract presentation with that already free deck action.

4.8 Exact equality of group elements

For a word w , let $M(w) \in \mathrm{SU}(1, 1)$ be the product of the fixed generator lifts. A different coherent lift convention can multiply a word matrix by $-I$, so equality must be tested projectively.

Criterion 4.8.1 (Exact equality). *For any presentation or geometric words u, v ,*

$$u = v \text{ in } \Gamma_B \iff M(u) = M(v) \text{ or } M(u) = -M(v). \quad (4.10)$$

Proof. If $u = v$, their projective images under the homomorphism agree. The kernel of the projection

$$\mathrm{SU}(1, 1) \longrightarrow \mathrm{PSU}(1, 1)$$

is $\{\pm I\}$, giving the two alternatives on the right. Conversely, either matrix alternative gives the same projective isometry. Faithfulness in Statement 4.7 then gives $u = v$ in the abstract surface group. Both directions use exact equality of all four matrix entries; no trace-only or numerical-tolerance test is sufficient. $\square$

The criterion handles $M \sim -M$ and nothing more. In particular, multiplying a matrix by an arbitrary nonzero scalar is not an allowed equivalence inside $\mathrm{SU}(1, 1)$.

5 Exact matrix arithmetic

5.1 Matrices of the two generator systems

The physical lifts are G_j from (2.8). In the oriented geometric basis,

$$H_0 = G_0, \quad H_1 = G_1^{-1}, \quad H_2 = G_2, \quad H_3 = G_3^{-1}. \quad (5.1)$$

The presentation lifts obtained from (4.6) are

$$A_1 = H_0 = G_0, \quad B_1 = H_1 = G_1^{-1}, \quad (5.2)$$

$$A_2 = H_1 H_0 H_2 = G_1^{-1} G_0 G_2, \quad B_2 = H_3 H_0^{-1} H_1^{-1} = G_3^{-1} G_0^{-1} G_1. \quad (5.3)$$

Proposition 5.1.1. *Equations (5.2)–(5.3) define a homomorphism from the presentation (4.2) to $\mathrm{SU}(1, 1)$, not only to its projective quotient, for the fixed coherent lifts. Its projectivization is the geometric action.*

Proof. Each G_j lies in $\mathrm{SU}(1, 1)$ by (2.7), and products and inverses remain in the group. Substitution of (5.2)–(5.3) into the relator repeats the word calculation (4.8)–(4.9), reducing it to the lifted boundary cycle. For clarity, (2.16) multiplied the cyclically shifted word

$$W = G_1 G_6 G_3 G_0 G_5 G_2 G_7 G_4.$$

With $X = G_1 G_6 G_3$, the lifted boundary word is exactly

$$G_0 G_5 G_2 G_7 G_4 G_1 G_6 G_3 = X^{-1} W X = I,$$

not merely $\pm I$. The universal property of a presented group therefore gives a matrix homomorphism. Passing to projective classes sends every generator to the side-pairing isometry already fixed in Statement 2.3, so it is precisely the geometric action. $\square$

Existence is proved; uniqueness holds relative to the four specified lift matrices by the presentation's universal property. Other coherent choices may differ by a $\{\pm I\}$ -valued character and are not silently identified at the linear level.

5.2 Coefficient field and integrality

Let

$$\beta = \sqrt{2 + 2\sqrt{2}}, \quad \xi = \zeta_{16} = e^{\pi i/8}, \quad E = \mathbb{Q}(\xi, \beta). \quad (5.4)$$

Then $\zeta = \xi^2$, $\sqrt{2} = \xi^2 + \xi^{-2}$, and $A = 1 + \sqrt{2}$.

Proposition 5.2.1. *The numbers $\xi, \beta, \sqrt{2}, A, \zeta$ are algebraic integers, every entry of every G_j lies in the ring of integers $\mathcal{O}_E$, and complex conjugation preserves both E and $\mathcal{O}_E$.*

Proof. The root of unity ξ satisfies the monic polynomial $x^8 + 1$. The element β satisfies

$$\beta^2 = 2 + 2\sqrt{2}, \quad (\beta^2 - 2)^2 = 8, \quad \beta^4 - 4\beta^2 - 4 = 0, \quad (5.5)$$

which is monic with integer coefficients. Thus ξ and β are integral. Sums, products, and powers of algebraic integers are integral; $\sqrt{2}$ also satisfies $x^2 - 2$, so $A = 1 + \sqrt{2}$ and $\zeta = \xi^2$ are integral. Formula (2.8) now places all entries in $\mathcal{O}_E$.

Complex conjugation sends ξ to ξ^{-1} and fixes the positive real number β , hence maps E to itself. A field automorphism carries a monic integral equation to another monic integral equation with the same integer coefficients, so it preserves $\mathcal{O}_E$. $\square$

No power basis is asserted to be an integral basis. Exact arithmetic only requires a fixed $\mathbb{Q}$ -basis and the defining relations; later ideal-theoretic arguments must use the actual ring of integers or a specified order with its index controlled.

5.3 Multiplication and inversion in pair coordinates

Write

$$M(a, b) = \begin{pmatrix} a & b \\ \bar{b} & \bar{a} \end{pmatrix}.$$

Entrywise multiplication gives

$$M(a, b)M(c, d) = \begin{pmatrix} ac + b\bar{d} & ad + b\bar{c} \\ \bar{b}c + \bar{a}\bar{d} & \bar{b}d + \bar{a}\bar{c} \end{pmatrix} = M(ac + b\bar{d}, ad + b\bar{c}). \quad (5.6)$$

If $|a|^2 - |b|^2 = 1$, the adjugate is

$$M(a, b)^{-1} = \begin{pmatrix} \bar{a} & -b \\ -\bar{b} & a \end{pmatrix} = M(\bar{a}, -b). \quad (5.7)$$

Lemma 5.3.1. *The set of pairs $(a, b) \in \mathcal{O}_E^2$ satisfying $|a|^2 - |b|^2 = 1$ is closed under (5.6) and (5.7). Both formulas are exact and preserve determinant one.*

Proof. Closure of the entries follows from ring closure and conjugation stability proved in Statement 5.2. Multiplicativity of determinant gives determinant one for (5.6); direct multiplication of (5.7) by $M(a, b)$ gives

$$\begin{pmatrix} |a|^2 - |b|^2 & 0 \\ 0 & |a|^2 - |b|^2 \end{pmatrix} = I.$$

Thus (5.7) is the inverse and also has determinant one. Conversely, the displayed inverse calculation shows why the norm equation is necessary. $\square$

5.4 Normalization modulo the central sign

Put

$$\mathrm{SU}(1, 1; E) = \mathrm{SU}(1, 1) \cap M_2(E).$$

Fix once and for all an ordered $\mathbb{Q}$ -basis $\mathcal{E} = (e_1, \dots, e_d)$ of E . Expanding the four entries of a matrix $M \in \mathrm{SU}(1, 1; E)$, in row-major entry order, gives a nonzero coordinate vector $q(M) \in \mathbb{Q}^{4d}$.

Definition 5.4.1 (Sign-normalized lift). Among $q(M)$ and $-q(M)$, choose the one whose first nonzero rational coordinate is positive. Denote the corresponding matrix by $\mathrm{sn}(M)$.

Lemma 5.4.2. *The map $[M] \mapsto \mathrm{sn}(M)$ is well defined on the projective classes arising from $\mathrm{SU}(1, 1; E)$, and*

$$\mathrm{sn}(M) = \mathrm{sn}(N) \iff M = N \text{ or } M = -N. \quad (5.8)$$

Proof. Every $\mathrm{SU}(1, 1; E)$ matrix is nonzero, so its coordinate vector has a first nonzero entry. Exactly one of that entry and its negative is positive, which proves existence and uniqueness of the chosen sign. If $N = \pm M$, the same normalized vector results. Conversely, equality of normalized coordinate vectors reconstructs equality of the normalized matrices, so the original matrices differ only by the sign used to normalize them. The kernel computation in Statement 1.5 shows that no other scalar equivalence is allowed. $\square$

The ordering is a serialization convention, not an order on E and not a choice of real embedding.

5.5 Lossless exact projective keys

For $M \in \mathrm{SU}(1, 1; E)$ and $q = (q_1, \dots, q_{4d}) = q(\mathrm{sn}(M))$, write $q_i = n_i/D$ using one common denominator $D > 0$, chosen so that

$$n_i \in \mathbb{Z}, \quad \gcd(D, n_1, \dots, n_{4d}) = 1. \quad (5.9)$$

Define

$$\mathrm{key}([M]) = (D; n_1, \dots, n_{4d}). \quad (5.10)$$

Theorem 5.5.1. *For $M, N \in \mathrm{SU}(1, 1; E)$, the key (5.10) is canonical relative to the fixed ordered $\mathbb{Q}$ -basis and fixed row-major entry order, and it is lossless:*

$$\mathrm{key}([M]) = \mathrm{key}([N]) \iff [M] = [N] \text{ in } \mathrm{PSU}(1, 1).$$

Proof. Taking the least common multiple of reduced positive denominators produces integers D, n_i ; dividing D and every n_i by their common gcd gives (5.9). If two primitive tuples represent the same rational vector, cross multiplication gives $D'n_i = Dn'_i$ for every i . Primitivity and positivity of the denominators force $D = D'$ and $n_i = n'_i$. Thus the tuple reconstructs every rational coefficient as n_i/D , hence reconstructs $\mathrm{sn}(M)$. Equality of keys is therefore equivalent to (5.8), which is projective equality. $\square$

It would be incorrect to discard D after clearing denominators or to divide only the numerator vector by its gcd: either operation can identify different field elements. Hashes may index (5.10), but a collision must be resolved by comparison of the full tuple.

5.6 Words, relations, and the matrix homomorphism

For a signed letter s^{-1} , set $M(s^{-1}) = M(s)^{-1}$, and for a word $w = s_1 \cdots s_m$ set

$$M(w) = M(s_1) \cdots M(s_m), \quad M(\emptyset) = I. \quad (5.11)$$

Proposition 5.6.1. *The assignment (5.11) is compatible with free reduction, the surface relator, concatenation, and inversion. It therefore defines the unique homomorphism $\rho_E : \Gamma_B \rightarrow \mathrm{SU}(1, 1)$ having the generator images (5.2)–(5.3).*

Proof. Associativity gives $M(uv) = M(u)M(v)$. Formula (5.7) gives $M(ss^{-1}) = I = M(s^{-1}s)$, so deleting adjacent inverse pairs leaves the product unchanged. Reversing a word and inverting every letter gives

$$M(s_m^{-1} \cdots s_1^{-1}) = M(s_m)^{-1} \cdots M(s_1)^{-1} = M(w)^{-1}.$$

Statement 5.1 computed the defining relator as I . Hence every element of its normal closure maps to I , which is precisely the well-definedness condition for descent from the free group to the presented quotient. Uniqueness follows because a homomorphism is determined by generator images. $\square$

The projectivization of ρ_E is faithful. The linear map ρ_E itself can contain a central sign only if a nontrivial group element maps to $-I$; faithfulness projectively excludes this, so ρ_E is also injective.

5.7 Exact orbit, displacement, and trace data

Let $M(\gamma) = M(a, b)$. Its action at the center is

$$\gamma o = \frac{b}{a}. \quad (5.12)$$

The norm equation makes $|a| \geq 1$ and gives

$$1 - \left| \frac{b}{a} \right|^2 = \frac{1}{|a|^2}, \quad 1 + \left| \frac{b}{a} \right|^2 = \frac{|a|^2 + |b|^2}{|a|^2}. \quad (5.13)$$

Proposition 5.7.1 (Exact geometric interface). *For every γ ,*

$$\cosh \frac{d_R(o, \gamma o)}{R} = |a|^2 + |b|^2 = 2|a|^2 - 1, \quad (5.14)$$

$$d_R(o, \gamma o) = 2R \operatorname{arccosh} |a|. \quad (5.15)$$

If $\gamma \neq 1$, then it is hyperbolic and

$$\ell_{\mathbb{H}}(\gamma) = 2R \operatorname{arccosh} \frac{|a + \bar{a}|}{2}. \quad (5.16)$$

Proof. Insert (5.12)–(5.13) into the centered case of (1.8). This gives the first equality in (5.14); $|a|^2 - |b|^2 = 1$ gives the second. Since $\cosh(2u) = 2\cosh^2 u - 1$ and both sides are nonnegative, inversion of the monotone $\cosh$ on $[0, \infty)$ gives (5.15). Statement 2.4 and compactness make every nonidentity group element hyperbolic, so Statement 1.7 applies and $\operatorname{tr} M(a, b) = a + \bar{a}$, proving (5.16). $\square$

An exact cutoff comparison must be posed in algebraic threshold coordinates. For example, $d_R(o, \gamma o) \leq D$ is equivalent to

$$2|a|^2 - 1 \leq C_D, \quad C_D = \cosh(D/R). \quad (5.17)$$

If C_D is supplied as a real algebraic number, (5.17) is a sign test in an ordered real algebraic extension. Merely declaring D algebraic does not make C_D algebraic.

5.8 Faithfulness of the exact representation

Theorem 5.8.1. *The homomorphism $\rho_E : \Gamma_B \rightarrow \mathrm{SU}(1,1)$ is injective. Consequently the key (5.10) decides equality in the abstract surface group.*

Proof. Projectivizing ρ_E gives ρ_{geom} , whose faithfulness was proved in Statement 4.7 from the Poincaré polygon presentation. If $\rho_E(\gamma) = I$, its projective image is the identity, so $\gamma = 1$. Thus

$$\ker \rho_E \subseteq \ker(\mathrm{proj} \circ \rho_E) = \ker \rho_{\mathrm{geom}} = \{1\}, \quad (5.18)$$

and equality holds because every homomorphism kills the identity. More generally, if $\rho_E(\gamma) = -I$, the projective image is again the identity, so $\gamma = 1$, but

$$\rho_E(1) = I \neq -I, \quad (5.19)$$

a contradiction. Thus neither identity nor a central sign hides a nontrivial kernel element.

For words u, v , equality of keys means $\rho_E(u) = \pm \rho_E(v)$, so their projective images agree. Faithfulness gives $u = v$. Conversely equal group elements have equal projective images and hence equal normalized keys. Existence of an exact equality test follows; its computational cost is a separate implementation question. $\square$

Corollary 5.8.2. *For every finite set of words W , grouping its members by (5.10) gives exactly the partition of W by equality in Γ_B .*

Proof. Two words occupy the same key class exactly when their matrices agree up to sign. By Statement 4.8 this is necessary and sufficient for group equality. Pairwise application proves equality of the two partitions; no assumption about how W was enumerated is needed. $\square$

6 Nielsen coordinates

6.1 Geometric free coordinates

Let $F_h = F(h_0, h_1, h_2, h_3)$ be the free group on the oriented geometric letters. Before imposing the polygon relation, an element is a freely reduced signed word. Let

$$\mathcal{B} = h_0 h_1 h_2 h_3 h_0^{-1} h_1^{-1} h_2^{-1} h_3^{-1}. \quad (6.1)$$

Then

$$\Gamma_B \cong F_h / \langle\langle \mathcal{B} \rangle\rangle. \quad (6.2)$$

Definition 6.1.1 (Geometric coordinate). A geometric word is a word in $h_j^{\pm 1}$, interpreted first in F_h and then under the quotient map in (6.2). Its exact geometric value is the matrix product obtained from (5.1).

Proposition 6.1.2. *The map $F_h \rightarrow \mathrm{SU}(1,1)$ defined by (5.1) factors through (6.2), and the induced projective representation is faithful.*

Proof. Free-group well-definedness follows from the universal property. The cycle calculation (2.16) gives

$$W = G_1 G_6 G_3 G_0 G_5 G_2 G_7 G_4 = I.$$

If $X = G_1 G_6 G_3$, then the lift determined by (5.1) satisfies

$$\begin{aligned} M(\mathcal{B}) &= G_0 G_5 G_2 G_7 G_4 G_1 G_6 G_3 \\ &= X^{-1} W X = I. \end{aligned}$$

Thus the normal closure of $\mathcal{B}$ lies in the kernel and the map descends. Conversely, the Poincaré polygon theorem presents the side-pairing group by this boundary cycle, so no additional projective relation occurs. Hence the descended projective map is injective. $\square$

This coordinate is adapted to sides and tile adjacency. It is not a unique word coordinate until the exact normal-form convention of Statement 4.5 is applied.

6.2 Standard surface coordinates

Let $F_a = F(a_1, b_1, a_2, b_2)$ and

$$\mathcal{R} = [a_1, b_1][a_2, b_2].$$

The presentation coordinate is the quotient

$$\Gamma_B \cong F_a / \langle\langle \mathcal{R} \rangle\rangle. \quad (6.3)$$

Proposition 6.2.1 (Abelianization). *The abelianization of Γ_B is freely generated by the four images of a_1, b_1, a_2, b_2 :*

$$\Gamma_B^{\text{ab}} = \Gamma_B / [\Gamma_B, \Gamma_B] \cong \mathbb{Z}^4. \quad (6.4)$$

Proof. The abelianization of F_a is $\mathbb{Z}^4$. Every commutator maps to zero in an abelian group, so the exponent-sum vector of $\mathcal{R}$ is $(0, 0, 0, 0)$. Quotienting by the normal closure of $\mathcal{R}$ therefore adds no relation to the free abelianization. Conversely, the four exponent sum maps $F_a \rightarrow \mathbb{Z}$ kill $\mathcal{R}$ and descend independently, so no hidden integral linear relation among the four classes exists. $\square$

These coordinates are adapted to topology and homology. They do not encode which eight transformations are physically nearest neighbors.

6.3 The Nielsen transformation

Define $N : F_a \rightarrow F_h$ on free generators by

$$\begin{aligned} N(a_1) &= h_0, & N(b_1) &= h_1, \\ N(a_2) &= h_1 h_0 h_2, & N(b_2) &= h_3 h_0^{-1} h_1^{-1}. \end{aligned} \quad (6.5)$$

Theorem 6.3.1. *The map N is a free-group isomorphism (a Nielsen change of basis). It carries the surface relator to the polygon boundary relator:*

$$N(\mathcal{R}) = \mathcal{B}. \quad (6.6)$$

It therefore descends to an isomorphism between the quotient presentations (6.3) and (6.2).

Proof. The universal property of F_a gives a unique homomorphism with the four images (6.5). Define

$$P(h_0) = a_1, \quad P(h_1) = b_1, \quad P(h_2) = a_1^{-1} b_1^{-1} a_2, \quad P(h_3) = b_2 b_1 a_1.$$

Then

$$\begin{aligned} P(N(a_1)) &= P(h_0) = a_1, & P(N(b_1)) &= P(h_1) = b_1, \\ P(N(a_2)) &= b_1 a_1 a_1^{-1} b_1^{-1} a_2 = a_2, & P(N(b_2)) &= b_2 b_1 a_1 a_1^{-1} b_1^{-1} = b_2, \\ N(P(h_0)) &= N(a_1) = h_0, & N(P(h_1)) &= N(b_1) = h_1, \\ N(P(h_2)) &= h_0^{-1} h_1^{-1} h_1 h_0 h_2 = h_2, & N(P(h_3)) &= h_3 h_0^{-1} h_1^{-1} h_1 h_0 = h_3. \end{aligned}$$

Every cancellation is adjacent and displayed. Hence both composites fix every free generator, so N is an isomorphism with inverse P .

Expanding the two commutators as in (4.8)–(4.9) gives

$$N(\mathcal{R}) = h_0 h_1 h_2 h_3 h_0^{-1} h_1^{-1} h_2^{-1} h_3^{-1} = \mathcal{B}.$$

Thus (6.6) is established inside this proof. Since N is invertible and takes the normal generator $\mathcal{R}$ to $\mathcal{B}$, it takes $\langle\langle \mathcal{R} \rangle\rangle$ onto $\langle\langle \mathcal{B} \rangle\rangle$. Coset independence follows: if $u\mathcal{R}^{F_a} = v\mathcal{R}^{F_a}$, then $N(u)N(v)^{-1}$ lies in the normal closure of $\mathcal{B}$, so $N(u)\mathcal{B}^{F_h} = N(v)\mathcal{B}^{F_h}$. The inverse argument uses P . Thus the descended maps are well defined and mutually inverse. $\square$

The existence of an automorphism, its inverse, and its descent are separate claims; the next three subsections supply the promised explicit checks.

6.4 Forward word substitution

For a signed letter, define

$$N(x^{-1}) = N(x)^{-1}.$$

For a word $w = x_1 \cdots x_m$, define

$$N(w) = N(x_1) \cdots N(x_m). \quad (6.7)$$

This is an algorithm: replace each letter by its word in (6.5), reverse and invert that word for an inverse letter, concatenate, and freely reduce.

Proposition 6.4.1. *Forward substitution is a homomorphism, preserves equality in the surface-group quotient, and agrees with exact matrix evaluation:*

$$M_a(w) = M_h(N(w)). \quad (6.8)$$

Proof. For words u, v , concatenation gives $N(uv) = N(u)N(v)$ before any reduction. The inverse convention gives $N(xx^{-1}) = N(x)N(x)^{-1} = 1$, so free reductions preserve the image. Equation (6.6) makes every conjugate of $\mathcal{R}^{\pm 1}$ map into the normal closure of $\mathcal{B}$; hence equality modulo the surface relation is preserved.

For (6.8), check the four positive generators. The presentation matrices (5.2)–(5.3) were defined by exactly the products (6.5). Inverse letters agree by (5.7), and concatenations agree by matrix associativity. Induction on word length proves (6.8) for every word. $\square$

Forward substitution need not preserve word length and need not output a normal form. It preserves the represented group element.

6.5 Inverse word substitution

Define $P : F_h \rightarrow F_a$ by

$$\begin{aligned} P(h_0) &= a_1, & P(h_1) &= b_1, \\ P(h_2) &= a_1^{-1}b_1^{-1}a_2, & P(h_3) &= b_2b_1a_1. \end{aligned} \quad (6.9)$$

Theorem 6.5.1. *The free-group maps P and N are mutually inverse.*

Proof. On the standard generators,

$$\begin{aligned} P(N(a_1)) &= a_1, & P(N(b_1)) &= b_1, \\ P(N(a_2)) &= b_1a_1(a_1^{-1}b_1^{-1}a_2) \longrightarrow a_2, \\ P(N(b_2)) &= (b_2b_1a_1)a_1^{-1}b_1^{-1} \longrightarrow b_2. \end{aligned}$$

The arrows cancel only adjacent inverse pairs. On the geometric generators,

$$\begin{aligned} N(P(h_0)) &= h_0, & N(P(h_1)) &= h_1, \\ N(P(h_2)) &= h_0^{-1}h_1^{-1}(h_1h_0h_2) \longrightarrow h_2, \\ N(P(h_3)) &= (h_3h_0^{-1}h_1^{-1})h_1h_0 \longrightarrow h_3. \end{aligned}$$

Homomorphisms between free groups are determined by their values on free generators. Hence $P \circ N = \text{id}_{F_a}$ and $N \circ P = \text{id}_{F_h}$. This proves existence and exactness of the inverse, not merely a one-sided substitution rule. $\square$

6.6 Relator invariance and descent

Theorem 6.6.1. *The Nielsen maps satisfy*

$$N(\mathcal{R}) = \mathcal{B}, \quad P(\mathcal{B}) = \mathcal{R}. \quad (6.10)$$

Consequently both descend to quotient homomorphisms and the descended maps are inverse isomorphisms of Γ_B .

Proof. Using (6.5),

$$N([a_1, b_1]) = h_0 h_1 h_0^{-1} h_1^{-1}. \quad (6.11)$$

Put $c = h_1 h_0 h_2$ and $d = h_3 h_0^{-1} h_1^{-1}$. Then

$$c^{-1} = h_2^{-1} h_0^{-1} h_1^{-1}, \quad d^{-1} = h_1 h_0 h_3^{-1},$$

and

$$\begin{aligned} [c, d] &= h_1 h_0 h_2 h_3 h_0^{-1} h_1^{-1} h_2^{-1} h_0^{-1} h_1^{-1} h_1 h_0 h_3^{-1} \\ &\longrightarrow h_1 h_0 h_2 h_3 h_0^{-1} h_1^{-1} h_2^{-1} h_3^{-1}. \end{aligned}$$

Multiplying by (6.11) cancels first $h_1^{-1} h_1$, then $h_0^{-1} h_0$, and yields $\mathcal{B}$. This proves the first identity. Applying $P = N^{-1}$ to it gives $P(\mathcal{B}) = P(N(\mathcal{R})) = \mathcal{R}$, proving the second without assuming quotient descent.

For well-definedness, if two free words differ by a product of conjugates of $\mathcal{R}^{\pm 1}$, their N -images differ by the corresponding product of conjugates of $\mathcal{B}^{\pm 1}$. The same argument applies to P . Their free-group composite identities then descend, so the quotient maps are mutually inverse. $\square$

6.7 The change of basis on abelianization

Use column exponent vectors. The ordered geometric row basis and standard column basis are, respectively,

$$(h_0, h_1, h_2, h_3), \quad (a_1, b_1, a_2, b_2).$$

Abelianizing (6.5) gives

$$[N]_{\text{ab}} = \begin{pmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (6.12)$$

Its inverse, obtained from (6.9), is

$$[P]_{\text{ab}} = \begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (6.13)$$

Proposition 6.7.1. *The matrices (6.12)–(6.13) lie in $\text{GL}_4(\mathbb{Z})$, multiply to I_4 , and preserve all integral abelian information.*

Proof. Both are upper triangular with diagonal entries one, so each determinant is one. Direct block multiplication gives

$$\begin{pmatrix} I_2 & B \\ 0 & I_2 \end{pmatrix} \begin{pmatrix} I_2 & -B \\ 0 & I_2 \end{pmatrix} = I_4, \quad B = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}.$$

Thus they are mutually inverse over $\mathbb{Z}$, not only over $\mathbb{Q}$. Exponent sums of a substituted word are obtained by the matching matrix multiplication. Since the relators have zero exponent vector, this calculation is independent of the chosen representative in the surface group. $\square$

6.8 Commuting symbolic, exact, and numerical coordinates

Let ρ_h and ρ_a be the exact matrix maps from the geometric and presentation free groups. Let $\sigma : E \hookrightarrow \mathbb{C}$ be a specified embedding satisfying $\sigma(\bar{x}) = \overline{\sigma(x)}$ for every $x \in E$, and let ev_σ apply it entrywise.

Theorem 6.8.1 (Coordinate interface). *The diagram*

$$\begin{array}{ccc} F_a/\langle\langle \mathcal{R} \rangle\rangle & \xrightarrow{\bar{N}} & F_h/\langle\langle \mathcal{B} \rangle\rangle \\ \rho_a \downarrow & & \downarrow \rho_h \\ \text{SU}(1, 1; \mathcal{O}_E) & = & \text{SU}(1, 1; \mathcal{O}_E) \end{array} \quad (6.14)$$

commutes. Applying ev_σ yields the same numerical matrix and orbit point along either path.

Proof. Equation (6.8) proves commutativity on every word. Quotient well-definedness is Statement 6.6, so (6.14) is independent of representatives. Field evaluation is a ring homomorphism: $\sigma(x + y) = \sigma(x) + \sigma(y)$ and $\sigma(xy) = \sigma(x)\sigma(y)$. It therefore commutes with every matrix product and inverse. The orbit map $M(a, b) \mapsto b/\bar{a}$ also commutes with the chosen complex embedding when conjugation compatibility is fixed. Thus both symbolic routes yield the same evaluated data. $\square$

Numerical evaluation is subsequent of the exact coordinate. It may approximate distances or draw tiles, but it does not replace the equality key (5.10) or choose a projective sign.

7 Eightfold rotation and equivariant quotients

7.1 The geometric rotation

Let

$$\xi = e^{\pi i/8} = \zeta_{16}, \quad C = \begin{pmatrix} \xi & 0 \\ 0 & \xi^{-1} \end{pmatrix} \in \text{SU}(1, 1; E), \quad c_8 = [C] \in \text{PSU}(1, 1). \quad (7.1)$$

Its disk action is

$$c_8(z) = \frac{\xi z}{\xi^{-1}} = \xi^2 z = \zeta z, \quad (7.2)$$

so it rotates the octagon through $\pi/4$, fixes $o = 0$, and maps s_j to s_{j+1} , with indices modulo 8.

Proposition 7.1.1 (Exact covariance). *For every $j \in \mathbb{Z}/8\mathbb{Z}$,*

$$CG_j C^{-1} = G_{j+1}. \quad (7.3)$$

Moreover c_8 has exact projective order 8.

Proof. Conjugation by C fixes the two diagonal entries in (2.8). It multiplies the upper-right entry by $\xi^2 = \zeta$ and the lower-left entry by $\xi^{-2} = \zeta^{-1}$, giving exactly G_{j+1} . Also

$$C^8 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I,$$

so $c_8^8 = 1$. If $0 < k < 8$ and $c_8^k = 1$, then $C^k = \pm I$, hence $\xi^k = \pm 1$. Since ξ has order 16, the least positive such k is 8, a contradiction. $\square$

Relative to the fixed field basis and entry order of Statement 5.4–Statement 5.5, the record

$$(\text{key}([C]), (\text{key}([G_j]))_{j=0}^7, (0\ 1\ 2\ 3\ 4\ 5\ 6\ 7)) \quad (7.4)$$

is a canonical serialization of the rotation action. Recomputing (7.3) gives the same exact keys, so no floating-point phase or branch choice enters this interface.

7.2 The induced group automorphism

Equation (7.3) shows that c_8 normalizes Γ_B . Define

$$\phi_8 : \Gamma_B \longrightarrow \Gamma_B, \quad \phi_8(\gamma) = c_8 \gamma c_8^{-1}. \quad (7.5)$$

Theorem 7.2.1. *The map ϕ_8 is an automorphism, satisfies*

$$\phi_8(g_j) = g_{j+1}, \quad (7.6)$$

and is compatible with both the action on $\mathbb{H}_R^2$ and matrix evaluation:

$$c_8(\gamma z) = \phi_8(\gamma) c_8(z), \quad CM(\gamma) C^{-1} = M(\phi_8(\gamma)). \quad (7.7)$$

Proof. Conjugation is a homomorphism and its inverse is conjugation by c_8^{-1} . Normalization follows from (7.3), because the g_j generate Γ_B ; hence its restriction is an automorphism of Γ_B . Equation (7.6) is (7.3) projectivized. Associativity gives

$$c_8 \gamma z = (c_8 \gamma c_8^{-1})(c_8 z),$$

which is the first identity in (7.7). The second follows generator by generator from (7.3), then for inverses and products by matrix multiplication. Since the coherent lifts satisfy (7.3) without a central sign, the equality is linear as displayed, not only projective. $\square$

7.3 Explicit action on every generator system

Combining (7.6), (4.3), and the Nielsen dictionary gives the following generator-image table:

$$\begin{array}{c|ccccc} x & g_j & h_0 & h_1 & h_2 & h_3 \\ \hline \phi_8(x) & g_{j+1} & h_1^{-1} & h_2^{-1} & h_3^{-1} & h_0 \end{array} \quad (7.8)$$

and

$$\begin{array}{c|c} x & \phi_8(x) \\ \hline a_1 & b_1^{-1} \\ b_1 & a_2^{-1} b_1 a_1 \\ a_2 & a_2^{-1} b_1 a_1 b_1^{-1} a_1^{-1} b_1^{-1} b_2^{-1} \\ b_2 & a_1 b_1 a_1^{-1} b_1^{-1} a_2. \end{array} \quad (7.9)$$

Proof. All four geometric rows are

$$\begin{aligned} \phi_8(h_0) &= \phi_8(g_0) = g_1 = h_1^{-1}, & \phi_8(h_1) &= \phi_8(g_5) = g_6 = h_2^{-1}, \\ \phi_8(h_2) &= \phi_8(g_2) = g_3 = h_3^{-1}, & \phi_8(h_3) &= \phi_8(g_7) = g_0 = h_0. \end{aligned}$$

The first two standard rows are

$$\begin{aligned} \phi_8(a_1) &= \phi_8(h_0) = h_1^{-1} = b_1^{-1}, \\ \phi_8(b_1) &= \phi_8(h_1) = h_2^{-1} = (a_1^{-1} b_1^{-1} a_2)^{-1} = a_2^{-1} b_1 a_1. \end{aligned}$$

For the last two, use (4.6) and (4.12):

$$\begin{aligned} \phi_8(a_2) &= h_2^{-1} h_1^{-1} h_3^{-1} \\ &= (a_2^{-1} b_1 a_1) b_1^{-1} (a_1^{-1} b_1^{-1} b_2^{-1}), \\ \phi_8(b_2) &= h_0 h_1 h_2 = a_1 b_1 a_1^{-1} b_1^{-1} a_2. \end{aligned}$$

The inverse geometric table is

$$\phi_8^{-1}(h_0) = h_3, \quad \phi_8^{-1}(h_1) = h_0^{-1}, \quad \phi_8^{-1}(h_2) = h_1^{-1}, \quad \phi_8^{-1}(h_3) = h_2^{-1};$$

its two compositions with (7.8) are displayed generator by generator:

x	h_0	h_1	h_2	h_3
$\phi_8 \phi_8^{-1}(x)$	$\phi_8(h_3)$	$\phi_8(h_0^{-1})$	$\phi_8(h_1^{-1})$	$\phi_8(h_2^{-1})$
	h_0	h_1	h_2	h_3
$\phi_8^{-1} \phi_8(x)$	$\phi_8^{-1}(h_1^{-1})$	$\phi_8^{-1}(h_2^{-1})$	$\phi_8^{-1}(h_3^{-1})$	$\phi_8^{-1}(h_0)$
	h_0	h_1	h_2	h_3

In standard generators the inverse is

$$\begin{array}{c|c} x & \phi_8^{-1}(x) \\ \hline a_1 & b_2 b_1 a_1 \\ b_1 & a_1^{-1} \\ a_2 & a_1^{-1} b_2 b_1 a_1 b_1^{-1} \\ b_2 & a_2^{-1} b_2^{-1} a_1. \end{array} \quad (7.9a)$$

$$\begin{aligned} \phi_8^{-1}(a_1) &= \phi_8^{-1}(h_0) = h_3 = b_2 b_1 a_1, \\ \phi_8^{-1}(b_1) &= \phi_8^{-1}(h_1) = h_0^{-1} = a_1^{-1}, \\ \phi_8^{-1}(a_2) &= \phi_8^{-1}(h_1 h_0 h_2) = h_0^{-1} h_3 h_1^{-1} = a_1^{-1} b_2 b_1 a_1 b_1^{-1}, \\ \phi_8^{-1}(b_2) &= \phi_8^{-1}(h_3 h_0^{-1} h_1^{-1}) = h_2^{-1} h_3^{-1} h_0 \\ &= (a_2^{-1} b_1 a_1)(a_1^{-1} b_1^{-1} b_2^{-1}) a_1 \longrightarrow a_2^{-1} b_2^{-1} a_1. \end{aligned}$$

Let $N : F_a \rightarrow F_h$ be (4.6), let $P = N^{-1} : F_h \rightarrow F_a$ be (4.12), and denote (7.8) by Φ_h . The forward table (7.9) is $\Phi_a = P\Phi_h N$, while (7.9a) is $\Phi_a^{-1} = P\Phi_h^{-1} N$. Since $NP = \text{id}_{F_h}$, $PN = \text{id}_{F_a}$, and the two geometric compositions were expanded above,

$$\begin{aligned} \Phi_a \Phi_a^{-1} &= P\Phi_h(NP)\Phi_h^{-1}N = PN = \text{id}_{F_a}, \\ \Phi_a^{-1} \Phi_a &= P\Phi_h^{-1}(NP)\Phi_h N = PN = \text{id}_{F_a}. \end{aligned}$$

Thus every standard generator returns to itself in both directions. Images of inverse letters are forced by $\phi_8(x^{-1}) = \phi_8(x)^{-1}$, with no additional choices. $\square$

For canonical serialization, scan a signed input word from left to right, replace every letter by its fixed row of (7.8) or (7.9), replace an inverse letter by the inverse word, freely reduce, and then apply the shortlex normal form of Statement 4.5. Fixed alphabet order makes the resulting word unique. Exact projective keys provide an independent consistency test: the serialized output and direct conjugation by C must have equal keys. Thus equal input elements give equal serialized outputs.

7.4 The eighth iterate

Theorem 7.4.1. *For every integer m and every shell generator,*

$$\phi_8^m(g_j) = g_{j+m}, \quad (7.10)$$

where the index is modulo 8. Consequently $\phi_8^8 = \text{id}_{\Gamma_B}$, and no positive power ϕ_8^k with $k < 8$ is the identity.

Proof. Induction on $m \geq 0$ using (7.6) proves (7.10); negative m follows from the inverse automorphism. At $m = 8$, every g_j is fixed, hence every word in the generating shell is fixed.

For $0 < k < 8$, (7.10) sends g_0 to g_k . These elements are different: by (2.21),

$$g_j o = (B/A)\zeta^j,$$

and the eight right-hand sides have distinct arguments. The orbit map is injective by Statement 2.6, so $g_k \neq g_0$. Hence ϕ_8^k cannot be the identity. $\square$

The relator check can also be performed wholly in the geometric presentation. If

$$u = h_0 h_1 h_2 h_3 h_0^{-1}, \quad v = h_1^{-1} h_2^{-1} h_3^{-1},$$

then $\mathcal{B} = uv$, whereas (7.8) gives

$$\phi_8(\mathcal{B}) = vu = u^{-1}\mathcal{B}u. \quad (7.11)$$

Thus the relator normal closure is preserved at each iteration, which is the generator-level relation test promised by the construction.

7.5 An exact order certificate

Definition 7.5.1. The order certificate for ϕ_8 is the triple consisting of the exact generator permutation (7.6), the equality $\phi_8^8(g_j) = g_j$ for all j , and the witness inequalities $\text{key}([G_k]) \neq \text{key}([G_0])$ for $1 \leq k < 8$.

Proposition 7.5.2. *The certificate is finite, canonical relative to the serialization conventions of Statement 5.5, and proves $\text{ord}(\phi_8) = 8$.*

Proof. All entries lie in the fixed number field E , so every comparison is an exact comparison of primitive integer tuples with their common denominators retained. Equation (7.10) verifies the eighth power on a generating set. For each smaller positive exponent k , the stored inequality gives $\phi_8^k(g_0) = g_k \neq g_0$. These two finite checks are respectively sufficient for the upper and lower bounds on the order. The fixed ordering $g_0, \dots, g_7$, fixed field basis, and fixed entry order make repeated serialization byte-for-byte identical. $\square$

7.6 Cyclic action on the physical shell

Proposition 7.6.1. *The rotation acts on the physical shell by the oriented 8-cycle*

$$\phi_8 : (g_0, g_1, \dots, g_7) \mapsto (g_1, g_2, \dots, g_0). \quad (7.12)$$

It induces an automorphism of the center-orbit adjacency graph by

$$\gamma o \mapsto \phi_8(\gamma) o$$

and sends the edge with right label g_j to the edge with right label g_{j+1} .

Proof. The first assertion is (7.6), including $g_7 \mapsto g_0$. Because $c_8 P = P$, equation (7.7) gives

$$c_8(\gamma P) = \phi_8(\gamma) P.$$

The adjacent tiles γP and $\gamma g_j P$ are therefore sent to

$$\phi_8(\gamma) P \quad \text{and} \quad \phi_8(\gamma) \phi_8(g_j) P = \phi_8(\gamma) g_{j+1} P.$$

This is again an adjacent pair by Statement 2.6, with exactly the stated right label. Bijectivity of ϕ_8 supplies the inverse graph map, so no vertices or edges are lost. The fixed cyclic index convention makes the action table a deterministic serialization. $\square$

7.7 Constructing the induced automorphism of a finite quotient

Let Q be a finite group with a fixed ordering, exact multiplication and inverse tables, identity e_Q , and an ordered generating tuple

$$q_j = \pi(g_j) \quad (0 \leq j < 8)$$

for a surjection $\pi : \Gamma_B \twoheadrightarrow Q$. The following finite procedure either constructs the descended rotation or returns a certificate that this quotient is not invariant under ϕ_8 .

Input: the ordered multiplication table of Q and $(q_0, \dots, q_7)$.

Output: either a table $\alpha : Q \rightarrow Q$, or two shell-index words u, v and exact values y, z, z' certifying $\pi(u) = \pi(v) = y$ but $\pi(\phi_8(u)) = z \neq z' = \pi(\phi_8(v))$.

Procedure: set $\alpha(e_Q) = e_Q$, store the empty predecessor word $p(e_Q)$, and enqueue e_Q . Repeatedly remove the earliest element x . For $j = 0, \dots, 7$, set $y = xq_j$, $u = p(x)j$, and $z = \alpha(x)q_{j+1}$. If $\alpha(y)$ is unset, set $\alpha(y) = z$, store $p(y) = u$, and enqueue y . If it is set to a different value z' , return $(u, p(y); y, z, z')$. When the queue is empty, return α .

Theorem 7.7.1 (Algorithm specification). *The procedure terminates. It returns a table precisely when $\phi_8(\ker \pi) = \ker \pi$; in that case the table is the unique automorphism satisfying*

$$\alpha(q_j) = q_{j+1}, \quad \alpha(\pi(\gamma)) = \pi(\phi_8(\gamma)), \quad (7.13)$$

and $\alpha^8 = \text{id}_Q$.

Proof. Each element is enqueued at most once, and Q is finite, so the procedure terminates. Since the q_j generate Q , the search reaches every element. Along a path labelled by a word w , the update rule forces

$$\alpha(\pi(w)) = \pi(\phi_8(w)). \quad (7.14)$$

Thus a conflict occurs exactly when two words u, v represent the same element of Q but their rotated words do not. Equivalently, $uv^{-1} \in \ker \pi$ while $\phi_8(uv^{-1}) \notin \ker \pi$. Absence of a conflict is therefore equivalent to $\phi_8(\ker \pi) \subseteq \ker \pi$. Applying the same inclusion eight times and using $\phi_8^8 = 1$ upgrades it to equality. The stored predecessor words make every negative output independently verifiable by multiplication in the supplied table; it is not merely a report that the deterministic run encountered two unequal values.

When the table exists, (7.14) gives, for $x = \pi(u)$ and $y = \pi(v)$,

$$\alpha(xy) = \pi(\phi_8(uv)) = \pi(\phi_8(u))\pi(\phi_8(v)) = \alpha(x)\alpha(y),$$

so it is a homomorphism. Its eighth power fixes the generating tuple and hence all of Q ; consequently $\alpha^{-1} = \alpha^7$, proving bijectivity. Finally, the fixed queue discipline, element order, and generator order make the output reproducible. $\square$

As minimal exact tests, the trivial quotient returns the identity table. For the nontrivial test take

$$Q = H_1(\Sigma_B; \mathbb{Z}/2) \cong (\mathbb{Z}/2)^4.$$

The shell images computed from (4.3) and (4.12) are

$$\begin{aligned} q_0 &= q_4 = \bar{a}_1, & q_1 &= q_5 = \bar{b}_1, \\ q_2 &= q_6 = \bar{a}_1 + \bar{b}_1 + \bar{a}_2, & q_3 &= q_7 = \bar{a}_1 + \bar{b}_1 + \bar{b}_2. \end{aligned}$$

They generate Q , and the procedure returns a valid linear automorphism. Explicitly, in the ordered column basis $(\bar{a}_1, \bar{b}_1, \bar{a}_2, \bar{b}_2)$, its matrix over $\mathbb{F}_2$ is

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (7.14a)$$

The first four shell columns multiply as

$$\begin{aligned} Aq_0 &= Ae_1 = e_2 = q_1, \\ Aq_1 &= Ae_2 = e_1 + e_2 + e_3 = q_2, \\ Aq_2 &= A(e_1 + e_2 + e_3) = e_1 + e_2 + e_4 = q_3, \\ Aq_3 &= A(e_1 + e_2 + e_4) = e_1 = q_4. \end{aligned}$$

Since $q_4 = q_0, q_5 = q_1, q_6 = q_2, q_7 = q_3$, the remaining four are

$$Aq_4 = Aq_0 = q_1 = q_5, \quad Aq_5 = Aq_1 = q_2 = q_6, \quad Aq_6 = Aq_2 = q_3 = q_7, \quad Aq_7 = Aq_3 = q_4 = q_0.$$

The intermediate square is

$$A^2 = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}, \quad A^4 = (A^2)^2 = I_4.$$

Also $A^2q_0 = q_2 \neq q_0$, so this test action has exact order 4. It therefore tests nontrivial shell-label collisions, the consistency branch of the algorithm, and the allowed drop from order 8 in a quotient.

7.8 C_8 -equivariant quotient maps

Theorem 7.8.1 (Descent and commuting square). *Let $\pi : \Gamma_B \twoheadrightarrow Q$ be a finite quotient. The following are equivalent:*

- (i) $\ker \pi$ is invariant under ϕ_8 ;
- (ii) there is a unique automorphism $\alpha \in \text{Aut}(Q)$ such that

$$\boxed{\pi \circ \phi_8 = \alpha \circ \pi}. \quad (7.15)$$

When they hold, $\alpha^8 = 1$. Moreover α is verified on the physical generators by the finite conditions

$$\alpha(\pi(g_j)) = \pi(g_{j+1}) \quad (0 \leq j < 8). \quad (7.16)$$

Proof. Assume (i). Define $\alpha(\pi(\gamma)) = \pi(\phi_8(\gamma))$. If $\pi(\gamma) = \pi(\delta)$, then $\gamma\delta^{-1} \in \ker \pi$; invariance puts $\phi_8(\gamma\delta^{-1})$ in the kernel, proving independence of the representative. Thus α is well defined. It is a homomorphism because ϕ_8 and π are, and applying the same construction to ϕ_8^{-1} provides its inverse. Definition gives (7.15).

Conversely, if (7.15) holds and $n \in \ker \pi$, then

$$\pi(\phi_8(n)) = \alpha(\pi(n)) = \alpha(e_Q) = e_Q,$$

so $\phi_8(n) \in \ker \pi$; as in Statement 7.7, finite order upgrades inclusion to equality. Surjectivity of π makes α unique and also gives $\alpha^8\pi = \pi\phi_8^8 = \pi$, hence $\alpha^8 = 1$. Finally, (7.15) implies (7.16), while (7.16) implies (7.15) because the g_j generate Γ_B . This proves that the generator table is a complete equivariance test rather than merely a sample. $\square$

The order of the descended α may be a proper divisor of 8; exact order 8 holds precisely when at least one quotient element is moved by each of $\alpha, \dots, \alpha^7$. The construction therefore preserves the C_8 -action law without falsely asserting that every finite quotient sees the action faithfully.

8 Physical and presentation shells

8.1 The geometric shell

Definition 8.1.1 (Ordered physical shell).

$$S_8^{\text{geo}} = (g_0, g_1, \dots, g_7), \quad g_{j+4} = g_j^{-1}. \quad (8.1)$$

The order is the counterclockwise order of the orbit points $g_j o = (B/A)\zeta^j$. A geometric shell word is serialized as a finite list of indices in $\mathbb{Z}/8\mathbb{Z}$, interpreted from left to right as a product of the corresponding g_j .

This is not an arbitrary generating tuple. Its entries are exactly the eight moves from a tile to the tiles sharing one side with it. Every entry has based displacement a_B , inversion is the index operation $j \mapsto j + 4$, and ϕ_8 acts by $j \mapsto j + 1$. Thus the fixed machine record

$$((\text{key}([G_j]))_{j=0}^7, (4, 5, 6, 7, 0, 1, 2, 3), (1, 2, 3, 4, 5, 6, 7, 0)) \quad (8.2)$$

stores respectively the entries, inverse table, and C_8 -action table.

Proposition 8.1.2. *The record (8.2) is lossless and reproducible. Its entries are eight distinct nonidentity elements and generate Γ_B .*

Proof. Losslessness follows from Statement 5.5 and the fixed index order. Equation (2.21) gives eight distinct nonzero orbit points, so no entry is the identity and no two entries coincide. The Poincaré polygon theorem defines Γ_B as the group generated by its side pairings, which are precisely $g_0, \dots, g_3$ and their inverses. Hence the shell generates. $\square$

As a positive test, index 0 has inverse index 4 and both exact matrix products $G_0 G_4, G_4 G_0$ equal I . As a boundary test, the empty index list represents the identity but is not a shell entry. A list with the same matrices in a different order represents the same underlying set but fails the ordered-shell specification because its inverse and rotation tables are no longer (8.2).

8.2 The presentation shell

Definition 8.2.1 (Ordered signed presentation shell).

$$S_8^{\text{pres}} = (a_1, a_1^{-1}, b_1, b_1^{-1}, a_2, a_2^{-1}, b_2, b_2^{-1}). \quad (8.3)$$

A presentation-shell word is a list of the indices $0, \dots, 7$, with inverse-index table

$$(1, 0, 3, 2, 5, 4, 7, 6). \quad (8.4)$$

Its group value is obtained in the presentation (4.2), and its canonical value is the shortlex representative fixed in Statement 4.5.

Proposition 8.2.2. *The tuple (8.3) is inverse closed, contains eight distinct nonidentity elements, and generates Γ_B . It is a word-coordinate shell, not a side-adjacency declaration.*

Proof. Inverse closure is built into adjacent pairs in (8.3). The four positive letters generate by presentation (4.2). If one were trivial or two signed letters equal, their images in the abelianization

$$H_1(\Sigma_B; \mathbb{Z}) \cong \mathbb{Z}^4$$

would make one of the eight distinct vectors $\{\pm e_1, \dots, \pm e_4\}$ equal to zero or to another; this is impossible. Thus the eight group elements are distinct and nonidentity. Nothing in this argument asserts that they take P to all eight side-sharing tiles; the adjacency-label uniqueness of Statement 2.6 together with Statement 4.3 proves that assertion false. $\square$

The one-letter word $[a_2]$ is a positive serialization test and evaluates, by (4.6), as $h_1 h_0 h_2$. The two-letter list $[a_2, a_2^{-1}]$ is the boundary test: free reduction and exact matrix evaluation both return the identity, while neither single letter does.

8.3 Exact conversion and the distinct roles

Using (4.3) and the inverse Nielsen dictionary (4.12), conversion of the physical shell to presentation words is

$$\frac{g_j}{S_8^{\text{pres}}\text{-word}} \left| \begin{array}{cccccccc} g_0 & g_1 & g_2 & g_3 & g_4 & g_5 & g_6 & g_7 \\ a_1 & b_1^{-1} & a_1^{-1}b_1^{-1}a_2 & a_1^{-1}b_1^{-1}b_2^{-1} & a_1^{-1} & b_1 & a_2^{-1}b_1a_1 & b_2b_1a_1. \end{array} \right. \quad (8.5)$$

Let w_j be the displayed word in column j . For an index word $J = j_1 \cdots j_m$, define the actual canonical conversion by

$$\text{Conv}(J) = \text{nf}(g_{j_1} \cdots g_{j_m}) = \text{nf}(\overline{w_{j_1}} \cdots \overline{w_{j_m}}).$$

Thus the canonical single-column output is $\text{nf}(\overline{w_j})$; the displayed spelling w_j itself is not assumed to be shortlex.

Proposition 8.3.1. *Table (8.5) is inverse compatible, evaluates exactly to the indicated g_j , and the map Conv is canonical. The underlying sets of S_8^{geo} and S_8^{pres} are unequal.*

Proof. The entries for g_0, g_5, g_2, g_7 are respectively h_0, h_1, h_2, h_3 ; the other four entries are their inverses, so every column follows from (4.3) and (4.12). Reversing each word and inverting each letter changes column j to column $j + 4$, proving inverse compatibility. Exact matrix evaluation followed by the projective key verifies every column independently. Applying Statement 4.5 makes the chosen output independent of the input spelling.

The first, second, fifth, and sixth columns are $a_1, b_1^{-1}, a_1^{-1}, b_1$, which lie in both signed shells. Statement 4.3 proves by exact upper-left matrix entries that none of $a_2^{\pm 1}, b_2^{\pm 1}$ is any g_j . Hence

$$S_8^{\text{geo}} \cap S_8^{\text{pres}} = \{a_1^{\pm 1}, b_1^{\pm 1}\},$$

and the two eight-element sets are unequal. The displayed presentation word lengths in (8.5) are 1, 1, 3, 3, 1, 1, 3, 3; this is a serialization statement, not an unproved claim of minimal word length. $\square$

For canonical words define $u \star v = \text{nf}(\overline{uv})$. The homomorphism property of evaluation and uniqueness of normal form give

$$\text{Conv}(JJ') = \text{Conv}(J) \star \text{Conv}(J'), \quad \text{Conv}(j + 4) = \text{nf}(\overline{w_j}^{-1}).$$

These are the executable product and inverse consistency tests for the conversion interface.

Thus conversion is legitimate, but substitution of one shell for the other is not: S_8^{geo} encodes physical side incidence, while S_8^{pres} encodes the chosen surface presentation.

8.4 The physical adjacency operator

Let $(\delta_\gamma)_{\gamma \in \Gamma_B}$ be the standard orthonormal basis of $\ell^2(\Gamma_B)$. Define first on finitely supported vectors

$$A_\infty \delta_\gamma = \sum_{j=0}^7 \delta_\gamma g_j. \quad (8.6)$$

Since each right translation is unitary, $\|A_\infty\| \leq 8$, so (8.6) extends uniquely to a bounded operator on all of $\ell^2(\Gamma_B)$.

Theorem 8.4.1 (Operator interface). *The operator A_∞ is the unweighted physical adjacency across sides. It is self-adjoint and has exactly eight nonzero entries in every matrix column. Its matrix-free action requires eight right-table lookups and eight accumulations per nonzero input coefficient.*

Proof. Statement 2.6 says that the tiles adjacent to γP are exactly $\gamma g_j P$. Thus (8.6) has precisely the required incidence pattern. For the matrix coefficient,

$$\langle \delta_\eta, A_\infty \delta_\gamma \rangle = \sum_{j=0}^7 \mathbf{1}_{\{\eta = \gamma g_j\}}. \quad (8.7)$$

The entries g_j are distinct, so the sum is either zero or one. If $\eta = \gamma g_j$, inverse closure gives $\gamma = \eta g_{j+4}$; hence (8.7) is symmetric in γ, η , proving self-adjointness. The same formula shows that there are eight entries per column. Direct evaluation of the sum gives the stated matrix-free cost. $\square$

The regression fixture on a finite carrier X with exact right-action tables $R_j : X \rightarrow X$ is fixed as follows:

$$(A_X)_{x,y} = \sum_{j=0}^7 \mathbf{1}_{\{x=R_j(y)\}}, \quad (A_X v)_x = \sum_{y,j: R_j(y)=x} v_y. \quad (8.8)$$

For every basis vector and then every test vector, explicit multiplication by the left matrix in (8.8) must equal the matrix-free accumulation on the right entry by entry. Dimension is $|X|$; symmetry is tested by $(A_X)_{x,y} = (A_X)_{y,x}$. At $X = \Gamma_B$, the fixture $A_\infty \delta_1 = \sum_j \delta_{g_j}$ checks all eight declared physical neighbors. Replacing S_8^{geo} by S_8^{pres} fails that fixture and defines a different operator.

8.5 The C_8 symmetry of physical adjacency

Define the permutation unitary

$$U_8 \delta_\gamma = \delta_{\phi_8(\gamma)}. \quad (8.9)$$

It is unitary because ϕ_8 is a bijection.

Theorem 8.5.1. *For the right-shift operation $T_g \delta_\gamma = \delta_{\gamma g}$,*

$$U_8 T_g U_8^{-1} = T_{\phi_8(g)}, \quad U_8 A_\infty U_8^{-1} = A_\infty. \quad (8.10)$$

Moreover U_8 has exact order 8.

Proof. On a basis vector,

$$\begin{aligned} U_8 T_g U_8^{-1} \delta_\gamma &= U_8 T_g \delta_{\phi_8^{-1}(\gamma)} = U_8 \delta_{\phi_8^{-1}(\gamma)g} \\ &= \delta_{\gamma \phi_8(g)} = T_{\phi_8(g)} \delta_\gamma. \end{aligned}$$

Therefore

$$U_8 A_\infty U_8^{-1} = \sum_{j=0}^7 T_{\phi_8(g_j)} = \sum_{j=0}^7 T_{g_{j+1}} = A_\infty,$$

where the last equality only reindexes the complete physical shell. Since the basis vectors span densely and all operators are bounded, the equality holds on all of $\ell^2(\Gamma_B)$. Equation $\phi_8^8 = 1$ gives $U_8^8 = I$. For every $1 \leq k < 8$,

$$U_8^k \delta_{g_0} = \delta_{\phi_8^k(g_0)} = \delta_{g_k} \neq \delta_{g_0},$$

because the eight physical generators are distinct. Thus no smaller positive power is the identity. $\square$

The generator-level covariance table $j \mapsto j+1$, the equality of the two exact operator sums, and $U_8^8 = I$ form a finite reproducible symmetry test. On an equivariant finite quotient, the identical proof uses α in place of ϕ_8 , but only $\text{ord}(U_\alpha) = \text{ord}(\alpha) \mid 8$ holds in general. If the eight-distinctness condition (8.12) holds, then

$$U_\alpha^k \delta_{q_0} = \delta_{q_k} \neq \delta_{q_0} \quad (1 \leq k < 8),$$

and the finite-quotient unitary also has exact order 8.

8.6 Inverse closure and self-adjointness

Lemma 8.6.1. *Both S_8^{geo} and S_8^{pres} are inverse closed. Consequently the unweighted right-adjacency sum defined by either shell is self-adjoint.*

Proof. For the geometric shell, (8.1) pairs index j with $j + 4$. For the presentation shell, (8.4) pairs each positive letter with its displayed inverse. If $S = S^{-1}$ and

$$A_S \delta_\gamma = \sum_{s \in S} \delta_{\gamma s},$$

then $T_s^* = T_{s^{-1}}$, so

$$A_S^* = \sum_{s \in S} T_{s^{-1}} = \sum_{s \in S} T_s = A_S.$$

□

Inverse closure alone does not identify the physical operator: $A_{S_8^{\text{pres}}}$ is self-adjoint but differs from A_∞ by Statement 8.3–Statement 8.4. It is also essential for Hermiticity. For the nonsymmetric half-shell $S^+ = \{g_0, g_1, g_2, g_3\}$,

$$\langle \delta_{g_0}, A_{S^+} \delta_1 \rangle = 1, \quad \langle A_{S^+} \delta_{g_0}, \delta_1 \rangle = 0,$$

because $g_0^{-1} = g_4 \notin S^+$. This is the required counterexample test against silently dropping the inverse half.

8.7 Descent of the physical shell

Let $K \triangleleft \Gamma_B$, let $Q = \Gamma_B/K$, and write $\pi = \pi_K$. The shell descends first as the ordered tuple

$$\mathbf{S}_Q = (q_0, \dots, q_7), \quad q_j = \pi(g_j), \quad (8.11)$$

and second as its deduplicated underlying set S_Q . Keeping both objects distinguishes a weighted quotient adjacency, which retains multiplicity, from the simple Cayley graph.

Input: an exact finite-group multiplication/inverse table with identity e_Q , and the four images $q_0, \dots, q_3$.

Construction: set $q_{j+4} = q_j^{-1}$; emit the ordered tuple $\mathbf{S}_Q$; scan it in index order to emit S_Q , an identity flag, and all collision pairs (i, j) with $i < j$ and $q_i = q_j$.

Output specification: exact table indices only; no hash equality is accepted without exact carrier equality.

Theorem 8.7.1. *The construction terminates, is reproducible, and returns an inverse-closed tuple. It has eight distinct nonidentity entries exactly when*

$$\begin{aligned} K \cap S_8^{\text{geo}} &= \emptyset, \\ K \cap \{g_i^{-1} g_j : 0 \leq i < j < 8\} &= \emptyset. \end{aligned} \quad (8.12)$$

If K is ϕ_8 -invariant and α is the descended automorphism, then $\alpha(q_j) = q_{j+1}$.

Proof. Only eight entries and twenty-eight pairs are tested, so termination is immediate. The fixed index order makes the result reproducible, and

$$q_{j+4} = \pi(g_{j+4}) = \pi(g_j^{-1}) = q_j^{-1}$$

proves inverse closure. An identity occurs precisely when $g_j \in K$. A collision $q_i = q_j$ occurs precisely when $\pi(g_i^{-1} g_j) = e_Q$, or $g_i^{-1} g_j \in K$. These equivalences prove both directions of (8.12). Finally Statement 7.8 gives

$$\alpha(q_j) = \alpha(\pi(g_j)) = \pi(\phi_8(g_j)) = \pi(g_{j+1}) = q_{j+1}.$$

□

The trivial quotient is the boundary test: all eight tuple entries collapse to e_Q , so the inverse and equivariance identities remain true but the simplicity flags fail. Any quotient injective on $\{1, g_0, \dots, g_7\}$ is a positive test with eight distinct nonidentity outputs. Thus the algorithm never confuses a legal descended multiset with a certified simple degree-eight shell.

8.8 The shell Cayley graph

Definition 8.8.1. Let the directed flags, or darts, be

$$D_Q = Q \times \mathbb{Z}/8\mathbb{Z}, \quad \text{tail}(q, j) = q, \quad \text{head}(q, j) = qq_j.$$

Inverse closure defines the fixed-point-free involution

$$\iota(q, j) = (qq_j, j + 4).$$

Indeed, $q_j q_{j+4} = e_Q$, so $\iota^2 = \text{id}$, while $j + 4 \neq j$. The undirected labelled multiedges are the two-element orbits $D_Q/\langle \iota \rangle$. Shell collisions produce parallel edges, and $q_j = e_Q$ produces a loop. Deleting loops and merging parallel edges gives the simple graph $\text{Cay}(Q, S_Q \setminus \{e_Q\})$.

Proposition 8.8.2. *If (8.12) holds, the graph is finite, simple, undirected, connected, and 8-regular. Its adjacency operator on $\mathbb{C}^Q$ is*

$$A_Q \delta_q = \sum_{j=0}^7 \delta_{qq_j}, \tag{8.13}$$

and is self-adjoint. If the quotient is C_8 -equivariant, the permutation $U_\alpha \delta_q = \delta_{\alpha(q)}$ commutes with A_Q .

Proof. Finiteness follows from $|D_Q| = 8|Q|$. The involution identifies every directed step $q \rightarrow qq_j$ with its reverse step labelled $j + 4$, so the graph is undirected. The operator (8.13) retains dart multiplicity; because the inverse dart is present, it is self-adjoint. The first condition in (8.12) excludes loops, and the second excludes two labels from reaching the same neighbor. Thus every vertex has exactly eight distinct neighbors.

The physical shell generates Γ_B , so its image generates $Q = \pi(\Gamma_B)$. A word in the q_j is precisely a path from the identity; left translation of such paths reaches every vertex, proving connectedness. Finally,

$$U_\alpha A_Q U_\alpha^{-1} \delta_q = \sum_{j=0}^7 \delta_{\alpha(\alpha^{-1}(q)q_j)} = \sum_{j=0}^7 \delta_{qq_{j+1}} = A_Q \delta_q,$$

using Statement 7.8 and reindexing the full shell. □

For a regression fixture, enumerate Q in its canonical order, build the explicit integer matrix

$$(A_Q)_{x,y} = \#\{j : x = yq_j\},$$

and compare every column with the matrix-free rule (8.13). In the simple case each entry is 0 or 1, every row sum is 8, the matrix is symmetric, and a breadth-first search from e_Q visits $|Q|$ vertices. The trivial quotient has eight darts and four undirected loops. Each loop has two inverse darts and hence contributes 2 to the diagonal of the weighted operator, giving the one-by-one matrix (8). This explicitly exposes the loop/multiplicity boundary convention.

9 Parity and bipartite structure

9.1 The parity homomorphism

Write $C_2 = \mathbb{Z}/2\mathbb{Z}$ additively. On the free presentation generators define

$$\chi(a_1) = \chi(b_1) = \chi(a_2) = \chi(b_2) = 1. \quad (9.1)$$

For a signed word $w = s_1^{\epsilon_1} \cdots s_m^{\epsilon_m}$, with $\epsilon_i \in \{\pm 1\}$, its directly computable value is

$$\chi_F(w) = \sum_{i=1}^m \epsilon_i \pmod{2} = m \pmod{2}. \quad (9.2)$$

Theorem 9.1.1. *The free-group map χ_F descends uniquely to a surjective homomorphism*

$$\chi : \Gamma_B \longrightarrow C_2. \quad (9.3)$$

Every physical generator is odd:

$$\chi(g_j) = 1 \quad (0 \leq j < 8). \quad (9.4)$$

Proof. The exponent-sum map from the free group to C_2 is a homomorphism. For any x, y ,

$$\chi_F([x, y]) = \chi_F(x) + \chi_F(y) - \chi_F(x) - \chi_F(y) = 0.$$

Hence $\chi_F(\mathcal{R}) = 0$, so the normal closure of the defining relator lies in the kernel and the presentation universal property gives (9.3). It is onto because $\chi(a_1) = 1$.

Using (8.5), the four independent physical images have parities

$$\frac{x}{\chi(x)} \mid \begin{array}{cccc} g_0 & g_1 & g_2 & g_3 \\ 1 & -1 & -1 & -1 \end{array} \equiv (1, 1, 1, 1) \pmod{2}.$$

Inversion does not change a value in C_2 , so (9.4) also holds for $g_4, \dots, g_7$. $\square$

The fixed serialization output is one bit, computed by XOR of one bit per signed input letter. The empty word is the boundary test and returns zero; each one-letter generator is a positive odd test. Relation words must return zero before a word evaluator is accepted.

9.2 Exact relation compatibility

The defining surface relator is

$$\mathcal{R} = a_1 b_1 a_1^{-1} b_1^{-1} a_2 b_2 a_2^{-1} b_2^{-1}. \quad (9.5)$$

Define the relation checker for proposed bits $(x_1, y_1, x_2, y_2) \in C_2^4$ by

$$\begin{aligned} \text{RelParity}(x_1, y_1, x_2, y_2) &= (x_1 + y_1 - x_1 - y_1) \\ &\quad + (x_2 + y_2 - x_2 - y_2). \end{aligned} \quad (9.6)$$

Proposition 9.2.1. *The checker (9.6) is identically zero. Consequently any assignment of four bits to a_1, b_1, a_2, b_2 defines a unique homomorphism $\Gamma_B \rightarrow C_2$; assignment (9.1) is the unique one sending all four positive presentation generators to 1.*

Proof. Every term in (9.6) cancels with its additive inverse. Equivalently, C_2 is abelian and every commutator maps to its identity. Thus the only defining relator maps to zero, which is necessary and sufficient for the free-generator map to factor through (4.2). A homomorphism from a presented group is determined by its generator images, proving uniqueness. The final assertion follows because all four coordinates have been prescribed. $\square$

A program-level certificate stores the eight signed letters of (9.5), the four input bits, the eight accumulator values, and the final zero. For (9.1) the accumulator sequence is

$$1, 0, 1, 0, 1, 0, 1, 0.$$

Changing the target from an abelian group to a nonabelian carrier is a counterexample to an unchecked generalization: cancellation in (9.6) then cannot be performed by merely commuting terms.

9.3 The parity kernel

Definition 9.3.1.

$$K_\chi = \ker \chi = \{\gamma \in \Gamma_B : \chi(\gamma) = 0\}. \quad (9.7)$$

Proposition 9.3.2. *The subgroup K_χ is invariant under every automorphism ψ satisfying $\chi \circ \psi = \chi$. It is normal of index 2 and consists exactly of the elements represented by even-length words in either declared shell.*

Proof. Every kernel is normal. Surjectivity of χ and the first isomorphism theorem give $\Gamma_B/K_\chi \cong C_2$, hence index two. If $\chi\psi = \chi$ for an automorphism ψ , then $\chi(\psi\gamma) = 0$ exactly when $\chi(\gamma) = 0$, proving $\psi(K_\chi) = K_\chi$.

Equation (9.2) proves the claim for presentation words. Equation (9.4) says that every physical-shell letter also contributes one, so a physical word has value equal to its length modulo two. Since χ is a group homomorphism, any two words for the same element have the same length parity. This is parity of a spelling, not a claim that an arbitrary nonminimal spelling realizes word length. $\square$

Thus $1 \in K_\chi$, no shell generator lies in K_χ , and every product $g_i g_j$ does. These give boundary, negative, and positive membership tests with a one-bit canonical output.

No unconditional characteristic claim is intended. For example, the standard Dehn twist

$$\psi(a_1) = a_1 b_1, \quad \psi(b_1) = b_1, \quad \psi(a_2) = a_2, \quad \psi(b_2) = b_2$$

preserves the surface relation, but $a_1 b_1^{-1} \in K_\chi$ whereas $\psi(a_1 b_1^{-1}) = a_1 \notin K_\chi$.

9.4 Compatibility with the eightfold rotation

Theorem 9.4.1. *Parity is C_8 -invariant:*

$$\boxed{\chi \circ \phi_8 = \chi}. \quad (9.8)$$

Consequently $\phi_8(K_\chi) = K_\chi$, and the rotation induces the identity automorphism on $\Gamma_B/K_\chi \cong C_2$.

Proof. It is enough to check a generating set. On the physical shell,

$$\chi(\phi_8(g_j)) = \chi(g_{j+1}) = 1 = \chi(g_j)$$

for all eight indices. Hence the homomorphisms $\chi\phi_8$ and χ agree on generators and therefore on every group element. Taking inverse images of 0 proves kernel invariance. For the induced map on C_2 ,

$$\bar{\phi}_8(\chi(\gamma)) = \chi(\phi_8(\gamma)) = \chi(\gamma),$$

and surjectivity of χ makes $\bar{\phi}_8$ the identity. $\square$

As an independent presentation-level consistency table, the four rows of (7.9) have respectively 1, 3, 7, 5 signed letters and are therefore all odd. This verifies (9.8) without using the geometric-shell table and makes the serialized compatibility test reproducible.

9.5 The parity factor of a finite quotient

Let $\pi : \Gamma_B \rightarrow Q$ have kernel K , and put $q_j = \pi(g_j)$. A parity factor is a homomorphism $\epsilon : Q \rightarrow C_2$ satisfying

$$\epsilon \circ \pi = \chi. \quad (9.9)$$

It can be constructed, or disproved, by the following finite algorithm.

Input: the exact ordered multiplication table of Q , its identity e_Q , and the ordered shell tuple $(q_0, \dots, q_7)$.

Procedure: set $\epsilon(e_Q) = 0$, store the empty predecessor word $p(e_Q)$, and enqueue e_Q . Remove elements in queue order. For each removed x and $j = 0, \dots, 7$, put $y = xq_j$ and propose $\epsilon(y) = \epsilon(x) + 1$. If unset, store the proposed bit and $p(y) = p(x)j$, then enqueue y . If the stored bit differs, return the two predecessor words and their evaluated parities.

Output: either the complete one-bit table ϵ , or, at a conflicting update from x through label j , the words

$$u = p(x)j, \quad v = p(y),$$

with $\pi(u) = \pi(v) = y$ and $\chi(u) \neq \chi(v)$.

Theorem 9.5.1. *The algorithm terminates and returns a parity table if and only if*

$$K \subseteq K_\chi. \quad (9.10)$$

When it returns a table, that table is the unique homomorphism satisfying (9.9).

Proof. The shell images generate Q , so the breadth-first search reaches every element, and each is enqueued once. Thus it terminates. Along a stored path labelled by a physical word w , the update rule gives

$$\epsilon(\pi(w)) = |w| \bmod 2 = \chi(w). \quad (9.11)$$

If a conflict supplies u, v , then $uv^{-1} \in K$ but

$$\chi(uv^{-1}) = \chi(u) - \chi(v) = 1,$$

so (9.10) fails. Conversely, if $k \in K$ has $\chi(k) = 1$, a physical word for k is an odd closed path at e_Q . Traversing this path alternates the proposed bit and returns with bit one, so no globally consistent table can pass all its edges; the finite search must expose a conflict. Thus every negative output contains the independently verifiable witness

$$\kappa = uv^{-1} \in K \setminus K_\chi.$$

Assume no conflict. Equation (9.11) is independent of the representing word. If $x = \pi(u)$ and $y = \pi(v)$, then

$$\epsilon(xy) = \epsilon(\pi(uv)) = \chi(uv) = \chi(u) + \chi(v) = \epsilon(x) + \epsilon(y),$$

so ϵ is a homomorphism and (9.9) holds. Surjectivity of π makes such a factor unique. $\square$

The trivial quotient fails: its shell entries are e_Q , so the first edge proposes $\epsilon(e_Q) = 1$ against the stored zero. The quotient $\Gamma_B/K_\chi \cong C_2$ is the minimal positive test and returns the identity bit table on C_2 . Fixed queue and shell orders make either output reproducible.

9.6 Bipartite Cayley graphs

Theorem 9.6.1 (Parity–bipartiteness equivalence). *For a finite quotient Q , the physical labelled Cayley multigraph is bipartite if and only if a parity factor $\epsilon : Q \rightarrow C_2$ exists. When it exists, the two parts are*

$$Q_0 = \epsilon^{-1}(0), \quad Q_1 = \epsilon^{-1}(1), \quad (9.12)$$

and every physical edge joins the two different parts.

Proof. If ϵ exists, then for every dart $q \rightarrow qq_j$,

$$\epsilon(qq_j) = \epsilon(q) + \epsilon(q_j) = \epsilon(q) + 1.$$

Thus no edge has both endpoints in the same part, proving bipartiteness.

Conversely, the shell images generate Q , so the graph is connected. Fix the color of e_Q as zero. In a bipartite graph, every path from e_Q to a fixed vertex has the same length parity: two paths of different parity would concatenate, after reversing one, to an odd closed walk, which a bipartite graph cannot contain. Define $\epsilon(q)$ to be this path parity. A word for x followed by a word for y , translated from e_Q to start at x , is a path to xy ; therefore

$$\epsilon(xy) = \epsilon(x) + \epsilon(y).$$

Each one-edge path gives $\epsilon(q_j) = 1$, so $\epsilon\pi = \chi$ on the physical generators and hence on all of Γ_B . $\square$

This equivalence includes loops correctly: a quotient shell element equal to e_Q creates a one-edge loop and makes bipartiteness impossible. The two-vertex quotient C_2 , with all eight labelled edges crossing between the vertices, is a positive multigraph test; the trivial quotient is the loop counterexample.

9.7 Uniform and staggered states

Assume a parity factor exists. Since $\epsilon(q_j) = 1$ and right multiplication by q_j is a bijection of Q , it maps Q_0 bijectively onto Q_1 , so

$$|Q_0| = |Q_1| = |Q|/2. \quad (9.13)$$

Define the normalized states

$$\mathbf{u} = |Q|^{-1/2} \sum_{q \in Q} \delta_q, \quad \mathbf{s} = |Q|^{-1/2} \sum_{q \in Q} (-1)^{\epsilon(q)} \delta_q. \quad (9.14)$$

Proposition 9.7.1. *The vectors $\mathbf{u}, \mathbf{s}$ are orthonormal. If the quotient is C_8 -equivariant, then both are fixed by its permutation unitary U_α .*

Proof. Both squared norms are $|Q|^{-1}|Q| = 1$. Their inner product is

$$\langle \mathbf{u}, \mathbf{s} \rangle = |Q|^{-1}(|Q_0| - |Q_1|) = 0$$

by (9.13). Equivariance and (9.8) give, for $q = \pi(\gamma)$,

$$\epsilon(\alpha(q)) = \epsilon(\pi(\phi_8(\gamma))) = \chi(\phi_8(\gamma)) = \chi(\gamma) = \epsilon(q).$$

Thus α permutes each parity part. Reindexing the sums in (9.14) proves $U_\alpha \mathbf{u} = \mathbf{u}$ and $U_\alpha \mathbf{s} = \mathbf{s}$. $\square$

On $Q = C_2$ these vectors are respectively $2^{-1/2}(1, 1)^T$ and $2^{-1/2}(1, -1)^T$. If (9.10) fails, the sign $(-1)^{\chi(w)}$ depends on the chosen word for a quotient element, so a purported staggered state is not well defined; the negative output of Statement 9.5 supplies an explicit ambiguity.

9.8 The two extremal adjacency modes

Order the basis of $\mathbb{C}^Q$ with Q_0 first and Q_1 second. For the weighted physical adjacency (8.13), every shell entry changes parity, hence

$$A_Q = \begin{pmatrix} 0 & B \\ B^* & 0 \end{pmatrix}, \quad J = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \quad JA_QJ = -A_Q. \quad (9.15)$$

Theorem 9.8.1. *The uniform and staggered states are exact adjacency eigenmodes:*

$$A_Q \mathbf{u} = 8\mathbf{u}, \quad A_Q \mathbf{s} = -8\mathbf{s}. \quad (9.16)$$

The spectrum is symmetric under $\lambda \mapsto -\lambda$. Because the physical Cayley multigraph is connected, each of the eigenvalues 8 and -8 is simple.

Proof. Every basis vertex has eight labelled outgoing darts, so the coefficient of δ_x in $A_Q \mathbf{u}$ is $8|Q|^{-1/2}$. For the staggered state, every predecessor xq_j^{-1} has parity $\epsilon(x) - 1$, and therefore

$$\begin{aligned} (A_Q \mathbf{s})_x &= |Q|^{-1/2} \sum_{j=0}^7 (-1)^{\epsilon(xq_j^{-1})} \\ &= -8|Q|^{-1/2} (-1)^{\epsilon(x)} = -8\mathbf{s}_x. \end{aligned}$$

Equation (9.15) implies that if $A_Q v = \lambda v$, then $A_Q(Jv) = -\lambda(Jv)$, proving spectral symmetry.

For simplicity at 8, let $A_Q v = 8v$ and choose x with $|v_x|$ maximal. The equality

$$8|v_x| = \left| \sum_j v_{xq_j^{-1}} \right| \leq \sum_j |v_{xq_j^{-1}}| \leq 8|v_x|$$

forces all eight neighboring values to equal v_x , including phase. Connectedness propagates this equality to all vertices, so v is a multiple of $\mathbf{u}$. Multiplication by J bijects the -8 -eigenspace with the 8 -eigenspace, proving simplicity at -8 and identifying its vector as $\mathbf{s}$. $\square$

The matrix-free interface stores the eight right-action tables and applies A_Q using $8|Q|$ indexed accumulations; the explicit regression matrix uses (8.13) and must be Hermitian with the block form (9.15). For $Q = C_2$, all eight labels swap the two vertices and

$$A_Q = \begin{pmatrix} 0 & 8 \\ 8 & 0 \end{pmatrix},$$

whose normalized eigenvectors are exactly (9.14) with eigenvalues ± 8 . This fixture simultaneously tests dimensions, multiplicities, Hermiticity, block structure, matrix-free agreement, and both extremal modes.

Part III

Finite quotients and their certification

10 Finite quotients and finite covers

10.1 Finite-index normal subgroups

Definition 10.1.1. A subgroup $K \leq \Gamma_B$ has finite index if the left-coset set $\Gamma_B/K = \{\gamma K : \gamma \in \Gamma_B\}$ has finite cardinality. It is normal if

$$\gamma K \gamma^{-1} = K \quad (\gamma \in \Gamma_B). \quad (10.1)$$

A finite-index normal subgroup is recorded by a finite coset carrier, a fixed representative of every coset, an exact coset-reduction map, and certificates for (10.1) on a generating set.

Proposition 10.1.2. *For a subgroup K , it is enough to check*

$$sKs^{-1} \subseteq K \quad (s \in S_8^{\text{geo}}) \quad (10.2)$$

to prove normality. A connected finite cover associated with K is regular precisely when K is normal; in that case its deck group is Γ_B/K .

Proof. Because S_8^{geo} is inverse closed and generates Γ_B , (10.2) also gives $s^{-1}Ks \subseteq K$. Conjugating the latter inclusion by s gives $K \subseteq sKs^{-1}$, hence equality for every shell generator. Equality is preserved under products and inverses, proving (10.1) for every γ .

For the left quotient $K \backslash \mathbb{H}_R^2$, put

$$T_\gamma(Kz) = K\gamma z.$$

If $Kz = Kz'$, write $z' = kz$. Then

$$T_\gamma(Kz') = K\gamma kz = K(\gamma k \gamma^{-1})\gamma z.$$

Thus T_γ is well defined exactly when $\gamma K \gamma^{-1} = K$. Composition and kernel are

$$T_\gamma T_\delta = T_{\gamma\delta}, \quad T_\gamma = \text{id} \iff \gamma \in K.$$

Consequently the full deck group is $N_{\Gamma_B}(K)/K$. It acts transitively on every fibre exactly when $N_{\Gamma_B}(K) = \Gamma_B$, which is exactly normality; in that case it is Γ_B/K . $\square$

The whole group $K = \Gamma_B$ is the index-one boundary example and is normal. Finite index does not imply normality. Define a surjection

$$\psi : \Gamma_B \longrightarrow S_3, \quad a_1 \mapsto (12), \quad b_1 \mapsto e, \quad a_2 \mapsto (123), \quad b_2 \mapsto e. \quad (10.3a)$$

Both commutators in the surface relator map to e , so the map is a homomorphism; (12) and (123) generate S_3 , so it is surjective. With $L = \langle (12) \rangle$ and $H = \psi^{-1}(L)$, one has

$$[\Gamma_B : H] = [S_3 : L] = 3.$$

The subgroup L is not normal in S_3 ; surjectivity then implies that H is not normal in Γ_B . This is an explicit finite-index nonnormal subgroup and prevents treating an arbitrary transitive coset action as a quotient group.

10.2 Constructing $Q = \Gamma_B/K$

Assume a certified finite-index normal subgroup and an ordered transversal

$$r_0 = 1, r_1, \dots, r_{n-1},$$

together with a total exact reduction function $\text{red}_K(w) \in \{0, \dots, n-1\}$ satisfying

$$\text{red}_K(u) = \text{red}_K(v) \iff u^{-1}v \in K. \quad (10.3)$$

Input: the ordered representatives r_i , the reduction oracle (10.3), and the signed presentation generators.

Construction: set

$$i \cdot j = \text{red}_K(r_i r_j), \quad i^{-1} = \text{red}_K(r_i^{-1}), \quad \pi_K(w) = \text{red}_K(w).$$

Evaluate all table entries in lexicographic pair order.

Output: the carrier $\{0, \dots, n-1\}$, multiplication and inverse tables, identity 0, generator images, and the table checks for associativity, identity, inverses, and the defining relator.

Theorem 10.2.1 (Quotient specification). *The construction is well defined and gives a finite group*

$$Q = \Gamma_B/K, \quad |Q| = [\Gamma_B : K], \quad (10.4)$$

with a surjective homomorphism $\pi_K : \Gamma_B \rightarrow Q$ whose kernel is exactly K . The fixed transversal and evaluation order make the output reproducible.

Proof. If $\gamma K = \gamma' K$ and $\delta K = \delta' K$, write $\gamma' = \gamma k$, $\delta' = \delta l$ with $k, l \in K$. Then

$$\gamma' \delta' = \gamma k \delta l = \gamma \delta (\delta^{-1} k \delta) l,$$

and normality puts $(\delta^{-1} k \delta) l$ in K . Hence $\gamma' \delta' K = \gamma \delta K$, proving representative independence. Associativity, identity, and inverses descend from Γ_B .

The map $\pi_K(\gamma) = \gamma K$ satisfies $\pi_K(\gamma \delta) = \pi_K(\gamma) \pi_K(\delta)$, and every coset has a representative, so it is onto. Finally, $\pi_K(\gamma) = K$ exactly when $\gamma \in K$, proving the kernel statement. The number of carrier entries is the definition of the index. Equation (10.3) and fixed evaluation orders make each serialized table entry unique. $\square$

For $K = \Gamma_B$, the output is the one-element group and all generator images are zero. For $K = K_\chi$, the output is C_2 , and the four positive presentation generators all map to its nonzero entry. These are the index-one boundary and the first nontrivial regression fixtures.

10.3 Finite permutation representations

Let $H \leq \Gamma_B$ have finite index d , without assuming normality. On the left cosets $\Omega = \{\gamma H\}$, define

$$\rho_H(x)(\gamma H) = x \gamma H. \quad (10.5)$$

Theorem 10.3.1. *The map $\rho_H : \Gamma_B \rightarrow \text{Sym}(\Omega)$ is a transitive permutation representation of degree d , and*

$$\ker \rho_H = \text{core}_{\Gamma_B}(H) := \bigcap_{\gamma \in \Gamma_B} \gamma H \gamma^{-1}. \quad (10.6)$$

Proof. If $\gamma H = \gamma' H$, then $\gamma' = \gamma h$ for $h \in H$, so $x \gamma' H = x \gamma h H = x \gamma H$; thus (10.5) is well defined. Furthermore,

$$\rho_H(xy)(\gamma H) = xy \gamma H = \rho_H(x) \rho_H(y)(\gamma H),$$

and $\rho_H(x^{-1})$ is the inverse permutation. Any coset γH is the image of H under $\rho_H(\gamma)$, proving transitivity.

An element x is in the kernel exactly when it fixes every coset:

$$\begin{aligned} x \gamma H = \gamma H &\iff \gamma^{-1} x \gamma \in H \\ &\iff x \in \gamma H \gamma^{-1}. \end{aligned}$$

Intersecting over all γ gives (10.6), proving both inclusions. $\square$

Computationally, choose ordered coset representatives r_i and emit for each signed generator s the permutation array

$$\sigma_s(i) = \text{red}_H(sr_i).$$

Check $\sigma_{s^{-1}} = \sigma_s^{-1}$ and evaluate the surface relator to the identity permutation. A breadth-first orbit from H must visit all d entries. These finite tests are the canonical I/O certificate. When H is normal, $\ker \rho_H = H$; the induced faithful action of $\Gamma_B/H = \text{im } \rho_H$ on Ω , rather than the original Γ_B -action, has trivial stabilizers and is regular.

10.4 Group order versus permutation degree

Put $C = \text{core}_{\Gamma_B}(H)$. The degree of ρ_H and the order of its image are different indices:

$$d = [\Gamma_B : H], \quad |\text{im } \rho_H| = [\Gamma_B : C]. \quad (10.7)$$

Proposition 10.4.1. *The stabilizer of the base coset H in $\text{im } \rho_H$ is isomorphic to H/C , and*

$$|\text{im } \rho_H| = d |H/C|. \quad (10.8)$$

Consequently the image order equals the permutation degree if and only if H is normal.

Proof. The first isomorphism theorem applied to ρ_H gives $\text{im } \rho_H \cong \Gamma_B/C$, proving the second formula in (10.7). An element $\rho_H(x)$ fixes H exactly when $xH = H$, or $x \in H$. Its stabilizer is therefore $\rho_H(H)$, and restriction of ρ_H to H has kernel C ; hence it is H/C . Orbit–stabilizer on the transitive degree- d action now gives (10.8). Equality with d holds exactly when H/C is trivial, that is $H = C$. Since C is the largest normal subgroup contained in H , this is equivalent to normality of H . $\square$

The canonical representation record stores separately

(carrier degree d , generated permutation-group order, base stabilizer order).

The invariant check is $|\text{im } \rho_H| = d |\text{Stab}(H)|$, never $|\text{im } \rho_H| = d$ by default. The index-one action has degree and image order one. A transitive S_3 -action on three points has degree 3, image order 6, and stabilizer order 2, providing the minimal standard counterexample to conflating degree with group order.

10.5 Regular and transitive actions

Definition 10.5.1. An action is transitive if it has one orbit. It is free if every point stabilizer is trivial, and regular if it is both transitive and free.

The right action of a finite group Q on itself,

$$q \cdot x = qx, \quad (10.9)$$

is regular: $e_Q \cdot x = x$ gives transitivity from e_Q , and $q \cdot x = q$ implies $x = e_Q$. If $R_x(q) = qx$, the exact composition convention is

$$(q \cdot x) \cdot y = q \cdot (xy), \quad R_y \circ R_x = R_{xy}. \quad (10.9a)$$

Thus $x \mapsto R_x$ is an antihomomorphism under the usual left-to-right notation for function products, whereas $x \mapsto R_{x^{-1}}$ is a homomorphism $Q \rightarrow \text{Sym}(Q)$. For $L \leq Q$, the left action on left cosets,

$$x \cdot (qL) = xqL, \quad (10.10)$$

is transitive of degree $[Q : L]$.

Proposition 10.5.2. *The kernel of (10.10) is*

$$\text{core}_Q(L) = \bigcap_{q \in Q} qLq^{-1}. \quad (10.11)$$

Thus the coset action is faithful exactly when this core is trivial, and it is regular exactly when $L = \{e_Q\}$.

Proof. The kernel calculation in Statement 10.3 applies verbatim with Q, L in place of Γ_B, H , giving (10.11) and the faithfulness equivalence. The stabilizer of the base coset L is L itself. In a transitive action all stabilizers are conjugate, so they are all trivial exactly when $L = \{e_Q\}$, which is regularity. $\square$

For exact serialization, order the carrier and write every generator permutation as the array of its images in that order. Composition is checked entry by entry using (10.9a), inverse arrays must compose to the identity array, and a breadth-first orbit must reach the full carrier. The one-point action of the trivial group is regular. For nontrivial Q , the one-point coset action on Q/Q is transitive but neither faithful nor free, hence not regular. The natural S_3 -action on three points is transitive and faithful but not regular because a point stabilizer has order two; these examples test all three notions.

10.6 The finite covering surface

Let $K \triangleleft \Gamma_B$ have finite index n , and define

$$X_K = K \backslash \mathbb{H}_R^2, \quad p_K : X_K \longrightarrow \Sigma_B, \quad Kz \longmapsto \Gamma_B z. \quad (10.12)$$

Theorem 10.6.1. *The space X_K is a compact smooth hyperbolic surface. The map p_K is a regular unbranched covering of degree $n = |Q|$, with deck group $Q = \Gamma_B/K$. Its area and genus are*

$$\text{area}(X_K) = 4\pi R^2 |Q|, \quad g(X_K) = |Q| + 1. \quad (10.13)$$

Proof. The group Γ_B acts freely and properly discontinuously on $\mathbb{H}_R^2$ by Statement 2.4. Its subgroup K inherits both properties, so the quotient is a smooth hyperbolic surface and the projection is locally an isometry. There are n cosets, hence n sheets over every point. A finite-sheeted cover of compact Σ_B is compact. Normality makes the cover regular, and Statement 10.1 identifies its deck group with Q .

The base surface has genus two, so Gauss–Bonnet at curvature $-R^{-2}$ gives

$$-R^{-2} \text{area}(\Sigma_B) = 2\pi(2 - 2 \cdot 2) = -4\pi,$$

or $\text{area}(\Sigma_B) = 4\pi R^2$. Area multiplies by the covering degree because p_K is locally isometric and has n sheets, giving the first formula in (10.13). Euler characteristic also multiplies:

$$2 - 2g(X_K) = n(2 - 2g(\Sigma_B)) = -2n.$$

Solving gives $g(X_K) = n + 1$. $\square$

At $n = 1$, the formulas recover Σ_B , area $4\pi R^2$, and genus two. For the parity cover $n = 2$, they give area $8\pi R^2$ and genus three. Any use of these formulas for a nonnormal H still gives a degree- $[\Gamma_B : H]$ cover and the same area/genus scaling, but its deck group is generally $N_{\Gamma_B}(H)/H$, not a quotient Γ_B/H .

10.7 The quotient Cayley graph

Let $\mathcal{G}_\infty = \text{Cay}(\Gamma_B, S_8^{\text{geo}})$ use the right-labelled darts $(\gamma, j) : \gamma \rightarrow \gamma g_j$. For $Q = \Gamma_B/K$, construct the finite dart graph of Statement 8.8 from

$$q_j = \pi_K(g_j), \quad D_Q = Q \times \mathbb{Z}/8, \quad \iota(q, j) = (qq_j, j + 4). \quad (10.14)$$

The quotient map is

$$\Pi_K(\gamma) = \pi_K(\gamma), \quad \Pi_K(\gamma, j) = (\pi_K(\gamma), j). \quad (10.15)$$

Theorem 10.7.1. *The map (10.15) preserves tails, heads, labels, and the inverse-dart involution, and is locally bijective on the eight labelled darts at every vertex. After loops are deleted and parallel edges merged, it is locally bijective on the simple neighbor sets if and only if the two exclusions (8.12) hold.*

Proof. For heads,

$$\Pi_K(\gamma g_j) = \pi_K(\gamma)\pi_K(g_j) = \pi_K(\gamma)q_j,$$

which is the head of $(\pi_K(\gamma), j)$. Also

$$\Pi_K(\gamma g_j, j+4) = (\pi_K(\gamma)q_j, j+4) = \iota(\pi_K(\gamma), j),$$

so inverse darts are preserved. At a fixed vertex, the eight outgoing darts are indexed by the same set $j = 0, \dots, 7$ in source and target; the label-preserving map is therefore a bijection even when some heads coincide.

For simple neighbor sets, a loop occurs exactly when $q_j = e_Q$, and two labels have the same head exactly when $q_i = q_j$. Statement 8.7 proves that absence of all such events is equivalent to (8.12). Under that condition the eight labelled darts have eight distinct other endpoints, so the simple local neighbor map is bijective. On the source graph, Statement 2.3 and Statement 8.4 give $g_j \neq 1$ and $g_i \neq g_j$ for $i \neq j$, so its simple neighbor set has exactly eight elements. If $q_j = e_Q$, the corresponding source neighbor maps to the base vertex and disappears on simplification; if $q_i = q_j$, two distinct source neighbors have the same image. Hence either failed exclusion destroys local bijectivity, which completes the converse. $\square$

Graph-construction specification: enumerate $q \in Q$ and $j = 0, \dots, 7$; emit the dart (q, j) , head qq_j , and inverse pointer $(qq_j, j+4)$. Verify every inverse pointer twice returns the original dart. Emit separately the loop list, collision list, labelled multigraph, and simplified graph. The cost is $8|Q|$ multiplications plus the fixed collision checks of Statement 8.7.

For the trivial quotient, the labelled graph has one vertex, eight darts, and four undirected loops; its simplified graph has no edges, so the two local notions visibly differ. Any quotient satisfying (8.12) yields the positive fixture: a connected simple 8-regular graph whose explicit adjacency agrees with (8.13).

10.8 Finite covers as physical finite systems

Index the $n = |Q|$ lifted tile centers in the regular cover X_K by $q \in Q$. With two layer labels, the one-orbital-per-center Hilbert space is

$$\mathcal{H}_K = \mathbb{C}^2 \otimes \ell^2(Q), \quad \dim \mathcal{H}_K = 2|Q|. \quad (10.16)$$

The certified intralayer physical adjacency is

$$H_{\text{intra}} = \begin{pmatrix} A_Q & 0 \\ 0 & A_Q \end{pmatrix}, \quad (10.17)$$

or the same block pattern with independently specified real hopping weights $t_{\ell,j}$ that satisfy

$$t_{\ell,j+4} = t_{\ell,j} \in \mathbb{R}. \quad (10.17a)$$

This inverse-direction condition guarantees Hermiticity. Full C_8 -invariance additionally requires $t_{\ell,j+1} = t_{\ell,j}$; without it, rotation gives covariance to the cyclically permuted parameter array rather than invariance at fixed parameters.

Proposition 10.8.1 (Finite-system interface). *Normality of K gives the left regular deck action*

$$L_x \delta_q = \delta_{xq} \quad (x, q \in Q)$$

on each layer, and $L_x A_Q = A_Q L_x$. Shell descent gives the exact right-neighbor tables used in (10.17). If the quotient is C_8 -equivariant or parity compatible, the corresponding operators U_α and J satisfy the covariance relations proved in C08–C09 after lifting them as

$$\tilde{L}_x = I_2 \otimes L_x, \quad \tilde{U}_\alpha = I_2 \otimes U_\alpha, \quad \tilde{J} = I_2 \otimes J_Q. \quad (10.18)$$

Here $J_Q \delta_q = (-1)^{\epsilon(q)} \delta_q$ is the single-layer parity operator J of (9.15). The two conditional statements are understood respectively: C_8 -equivariance supplies $\tilde{U}_\alpha$, while parity compatibility supplies $\tilde{J}$.

Proof. The deck group is Q by Statement 10.6, so $L_x L_y = L_{xy}$ and every L_x is a permutation unitary. Left and right multiplication commute:

$$L_x A_Q \delta_q = \sum_j \delta_{x(qq_j)} = A_Q L_x \delta_q.$$

Thus translations are represented exactly. The remaining statements are the finite-shell construction of Statement 8.7, its unweighted C_8 covariance from Statement 8.8, and the parity block relation (9.15), tensored with the two-layer identity as displayed in (10.18). $\square$

This interface does not by itself authorize interlayer edges chosen from geometric cutoffs. Such a term needs a separately declared displacement rule, proof that it is independent of lifts, and an injectivity-radius bound preventing unintended wraparound. Likewise, eight distinct physical directions require (8.12); normality and finiteness alone do not imply it.

As a dimension fixture, the index-one cover gives $\dim \mathcal{H}_K = 2$ but collapses every physical shell direction to a loop, so it is a boundary example rather than a valid eight-neighbor sample. A quotient passing (8.12) gives the positive regression: two $|Q| \times |Q|$ self-adjoint adjacency blocks, each matching the matrix-free rule entry by entry, and total dimension exactly $2|Q|$.

11 Admissibility contracts for finite quotients

11.1 Preservation of the surface relation

Let Q be a finite group with exact multiplication and inverse tables, and let $A_1, B_1, A_2, B_2 \in Q$. Define

$$\text{Rel}_Q(A_1, B_1, A_2, B_2) = [A_1, B_1][A_2, B_2]. \quad (11.1)$$

Criterion 11.1.1 (Marked-quotient relation). *The assignment*

$$a_i \mapsto A_i, \quad b_i \mapsto B_i$$

extends to a unique homomorphism $\pi : \Gamma_B \rightarrow Q$ if and only if

$$\text{Rel}_Q(A_1, B_1, A_2, B_2) = e_Q. \quad (11.2)$$

The homomorphism is surjective if and only if $\langle A_1, B_1, A_2, B_2 \rangle = Q$.

Proof. The free group on a_1, b_1, a_2, b_2 admits a unique homomorphism with the four prescribed images. It factors through the quotient presentation (4.2) exactly when its defining relator belongs to the kernel, which is precisely (11.2). This proves necessity, sufficiency, and uniqueness. The image of the resulting homomorphism is the subgroup generated by the four generator images, proving the separate surjectivity equivalence. $\square$

The exact checker evaluates, in this order,

$$\begin{aligned} C_1 &= A_1 B_1 A_1^{-1} B_1^{-1}, \\ C_2 &= A_2 B_2 A_2^{-1} B_2^{-1}, \\ R_Q &= C_1 C_2, \end{aligned}$$

and accepts the relation exactly when the carrier index of R_Q is that of e_Q . It then performs closure from e_Q under the eight signed images and accepts surjectivity exactly when every carrier element is visited. The all-identity tuple passes (11.2) but fails surjectivity for every nontrivial Q , which is the minimal counterexample to conflating the two tests.

11.2 Eight distinct physical directions

For the homomorphism in Statement 11.1, evaluate the physical generators by (8.5) and write $q_j = \pi(g_j)$. Define the finite dangerous set

$$D_{\text{shell}} = \{g_j : 0 \leq j < 8\} \cup \{g_i^{-1} g_j : 0 \leq i < j < 8\}. \quad (11.3)$$

Exact group keys are used to deduplicate this set, while one stored word and its source indices are retained for every key.

Criterion 11.2.1. *The quotient has eight distinct nonidentity physical directions if and only if*

$$K \cap D_{\text{shell}} = \emptyset, \quad K = \ker \pi. \quad (11.4)$$

Proof. If $g_j \in K$, then $q_j = e_Q$. Conversely $q_j = e_Q$ implies $g_j \in \ker \pi = K$. Likewise,

$$\begin{aligned} q_i = q_j &\iff q_i^{-1} q_j = e_Q \\ &\iff \pi(g_i^{-1} g_j) = e_Q \\ &\iff g_i^{-1} g_j \in K. \end{aligned}$$

Thus avoiding the first part of (11.3) is equivalent to nonidentity, and avoiding the second is equivalent to pairwise distinctness. This proves both directions of (11.4). $\square$

The verifier emits the ordered eight carrier indices, eight identity flags, and all twenty-eight equality flags. On failure it emits the corresponding word g_j or $g_i^{-1} g_j$, whose exact nonidentity in Γ_B is certified by Statement 2.7 and Statement 4.8, and whose quotient value is e_Q . The trivial quotient fails every identity flag; a quotient satisfying (8.12) is the positive fixture.

11.3 Relations between inverse directions

Theorem 11.3.1. *Every homomorphism $\pi : \Gamma_B \rightarrow Q$ satisfies*

$$q_{j+4} = q_j^{-1} \quad (j \in \mathbb{Z}/8). \quad (11.5)$$

If Statement 11.2 holds, the eight directed labels form four disjoint inverse pairs, none of which is an involution:

$$\{q_0, q_4\}, \{q_1, q_5\}, \{q_2, q_6\}, \{q_3, q_7\}. \quad (11.6)$$

Proof. Equation $g_{j+4} = g_j^{-1}$ holds in Γ_B . Applying the homomorphism and using preservation of inverses gives

$$q_{j+4} = \pi(g_{j+4}) = \pi(g_j^{-1}) = \pi(g_j)^{-1} = q_j^{-1}.$$

If two pairs in (11.6) met, two shell entries would be equal, contradicting Statement 11.2. If q_j were an involution, then $q_j = q_j^{-1} = q_{j+4}$, the same contradiction. Nonidentity is already part of Statement 11.2. $\square$

The machine specification checks all eight equations (11.5), not only the first four, and checks that the inverse-index array is $(4, 5, 6, 7, 0, 1, 2, 3)$. An arbitrary eight-tuple satisfying the surface relator but not (11.5) is not the image of the physical shell, even if it generates Q ; this is the boundary between presentation data and geometric-shell data.

11.4 C_8 -equivariance

Criterion 11.4.1. *Let $\pi : \Gamma_B \twoheadrightarrow Q$ be surjective and put $K = \ker \pi$. There exists an automorphism $\alpha \in \text{Aut}(Q)$ satisfying*

$$\alpha\pi = \pi\phi_8 \quad (11.7)$$

if and only if $\phi_8(K) = K$. It is then unique, $\alpha^8 = \text{id}_Q$, and

$$\alpha(q_j) = q_{j+1}. \quad (11.8)$$

Proof. This is the descent theorem Statement 7.8: kernel invariance makes $\alpha(\pi(\gamma)) = \pi(\phi_8(\gamma))$ independent of the lift; the inverse is induced by ϕ_8^{-1} . Conversely, (11.7) sends every $k \in K$ back into K , and finite order upgrades inclusion to equality. Surjectivity of π proves uniqueness. Applying (11.7) eight times gives $\alpha^8\pi = \pi$, hence $\alpha^8 = 1$, and evaluation on g_j gives (11.8). $\square$

The certificate stores the complete permutation array of α in the fixed carrier order. It checks multiplication preservation on every ordered pair, bijectivity, the eight identities (11.8), and $\alpha^8(q) = q$ for every q . Cycle decomposition gives

$$\text{ord}(\alpha) = \text{lcm}\{\text{cycle lengths of } \alpha\} \mid 8. \quad (11.9)$$

Thus exact order eight is equivalent to the presence of an eight-cycle. If Statement 11.2 holds, $q_0, \dots, q_7$ themselves form such a cycle. The trivial quotient is the boundary example: equivariance descends, but the induced order is one.

11.5 Parity factorization

Criterion 11.5.1. *Let $\pi : \Gamma_B \twoheadrightarrow Q$ be surjective and put $K = \ker \pi$. A homomorphism $\epsilon : Q \rightarrow C_2$ with*

$$\chi = \epsilon\pi \quad (11.10)$$

exists if and only if

$$K \subseteq K_\chi. \quad (11.11)$$

When it exists, it is unique and satisfies $\epsilon(q_j) = 1$ for all physical directions.

Proof. If (11.10) holds and $k \in K$, then $\chi(k) = \epsilon(e_Q) = 0$, proving (11.11). Conversely, under (11.11), define $\epsilon(\pi(\gamma)) = \chi(\gamma)$. If two lifts γ, δ represent the same quotient element, then $\gamma\delta^{-1} \in K \subseteq K_\chi$, so $\chi(\gamma) = \chi(\delta)$. Hence the definition is well defined. It is a homomorphism because χ is, and it satisfies (11.10). Surjectivity of π proves uniqueness; (9.4) gives the shell values. $\square$

The deterministic verifier is the parity breadth-first search of Statement 9.5. Its positive output is the complete bit table ϵ ; its negative output contains

$$\kappa = uv^{-1} \in K, \quad \chi(\kappa) = 1,$$

which directly disproves (11.11). Parity compatibility and C_8 -equivariance are logically independent kernel conditions, although when both hold (9.8) implies $\epsilon\alpha = \epsilon$.

11.6 Injectivity on a finite local ball

Let $\emptyset \neq B \subset \Gamma_B$ be a finite set supplied in canonical exact-key order. Examples include a closed word ball for a declared shell or the closed based-geometric ball enumerated in Statement 3.7. Define

$$B^{-1}B = \{y^{-1}x : x, y \in B\}. \quad (11.12)$$

Criterion 11.6.1 (Difference-set test). *The restriction $\pi|_B$ is injective if and only if*

$$K \cap (B^{-1}B) = \{1\}. \quad (11.13)$$

Proof. For $x, y \in B$,

$$\begin{aligned} \pi(x) = \pi(y) &\iff \pi(y)^{-1}\pi(x) = e_Q \\ &\iff \pi(y^{-1}x) = e_Q \\ &\iff y^{-1}x \in K. \end{aligned}$$

If the restriction is injective, a difference in K forces $x = y$ and hence $y^{-1}x = 1$, proving (11.13). Conversely, (11.13) turns every equality of quotient images into $y^{-1}x = 1$, so $x = y$. $\square$

Complete scan: enumerate all ordered pairs $(y, x) \in B^2$; compute the exact key of $y^{-1}x$; deduplicate by the full lossless key while retaining the complete source-pair list for each key; sort keys canonically; evaluate each distinct element in Q . SATISFIED exactly when the only key mapping to e_Q is the identity key. On VIOLATED, emit the least offending key and the least pair $x \neq y$ that produced it.

The record records the input boundary convention, $|B|$, the $|B|^2$ ordered-pair count, the number of distinct difference keys, all source-pair multiplicities, and the quotient result of every key. These data prove coverage, absence of duplicate evaluations, and recoverability. For $B = \{1\}$, the scan passes every quotient. If $B = \{1, g_0\}$ and $g_0 \in K$, the element g_0 is a independently verifiable collision witness and the scan fails.

11.7 The global systole condition

Let $K \triangleleft \Gamma_B$ have finite index, set $Q = \Gamma_B/K$, and let $\pi : \Gamma_B \twoheadrightarrow Q$ be the canonical projection. Then $X_K = K \backslash \mathbb{H}_R^2$ is a compact hyperbolic surface. Every nonidentity element of K is hyperbolic, and

$$\ell_h(k) = \inf_{z \in \mathbb{H}_R^2} d_{\mathbb{H}_R^2}(z, kz) = 2R \operatorname{arcosh} \frac{|\operatorname{tr} k^\sharp|}{2}, \quad (11.14)$$

where $k^\sharp$ is either $\operatorname{SU}(1, 1)$ lift.

Definition 11.7.1. Let

$$D_P = \max_{x \in P} d_{\mathbb{H}_R^2}(o, x) = r_{\text{out}}$$

be the fundamental-polygon outer-radius constant from Statement 3.4. For $L \geq 0$, define the finite covering list

$$\begin{aligned} \operatorname{sys}(X_K) &:= \min_{1 \neq k \in K} \ell_h(k), \\ \tilde{\mathcal{D}}_{\text{sys}}(L) &:= \{\gamma \in B_{\text{geo}}(L + 2D_P) : \gamma \neq 1, \ell_h(\gamma) \leq L\}. \end{aligned} \quad (11.15)$$

The minimum exists because closed geodesic lengths on the compact surface X_K form a discrete positive set. The based ball in (11.15) is finite by proper discontinuity, hence $\tilde{\mathcal{D}}_{\text{sys}}(L)$ is finite. It is intentionally a covering list, not an unproved canonical list of conjugacy classes.

Criterion 11.7.2 (Complete dangerous-class test).

$$\boxed{\operatorname{sys}(X_K) > L \iff K \cap \tilde{\mathcal{D}}_{\text{sys}}(L) = \emptyset.} \quad (11.16)$$

Proof. If the intersection contains d , then $d \neq 1$ and

$$\text{sys}(X_K) \leq \ell_h(d) \leq L,$$

so the left side fails. Conversely, suppose $\text{sys}(X_K) \leq L$, and choose $1 \neq k \in K$ attaining the minimum. The conjugation argument of Statement 3.4 produces $\eta \in \Gamma_B$ such that $d = \eta k \eta^{-1}$ obeys

$$d_{\mathbb{H}_R^2}(o, do) \leq \ell_h(k) + 2D_P \leq L + 2D_P. \quad (11.16a)$$

Translation length is conjugacy invariant, so $\ell_h(d) = \ell_h(k) \leq L$, and $d \neq 1$. Hence $d \in \tilde{D}_{\text{sys}}(L)$. Normality gives $\eta K \eta^{-1} = K$, so $d \in K$. The intersection is nonempty, which proves the converse and the criterion. $\square$

The inequality is strict: a class with $\ell_h(d) = L$ is dangerous. A proof-producing scan first enumerates the complete exact-key-ordered ball $B_{\text{geo}}(L + 2D_P)$ by Statement 3.7, then filters every entry using $\gamma \neq 1$ and $\ell_h(\gamma) \leq L$. It stores each retained key and exact matrix, both comparison certificates, every quotient value $\pi(\gamma)$, and a hash of the ordered list. SATISFIED means all values are nonidentity; VIOLATED emits the least retained γ mapping to e_Q . Completeness follows from (11.16a), with no conjugacy decision or selector required.

Two exact-comparison interfaces are declared separately: one decides $d_{\mathbb{H}_R^2}(o, \gamma o) \leq L + 2D_P$ during the based-ball enumeration, and one decides $\ell_h(\gamma) \leq L$ during filtering. Each threshold is represented in an effectively ordered exact extension as in Statement 3.7. If either boundary comparison is undecided, the result is UNDECIDED, not SATISFIED or VIOLATED.

11.8 The no-wraparound condition

Let $\rho_c \geq 0$ be the largest closed-ball radius used by a declared local observable, coupling stencil, or correlation calculation. For a compact hyperbolic surface,

$$r_{\text{inj}}(X_K) = \frac{1}{2} \text{sys}(X_K). \quad (11.17)$$

Let

$$p : \mathbb{H}_R^2 \longrightarrow X_K, \quad p(y) = Ky,$$

be the universal quotient covering projection.

Criterion 11.8.1 (No wraparound).

$$p|_{\overline{B}(x, \rho_c)} \text{ is injective for every } x \in \mathbb{H}_R^2 \iff \text{sys}(X_K) > 2\rho_c. \quad (11.18)$$

Proof. Assume first that the right side holds. Take $y, z \in \overline{B}(x, \rho_c)$ and suppose $p(y) = p(z)$. By the definition of the quotient, $z = ky$ for some $k \in K$. The triangle inequality gives

$$\begin{aligned} d_{\mathbb{H}_R^2}(y, ky) &= d_{\mathbb{H}_R^2}(y, z) \\ &\leq d_{\mathbb{H}_R^2}(y, x) + d_{\mathbb{H}_R^2}(x, z) \\ &\leq \rho_c + \rho_c = 2\rho_c. \end{aligned} \quad (11.19)$$

If $k \neq 1$, then

$$\text{sys}(X_K) \leq \ell_h(k) \leq d_{\mathbb{H}_R^2}(y, ky) \leq 2\rho_c,$$

contradicting (11.18). Hence $k = 1$, and $y = z$.

Conversely, suppose $\text{sys}(X_K) \leq 2\rho_c$. Choose a systolic element $1 \neq k \in K$, take y on its axis, put $z = ky$, and let x be the midpoint of the axis segment $[y, z]$. Then

$$d_{\mathbb{H}_R^2}(x, y) = d_{\mathbb{H}_R^2}(x, z) = \frac{1}{2} \ell_h(k) = \frac{1}{2} \text{sys}(X_K) \leq \rho_c. \quad (11.19a)$$

Thus $y, z \in \overline{B}(x, \rho_c)$, while $y \neq z$ and $p(y) = p(z)$. Injectivity on every such closed ball fails. $\square$

In particular, equality is unsafe for closed balls: the endpoints in (11.19a) lie exactly on the boundary.

An independent verifier runs Statement 11.7 with the complete list

$$\tilde{\mathcal{D}}_{\text{wrap}}(\rho_c) := \tilde{\mathcal{D}}_{\text{sys}}(2\rho_c). \quad (11.20)$$

It returns SATISFIED exactly when every listed representative has nonidentity image in Q ; it returns VIOLATED with a kernel representative otherwise; and it returns UNDECIDED if an exact boundary comparison cannot be certified. Its certificate serializes ρ_c , the closed-boundary convention, the ordered dangerous list, the certified based-displacement comparisons regenerating $B_{\text{geo}}(2\rho_c + 2D_P)$, the certified translation-length comparisons used in filtering, every quotient value, and hashes of the input and output. Independent independent verification regenerates the list, checks coverage, and recomputes every quotient value.

Statement 11.6 and Statement 11.8 are different. The former tests one declared finite set of group elements. The latter excludes every short closed geodesic and proves injectivity of every radius- ρ_c metric ball, regardless of its centre.

11.9 Completeness of a finite-group representation list

Fix the complete multiplication table of the finite quotient Q . A candidate record contains d_ρ , matrices $\rho(q) \in \text{Mat}_{d_\rho}(\mathbb{C})$ in carrier order, and the character $\chi_\rho(q) = \text{tr } \rho(q)$. The checker first verifies

$$\rho(e_Q) = I, \quad \rho(qr) = \rho(q)\rho(r) \quad (q, r \in Q). \quad (11.21)$$

Every finite-group representation over $\mathbb{C}$ can be unitarized by averaging an inner product. Therefore it is irreducible exactly when its commutant

$$\mathcal{C}_\rho = \{T : T\rho(q) = \rho(q)T \text{ for every } q \in Q\} \quad (11.22)$$

has complex dimension one. One implication is Schur's lemma. For the other, a nonzero proper invariant subspace in a unitary realization has a nonscalar orthogonal projection commuting with every $\rho(q)$.

For class functions set

$$\langle \chi, \psi \rangle_Q := \frac{1}{|Q|} \sum_{q \in Q} \chi(q) \overline{\psi(q)}. \quad (11.23)$$

Criterion 11.9.1 (Irreducible-list completeness). *Let $\mathcal{R}$ be a finite list satisfying (11.21) and*

$$\dim_{\mathbb{C}} \mathcal{C}_\rho = 1 \quad (\rho \in \mathcal{R}). \quad (11.22a)$$

It contains exactly one representative of every equivalence class of irreducible complex representations of Q if and only if

$$\langle \chi_\rho, \chi_\sigma \rangle_Q = \delta_{\rho\sigma} \quad (\rho, \sigma \in \mathcal{R}), \quad (11.24)$$

and

$$\boxed{\sum_{\rho \in \mathcal{R}} d_\rho^2 = |Q|.} \quad (11.25)$$

Proof. For a complete list, character orthogonality gives (11.24). The left regular representation has character

$$\chi_{\text{reg}}(q) = \begin{cases} |Q|, & q = e_Q, \\ 0, & q \neq e_Q. \end{cases} \quad (11.26)$$

For every irreducible ρ ,

$$\begin{aligned}\langle \chi_{\text{reg}}, \chi_\rho \rangle_Q &= \frac{1}{|Q|} \left(|Q| \overline{\chi_\rho(e_Q)} + \sum_{q \neq e_Q} 0 \cdot \overline{\chi_\rho(q)} \right) \\ &= \overline{d_\rho} = d_\rho.\end{aligned}\tag{11.27}$$

Thus ρ occurs in the regular representation with multiplicity d_ρ , and comparison of dimensions gives (11.25).

Conversely, assume (11.24)–(11.25). Equation (11.24) makes the listed irreducibles pairwise inequivalent. Complete reducibility decomposes the regular representation. By (11.27), the listed isotypic summands have total dimension

$$\sum_{\rho \in \mathcal{R}} d_\rho \dim \rho = \sum_{\rho \in \mathcal{R}} d_\rho^2 = |Q| = \dim \lambda_Q.\tag{11.28}$$

No dimension remains for an unlisted irreducible. Every irreducible occurs in the regular representation, so the list is complete. $\square$

The reconstruction certificate supplies an explicit square change-of-basis matrix S , verifies $\det S \neq 0$, and then verifies, for every $q \in Q$,

$$S^{-1} \lambda_Q(q) S = \bigoplus_{\rho \in \mathcal{R}} \underbrace{\rho(q) \oplus \cdots \oplus \rho(q)}_{d_\rho \text{ copies}}.\tag{11.29}$$

Canonical serialization sorts conjugacy classes by their least carrier element. Each exact field element has a declared canonical byte encoding ser . A character is serialized in carrier order as

$$(\text{ser}(\chi_\rho(q_0)), \dots, \text{ser}(\chi_\rho(q_{n-1}))),$$

and records are sorted lexicographically by that tuple preceded by d_ρ ; matrix entries use row-major order. For proof classification, all matrices and S are represented by exact algebraic data in an effectively presented field closed under complex conjugation. Alternatively, interval data must carry a validated existence-and-uniqueness or exact-reconstruction certificate identifying that exact representation. A finite backward-error bound by itself certifies only a nearby approximate decomposition; it does not certify the exact equalities (11.21)–(11.29), the commutant rank, or character orthogonality.

For the trivial group, its one-dimensional trivial representation is the minimal positive test. Omitting the sign representation from the two irreducibles of C_2 is a boundary failure because $\sum d_\rho^2 = 1 < 2$.

11.10 A reproducible numerical-tractability specification

Numerical tractability is a property of a declared computation and resource envelope, not a consequence of the finiteness of Q . Fix

$$\mathfrak{W} = (n, s, z, \mathcal{P}, M_*, W_*, T_*, \varepsilon_{\text{asm}}, \varepsilon_{\text{solve}}, \varepsilon_{\text{obs}}),\tag{11.30}$$

where $n = |Q|$, s is the number of internal degrees of freedom per quotient vertex, $N = sn$, z bounds the stored nonzeros per row, $\mathcal{P}$ is the canonically ordered parameter grid, M_*, W_*, T_* are memory, operation, and wall-time budgets, and the three ε 's are certified error limits. The input validator requires

$$n, s \in \mathbb{Z}_{\geq 1}, \quad z, |\mathcal{P}| \in \mathbb{Z}_{\geq 0}, \quad n = |Q|,$$

and every budget and error limit to be a nonnegative exactly comparable real number.

Definition 11.10.1. The pair $(Q, \mathfrak{W})$ is *certifiably tractable* for a declared algorithm $\mathcal{A}$ if a machine-readable estimate proves

$$\begin{aligned}\widehat{M}_{\mathcal{A}}(N, z, |\mathcal{P}|) &\leq M_*, \\ \widehat{W}_{\mathcal{A}}(N, z, |\mathcal{P}|) &\leq W_*, \\ \widehat{T}_{\mathcal{A}}(N, z, |\mathcal{P}|) &\leq T_*, \\ E_{\text{asm}} &\leq \varepsilon_{\text{asm}}, \\ E_{\text{solve}} &\leq \varepsilon_{\text{solve}}, \\ E_{\text{obs}} &\leq \varepsilon_{\text{obs}}.\end{aligned}\tag{11.31}$$

where the three hatted quantities are certified upper bounds for the declared implementation and execution schedule.

The record includes all units, scalar formats, hardware assumptions, library versions, and worst-case constants. For the eight-direction quotient graph, constructing the directed adjacency table uses exactly $8n$ generator multiplications and stores $8n$ destination indices. For a sparse $N \times N$ operator with at most zN nonzeros, one matrix–vector product uses at most zN scalar multiplications and zN scalar additions, hence $2zN$ scalar arithmetic operations, or zN fused multiply-add instructions when that is the declared unit. For one declared dense eigensolver and allocation model, valid certified upper estimates may have the form

$$\widehat{M}_{\text{dense}} \leq c_0 + (b_{\text{sc}} + b_{\text{work}})N^2 + c_1N, \quad \widehat{W}_{\text{dense}} \leq c_2N^3 + c_3N^2, \tag{11.32}$$

with nonnegative declared constants justified from that implementation. These are estimates, not universal equalities for every eigensolver. For $|\mathcal{P}|$ independent dense solves, the certified work upper bound is multiplied by $|\mathcal{P}|$. The memory upper bound is multiplied by $|\mathcal{P}|$ only if all matrices are simultaneously resident; a serial procedure may reuse one allocation. The execution schedule is therefore part of the serialized input.

The deterministic checker evaluates every left side of (11.31) from integer dimensions and certified bounds, then compares it with its budget in fixed field order. SATISFIED requires all six inequalities. If a certified upper bound exceeds its budget, the result is NOT ESTABLISHED UNDER THIS ESTIMATE, with the first such field, bound, and budget as witness; this does not by itself prove that the actual execution exceeds the budget. An independently certified lower bound above a budget may instead support a definitive INTRACTABLE FOR THIS DECLARED ALGORITHM result. Identical inputs produce byte-identical outputs.

For a positive minimal test, take $Q = \{e\}$, $s = 1$, one parameter point, and exact scalar arithmetic. The adjacency table has eight entries, $N = 1$, and (11.31) passes whenever the declared budgets meet or exceed the explicitly evaluated constants and the exact errors are zero. This tests only the tractability specification; it does not meet the eight-distinct-directions condition Statement 11.2.

For a rigorous boundary counterexample, let

$$\pi_m : \Gamma_{\text{B}} \twoheadrightarrow C_m, \quad a_1 \mapsto 1, \quad b_1, a_2, b_2 \mapsto 0. \tag{11.33}$$

Both commutators map to zero, and 1 generates C_m , so this is a surjection and $n = m$. Fix $s \geq 1$, a positive per-entry storage cost b , and a finite memory budget M_* . Any declared algorithm that explicitly stores a full dense $sm \times sm$ scalar matrix has the certified lower bound

$$M_{\text{actual}} \geq b(sm)^2. \tag{11.34}$$

For every $m > \sqrt{M_*/b}/s$, this lower bound exceeds M_* . Thus that declared dense-storage procedure is genuinely intractable under the fixed envelope although every C_m is finite. Finiteness and certified numerical tractability are different predicates.

12 Kernel-first construction and certified optimization

12.1 Construct the kernel, then read off the quotient

Fix a finite dangerous set $D \subset \Gamma_B \setminus \{1\}$. A kernel-first candidate is encoded by a finite group table Q and four marked elements A_1, B_1, A_2, B_2 . After Statement 11.1 verifies the relator and generation, they define a surjection $\pi : \Gamma_B \twoheadrightarrow Q$, and the primary mathematical object is

$$K := \ker \pi = \{\gamma \in \Gamma_B : \pi(\gamma) = e_Q\}. \quad (12.1)$$

This definition makes every unwanted quotient collision a concrete kernel-membership statement.

Deterministic algorithm KernelFirst.

Input: the canonical group table of Q ; the four marked generator indices; the complete exact-key-ordered list D ; the generator tables for ϕ_8 ; and flags requesting C_8 , parity, or both.

Step 0: verify that the supplied carrier operation, identity index, and inverse array satisfy all finite-group axioms, and verify the declared canonical byte encoding. All later steps are conditional on this validation.

Steps:

1. evaluate the two commutators and their product as in Statement 11.1;
2. close the marked images under multiplication and inverse, rejecting unless all of Q is reached;
3. evaluate every $d \in D$, rejecting if any value is e_Q ;
4. if requested, construct and verify the full descended rotation table α as in Statement 11.4;
5. if requested, construct and verify the full parity table ϵ as in Statement 11.5;
6. serialize the finite tuple consisting of the validated table, marked generator indices, and fixed word-evaluation algorithm. This tuple represents the membership decision $w \mapsto [\pi(w) = e_Q]$; the infinite oracle is not itself treated as a finite byte string.

Output: REJECT with the first independently verifiable failed equation; or, on a new record line, ACCEPT with $(K, Q, \pi, \alpha, \epsilon, D)$, omitting unrequested fields.

Theorem 12.1.1 (Correctness and reproducibility). *Among validated canonical finite-group-table inputs, the algorithm accepts exactly the marked finite-index normal kernels that avoid D and satisfy every requested symmetry condition. Its quotient is canonically isomorphic to Q , and identical canonical inputs give byte-identical output.*

Proof. Steps 1–2 are necessary and sufficient for a surjection by Statement 11.1. Its kernel is normal, and the first isomorphism theorem gives $\Gamma_B/K \cong Q$, hence $[\Gamma_B : K] = |Q| < \infty$. Step 3 accepts exactly when $K \cap D = \emptyset$, because

$$\pi(d) = e_Q \iff d \in \ker \pi = K.$$

By Statement 11.4, step 4 accepts exactly when $\phi_8(K) = K$; by Statement 11.5, step 5 accepts exactly when $K \subseteq K_\chi$. These equivalences prove both soundness and completeness. Every loop traverses a fixed finite carrier or a supplied finite ordered list, and every tie is broken by carrier or exact-key order, proving termination and reproducibility. $\square$

With $D = \emptyset$, the trivial table and all-identity marking give the boundary kernel $K = \Gamma_B$. With parity requested, $Q = C_2$ and all four presentation generators marked by 1 give $K = K_\chi$, a minimal positive regression for steps 1, 2, and 5.

12.2 ϕ_8 -stable normal subgroups

Definition 12.2.1. A normal subgroup $K \triangleleft \Gamma_B$ is C_8 -stable if

$$\phi_8(K) = K. \quad (12.2)$$

Because $\phi_8^8 = \text{id}$, the one-sided condition $\phi_8(K) \subseteq K$ is already sufficient. Indeed, iteration gives

$$K = \phi_8^8(K) \subseteq \phi_8^7(K) \subseteq \cdots \subseteq \phi_8(K) \subseteq K, \quad (12.3)$$

so every inclusion is equality. Conversely, (12.2) plainly implies the inclusion.

For a kernel encoded by a surjection $\pi : \Gamma_B \twoheadrightarrow Q$, Statement 11.4 turns (12.2) into the finite condition that the rule

$$\alpha(\pi(\gamma)) = \pi(\phi_8(\gamma)) \quad (12.4)$$

be well defined. The certificate is not a generator-only assertion: it stores the entire carrier permutation, checks multiplication on every ordered pair, and checks

$$\alpha(q_j) = q_{j+1}, \quad \alpha^8(q) = q. \quad (12.5)$$

If the shell images are distinct, (12.5) contains an eight-cycle and therefore $\text{ord}(\alpha) = 8$. For $K = \Gamma_B$, (12.2) still holds but the descended automorphism on the trivial quotient has order one; this is the boundary case distinguishing stable kernel from faithful rotation order.

12.3 Parity-compatible normal subgroups

Definition 12.3.1. A normal subgroup $K \triangleleft \Gamma_B$ is parity compatible if

$$K \subseteq K_\chi = \ker \chi. \quad (12.6)$$

For $K = \ker \pi$ with π surjective, (12.6) is equivalent by Statement 11.5 to the existence of the unique quotient homomorphism

$$\epsilon : Q \rightarrow C_2, \quad \epsilon\pi = \chi. \quad (12.7)$$

It follows without an extra assumption that every physical shell image is odd:

$$\epsilon(q_j) = \epsilon(\pi(g_j)) = \chi(g_j) = 1. \quad (12.8)$$

The surface relator is consistent in C_2 , since every commutator maps to zero and hence their product maps to zero.

If each K_λ is parity compatible, then

$$\bigcap_{\lambda} K_\lambda \subseteq K_\chi, \quad (12.9)$$

so arbitrary intersections preserve compatibility. Enlargement does not: $K_\chi \subsetneq \Gamma_B$, but $\Gamma_B \not\subseteq K_\chi$. Consequently quotient compression, which enlarges a kernel, must rerun the complete parity test. The C_2 quotient of Statement 12.1 is the positive fixture; the trivial quotient is the smallest failure.

12.4 Avoiding dangerous elements

Definition 12.4.1. For finite $D \subset \Gamma_B \setminus \{1\}$, a subgroup K is D -separating if

$$K \cap D = \emptyset. \quad (12.10)$$

Proposition 12.4.2 (Master avoidance criterion). *Let $\pi : \Gamma_B \rightarrow Q$ have kernel K . Then*

$$K \text{ is } D\text{-separating} \iff \pi(d) \neq e_Q \text{ for every } d \in D. \quad (12.11)$$

If $D = \bigcup_{r=1}^m D_r$, then K is D -separating if and only if it is D_r -separating for every r .

Proof. For each d , the equivalence $\pi(d) = e_Q \iff d \in \ker \pi$ proves (12.11). Set distributivity gives

$$K \cap \left(\bigcup_r D_r \right) = \bigcup_r (K \cap D_r),$$

and this union is empty exactly when every term is empty. $\square$

The sets are requirement-specific. Shell distinctness uses D_{shell} from Statement 11.2; injectivity on a finite set B uses $(B^{-1}B) \setminus \{1\}$; based injectivity uses a closed based displacement cutoff; and global systole uses the complete covering list $\tilde{\mathcal{D}}_{\text{sys}}(L)$ of Statement 11.7. Replacing one by another is valid only after proving an inclusion. If $K' \subseteq K$ and $K \cap D = \emptyset$, then also $K' \cap D = \emptyset$; enlarging a kernel can introduce a dangerous element.

The certificate evaluates all of the deduplicated union in canonical exact-key order, retains every source-set tag, and records the expected input count, duplicate multiplicities, and output count. These data prove only that deduplication loses no tagged record relative to the supplied source lists. They do not prove that a source list is mathematically complete. Completeness is either an explicit input assumption or is certified by replaying its upstream generator; in particular, $\tilde{\mathcal{D}}_{\text{sys}}(L)$ requires the complete based-ball enumeration and both exact boundary comparisons of Statement 11.7.

12.5 The finite-index condition

Criterion 12.5.1. *For a normal subgroup $K \triangleleft \Gamma_B$, the following are equivalent:*

- (i) $[\Gamma_B : K] < \infty$;
- (ii) K is the kernel of a homomorphism from Γ_B with finite image;
- (iii) K is the kernel of a surjection onto a finite group.

Proof. If (i) holds, the quotient map $\pi_K : \Gamma_B \twoheadrightarrow \Gamma_B/K$ has kernel K and finite target, proving (iii). Statement (iii) implies (ii). If $\pi : \Gamma_B \rightarrow F$ has finite image and kernel K , the first isomorphism theorem gives

$$\Gamma_B/K \cong \text{im } \pi, \quad [\Gamma_B : K] = |\text{im } \pi| < \infty, \quad (12.12)$$

proving (i). $\square$

Thus a finite carrier with an unchecked marking is not a finite-index certificate: the group-table axioms and surface relation must first prove a homomorphism. Surjectivity is additionally required only to prove

$$Q \cong \Gamma_B/K, \quad [\Gamma_B : K] = |Q|.$$

Without surjectivity, the same homomorphism still proves

$$\Gamma_B/K \cong \text{im } \pi, \quad [\Gamma_B : K] = |\text{im } \pi| \leq |Q|. \quad (12.12a)$$

Conversely, a terminating complete coset enumeration with n cosets proves finite index n ; if it also verifies normality, its quotient table must have n elements by (12.12). The subgroup Γ_B has index one, whereas the trivial subgroup has infinite index because Γ_B is infinite.

12.6 Minimizing quotient order

Fix a finite dangerous set D . Normality and finite index are mandatory below; let $\mathfrak{F}$ list only additional requested conditions such as C_8 -stability and parity compatibility. Define the feasible family

$$\mathcal{K}(D, \mathfrak{F}) := \{K \triangleleft \Gamma_B : [\Gamma_B : K] < \infty, K \cap D = \emptyset, K \text{ satisfies } \mathfrak{F}\}. \quad (12.13)$$

If this family is nonempty, its quotient-order optimum is

$$N(D, \mathfrak{F}) := \min\{[\Gamma_B : K] : K \in \mathcal{K}(D, \mathfrak{F})\}. \quad (12.14)$$

Proposition 12.6.1. *The minimum in (12.14) exists. A feasible candidate of order U , together with a complete rejection certificate for every marked finite quotient of orders $1, \dots, U-1$, proves $N(D, \mathfrak{F}) = U$.*

Proof. The displayed set of indices is a nonempty subset of $\mathbb{Z}_{\geq 1}$, so well-ordering supplies a least element. For fixed m , there are finitely many binary tables on the carrier $\{0, \dots, m-1\}$, finitely many choices of identity and inverse arrays, and finitely many marked four-tuples. Exhaustively checking the group axioms, the surface relator, generation, dangerous elements, and the requested finite conditions of C11 enumerates every marked quotient of order m , possibly with isomorphic repetitions. Any feasible kernel of index m yields one of these tables after labelling its quotient carrier. Therefore rejection of every entry for every $m < U$ proves the lower bound $N \geq U$, while the candidate proves $N \leq U$. $\square$

The certificate contains the candidate table and a proof-producing exhaustive transcript. An independent verifier reconstructs the complete declared table-and-marking enumeration and replays the first failed predicate for every candidate; an equivalent exhaustive proof object is also allowed. Aggregate rejection counts and checksums are analysis aids, not by themselves proofs of coverage or rejection correctness. All order comparisons are exact integers, stronger than outward-rounded real bounds.

Useful analytic lower bounds precede enumeration. If Statement 11.2 is required, the identity and eight distinct nonidentity shell images give

$$|Q| \geq 9. \quad (12.15)$$

If parity compatibility is also required, $\epsilon : Q \rightarrow C_2$ makes $|Q|$ even, so (12.15) strengthens to $|Q| \geq 10$. For the boundary instance $D = \emptyset$ with no requested symmetry, $K = \Gamma_B$ proves $N = 1$. If parity compatibility alone is requested, the trivial quotient fails and K_χ gives order two, proving $N = 2$; this independently checks both sides of (12.14).

12.7 Minimizing transitive permutation degree

For a finite group Q , define its least faithful transitive permutation degree by

$$\mu_{\text{tr}}(Q) := \min\{[Q : L] : L \leq Q, \text{core}_Q(L) = \{e_Q\}\}. \quad (12.16)$$

This is finite because $L = \{e_Q\}$ is admissible, so

$$1 \leq \mu_{\text{tr}}(Q) \leq |Q|, \quad (12.17)$$

with equality $\mu_{\text{tr}}(Q) = 1$ exactly for the trivial group. Statement 10.5 proves that (12.16) is precisely the minimum degree among faithful transitive actions.

For the kernel-first feasible family, the global objective is

$$\Delta(D, \mathfrak{F}) := \min\{\mu_{\text{tr}}(\Gamma_B/K) : K \in \mathcal{K}(D, \mathfrak{F})\}, \quad (12.18)$$

whenever the feasible family is nonempty. This global minimum is then attained by well-ordering of $\mathbb{Z}_{\geq 1}$. Whenever the feasible family under a declared order cap U is nonempty, define separately

$$\Delta_U(D, \mathfrak{F}) := \min\{\mu_{\text{tr}}(\Gamma_B/K) : K \in \mathcal{K}(D, \mathfrak{F}), [\Gamma_B : K] \leq U\}. \quad (12.18a)$$

Proposition 12.7.1 (Finite certificate at a declared order cap). *Fix U . If the capped feasible family is nonempty, then among its quotients both $\mu_{\text{tr}}(Q)$ and $\Delta_U(D, \mathfrak{F})$ are exactly computable. If it is empty, exhaustive enumeration certifies INFEASIBLE UNDER CAP U . A claimed value d is certified by one core-free subgroup of index d , plus exhaustive rejection of every subgroup of smaller index in every feasible quotient under the cap.*

Proof. Every subgroup of a finite carrier can be enumerated as a subset and checked for identity, products, and inverses. For each accepted L , the finite intersection

$$\text{core}_Q(L) = \bigcap_{q \in Q} qLq^{-1}$$

is computed exactly. Hence (12.16) is decided by a finite scan. Statement 12.6 enumerates all feasible marked quotients under U . Combining the two finite scans either proves emptiness or proves the stated upper and lower bounds. If no order cap or other exhaustive global enumeration is supplied, a result under U is not promoted to the unrestricted value (12.18). $\square$

There is also a finite certificate for a proposed unrestricted value. If a feasible candidate has $\mu_{\text{tr}} = d > 1$, every strictly better competitor has a faithful action of some degree $m < d$, hence embeds in S_m and satisfies

$$|Q| \leq m! \leq (d-1)!. \quad (12.18b)$$

Exhaustive enumeration through order $(d-1)!$, with rejection of every competitor of degree below d , therefore proves $\Delta(D, \mathfrak{F}) = d$. For $d = 1$, the positive lower bound in (12.17) is already conclusive.

For $Q = C_3$, the only proper subgroup is trivial, so $\mu_{\text{tr}}(C_3) = 3$. For $Q = S_3$, a subgroup generated by a transposition has index three and trivial core, giving $\mu_{\text{tr}}(S_3) \leq 3$; degrees one and two cannot be faithful because they would embed S_3 into S_1 or S_2 , so $\mu_{\text{tr}}(S_3) = 3 < 6 = |S_3|$. This is the minimal standard check that permutation degree and quotient order are distinct objectives. The cap distinction is real: a cap may exclude a larger quotient having a smaller faithful degree, so Δ_U need not equal Δ .

12.8 Minimizing certified numerical cost

Fix a mathematical feasibility specification $(D, \mathfrak{F})$, an order cap U , one algorithm $\mathcal{A}$, one execution schedule, and one resource scalarization

$$C_{\mathcal{A}}(X) = w_M \widehat{M}_{\mathcal{A}}(X) + w_W \widehat{W}_{\mathcal{A}}(X) + w_T \widehat{T}_{\mathcal{A}}(X), \quad w_M, w_W, w_T \in \mathbb{Q}_{\geq 0}. \quad (12.19)$$

Here $X = (Q, \pi)$ is a marked quotient. Two records are identified only when a carrier isomorphism intertwines their marking maps. At least one weight is positive, and only mathematically feasible marked quotients enter

$$\mathcal{X}_U := \{X = (Q, \pi) \text{ modulo marked equivalence} : \ker \pi \in \mathcal{K}(D, \mathfrak{F}), |Q| \leq U\}. \quad (12.20)$$

Only after proving that the cost is invariant under the marking may it be written as a function of the abstract group Q alone.

When costs contain transcendental or measured real inputs, the evaluator returns outward-rounded certified intervals

$$C_{\mathcal{A}}(X) \in [\underline{C}(X), \overline{C}(X)]. \quad (12.21)$$

All lower endpoints are rounded downward and all upper endpoints upward. For measured inputs, containment additionally requires a declared deterministic measured quantity and a validated uncertainty model. Outward arithmetic rounding alone does not certify measurement error, run-to-run variation, or model error.

Criterion 12.8.1 (Certified cost optimum). *A candidate $X_* \in \mathcal{X}_U$ is certified optimal if*

$$\overline{C}(X_*) \leq \underline{C}(X) \quad \text{for every } X \in \mathcal{X}_U \setminus \{X_*\}. \quad (12.22)$$

Then $C_A(X_) \leq C_A(X)$ for every competitor.*

Proof. For each competitor, interval containment and (12.22) give

$$C_A(X_*) \leq \overline{C}(X_*) \leq \underline{C}(X) \leq C_A(X).$$

This proves the assertion without using midpoint values. If intervals overlap, the comparison is INDETERMINATE until precision is increased or a symbolic comparison is supplied. $\square$

Statement 12.6 makes $\mathcal{X}_U$ a finite exactly enumerable set. The output stores the feasible-table hashes, all interval endpoints, rounding mode, precision, weights, and the comparison matrix. Removing a physical edge, discarding a symmetry sector, or weakening D changes the feasible problem and cannot be reported as a cheaper solution to (12.19).

As a minimal independent check, take two declared marked records with underlying groups C_2, C_3 , put $s = 1$, and use exact cost $C(X) = |Q_X|^2$. The intervals $[4, 4]$ and $[9, 9]$ certify C_2 . Replacing them by $[4, 10]$ and $[9, 9]$ yields INDETERMINATE, which checks the outward-bound direction at the boundary.

12.9 A deterministic multiobjective construction

Let every feasible marked quotient $X = (Q, \pi)$ carry the objective vector

$$F(X) = (|Q|, \mu_{\text{tr}}(Q), \widehat{M}_A(X), \widehat{W}_A(X)). \quad (12.23)$$

For exact components, X dominates Y , written $X \prec Y$, if every component of $F(X)$ is at most the corresponding component of $F(Y)$, and at least one is strictly smaller. For interval components, certified dominance requires upper bounds for X to be no larger than lower bounds for Y , with one certified strict inequality.

Algorithm ParetoKernel.

1. Enumerate all marked tables of order at most the declared cap U .
2. Run KernelFirst and retain exactly the feasible records.
3. Canonically deduplicate marked-quotient equivalence classes.
4. Compute μ_{tr} by Statement 12.7 and resource intervals by Statement 12.8.
5. Compare every ordered pair. Delete a record only with a certified dominance witness; retain unresolved pairs.
6. Sort survivors by the canonical marked-quotient byte string and emit the comparison matrix.

Theorem 12.9.1. *Relative to the finite capped set $\mathcal{X}_U$, every deleted record is dominated by a retained record and every actually nondominated record is retained. The survivor set is a certified outer approximation to the capped Pareto frontier; it equals that frontier when every relevant component comparison is exact or resolved. Repeated execution on identical inputs returns the same ordered survivor set.*

Proof. The candidate list is finite. A deletion uses a componentwise certified inequality, so it never removes a nondominated record. Dominance is transitive: composing componentwise weak inequalities preserves them, and a strict component remains strict. Thus if a deleting witness is itself later deleted, following the finite chain ends at a retained record that dominates the original. Canonical enumeration and sorting prove reproducibility. An unresolved interval pair is retained, so uncertainty cannot cause a false deletion, although it can leave a dominated record in the outer approximation. If all comparisons are resolved, precisely the nondominated records remain. $\square$

For a declared two-record example with exact vectors

$$F(X) = (4, 4, 2, 6), \quad F(Y) = (6, 3, 3, 4),$$

neither record dominates the other, so both form the frontier of this two-record set. Membership in a larger frontier requires comparison with every other candidate. A weighted sum may choose only one, showing why scalar optimization and Pareto construction are distinct. The single-record input is the minimal positive fixture.

12.10 What it means for a quotient to be optimal

Optimization is performed on *marked quotients*, not merely abstract groups. Two surjections $\pi : \Gamma_B \twoheadrightarrow Q$ and $\pi' : \Gamma_B \twoheadrightarrow Q'$ are equivalent if there is an isomorphism $\theta : Q \rightarrow Q'$ such that

$$\theta\pi = \pi'. \quad (12.24)$$

Equation (12.24) holds if and only if $\ker \pi = \ker \pi'$: the forward implication is immediate; for the reverse implication define $\theta(\pi(\gamma)) = \pi'(\gamma)$, whose well-definedness, bijectivity, and homomorphism property follow from the common kernel and surjectivity.

Fix a feasible set $\mathcal{X}$ and an objective map F into a declared totally ordered exact set. Define $\text{ser}(X)$ on marked-quotient equivalence classes as the lexicographically least byte string over all valid carrier relabellings of the complete group table and marked generator array. This string is invariant under relabelling. Equality of two such complete canonical strings supplies an isomorphism intertwining the markings, so the serialization is injective on equivalence classes. Use the total key

$$\text{Key}(X) = (F(X), \text{ser}(X)). \quad (12.25)$$

Definition 12.10.1. A marked quotient $X_* \in \mathcal{X}$ is *certifiably optimal* if

$$\text{Key}(X_*) \leq \text{Key}(X) \quad (X \in \mathcal{X}), \quad (12.26)$$

and the certificate contains both the feasible witness for X_* and a complete lower-bound or exhaustive-rejection proof for every strictly smaller key.

For every nonempty finite capped set $\mathcal{X}_U$, a unique canonical least equivalence class exists because a finite total order has a unique least member. Without a cap, existence follows only after proving that the feasible key set attains a minimum; an observed best candidate is merely an upper bound. Statement 12.6 supplies such attainment automatically for quotient order once feasibility is nonempty, but arbitrary real-valued costs need a separate compactness, discreteness, or coercivity argument.

For $D = \emptyset$ and objective $|Q|$, the all-identity marked trivial quotient is the unique optimum with value one. With parity compatibility required, it is infeasible and the marked C_2 quotient of Statement 12.1 is optimal with value two. These cases verify that feasibility is applied before optimization and that equality at the lower bound is accepted.

Part IV

Local and global injectivity

13 Exact local balls and local injectivity

13.1 The word ball $B_\Gamma(e, r)$

Use the ordered inverse-closed physical shell $S_8^{\text{geo}} = (g_0, \dots, g_7)$. For $r \in \mathbb{Z}_{\geq 0}$, define

$$B_r := B_{\Gamma_B, S_8^{\text{geo}}}(e, r) = \{\gamma \in \Gamma_B : |\gamma|_{S_8^{\text{geo}}} \leq r\}. \quad (13.1)$$

The identity and the boundary $|\gamma| = r$ are included.

Complete breadth-first enumerator. Start with $V_0 = F_0 = \{1\}$. Having built the visited set V_t and frontier F_t , form all ordered candidates xg_j with $x \in F_t$, $0 \leq j < 8$. Compute the lossless exact group key, sort by $(\text{key}, \text{key}(x), j)$, and add each previously unseen key once to F_{t+1} . Set $V_{t+1} = V_t \sqcup F_{t+1}$. Stop after constructing F_r . For every new key retain the least predecessor pair and word, but also record all duplicate source pairs.

Theorem 13.1.1. *For every $0 \leq t \leq r$,*

$$V_t = B_t, \quad F_t = \{\gamma : |\gamma|_{S_8^{\text{geo}}} = t\}. \quad (13.2)$$

Thus the enumerator is complete, duplicate-free, and terminating.

Proof. At $t = 0$, both claims say $\{1\}$. Assume them at t . Every new candidate has length at most $t + 1$, so $V_{t+1} \subseteq B_{t+1}$. Conversely, let $\gamma \in B_{t+1} \setminus B_t$. A shortest word has length $t + 1$, say $\gamma = xg_j$, where its prefix x has length exactly t . By induction $x \in F_t$, so the candidate is generated. Its exact key is either already in V_t , which would contradict $|\gamma| = t + 1$, or is inserted into F_{t+1} . Hence $B_{t+1} \subseteq V_{t+1}$, and set difference gives the frontier identity. Finiteness follows from at most $1 + 8 + \dots + 8^t$ candidates before deduplication. $\square$

The record stores r , the generator-key array, $|V_t|$, $|F_t|$, and the candidate count $8|F_t|$. It separately stores the new, previously visited, and same-layer duplicate counts, all predecessor records, and a rolling hash after each layer. A checkpoint is the complete V_t, F_t plus this hash; independent verification verifies it before continuing. At $r = 0$ the output is exactly $\{1\}$.

13.2 Injectivity of a quotient map on a local ball

Let $\pi : \Gamma_B \rightarrow Q$ have kernel K . For the symmetric shell,

$$B_r^{-1}B_r = B_{2r}. \quad (13.3)$$

The inclusion from left to right follows from $|y^{-1}x| \leq |y| + |x| \leq 2r$. For the reverse inclusion, take a shortest word $w = s_1 \dots s_m$, $m \leq 2r$, split it as $w = uv$ after $\min(m, r)$ letters, and put $y = u^{-1}$, $x = v$. Then $|x|, |y| \leq r$ and $w = y^{-1}x$, proving equality.

Criterion 13.2.1.

$$\pi|_{B_r} \text{ is injective} \iff K \cap (B_{2r} \setminus \{1\}) = \emptyset. \quad (13.4)$$

Proof. Statement 11.6 gives injectivity exactly when $K \cap (B_r^{-1}B_r) = \{1\}$. Substitute (13.3) and remove the identity from both sides to obtain (13.4). $\square$

The direct algorithm enumerates B_r , evaluates its members in canonical order, and inserts quotient indices into a table. A repeated index emits the least pair $x \neq y$ and the nontrivial kernel witness $y^{-1}x$. The independent difference algorithm enumerates B_{2r} and checks (13.4). Agreement of both methods is part of the certificate. At $r = 0$ every quotient passes. If $g_0 \in K$, the radius-one scan fails at the collision of 1 and g_0 .

13.3 Short kernel relations and local collisions

Define the shortest nontrivial kernel word length by

$$\text{rel}_{S_8^{\text{geo}}}(K) := \min\{|k|_{S_8^{\text{geo}}} : 1 \neq k \in K\}, \quad (13.5)$$

with value $+\infty$ for $K = \{1\}$. For finite-index $K \neq \{1\}$ the minimum exists because word length is integer-valued and K has nonidentity elements.

Theorem 13.3.1. *For every integer $r \geq 0$,*

$$\pi|_{B_r} \text{ is injective} \iff \text{rel}_{S_8^{\text{geo}}}(K) > 2r. \quad (13.6)$$

Proof. If injectivity fails, choose $x \neq y$ in B_r with $\pi(x) = \pi(y)$. Then $1 \neq y^{-1}x \in K$ and

$$\text{rel}_{S_8^{\text{geo}}}(K) \leq |y^{-1}x|_{S_8^{\text{geo}}} \leq |y|_{S_8^{\text{geo}}} + |x|_{S_8^{\text{geo}}} \leq 2r.$$

Conversely, if $1 \neq k \in K$ has a shortest spelling of length $m \leq 2r$, the splitting argument for (13.3) writes $k = y^{-1}x$ with $x, y \in B_r$. Since $k \neq 1$, one has $x \neq y$, while $\pi(x) = \pi(y)$. Thus injectivity fails. $\square$

The strict boundary is essential: a kernel element of length exactly $2r$ produces a collision. These are relations of the quotient relative to elements already distinguished in Γ_B , not free-word relators that already equal the identity in Γ_B .

13.4 The B_1, B_2, B_3 hierarchy

The enumerator gives nested, disjoint-layer decompositions

$$\{1\} = B_0 \subseteq B_1 \subseteq B_2 \subseteq B_3, \quad B_t = \bigsqcup_{j=0}^t F_j. \quad (13.7)$$

Because the physical generators are eight distinct nonidentity elements,

$$|F_0| = 1, \quad |F_1| = 8, \quad |B_1| = 9. \quad (13.8)$$

Every nonidentity frontier vertex has at most seven outward non-backtracking extensions, so

$$|F_t| \leq 8 \cdot 7^{t-1} \quad (t \geq 1), \quad |B_t| \leq 1 + 8 \sum_{j=0}^{t-1} 7^j. \quad (13.9)$$

In particular,

$$|B_2| \leq 65, \quad |B_3| \leq 457. \quad (13.10)$$

If the exact scan proves that no two reduced words of length at most t have the same group key, equality holds in (13.9) through that layer. This is a checkable condition, not an assumption smuggled into the counts. The canonical benchmark is the layer-count vector

$$(|F_0|, |F_1|, |F_2|, |F_3|)$$

recorded by the complete Statement 13.1 record. The identity-only B_0 tests initialization; deleting an inverse edge from the generator array must be rejected before the hierarchy is generated.

13.5 Isomorphism of labelled local Cayley graphs

Let $\mathcal{C}_r$ be the labelled directed subgraph of $\text{Cay}(\Gamma_B, S_8^{\text{geo}})$ induced by B_r . Define the finite nonidentity obstruction set

$$\begin{aligned} D_{\text{vert}}(r) &:= \{y^{-1}x : x, y \in B_r, x \neq y\}, \\ D_{\text{edge}}(r) &:= \left\{ y^{-1}xg_j : \begin{array}{l} x, y \in B_r, 0 \leq j < 8 \\ y \neq xg_j \end{array} \right\}, \\ D_{\text{graph}}(r) &:= D_{\text{vert}}(r) \cup D_{\text{edge}}(r). \end{aligned} \quad (13.11)$$

The second line records precisely the possible extra labelled edges among quotient images.

Criterion 13.5.1. *The vertex map $x \mapsto \pi(x)$ is an isomorphism from C_r to the labelled directed subgraph induced by $\pi(B_r)$ if and only if*

$$K \cap D_{\text{graph}}(r) = \emptyset. \quad (13.12)$$

Proof. Avoiding the first line of (13.11) is exactly vertex injectivity. A source edge labelled j from x has head xg_j , and homomorphicity always sends it to the target edge from $\pi(x)$ with head $\pi(xg_j)$. Conversely, that target edge lands at $\pi(y)$ for some $y \in B_r$ exactly when

$$\begin{aligned} \pi(xg_j) &= \pi(y) \\ \iff \pi(y^{-1}xg_j) &= e_Q \\ \iff y^{-1}xg_j &\in K. \end{aligned} \quad (13.13)$$

If $y = xg_j$, this is the genuine source edge and the group element is the identity, excluded from (13.11). Every other occurrence is a spurious induced edge and is excluded exactly by the second line. Together with vertex injectivity, this proves both directions. $\square$

A convenient stronger sufficient test is injectivity on B_{r+1} : both xg_j and y in (13.13) then lie in that ball, so equality of their images forces equality in the group. Injectivity only on B_r does not exclude a new chord whose intermediate endpoint xg_j lies outside B_r , which is the required boundary counterexample pattern.

13.6 Preservation of the local geometric structure

Let

$$U_r := \bigcup_{\gamma \in B_r} \gamma P \subset \mathbb{H}_R^2, \quad \Delta_r := \text{diam}_{\mathbb{H}_R^2}(U_r). \quad (13.14)$$

This finite union is compact, so $\Delta_r < \infty$. Its marked geometric structure consists of the tile polygons, side labels, vertex incidences, and the hyperbolic metric restricted to each tile.

Theorem 13.6.1. *Let $K \triangleleft \Gamma_B$ have finite index and let $p : \mathbb{H}_R^2 \rightarrow K \backslash \mathbb{H}_R^2$ be the quotient projection.*

- (i) *If $K \cap D_{\text{graph}}(r) = \emptyset$, the tile-centre labelled Cayley structure on B_r is preserved exactly.*
- (ii) *If $\text{sys}(K \backslash \mathbb{H}_R^2) > \Delta_r$, then $p|_{U_r}$ is injective and is an isometry onto its image on every polygonal piece. Thus the entire marked local geometric structure is preserved.*

Proof. Part (i) is Statement 13.5 together with the fixed correspondence between a physical label j and the paired side s_j . For (ii), suppose $y, z \in U_r$ and $p(y) = p(z)$. Then $z = ky$ for some $k \in K$, and

$$\ell_h(k) \leq d_{\mathbb{H}_R^2}(y, ky) = d_{\mathbb{H}_R^2}(y, z) \leq \Delta_r. \quad (13.15)$$

If $k \neq 1$, (13.15) contradicts the strict systole hypothesis. Hence $k = 1$ and $y = z$. The covering projection is a local hyperbolic isometry; injectivity on U_r therefore preserves lengths, angles, side pairings, and incidences on the marked pieces. $\square$

The continuous condition in (ii) is stronger than purely combinatorial local injectivity and must not be inferred from (i). An alternative certificate encloses U_r in a ball $\bar{B}(x, \rho)$ and applies Statement 11.8 with $\text{sys} > 2\rho$.

13.7 Local-ball size as a quotient-order lower bound

Theorem 13.7.1. *If $\pi : \Gamma_B \rightarrow Q$ is injective on B_r , then*

$$|Q| \geq |B_r|. \quad (13.16)$$

If, in addition, the source Cayley graph has no nontrivial reduced closed walk of length at most $2r$, then

$$|Q| \geq 1 + 8 \sum_{j=0}^{r-1} 7^j = 1 + \frac{4}{3}(7^r - 1). \quad (13.17)$$

Proof. An injection from the finite set B_r into Q gives (13.16). Under the extra girth condition, two distinct reduced paths of length at most r cannot have the same endpoint, because following one and then the inverse of the other would reduce to a nontrivial closed walk of length at most $2r$. Hence the non-backtracking tree count is exact: one root, eight vertices at distance one, and $8 \cdot 7^{j-1}$ at distance j . Summing gives (13.17). $\square$

The exact BFS count $|B_r|$, not the conditional tree formula, is the unconditional numerical certificate. It is an integer, so no rounding is needed; if a real surrogate bound is also reported, its lower endpoint is rounded downward. At $r = 0$, (13.16) is $|Q| \geq 1$ and the trivial quotient attains equality. At $r = 1$, (13.8) gives the unconditional bound $|Q| \geq 9$; a quotient with only eight elements cannot have eight distinct nonidentity directions.

13.8 An independently independently verifiable local-injectivity certificate

A certificate has four hash-linked blocks:

- (i) the presentation, ordered physical shell, exact-key specification, radius r , and their input hash;
- (ii) every BFS layer from Statement 13.1, predecessor and duplicate records, layer counts, checkpoints, and the final ball hash;
- (iii) the canonical finite group table and marked generator images, with the Statement 11.1 relation and surjectivity checks;
- (iv) the ordered image of every member of B_r , plus the independent $B_{2r} \setminus \{1\}$ kernel scan from (13.4).

Criterion 13.8.1 (Verifier specification). *The independent verifier returns SATISFIED if and only if it regenerates the same complete ball and hashes, validates the quotient, finds pairwise distinct images of B_r , and finds no nonidentity member of B_{2r} mapping to e_Q . Any mismatch returns VIOLATED with the first canonical witness.*

Proof. The BFS proof Statement 13.1 establishes coverage and uniqueness of the regenerated ball. Statement 11.1 establishes that the finite table data define the declared surjection. Statement 13.2 proves that either of the last two scans is individually equivalent to local injectivity, so their joint SATISFIED is sound, and a genuinely injective quotient makes both pass. Fixed ordering makes the first mismatch reproducible. $\square$

The certificate metadata includes the verifier source hash, exact arithmetic backend version, input hash, and output hash. A corrupted predecessor, omitted boundary element, duplicate quotient image, or kernel element in $B_{2r} \setminus \{1\}$ is a mandatory negative test. The $r = 0$ trivial quotient is the minimal SATISFIED fixture.

14 Based geometric injectivity

14.1 Based displacement

Fix the disk centre $o = 0$. For $\gamma \in \Gamma_B$, define

$$\delta_o(\gamma) := d_{\mathbb{H}_R^2}(o, \gamma o). \quad (14.1)$$

This is not translation length: it depends on the declared basepoint. It satisfies

$$\delta_o(1) = 0, \quad \delta_o(\gamma^{-1}) = \delta_o(\gamma), \quad \delta_o(\gamma\eta) \leq \delta_o(\gamma) + \delta_o(\eta). \quad (14.2)$$

For example, the inverse identity follows by applying the isometry γ :

$$d(o, \gamma^{-1}o) = d(\gamma o, o) = d(o, \gamma o).$$

The triangle inequality gives the last formula after inserting γo .

If a lift of γ is

$$G = \begin{pmatrix} a & b \\ \bar{b} & \bar{a} \end{pmatrix} \in \mathrm{SU}(1, 1),$$

then $Go = b/\bar{a}$ and $|a|^2 - |b|^2 = 1$. Substitution in the disk distance formula gives, without skipping the denominator,

$$\begin{aligned} \cosh \frac{\delta_o(\gamma)}{R} &= 1 + \frac{2|b/\bar{a}|^2}{1 - |b/\bar{a}|^2} \\ &= 1 + \frac{2|b|^2}{|a|^2 - |b|^2} \\ &= 1 + 2|b|^2 = 2|a|^2 - 1, \end{aligned} \quad (14.3)$$

and hence

$$\delta_o(\gamma) = 2R \operatorname{arcosh} |a|. \quad (14.4)$$

The absolute value makes this independent of the lift sign.

The identity is the zero-displacement positive check. If h is hyperbolic and u moves its axis away from o , then $\delta_o(uhu^{-1})$ generally changes although translation length does not; this is the basic boundary test against accidental conjugacy invariance.

14.2 The based injectivity radius

For a discrete subgroup $K \leq \Gamma_B$, define

$$r_{\text{based}}(K; o) := \frac{1}{2} \inf_{1 \neq k \in K} \delta_o(k), \quad (14.5)$$

with $r_{\text{based}}(\{1\}; o) = +\infty$. If $K \neq \{1\}$, proper discontinuity makes every set $\{k \in K : \delta_o(k) \leq T\}$ finite. Choosing one nonidentity k_0 and taking $T = \delta_o(k_0)$, the nonempty finite set below T has a minimum. Freeness makes every nonidentity displacement strictly positive. Thus the infimum in (14.5) is attained and is positive.

For the quotient $p : \mathbb{H}_R^2 \rightarrow K \backslash \mathbb{H}_R^2$, the free, properly discontinuous deck action gives the standard formula

$$r_{\text{based}}(K; o) = \inf_{K \backslash \mathbb{H}_R^2} (p(o)). \quad (14.6)$$

Indeed, for each $k \neq 1$, the point of the bisector of o and ko nearest to o is their geodesic midpoint, at distance $\frac{1}{2}\delta_o(k)$. Minimizing these distances over $k \neq 1$ gives (14.6).

Since $\ell_h(k) \leq \delta_o(k)$ for every k ,

$$\frac{1}{2} \operatorname{sys}(K \backslash \mathbb{H}_R^2) \leq r_{\text{based}}(K; o). \quad (14.7)$$

Equality need not hold unless a systolic axis passes through a lift of the basepoint.

14.3 The based dangerous set

For $D \geq 0$, define the closed based dangerous set

$$D_{\text{based}}(D) := \{\gamma \in \Gamma_B \setminus \{1\} : \delta_o(\gamma) \leq 2D\}. \quad (14.8)$$

It is finite by proper discontinuity. The factor two and the closed cutoff are part of the definition.

Criterion 14.3.1.

$$r_{\text{based}}(K; o) > D \iff K \cap D_{\text{based}}(D) = \emptyset. \quad (14.9)$$

Proof. Expanding (14.5), the left side says $\delta_o(k) > 2D$ for every $1 \neq k \in K$. Expanding (14.8), the right side says there is no nonidentity $k \in K$ with $\delta_o(k) \leq 2D$. These statements are logical negations of the same closed inequality and are therefore equivalent. $\square$

At $D = 0$, freeness of the Γ_B -action makes the dangerous set empty and every subgroup passes. If some $k \in K$ has displacement exactly $2D$, both sides fail; equality is not silently rounded into SATISFIED.

14.4 Exact enumeration of a based geometric ball

For $T \geq 0$, put

$$B_{\text{geo}}(T) := \{\gamma \in \Gamma_B : \delta_o(\gamma) \leq T\}. \quad (14.10)$$

The identity and closed boundary are included. Apply the finite word-ball enumeration of Statement 3.7 with the explicit radius

$$N(T) = \left\lfloor M \left(\frac{T}{R} + 1 \right) \right\rfloor. \quad (14.10a)$$

It lists every word of length at most $N(T)$, multiplies exact matrices, and deduplicates their projective keys before applying the metric cutoff.

For each candidate lift G , (14.3) reduces membership to

$$2|a|^2 - 1 \leq \cosh(T/R). \quad (14.11)$$

The input specification supplies an exact representation of the threshold in an effectively ordered extension, or a decision procedure for every comparison in (14.11). An undecided equality produces UNDECIDED, not a partial ball.

Theorem 14.4.1. *Under that comparison hypothesis, the word-ball procedure terminates and returns every element of $B_{\text{geo}}(T)$ exactly once in lossless-key order.*

Proof. The search has only $\sum_{j=0}^{N(T)} 8^j$ raw words. By Statement 3.7, every element of the target ball has at least one word among them. Equation (14.11) is equivalent to the metric cutoff by the monotonicity of $\cosh$ on $[0, \infty)$, so filtering includes all and only the closed ball. The canonical projective key is lossless by Statement 5.5, while faithfulness is Statement 5.8; hence it identifies equal group elements and no unequal ones. Deduplication therefore proves exact-once output. $\square$

The record records $N(T)$, every enumerated word, exact product and key, each duplicate predecessor list, every accepted or rejected comparison certificate, and the final sorted key list. A checkpoint contains the complete word index, deduplication map, comparison state, and rolling hash, so a restart is independently recoverable. At $T = 0$, the output is $\{1\}$.

14.5 Separating the kernel from the based dangerous set

Let Q be a finite group with validated exact multiplication and inverse tables, and let $\pi : \Gamma_B \twoheadrightarrow Q$ be the validated surjection with kernel K . Run Statement 14.4 at threshold $2D$, remove the identity key, and obtain exactly

$$D_{\text{based}}(D) = B_{\text{geo}}(2D) \setminus \{1\}. \quad (14.12)$$

Evaluate every remaining word in Q .

Criterion 14.5.1 (Finite separation scan). *The scan returns SATISFIED exactly when*

$$\pi(\gamma) \neq e_Q \quad \text{for every } \gamma \in D_{\text{based}}(D), \quad (14.13)$$

and this is equivalent to $r_{\text{based}}(K; o) > D$.

Proof. Because $K = \ker \pi$, (14.13) is equivalent to $K \cap D_{\text{based}}(D) = \emptyset$. Criterion (14.9) makes that equivalent to the based-radius inequality. $\square$

The complete scan uses canonical exact-key order, stores one quotient evaluation per distinct key, and retains every geometric-ball predecessor and comparison certificate. Its record checks

$$|D_{\text{based}}(D)| = |B_{\text{geo}}(2D)| - 1, \quad (14.14)$$

the one removed key being exactly the identity. VIOLATED returns the least kernel key and its based displacement; an undecided boundary comparison returns UNDECIDED. Empty $D_{\text{based}}(0)$ gives the minimal SATISFIED fixture, while a validated quotient killing a listed element supplies the minimal negative witness.

14.6 Based no-wraparound for orbit samples

Assume now that $K \triangleleft \Gamma_B$ has finite index, and put

$$\mathcal{O}_D := \{\gamma o : \gamma \in \Gamma_B, \delta_o(\gamma) \leq D\}. \quad (14.15)$$

Let $p : \mathbb{H}_R^2 \rightarrow K \backslash \mathbb{H}_R^2$ be the quotient projection.

Theorem 14.6.1 (Based orbit no-wraparound). *If $r_{\text{based}}(K; o) > D$, then $p|_{\mathcal{O}_D}$ is injective.*

Proof. Suppose $p(\gamma o) = p(\delta o)$ for two points in $\mathcal{O}_D$. There is a $k \in K$ with $\delta o = k\gamma o$. The Γ_B -action is free, so $\delta = k\gamma$. Normality gives

$$h := \gamma^{-1}\delta = \gamma^{-1}k\gamma \in K. \quad (14.16)$$

Moreover,

$$\begin{aligned} \delta_o(h) &= d_{\mathbb{H}_R^2}(o, \gamma^{-1}\delta o) \\ &= d_{\mathbb{H}_R^2}(\gamma o, \delta o) \\ &\leq d_{\mathbb{H}_R^2}(\gamma o, o) + d_{\mathbb{H}_R^2}(o, \delta o) \\ &\leq D + D = 2D. \end{aligned} \quad (14.17)$$

If the two orbit points were distinct, freeness would give $h \neq 1$, contradicting (14.9). Hence they coincide. $\square$

This is a statement about the declared orbit sample, not every point of the continuous ball. Its independent SATISFIED condition is the complete Statement 14.5 scan. Finite index is needed for the finite quotient table and its certificate, not for the geometric implication once the stated normal subgroup and radius condition are given. The certificate contains the complete exact ball $B_{\text{geo}}(2D)$, its identity-removed dangerous sublist, a verified nonidentity quotient image for every dangerous key, and a independent verification of (14.16)–(14.17). Thus strictness is certified by avoiding the closed dangerous set; equality $\delta_o(h) = 2D$ is rejected.

14.7 Finite proof certificates for the based condition

Although K is infinite, (14.9) reduces its based-radius claim to a finite exact computation. A proof certificate contains:

- (i) the exact presentation, matrix field, basepoint o , radius D , threshold encoding, and input hash;
- (ii) a independently verifiable execution of the Statement 14.4 enumeration, with output $B_{\text{geo}}(2D)$ and all closed-boundary comparisons;
- (iii) the identity-removal count (14.14) and the ordered dangerous-list hash;
- (iv) the validated finite quotient table, marking, and every image of the dangerous list;
- (v) the verifier source hash, arithmetic-backend version, and final output hash.

Criterion 14.7.1 (Independent verifier). *Return SATISFIED if and only if the regenerated ball and all hashes match, the quotient relation and generation checks pass, the identity is the only removed key, and no dangerous key maps to e_Q . If all prerequisite structural, independent verification, hash, and quotient checks pass but a regenerated dangerous key maps to e_Q , return VIOLATED with that kernel witness. Return INVALID for malformed data, a hash or independent verification mismatch, or a failed quotient relation, table, surjectivity, or generation check. Return UNDECIDED when an exact threshold comparison is unavailable.*

Proof. The ball independent verification proves complete coverage by Statement 14.4. Removing exactly the identity produces (14.12). The quotient independent verification proves that table evaluation is the declared homomorphism. Criterion (14.9) then proves the SATISFIED conclusion. A VIOLATED supplies an element of $K \cap D_{\text{based}}(D)$, and hence proves that the strict radius claim is false. INVALID only rejects the proposed certificate and makes no assertion about the true radius; UNDECIDED likewise makes no radius claim. $\square$

Independent implementations rebuild the word ball through $N(2D)$ and their own deduplication map rather than trusting serialized completeness counts. Structural, independent verification, and hash validation occurs first. Corrupting one boundary flag, omitting one enumerated word, or altering a serialized quotient image without changing its validated table evaluation must produce INVALID. For a separately validated quotient in which a dangerous key genuinely evaluates to e_Q , the verifier must produce VIOLATED with that kernel witness. UNDECIDED is returned only after the preceding structural checks pass and a required exact comparison remains unavailable.

14.8 Based certificates are not global systole certificates

For every nonidentity $k \in K$,

$$\ell_h(k) \leq \delta_o(k).$$

After taking minima,

$$r_{\text{global}}(K) := \frac{1}{2} \text{sys}(K \setminus \mathbb{H}_R^2) \leq r_{\text{based}}(K; o). \quad (14.18)$$

Therefore the global condition $\text{sys}(K \setminus \mathbb{H}_R^2) > 2D$ implies the based condition $r_{\text{based}}(K; o) > D$, but the converse is not a formal consequence.

The dangerous lists display the difference:

$$\begin{aligned} D_{\text{based}}(D) &= \{\gamma \neq 1 : \delta_o(\gamma) \leq 2D\}, \\ \tilde{D}_{\text{sys}}(2D) &= \{\gamma \in B_{\text{geo}}(2D + 2D_P) : \gamma \neq 1, \ell_h(\gamma) \leq 2D\}. \end{aligned} \quad (14.19)$$

The second list covers conjugacy classes and generally searches a larger based ball.

The inequality alone cannot be reversed in the unrestricted class of discrete torsion-free hyperbolic quotients. The following cyclic example is outside the standing finite-index Bolza-kernel setting. Fix $D > 0$, let $K = \langle h \rangle$, where h is hyperbolic with $0 < \ell_h(h) < 2D$, and choose o at distance ρ from its axis. Statement 1.6 gives

$$\sinh \frac{\delta_o(h)}{2R} = \cosh \frac{\rho}{R} \sinh \frac{\ell_h(h)}{2R}. \quad (14.20)$$

For sufficiently large ρ , the right side exceeds $\sinh(D/R)$, so $\delta_o(h) > 2D$. For the cyclic group $\langle h \rangle$, the same formula with $\ell_h(h^m) = |m|\ell_h(h)$ shows every nonzero power has based displacement at least that of h . Thus the based test passes while the global systole test fails. Thus the converse fails in the unrestricted class. This does not establish, and does not claim, a BASED SATISFIED/GLOBAL VIOLATED example among finite-index normal subgroups of the fixed Γ_B .

The comparison verifier runs the based certificate of Statement 14.7 and the global certificate of Statement 11.7 independently and reports the ordered pair of classifications. GLOBAL SATISFIED/BASED VIOLATED is rejected as inconsistent with (14.18) whenever both certificates are valid and decisive. BASED SATISFIED/GLOBAL VIOLATED is logically permitted by (14.18), but existence in the finite-index Bolza class is not claimed. INVALID or UNDECIDED makes no mathematical assertion. Both input/code/output hash triples are retained, so neither result can be substituted for the other.

15 Global systole: conjugacy, exact cutoffs, and certificates

Throughout this chapter $K \triangleleft \Gamma_B$ has finite index, $Q = \Gamma_B/K$, and $\pi : \Gamma_B \twoheadrightarrow Q$ is represented by validated exact multiplication and inverse tables together with the eight marked generator images. Thus $X_K = K \backslash \mathbb{H}_R^2$ is a compact hyperbolic surface. The radius

$$D_P = \max_{x \in P} d_{\mathbb{H}_R^2}(o, x)$$

is the exact outer radius of the fixed Bolza fundamental polygon.

15.1 Closed geodesics and conjugacy classes

Put $X_B = \Gamma_B \backslash \mathbb{H}_R^2$. Every $1 \neq \gamma \in \Gamma_B$ is hyperbolic, and the projection of its invariant axis $\mathcal{A}_\gamma$ to X_B , traversed from x to γx , is an oriented closed geodesic of length $\ell_h(\gamma)$. Replacing the entire lifted axis by $u\mathcal{A}_\gamma$, and x by ux , replaces the deck element by $u\gamma u^{-1}$. Moving the initial point along the same lifted axis does not change γ .

Conversely, lift an oriented closed geodesic on X_B together with one full traversal. Its terminal lift is γ times its initial lift for a unique $\gamma \neq 1$. Replacing the lift by its u -translate conjugates γ by u . Thus nontrivial Γ_B -conjugacy classes correspond to oriented closed geodesics on the fixed surface X_B , counted with their traversal multiplicity. Primitive classes correspond to primitive geodesics; forgetting orientation identifies $[\gamma]_{\Gamma_B}$ with $[\gamma^{-1}]_{\Gamma_B}$.

For $L \geq 0$, define the finite covering set

$$E_L := \{\gamma \in B_{\text{geo}}(L + 2D_P) : \gamma \neq 1, \ell_h(\gamma) \leq L\}. \quad (15.1)$$

Every conjugacy class of translation length at most L meets E_L . Indeed, choose $x \in \mathcal{A}_\gamma$, and choose $\eta \in \Gamma_B$ with $y = \eta x \in P$. For $\gamma' = \eta\gamma\eta^{-1}$, the point y lies on $\mathcal{A}_{\gamma'}$, so

$$\begin{aligned} d_{\mathbb{H}_R^2}(o, \gamma' o) &\leq d_{\mathbb{H}_R^2}(o, y) + d_{\mathbb{H}_R^2}(y, \gamma' y) + d_{\mathbb{H}_R^2}(\gamma' y, \gamma' o) \\ &= d_{\mathbb{H}_R^2}(o, y) + \ell_h(\gamma') + d_{\mathbb{H}_R^2}(y, o) \\ &\leq D_P + L + D_P = L + 2D_P. \end{aligned} \quad (15.2)$$

Conjugacy invariance gives $\ell_h(\gamma') = \ell_h(\gamma)$, proving the claim.

To obtain a duplicate-free class list, the input specification includes a verified conjugacy solver for the fixed presentation of Γ_B . On two exact words it returns either a word u satisfying

$$u\alpha u^{-1} = \beta \quad (15.3)$$

with equality checked by the lossless matrix key, or a independently verifiable negative certificate from a fixed complete conjugacy procedure. Such a procedure is effective for the fixed closed-surface group; completeness of the chosen procedure is part of the verifier specification and is not inferred from failure to find a short conjugator.

Enumerate E_L in projective-key order and partition it by the certified equivalence relation

$$\alpha \sim_{\Gamma_B} \beta \iff \beta = u\alpha u^{-1} \text{ for some } u \in \Gamma_B. \quad (15.4)$$

The canonical oriented representative of a block is its least lossless key; the canonical unoriented representative is the smaller of the oriented blocks of γ and γ^{-1} . Equation (15.2) proves coverage. Equivalence-relation partitioning proves that no class occurs twice, and choosing exactly one least key proves uniqueness. The record records the raw word count, distinct element count, all comparison certificates, block sizes, their sum $|E_L|$, canonical ordering, and rolling hash. A checkpoint stores the next pair index and the complete union-find state, so resumption neither omits nor repeats a pair.

15.2 Nontrivial conjugacy classes in the kernel

There are two distinct equivalence relations. Ambient Γ_B -conjugacy is sufficient for a systole test because K is normal, but closed geodesics on X_K are indexed by conjugacy inside K . We now enumerate the latter without conflating the two.

Let $[\gamma]_{\Gamma_B}$ be a canonical class from Statement 15.1 with $\pi(\gamma) = e_Q$. The input specification includes a verified root and centralizer procedure returning

$$\gamma = r^m, \quad m \in \mathbb{Z} \setminus \{0\}, \quad r \text{ not a proper power}, \quad C_{\Gamma_B}(\gamma) = \langle r \rangle. \quad (15.5)$$

The last equality is the centralizer theorem for a nontrivial element of a torsion-free cocompact Fuchsian group: a commuting element preserves the two ideal endpoints of $\mathcal{A}_\gamma$, hence preserves the axis; its restriction is a translation. Discreteness makes the subgroup of signed translation distances discrete in $\mathbb{R}$, so it is generated by one least positive translation, represented by r .

The Γ_B -class of γ splits into K -classes indexed by

$$K \backslash \Gamma_B / C_{\Gamma_B}(\gamma) \cong Q / \langle \pi(r) \rangle, \quad (15.6)$$

where the right side denotes right cosets. To verify (15.6), two conjugates $t_1\gamma t_1^{-1}$ and $t_2\gamma t_2^{-1}$ are K -conjugate exactly when some $k \in K$ satisfies

$$kt_1\gamma t_1^{-1}k^{-1} = t_2\gamma t_2^{-1}.$$

After multiplying on the left by t_2^{-1} and on the right by t_2 , this is equivalent to

$$t_2^{-1}kt_1 \in C_{\Gamma_B}(\gamma),$$

which is exactly $Kt_1C_{\Gamma_B}(\gamma) = Kt_2C_{\Gamma_B}(\gamma)$. Normality identifies the double cosets with the right cosets in (15.6).

Enumerate Q from its validated table, form $H_\gamma = \langle \pi(r) \rangle$, and choose the least serialized member of every right coset qH_γ . Lift each chosen q by the least shortlex word found by quotient BFS and output

$$([\gamma]_{\Gamma_B}, qH_\gamma, [t_q\gamma t_q^{-1}]_K). \quad (15.7)$$

The first two fields form a canonical key. Equations (15.5)–(15.6) prove coverage and duplicate freedom. The count contributed by the ambient class is

$$\frac{|Q|}{|\langle \pi(r) \rangle|}. \quad (15.8)$$

Summing (15.8) over all retained ambient classes is an independent coverage invariant. A failed root, centralizer, quotient, or conjugacy certificate is INVALID, not evidence that a class is absent.

15.3 Translation length

For an orientation-preserving isometry g , its translation length is

$$\ell_h(g) := \inf_{x \in \mathbb{H}_R^2} d_{\mathbb{H}_R^2}(x, gx). \quad (15.9)$$

For $1 \neq \gamma \in \Gamma_B$, the infimum is attained precisely on $\mathcal{A}_\gamma$, is positive, and has the computable form

$$\ell_h(\gamma) = 2R \operatorname{arccosh} \frac{|\operatorname{tr} \Gamma^\sharp|}{2}, \quad (15.10)$$

where $\Gamma^\sharp$ is either of the two $\operatorname{SU}(1, 1)$ lifts of γ . The lift sign disappears under absolute value.

For $u \in \Gamma_B$ and $n \in \mathbb{Z}$,

$$\begin{aligned} \ell_h(u\gamma u^{-1}) &= \inf_x d_{\mathbb{H}_R^2}(x, u\gamma u^{-1}x) = \inf_y d_{\mathbb{H}_R^2}(uy, u\gamma y) = \ell_h(\gamma), \\ \ell_h(\gamma^n) &= |n|\ell_h(\gamma). \end{aligned} \quad (15.11)$$

For $n = 0$, the second equality is the identity case. For $n \neq 0$, γ^n has the same axis as γ and translates it by $|n|\ell_h(\gamma)$; the axis theorem shows this is the global minimum.

For a reproducible formula test, take

$$H_t = \begin{pmatrix} \cosh(t/2) & \sinh(t/2) \\ \sinh(t/2) & \cosh(t/2) \end{pmatrix}, \quad t > 0.$$

Then $|\operatorname{tr} H_t| = 2 \cosh(t/2)$, so (15.10) gives $\ell_h([H_t]) = Rt$. Thus $t = L/R$ is the exact dangerous boundary $\ell_h = L$, whereas $t = (L + \varepsilon)/R$, $\varepsilon > 0$, lies strictly outside. The identity is excluded before applying a dangerous length test.

15.4 Definition and realization of the systole

Define

$$\operatorname{sys}(X_K) := \min_{1 \neq k \in K} \ell_h(k). \quad (15.12)$$

This minimum exists without an appeal to numerical sampling. Choose any $1 \neq k_0 \in K$ and put $U = \ell_h(k_0)$. Equation (15.2) places a conjugate of every class of length at most U in the finite set E_U . Normality keeps those conjugates in K . The nonempty finite set

$$\{\ell_h(\gamma) : \gamma \in E_U, \pi(\gamma) = e_Q\}$$

therefore has a minimum, and every shorter kernel class would also occur in it. Its minimum is exactly (15.12).

Theorem 15.4.1 (Geometric meaning of the systole). *$\operatorname{sys}(X_K)$ is the length of the shortest noncontractible closed geodesic on X_K .*

Proof. For $1 \neq k \in K$, the quotient $\mathcal{A}_k/\langle k \rangle$ is a circle of length $\ell_h(k)$, and the covering projection induces from it a closed geodesic on X_K , possibly multiply covered. Conversely, lift any noncontractible closed curve c from a point $x \in \mathbb{H}_R^2$. Its terminal point is kx for a unique $1 \neq k \in K$. Length is preserved by the covering map, so

$$\text{length}(c) = \text{length}(\tilde{c}) \geq d_{\mathbb{H}_R^2}(x, kx) \geq \ell_h(k) \geq \text{sys}(X_K). \quad (15.13)$$

The axis quotient for a minimizing k attains the final bound. $\square$

Consequently, for every $L \geq 0$,

$$\text{sys}(X_K) > L \iff \ell_h(k) > L \text{ for all } 1 \neq k \in K. \quad (15.14)$$

The inequality is strict. A kernel element with length exactly L causes the condition to fail.

15.5 An exact trace cutoff

Let A be either lift of $\gamma \in \Gamma_B$, and define the projectively invariant exact quantity

$$\tau(\gamma) := (\text{tr } A)^2. \quad (15.15)$$

Changing A to $-A$ leaves τ unchanged. Since the action of Γ_B is free and cocompact, every $\gamma \neq 1$ is hyperbolic, so $\tau(\gamma) > 4$. For $L \geq 0$, put

$$T_L := 4 \cosh^2 \frac{L}{2R} = 2 \left(\cosh \frac{L}{R} + 1 \right). \quad (15.16)$$

The second expression follows from $2 \cosh^2 x = \cosh(2x) + 1$.

Criterion 15.5.1 (Trace cutoff). *For every $1 \neq \gamma \in \Gamma_B$,*

$$\ell_h(\gamma) \leq L \iff \tau(\gamma) \leq T_L. \quad (15.17)$$

Proof. By (15.10),

$$\begin{aligned} \ell_h(\gamma) \leq L &\iff 2R \operatorname{arcosh} \frac{|\operatorname{tr} A|}{2} \leq L \\ &\iff \operatorname{arcosh} \frac{|\operatorname{tr} A|}{2} \leq \frac{L}{2R} \\ &\iff \frac{|\operatorname{tr} A|}{2} \leq \cosh \frac{L}{2R} \\ &\iff (\operatorname{tr} A)^2 \leq 4 \cosh^2 \frac{L}{2R}. \end{aligned} \quad (15.18)$$

The third equivalence uses that $\cosh : [0, \infty) \rightarrow [1, \infty)$ and arcosh are increasing inverse functions. Both sides are nonnegative before squaring, so the last equivalence is reversible. $\square$

An exact verifier is given T_L in an effectively ordered extension of the coefficient field, or a certified decision procedure for every sign of $\tau(\gamma) - T_L$. Supplying only a floating approximation to L does not authorize a boundary decision. The canonical record is

$$(\text{element key}, \tau(\gamma), T_L, \text{sign}(\tau(\gamma) - T_L), \text{comparison certificate}). \quad (15.19)$$

For H_t from Statement 15.3, $t = L/R$ gives equality in (15.17), while $t = (L + \varepsilon)/R$ gives strict rejection. At $L = 0$, no nonidentity element of Γ_B passes, because $\tau > 4 = T_0$.

15.6 Axis distance and based displacement

For a hyperbolic γ , set

$$\rho_o(\gamma) := d_{\mathbb{H}_R^2}(o, \mathcal{A}_\gamma), \quad \delta_o(\gamma) := d_{\mathbb{H}_R^2}(o, \gamma o).$$

The Fermi-coordinate displacement identity gives

$$\sinh \frac{\delta_o(\gamma)}{2R} = \cosh \frac{\rho_o(\gamma)}{R} \sinh \frac{\ell_h(\gamma)}{2R}. \quad (15.20)$$

Every factor is positive for $\gamma \neq 1$. Therefore

$$\rho_o(\gamma) = R \operatorname{arcosh} \left(\frac{\sinh(\delta_o(\gamma)/(2R))}{\sinh(\ell_h(\gamma)/(2R))} \right), \quad (15.21)$$

and

$$\delta_o(\gamma) \geq \ell_h(\gamma), \quad \text{with equality exactly when } o \in \mathcal{A}_\gamma. \quad (15.22)$$

Indeed, the factor $\cosh(\rho_o/R)$ in (15.20) is at least one, with equality exactly at $\rho_o = 0$, and $\sinh$ is strictly increasing.

The global search bound is obtained by moving an axis, not by replacing translation length with based displacement. If $\ell_h(\gamma) \leq L$, choose $x \in \mathcal{A}_\gamma$ and $\eta \in \Gamma_B$ with $y = \eta x \in P$. Then $\gamma' = \eta \gamma \eta^{-1}$ satisfies

$$\rho_o(\gamma') \leq d_{\mathbb{H}_R^2}(o, y) \leq D_P \quad (15.23)$$

and, by the three segments displayed in (15.2),

$$\delta_o(\gamma') \leq \ell_h(\gamma') + 2D_P \leq L + 2D_P. \quad (15.24)$$

Thus a short global class can lie arbitrarily far from the chosen basepoint in one representative, but it always has a conjugate in the finite based ball of (15.24).

The formula tests separate the two notions. If $\rho_o = 0$, (15.20) gives $\delta_o = \ell_h$. If $\rho_o > 0$, it gives $\delta_o > \ell_h$. Holding $\ell_h < L$ fixed and letting $\rho_o \rightarrow \infty$ along conjugate representatives makes $\delta_o \rightarrow \infty$. Thus the cutoff $\delta_o(\gamma) \leq L$ does not enumerate all short elements and, by itself, supplies no uniform class-coverage theorem. The guaranteed class cover is (15.24), obtained after conjugation with the $2D_P$ margin.

15.7 The global dangerous set

Define the finite element covering list and the finite class set

$$\begin{aligned} \tilde{\mathcal{D}}_{\text{sys}}(L) &:= \{\gamma \in B_{\text{geo}}(L + 2D_P) : \gamma \neq 1, \ell_h(\gamma) \leq L\} = E_L, \\ \mathcal{C}_{\text{sys}}(L) &:= \{[\gamma]_{\Gamma_B} : \gamma \neq 1, \ell_h(\gamma) \leq L\}. \end{aligned} \quad (15.25)$$

The first is computable without a conjugacy solver. The second is the duplicate-free canonical list of Statement 15.1 when its verified conjugacy interface is available. Equation (15.2) proves that the projection

$$\tilde{\mathcal{D}}_{\text{sys}}(L) \longrightarrow \mathcal{C}_{\text{sys}}(L), \quad \gamma \longmapsto [\gamma]_{\Gamma_B} \quad (15.26)$$

is surjective.

There is also an axis-normalized covering family

$$\mathcal{A}_{\text{sys}}^P(L) := \{\mathcal{A}_\gamma : \gamma \neq 1, \ell_h(\gamma) \leq L, \mathcal{A}_\gamma \cap P \neq \emptyset\}. \quad (15.27)$$

Every short ambient class has an axis in this family after conjugation. Moreover (15.24) places every corresponding element in the finite ball $B_{\text{geo}}(L + 2D_P)$, so (15.27) is finite after equal axes are deduplicated. Computing (15.27) directly requires an exact closed axis–polygon intersection predicate; the element list (15.25) does not.

Criterion 15.7.1 (Complete global dangerous-set test).

$$\boxed{\begin{aligned} \text{sys}(X_K) > L &\iff K \cap \tilde{\mathcal{D}}_{\text{sys}}(L) = \emptyset \\ &\iff \pi(c) \neq e_Q \quad \text{for every canonical } [c]_{\Gamma_B} \in \mathcal{C}_{\text{sys}}(L). \end{aligned}} \quad (15.28)$$

Proof. If $d \in K \cap \tilde{\mathcal{D}}_{\text{sys}}(L)$, then $d \neq 1$ and

$$\text{sys}(X_K) \leq \ell_h(d) \leq L,$$

so the strict systole condition fails.

Conversely, if $\text{sys}(X_K) \leq L$, choose a minimizing $1 \neq k \in K$. Equation (15.2) supplies $d = \eta k \eta^{-1} \in \tilde{\mathcal{D}}_{\text{sys}}(L)$. Normality gives $d \in K$, so the intersection is nonempty. This proves the first equivalence.

Kernel membership is constant on an ambient conjugacy class because

$$\pi(\eta \gamma \eta^{-1}) = \pi(\eta) \pi(\gamma) \pi(\eta)^{-1}. \quad (15.29)$$

It equals e_Q after conjugation exactly when it did before. Surjectivity (15.26) and one representative per class therefore prove the second equivalence. $\square$

The cutoff in (15.25) is closed. A class of length exactly L is dangerous, so a SATISFIED certificate proves the requested strict inequality. If the conjugacy solver is unavailable, the covering-list version of (15.28) remains complete; it performs more quotient evaluations but makes no unproved class-uniqueness claim.

15.8 A streaming exact global kernel scan

The scan for the strict target $\text{sys}(X_K) > L$ uses the following finite stages.

- S1.** Validate the presentation, exact matrix field, polygon constants, finite group table, marked generator images, relations, and surjectivity.
- S2.** Put $S = L + 2D_P$. Invoke Statement 3.7 with the exact based-distance comparison interface for the closed threshold S , obtaining the projective-key-ordered ball $B_{\text{geo}}(S)$.
- S3.** Remove exactly the identity key. For every remaining element, compute $\tau(\gamma)$ and apply the exact closed trace comparison $\tau(\gamma) \leq T_L$. The retained list is exactly $\tilde{\mathcal{D}}_{\text{sys}}(L)$.
- S4.** Evaluate the stored canonical word of every retained distinct key in Q . Return VIOLATED with the first key whose value is e_Q , or SATISFIED after the final key has a verified nonidentity value.

Theorem 15.8.1 (Termination, coverage, and absence of duplicates). *If both exact threshold interfaces decide every required comparison, the scan terminates. It evaluates each element of $\tilde{\mathcal{D}}_{\text{sys}}(L)$ exactly once, misses none, and returns SATISFIED exactly when $\text{sys}(X_K) > L$.*

Proof. Statement 3.7 bounds Stage S2 by the finite raw word ball through

$$N(S) = \left\lfloor M \left(\frac{S}{R} + 1 \right) \right\rfloor. \quad (15.30)$$

It proves that the deduplicated output is exactly $B_{\text{geo}}(S)$. The lossless projective key makes distinct output keys distinct group elements. Stage S3 applies the definition (15.25) to every such element, so it retains the complete dangerous list and no other element. It does not deduplicate by trace, since distinct elements may have equal trace. Stage S4 has one table evaluation per retained key and is finite. Criterion (15.28) proves the final equivalence. $\square$

The covering record records

$$\begin{aligned}
n_{\text{raw}} &= \sum_{j=0}^{N(S)} 8^j, \\
n_{\text{ball}} &= |B_{\text{geo}}(S)|, \\
n_{\text{nonid}} &= n_{\text{ball}} - 1, \\
n_{\text{danger}} &= |\tilde{\mathcal{D}}_{\text{sys}}(L)|, \\
n_Q &= n_{\text{danger}}.
\end{aligned} \tag{15.31}$$

It also records every raw-word predecessor list, exact product and projective key, duplicate multiplicity, based comparison, trace comparison, quotient value, and the hash of each ordered stage. The equalities in (15.31), together with the predecessor multiplicities summing to n_{raw} , are checked during independent verification.

A checkpoint is committed only after an atomic record consisting of the raw word index, deduplication map, filter index, quotient index, rolling hashes, and previous checkpoint hash. Resumption replays the last complete record and continues at its successor. It cannot convert an unfinished prefix into SATISFIED.

All structural and independent verification checks precede mathematical classification. Malformed input, a count/hash mismatch, an invalid quotient, or an altered stored evaluation returns INVALID. An unavailable exact cutoff comparison returns UNDECIDED. VIOLATED is reserved for a fully regenerated dangerous element that genuinely evaluates to e_Q ; it includes the element word, matrix key, trace certificate, and quotient multiplication trace.

An optional class mode first applies the verified Statement 15.1 partition and evaluates one canonical representative per ambient conjugacy class. Normality and (15.29) make this reduction sound. Without that verified partition, the element covering mode above remains the required complete algorithm.

15.9 Systole certificates

A threshold certificate for the assertion $\text{sys}(X_K) > L$ contains:

- (i) the exact presentation, matrix field, polygon data R, D_P , the nonnegative threshold L , its exact comparison interfaces, and input hash;
- (ii) the validated quotient table, inverse table, marking, relation checks, surjectivity transcript, and canonical quotient hash;
- (iii) the complete Statement 15.8 records through $B_{\text{geo}}(L + 2D_P)$, identity removal, trace filtering, and quotient evaluation;
- (iv) every comparison certificate, ordered dangerous-list hash, verifier source hash, arithmetic-backend version, and final hash.

Criterion 15.9.1 (Independent threshold verifier). *After independently regenerating all finite stages, return:*

- (a) *SATISFIED if every structural, independent verification, threshold, and quotient check is valid and no dangerous key evaluates to e_Q ;*
- (b) *VIOLATED if all prerequisite checks are valid and a regenerated dangerous key evaluates to e_Q ;*
- (c) *INVALID for malformed data, a failed presentation or quotient check, or any independent verification, count, ordering, evaluation, or hash mismatch;*

(d) *UNDECIDED* if the structural checks pass but a required exact boundary comparison cannot be decided.

Proof. On SATISFIED, Statement 15.8 has regenerated every element of $\tilde{\mathcal{D}}_{\text{sys}}(L)$, and none belongs to $\ker \pi = K$. Criterion (15.28) proves $\text{sys}(X_K) > L$. On VIOLATED, the emitted element d satisfies, by independently checked records,

$$d \neq 1, \quad \pi(d) = e_Q, \quad \tau(d) \leq T_L.$$

Thus $d \in K \setminus \{1\}$, and (15.17) gives

$$\text{sys}(X_K) \leq \ell_h(d) \leq L, \quad (15.32)$$

which is a mathematical counter-witness. INVALID and UNDECIDED make no systole assertion. $\square$

The same finite mechanism can certify the exact systole rather than only a threshold. Let

$$m = \text{ord}_Q(\pi(g_0)), \quad k_0 = g_0^m.$$

Then $m \geq 1$, $\pi(k_0) = e_Q$, and $k_0 \neq 1$ because g_0 is hyperbolic and has infinite order. Put $U = \ell_h(k_0)$, run the complete global scan through the closed threshold U , and form

$$F_U := \{\gamma \in \tilde{\mathcal{D}}_{\text{sys}}(U) : \pi(\gamma) = e_Q\}. \quad (15.33)$$

This finite set is nonempty because a conjugate of k_0 occurs in it. Let $d_{\min}$ be the least projective key among the elements minimizing τ on F_U . Since (15.10) is strictly increasing in $|\text{tr}|$ on the hyperbolic range,

$$\boxed{\text{sys}(X_K) = 2R \text{arcosh} \frac{\sqrt{\tau(d_{\min})}}{2}}. \quad (15.34)$$

Indeed, $d_{\min} \in K \setminus \{1\}$, so the right side is at least the systole. If a kernel element had smaller translation length, its ambient conjugacy class would have a representative in F_U by (15.2), with a strictly smaller τ , contradicting the choice of $d_{\min}$. Equality follows.

An exact-value certificate adds m, k_0, U , the complete set F_U , all pairwise minimum comparisons, and the tie-breaking key. It stores $\tau(d_{\min})$ and the symbolic expression (15.34), rather than rounding the inverse hyperbolic function. Equality of several retained elements at the minimum is recorded as a tie list, not discarded. If multiplicities of shortest ambient conjugacy classes are claimed, the verified Statement 15.1 partition is additionally applied and one canonical representative per block is retained. Multiplicity of K -conjugacy classes additionally requires the Statement 15.2 double-coset splitting.

15.10 The global no-wraparound theorem

Let $\rho \geq 0$, and let

$$p : \mathbb{H}_R^2 \longrightarrow X_K, \quad p(y) = Ky$$

be the quotient covering. The closed-ball convention is fixed throughout this subsection.

Theorem 15.10.1 (Exact no-wraparound criterion).

$$\boxed{p|_{\overline{B}_{\mathbb{H}_R^2}(x, \rho)} \text{ is injective for every } x \in \mathbb{H}_R^2 \iff \text{sys}(X_K) > 2\rho}. \quad (15.35)$$

Proof. Assume first that $\text{sys}(X_K) > 2\rho$. Let $y, z \in \overline{B}_{\mathbb{H}_R^2}(x, \rho)$ and suppose $p(y) = p(z)$. Then $z = ky$ for some $k \in K$. If $k \neq 1$, then

$$\begin{aligned}
\text{sys}(X_K) &\leq \ell_h(k) \\
&\leq d_{\mathbb{H}_R^2}(y, ky) \\
&= d_{\mathbb{H}_R^2}(y, z) \\
&\leq d_{\mathbb{H}_R^2}(y, x) + d_{\mathbb{H}_R^2}(x, z) \\
&\leq \rho + \rho = 2\rho,
\end{aligned} \tag{15.36}$$

contradicting the strict assumption. Hence $k = 1$, and $y = z$. This proves injectivity for every center.

Conversely, suppose $\text{sys}(X_K) \leq 2\rho$. Choose a systolic element $1 \neq k \in K$, choose $y \in \mathcal{A}_k$, put $z = ky$, and let x be the midpoint of the axis segment from y to z . Since displacement on the axis realizes translation length,

$$\begin{aligned}
d_{\mathbb{H}_R^2}(y, z) &= \ell_h(k) = \text{sys}(X_K), \\
d_{\mathbb{H}_R^2}(x, y) &= d_{\mathbb{H}_R^2}(x, z) = \frac{1}{2} \text{sys}(X_K) \leq \rho.
\end{aligned} \tag{15.37}$$

Thus $y, z \in \overline{B}_{\mathbb{H}_R^2}(x, \rho)$, while $y \neq z$ and $p(y) = p(z)$. The restriction is not injective for this center, proving the converse. $\square$

At equality $\text{sys}(X_K) = 2\rho$, the two points in (15.37) lie on the boundary of the closed ball and give wraparound. Therefore replacing the strict sign in (15.35) by a non-strict one is false for the declared closed-ball semantics.

The independent no-wrap verifier invokes the Statement 15.9 threshold verifier with $L = 2\rho$. A valid SATISFIED certifies the right side of (15.35) and therefore the left side. A valid VIOLATED supplies $d \in K \setminus \{1\}$ with $\ell_h(d) \leq 2\rho$; choosing its axis point and midpoint as in (15.37) yields a independently verifiable geometric wraparound witness. INVALID and UNDECIDED retain their Statement 15.9 meanings and make no no-wrap assertion. The certificate records the exact multiplication by two, the closed-ball flag, the systole-threshold record, and the axis/midpoint construction for every VIOLATED witness.

Part V

Residual and characteristic constructions

16 Residual finiteness and normal residual constructions

16.1 Residually finite groups

Definition 16.1.1. A group G is residually finite if, for every $1 \neq g \in G$, there is a finite group F_g and a homomorphism $\varphi_g : G \rightarrow F_g$ such that $\varphi_g(g) \neq e_{F_g}$.

Theorem 16.1.2 (Equivalent residual criteria). *For a group G , the following are equivalent:*

- (i) G is residually finite;
- (ii) the intersection of all finite-index normal subgroups is trivial;
- (iii) the diagonal map from G to the product of all its finite quotients is injective.

Proof. Assume (i), and let $1 \neq g \in G$. A separating map φ_g has finite-index normal kernel and does not contain g . Thus no nonidentity element lies in the intersection in (ii).

Assume (ii). For $g \neq 1$, choose a finite-index normal subgroup N with $g \notin N$. The quotient map

$$G \longrightarrow G/N \quad (16.1)$$

has finite target and sends g to a nonidentity coset, proving (i).

Finally, the kernel of the diagonal map in (iii) is exactly the intersection of the kernels of all maps to finite groups. Every such kernel is finite-index and normal; conversely every finite-index normal subgroup occurs as the kernel of its finite quotient map. Hence this kernel is the intersection in (ii), proving (ii) $\iff$ (iii). $\square$

The definition requires a homomorphism, not merely a finite set on which one group element acts nontrivially. A canonical separation record has the form

$$(\text{input word, lossless input key, finite group table, generator marking, image, validation hash}). \quad (16.2)$$

For $G = \mathbb{Z}$ and $m \neq 0$, reduction modulo any prime $p \nmid m$ is a positive test. Every finite group is residually finite by the identity map. In contrast, the additive group $\mathbb{Q}$ is not residually finite: if $f : \mathbb{Q} \rightarrow F$ and $n = |F|$, write $x = ny$; then $f(x) = f(y)^n = e_F$. Thus every finite image is trivial.

16.2 Residual finiteness of the Bolza surface group

Use the exact field and integral model of Statement 5.2:

$$E = \mathbb{Q}(\zeta_{16}, \sqrt{2 + 2\sqrt{2}}), \quad \rho : \Gamma_B \longrightarrow \mathrm{SL}_2(\mathcal{O}_E). \quad (16.3)$$

The entries are algebraic integers, the presentation relator maps to I , and the projectivization of ρ is the faithful deck action. Consequently,

$$\gamma \neq 1 \implies \rho(\gamma) \neq I \text{ and } \rho(\gamma) \neq -I. \quad (16.4)$$

Indeed, either scalar matrix would project to the identity isometry, and faithfulness would force $\gamma = 1$; the homomorphism sends the identity to I , not to $-I$.

Let

$$G = \rho(\gamma) = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \text{for } \gamma \neq 1.$$

At least one member of

$$\{b, c, a - d\} \quad (16.5)$$

is nonzero. Otherwise $G = aI$, and $\det G = 1$ gives $a^2 = 1$. Since E has characteristic zero, $a = \pm 1$, contradicting (16.4). Choose the first nonzero member λ in the fixed order shown in (16.5).

Its field norm $N_{E/\mathbb{Q}}(\lambda)$ is a nonzero integer. Choose a rational prime

$$p \nmid 2N_{E/\mathbb{Q}}(\lambda), \quad (16.6)$$

and a prime ideal $\mathfrak{p}$ of $\mathcal{O}_E$ above p . The residue field $k_{\mathfrak{p}} = \mathcal{O}_E/\mathfrak{p}$ is finite and has odd characteristic. Since

$$N_{E/\mathbb{Q}}(\lambda) \not\equiv 0 \pmod{p},$$

$\lambda \notin \mathfrak{p}$: if $\lambda \in \mathfrak{p}$, then $\mathfrak{p}$ would divide the principal ideal (λ) , forcing p to divide its ideal norm and hence the absolute field norm.

Entrywise reduction therefore defines

$$\bar{\rho}_{\mathfrak{p}} : \Gamma_B \longrightarrow \mathrm{PSL}_2(k_{\mathfrak{p}}), \quad \gamma \longmapsto [\rho(\gamma) \bmod \mathfrak{p}]. \quad (16.7)$$

Determinant one is preserved. The image of the witness in (16.5) is nonzero, so the reduced matrix is not scalar. In odd characteristic the scalar matrices in $\mathrm{SL}_2(k_{\mathfrak{p}})$ are precisely $\pm I$; hence $\bar{\rho}_{\mathfrak{p}}(\gamma) \neq 1$. The target in (16.7) is finite. We have proved:

Theorem 16.2.1. *The Bolza surface group Γ_B is residually finite.*

For a generator g_j , the off-diagonal entry $\sqrt{2 + 2\sqrt{2}}\zeta^j$ is a nonzero witness, giving the minimal positive test. For the identity, every member of (16.5) is zero; the separation precondition fails rather than producing a false finite witness.

16.3 A constructive finite-quotient separation theorem

The preceding proof yields a deterministic algorithm. Its input is a word w in the fixed presentation alphabet with a validated nonidentity projective key. Its output is a finite multiplication table Q_w , a marked surjection $\pi_w : \Gamma_B \rightarrow Q_w$, and a certificate that $\pi_w(w) \neq e_{Q_w}$.

- Step I.** Evaluate w as the exact matrix $G = \rho(w) \in \mathrm{SL}_2(\mathcal{O}_E)$, verify $G \neq \pm I$, and select the first nonzero witness λ from (16.5).
- Step II.** Compute the nonzero integer $n_{\lambda} = |N_{E/\mathbb{Q}}(\lambda)|$. Enumerate rational primes in increasing order and select the first $p \nmid 2n_{\lambda}$.
- Step III.** Factor $p\mathcal{O}_E$ with a certified number-field routine, serialize its prime factors canonically, and choose the least $\mathfrak{p}$. Reduce the four presentation-generator matrices modulo $\mathfrak{p}$ and then project to $\mathrm{PSL}_2(k_{\mathfrak{p}})$.
- Step IV.** Enumerate, by marked-generator BFS, the subgroup generated by those four images. Canonically relabel its carrier, construct exact multiplication and inverse tables, and call the resulting image Q_w .
- Step V.** Re-evaluate the surface relator and w in the table. Accept only if the relator is the identity, the marking generates all of Q_w , and the image of w is nonidentity.

Theorem 16.3.1 (Finite-quotient separation). *For every $1 \neq \gamma \in \Gamma_B$, Steps I–V terminate and return a finite marked quotient separating γ from the identity.*

Proof. Exact matrix evaluation terminates in the fixed number field. By (16.4) Step I finds a nonzero witness. Its norm is a nonzero integer, so only finitely many rational primes divide $2n_{\lambda}$; prime enumeration therefore reaches Step II's choice. Ideal factorization in a fixed number field is finite, and the residue field is finite, so the subgroup search in Step IV terminates.

Reduction is a ring homomorphism, hence preserves multiplication, determinants, and the exact relator. Therefore the marked table defines a homomorphism from the presentation, and using its generated image makes that homomorphism surjective. Condition (16.6) preserves the selected nonscalar witness, so the image of γ is nonidentity by the proof of Statement 16.2. Step V independently verifies all three claims. $\square$

The record stores the exact input matrix and key, witness coordinate, signed norm, rejected-prime divisibility checks, defining data for $\mathfrak{p}$ and its residue field, reduced matrices, BFS predecessor of every group element, table and marking, relation evaluations, image of w , and all hashes. Fixed prime order, ideal serialization, BFS order, and carrier relabeling make identical inputs produce identical outputs.

16.4 Simultaneous separation of a finite set

Let $S \subseteq G \setminus \{1\}$ be finite and let G be residually finite. For each $s \in S$, choose $\varphi_s : G \rightarrow F_s$ with $\varphi_s(s) \neq 1$, and define

$$\Phi_S : G \longrightarrow \prod_{s \in S} F_s, \quad \Phi_S(g) = (\varphi_s(g))_{s \in S}. \quad (16.8)$$

For a fixed s , the s -coordinate of $\Phi_S(s)$ is nonidentity, so $\Phi_S(s) \neq 1$. Its generated image

$$Q_S := \Phi_S(G)$$

is finite, the induced map $G \twoheadrightarrow Q_S$ is surjective, and

$$\ker \Phi_S = \bigcap_{s \in S} \ker \varphi_s \quad (16.9)$$

is finite-index and normal. This proves simultaneous finite separation in every residually finite group.

For Γ_B , the integral representation gives a single-residue-field version. Deduplicate S by the lossless projective key. For each $s \in S$, select a nonzero witness λ_s from (16.5), and put

$$\Lambda_S := \prod_{s \in S} \lambda_s \in \mathcal{O}_E. \quad (16.10)$$

Because $\mathcal{O}_E$ is an integral domain, $\Lambda_S \neq 0$. Choose

$$p \nmid 2N_{E/\mathbb{Q}}(\Lambda_S) \quad (16.11)$$

and any prime ideal $\mathfrak{p} \mid p\mathcal{O}_E$. No λ_s lies in $\mathfrak{p}$: their product has nonzero residue in the field $k_{\mathfrak{p}}$. Thus the single reduction

$$\Gamma_B \longrightarrow \mathrm{PSL}_2(k_{\mathfrak{p}}) \quad (16.12)$$

separates every member of S simultaneously.

The empty set is separated by the trivial quotient; the empty product in (16.10) is one. A requested set containing the identity is INFEASIBLE, because every homomorphism sends the identity to the identity. This is a bad separation specification, not a failure of residual finiteness. The record records the input count, distinct-key count, duplicate predecessors, every λ_s , the product and norm checks, all reduced images, and the equality

$$|\{s \in S : \bar{\rho}_{\mathfrak{p}}(s) \neq 1\}| = |S|. \quad (16.13)$$

16.5 A normal residual tower

Enumerate the nonidentity elements of Γ_B exactly once as

$$\gamma_1, \gamma_2, \dots \quad (16.14)$$

by shortlex word BFS, exact projective-key deduplication, and identity removal. Every element appears because it has a finite word; every BFS layer is finite, so each fixed element appears after finitely many steps.

For each i , let $\psi_i : \Gamma_B \twoheadrightarrow F_i$ be the deterministic separating quotient from Statement 16.3. Define

$$\begin{aligned} \Phi_n &= (\psi_1, \dots, \psi_n) : \Gamma_B \longrightarrow \prod_{i=1}^n F_i, \\ Q_n &:= \Phi_n(\Gamma_B), \\ N_n &:= \ker \Phi_n = \bigcap_{i=1}^n \ker \psi_i, \end{aligned} \quad (16.15)$$

with $N_0 = \Gamma_B$ and $Q_0 = \{1\}$.

Theorem 16.5.1 (Normal residual tower). *Every N_n is normal and finite-index,*

$$\Gamma_B = N_0 \supseteq N_1 \supseteq N_2 \supseteq \cdots, \quad \bigcap_{n \geq 0} N_n = \{1\}. \quad (16.16)$$

Moreover the coordinate projection induces a surjection

$$q_{n+1,n} : Q_{n+1} \twoheadrightarrow Q_n. \quad (16.17)$$

Proof. Each N_n is the kernel of a homomorphism to a finite direct product, so it is normal and finite-index. Adding one kernel to an intersection gives $N_{n+1} \subseteq N_n$. If $1 \neq \gamma \in \Gamma_B$, then $\gamma = \gamma_j$ for a unique j , and $\psi_j(\gamma_j) \neq 1$; hence $\gamma \notin N_j$. No nonidentity element lies in the total intersection, proving (16.16).

For any $\Phi_n(g) \in Q_n$, the element $\Phi_{n+1}(g) \in Q_{n+1}$ projects to it. Thus the restricted coordinate projection is onto, proving (16.17). $\square$

The inclusions need not all be strict: an already separated γ_{n+1} may leave the kernel unchanged. The tower certificate therefore records equality or strictness rather than assuming growth. At stage n , the canonical image subgroup is enumerated by BFS in the product, and its transition map is stored and checked on every carrier element. A checkpoint contains the last emitted element key, all individual quotient hashes, the product-image table, and the transition hash; independent verification starts from the preceding complete stage.

16.6 Asymptotic and quantitative injectivity radius

Put

$$X_n = N_n \setminus \mathbb{H}_R^2, \quad q_n = [\Gamma_B : N_n] = |Q_n|, \quad r_n = \frac{1}{2} \text{sys}(X_n). \quad (16.18)$$

The sequence r_n is nondecreasing because $N_{n+1} \subseteq N_n$, so the minimum translation length over $N_{n+1} \setminus \{1\}$ cannot be smaller than the corresponding minimum over $N_n \setminus \{1\}$.

Theorem 16.6.1 (Divergence along the residual tower).

$$r_n \longrightarrow +\infty. \quad (16.19)$$

Proof. Fix $L \geq 0$. The complete dangerous element list $\tilde{\mathcal{D}}_{\text{sys}}(2L)$ of Statement 15.7 is finite. Every one of its elements occurs in (16.14), so there is an index n_L not smaller than all their enumeration indices. For $n \geq n_L$, each listed element γ_i is excluded from N_n by its own factor ψ_i . Criterion (15.28) gives

$$\text{sys}(X_n) > 2L, \quad r_n > L.$$

This holds for every L , which proves (16.19). $\square$

There is an exact order-dependent upper bound. Gauss–Bonnet gives

$$\text{area}(X_B) = 4\pi R^2, \quad \text{area}(X_n) = 4\pi R^2 q_n. \quad (16.20)$$

Every disk of radius $r < r_n$ embeds in X_n , and a hyperbolic disk has area

$$A_{\text{disk}}(r) = 2\pi R^2 (\cosh(r/R) - 1).$$

Comparing areas and taking $r \uparrow r_n$ gives

$$2\pi R^2 (\cosh(r_n/R) - 1) \leq 4\pi R^2 q_n,$$

hence

$$\boxed{r_n \leq R \operatorname{arcosh}(2q_n + 1)}. \quad (16.21)$$

For $L \geq 0$, define the residual-finiteness growth target

$$\text{RF}_{\Gamma_B}(L) := \min\{|Q| : \pi : \Gamma_B \twoheadrightarrow Q, Q \text{ finite, } \text{sys}(\ker \pi \backslash \mathbb{H}_R^2) > L\}. \quad (16.22)$$

Statement 16.4 applied to $\tilde{\mathcal{D}}_{\text{sys}}(L)$ proves the feasible set is nonempty. Let Q_L^{prod} be the image of the product of the individual Statement 16.3 separators Q_d . Then

$$\max\left\{1, \left\lfloor \frac{\cosh(L/(2R)) - 1}{2} \right\rfloor + 1\right\} \leq \text{RF}_{\Gamma_B}(L) \leq |Q_L^{\text{prod}}| \leq \prod_{d \in \tilde{\mathcal{D}}_{\text{sys}}(L)} |Q_d|. \quad (16.23)$$

The left inequality follows from (16.21): admissibility gives the strict inequality

$$|Q| > \frac{\cosh(L/(2R)) - 1}{2},$$

and integrality gives the displayed floor-plus-one bound. The last inequality in (16.23) follows because an image subgroup of a direct product has order at most the product of the factor orders.

A certified passing threshold $\underline{L}_n$ from Statement 15.9 yields the strict lower estimate

$$r_n > \frac{\underline{L}_n}{2},$$

while (16.21) is the certified upper estimate. Numerical output rounds $\underline{L}_n/2$ downward and $R \text{arcosh}(2q_n + 1)$ upward. The exact order q_n , input intervals, rounding mode, and directed endpoints are stored. Residual finiteness proves divergence but no universal rate in the stage number n ; such a rate depends on the element order and separator sizes.

As a small boundary check, $L = 0$ has an empty dangerous list, so the trivial quotient of order one is feasible and $\text{RF}_{\Gamma_B}(0) = 1$. Formula (16.21) at $q = 1$ gives the independently checkable area bound $r \leq R \text{arcosh } 3$.

16.7 Existence of finite quotients with any target systole

Theorem 16.7.1 (Finite quotient at every global scale). *For every $L \geq 0$, there is a finite marked quotient*

$$\pi_L : \Gamma_B \twoheadrightarrow Q_L$$

whose kernel K_L satisfies

$$\text{sys}(K_L \backslash \mathbb{H}_R^2) > L. \quad (16.24)$$

Proof. Enumerate the finite closed dangerous list $\tilde{\mathcal{D}}_{\text{sys}}(L)$ by Statement 15.8 without performing its quotient stage. Every member is nonidentity. Apply simultaneous separation Statement 16.4 to obtain a finite quotient in which every listed element has nonidentity image. Let K_L be its kernel. Criterion (15.28) gives (16.24). If the list is empty, the trivial quotient has kernel Γ_B , and the same criterion proves (16.24). $\square$

This proof is an explicit construction:

- F1.** Validate $L \geq 0$, R , D_P , the two exact threshold interfaces, and the integral matrix model.
- F2.** Enumerate $B_{\text{geo}}(L + 2D_P)$, remove the identity, and apply the closed trace cutoff $\tau \leq T_L$.
- F3.** If nothing remains, output the canonical trivial marked quotient. Otherwise form the product Λ of the first nonzero nonscalar witness for every retained key.
- F4.** Choose the least rational prime satisfying (16.11), the least canonical prime ideal above it, reduce all generators, and enumerate their image Q_L .

F5. Verify the table, marking, relator, surjectivity, and nonidentity image of every dangerous key. Output Q_L, π_L , and the combined Statement 15.8/Statement 16.4 certificate.

Every finite stage terminates by Statement 15.8 and Statement 16.4. The choices are made in fixed orders, so the output is reproducible. Equality $\ell_h(\gamma) = L$ is included in the dangerous list and separated, which is why the conclusion is strict. The construction proves existence and supplies an upper bound on quotient order; it does not claim that the returned quotient is order-optimal, characteristic, C_8 -stable, or parity compatible. Those are additional constraints.

16.8 A constructive residual-finiteness interface

The complete procedure accepts one of two input modes:

ELEMENTS : a finite serialized list of exact words required to survive,
RADIUS : a nonnegative L and exact comparison interfaces. (16.25)

In ELEMENTS mode, evaluate and deduplicate projective keys and reject the identity as INFEASIBLE. In RADIUS mode, regenerate $\tilde{\mathcal{D}}_{\text{sys}}(L)$; an unavailable exact boundary comparison returns UNDECIDED. Call the resulting identity-free finite set S .

The construction then executes the following pseudocode specification:

1. $S = \emptyset$: return the canonical trivial quotient;
2. for each $s \in S$, compute $\rho(s)$ and its first nonzero witness λ_s ;
3. $\Lambda = \prod_{s \in S} \lambda_s$, $p = \min\{q \text{ rational prime} : q \nmid 2N_{E/\mathbb{Q}}(\Lambda)\}$;
4. $\mathfrak{p}$ = least canonical prime ideal above p ;
5. $Q = \langle \rho(\Gamma_B) \bmod \mathfrak{p} \rangle \leq \text{PSL}_2(\mathcal{O}_E/\mathfrak{p})$;
6. canonically table Q , re-evaluate every s , and return the marked quotient and certificate. (16.26)

Theorem 16.8.1 (Constructive residual finiteness). *For every valid finite $S \subseteq \Gamma_B \setminus \{1\}$, (16.26) terminates and returns a finite marked quotient in which every member of S is nonidentity. In RADIUS mode its kernel has systole strictly greater than L .*

Proof. Exact word evaluation and finite-key deduplication terminate. Each $s \neq 1$ has a nonzero witness by (16.4)–(16.5), so the product Λ is nonzero. Its norm is a nonzero integer; consequently the prime search in Step 3 terminates. The residue field is finite, so image BFS and table construction terminate.

Condition (16.11) leaves every λ_s nonzero after reduction. Thus no reduced matrix for s is scalar, and every projective image is nonidentity. The generated image makes the marked map surjective. In RADIUS mode, S is the complete dangerous list, so (15.28) proves the strict systole conclusion. □

The independent verifier first reconstructs the field, integral basis, ideal, residue operations, and reduced generator matrices. It then rebuilds the image table, checks all presentation relations and generation, and re-evaluates all input keys. SATISFIED is returned only when every structural check succeeds and all required images are nonidentity. INVALID covers malformed inputs, arithmetic/table/independent verification failures, and hash mismatches. INFEASIBLE is reserved for a required identity element; UNDECIDED is reserved for an unresolved exact radius comparison. None of these three classifications is a counterexample to residual finiteness.

The certificate includes input and deduplication records, every matrix and witness, the exact product Λ , norm and rejected-prime transcript, prime-ideal factorization certificate, residue-field data, canonical table and marking, per-element evaluation traces, and input/code/output hashes.

Required regression fixtures are:

S	expected result
$\emptyset$	SATISFIED with the trivial quotient
$\{g_0\}$	SATISFIED with a nontrivial finite image
$\{g_0, g_0\}$	the same result after key deduplication
$\{1\}$	INFEASIBLE
a valid certificate with one altered table entry	INVALID.

(16.27)

17 Canonical characteristic residual towers

17.1 Characteristic subgroups

Definition 17.1.1. A subgroup $H \leq G$ is characteristic, written $H \text{ char } G$, if

$$\alpha(H) = H \quad \text{for every } \alpha \in \text{Aut}(G). \quad (17.1)$$

Every characteristic subgroup is normal because each inner automorphism is an automorphism. The converse is false: in $C_2 \times C_2$, every subgroup is normal, but an automorphism permuting the three nonzero elements permutes the three order-two subgroups, so none of those three is characteristic. The trivial subgroup, the whole group, the center, and the commutator subgroup are standard characteristic examples. Arbitrary intersections of characteristic subgroups are characteristic, since

$$\alpha \left(\bigcap_i H_i \right) = \bigcap_i \alpha(H_i) = \bigcap_i H_i. \quad (17.2)$$

If $H \text{ char } G$, every automorphism descends by

$$\bar{\alpha} : G/H \longrightarrow G/H, \quad \bar{\alpha}(gH) = \alpha(g)H. \quad (17.3)$$

This is well defined: $gH = g'H$ means $g'^{-1}g \in H$, so $\alpha(g')^{-1}\alpha(g) \in \alpha(H) = H$. It is a homomorphism, and the map induced by α^{-1} is its inverse. Thus $\bar{\alpha} \in \text{Aut}(G/H)$.

A canonical subgroup record stores membership by a kernel map rather than by generator spelling:

$$(\text{ambient presentation, finite quotient table, marking, kernel certificate, hash}). \quad (17.4)$$

The boundary examples $H = \{1\}$ and $H = G$ must both pass (17.1). A merely normal order-two subgroup of C_2^2 is the minimal negative test against replacing *characteristic* by *normal*.

17.2 Finite-index characteristic subgroups

Let G be generated by $d < \infty$ elements and let $N \leq G$ have finite index n . Define its characteristic core

$$\text{CharCore}_G(N) := \bigcap_{\alpha \in \text{Aut}(G)} \alpha(N). \quad (17.5)$$

Although the automorphism group can be infinite, the orbit in (17.5) is finite. Indeed, an index- n subgroup gives a transitive action on its left cosets and hence a homomorphism $G \rightarrow S_n$. A homomorphism is determined by the images of the d generators, so there are at most $(n!)^d$ such action tables and therefore finitely many index- n subgroups.

Theorem 17.2.1. *CharCore_G(N) is characteristic, has finite index, and is contained in N. If M_n is the number of distinct subgroups in the automorphism orbit of N, then*

$$[G : \text{CharCore}_G(N)] \leq n^{M_n} \leq n^{(n!)^d}. \quad (17.6)$$

Proof. For $\beta \in \text{Aut}(G)$,

$$\beta \left(\bigcap_{\alpha} \alpha(N) \right) = \bigcap_{\alpha} (\beta\alpha)(N) = \bigcap_{\alpha} \alpha(N),$$

because left multiplication by β permutes all automorphisms. Thus the core is characteristic. The identity automorphism occurs, so the intersection is contained in N .

List the distinct orbit subgroups as $N_1, \dots, N_{M_n}$, and put $H = \bigcap_i N_i$. There is an injective map of finite sets

$$G/H \longrightarrow \prod_{i=1}^{M_n} G/N_i, \quad gH \longmapsto (gN_i)_i.$$

It is well defined because $H \subseteq N_i$ for every i , and it is injective because equality of every left coset gives $g'^{-1}g \in \bigcap_i N_i = H$. Therefore

$$[G : H] \leq \prod_{i=1}^{M_n} [G : N_i] = n^{M_n},$$

and the final bound follows from $M_n \leq (n!)^d$. $\square$

If G is finitely generated and residually finite, then characteristic finite-index subgroups also separate points. For $g \neq 1$, choose a finite-index normal N with $g \notin N$. Since $\text{CharCore}_G(N) \subseteq N$, the characteristic core also excludes g . This proves existence; the next section gives a canonical construction that does not require enumerating $\text{Aut}(G)$.

17.3 Intersections of kernels of finite-group homomorphisms

For $m \geq 1$, let $\mathfrak{G}_m$ contain one canonical multiplication table from every isomorphism class of finite groups of order at most m . For a finitely generated group G , define

$$K_m^{\text{char}} := \bigcap_{F \in \mathfrak{G}_m} \bigcap_{f \in \text{Hom}(G, F)} \ker f. \quad (17.7)$$

This equals the intersection of all normal subgroups of index at most m :

$$K_m^{\text{char}} = \bigcap_{\substack{N \triangleleft G \\ [G:N] \leq m}} N. \quad (17.8)$$

To prove both inclusions, every displayed N is the kernel of the quotient map to G/N , which is represented in $\mathfrak{G}_m$ after a table isomorphism. Conversely, for any $f : G \rightarrow F$, its kernel is the kernel of the surjection $G \twoheadrightarrow f(G)$, whose image has order at most m ; that kernel occurs on the right side of (17.8).

The family of kernels occurring in (17.7) is finite, so (17.7) is a finite intersection. Indeed, there are only finitely many binary tables on carriers of size at most m , hence finitely many group tables. If G has d fixed generators, a map to F is determined by a d -tuple, so

$$|\text{Hom}(G, F)| \leq |F|^d. \quad (17.9)$$

Therefore (17.7) is a finite intersection of finite-index normal subgroups and itself has finite index.

Theorem 17.3.1. K_m^{char} is characteristic in G .

Proof. Let $\alpha \in \text{Aut}(G)$. An element x lies in K_m^{char} exactly when $f(x) = 1$ for every homomorphism to every $F \in \mathfrak{G}_m$. But

$$f(\alpha(x)) = (f \circ \alpha)(x),$$

and $f \mapsto f \circ \alpha$ is a permutation of $\text{Hom}(G, F)$, with inverse $f \mapsto f \circ \alpha^{-1}$. Hence $x \in K_m^{\text{char}}$ exactly when $\alpha(x) \in K_m^{\text{char}}$, proving $\alpha(K_m^{\text{char}}) = K_m^{\text{char}}$. $\square$

The canonical diagonal homomorphism

$$\Theta_m : G \longrightarrow \prod_{F \in \mathfrak{G}_m} \prod_{f \in \text{Hom}(G, F)} F, \quad \Theta_m(g) = (f(g))_{F, f} \quad (17.10)$$

has kernel K_m^{char} . Its generated image is a finite canonical model of G/K_m^{char} , once table, homomorphism, coordinate, and carrier orders are fixed.

17.4 The canonical characteristic residual tower

Apply (17.7) to $G = \Gamma_B$, and put

$$K_m = K_m^{\text{char}}, \quad Q_m = \Gamma_B / K_m, \quad \pi_m : \Gamma_B \twoheadrightarrow Q_m. \quad (17.11)$$

The canonical construction at stage m is:

- C1.** enumerate all carrier tables of orders $1, \dots, m$, retain exactly the group tables, and choose the lexicographically least table in every isomorphism class;
- C2.** for each retained F , enumerate all four-tuples $(x_1, y_1, x_2, y_2) \in F^4$ and retain exactly those satisfying $[x_1, y_1][x_2, y_2] = e_F$; by presentation (4.2), these are exactly the homomorphisms $\Gamma_B \rightarrow F$;
- C3.** evaluate the four presentation generators in every retained coordinate of (17.10), enumerate the subgroup they generate in the finite product, and canonically table that image as Q_m ;
- C4.** store all coordinate projections, the marked physical shell obtained from (8.5), and a record of table, isomorphism, tuple, relation, image, and hash counts.

Every stage terminates because all carriers, tuples, and products are finite. It is not claimed computationally small.

Theorem 17.4.1 (Characteristic residual tower).

$$\Gamma_B = K_1 \supseteq K_2 \supseteq K_3 \supseteq \dots, \quad \bigcap_{m \geq 1} K_m = \{1\}. \quad (17.12)$$

Each K_m is characteristic and finite-index, and every automorphism of Γ_B descends to Q_m .

Proof. Characteristic finite index was proved in Statement 17.3. Increasing m adds homomorphism kernels to (17.7), so $K_{m+1} \subseteq K_m$. For $m = 1$, only the trivial group occurs, hence $K_1 = \Gamma_B$.

Let $1 \neq \gamma \in \Gamma_B$. By Statement 16.2 there is a finite quotient $\varphi : \Gamma_B \rightarrow F$ with $\varphi(\gamma) \neq 1$. If $|F| = r$, then the corresponding homomorphism occurs in (17.7) for every $m \geq r$. Thus $\gamma \notin K_m$ for $m \geq r$, proving the trivial intersection. Descent of automorphisms is (17.3). $\square$

The quotient tables are canonical relative to the declared exhaustive orders. A complete checkpoint contains the last carrier table, isomorphism class map, homomorphism tuple, product-image BFS state, and rolling hashes. Replaying from the preceding complete record proves recovery without silently omitting a table or tuple.

17.5 Automatic preservation of the eightfold rotation

Because K_m is characteristic, it is invariant under the exact automorphism ϕ_8 of Statement 7.2. Define

$$\alpha_m : Q_m \longrightarrow Q_m, \quad \alpha_m(\pi_m(\gamma)) = \pi_m(\phi_8(\gamma)). \quad (17.13)$$

If $\pi_m(\gamma) = \pi_m(\delta)$, then $\delta^{-1}\gamma \in K_m$, and characteristic invariance gives $\phi_8(\delta)^{-1}\phi_8(\gamma) \in K_m$; hence (17.13) is well defined. It is a homomorphism, and the map induced by ϕ_8^{-1} is its inverse.

With $q_j^{(m)} = \pi_m(g_j)$, the generator action and covariance are

$$\begin{aligned} \alpha_m(q_j^{(m)}) &= q_{j+1}^{(m)}, \\ \alpha_m(xq_j^{(m)}) &= \alpha_m(x)q_{j+1}^{(m)} \quad (x \in Q_m). \end{aligned} \quad (17.14)$$

Also $\phi_8^8 = \text{id}_{\Gamma_B}$ gives

$$\alpha_m^8 = \text{id}_{Q_m}, \quad \text{ord}(\alpha_m) \mid 8. \quad (17.15)$$

Exact order eight is not automatic at every finite stage.

It is, however, automatic eventually. For $1 \leq k < 8$, put

$$d_k = g_k^{-1}g_0 \neq 1.$$

By (17.12), there is m_k with $d_k \notin K_{m_k}$. Nestedness keeps it outside every later K_m . Thus for

$$m \geq m_8 := \max_{1 \leq k < 8} m_k, \quad (17.16)$$

one has $q_k^{(m)} \neq q_0^{(m)}$ for $1 \leq k < 8$. In fact all eight elements are pairwise distinct. If $q_i^{(m)} = q_j^{(m)}$ for $0 \leq i < j < 8$, applying α_m^{8-i} to both sides and using (17.14)–(17.15) gives

$$q_0^{(m)} = q_{j-i}^{(m)},$$

contradicting $1 \leq j - i < 8$. In particular,

$$\alpha_m^k(q_0^{(m)}) = q_k^{(m)} \neq q_0^{(m)} \quad (1 \leq k < 8),$$

while (17.15) gives the eighth power identity. Therefore

$$\text{ord}(\alpha_m) = 8 \quad (m \geq m_8). \quad (17.17)$$

The finite certificate stores the permutation table of α_m , the eight generator rows (17.14), all multiplication covariance checks, the eighth iterate, and either the seven distinctness witnesses proving (17.17) or the actual smaller order. Canonical carrier order and the fixed shell order make serialization reproducible.

17.6 Containment in the parity kernel

Let $\chi : \Gamma_B \twoheadrightarrow C_2$ be the parity homomorphism of Statement 9.1 and $K_\chi = \ker \chi$. For every $m \geq 2$, the target C_2 occurs in $\mathfrak{G}_m$, and χ is one of the homomorphisms in (17.7). Therefore

$$\boxed{K_m \subseteq K_\chi \quad (m \geq 2).} \quad (17.18)$$

At $m = 1$, $K_1 = \Gamma_B \not\subseteq K_\chi$, so the threshold two is sharp for this tower indexing.

Containment (17.18) is necessary and sufficient for parity to descend:

$$\varepsilon_m : Q_m \longrightarrow C_2, \quad \varepsilon_m(\pi_m(\gamma)) = \chi(\gamma). \quad (17.19)$$

If the two representatives differ by $k \in K_m$, (17.18) gives $\chi(k) = 0$, proving well-definedness. Conversely, any factorization $\chi = \varepsilon_m \circ \pi_m$ sends every $k \in K_m$ to zero, so $K_m \subseteq K_\chi$. The factor is unique because π_m is surjective and is itself surjective because χ is.

Every physical generator remains odd:

$$\varepsilon_m(q_j^{(m)}) = \chi(g_j) = 1. \quad (17.20)$$

The defining surface relator maps to zero because it is a product of commutators, as checked in (9.6). Finally, (9.8) and (17.13) give

$$\varepsilon_m \circ \alpha_m = \varepsilon_m. \quad (17.21)$$

Thus the quotient marking, relation, and C_8 covariance all obey the same one-bit specification. The $m = 1$ trivial quotient is the boundary counterexample: it cannot support a map sending its sole element to one.

17.7 Nested quotient maps

Let $m' \geq m$. The family of homomorphisms occurring in the intersection (17.7) for m is contained in the family occurring for m' . Hence

$$K_{m'} \subseteq K_m. \quad (17.22)$$

This inclusion defines a transition map

$$p_{m',m} : Q_{m'} \longrightarrow Q_m, \quad p_{m',m}(\pi_{m'}(\gamma)) = \pi_m(\gamma). \quad (17.23)$$

We verify every part of this definition. If $\pi_{m'}(\gamma) = \pi_{m'}(\delta)$, then $\delta^{-1}\gamma \in K_{m'}$. By (17.22) it lies in K_m , so $\pi_m(\gamma) = \pi_m(\delta)$; thus (17.23) is well defined. For $\gamma, \delta \in \Gamma_B$,

$$\begin{aligned} p_{m',m}(\pi_{m'}(\gamma)\pi_{m'}(\delta)) &= p_{m',m}(\pi_{m'}(\gamma\delta)) \\ &= \pi_m(\gamma\delta) = \pi_m(\gamma)\pi_m(\delta), \end{aligned}$$

so it is a homomorphism. Every $x \in Q_m$ has the form $x = \pi_m(\gamma)$, and then $x = p_{m',m}(\pi_{m'}(\gamma))$; hence $p_{m',m}$ is surjective.

The kernel is obtained without suppressing the representative calculation:

$$\begin{aligned} \pi_{m'}(\gamma) \in \ker p_{m',m} &\iff \pi_m(\gamma) = 1 \\ &\iff \gamma \in K_m \\ &\iff \pi_{m'}(\gamma) \in K_m/K_{m'}. \end{aligned}$$

Consequently

$$\ker p_{m',m} = K_m/K_{m'}, \quad |Q_{m'}| = |K_m : K_{m'}| |Q_m|. \quad (17.24)$$

Formula (17.24) permits equality: different parameters can define the same kernel. Thus the proven statement is nestedness, not strict decrease at every step.

If $\ell \geq m' \geq m$, then both maps below send $\pi_\ell(\gamma)$ to $\pi_m(\gamma)$. Surjectivity of π_ℓ therefore gives

$$p_{\ell,m} = p_{m',m} \circ p_{\ell,m'}, \quad p_{m,m} = \text{id}_{Q_m}. \quad (17.25)$$

The rotations are compatible with every transition. Indeed,

$$\begin{aligned} p_{m',m} \alpha_{m'}(\pi_{m'}(\gamma)) &= p_{m',m}(\pi_{m'}(\phi_8(\gamma))) \\ &= \pi_m(\phi_8(\gamma)) \\ &= \alpha_m(\pi_m(\gamma)) \\ &= \alpha_m p_{m',m}(\pi_{m'}(\gamma)), \end{aligned}$$

and hence

$$p_{m',m} \circ \alpha_{m'} = \alpha_m \circ p_{m',m}. \quad (17.26)$$

When $m' \geq m \geq 2$, parity is compatible as well:

$$\begin{aligned} \varepsilon_m p_{m',m}(\pi_{m'}(\gamma)) &= \varepsilon_m(\pi_m(\gamma)) \\ &= \chi(\gamma) = \varepsilon_{m'}(\pi_{m'}(\gamma)), \end{aligned} \quad \varepsilon_m \circ p_{m',m} = \varepsilon_{m'}. \quad (17.27)$$

An executable transition certificate lists the two quotient tables, the row map induced by (17.23), and exhaustive checks of well-defined row assignment, multiplication, surjectivity, (17.26), and, when applicable, (17.27). For three stored levels it additionally compares the two sides of (17.25) row by row. At $m' = m$ this procedure returns the identity table, including the one-level boundary case.

17.8 Finite-set separation and geometric exhaustion

Theorem 17.8.1 (Uniform separation by the characteristic tower). *Let $S \subset \Gamma_B \setminus \{1\}$ be finite. There is an effectively obtainable integer $m(S)$ such that*

$$S \cap K_m = \emptyset \quad (m \geq m(S)). \quad (17.28)$$

Proof. Write $S = \{\gamma_1, \dots, \gamma_s\}$. If $s = 0$, take $m(S) = 1$; then (17.28) is vacuous. Suppose $s > 0$. The simultaneous separation construction of Statement 16.4 produces a finite group F and a homomorphism $\Phi : \Gamma_B \rightarrow F$ such that

$$\Phi(\gamma_i) \neq 1 \quad (1 \leq i \leq s).$$

Put $r = |F|$ and $m(S) = r$. For every $m \geq r$, the group F occurs among the tables of order at most m , and Φ is one of the homomorphisms whose kernel occurs in (17.7). Therefore

$$K_m \subseteq \ker \Phi. \quad (17.29)$$

If some γ_i lay in K_m , then (17.29) would give $\Phi(\gamma_i) = 1$, contradicting its construction. Hence $S \cap K_m = \emptyset$. Finally, Statement 16.4 is effective: it searches exact integral reductions until all finitely many nonidentity witnesses survive. Thus $m(S)$ is effectively obtainable. $\square$

Taking $S = \{\gamma\}$ recovers point separation and, by quantifying over all $\gamma \neq 1$, gives again

$$\bigcap_{m \geq 1} K_m = \{1\}. \quad (17.30)$$

The finite-set statement is stronger than merely repeating (17.30): one common level works for all elements of the prescribed finite set.

There is also a quantitative interface with the residual-finiteness growth function of Statement 16.6. Fix $L > 0$, and let

$$n = \text{RF}_{\Gamma_B}(L).$$

By the definition and attainment statement in Statement 16.6, there is a finite quotient $\rho : \Gamma_B \twoheadrightarrow F_L$, with $|F_L| = n$, whose kernel contains no nonidentity element of geometric translation length at most L . Since F_L is among the target groups used to define K_m whenever $m \geq n$,

$$K_m \subseteq \ker \rho \quad (m \geq \text{RF}_{\Gamma_B}(L)). \quad (17.31)$$

If $1 \neq k \in K_m$, then $k \in \ker \rho$; the defining property of ρ forces

$$\ell_h(k) > L.$$

Therefore the closed hyperbolic surface

$$X_m = K_m \backslash \mathbb{H}^2$$

satisfies

$$\text{sys}(X_m) > L, \quad \text{inj}(X_m) > \frac{L}{2} \quad \text{whenever } m \geq \text{RF}_{\Gamma_B}(L). \quad (17.32)$$

As L was arbitrary,

$$\text{sys}(X_m) \longrightarrow \infty, \quad \text{inj}(X_m) \longrightarrow \infty. \quad (17.33)$$

Here no monotonicity of a chosen systolic witness is assumed: (17.31) places every sufficiently late characteristic kernel inside one fixed separating kernel.

For the word-metric version, fix an integer $r \geq 0$, use the word length of Statement 3.1, and put

$$B_{\Gamma_B}^{\text{word}}(r) := \{\gamma \in \Gamma_B : |\gamma|_{S_8^{\text{geo}}} \leq r\}, \quad S_r = B_{\Gamma_B}^{\text{word}}(2r) \setminus \{1\}.$$

This ball is finite because there are only finitely many words of length at most r over the finite symmetric alphabet S_8^{geo} . The theorem supplies m_r such that $S_r \cap K_m = \emptyset$ for $m \geq m_r$. If $\gamma, \delta \in B_{\Gamma_B}^{\text{word}}(r)$ and $\pi_m(\gamma) = \pi_m(\delta)$, then $\delta^{-1}\gamma \in K_m$ and

$$|\delta^{-1}\gamma|_{S_8^{\text{geo}}} \leq |\delta|_{S_8^{\text{geo}}} + |\gamma|_{S_8^{\text{geo}}} \leq 2r.$$

Thus $\delta^{-1}\gamma \in S_r \cup \{1\}$; separation of S_r forces $\gamma = \delta$. Hence

$$\pi_m|_{B_{\Gamma_B}^{\text{word}}(r)} \text{ is injective} \quad (m \geq m_r). \quad (17.34)$$

The verifier accepts either a terminal record or a checkpoint. A terminal record is independently verified completely before any separation verdict; the verifier then evaluates every word in S in the stored quotient table. It returns PASS only when all values are nonidentity and FAIL when a listed value is the identity in a valid complete table. A malformed or internally inconsistent record or checkpoint returns INVALID. A structurally valid checkpoint lacking the terminal exhaustive-completion marker returns UNDECIDED and makes no separation assertion. These four outcomes are disjoint. In particular, an unfinished search is never promoted to a proof of separation.

Combining (17.17), (17.18), (17.22), and (17.33), every sufficiently late stage is a parity-compatible finite quotient carrying an exact order-eight rotation, and its associated closed hyperbolic cover has arbitrarily large injectivity radius. This is precisely the canonical characteristic tower specification required for the later equivariant construction.

18 Symmetry cores and parity cores

18.1 The normal core

Let G be a finitely generated group and let $H \leq G$ have finite index

$$n = [G : H].$$

The normal core of H in G is

$$\text{Core}_G(H) := \bigcap_{g \in G} gHg^{-1}. \quad (18.1)$$

It is contained in H , because the factor with $g = 1$ occurs in the intersection. It is normal: for $x \in G$,

$$\begin{aligned} x \text{Core}_G(H) x^{-1} &= \bigcap_{g \in G} xgHg^{-1}x^{-1} \\ &= \bigcap_{u \in G} uHu^{-1} = \text{Core}_G(H), \end{aligned} \quad (18.2)$$

where $u = xg$ ranges over all of G . Moreover it is the largest normal subgroup of G contained in H . Indeed, if $N \trianglelefteq G$ and $N \subseteq H$, then

$$N = gNg^{-1} \subseteq gHg^{-1} \quad (g \in G),$$

so $N \subseteq \text{Core}_G(H)$.

Although (18.1) is written as an infinite intersection, finite-index input makes it a finite and computable object. Let G/H denote the finite set of left cosets and let G act on it by left multiplication:

$$\rho_H : G \longrightarrow \text{Sym}(G/H), \quad \rho_H(y)(xH) = yxH. \quad (18.3)$$

For $y \in G$,

$$\begin{aligned} y \in \ker \rho_H &\iff yxH = xH \quad \text{for every } x \in G \\ &\iff x^{-1}yx \in H \quad \text{for every } x \in G \\ &\iff y \in xHx^{-1} \quad \text{for every } x \in G \\ &\iff y \in \text{Core}_G(H). \end{aligned} \quad (18.4)$$

Thus

$$\boxed{\text{Core}_G(H) = \ker \rho_H.} \quad (18.5)$$

The first isomorphism theorem and $\rho_H(G) \leq \text{Sym}(n)$ give

$$\begin{aligned} [G : \text{Core}_G(H)] &= |\rho_H(G)| \leq n!, \\ [G : \text{Core}_G(H)] &= [G : H][H : \text{Core}_G(H)] \geq n. \end{aligned} \quad (18.6)$$

In particular,

$$n \leq [G : \text{Core}_G(H)] \leq n!. \quad (18.7)$$

Here is a deterministic computational form of (18.5). The input is a finite presentation

$$G = \langle x_1, \dots, x_d \mid r_1, \dots, r_s \rangle$$

together with a certified complete coset table for H . Row i represents a left coset, and the column for x_a gives the permutation σ_a induced by left multiplication. The procedure is:

1. Check that every generator column is a permutation of the n rows and that its declared inverse column is inverse.
2. Evaluate every relator r_b in the permutations and require the identity. This proves that $x_a \mapsto \sigma_a$ extends to ρ_H .
3. Enumerate $P = \langle \sigma_1, \dots, \sigma_d \rangle$ by breadth-first search, using the declared generator and row orders.
4. Return the finite map $G \twoheadrightarrow P$. Its kernel is the normal core by (18.5).
5. If kernel generators are requested, store for every $p \in P$ the least breadth-first word u_p with $\rho_H(u_p) = p$. For every p and x_a , put $p' = p\sigma_a$ and output

$$u_p x_a u_{p'}^{-1}. \quad (18.8)$$

The Schreier rewriting argument expresses every kernel word as a product of the words (18.8), so they generate $\ker \rho_H$.

The output record is

$$(\text{presentation hash, coset-table hash, } \{\sigma_a\}, P, \text{kernel generators, independent verification hash}). \quad (18.9)$$

All orders are fixed, so repeating the computation gives identical bytes.

For a positive test, take $G = S_3$ and $H = \langle (12) \rangle$. Then $n = 3$; the action on the three cosets is faithful, so $\text{Core}_{S_3}(H) = \{1\}$ and (18.7) reads $3 \leq 6 \leq 6$. For the normal boundary, if $H \trianglelefteq G$, every conjugate equals H , so $\text{Core}_G(H) = H$. The finite-index assumption cannot be dropped from the conclusion: for $G = \mathbb{Z}$ and $H = \{0\}$, the core is again H , but it has infinite index and there is no finite coset permutation certificate.

18.2 The complete C_8 -orbit core

Let $\phi \in \text{Aut}(G)$ satisfy $\phi^8 = \text{id}_G$, and let $H \leq G$ have index n . Define the full orbit core

$$\text{OrbCore}_\phi(H) := \bigcap_{j=0}^7 \phi^j(H). \quad (18.10)$$

If d is the size of the orbit of H under ϕ , then $d \mid 8$, the subgroups

$$H, \phi(H), \dots, \phi^{d-1}(H)$$

are distinct, and (18.10) is their intersection. Applying ϕ cyclically permutes the eight factors:

$$\begin{aligned} \phi(\text{OrbCore}_\phi(H)) &= \bigcap_{j=0}^7 \phi^{j+1}(H) \\ &= \bigcap_{j=1}^8 \phi^j(H) = \bigcap_{j=0}^7 \phi^j(H). \end{aligned} \quad (18.11)$$

Thus the orbit core is ϕ -stable. It is also the largest ϕ -stable subgroup contained in H : if $L \subseteq H$ and $\phi(L) = L$, then

$$L = \phi^j(L) \subseteq \phi^j(H) \quad (0 \leq j < 8),$$

and hence $L \subseteq \text{OrbCore}_\phi(H)$.

Every $\phi^j(H)$ has index n . Set $H_j = \phi^j(H)$ for $0 \leq j < d$ and define a map of finite sets

$$\begin{aligned} \iota : G / \text{OrbCore}_\phi(H) &\longrightarrow \prod_{j=0}^{d-1} G / H_j, \\ g \text{OrbCore}_\phi(H) &\longmapsto (gH_0, \dots, gH_{d-1}). \end{aligned} \quad (18.12)$$

If two representatives determine the same left coset of the orbit core, their quotient lies in every H_j , so (18.12) is well defined. If their images in every factor agree, their quotient lies in $\bigcap_j H_j$, so the original orbit-core cosets agree. Therefore ι is injective and

$$n \leq [G : \text{OrbCore}_\phi(H)] \leq n^d \leq n^8. \quad (18.13)$$

No quotient-group structure was used in (18.12); the H_j need not be normal.

If $H \trianglelefteq G$, every $\phi^j(H)$ is normal and the orbit core is normal. Only in this case do we form the quotient group. The map

$$\overline{\phi}(g \text{OrbCore}_\phi(H)) = \phi(g) \text{OrbCore}_\phi(H) \quad (18.14)$$

is then well defined by (18.11), is an automorphism, and satisfies $\overline{\phi}^8 = \text{id}$. Its actual order can be a proper divisor of eight and must be computed rather than assumed. Every defining relator remains true after quotienting because (18.14) is induced by the automorphism ϕ .

For computation, first derive a certified finite Schreier generating set for H from its input coset table. Apply ϕ^j to those generator words and run deterministic coset enumeration, with expected index n , to construct and verify the table for each H_j . Discard exact duplicates after subgroup-equality tests, and let G act diagonally on

$$\prod_{j=0}^{d-1} G/H_j.$$

The stabilizer of the base tuple $(H_0, \dots, H_{d-1})$ is exactly $\bigcap_j H_j$. A deterministic breadth-first traversal of the orbit of this tuple therefore produces the coset table of the orbit core without enumerating irrelevant tuples. The input/output specification is

$$\begin{aligned} \text{input: } & (\text{presentation, } H \text{ coset table, } \phi(x_1), \dots, \phi(x_d), \phi^8 \text{ certificate}), \\ \text{output: } & (d, \{H_j\}, \text{product-orbit table, } \bar{\phi} \text{ table when normal, hash}). \end{aligned} \quad (18.15)$$

Replay checks the generator images, all relators, (18.11), the product stabilizer, and every row of (18.14). Canonical group, generator, coset, and breadth-first orders make repeated output identical.

As a sharp small example, let $G = C_3 \times C_3$, let ϕ swap the two coordinates, and let $H = C_3 \times \{0\}$. Then $d = 2$, $\text{OrbCore}_\phi(H) = \{0\}$, and $[G : \text{OrbCore}_\phi(H)] = 9 = 3^2$. If H is already ϕ -stable, the boundary value is $d = 1$ and the orbit core is H . Completeness of all eight powers matters: on $G = C_2^8$, cyclic coordinate shift and the coordinate-zero subgroup produce an eight-element subgroup orbit; intersecting only seven orbit members is not shift-stable.

18.3 The parity core

Let

$$\chi : G \longrightarrow C_2, \quad K_\chi = \ker \chi \quad (18.16)$$

be a homomorphism, and let $H \leq G$ have index n . The parity core of H is

$$\text{ParCore}_\chi(H) := H \cap K_\chi. \quad (18.17)$$

It is the largest subgroup $L \leq H$ on which χ vanishes: $\chi|_L = 0$ is equivalent to $L \subseteq K_\chi$, and therefore to $L \subseteq H \cap K_\chi$.

Restriction of χ to H has kernel (18.17). The first isomorphism theorem gives the exact relative index

$$[H : \text{ParCore}_\chi(H)] = |\chi(H)| \in \{1, 2\}. \quad (18.18)$$

Multiplicativity of finite indices now gives

$$\boxed{[G : \text{ParCore}_\chi(H)] = n |\chi(H)| \in \{n, 2n\}.} \quad (18.19)$$

The factor is one exactly when $H \subseteq K_\chi$, and it is two exactly when H contains an odd element.

For an explicit coset construction, let G act on

$$(G/H) \times C_2$$

by

$$y \cdot (gH, b) = (ygH, b + \chi(y)). \quad (18.20)$$

This is an action because

$$\begin{aligned} y \cdot (z \cdot (gH, b)) &= y \cdot (zgH, b + \chi(z)) \\ &= (yzgH, b + \chi(z) + \chi(y)) \\ &= (yzgH, b + \chi(yz)) \\ &= (yz) \cdot (gH, b). \end{aligned}$$

The stabilizer of the base state $(H, 0)$ consists of those y with $yH = H$ and $\chi(y) = 0$, namely

$$\text{Stab}_G(H, 0) = H \cap K_\chi. \quad (18.21)$$

Thus breadth-first traversal of the orbit of $(H, 0)$ gives the parity core coset table, and the number of visited states verifies (18.19).

If $H \leq G$, then (18.17) is normal and the group quotient

$$Q = G / \text{ParCore}_\chi(H)$$

is defined. Parity descends by

$$\varepsilon(g \text{ParCore}_\chi(H)) = \chi(g). \quad (18.22)$$

If the two representatives differ by an element of the parity core, their χ -values agree; this proves well-definedness. Conversely, any factorization of χ through G/L requires $L \subseteq K_\chi$, so (18.17) is the largest subkernel of H that permits the factorization.

Suppose additionally that $\phi(H) = H$ and

$$\chi \circ \phi = \chi. \quad (18.23)$$

Then $\phi(K_\chi) = K_\chi$, because

$$\chi(k) = 0 \iff \chi(\phi(k)) = 0.$$

Consequently

$$\phi(\text{ParCore}_\chi(H)) = \phi(H) \cap \phi(K_\chi) = H \cap K_\chi, \quad (18.24)$$

so ϕ descends to Q , and (18.22)–(18.23) give

$$\varepsilon \circ \overline{\phi} = \varepsilon. \quad (18.25)$$

For Γ_B , take the parity map of Statement 9.1. Its physical generators satisfy

$$\chi(g_j) = 1 \quad (0 \leq j < 8), \quad (18.26)$$

whereas the surface relator has parity zero because each commutator does. Whenever the parity core is normal, its quotient marking $q_j = g_j \text{ParCore}_\chi(H)$ therefore satisfies

$$\varepsilon(q_j) = 1,$$

and every relator evaluation remains valid in both the quotient and C_2 .

The deterministic input is a verified H -coset table and the bits $\chi(x_a)$ on the ambient generators. The algorithm verifies every relator in C_2 , runs (18.20) in fixed order, outputs the stabilizer table (18.21), and, when H is normal, emits the quotient multiplication table and ε -column. It also records $|\chi(H)|$, obtained from the parity values of certified Schreier generators for H . The equality of that value with the orbit-size factor in (18.19) is an independent internal check.

For $G = C_4$, $\chi(x) = x \bmod 2$, and $H = G$, the parity core is $2C_4$ and has index two. If $H = 2C_4$, then $H \subseteq K_\chi$, the factor in (18.18) is one, and the core is unchanged. Omitting the intersection fails even in the first example: parity cannot descend through the trivial quotient G/H , because its kernel $H = G$ is not contained in K_χ .

18.4 The joint C_8 -parity core

Assume henceforth that

$$\phi \in \text{Aut}(G), \quad \phi^8 = \text{id}_G, \quad \chi : G \rightarrow C_2, \quad \chi \circ \phi = \chi. \quad (18.27)$$

For a finite-index subgroup $H \leq G$, set

$$\begin{aligned} N(H) &= \text{Core}_G(H), \\ O_\phi(H) &= \bigcap_{j=0}^7 \phi^j(N(H)), \\ J_{\phi,\chi}(H) &= O_\phi(H) \cap K_\chi. \end{aligned} \tag{18.28}$$

We call $J_{\phi,\chi}(H)$ the joint C_8 -parity core.

Theorem 18.4.1 (Universal property of the joint core). *$J_{\phi,\chi}(H)$ is the largest subgroup L satisfying*

$$L \trianglelefteq G, \quad \phi(L) = L, \quad L \subseteq K_\chi, \quad L \subseteq H. \tag{18.29}$$

Proof. The subgroup $N(H)$ is normal by Statement 18.1. Every $\phi^j(N(H))$ is normal because automorphisms preserve normality. Their intersection $O_\phi(H)$ is therefore normal, and Statement 18.2 proves that it is ϕ -stable. The kernel K_χ is normal, and (18.27) implies $\phi(K_\chi) = K_\chi$. Hence $J_{\phi,\chi}(H)$ is normal and ϕ -stable. Definition (18.28) gives

$$J_{\phi,\chi}(H) \subseteq K_\chi, \quad J_{\phi,\chi}(H) \subseteq N(H) \subseteq H,$$

so it satisfies (18.29).

Conversely, suppose L satisfies (18.29). Since $L \trianglelefteq G$ and $L \subseteq H$, maximality of the normal core gives $L \subseteq N(H)$. Since $\phi(L) = L$,

$$L = \phi^j(L) \subseteq \phi^j(N(H)) \quad (0 \leq j < 8).$$

Thus $L \subseteq O_\phi(H)$. The parity condition in (18.29) also gives $L \subseteq K_\chi$, and hence $L \subseteq J_{\phi,\chi}(H)$. $\square$

Because the joint core is normal, define

$$Q_H = G/J_{\phi,\chi}(H), \quad \pi_H : G \twoheadrightarrow Q_H. \tag{18.30}$$

Its ϕ -stability and containment in K_χ give the two well-defined maps

$$\begin{aligned} \alpha_H(\pi_H(g)) &= \pi_H(\phi(g)), \\ \varepsilon_H(\pi_H(g)) &= \chi(g). \end{aligned} \tag{18.31}$$

For the first map, equality of representatives means $g'^{-1}g \in J_{\phi,\chi}(H)$; applying ϕ keeps this difference in the joint core. For the second, applying χ to the difference gives zero. The inverse of α_H is induced by ϕ^{-1} , so α_H is an automorphism. Direct substitution yields

$$\alpha_H^8 = \text{id}_{Q_H}, \quad \varepsilon_H \circ \alpha_H = \varepsilon_H. \tag{18.32}$$

In the Bolza case, $\phi = \phi_8$ and $\phi_8(g_j) = g_{j+1}$, with indices modulo eight. Therefore

$$\alpha_H(q_j) = q_{j+1}, \quad \varepsilon_H(q_j) = 1, \quad q_j = \pi_H(g_j). \tag{18.33}$$

The surface relator evaluates to one because π_H is a homomorphism. The order of α_H divides eight; a claim of exact order eight is made only after its permutation table has no smaller positive identity power.

The executable construction applies Statement 18.1, Statement 18.2, and Statement 18.3 in that order:

1. validate the presentation, H -coset table, ϕ -generator words, ϕ^8 , the parity bits, and (18.27);
2. compute $N(H) = \ker \rho_H$;

3. compute the complete, duplicate-reduced orbit of $N(H)$ and its intersection $O_\phi(H)$;
4. intersect with K_χ using the two-state product action;
5. enumerate Q_H , then verify every multiplication row, relator, generator equation in (18.33), covariance equation in (18.32), and the actual order of α_H .

The output is

$$(Q_H, \pi_H, \alpha_H, \varepsilon_H, \text{normal-core proof, orbit-core proof, parity proof, hash}). \quad (18.34)$$

Fixed enumeration orders and canonical carrier relabelling make (18.34) reproducible.

A minimal exact-symmetry test is the additive group $G = C_2^8$, cyclic coordinate shift ϕ , coordinate-sum parity χ , and $H = \{0\}$. Then $J_{\phi, \chi}(H) = \{0\}$, the vectors $q_j = e_j$ form an eight-cycle, α_H has exact order eight, and all q_j are odd. For the parity boundary $H = G$, the normal and orbit stages do nothing and $J_{\phi, \chi}(G) = K_\chi$.

18.5 Finite index of the joint core

Let

$$\begin{aligned} n &= [G : H], & c &= [G : N(H)], \\ d &= |\{\phi^j(N(H)) : 0 \leq j < 8\}|, & a &= [G : O_\phi(H)], \\ e &= [O_\phi(H) : J_{\phi, \chi}(H)], & q &= [G : J_{\phi, \chi}(H)]. \end{aligned} \quad (18.35)$$

The orbit size satisfies $d \mid 8$. From (18.7),

$$n \leq c \leq n!. \quad (18.36)$$

Applying the injective finite-set map (18.12) to the d distinct normal subgroups $\phi^j(N(H))$, each of index c , gives

$$c \leq a \leq c^d. \quad (18.37)$$

Restriction of parity to $O_\phi(H)$ gives

$$e = |\chi(O_\phi(H))| \in \{1, 2\}. \quad (18.38)$$

Finally, multiplicativity of indices gives the exact factorization

$$\boxed{q = a e.} \quad (18.39)$$

Combining (18.36)–(18.39) proves the explicit finite-index bounds

$$n \leq q \leq 2c^d \leq 2(n!)^d \leq 2(n!)^8. \quad (18.40)$$

This proves finiteness without assuming that any of the inclusions between the core stages is strict.

There is a more diagnostic exact product. Let

$$O_t = \bigcap_{j=0}^{t-1} \phi^j(N(H)) \quad (1 \leq t \leq d),$$

so $O_1 = N(H)$ and $O_d = O_\phi(H)$. Put

$$r_t = [O_{t-1} : O_t] \quad (2 \leq t \leq d).$$

Since $O_t = O_{t-1} \cap \phi^{t-1}(N(H))$, the map

$$O_{t-1}/O_t \longrightarrow G/\phi^{t-1}(N(H)), \quad xO_t \longmapsto x\phi^{t-1}(N(H)) \quad (18.41)$$

is a well-defined injection of finite sets. Thus $1 \leq r_t \leq c$, and repeated index multiplication yields

$$q = c \left(\prod_{t=2}^d r_t \right) e. \quad (18.42)$$

Empty products are one, so (18.42) includes the boundary $d = 1$.

The algorithm computes c, a, e, q directly from the certified finite tables and verifies both (18.39) and (18.42) as exact integer identities. For $G = C_4$, $H = G$, $\phi(x) = -x$, and $\chi(x) = x \bmod 2$, one gets

$$(n, c, d, a, e, q) = (1, 1, 1, 1, 2, 2).$$

For $G = S_3$, $H = \langle (12) \rangle$, $\phi = \text{id}$, and $\chi = \text{sgn}$, one gets

$$(n, c, d, a, e, q) = (3, 6, 1, 6, 1, 6).$$

Both table enumerations independently reproduce (18.39), test the two possible values of e , and attain opposite ends of the normal-core bound (18.36).

18.6 Nestedness of the core construction

Theorem 18.6.1 (Monotonicity). *If $H' \subseteq H$ are finite-index subgroups of G , then*

$$\begin{aligned} N(H') &\subseteq N(H), \\ O_\phi(H') &\subseteq O_\phi(H), \\ J_{\phi, \chi}(H') &\subseteq J_{\phi, \chi}(H). \end{aligned} \quad (18.43)$$

Proof. Conjugation preserves inclusion, so for every $g \in G$,

$$gH'g^{-1} \subseteq gHg^{-1}.$$

Intersecting these inclusions over all g gives

$$N(H') = \bigcap_g gH'g^{-1} \subseteq \bigcap_g gHg^{-1} = N(H).$$

Applying ϕ^j preserves this inclusion for each $0 \leq j < 8$. Intersecting the eight resulting inclusions gives

$$O_\phi(H') = \bigcap_{j=0}^7 \phi^j(N(H')) \subseteq \bigcap_{j=0}^7 \phi^j(N(H)) = O_\phi(H).$$

Finally, intersection with the same subgroup K_χ preserves inclusion and gives the third line of (18.43). $\square$

Consequently, if

$$H_1 \supseteq H_2 \supseteq H_3 \supseteq \cdots \quad (18.44)$$

is a nested finite-index sequence and

$$J_m = J_{\phi, \chi}(H_m), \quad Q_m = G/J_m,$$

then

$$J_1 \supseteq J_2 \supseteq J_3 \supseteq \cdots. \quad (18.45)$$

For $m' \geq m$, define

$$p_{m', m} : Q_{m'} \longrightarrow Q_m, \quad p_{m', m}(gJ_{m'}) = gJ_m. \quad (18.46)$$

If $gJ_{m'} = g'J_{m'}$, then $g'^{-1}g \in J_{m'} \subseteq J_m$, so (18.46) is well defined. It is a homomorphism because

$$\begin{aligned} p_{m',m}((gJ_{m'})(hJ_{m'})) &= p_{m',m}(ghJ_{m'}) \\ &= ghJ_m = (gJ_m)(hJ_m), \end{aligned}$$

and it is surjective because every gJ_m is the image of $gJ_{m'}$. Exactly as in Statement 17.7,

$$\begin{aligned} \ker p_{m',m} &= J_m/J_{m'}, \\ p_{\ell,m} &= p_{m',m} \circ p_{\ell,m'} \quad (\ell \geq m' \geq m), \\ p_{m,m} &= \text{id}_{Q_m}. \end{aligned} \tag{18.47}$$

The induced symmetry and parity maps satisfy

$$\begin{aligned} p_{m',m} \circ \alpha_{m'} &= \alpha_m \circ p_{m',m}, \\ \varepsilon_m \circ p_{m',m} &= \varepsilon_{m'}. \end{aligned} \tag{18.48}$$

For example, both sides of the first identity send $gJ_{m'}$ to $\phi(g)J_m$, and both sides of the second send it to $\chi(g)$.

No strictness is asserted. If $H' = H$, all three lines of (18.43) are equalities. Distinct inputs can also have the same joint core: in S_3 , two distinct order-two subgroups have trivial normal core, so with $\phi = \text{id}$ and sign parity their joint cores both equal $\{1\}$. Thus equality of joint cores does not imply equality or comparability of the input subgroups.

The paired deterministic verifier first checks the certified inclusion $H' \subseteq H$ on coset tables. It independently rebuilds the three core stages, checks each inclusion in (18.43) on subgroup generators, constructs the row map (18.46), and exhaustively verifies multiplication, surjectivity, (18.47), and (18.48). Equal consecutive kernels are stored as a valid zero-growth step, not discarded. Fixed row order makes a second run byte-for-byte identical.

18.7 Preservation of residual separation

Theorem 18.7.1 (Residual joint-core tower). *Suppose the finite-index sequence (18.44) satisfies*

$$\bigcap_{m \geq 1} H_m = \{1\}. \tag{18.49}$$

Then its joint cores are finite-index normal subgroups satisfying

$$\begin{aligned} J_1 \supseteq J_2 \supseteq \cdots, \quad \phi(J_m) &= J_m, \quad J_m \subseteq K_\chi, \\ \bigcap_{m \geq 1} J_m &= \{1\}. \end{aligned} \tag{18.50}$$

Proof. Finite index is Statement 18.5, and normality, symmetry stability, and parity containment are Statement 18.4. Nestedness is Statement 18.6. Since

$$J_m \subseteq N(H_m) \subseteq H_m$$

for every m ,

$$\bigcap_{m \geq 1} J_m \subseteq \bigcap_{m \geq 1} H_m = \{1\}.$$

The identity lies in every subgroup, so equality follows. $\square$

The conclusion has a uniform finite-set form. Let $S = \{\gamma_1, \dots, \gamma_s\} \subseteq G \setminus \{1\}$. For each i , (18.49) gives an index m_i with $\gamma_i \notin H_{m_i}$. By nestedness, $\gamma_i \notin H_m$ for every $m \geq m_i$. With

$$m(S) = \max_{1 \leq i \leq s} m_i$$

and the empty-set convention $m(\emptyset) = 1$, one gets

$$S \cap H_m = \emptyset \implies S \cap J_m = \emptyset \quad (m \geq m(S)). \quad (18.51)$$

Thus passing to the joint core never delays a separation already certified at the same stage of the input tower.

For $G = \Gamma_B$, the joint-core tower also has diverging global systole. Fix $L > 0$. By the finite short-class enumeration of Statement 15.1 and Statement 15.7, there are finitely many nontrivial Γ_B -conjugacy classes with translation length at most L . Choose one representative from each:

$$S_L = \{\gamma_1, \dots, \gamma_t\}, \quad \ell_h(\gamma_i) \leq L. \quad (18.52)$$

Equation (18.51) supplies m_L for which $S_L \cap J_m = \emptyset$ whenever $m \geq m_L$. Suppose that a nonidentity $k \in J_m$ had $\ell_h(k) \leq L$. Then $k = u\gamma_i u^{-1}$ for some $u \in \Gamma_B$ and some i . Normality of J_m would imply

$$\gamma_i = u^{-1}ku \in J_m,$$

contradicting the separation of S_L . Hence

$$\text{sys}(J_m \backslash \mathbb{H}^2) > L, \quad \text{inj}(J_m \backslash \mathbb{H}^2) > \frac{L}{2} \quad (m \geq m_L). \quad (18.53)$$

Because L is arbitrary, both quantities tend to infinity. This argument needs normality: it converts exclusion of one representative into exclusion of its entire ambient conjugacy class.

The executable residual certificate for a finite request (S, M) contains the first M input coset tables, their inclusion maps, all joint-core tables, and the first stage after which every element of S stays excluded. A complete independent verification yields PASS when (18.50)–(18.51) hold and FAIL when a valid completed table contains a listed element at or after the claimed stage. Malformed or inconsistent evidence is INVALID; a valid nonterminal checkpoint is UNDECIDED and makes no residual claim. No finite prefix file is presented as a proof of the infinite intersection (18.49); that hypothesis must come from a symbolic theorem or from a separately verified generating construction such as Statement 17.4.

For a boundary test, take $G = \mathbb{Z}$, $H_m = 2^m \mathbb{Z}$, $\phi(x) = -x$, and $\chi(x) = x \bmod 2$. Every H_m is already normal, ϕ -stable, and contained in K_χ , so $J_m = H_m$ and the intersection is zero. In contrast, the constant sequence $H_m = G$ is not residual; its joint core is the constant subgroup K_χ , showing that (18.49) is essential.

18.8 Quotient-order growth and the optimal core problem

The exact quotient order caused by the joint construction is

$$q_{\text{core}}(H) := |G/J_{\phi, \chi}(H)| = c \left(\prod_{t=2}^d r_t \right) e, \quad (18.54)$$

with the integer factors from (18.35), (18.38), and (18.42). Relative to the input coset degree $n = [G : H]$, write

$$h = [H : N(H)] = \frac{c}{n}.$$

Then the exact overhead factor is

$$\Omega_{\text{core}}(H) := \frac{q_{\text{core}}(H)}{n} = h \left(\prod_{t=2}^d r_t \right) e. \quad (18.55)$$

All factors are positive integers. In particular,

$$\begin{aligned} 1 &\leq h \leq (n-1)!, \\ 1 &\leq r_t \leq c \leq n!, \quad 1 \leq \Omega_{\text{core}}(H) \leq 2(n-1)!(n!)^{d-1}, \\ e &\in \{1, 2\}, \end{aligned} \quad (18.56)$$

Multiplication by n recovers the proven bound (18.40). If χ is surjective, then $J_{\phi,\chi}(H) \subseteq K_\chi$ also shows that $q_{\text{core}}(H)$ is even. Since it is a multiple of n , one obtains the certified divisibility lower bound

$$\text{lcm}(n, 2) \mid q_{\text{core}}(H), \quad q_{\text{core}}(H) \geq \text{lcm}(n, 2). \quad (18.57)$$

The core has an exact optimization interpretation. Define the feasible set

$$\mathcal{F}(H) = \left\{ K \leq G \mid \begin{array}{l} [G : K] < \infty, K \trianglelefteq G, K \subseteq H, \\ \phi(K) = K, K \subseteq K_\chi \end{array} \right\}. \quad (18.58)$$

It is nonempty because $J_{\phi,\chi}(H) \in \mathcal{F}(H)$. The universal property in Statement 18.4 says that every $K \in \mathcal{F}(H)$ satisfies

$$K \subseteq J_{\phi,\chi}(H).$$

Therefore

$$[G : K] = [G : J_{\phi,\chi}(H)] [J_{\phi,\chi}(H) : K] \geq q_{\text{core}}(H), \quad (18.59)$$

and equality is attained by the joint core. Hence the optimization problem is solved exactly:

$$\boxed{\min_{K \in \mathcal{F}(H)} [G : K] = q_{\text{core}}(H).} \quad (18.60)$$

Thus the staged construction does not merely supply an admissible quotient; it supplies the smallest quotient order among all normal, C_8 -stable, parity-compatible kernels contained in the prescribed H .

The numerical certificate stores

$$(n, c, d, h, \{r_t\}, a, e, q_{\text{core}}, \Omega_{\text{core}}) \quad (18.61)$$

as arbitrary-precision integers and rational numbers. It verifies

$$c = nh, \quad a = c \prod_{t=2}^d r_t, \quad q_{\text{core}} = ae = n\Omega_{\text{core}}$$

before evaluating any bound. Factorials and products in (18.56) are computed exactly, so they require no floating-point rounding. If decimal logarithms are reported, an interval implementation evaluates the lower endpoint with directed downward rounding and the upper endpoint with directed upward rounding; a displayed decimal is accepted only when the resulting interval proves the claimed inequality.

For an independent finite lower certificate, enumerate canonical group tables of orders $r < q_{\text{core}}(H)$ and all surjective homomorphisms $\psi : G \twoheadrightarrow F$. A candidate kernel lies in H exactly when the H -coset action factors through ψ ; it is ϕ -stable exactly when an automorphism β of F satisfies

$$\beta \circ \psi = \psi \circ \phi; \quad (18.62)$$

and it lies in K_χ exactly when a map $\eta : F \rightarrow C_2$ satisfies

$$\eta \circ \psi = \chi. \quad (18.63)$$

All three are finite table checks. Exhaustion with no passing candidate independently certifies the strict lower bound $[G : K] \geq q_{\text{core}}(H)$; the explicit joint-core table certifies the matching upper bound. A checkpoint without an exhaustive terminal marker is UNDECIDED, not a lower-bound proof.

For $G = C_4$, $H = G$, inversion ϕ , and mod-two parity, (18.61) gives $q_{\text{core}} = 2$. Exhaustion of the sole smaller order excludes the trivial quotient because parity cannot factor through it. For the S_3 transposition example of Statement 18.5, the only normal subgroups are $\{1\}$, A_3 , and S_3 ;

among them only $\{1\}$ is contained in both H and $K_\chi = A_3$. Hence the independent optimum is six, agreeing with (18.54). These two examples cross-check respectively the parity factor and the normal-core factor.

Part VI

p -quotients and nilpotent constructions

19 Lower central series and finite nilpotent quotients

19.1 Commutator structure

We retain the convention fixed in Statement 4.1:

$$[x, y] = xyx^{-1}y^{-1}. \quad (19.1)$$

For subgroups $A, B \leq G$, define

$$[A, B] = \langle [a, b] : a \in A, b \in B \rangle. \quad (19.2)$$

This is a generated subgroup, not merely the set of single commutators. Since

$$[y, x] = [x, y]^{-1}, \quad (19.3)$$

one has $[A, B] = [B, A]$. Direct multiplication gives the two identities

$$\begin{aligned} [xy, z] &= x[y, z]x^{-1}[x, z], \\ [x, yz] &= [x, y]y[x, z]y^{-1}. \end{aligned} \quad (19.4)$$

For example,

$$\begin{aligned} x[y, z]x^{-1}[x, z] &= xyzzy^{-1}z^{-1}x^{-1}xzx^{-1}z^{-1} \\ &= xyzzy^{-1}x^{-1}z^{-1} = [xy, z], \end{aligned}$$

and the second identity follows by the analogous cancellation

$$\begin{aligned} [x, y]y[x, z]y^{-1} &= xyx^{-1}y^{-1}yxzx^{-1}z^{-1}y^{-1} \\ &= xyzx^{-1}z^{-1}y^{-1} = [x, yz]. \end{aligned}$$

If $A \trianglelefteq G$ and $B \trianglelefteq G$, then $[A, B] \trianglelefteq G$. Indeed,

$$g[a, b]g^{-1} = [gag^{-1}, gbg^{-1}] \in [A, B], \quad (19.5)$$

and conjugation therefore preserves the generating set in (19.2). Normality also gives the useful containment

$$[A, B] \subseteq A \cap B. \quad (19.6)$$

To see the first inclusion, write

$$[a, b] = a(ba^{-1}b^{-1}) \in A$$

because A is normal. Interchanging A and B , and using (19.3), gives membership in B . The normality hypothesis matters: for $G = S_3$, $A = \langle (12) \rangle$, and $b = (123)$, the commutator $[(12), b]$ is a nontrivial three-cycle and does not lie in A .

The derived subgroup is

$$G' = [G, G]. \quad (19.7)$$

It is characteristic. If $\alpha \in \text{Aut}(G)$, then

$$\alpha([x, y]) = [\alpha(x), \alpha(y)],$$

so $\alpha(G') \subseteq G'$; applying the same argument to α^{-1} gives equality. The quotient G/G' is abelian because

$$(xG')(yG')(xG')^{-1}(yG')^{-1} = [x, y]G' = G'. \quad (19.8)$$

It has the universal abelian property: if $f : G \rightarrow A$ and A is abelian, then $f([x, y]) = 1$, hence $G' \subseteq \ker f$, and the unique factor

$$\bar{f} : G/G' \rightarrow A, \quad \bar{f}(gG') = f(g) \quad (19.9)$$

is well defined. For the Bolza surface group, Statement 6.2 identifies $\Gamma_B/\Gamma'_B \cong \mathbb{Z}^4$.

For a certified finite multiplication table, (19.2) is directly computable. Enumerate the elements of A and B , evaluate every $aba^{-1}b^{-1}$, and close the resulting list under multiplication and inversion. The output record

$$(\text{ambient-table hash}, A \text{ rows}, B \text{ rows}, [A, B] \text{ rows}, \text{generator witnesses}, \text{hash}) \quad (19.10)$$

uses increasing carrier order at every step. Replay checks subgroup closure, every commutator witness, and that every output generator lies in the returned closure. The abelian boundary returns $[G, G] = \{1\}$; for S_3 , the full computation returns $G' = A_3$. These examples separate the trivial and nontrivial outcomes.

19.2 The lower central series

Define recursively

$$\gamma_1(G) = G, \quad \gamma_{r+1}(G) = [\gamma_r(G), G] \quad (r \geq 1). \quad (19.11)$$

Every term is characteristic. The proof is by induction. The first term is characteristic. If $\alpha(\gamma_r(G)) = \gamma_r(G)$, then

$$\begin{aligned} \alpha(\gamma_{r+1}(G)) &= \alpha([\gamma_r(G), G]) \\ &= [\alpha(\gamma_r(G)), \alpha(G)] \\ &= [\gamma_r(G), G] = \gamma_{r+1}(G). \end{aligned} \quad (19.12)$$

In particular every term is normal. Applying (19.6) with $A = \gamma_r(G)$ and $B = G$ gives the descending chain

$$G = \gamma_1(G) \supseteq \gamma_2(G) \supseteq \gamma_3(G) \supseteq \cdots. \quad (19.13)$$

The r -th central layer is

$$L_r(G) = \gamma_r(G)/\gamma_{r+1}(G). \quad (19.14)$$

It is central in $G/\gamma_{r+1}(G)$. For $x \in \gamma_r(G)$ and $g \in G$,

$$[x, g] \in [\gamma_r(G), G] = \gamma_{r+1}(G),$$

so

$$(x\gamma_{r+1})(g\gamma_{r+1}) = (g\gamma_{r+1})(x\gamma_{r+1}). \quad (19.15)$$

Equivalently, the formal centrality specification is

$$[x\gamma_{r+1}(G), g\gamma_{r+1}(G)] = 1. \quad (19.16)$$

It follows in particular that $L_r(G)$ is abelian.

The construction is functorial. If $f : G \rightarrow H$ is a homomorphism, then

$$f(\gamma_r(G)) \subseteq \gamma_r(H) \quad (r \geq 1). \quad (19.17)$$

For $r = 1$ this is $f(G) \subseteq H$. If it holds at r , then

$$\begin{aligned} f(\gamma_{r+1}(G)) &= f([\gamma_r(G), G]) \\ &\subseteq [f(\gamma_r(G)), f(G)] \\ &\subseteq [\gamma_r(H), H] = \gamma_{r+1}(H). \end{aligned}$$

When f is surjective, both inclusions in the last two lines are equalities by induction, so

$$f(\gamma_r(G)) = \gamma_r(H). \quad (19.18)$$

For a finite group table Q , (19.11) is an exact terminating algorithm. Begin with the full carrier $C_1 = Q$. Given C_r , evaluate $[x, y]$ for every $x \in C_r$, $y \in Q$, close the results under multiplication and inversion, and call the result C_{r+1} . Equation (19.11) proves $C_r = \gamma_r(Q)$ inductively. The cardinalities form a nonincreasing sequence of positive integers, so eventually either

$$C_{r+1} = \{1\} \quad \text{or} \quad C_{r+1} = C_r \neq \{1\}. \quad (19.19)$$

Once equality occurs, the recursion keeps the same subgroup forever. The serialized result lists each distinct subgroup row set, its commutator witnesses, the layer coset table, and the terminal condition.

For the unitriangular group

$$U_3(\mathbb{F}_p) = \left\{ \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in \mathbb{F}_p \right\},$$

direct multiplication gives

$$\gamma_2(U_3) = \left\{ \begin{pmatrix} 1 & 0 & c \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} : c \in \mathbb{F}_p \right\}, \quad \gamma_3(U_3) = \{1\}. \quad (19.20)$$

An abelian group reaches $\gamma_2 = \{1\}$. By contrast, $\gamma_2(S_3) = A_3$ and $\gamma_3(S_3) = [A_3, S_3] = A_3$; the nontrivial stabilization in (19.19) is the negative test for nilpotence.

19.3 The universal nilpotent quotient

For an integer $c \geq 0$, define

$$\mathcal{N}_c(G) := G/\gamma_{c+1}(G), \quad \nu_c : G \twoheadrightarrow \mathcal{N}_c(G). \quad (19.21)$$

The boundary $c = 0$ gives $\mathcal{N}_0(G) = G/\gamma_1(G) = \{1\}$. For $c = 1$, $\mathcal{N}_1(G) = G/G'$ is the abelianization.

Surjectivity and (19.18) give

$$\gamma_r(\mathcal{N}_c(G)) = \nu_c(\gamma_r(G)) = \frac{\gamma_r(G)\gamma_{c+1}(G)}{\gamma_{c+1}(G)}. \quad (19.22)$$

When $1 \leq r \leq c + 1$, the descending chain (19.13) gives $\gamma_{c+1}(G) \subseteq \gamma_r(G)$, so

$$\gamma_r(\mathcal{N}_c(G)) = \gamma_r(G)/\gamma_{c+1}(G).$$

At $r = c + 1$ this quotient is trivial. Therefore

$$\gamma_{c+1}(\mathcal{N}_c(G)) = \{1\}, \quad (19.23)$$

and $\mathcal{N}_c(G)$ is nilpotent of class at most c .

Theorem 19.3.1 (Universal property). *If $f : G \rightarrow P$ is a homomorphism and $\gamma_{c+1}(P) = \{1\}$, there is a unique homomorphism $\bar{f} : \mathcal{N}_c(G) \rightarrow P$ such that*

$$f = \bar{f} \circ \nu_c. \quad (19.24)$$

Proof. Functoriality (19.17) gives

$$f(\gamma_{c+1}(G)) \subseteq \gamma_{c+1}(P) = \{1\}.$$

Thus $\gamma_{c+1}(G) \subseteq \ker f$, and

$$\bar{f}(g\gamma_{c+1}(G)) := f(g) \quad (19.25)$$

is well defined. It is a homomorphism and satisfies (19.24). Surjectivity of ν_c makes the factor unique. $\square$

Equivalently, if $K \trianglelefteq G$ and G/K has class at most c , then

$$\gamma_{c+1}(G) \subseteq K. \quad (19.26)$$

Indeed, apply the theorem to the quotient map. Thus $\mathcal{N}_c(G)$ is the largest quotient of G , in the factorization order, having nilpotency class at most c .

The quotients form a compatible tower. If $c' \geq c$, then $\gamma_{c'+1}(G) \subseteq \gamma_{c+1}(G)$, so

$$\begin{aligned} p_{c',c} : \mathcal{N}_{c'}(G) &\rightarrow \mathcal{N}_c(G), \\ p_{c',c}(g\gamma_{c'+1}(G)) &= g\gamma_{c+1}(G) \end{aligned} \quad (19.27)$$

is well defined and surjective. Representative multiplication gives

$$p_{c'',c} = p_{c',c} \circ p_{c'',c'} \quad (c'' \geq c' \geq c), \quad p_{c,c} = \text{id}. \quad (19.28)$$

For a certified finite group table Q , the quotient is explicitly computable:

1. validate the multiplication table, identity, inverses, and associativity;
2. run Statement 19.2 through $C_{c+1} = \gamma_{c+1}(Q)$;
3. verify normality of C_{c+1} by exhaustive conjugation;
4. choose the least carrier element in each left coset as its representative;
5. define

$$(xC_{c+1})(yC_{c+1}) = (xy)C_{c+1} \quad (19.29)$$

and fill the quotient table;

6. independent verification the lower central series of the output and require (19.23).

Normality makes (19.29) independent of representatives. Every step is finite and the fixed carrier order makes the result reproducible. The I/O record is

$$(Q \text{ table hash}, c, \{C_r\}_{r \leq c+1}, \mathcal{N}_c(Q) \text{ table}, \nu_c \text{ row map, hash}). \quad (19.30)$$

For $U_3(\mathbb{F}_p)$, the class-one output has order p^2 , while the class-two output is the full group of order p^3 . For S_3 , both $\mathcal{N}_1(S_3)$ and $\mathcal{N}_2(S_3)$ are C_2 , because $\gamma_2(S_3) = \gamma_3(S_3) = A_3$. These tests detect both genuine growth and a repeated quotient level.

For an arbitrary finite presentation, $\mathcal{N}_c(G)$ can be infinite; a finite multiplication table is then not an admissible output. A presentation-mode implementation may instead return a consistent nilpotent polycyclic presentation, a certified collection procedure, and mutually verified generator maps implementing (19.24). Without a completed consistency and kernel certificate, its classification is UNDECIDED; a partial collector is not a proof of the universal quotient.

19.4 Nilpotency class

A group G is nilpotent if $\gamma_{c+1}(G) = \{1\}$ for some integer $c \geq 0$. Its nilpotency class is

$$\text{cl}(G) := \min\{c \geq 0 : \gamma_{c+1}(G) = \{1\}\}. \quad (19.31)$$

Thus the trivial group has class zero, every nontrivial abelian group has class one, and a class- c group with $c \geq 1$ satisfies

$$\gamma_c(G) \neq \{1\}, \quad \gamma_{c+1}(G) = \{1\}. \quad (19.32)$$

For $c \geq 1$, the universal quotient satisfies, by (19.22),

$$\gamma_c(\mathcal{N}_c(G)) = \gamma_c(G)/\gamma_{c+1}(G). \quad (19.33)$$

Therefore

$$\text{cl}(\mathcal{N}_c(G)) = c \iff \gamma_c(G) \neq \gamma_{c+1}(G), \quad (19.34)$$

while equality of the two terms means that the quotient class is strictly smaller than c . This prevents a requested class bound from being misreported as the attained class.

For a finite table Q , the Statement 19.2 recursion decides nilpotence. If it first reaches $\{1\}$ at C_{c+1} , return class c . If instead it first reaches

$$C_{r+1} = C_r \neq \{1\},$$

then

$$C_{r+2} = [C_{r+1}, Q] = [C_r, Q] = C_{r+1},$$

and induction gives $C_{r+k} = C_r$ for every $k \geq 0$. The series can never reach the identity, so Q is not nilpotent. Since subgroup cardinalities strictly decrease until one of these terminal states occurs, the finite-table decision procedure terminates.

The fixed output specification is

$$(\text{table hash}, \{|C_r|\}, \text{terminal index}, \text{NILPOTENT}(c) \text{ or NOT-NILPOTENT, witness rows, hash}). \quad (19.35)$$

For $U_3(\mathbb{F}_p)$, (19.20) gives class two. A nontrivial cyclic group has class one, and the trivial group tests class zero. The stabilization $S_3 \supset A_3 = A_3 = \dots$ supplies the nonnilpotent boundary.

19.5 C_8 -actions on the central layers

Let $\phi \in \text{Aut}(G)$ satisfy

$$\phi^8 = \text{id}_G. \quad (19.36)$$

Characteristic invariance (19.12) gives

$$\phi(\gamma_r(G)) = \gamma_r(G) \quad (r \geq 1). \quad (19.37)$$

Hence ϕ induces an automorphism on every nilpotent quotient:

$$\begin{aligned} \phi^{(c)} : \mathcal{N}_c(G) &\longrightarrow \mathcal{N}_c(G), \\ \phi^{(c)}(g\gamma_{c+1}(G)) &= \phi(g)\gamma_{c+1}(G). \end{aligned} \quad (19.38)$$

If two representatives differ by an element of $\gamma_{c+1}(G)$, their images differ by an element of the same characteristic subgroup, so (19.38) is well defined. Its inverse is induced by ϕ^{-1} , and

$$(\phi^{(c)})^8 = \text{id}_{\mathcal{N}_c(G)}. \quad (19.39)$$

Likewise, the map on the r -th central layer is

$$\begin{aligned}\phi_r : L_r(G) &\longrightarrow L_r(G), \\ \phi_r(x\gamma_{r+1}(G)) &= \phi(x)\gamma_{r+1}(G).\end{aligned}\tag{19.40}$$

The same representative argument proves well-definedness, and

$$\phi_r^8 = \text{id}_{L_r(G)}.\tag{19.41}$$

The actual order of either induced automorphism need only divide eight. For instance, ϕ can act trivially on an abelian layer even when it has exact order eight on G .

The quotient transition maps (19.27) are C_8 -equivariant. On a representative $g\gamma_{c'+1}(G)$, both composites below have value $\phi(g)\gamma_{c+1}(G)$, so

$$p_{c',c} \circ \phi^{(c')} = \phi^{(c)} \circ p_{c',c} \quad (c' \geq c).\tag{19.42}$$

Commutators are compatible on representatives:

$$\phi([x, y]) = [\phi(x), \phi(y)].\tag{19.43}$$

Thus applying ϕ before one lower-central step gives the same subgroup and the same element images as applying it afterward. Equations (19.37), (19.42), and (19.43) are the covariance specification.

For a finite group table Q , set

$$C_r = \gamma_r(Q), \quad L_r = C_r/C_{r+1}.$$

The layer is a finite abelian group by Statement 19.2. A reproducible representation is constructed as follows:

1. compute the canonical coset table of C_{r+1} in C_r ;
2. choose a deterministic invariant-factor decomposition

$$L_r \cong \mathbb{Z}/d_{r,1}\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/d_{r,k_r}\mathbb{Z}, \quad d_{r,1} \mid \cdots \mid d_{r,k_r};\tag{19.44}$$

3. store canonical generators $z_{r,i}$ and express

$$\phi_r(z_{r,i}) = \sum_{j=1}^{k_r} A_{r,ji} z_{r,j};\tag{19.45}$$

4. verify each generator order relation after applying A_r , verify every row against the original group table, construct the inverse matrix from ϕ^{-1} , and check

$$A_r^8 = I\tag{19.46}$$

in the invariant-factor group;

5. test the divisors 1, 2, 4, 8 in increasing order and record the actual order of A_r .

The matrix arithmetic in (19.45)–(19.46) is interpreted componentwise modulo the declared invariant factors, with relation preservation checked explicitly; it is not silently treated as linear algebra over a field.

The output record is

$$(r, \{|C_r|, |C_{r+1}|\}, \{d_{r,i}\}, \{z_{r,i}\}, A_r, A_r^{-1}, \text{ord}(A_r), \text{hash}).\tag{19.47}$$

It is independently verified both in invariant-factor coordinates and in the original carrier table. Canonical carrier, coset, Smith-reduction, and generator tie-break orders make repeated serialization identical.

For $Q = C_2^8$ with cyclic coordinate shift, $L_1 = Q$, the matrix A_1 is the eight-cycle permutation matrix and has exact order eight. For $Q = U_3(\mathbb{F}_p)$, an automorphism can act nontrivially on $L_1 \cong C_p^2$ while its action on the central layer $L_2 \cong C_p$ is the induced determinant action. The identity automorphism tests the boundary where every recorded layer order is one.

19.6 Preservation of parity

Let $\chi : G \rightarrow C_2$ be a homomorphism. Since C_2 is abelian, the universal property (19.9) gives

$$\gamma_2(G) = G' \subseteq K_\chi. \quad (19.48)$$

For every $c \geq 1$, descendingness gives $\gamma_{c+1}(G) \subseteq \gamma_2(G)$, and hence

$$\gamma_{c+1}(G) \subseteq K_\chi. \quad (19.49)$$

Therefore parity descends to every positive-class nilpotent quotient:

$$\begin{aligned} \varepsilon_c : \mathcal{N}_c(G) &\longrightarrow C_2, \\ \varepsilon_c(g\gamma_{c+1}(G)) &= \chi(g). \end{aligned} \quad (19.50)$$

If two representatives differ by an element of $\gamma_{c+1}(G)$, (19.49) makes their parity equal, proving well-definedness. Surjectivity of ε_c is equivalent to surjectivity of χ .

The class-zero boundary is different: $\mathcal{N}_0(G) = \{1\}$, so a factorization exists only when χ is trivial. In particular, the nontrivial Bolza parity does not descend to $\mathcal{N}_0(\Gamma_B)$.

If $\chi \circ \phi = \chi$, then on every representative

$$\begin{aligned} \varepsilon_c \phi^{(c)}(g\gamma_{c+1}) &= \varepsilon_c(\phi(g)\gamma_{c+1}) \\ &= \chi(\phi(g)) \\ &= \chi(g) = \varepsilon_c(g\gamma_{c+1}), \end{aligned}$$

so

$$\varepsilon_c \circ \phi^{(c)} = \varepsilon_c. \quad (19.51)$$

For $G = \Gamma_B$, the physical marking

$$q_j = \nu_c(g_j)$$

satisfies

$$\varepsilon_c(q_j) = 1, \quad \phi^{(c)}(q_j) = q_{j+1}. \quad (19.52)$$

The surface relation remains valid under ν_c , and its image in C_2 is zero because it is a product of commutators.

For any further quotient $\theta : \mathcal{N}_c(G) \rightarrow Q$, parity descends to Q if and only if

$$\ker \theta \subseteq \ker \varepsilon_c. \quad (19.53)$$

Necessity follows by evaluating a proposed factorization on $\ker \theta$; sufficiency follows by defining the parity of $\theta(x)$ to be $\varepsilon_c(x)$. The same criterion is checked on finite tables by verifying that every identity-fibre row has parity zero.

There is an immediate obstruction for p -groups.

Theorem 19.6.1 (Parity obstruction). *If a finite group Q admits a surjection $Q \twoheadrightarrow C_2$, then $|Q|$ is even. Consequently no finite p -group for odd p can preserve a nontrivial parity map.*

Proof. The kernel has index two, so

$$|Q| = 2|\ker \varepsilon|.$$

An odd-order p -group has odd cardinality, giving a contradiction. $\square$

Thus a single finite p -group quotient compatible with the Bolza parity must have $p = 2$. Odd- p quotients may still be used for separation only after pairing them with an independent even factor.

The finite verifier stores the quotient row map, one parity bit per row, the images of the physical generators, the surface-relator evaluation, the covariance table (19.51), and the kernel-inclusion result (19.53). It returns INVALID for inconsistent tables, not a parity FAIL; a valid quotient with an odd identity-fibre element is the mathematical FAIL case.

19.7 Images of short words

Let $S = S^{-1}$ be a finite ordered generating set of a group G , and write

$$B_G^{\text{word}}(r) = \{g \in G : |g|_S \leq r\} \quad (r \in \mathbb{Z}_{\geq 0}). \quad (19.54)$$

Let

$$\theta : G \twoheadrightarrow P \quad (19.55)$$

be a certified finite nilpotent quotient, represented by a multiplication table and the rows $\theta(s)$ for $s \in S$. A word

$$w = s_1^{\epsilon_1} \cdots s_\ell^{\epsilon_\ell}$$

is evaluated by the unambiguous recurrence

$$v_0 = 1_P, \quad v_i = v_{i-1}\theta(s_i)^{\epsilon_i}, \quad \theta(w) = v_\ell. \quad (19.56)$$

Relator validation proves that different words representing the same element of G have the same value.

The algorithm requires a certified canonical-normal-form oracle that decides equality and returns canonical representatives for every ambient element reached through depth $2r$. It proceeds as follows:

1. start with the ambient normal form of 1 at depth zero;
2. in breadth-first order, multiply every depth- $< 2r$ element by the ordered letters of S , reduce to ambient canonical normal form, and retain a form only on its first occurrence;
3. evaluate every retained representative through depth $2r$ by (19.56);
4. restrict to the depth- $\leq r$ elements and sort them into fibres

$$\mathcal{B}_y(r) = \{g \in B_G^{\text{word}}(r) : \theta(g) = y\} \quad (y \in P); \quad (19.57)$$

5. return every nonidentity kernel element in $B_G^{\text{word}}(2r)$ and every radius- r collision fibre of size greater than one.

Induction on word length proves that the breadth-first traversal reaches every element of (19.54); canonical reduction removes duplicates without removing any group element. The homomorphism property proves that (19.57) is exactly the desired image partition.

There is an exact injectivity criterion:

$$\theta|_{B_G^{\text{word}}(r)} \text{ is injective} \iff \ker \theta \cap B_G^{\text{word}}(2r) = \{1\}. \quad (19.58)$$

If $\theta(\gamma) = \theta(\delta)$ for two elements of the radius- r ball, then

$$\delta^{-1}\gamma \in \ker \theta, \quad |\delta^{-1}\gamma|_S \leq |\delta|_S + |\gamma|_S \leq 2r. \quad (19.59)$$

Thus the right side of (19.58) implies the left. Conversely, suppose $1 \neq k \in \ker \theta$ has a shortest word of length $\ell \leq 2r$. Split this word as $k = ab$, with the first factor of length at most r and the remaining factor also of length at most r . Put

$$\delta = a^{-1}, \quad \gamma = b.$$

Then $\delta, \gamma \in B_G^{\text{word}}(r)$, $\delta^{-1}\gamma = ab = k$, and $\theta(\delta) = \theta(\gamma)$. Since $k \neq 1$, one has $\delta \neq \gamma$, so the restriction is not injective. This proves the converse without identifying raw words with ambient group elements.

If an automorphism ϕ permutes S and descends to $\bar{\phi} \in \text{Aut}(P)$, then it preserves every word ball and

$$\theta(\phi(g)) = \bar{\phi}(\theta(g)). \quad (19.60)$$

Hence ϕ carries the fibre $\mathcal{B}_y(r)$ bijectively onto $\mathcal{B}_{\bar{\phi}(y)}(r)$. If parity descends as $\varepsilon : P \rightarrow C_2$, every recorded image also satisfies

$$\varepsilon(\theta(g)) = \chi(g). \quad (19.61)$$

Equations (19.60)–(19.61) are checked row by row on the finite output.

The reproducible record is

$$(r, S \text{ order, ambient normal forms through depth } 2r, \\ \text{quotient table hash, image rows, fibres, collisions, hash}). \quad (19.62)$$

A valid nontrivial element in the right side of (19.58) is FAIL; malformed normal-form or quotient evidence is INVALID; a structurally valid traversal stopped before depth $2r$ is UNDECIDED.

For $m \geq 3$, consider the finite class-one quotient

$$F(x, y) \twoheadrightarrow C_m \times C_m, \quad x \mapsto (1, 0), \quad y \mapsto (0, 1),$$

the distinct length-two elements xy and yx collide, and the length-four commutator $[x, y]$ maps to the identity. The radius-zero ball contains only the identity and is always injected. These are the first-radius commutativity-collision and boundary tests. The restriction $m \geq 3$ avoids the unrelated radius-one collision $x = x^{-1}$ present when $m = 2$.

19.8 Orders of finite nilpotent quotients

Let Q be finite and $c \geq 0$. Put

$$P_c = \mathcal{N}_c(Q) = Q/\gamma_{c+1}(Q), \quad m_r = [\gamma_r(Q) : \gamma_{r+1}(Q)]. \quad (19.63)$$

Repeated index multiplication gives

$$\begin{aligned} |P_c| &= [Q : \gamma_{c+1}(Q)] \\ &= [\gamma_1(Q) : \gamma_2(Q)][\gamma_2(Q) : \gamma_3(Q)] \cdots [\gamma_c(Q) : \gamma_{c+1}(Q)] \\ &= \prod_{r=1}^c m_r. \end{aligned} \quad (19.64)$$

For $c = 0$, the product is empty and equals one, agreeing with the trivial quotient. Terms with $\gamma_r(Q) = \gamma_{r+1}(Q)$ contribute the factor one and must be retained in a fixed-length certificate.

If P_c is a finite p -group, every central layer is a finite abelian p -group. Write its invariant factors as

$$\gamma_r(P_c)/\gamma_{r+1}(P_c) \cong \bigoplus_{i=1}^{k_r} C_{p^{a_{r,i}}}. \quad (19.65)$$

For $1 \leq r \leq c$, formula (19.22) and the third isomorphism theorem give

$$\frac{\gamma_r(P_c)}{\gamma_{r+1}(P_c)} \cong \frac{\gamma_r(Q)/\gamma_{c+1}(Q)}{\gamma_{r+1}(Q)/\gamma_{c+1}(Q)} \cong \frac{\gamma_r(Q)}{\gamma_{r+1}(Q)}.$$

Thus the order of (19.65) is exactly the previously defined m_r , and

$$m_r = p^{\sum_i a_{r,i}}, \quad |P_c| = p^{\sum_{r=1}^c \sum_{i=1}^{k_r} a_{r,i}}. \quad (19.66)$$

No assumption that the layers are elementary abelian is made; exponents $a_{r,i} > 1$ are allowed. For a general finite nilpotent group, the same computation may be performed independently on its characteristic Sylow factors and multiplied, but (19.64) already gives the order without choosing a Sylow decomposition.

The deterministic finite-table algorithm is:

1. compute the row sets $\gamma_1(Q), \dots, \gamma_{c+1}(Q)$ by Statement 19.2;
2. verify every containment and compute the exact integer quotient

$$m_r = \frac{|\gamma_r(Q)|}{|\gamma_{r+1}(Q)|};$$

3. reject if any division has a remainder;
4. for every $1 \leq r \leq c$, construct the quotient multiplication table

$$\gamma_r(Q)/\gamma_{r+1}(Q)$$

using least coset representatives and verify that its carrier has cardinality m_r ;

5. if a p -primary decomposition is requested, apply the deterministic invariant-factor procedure of Statement 19.5 to each layer table. For every returned generator $z_{r,i}$, verify its exact order $p^{a_{r,i}}$, and exhaustively verify that

$$(u_1, \dots, u_{k_r}) \mapsto \sum_i u_i z_{r,i} \quad (19.67)$$

is a bijective homomorphism from $\prod_i C_{p^{a_{r,i}}}$ onto the layer table;

6. multiply the m_r using arbitrary-precision integers;
7. independently enumerate the cosets of $\gamma_{c+1}(Q)$ and compare their count with the product;
8. build the quotient table and compare its carrier size with both counts.

The three equal integers certify (19.64). If a prime p is declared, the verifier also factors every m_r , requires a pure power of p , and checks the exponent sum (19.66) against the independently verified decomposition (19.67). The invariant factors are never inferred from the layer order alone. A stored C_8 -action is then checked to permute each lower-central row set and to act invertibly on every layer; it cannot change any factor m_r . A stored surjective parity map additionally requires the product in (19.64) to be even, by Statement 19.6.

The output specification is

$$(c, \{|\gamma_r(Q)|\}, \{m_r\}, \{a_{r,i}\} \text{ when } p\text{-primary, } |P_c|, \text{coset-count check, hash}). \quad (19.68)$$

All arithmetic is exact. Optional decimal logarithms use directed outward rounding, but they are never used to decide an equality or an integer bound.

For $Q = U_3(\mathbb{F}_p)$ and $c = 2$,

$$m_1 = p^2, \quad m_2 = p, \quad |P_2| = p^3.$$

For $Q = C_4$ and $c = 1$, the single layer has invariant factor 4 and the output order is four, illustrating why central layers need not be elementary abelian. For $c = 0$, every input produces the independently verified order one. These three cases cross-check the multi-layer, non-elementary, and empty-product branches.

20 p -central series and certified finite p -quotients

Fix a prime p . For a subgroup $H \leq G$, the notation

$$H^p = \langle h^p : h \in H \rangle \quad (20.1)$$

always means the subgroup generated by the p -th powers; it does not mean the set of p -th powers, which need not itself be a subgroup. All vector spaces in this section are over $\mathbb{F}_p$.

20.1 The lower p -central series

Define

$$P_1(G) = G, \quad P_{n+1}(G) = P_n(G)^p [P_n(G), G] \quad (n \geq 1). \quad (20.2)$$

The product in (20.2) is a subgroup because both factors are normal. We prove this and the structural properties needed below rather than treating the recursion as formal notation.

Proposition 20.1.1. *For every $n \geq 1$, $P_n(G)$ is characteristic in G , the series is descending, and*

$$[P_n(G), G] \leq P_{n+1}(G), \quad P_n(G)^p \leq P_{n+1}(G). \quad (20.3)$$

Consequently $P_n(G)/P_{n+1}(G)$ is a central elementary abelian p -group in $G/P_{n+1}(G)$.

Proof. The group $P_1(G) = G$ is characteristic. Suppose $P_n(G)$ is characteristic and let $\alpha \in \text{Aut}(G)$. Since $\alpha(h^p) = \alpha(h)^p$,

$$\alpha(P_n(G)^p) = P_n(G)^p. \quad (20.4)$$

The commutator calculation from Statement 19.1 gives

$$\alpha([P_n(G), G]) = [\alpha(P_n(G)), \alpha(G)] = [P_n(G), G]. \quad (20.5)$$

Thus $\alpha(P_{n+1}(G)) = P_{n+1}(G)$, completing the induction. In particular $P_n(G) \trianglelefteq G$. Every p -th power of an element of $P_n(G)$ belongs to $P_{n+1}(G)$, and Statement 19.1 gives $[P_n(G), G] \leq P_{n+1}(G)$. Hence (20.2) implies

$$P_{n+1}(G) \leq P_n(G). \quad (20.6)$$

The two containments in (20.3) hold directly from the definition.

Put $V_n = P_n(G)/P_{n+1}(G)$. If $x \in P_n(G)$ and $g \in G$, then $[x, g] \in P_{n+1}(G)$, so $xP_{n+1}(G)$ commutes with $gP_{n+1}(G)$. Therefore V_n is central, hence abelian. Moreover

$$(xP_{n+1}(G))^p = x^p P_{n+1}(G) = P_{n+1}(G), \quad (20.7)$$

so every element has order dividing p . Writing the group operation additively and defining

$$a(xP_{n+1}) = x^a P_{n+1} \quad (a \in \mathbb{F}_p) \quad (20.8)$$

makes V_n an $\mathbb{F}_p$ -vector space. Independence of the integer representative of a follows from (20.7), and the vector space axioms are precisely the exponent laws in the abelian group V_n . $\square$

The construction is functorial. If $f : G \rightarrow H$ is a homomorphism, then

$$f(P_n(G)) \leq P_n(H) \quad (n \geq 1). \quad (20.9)$$

Indeed, the assertion is immediate for $n = 1$. If it holds for n , then

$$\begin{aligned} f(P_{n+1}(G)) &= f(P_n(G)^p [P_n(G), G]) \\ &\leq f(P_n(G))^p [f(P_n(G)), f(G)] \\ &\leq P_n(H)^p [P_n(H), H] = P_{n+1}(H). \end{aligned} \quad (20.10)$$

When f is onto, every generator on the last line has a preimage of the required form, so both inclusions become equalities. Thus

$$f(P_n(G)) = P_n(H) \quad \text{for surjective } f. \quad (20.11)$$

For a certified finite group table, (20.2) is an exact algorithm. Starting with the whole carrier P_1 , enumerate all x^p with $x \in P_n$, all $[x, g]$ with $x \in P_n$, $g \in G$, and close these elements under multiplication and inversion. Equality (20.2) proves inductively that the returned row set is

exactly P_{n+1} . Because the carrier is finite and the row sets descend, the calculation reaches either 1 or a nontrivial fixed point. The reproducible output is

$$(p, \text{table hash}, (P_n \text{ row sets}), \\ (\text{power witnesses}), (\text{commutator witnesses}), (\dim_{\mathbb{F}_p} P_n/P_{n+1}), \text{hash}). \quad (20.12)$$

Replay recomputes every closure and the elementary-abelian test (20.7).

For $G = C_{p^2} = \langle x \mid x^{p^2} = 1 \rangle$,

$$P_1(G) = G, \quad P_2(G) = \langle x^p \rangle, \quad P_3(G) = 1. \quad (20.13)$$

This is a positive two-layer example. The boundary $G = C_q$, where $q \neq p$ is prime, behaves differently. Choose a with $ap \equiv 1 \pmod{q}$. Then $x = (x^a)^p$, so $G^p = G$ and

$$P_n(C_q) = C_q \quad \text{for every } n. \quad (20.14)$$

Thus a p -central series need not separate points in a group having prime-to- p torsion.

20.2 The Frattini quotient

Let Q be a finite p -group. Its Frattini subgroup is the intersection of its maximal proper subgroups:

$$\Phi(Q) = \bigcap_{M < Q \text{ maximal}} M. \quad (20.15)$$

The following equality is both the structural theorem and the computational formula:

$$\boxed{\Phi(Q) = Q^p[Q, Q]}. \quad (20.16)$$

Proof. Set $N = Q^p[Q, Q]$. We first prove that every maximal subgroup $M < Q$ is normal of index p . Let M act by left multiplication on the finite set Q/M . An orbit has size a power of p . Hence the number of fixed cosets is congruent to

$$|Q/M| = [Q : M] \equiv 0 \pmod{p}. \quad (20.17)$$

The coset M is fixed, so there is another fixed coset gM . Fixedness means $g^{-1}Mg \leq M$; equal finite orders make this an equality. Thus $g \in N_Q(M) \setminus M$. Maximality gives $N_Q(M) = Q$, so $M \trianglelefteq Q$.

Now Q/M is a nontrivial finite p -group having no nontrivial proper subgroup. If $1 \neq yM$, then $\langle yM \rangle$ must be the whole quotient. If its order were p^a with $a > 1$, its unique subgroup of order p would be nontrivial and proper. Therefore $Q/M \cong C_p$. Every commutator and every p -th power maps to 1, so $N \leq M$. Since M was arbitrary,

$$N \leq \Phi(Q). \quad (20.18)$$

Conversely, $V = Q/N$ is abelian of exponent p , so it is a finite $\mathbb{F}_p$ -vector space. If $x \notin N$, extend the nonzero vector xN to a basis of V , and define a linear functional $\lambda : V \rightarrow \mathbb{F}_p$ that takes xN to 1. Then

$$M_x = \{q \in Q : \lambda(qN) = 0\} \quad (20.19)$$

has index p , hence is maximal, and $x \notin M_x$. Thus $x \notin \Phi(Q)$. This proves $\Phi(Q) \leq N$, which together with (20.18) proves (20.16). $\square$

The proof also gives the Burnside basis criterion. For $X = \{x_1, \dots, x_d\} \subseteq Q$,

$$\langle X \rangle = Q \iff \{x_i \Phi(Q)\}_{i=1}^d \text{ spans } Q/\Phi(Q). \quad (20.20)$$

The forward implication is immediate. For the reverse implication, put $H = \langle X \rangle$. If $H < Q$, finiteness lets us choose a maximal subgroup M containing H . Equation (20.16) gives $\Phi(Q) \leq M$, hence $H\Phi(Q) \leq M < Q$, contrary to spanning.

The deterministic construction from a finite p -group table is:

1. verify $|Q| = p^a$ by repeated exact division;
2. close the set of all q^p and $[q, r]$ to obtain $\Phi(Q)$;
3. construct the quotient table $V = Q/\Phi(Q)$;
4. scan quotient rows in carrier order, retaining a row exactly when it increases the span, and record its least carrier lift;
5. verify addition, scalar multiplication, independence, spanning, and (20.20) by exhaustive table calculations.

Its fixed output specification is

$$(p, \text{table hash}, \Phi(Q) \text{ rows}, V \text{ table, ordered basis}, \\ \text{least lifts, } \dim_{\mathbb{F}_p} V, \text{generation independent verification, hash}). \quad (20.21)$$

Each loop ranges over a finite carrier, and the subgroup closure grows strictly until it stops, proving termination. Formula (20.16) and the basis criterion prove correctness.

For the quaternion group

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\},$$

squares and commutators generate $\{\pm 1\}$, so

$$\Phi(Q_8) = \{\pm 1\}, \quad Q_8/\Phi(Q_8) \cong \mathbb{F}_2^2. \quad (20.22)$$

The trivial group is the dimension-zero boundary. The p -group hypothesis cannot be dropped: for S_3 and $p = 2$,

$$S_3^2[S_3, S_3] = A_3, \quad \Phi(S_3) = 1. \quad (20.23)$$

Indeed the squares generate A_3 , while the three order-two maximal subgroups have trivial intersection.

20.3 The Zassenhaus filtration

Let $A = \mathbb{F}_p[G]$ be the group algebra, let

$$\varepsilon : A \longrightarrow \mathbb{F}_p, \quad \varepsilon\left(\sum_{g \in G} a_g g\right) = \sum_{g \in G} a_g,$$

and let $I_G = \ker \varepsilon$ be its augmentation ideal. The Zassenhaus filtration is the dimension-subgroup filtration

$$D_n(G) = \{g \in G : g - 1 \in I_G^n\} \quad (n \geq 1). \quad (20.24)$$

This definition will be used directly; it avoids placing any unproved Hall–Petrescu collection assertion inside the construction.

The set in (20.24) is a characteristic subgroup. If $g - 1, h - 1 \in I_G^n$, then

$$gh - 1 = (g - 1) + g(h - 1) \in I_G^n, \quad g^{-1} - 1 = -g^{-1}(g - 1) \in I_G^n. \quad (20.25)$$

Thus it is closed under products and inverses. An automorphism of G extends linearly to an algebra automorphism of A , preserves ε , hence preserves I_G^n ; characteristicity follows. Since $I_G^{n+1} \leq I_G^n$,

$$D_1(G) = G, \quad D_{n+1}(G) \leq D_n(G). \quad (20.26)$$

The two weight laws now follow by exact algebra identities. If $x \in D_m(G)$ and $y \in D_n(G)$, then

$$[x, y] - 1 = (xy - yx)x^{-1}y^{-1}, \\ xy - yx = (x - 1)(y - 1) - (y - 1)(x - 1). \quad (20.27)$$

The second line belongs to I_G^{m+n} , and multiplication by the units $x^{-1}y^{-1}$ preserves that two-sided ideal. Therefore

$$[D_m(G), D_n(G)] \leq D_{m+n}(G). \quad (20.28)$$

If $x \in D_n(G)$, the elements x and 1 commute, so the ordinary binomial formula is valid in A . Every intermediate binomial coefficient is divisible by p , and in characteristic p

$$x^p - 1 = (x - 1)^p \in (I_G^n)^p = I_G^{pn}.$$

Hence

$$D_n(G)^p \leq D_{pn}(G). \quad (20.29)$$

Equations (20.28)–(20.29) imply

$$[D_n(G), G] \leq D_{n+1}(G), \quad D_n(G)^p \leq D_{n+1}(G), \quad (20.30)$$

because $D_1(G) = G$, $pn \geq n+1$, and the filtration descends. It follows exactly as in (20.7)–(20.8) that every D_n/D_{n+1} is a central elementary abelian p -group. At the first step one has the exact identity

$$D_2(G) = G^p[G, G]. \quad (20.30a)$$

To prove the reverse inclusion that is not already supplied by (20.30), consider

$$\theta : G \longrightarrow I_G/I_G^2, \quad \theta(g) = (g - 1) + I_G^2.$$

Since $gh - 1 = (g - 1) + (h - 1) + (g - 1)(h - 1)$, this is a homomorphism to the additive elementary abelian group I_G/I_G^2 , with kernel $D_2(G)$. It induces a map from $G/G^p[G, G]$. Conversely define, on the augmentation ideal,

$$L\left(\sum_g a_g g\right) = \sum_g a_g \bar{g} \quad \text{in } G/G^p[G, G],$$

where the last quotient is written additively. For group elements u, v ,

$$L((u - 1)(v - 1)) = \overline{uv} - \bar{u} - \bar{v} = 0,$$

so L kills I_G^2 , and $L(g - 1) = \bar{g}$. Thus the two induced maps are inverse. Their kernels give (20.30a).

For a certified finite group table, (20.24) is a finite linear-algebra algorithm. Represent A on the ordered basis $\{e_g : g \in G\}$, with $e_g e_h = e_{gh}$. The vectors

$$e_g - e_1 \quad (g \neq 1) \quad (20.31)$$

form a basis of I_G . Starting from this basis, obtain I_G^{n+1} as the row span of all products of a basis vector of I_G^n with a basis vector of I_G . Gaussian elimination over $\mathbb{F}_p$ gives a unique reduced basis. Finally retain precisely those carrier rows g for which $e_g - e_1$ reduces to zero against the basis of I_G^n . By (20.24), that row set is exactly $D_n(G)$. The input may prescribe a finite terminal height. In complete mode, iterate until the first index t with

$$I_G^{t+1} = I_G^t. \quad (20.31a)$$

The ideal powers form a descending chain in the finite-dimensional algebra. Before equality their dimensions strictly decrease, so such a t occurs after finitely many steps. Multiplying (20.31a) by I_G gives $I_G^{t+2} = I_G^{t+1}$; induction makes every later ideal power, dimension subgroup, and layer fixed. Thus the complete algorithm terminates. Replay verifies the algebra multiplication, the

ideal products, all membership decisions, (20.28)–(20.30), and the terminal fixed-point equality. The fixed output is

$$(p, \text{table hash, requested height or terminal } t, \\ (I_G^n \text{ reduced bases}), (D_n \text{ row sets}), (\dim_{\mathbb{F}_p} D_n/D_{n+1}), \\ \text{fixed-point witness, commutator and power independent verification, hash}). \quad (20.32)$$

For $G = C_{p^a} = \langle x \rangle$, put $T = x - 1$. In characteristic p ,

$$\mathbb{F}_p[C_{p^a}] \cong \mathbb{F}_p[T]/(T^{p^a}), \quad (1 + T)^{p^a} - 1 = T^{p^a}. \quad (20.33)$$

For a nonidentity element choose its unique exponent $1 \leq k < p^a$, and write $k = p^v u$ with $p \nmid u$. Then

$$(1 + T)^k - 1 = (1 + T^{p^v})^u - 1 = uT^{p^v} + \text{terms of higher degree}. \quad (20.34)$$

Thus, for these nonidentity representatives, $x^k - 1 \in I_G^n$ exactly when $p^v \geq n$. The identity element belongs to every D_n separately. It follows that

$$D_n(C_{p^a}) = \langle x^{p^{e(n)}} \rangle, \quad e(n) = \min\{a, \lceil \log_p n \rceil\}. \quad (20.34a)$$

For the prime-to- p boundary $G = C_q$, write $t^q - 1 = (t - 1)h(t)$. Since $h(1) = q \neq 0$ in $\mathbb{F}_p$, $\gcd(t - 1, h) = 1$. A Bézout identity, multiplied by $t - 1$ in $\mathbb{F}_p[t]/(t^q - 1)$, gives $I_G = I_G^2$. Hence

$$D_n(C_q) = C_q \quad (n \geq 1), \quad (20.34b)$$

the required nonseparating boundary example.

20.4 Universal finite p -quotients

For $c \geq 0$, define the Zassenhaus p -class- c quotient

$$Q_{p,c}(G) = G/D_{c+1}(G), \quad \pi_{p,c} : G \longrightarrow Q_{p,c}(G). \quad (20.35)$$

Here $Q_{p,0}(G) = 1$. A homomorphism $f : G \rightarrow H$ extends linearly to an algebra homomorphism $f_* : \mathbb{F}_p[G] \rightarrow \mathbb{F}_p[H]$. It preserves augmentation, so $f_*(I_G^n) \leq I_H^n$. Therefore

$$f(D_n(G)) \leq D_n(H) \quad (20.36)$$

for every homomorphism $f : G \rightarrow H$.

There is also a faithful description of the quotient in (20.35). The map

$$gD_{c+1}(G) \longmapsto g + I_G^{c+1} \quad (20.37)$$

embeds $Q_{p,c}(G)$ in the unit group of $\mathbb{F}_p[G]/I_G^{c+1}$, because its kernel is exactly (20.24). If $J = I_G/I_G^{c+1}$, the image lies in $1 + J$ and $J^{c+1} = 0$. Applying the group-algebra map induced by (20.37) to the augmentation ideal of $Q_{p,c}$ shows that an element of $D_{c+1}(Q_{p,c})$ maps to $1 + J^{c+1} = 1$. Faithfulness then gives

$$D_{c+1}(Q_{p,c}(G)) = 1. \quad (20.37a)$$

Theorem 20.4.1 (universality). *Let $f : G \rightarrow R$ be a homomorphism and suppose*

$$D_{c+1}(f(G)) = 1.$$

Then there is a unique homomorphism $\bar{f} : Q_{p,c}(G) \rightarrow R$ such that

$$f = \bar{f} \circ \pi_{p,c}. \quad (20.38)$$

If G is generated by $d < \infty$ elements, then $Q_{p,c}(G)$ is a finite p -group.

Proof. Applying (20.36) to $G \rightarrow f(G)$ gives

$$f(D_{c+1}(G)) \leq D_{c+1}(f(G)) = 1.$$

Hence $D_{c+1}(G) \leq \ker f$, and the quotient formula

$$\bar{f}(gD_{c+1}(G)) = f(g) \quad (20.39)$$

is well defined. It is a homomorphism, and surjectivity of $\pi_{p,c}$ proves uniqueness.

It remains to prove finiteness. Let $X = \{x_1, \dots, x_d\}$ generate G , put $X_i = x_i - 1$, and set

$$\bar{A} = \mathbb{F}_p[G]/I_G^{c+1}. \quad (20.40)$$

Modulo I_G^{c+1} , each inverse has the finite expansion

$$x_i^{-1} = (1 + X_i)^{-1} \equiv \sum_{j=0}^c (-X_i)^j, \quad (20.41)$$

because multiplication by $1 + X_i$ leaves 1 plus a term in I_G^{c+1} . Expanding a word in the $x_i^{\pm 1}$ now shows that $\bar{A}$ is spanned by the finitely many noncommutative monomials

$$X_{i_1} \cdots X_{i_r} \quad (0 \leq r \leq c). \quad (20.41a)$$

It is therefore a finite set.

Let $J = I_G/I_G^{c+1}$. Since $J^{c+1} = 0$, every $1 + a$, $a \in J$, is a unit. The filtration

$$1 + J \geq 1 + J^2 \geq \cdots \geq 1 + J^{c+1} = 1 \quad (20.41b)$$

has factors embedding by $1 + a \mapsto a$ into the additive $\mathbb{F}_p$ -spaces J^r/J^{r+1} . Consequently $1 + J$ is a finite p -group. By (20.37), $Q_{p,c}(G)$ is a subgroup of $1 + J$, and hence is a finite p -group. $\square$

There are two exact construction modes, and their termination claims must not be conflated.

- (A) For a certified finite group table, compute D_{c+1} by the ideal-power algorithm (20.31)–(20.32), construct its coset table, and exhaustively verify all products. This always terminates.
- (B) For a finite presentation $G = \langle x_1, \dots, x_d \mid \mathcal{R} \rangle$, a p -quotient engine may return a consistent power-commutator presentation for a candidate. Relator evaluation and $D_{c+1} = 1$ prove only that the candidate is a factor of $Q_{p,c}$, not that it is universal.

Universal verification requires an augmentation certificate. Form

$$T_c = \mathbb{F}_p[G]/I_G^{c+1}. \quad (20.41c)$$

A finite normal-form certificate for T_c contains algebra generators u_i, v_i , the inverse relations $u_i v_i = v_i u_i = 1$, every group relator in $\mathcal{R}$, all products of $c + 1$ augmentation generators $u_i - 1, v_i - 1$, an oriented terminating reduction system, and joined reductions for every critical overlap. Replaying termination, the defining reductions, and all critical pairs certifies the stated canonical basis and multiplication table for T_c , rather than for an unspecified quotient algebra. Indeed, termination permits induction on the maximum remaining reduction height. Two different first reductions either occur on disjoint subwords and commute, or form a recorded critical overlap; after that first join, the induction joins their remaining tails. Hence every word has a unique normal form.

In that table compute

$$U_c = \langle u_1, \dots, u_d \rangle \leq T_c^\times. \quad (20.41d)$$

By (20.24), the map $G \rightarrow U_c$ has kernel exactly $D_{c+1}(G)$. The candidate is accepted as $Q_{p,c}(G)$ only after an explicit bijective table homomorphism from the candidate to U_c , together with its inverse, is checked on every carrier row and shown to match every generator image. This proves both kernel containments. Without the algebra certificate and bijection, the output is labelled ESTABLISHED FACTOR, or UNDECIDED if even the factor certificate is unfinished; neither classification is SATISFIED.

The fixed output record is

$$(p, c, \text{source hash}, Q_{p,c} \text{ pc data}, \pi_{p,c}(X), \\ D_i/D_{c+1} \text{ rows and dimensions}, T_c \text{ normal-form table}, U_c \text{ rows}, \\ \text{candidate-}U_c \text{ bijection and inverse, relator and critical-pair independent verification, hash}). \quad (20.42)$$

For the positive test $G = \mathbb{Z} = \langle x \rangle$, the calculation in (20.33)–(20.34a) gives

$$Q_{p,c}(\mathbb{Z}) \cong C_{p^{\lceil \log_p(c+1) \rceil}}. \quad (20.43)$$

For the boundary $c = 0$, the output is trivial. For $G = C_q$, $q \neq p$, (20.34b) gives only the trivial p -quotient, showing why a nontrivial output cannot be promised.

Parity has a sharp prime boundary. Since

$$D_2(G) = G^p[G, G], \quad (20.44)$$

the parity map $\chi : \Gamma_B \rightarrow C_2$ vanishes on $D_2(\Gamma_B)$ when $p = 2$, and hence descends to every $Q_{2,c}$ with $c \geq 1$. If p is odd, every homomorphism from a finite p -group to C_2 is trivial, so no nontrivial parity factor can descend.

20.5 The associated graded Lie algebra

Set

$$\text{gr}_p(G) = \bigoplus_{n \geq 1} L_n^p(G), \quad L_n^p(G) = D_n(G)/D_{n+1}(G). \quad (20.45)$$

For homogeneous classes define

$$[xD_{m+1}, yD_{n+1}]_{\text{gr}} = [x, y]D_{m+n+1}. \quad (20.46)$$

We verify that this is a graded Lie bracket.

For each n , define

$$\iota_n : L_n^p(G) \longrightarrow I_G^n/I_G^{n+1}, \quad \iota_n(gD_{n+1}) = (g - 1) + I_G^{n+1}. \quad (20.47)$$

If $g' = gh$ with $h \in D_{n+1}$, then $g' - g = g(h - 1) \in I_G^{n+1}$, proving well-definedness. Its kernel is zero by the definition (20.24). Moreover

$$gh - 1 = (g - 1) + (h - 1) + (g - 1)(h - 1) \equiv (g - 1) + (h - 1) \pmod{I_G^{n+1}} \quad (20.48)$$

for $g, h \in D_n$, because $I_G^{2n} \leq I_G^{n+1}$. Thus ι_n is an injective linear map.

The associated graded algebra

$$\text{gr}_{I_G}(A) = \bigoplus_{n \geq 1} I_G^n/I_G^{n+1} \quad (20.49)$$

is associative. Its commutator $uv - vu$ is bilinear, alternating, and satisfies Jacobi by the direct expansion

$$\begin{aligned} & [uv - vu, w] + [vw - wv, u] + [wu - uw, v] \\ &= uvw - vwu - wuv + wvu \\ &+ vwu - wvu - uvw + uvw \\ &+ wuv - uwv - vwu + vuw = 0, \end{aligned} \quad (20.50)$$

where equal monomials cancel pairwise.

For $x \in D_m$, $y \in D_n$, equation (20.27) and the fact that $x^{-1}y^{-1} - 1 \in I_G$ give

$$[x, y] - 1 \equiv (x - 1)(y - 1) - (y - 1)(x - 1) \pmod{I_G^{m+n+1}}. \quad (20.50a)$$

Hence (20.46) is carried by the injective maps ι_n to the associative-algebra commutator. It is therefore well defined, bilinear, alternating, and Jacobi. Consequently $\text{gr}_p(G)$ is a graded Lie algebra and

$$[L_m^p, L_n^p] \subseteq L_{m+n}^p. \quad (20.51)$$

The filtration also records the homogeneous p -power operation

$$(xD_{n+1})^{[p]} = x^p D_{pn+1} \in L_{pn}^p. \quad (20.52)$$

It is well defined by another explicit ideal calculation. Write $x = 1 + a$, with $a \in I_G^n$, and $y = 1 + b$, with $b \in I_G^{n+1}$. Then $xy = 1 + a + c$, where $c = b + ab \in I_G^{n+1}$. Characteristic p gives

$$(xy)^p = (1 + a + c)^p = 1 + (a + c)^p, \quad x^p = 1 + a^p. \quad (20.52a)$$

In the noncommutative expansion of $(a + c)^p - a^p$, every word contains at least one c and at most $p - 1$ copies of a ; it therefore lies in

$$I_G^{(p-1)n+(n+1)} = I_G^{pn+1}. \quad (20.52b)$$

Thus $(xy)^p D_{pn+1} = x^p D_{pn+1}$. The ordinary Lie algebra therefore retains the extra homogeneous p -power data needed by a p -quotient implementation.

For a finite table, choose the deterministic bases of the elementary abelian layers as in Statement 20.2. Evaluate all pairs of basis lifts, locate their commutators in the appropriate quotient table, and record the coordinate tensors

$$c_{m,n;i,j}^k \in \mathbb{F}_p, \quad [e_{m,i}, e_{n,j}] = \sum_k c_{m,n;i,j}^k e_{m+n,k}. \quad (20.53)$$

Replay checks (20.47)–(20.50a) on basis triples; bilinearity then extends the checks to all vectors. It also records the coordinates of (20.52). This finite tensor package is the canonical computational expression of the graded algebra.

For $G = C_{p^2}$, the only nonzero degrees occur at the jumps described by (20.33)–(20.34a); every bracket tensor is zero. A nonabelian positive bracket test is the Heisenberg group $U_3(\mathbb{F}_p)$: the classes of

$$X = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad Y = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

have bracket represented by

$$[X, Y] = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (20.54)$$

the central degree-two generator. The trivial group gives the zero-dimensional boundary algebra.

20.6 C_8 -module decomposition of the layers

Every $D_n(G)$ is characteristic by (20.24). Hence the order-eight automorphism ϕ_8 induces

$$A_n : L_n^p(\Gamma_B) \longrightarrow L_n^p(\Gamma_B), \quad A_n(xD_{n+1}) = \phi_8(x)D_{n+1}. \quad (20.55)$$

Well-definedness follows from characteristicity; multiplication and p -th powers show linearity. Moreover

$$A_n^8 = I. \quad (20.56)$$

The actual order may be a proper divisor of eight and must be computed, not assumed.

Let V be any finite-dimensional layer and $A = A_n$. If p is odd, $F(t) = t^8 - 1$ is square-free because

$$\gcd(F, F') = \gcd(t^8 - 1, 8t^7) = 1. \quad (20.57)$$

Factor $F = \prod_f f$ into distinct monic irreducibles over $\mathbb{F}_p$. Put $h_f = F/f$, and use the extended Euclidean algorithm to choose a_f, b_f with

$$a_f(t)f(t) + b_f(t)h_f(t) = 1. \quad (20.58)$$

Then

$$E_f = b_f(A)h_f(A) \quad (20.59)$$

is a projection. Indeed E_f is the identity on $\ker f(A)$, its image is killed by $f(A)$, different projections have zero product, and their sum is I after reducing the corresponding polynomial identity modulo F . Therefore

$$V = \bigoplus_{f|t^8-1} \ker f(A). \quad (20.60)$$

The summand $\ker f(A)$ is a vector space over $\mathbb{F}_p[t]/(f)$, with t acting as A , so its multiplicity is $\dim_{\mathbb{F}_p} \ker f(A) / \deg f$.

If $p = 2$, semisimplicity fails:

$$t^8 - 1 = (t - 1)^8.$$

Put $N = A - I$. The binomial coefficients between the endpoints of $(A - I)^8$ are even, so (20.56) gives $N^8 = 0$. Let

$$d_k = \dim \ker N^k \quad (0 \leq k \leq 8), \quad d_0 = 0, \quad d_9 = d_8. \quad (20.61)$$

The number of nilpotent Jordan blocks of size at least k is $d_k - d_{k-1}$: each block contributes one new kernel vector at every step until its length is reached. Hence the number of blocks of size exactly k is

$$e_k = (d_k - d_{k-1}) - (d_{k+1} - d_k) = 2d_k - d_{k-1} - d_{k+1}. \quad (20.62)$$

These integers are nonnegative, and

$$V \cong \bigoplus_{k=1}^8 \left(\mathbb{F}_2[t]/((t-1)^k) \right)^{e_k}. \quad (20.63)$$

The algorithm forms the matrix of A in the deterministic layer basis from Statement 20.5, verifies $A^8 = I$, and computes its actual order by testing the divisors 1, 2, 4, 8. For odd p it factors F , computes (20.58)–(20.60), and verifies directness and spanning by row reduction. For $p = 2$ it computes the eight kernels, the integers (20.62), and a block basis by deterministic pivot choices. The output is

$$(p, n, \text{layer-basis hash}, A, \text{ord}(A), \text{factor or block data, change-of-basis matrix, inverse matrix, independent verification hash}). \quad (20.64)$$

The one-dimensional trivial action is the boundary test. For an odd prime with $8 \mid p - 1$, a diagonal action by eight distinct eighth roots supplies the split positive test; over $\mathbb{F}_2$, a single block of length eight supplies the nonsemisimple positive test.

20.7 Canonical C_8 -stable submodules

Let V be a finite-dimensional $\mathbb{F}_p$ -space and let $A \in \text{GL}(V)$ satisfy $A^8 = I$. For a finite seed set $S \subseteq V$, define its orbit closure

$$\langle S \rangle_{C_8} = \text{span}_{\mathbb{F}_p} \{A^j s : s \in S, 0 \leq j < 8\}. \quad (20.65)$$

This is the unique least A -stable subspace containing S . It contains the seeds by taking $j = 0$. Applying A cycles the listed orbit because $A^8 = I$, so it is stable. If U is any stable subspace containing S , induction gives $A^j s \in U$ for every listed vector; hence (20.65) lies in U .

If $W \leq V$ is A -stable, the quotient action is

$$\bar{A} : V/W \longrightarrow V/W, \quad \bar{A}(v + W) = Av + W. \quad (20.66)$$

If $v - v' \in W$, then $A(v - v') \in W$, proving well-definedness. Linearity is inherited from A , and

$$\bar{A}^8(v + W) = A^8 v + W = v + W. \quad (20.67)$$

Thus the quotient is again an $\mathbb{F}_p[C_8]$ -module. Its actual action order is the least member of $\{1, 2, 4, 8\}$ whose matrix power is the identity.

There are two complementary deterministic algorithms.

- (A) To construct the least stable submodule generated by constraints, scan S in its serialized order, scan $A^j s$ in increasing j , and retain a vector precisely when it increases the row rank. Reduced row echelon form gives a unique matrix for the resulting subspace.
- (B) To enumerate all stable submodules, enumerate all reduced row echelon matrices with $\dim V$ columns. Each subspace occurs exactly once. Retain a row space W exactly when reducing every Aw against its basis gives zero remainder. Since A is invertible, $A(W) \leq W$ then implies $A(W) = W$.

Both algorithms terminate because the vector space and the set of its subspaces are finite. The first is correct by the minimality proof after (20.65); the second is complete by uniqueness of reduced row echelon form.

For odd p , the decomposition (20.60) can accelerate the second algorithm. On the f -summand, A -stable subspaces are precisely the subspaces over the field $K_f = \mathbb{F}_p[t]/(f)$. This statement includes multiplicities: when the f -summand is K_f^m , every K_f -subspace is allowed, not merely 0 and the full isotypic summand. For $p = 2$, the nilpotent block structure (20.63) replaces this semisimple reduction; arbitrary direct sums of whole blocks do not exhaust the submodules, so the exact fallback remains the RREF enumeration.

The serialized record is

$$(p, n, A, \text{ord}(A), S, \text{RREF}(\langle S \rangle_{C_8}), \text{orbit provenance, stable-submodule RREF list, quotient actions, hash}). \quad (20.68)$$

Replay checks the seed containment, every orbit vector, row-space minimality, stability, quotient well-definedness, and the exact order of every induced action.

For $p = 2$, parity supplies a distinguished positive example in the first layer. By (20.44), χ induces a nonzero linear functional

$$\lambda : L_1^2(\Gamma_B) = \Gamma_B/D_2(\Gamma_B) \longrightarrow \mathbb{F}_2. \quad (20.69)$$

The identity $\chi\phi_8 = \chi$ from Statement 9.5 gives $\lambda A_1 = \lambda$, and therefore $\ker \lambda$ is A_1 -stable. No analogous nonzero parity functional is asserted on higher layers: for $n \geq 2$, $D_n \leq D_2 \subseteq \ker \chi$. The zero seed set in (20.65) returns 0, while a cyclic vector for a single length-eight block returns all of V ; these are the boundary and positive closure tests.

20.8 The certified p -quotient tower

The characteristic subgroups

$$G = D_1(G) \geq D_2(G) \geq D_3(G) \geq \cdots \quad (20.70)$$

produce the nested quotient tower

$$\cdots \longrightarrow Q_{p,c+1}(G) \xrightarrow{\kappa_{c+1,c}} Q_{p,c}(G) \longrightarrow \cdots \longrightarrow Q_{p,0}(G) = 1, \quad (20.71)$$

where

$$\kappa_{c+1,c}(gD_{c+2}) = gD_{c+1}. \quad (20.72)$$

This is well defined because $D_{c+2} \leq D_{c+1}$, and it is surjective. Its kernel is computed without an order argument:

$$\begin{aligned} gD_{c+2} \in \ker \kappa_{c+1,c} &\iff gD_{c+1} = D_{c+1} \\ &\iff g \in D_{c+1}. \end{aligned}$$

Thus there is an exact sequence

$$1 \longrightarrow D_{c+1}/D_{c+2} \longrightarrow Q_{p,c+1} \xrightarrow{\kappa_{c+1,c}} Q_{p,c} \longrightarrow 1, \quad (20.73)$$

and consequently

$$|Q_{p,c+1}| = |Q_{p,c}| p^{\dim_{\mathbb{F}_p} L_{c+1}^p(G)}. \quad (20.74)$$

A zero-dimensional layer is allowed and gives an isomorphism between two consecutive levels.

For $G = \Gamma_B$, characteristicity makes the action

$$\alpha_c(gD_{c+1}) = \phi_8(g)D_{c+1} \quad (20.75)$$

well defined. It satisfies

$$\alpha_c^8 = 1, \quad \kappa_{c+1,c}\alpha_{c+1} = \alpha_c\kappa_{c+1,c}. \quad (20.76)$$

The first equality follows from $\phi_8^8 = 1$; evaluating both sides of the second on gD_{c+2} gives the same coset $\phi_8(g)D_{c+1}$. Again, the actual order of α_c can be less than eight.

To state the local injection radius, fix a finite symmetric generating set S of G , and let

$$B_G(r) = \{g \in G : |g|_S \leq r\}. \quad (20.77)$$

For a quotient $\pi : G \rightarrow Q$, define

$$\lambda(\pi) = \min\{|g|_S : 1 \neq g \in \ker \pi\}, \quad (20.78)$$

with value $+\infty$ when the kernel is trivial. The exact largest integer radius on which π is injective is

$$\rho(\pi) = \begin{cases} +\infty, & \lambda(\pi) = +\infty, \\ \left\lfloor \frac{\lambda(\pi) - 1}{2} \right\rfloor, & \lambda(\pi) < +\infty. \end{cases} \quad (20.79)$$

Indeed Statement 19.7 proves that injectivity on $B_G(r)$ is equivalent to $\ker \pi \cap B_G(2r) = \{1\}$. If $\lambda < \infty$, this is equivalent to $2r < \lambda$, whose largest integer solution is exactly (20.79).

The tower algorithm takes p , a terminal height C , the ordered generators S , certified quotient records (20.42), the generator table for ϕ_8 , and a deterministic canonical-normal-form oracle for source words. It first performs the following level operations for each $c = 0, \dots, C$:

1. independent verification the presentation or finite-table certificate for $Q_{p,c}$, and verify that its order is a power of p ;

2. construct α_c , check its defining covariance on every generator, and compute its actual order;
3. enumerate canonical source elements by nondecreasing word length and increasing shortlex word, evaluate them in $Q_{p,c}$, and stop at the first nonidentity source element mapping to 1. This gives λ_c and then ρ_c by (20.79).

It then performs the following transition operations only for $c = 0, \dots, C - 1$:

4. construct $\kappa_{c+1,c}$ from the generator images, verify all defining relations, exhaustively verify the map on the finite normal-form range, and compare its kernel with D_{c+1}/D_{c+2} ;
5. verify the order recursion (20.74) and the action compatibility (20.76).

Step 3 terminates whenever the kernel is nontrivial. For an infinite finitely generated source and a finite quotient, the kernel is necessarily nontrivial. For a certified finite source table, exhaustive enumeration instead detects a trivial kernel and returns $+\infty$. Without either condition, a search that has not found a kernel element is UNDECIDED; it is not a certificate of infinite radius.

The fixed tower artifact is

$$\begin{aligned}
& (p, C, \text{source and generator hashes}, \\
& \{Q_{p,c}, \pi_{p,c}, D_{c+1}, |Q_{p,c}|, \alpha_c, \text{ord}(\alpha_c), \lambda_c, \rho_c\}_{c=0}^C, \\
& \{\kappa_{c+1,c}, \ker \kappa_{c+1,c}, \text{compatibility witnesses}\}_{c=0}^{C-1}, \text{terminal hash}).
\end{aligned} \tag{20.80}$$

All carrier lists, bases, and matrices use the deterministic orders fixed above, so identical input produces byte-identical serialization.

For $G = \mathbb{Z} = \langle x \rangle$, put

$$e_c = \lceil \log_p(c+1) \rceil.$$

Then

$$Q_{p,c}(\mathbb{Z}) \cong C_{p^{e_c}}, \quad \lambda_c = p^{e_c}, \quad \rho_c = \left\lfloor \frac{p^{e_c} - 1}{2} \right\rfloor. \tag{20.81}$$

The transition is an isomorphism whenever $e_{c+1} = e_c$, which tests the zero-layer branch of (20.74). The action $x \mapsto x$ has actual order one, testing the proper-divisor branch of (20.76). For the prime-to- p boundary $G = C_q$, every level is trivial by (20.34b) and the first generator is a length-one kernel witness, so $\rho_c = 0$.

21 Two-step and higher nilpotent extensions

21.1 Two-step nilpotent groups

For a group N , the following conditions are equivalent:

$$\gamma_3(N) = 1, \quad [N, N] \leq Z(N), \quad [[x, y], z] = 1 \quad (x, y, z \in N). \tag{21.1}$$

The first and third are equivalent because $\gamma_3(N) = [[N, N], N]$. The third says exactly that every generator $[x, y]$ of $[N, N]$ commutes with every z , which is the middle condition. A group satisfying (21.1) is nilpotent of class at most two; it has class exactly two when $[N, N] \neq 1$.

In such a group, all commutators are central. The identities of Statement 19.1 therefore lose their conjugation terms and become

$$\begin{aligned}
[xy, z] &= [x, z][y, z], & [x, yz] &= [x, y][x, z], \\
[x^a, y^b] &= [x, y]^{ab}, & yx &= [x, y]^{-1}xy.
\end{aligned} \tag{21.2}$$

The first two follow directly from (19.4), because the commutator factors are central. Induction on $a, b \geq 0$ gives the third; inverse exponents follow from $[x^{-1}, y] = [x, y]^{-1} = [x, y^{-1}]$, again using centrality. The last identity is a rearrangement of $[x, y] = xyx^{-1}y^{-1}$.

Repeatedly using the last rule in (21.2) gives the power formula

$$(xy)^n = x^n y^n [y, x]^{\binom{n}{2}} \quad (n \geq 0). \quad (21.3)$$

For the induction step, assume the formula at n . From $y^n x = xy^n [y, x]^n$ and centrality,

$$\begin{aligned} (xy)^{n+1} &= x^n y^n [y, x]^{\binom{n}{2}} xy \\ &= x^{n+1} y^{n+1} [y, x]^{n+\binom{n}{2}} = x^{n+1} y^{n+1} [y, x]^{\binom{n+1}{2}}. \end{aligned} \quad (21.4)$$

For a certified finite table, compute $[N, N]$ by Statement 19.1 and

$$Z(N) = \{z \in N : zg = gz \text{ for every } g \in N\}. \quad (21.5)$$

The class test returns

$$\begin{cases} 0, & N = 1, \\ 1, & [N, N] = 1 \text{ and } N \neq 1, \\ 2, & 1 \neq [N, N] \leq Z(N), \\ > 2, & [N, N] \not\leq Z(N). \end{cases} \quad (21.6)$$

The last symbol is a rejection of the two-step specification, not a computed exact class. The output lists the table hash, center rows, commutator rows and witnesses, the inclusion test, the class branch, and a independent verification hash. All loops are finite.

The Heisenberg group $U_3(\mathbb{F}_p)$ is the positive exact-class-two test: its commutator subgroup and center are both the matrices having only the upper-right entry possibly nonzero. A nontrivial abelian group is the class-one boundary. The group S_3 is a negative test because $[S_3, S_3] = A_3 \not\leq Z(S_3) = 1$.

21.2 The abelianization module

Let N be a finite p -group of class at most two and let $\alpha \in \text{Aut}(N)$ satisfy $\alpha^8 = 1$. Put

$$A = N_{\text{ab}} = N/[N, N]. \quad (21.7)$$

It is a finite abelian p -group. Since the commutator subgroup is characteristic, the formula

$$P(x[N, N]) = \alpha(x)[N, N] \quad (21.8)$$

is well defined and defines an automorphism with $P^8 = 1$. Thus A is a module over

$$\mathbb{Z}[C_8] \cong \mathbb{Z}[t]/(t^8 - 1), \quad (21.9)$$

where integers act by repeated addition and t acts as P . The action need not have exact order eight.

Write the invariant-factor decomposition

$$A \cong C_{p^{a_1}} \oplus \cdots \oplus C_{p^{a_d}}, \quad 1 \leq a_1 \leq \cdots \leq a_d. \quad (21.10)$$

An action matrix $P = (P_{ij})$ relative to chosen generators $\bar{x}_j$ means

$$P(\bar{x}_j) = \sum_{i=1}^d P_{ij} \bar{x}_i. \quad (21.11)$$

Its entries are interpreted modulo p^{a_i} , and well-definedness requires

$$p^{a_j} P_{ij} \equiv 0 \pmod{p^{a_i}} \quad \text{for every } i, j. \quad (21.12)$$

This condition is necessary because $p^{a_j} \bar{x}_j = 0$, and it is sufficient because it makes the homomorphism from the direct-sum presentation respect every order relation.

The deterministic finite-table algorithm computes $[N, N]$, constructs the quotient table, and applies the verified invariant-factor procedure of Statement 19.8. It chooses the least carrier lift of each invariant-factor generator, evaluates α on those lifts, and solves (21.11) by exhaustive coordinates in the canonical direct sum. Replay checks (21.12), every generator image, bijectivity, and $P^8 = I$. The record is

$$(p, \text{table and action hashes}, [N, N] \text{ rows}, (p^{a_i})_{i=1}^d, \\ (\bar{x}_i), (\text{least lifts}), P, \text{ord}(P), \text{coordinate bijection, hash}). \quad (21.13)$$

For $U_3(\mathbb{F}_p)$, $A \cong \mathbb{F}_p^2$. For a cyclic group C_{p^a} , one has $A = N$, while the trivial group gives the zero-generator boundary.

21.3 The commutator module

Set

$$C = [N, N] \leq Z(N). \quad (21.14)$$

This is a finite abelian p -group. The restriction

$$R = \alpha|_C : C \longrightarrow C \quad (21.15)$$

is well defined because C is characteristic, and $R^8 = 1$. Consequently C is a second $\mathbb{Z}[C_8]$ -module.

If $x_1, \dots, x_d$ lift generators of A , then

$$C = \langle [x_i, x_j] : 1 \leq i < j \leq d \rangle. \quad (21.16)$$

To prove this, write arbitrary elements as

$$x = x_1^{a_1} \cdots x_d^{a_d} c, \quad y = x_1^{b_1} \cdots x_d^{b_d} c', \quad c, c' \in C.$$

The central factors disappear from the commutator, and repeated use of (21.2) expresses $[x, y]$ as a product of powers of the $[x_i, x_j]$. Since all commutators generate C , (21.16) follows.

Write

$$C \cong C_{p^{b_1}} \oplus \cdots \oplus C_{p^{b_s}}, \quad b_1 \leq \cdots \leq b_s. \quad (21.17)$$

The finite algorithm constructs and verifies this decomposition exactly as in Statement 19.8, expresses every $[x_i, x_j]$ in its coordinates, and computes the matrix of R subject to the analogue of (21.12). It also verifies

$$R([x_i, x_j]) = [\alpha(x_i), \alpha(x_j)] \quad (21.18)$$

directly in the table. Its fixed output is

$$(p, C \text{ rows}, (p^{b_k}), (\bar{z}_k), R, \text{ord}(R), \\ ([x_i, x_j] \text{ coordinate vectors}), \text{generation and covariance independent verification, hash}). \quad (21.19)$$

For an abelian N , this returns $C = 0$, the required boundary. For $U_3(\mathbb{F}_p)$, it returns a one-dimensional commutator module.

21.4 The alternating commutator pairing

Define

$$\beta : A \times A \longrightarrow C, \quad \beta(xC, yC) = [x, y]. \quad (21.20)$$

This is well defined. If $c, d \in C$, then c, d are central and (21.2) gives

$$[xc, yd] = [x, y][x, d][c, y][c, d] = [x, y]. \quad (21.21)$$

The first two identities in (21.2) prove additivity in each variable, and

$$\beta(a, a) = 0, \quad \beta(b, a) = -\beta(a, b) \quad (21.22)$$

in additive notation. Hence β is an alternating biadditive map. It induces a homomorphism

$$\tilde{\beta} : \bigwedge_{\mathbb{Z}}^2 A \longrightarrow C, \quad a \wedge b \mapsto \beta(a, b). \quad (21.23)$$

Equation (21.16) proves that this homomorphism is onto.

The radical is exactly

$$\text{rad } \beta = \{a : \beta(a, b) = 0 \ \forall b \in A\} = Z(N)/C. \quad (21.24)$$

Indeed xC is in the radical precisely when $[x, y] = 1$ for every $y \in N$, which is precisely $x \in Z(N)$.

The pairing is C_8 -equivariant:

$$\boxed{\beta(Pa, Pb) = R\beta(a, b)}. \quad (21.25)$$

For lifts x, y , its left side is $[\alpha(x), \alpha(y)] = \alpha([x, y])$, which is the right side.

In invariant-factor coordinates, the tensor entries

$$B_{ij,k} \in \mathbb{Z}/p^{b_k}\mathbb{Z}, \quad \beta(\bar{x}_i, \bar{x}_j) = \sum_k B_{ij,k} \bar{z}_k \quad (21.26)$$

must satisfy alternation, the order constraints

$$p^{a_i} B_{ij,k} \equiv p^{a_j} B_{ij,k} \equiv 0 \pmod{p^{b_k}}, \quad (21.27)$$

surjectivity of (21.23), and the coordinate form of (21.25). These conditions are necessary by bilinearity and equivariance. They are sufficient to define an alternating pairing on the presented direct sums, because (21.27) makes every change by an order relation vanish.

The algorithm evaluates all pairs of least lifts from Statement 21.2, expresses their commutators in the Statement 21.3 basis, and verifies (21.21)–(21.27) exhaustively. The output records both module decompositions, P, R , the tensor B , a row-reduced matrix for $\tilde{\beta}$, its kernel and image, the radical, and a independent verification hash. The zero pairing of an abelian group is the boundary. For $U_3(\mathbb{F}_p)$,

$$\beta((a, b), (a', b')) = ab' - a'b \quad (21.28)$$

is the standard nondegenerate alternating form with one-dimensional target.

21.5 C_8 -stable commutator quotients

Let $W \leq C = [N, N]$. Because $C \leq Z(N)$, the subgroup W is normal in N . The automorphism α descends to N/W if and only if

$$R(W) = W. \quad (21.29)$$

Necessity follows by applying a descended map to the identity coset. For sufficiency, define

$$\alpha_W(xW) = \alpha(x)W. \quad (21.30)$$

If $xW = yW$, then $x^{-1}y \in W$, so $\alpha(x)^{-1}\alpha(y) \in R(W) = W$. Thus (21.30) is well defined. It is an automorphism and $\alpha_W^8 = 1$.

The quotient remains class at most two, and its structural data are

$$\begin{aligned} (N/W)_{\text{ab}} &\cong N/C = A, \\ [N/W, N/W] &= C/W, \\ \beta_W(a, b) &= \beta(a, b) + W. \end{aligned} \tag{21.31}$$

Every quotient commutator is $[xW, yW] = [x, y]W$, so the second line holds. The first then follows from $(N/W)/(C/W) \cong N/C$. The last formula is the quotient of the well-defined pairing in Statement 21.4, and remains alternating and equivariant.

For a constraint set $S \subseteq C$, the least stable subgroup killing those constraints is

$$W_S = \langle R^j s : s \in S, 0 \leq j < 8 \rangle. \tag{21.32}$$

It is stable because $R^8 = 1$. Any stable subgroup containing S contains every displayed orbit, which proves minimality. This is the finite-abelian-group analogue of Statement 20.7 and does not assume exponent p .

The deterministic algorithm scans the serialized seeds and their eight orbits, closes under the abelian group operation, and returns the canonical row set W_S . It constructs the tables of N/W_S and C/W_S , their induced actions, and the tensor obtained by reducing (21.26). Replay checks subgroup closure, every orbit, minimality by generator containment, (21.29)–(21.31), and the exact order of the descended action. The record is

$$\begin{aligned} &(\text{source hashes}, S, \text{orbit provenance}, W_S \text{ rows}, \\ &N/W_S, C/W_S, P, R_W, B_W, \text{ord}(\alpha_W), \text{independent verification hash}). \end{aligned} \tag{21.33}$$

The zero seed gives $W = 0$ and changes nothing. Taking $W = C$ makes the quotient abelian. For the Heisenberg positive test, any nonzero central seed generates its one-dimensional center and produces $\mathbb{F}_p^2$.

21.6 Higher commutator layers

For a finite p -quotient Q , define

$$L_n^p(Q) = D_n(Q)/D_{n+1}(Q), \quad \text{gr}_p(Q) = \bigoplus_{n \geq 1} L_n^p(Q). \tag{21.34}$$

Statement 20.3 proves that every $D_n(Q)$ is characteristic and every layer is a finite-dimensional $\mathbb{F}_p$ -space. Statement 20.5 proves that

$$\begin{aligned} [xD_{m+1}, yD_{n+1}] &= [x, y]D_{m+n+1}, \\ (xD_{n+1})^{[p]} &= x^p D_{pn+1} \end{aligned} \tag{21.35}$$

are well defined. The first operation is bilinear, alternating, and satisfies Jacobi; the second raises degree from n to pn .

For homogeneous $v_i \in L_{n_i}^p(Q)$, the left-normed higher commutator

$$[v_1, v_2, \dots, v_r] = [\dots [[v_1, v_2], v_3], \dots, v_r] \tag{21.36}$$

lies in $L_{n_1 + \dots + n_r}^p(Q)$. This follows by induction from the degree rule in (21.35). Thus if $D_{c+1}(Q) = 1$, every bracket of total degree greater than c vanishes. The converse need not follow from brackets of degree-one elements alone, because the homogeneous $[p]$ -operation can create additional layers.

If $f : Q \rightarrow Q'$ is a homomorphism, Statement 20.3 gives $f(D_n(Q)) \leq D_n(Q')$, so

$$f_n(xD_{n+1}(Q)) = f(x)D_{n+1}(Q') \quad (21.37)$$

is well defined. Direct substitution proves

$$f_{m+n}([u, v]) = [f_m(u), f_n(v)], \quad f_{pn}(u^{[p]}) = f_n(u)^{[p]}. \quad (21.38)$$

No layerwise surjectivity is claimed without a separate proof.

An automorphism α with $\alpha^8 = 1$ gives linear maps A_n on every layer, satisfying

$$A_{m+n}[u, v] = [A_m u, A_n v], \quad A_{pn}(u^{[p]}) = (A_n u)^{[p]}, \quad A_n^8 = I. \quad (21.39)$$

These follow from (21.38) with $f = \alpha$ and from $\alpha(x^p) = \alpha(x)^p$.

For a finite table, compute the augmentation ideal powers and deterministic layer bases as in Statement 20.3. Evaluate all basis brackets and $[p]$ -powers and serialize their coordinate tensors, together with the matrices A_n . Replay checks alternation, bilinearity, Jacobi, the local bracket and $[p]$ -power tables, and (21.39) on bases.

Let $c_Q, c_{Q'}$ be the certified terminal heights stored with the source and target filtrations, and put

$$c_* = \min(c_Q, c_{Q'}).$$

Only identities whose source and target degrees are at most c_* are requested. The optional functoriality input is the closed object

$$\mathcal{F} = (c_*, h_{Q'}, h_{\alpha'}, f_{\text{tab}}, (F_n)_{1 \leq n \leq c_*}), \quad (21.40)$$

where the first two entries hash the target multiplication and action tables, f_{tab} lists every value of $f : Q \rightarrow Q'$, and F_n is the claimed matrix of (21.37) in the stored source and target bases. Replay first verifies $f(1) = 1$ and $f(xy) = f(x)f(y)$ for every table pair. It then verifies $f(D_n(Q)) \subseteq D_n(Q')$ and recomputes every F_n from f_{tab} . The bracket identity in (21.38) is checked on every pair of stored basis vectors with $m + n \leq c_*$; bilinearity then proves it for the whole pair of layers. The $[p]$ -operation is not assumed linear. For every n with $pn \leq c_*$, independent verification therefore enumerates every vector of the finite layer $L_n^p(Q)$, chooses its stored canonical representative $x \in D_n(Q)$, and directly compares the coordinates of $f(x^p)$ and $f(x)^p$ in $L_{pn}^p(Q')$. Independence of the representative follows from the well-definedness proof of (21.35) and (21.37). Its optional certificate is

$$\mathcal{C}_f = (c_*, h_{Q'}, h_{\alpha'}, f_{\text{tab}}, (F_n), w_{\text{hom}}, w_D, w_{[\cdot, \cdot]}, w_{[p]}, \text{SATISFIED}), \quad (21.41)$$

with the four witnesses respectively recording the exhaustive homomorphism and filtration tables, all basis-pair bracket comparisons, and all-vector power comparisons. If $\mathcal{F}$ is absent, the literal tag **NONE** replaces $\mathcal{C}_f$; no functoriality independent verification is then claimed. Thus every performed optional check is recoverable from the output rather than from unstored run-time data.

A requested finite height terminates directly; complete mode stops at the first stable ideal power. The fixed output is

$$\begin{aligned} & (p, \text{source table and action hashes}, (D_n), (L_n^p \text{ bases}), \\ & (\text{bracket tensors}), ([p]\text{-coordinates}), (A_n), (\text{ord } A_n), \text{terminal witness}, \mathcal{C}_f \text{ or NONE}, \text{hash}). \end{aligned} \quad (21.42)$$

An abelian group of exponent p has only degree one and zero bracket, the boundary test. The group $U_3(\mathbb{F}_p)$ has the nonzero degree-(1, 1) bracket from (20.54), providing the first higher-layer positive test.

21.7 A Lyndon–Shirshov Hall basis

Fix the alphabet $X_1 < \cdots < X_d$ and dictionary lexicographic order on nonempty words: a proper prefix is smaller, and otherwise the first unequal letter decides. A word ℓ is *Lyndon* when

$$\ell < s \quad \text{for every nonempty proper suffix } s \text{ of } \ell. \quad (21.43)$$

The word arguments and the Lie collection argument are separated below; in particular, no induction attempts to factor a one-letter right word.

Lemma 21.7.1 (Lexicographic comparisons). *For words $a, b, c \neq 1$ and arbitrary words x, y :*

- (i) *if $a < b$ and a is not a prefix of b , then $ax < by$;*
- (ii) *if $a < b < ac$, then $b = ab'$ for a nonempty word $b' < c$.*

Proof. In (i), a and b differ before either word ends. Appending letters does not alter that first unequal position, so the same smaller letter occurs in ax . For (ii), if a were not a prefix of b , part (i), with $x = c$ and $y = 1$, would give $ac < b$, contrary to the hypothesis. Thus $b = ab'$, with $b' \neq 1$ because $a < b$. Removing the common prefix a from $b < ac$ gives $b' < c$. $\square$

Lemma 21.7.2 (Lyndon concatenation and standard factorization). (i) *If u, v are Lyndon and $u < v$, then uv is Lyndon.*

- (ii) *If ℓ is Lyndon of length greater than one and v is its lexicographically least nonempty proper suffix, then $\ell = uv$ with u, v Lyndon and $u < v$. The word v is also the longest proper Lyndon suffix; this is the standard factorization.*

- (iii) *If u, v are Lyndon with $u < v$, then uv has standard factorization $u \mid v$ exactly when u is a letter or, writing the standard factorization $u = u_1u_2$, one has $u_2 \geq v$.*

Proof. For (i), let s be a nonempty proper suffix of uv . If $|s| > |v|$, write $s = u'v$ with u' a nonempty proper suffix of u . Then $u < u'$; because u cannot be a prefix of the shorter word u' , the comparison lemma gives $uv < u'v = s$. If $s = v$, there are two cases. When u is not a prefix of v , the same comparison lemma gives $uv < v$. When $v = uv'$, the Lyndon inequality $v < v'$, after the common initial u is restored, gives $uv = u(uv') < uv' = v$. Finally, if $|s| < |v|$, then s is a proper suffix of v , whence $v < s$; the preceding case gives $uv < v < s$. Thus (21.43) holds for every s .

For (ii), minimality makes v Lyndon: otherwise a proper suffix s of v would satisfy $s < v$, and s would be a proper suffix of ℓ , contradicting the choice of v . Since u is a proper prefix of $\ell = uv$, one has $u < \ell < v$, hence $u < v$. Let y be a nonempty proper suffix of u . The word yv is a proper suffix of ℓ , so $v \leq yv$. If $y < v$, then $y < v < yv$, and the comparison lemma would write $v = yv'$ with $v' < v$. This v' is another proper suffix of ℓ , again a contradiction. Hence $u < v \leq y$ for every such y , proving that u is Lyndon. If a longer proper suffix t were Lyndon, then v would be a proper suffix of t , so $t < v$, contradicting the minimality $v \leq t$. Thus v is the longest Lyndon suffix.

For (iii), suppose first that $u = u_1u_2$ is standard and $u_2 < v$. Then u_2v is Lyndon by (i), is a proper suffix of uv , and is lexicographically smaller than v ; hence v cannot be the least proper suffix. Conversely assume u is a letter, or its standard right factor satisfies $u_2 \geq v$. A proper suffix contained in v is at least v . Any longer proper suffix has the form tv , where t is a nonempty proper suffix of u . Part (ii), applied inside u , says u_2 is the least such suffix, so $t \geq u_2 \geq v$, and therefore $tv > v$. If u is a letter there is no longer proper suffix. Thus v is precisely the least proper suffix of uv . $\square$

Theorem 21.7.3 (Chen–Fox–Lyndon factorization). *Every nonempty word has a unique factorization*

$$w = \ell_1 \ell_2 \cdots \ell_r, \quad \ell_1 \geq \ell_2 \geq \cdots \geq \ell_r. \quad (21.44)$$

into Lyndon words.

Proof. Existence starts with the factorization of w into letters. Among all factorizations into Lyndon words choose one with the fewest factors. If adjacent factors had $\ell_i < \ell_{i+1}$, the concatenation lemma would replace them by the single Lyndon word $\ell_i \ell_{i+1}$, contradicting minimality. Hence the chosen factorization is nonincreasing.

For uniqueness, first suppose $r > 1$. The last factor is then the least nonempty proper suffix of w . A proper suffix starting at a factor boundary begins with some $\ell_i \geq \ell_r$; if equality holds, ℓ_r is a proper prefix of that suffix, and if the inequality is strict the first unequal letter or prefix rule again puts ℓ_r first. A suffix beginning strictly inside ℓ_i begins with a proper suffix t of ℓ_i . Since ℓ_i is Lyndon, $\ell_i < t$, and therefore $\ell_r \leq \ell_i < t$; the same lexicographic comparison gives ℓ_r smaller than that suffix. Thus the last factor is determined by w . If $r = 1$, then w itself is Lyndon and no factorization with two or more nonincreasing factors is possible: its last factor would be a proper suffix ℓ_r , whereas the comparison just given would yield $\ell_r < w$, contrary to the Lyndon inequality $w < \ell_r$. Consequently two nonincreasing factorizations either both have one factor, or both have the same last factor. In the latter case, cancelling that factor on the right and inducting on $|w|$ proves uniqueness. $\square$

For a Lyndon word ℓ , define its standard tree and polynomial recursively by

$$P_{X_i} = X_i, \quad P_\ell = [P_u, P_v] \quad \text{when } \ell = uv \text{ is standard.} \quad (21.45)$$

The preceding lemma verifies the Hall condition exactly: for Lyndon $u < v$, the node (u, v) is standard if and only if u is a letter or the standard right child of u is at least v .

Lemma 21.7.4 (Integral Hall collection). *For Lyndon words h, k , the commutator $[P_h, P_k]$ is an integral linear combination of polynomials P_ℓ of degree $|h| + |k|$. More precisely, every output ℓ of degree at least two has standard right factor at most $\max(h, k)$.*

Proof. Order the induction pairs

$$(|h| + |k|, \max(h, k)) \quad (21.46)$$

lexicographically. At fixed total degree only finitely many Lyndon words occur, so this is a well-founded induction. Antisymmetry reduces to $h < k$, while $h = k$ gives zero. If h is a letter, or if $h = h'h''$ is standard with $h'' \geq k$, the Hall condition gives $[P_h, P_k] = P_{hk}$; its standard right factor is k .

It remains to treat the nonstandard case. Here it is the *left* word which is decomposed:

$$h = h'h'', \quad h' < h'' < k.$$

The second inequality is exactly the failure of the Hall condition. Jacobi, with signs displayed, gives

$$[P_h, P_k] = [[P_{h'}, P_k], P_{h''}] + [P_{h'}, [P_{h''}, P_k]]. \quad (21.47)$$

Both inner commutators have smaller total degree. By induction write them as $\sum_i a_i P_{u_i}$ and $\sum_j b_j P_{v_j}$, with the standard right factors of u_i, v_j at most k . These output words have degree at least two. Each is Lyndon and hence is strictly smaller than its proper standard right suffix. Therefore

$$u_i < (u_i)'' \leq k, \quad v_j < (v_j)'' \leq k,$$

so in particular $u_i < k$ and $v_j < k$. In the two families of outer brackets in (21.47), respectively

$$\max(u_i, h'') < k, \quad \max(h', v_j) < k.$$

They have the original total degree but strictly smaller second coordinate in (21.46), so the induction applies once more. For either outer bracket the inductive conclusion bounds every final standard right factor by its new maximum, which is $< k = \max(h, k)$. This is the promised stronger range statement and proves both closure and termination, including the case k is a single letter. $\square$

Theorem 21.7.5 (Lyndon–Shirshov Hall and PBW bases). *The P_ℓ with $|\ell| = n$ form a $\mathbb{Z}$ -basis of the degree- n free Lie ring on $X_1, \dots, X_d$. Their reductions modulo p form an $\mathbb{F}_p$ -basis of the free Lie algebra over $\mathbb{F}_p$. Moreover, the products*

$$P_{\ell_1} P_{\ell_2} \cdots P_{\ell_r}, \quad \ell_1 \geq \ell_2 \geq \cdots \geq \ell_r, \quad (21.48)$$

together with 1, form a $\mathbb{Z}$ -basis of $\mathbb{Z}\langle X_1, \dots, X_d \rangle$.

Proof. Start with a word $X_{i_1} \cdots X_{i_n}$, viewed as a product of one-letter Hall polynomials. Whenever adjacent factors form a rise $h < k$, use

$$P_h P_k = P_k P_h + [P_h, P_k]$$

and expand the commutator by the collection lemma. The swapped term has one fewer inversion, while every commutator term has one fewer factor. Induction first on the number of factors and then on the inversion number terminates and expresses every word integrally as a sum of products (21.48). In degree n , the factorization theorem puts these products in bijection with the d^n associative words. They are therefore a spanning family of d^n elements in the free abelian group of rank d^n , and hence a basis.

The individual P_ℓ are a subset of that associative basis, so they are integrally independent. Their integral span is closed under brackets by the collection lemma and contains every generator X_i . It is therefore the Lie subring generated by the alphabet, namely the free Lie ring in its canonical associative realization. This proves the integral basis assertion. Tensoring this direct-summand basis with $\mathbb{F}_p$ preserves linear independence and spanning, proving the modular assertion. $\square$

The cited Hall-basis results supply the stated existence and uniqueness properties. The integral Hall collection and PBW conclusions above are the specialization of Reutenauer, *Free Lie Algebras*, Theorem 4.9, to the commutative coefficient ring $\mathbb{Z}$; the identification of lexicographic Lyndon trees as the required Hall set is Theorem 5.1 there. The hypotheses used by those results have been verified above: the alphabet is totally ordered and finite, the coefficient ring is commutative with identity, the standard factor exists, and the Hall node condition is precisely part (iii) of the factorization lemma. Thus no field or characteristic-zero hypothesis is silently imposed.

Let $h_n(d)$ be the number of length- n Lyndon words. Counting arbitrary multiplicities in the unique ordered basis (21.48), then taking formal logarithms and comparing coefficients, gives

$$\begin{aligned} \frac{1}{1-dt} &= \prod_{n \geq 1} (1-t^n)^{-h_n(d)}, \\ d^m &= \sum_{n|m} n h_n(d), \\ h_n(d) &= \frac{1}{n} \sum_{r|n} \mu(r) d^{n/r}. \end{aligned} \quad (21.49)$$

The last line is Möbius inversion. For $d = 2$, the first brackets are

$$X_1, X_2; \quad [X_1, X_2]; \quad [X_1, [X_1, X_2]], [[X_1, X_2], X_2],$$

and (21.49) gives 2, 1, 2. For $d = 1$, the formula gives one basis word in degree one and none in larger degree.

The deterministic algorithm enumerates Lyndon words in lexicographic order, finds the least proper suffix (equivalently the longest proper Lyndon suffix), and serializes the trees (21.45). To collect a nonstandard bracket, it applies the left-factor recursion (21.47), using the induction key (21.46); the proof establishes termination and integrality. It checks (21.49) at every requested weight. In a finite graded Lie algebra, it evaluates these trees using Statement 21.6 and row-reduces the coordinates. The retained vectors span the ordinary Lie subalgebra generated in degree one; homogeneous $[p]$ -powers are stored separately. The fixed output is

$$(d, p, \text{terminal weight, Lyndon words, factors, and trees, } (h_n(d)), \\ \text{collection traces, evaluated coordinates, pivot basis, relation rows, } [p]\text{-rows, independent verification hash}). \quad (21.50)$$

21.8 Finite presentations of nilpotent extensions

The commutator pairing alone does not determine a two-step group. The missing power and section data are encoded by a normalized two-cocycle. Let A, C be finite abelian groups, written additively, and let

$$f : A \times A \longrightarrow C \quad (21.51)$$

satisfy

$$f(0, a) = f(a, 0) = 0, \\ f(a, b) + f(a + b, c) = f(b, c) + f(a, b + c). \quad (21.52)$$

On $E = A \times C$, define

$$(a, u)(b, v) = (a + b, u + v + f(a, b)). \quad (21.53)$$

The two products of three elements have central coordinates

$$((a, u)(b, v))(c, w) = u + v + w + f(a, b) + f(a + b, c), \\ (a, u)((b, v)(c, w)) = u + v + w + f(b, c) + f(a, b + c),$$

so (21.52) is exactly associativity. Normalization makes $(0, 0)$ the identity. The cocycle equation at $(a, -a, a)$ gives $f(a, -a) = f(-a, a)$, and hence

$$(a, u)^{-1} = (-a, -u - f(a, -a)) \quad (21.54)$$

is both a left and right inverse. Thus (21.53) is a group. The subgroup $0 \times C$ is central, the quotient is A , and

$$[(a, u), (b, v)] = (0, f(a, b) - f(b, a)). \quad (21.55)$$

Indeed $xy = [x, y]yx$, and the two products xy, yx have identical A -coordinates and central-coordinate difference displayed in (21.55). The extension has class exactly two precisely when the alternating part of f is nonzero.

Suppose

$$A = \bigoplus_{i=1}^d C_{m_i}, \quad C = \bigoplus_{j=1}^s C_{n_j}. \quad (21.56)$$

Let $x_i = (e_i, 0)$ and $z_j = (0, \epsilon_j)$. They give a finite presentation with the central relations

$$z_j^{n_j} = 1, \quad [z_j, z_k] = [x_i, z_j] = 1, \quad (21.57)$$

the commutator relations obtained from (21.55), and the power relations

$$x_i^{m_i} = \left(0, \sum_{r=0}^{m_i-1} f(re_i, e_i) \right), \quad (21.58)$$

expressed in the z_j . The sum in (21.58) follows by induction from (21.53).

These relations present E , not merely a quotient of it. Using (21.57), move every z_j to the right. Using $yx = [x, y]^{-1}xy$, order the x_i , and use (21.58) to reduce their exponents. Every word reaches one of at most

$$\prod_i m_i \prod_j n_j = |A||C| \quad (21.59)$$

normal forms

$$x_1^{r_1} \cdots x_d^{r_d} z_1^{s_1} \cdots z_s^{s_s}, \quad 0 \leq r_i < m_i, \quad 0 \leq s_j < n_j. \quad (21.60)$$

Mapping the presented generators to (21.53) is onto E , which has exactly the number of elements in (21.59). A surjection from a group having at most $|A||C|$ elements onto a group having exactly that many is bijective. Hence the normal forms are unique and the presentation is exact.

Now let $P \in \text{Aut}(A)$, $R \in \text{Aut}(C)$, and $t : A \rightarrow C$ with $t(0) = 0$. The formula

$$\alpha(a, u) = (Pa, Ru + t(a)) \quad (21.61)$$

is a homomorphism exactly when

$$Rf(a, b) + t(a + b) = t(a) + t(b) + f(Pa, Pb) \quad (21.62)$$

for all a, b . This follows by evaluating (21.61) on the two sides of (21.53) and comparing central coordinates. If P, R are bijective, then so is α , with inverse obtained by first solving $a = P^{-1}a'$ and then $u = R^{-1}(u' - t(a))$.

Assume $P^8 = R^8 = 1$. Direct induction gives

$$\alpha^k(a, u) = \left(P^k a, R^k u + \sum_{j=0}^{k-1} R^{k-1-j} t(P^j a) \right). \quad (21.63)$$

Therefore $\alpha^8 = 1$ exactly when

$$\sum_{j=0}^7 R^{7-j} t(P^j a) = 0 \quad (a \in A). \quad (21.64)$$

Equations (21.62) and (21.64) are the complete covariance and order conditions; checking only the alternating part (21.55) would miss section and power obstructions.

The finite construction enumerates the tables of A, C, f , verifies (21.52) on every triple, builds (21.53), and exhaustively checks the group table and presentation normal forms. If symmetry data are present, it verifies (21.62)–(21.64) on all elements and computes the actual order of α . Its fixed record is

(A, C , invariant factors, f table, P, R, t ,
presentation relators, normal-form table, cocycle/covariance/order independent verification, hash).
(21.65)

Higher nilpotent quotients are assembled by the same finite operation. At stage r , let H be the already constructed quotient and let C_r be the next central layer. Replace addition $a + b$ in (21.52)–(21.53) by multiplication hk in H :

$$\begin{aligned} f(h, k) + f(hk, \ell) &= f(k, \ell) + f(h, k\ell), \\ (h, u)(k, v) &= (hk, u + v + f(h, k)). \end{aligned} \quad (21.66)$$

The same three-element calculation proves associativity, and $1 \times C_r$ is central. Iterating certified finite tables and recording a cocycle at every layer produces a finite presentation of the full nilpotent extension. A zero layer gives the split boundary $H \times 0$; the Heisenberg cocycle $f((a, b), (a', b')) = ab'$ gives (21.28) as its nontrivial positive test.

Part VII

Group cohomology and central extensions

22 Foundations of group extensions

22.1 Short exact sequences

Let A, E, G be groups. An extension of G by A is a diagram

$$1 \longrightarrow A \xrightarrow{i} E \xrightarrow{\pi} G \longrightarrow 1 \quad (22.1)$$

which is exact at every term. Written without shorthand, the hypotheses are

$$i \text{ is a homomorphism, } \pi \text{ is a homomorphism, } \ker i = 1, \quad \text{im } i = \ker \pi, \quad \text{im } \pi = G. \quad (22.2)$$

Thus exactness contains injectivity, kernel equality, and surjectivity as three separate obligations.

Proposition 22.1.1 (Kernel, quotient, and reconstruction). *Conditions (22.2) hold if and only if i identifies A with a normal subgroup $K \trianglelefteq E$, π is onto, and the rule*

$$\bar{\pi} : E/K \longrightarrow G, \quad eK \longmapsto \pi(e) \quad (22.3)$$

is an isomorphism. Under these conditions the extension determines, and is determined by, the pair consisting of $K = i(A) \trianglelefteq E$ and the chosen identifications $A \cong K$, $E/K \cong G$.

Proof. Assume (22.2). The image $K = i(A) = \ker \pi$ is normal because kernels are normal:

$$\pi(eke^{-1}) = \pi(e)1\pi(e)^{-1} = 1 \quad (e \in E, k \in K).$$

If $eK = e'K$, then $e^{-1}e' \in K = \ker \pi$, hence $\pi(e) = \pi(e')$; this proves that (22.3) is well defined. It is a homomorphism because

$$\bar{\pi}((eK)(e'K)) = \pi(ee') = \pi(e)\pi(e') = \bar{\pi}(eK)\bar{\pi}(e'K).$$

Surjectivity follows from that of π . If $\bar{\pi}(eK) = 1$, then $e \in \ker \pi = K$, so $eK = K$; therefore $\bar{\pi}$ is injective.

Conversely, suppose $i : A \rightarrow K$ and $\bar{\pi} : E/K \rightarrow G$ are isomorphisms and let π be the composite $E \rightarrow E/K \xrightarrow{\bar{\pi}} G$. Then i is injective, π is onto, and

$$\ker \pi = \ker(E \rightarrow E/K) = K = i(A).$$

This is precisely (22.2). The two constructions retain the maps i, π , so they are inverse as constructions with chosen identifications. $\square$

A *normalized set-theoretic section* is a map

$$s : G \longrightarrow E, \quad \pi s = \text{id}_G, \quad s(1) = 1. \quad (22.4)$$

When π is onto, such a section exists for finite G : choose one element in each fiber and choose 1 in the identity fiber. It need not be a homomorphism. Every $e \in E$ nevertheless has a unique normal form

$$e = i(a)s(g), \quad g = \pi(e), \quad (22.5)$$

because $es(g)^{-1} \in \ker \pi = i(A)$, and injectivity of i makes a unique.

Theorem 22.1.2 (Three equivalent splitting criteria). *For an extension (22.1), the following are equivalent:*

- (i) *there is a homomorphic section $s : G \rightarrow E$;*
- (ii) *there is a subgroup $H \leq E$ such that $E = i(A)H$ and $i(A) \cap H = 1$;*
- (iii) *there is a subgroup $H \leq E$ for which $\pi|_H : H \rightarrow G$ is an isomorphism.*

When these conditions hold, E is an internal semidirect product $i(A) \rtimes H$. No direct-product conclusion follows unless H centralizes $i(A)$.

Proof. If (i) holds, put $H = s(G)$. For $e \in E$, take $g = \pi(e)$. Then $es(g)^{-1} \in i(A)$, proving $E = i(A)H$. If $h = s(g) \in i(A) \cap H$, then $g = \pi(s(g)) = 1$, so $h = s(1) = 1$. Thus (ii) holds.

Under (ii), $\pi(H) = G$: write a lift e of g as kh with $k \in i(A)$, $h \in H$, and obtain $\pi(h) = g$. Moreover $\ker(\pi|_H) = H \cap i(A) = 1$, proving (iii). Under (iii), define $s = (\pi|_H)^{-1}$. It is a homomorphism and $\pi s = \text{id}_G$, proving (i). Normality of $i(A)$, the product equality, and trivial intersection are exactly the internal semidirect-product conditions. A direct product additionally requires the conjugation action of H on $i(A)$ to be trivial. $\square$

For finite multiplication tables, the verification is exhaustive and terminating. The input consists of canonical row orders for A, E, G and arrays for i, π . Replay performs:

$$\begin{aligned}
 & \text{(a) } i(aa') = i(a)i(a') \quad \text{for every } (a, a') \in A^2, \\
 & \text{(b) } \pi(ee') = \pi(e)\pi(e') \quad \text{for every } (e, e') \in E^2, \\
 & \text{(c) } i(a) = i(a') \implies a = a', \\
 & \text{(d) } \{\pi(e) : e \in E\} = G, \\
 & \text{(e) } \{i(a) : a \in A\} = \{e \in E : \pi(e) = 1\}.
 \end{aligned} \tag{22.6}$$

When a proposed section is supplied, independent verification checks every equality in (22.4); a separate flag records whether it also preserves every product. The fixed certificate is

$$\begin{aligned}
 & (\mathbf{h}_A, \mathbf{h}_E, \mathbf{h}_G, i_{\text{tab}}, \pi_{\text{tab}}, \text{im } i, \ker \pi, \text{im } \pi, \\
 & \text{whom}, \mathbf{w}_{\text{inj}}, \mathbf{w}_{\text{surj}}, \mathbf{w}_{\text{exact}}, s_{\text{tab}} \text{ or NONE, split_flag, h}_{\text{independentverification}}).
 \end{aligned} \tag{22.7}$$

The equality of finite row sets in (e), rather than equality of their cardinalities alone, certifies exactness.

As a positive nonsplit test, take

$$1 \longrightarrow C_2 \xrightarrow{i} C_4 \xrightarrow{\pi} C_2 \longrightarrow 1, \quad i(\bar{1}) = \bar{2}, \quad \pi(\bar{x}) = \bar{x} \pmod{2}. \tag{22.8}$$

It is exact, but the only elements above the nonidentity of C_2 have order four, so no homomorphic section exists. The boundary $1 \rightarrow 1 \rightarrow G \xrightarrow{\text{id}} G \rightarrow 1$ is split. A negative test which sends the nonidentity of C_2 trivially into C_4 fails injectivity in (c), and the certificate identifies that exact row.

22.2 Central extensions and cocycle construction

An extension (22.1) is *central* when

$$i(A) \leq Z(E). \tag{22.9}$$

Injectivity then identifies A with a subgroup of an abelian group, so A itself must be abelian. We therefore write its operation additively. Fix a normalized section $s : G \rightarrow E$ as in (22.4). Because

$$\pi(s(g)s(h)s(gh)^{-1}) = gh(gh)^{-1} = 1,$$

exactness and injectivity give a unique element $f(g, h) \in A$ such that

$$s(g)s(h) = i(f(g, h))s(gh). \tag{22.10}$$

This proves existence and uniqueness of the factor set; it is not an extra choice once s is fixed.

Theorem 22.2.1 (Central cocycle construction, both directions). *For a central extension with normalized section, the factor set $f : G^2 \rightarrow A$ satisfies*

$$\begin{aligned} f(1, g) &= f(g, 1) = 0, \\ f(g, h) + f(gh, k) &= f(h, k) + f(g, hk) \quad (g, h, k \in G). \end{aligned} \quad (22.11)$$

Conversely, if A is abelian, G is a group, and f satisfies (22.11), then

$$(a, g)(b, h) = (a + b + f(g, h), gh) \quad (22.12)$$

defines a group $E_f = A \times G$, and

$$1 \longrightarrow A \xrightarrow{i_f} E_f \xrightarrow{\pi_f} G \longrightarrow 1, \quad i_f(a) = (a, 1), \quad \pi_f(a, g) = g \quad (22.13)$$

is a central extension. Its canonical section $s_f(g) = (0, g)$ has factor set f . The original extension is isomorphic to the extension reconstructed from its factor set through

$$\Theta_f : E_f \longrightarrow E, \quad \Theta_f(a, g) = i(a)s(g). \quad (22.14)$$

Proof. Normalization follows from $s(1) = 1$ in (22.10). Associate a product of three section values in the two possible ways. Centrality of $i(A)$ gives

$$\begin{aligned} (s(g)s(h))s(k) &= i(f(g, h) + f(gh, k))s(ghk), \\ s(g)(s(h)s(k)) &= i(f(h, k) + f(g, hk))s(ghk). \end{aligned}$$

Associativity in E and injectivity of i imply the second line of (22.11).

For the converse, the two associations in E_f are

$$\begin{aligned} ((a, g)(b, h))(c, k) &= (a + b + c + f(g, h) + f(gh, k), ghk), \\ (a, g)((b, h)(c, k)) &= (a + b + c + f(h, k) + f(g, hk), ghk). \end{aligned} \quad (22.15)$$

They agree exactly by (22.11), proving associativity. Normalization makes $(0, 1)$ a two-sided identity. Substitution (g^{-1}, g, g^{-1}) in (22.11) shows

$$f(g^{-1}, g) = f(g, g^{-1}),$$

and direct multiplication gives the two-sided inverse

$$(a, g)^{-1} = (-a - f(g^{-1}, g), g^{-1}). \quad (22.16)$$

Thus E_f is a group.

The maps in (22.13) are homomorphisms:

$$\begin{aligned} i_f(a)i_f(b) &= (a + b + f(1, 1), 1) = i_f(a + b), \\ \pi_f((a, g)(b, h)) &= gh = \pi_f(a, g)\pi_f(b, h). \end{aligned}$$

The first is injective, the second is surjective, and

$$\ker \pi_f = A \times \{1\} = \operatorname{im} i_f.$$

Moreover, for all $(b, g) \in E_f$,

$$(a, 1)(b, g) = (a + b, g) = (b + a, g) = (b, g)(a, 1), \quad (22.17)$$

so the extension is central. Finally

$$s_f(g)s_f(h) = (f(g, h), gh) = i_f(f(g, h))s_f(gh),$$

which recovers f .

Every element of the original E has a unique form $i(a)s(g)$ by (22.5), so Θ_f is bijective. Using centrality and (22.10),

$$\begin{aligned}\Theta_f(a, g)\Theta_f(b, h) &= i(a)s(g)i(b)s(h) \\ &= i(a + b + f(g, h))s(gh) = \Theta_f((a, g)(b, h)).\end{aligned}\tag{22.18}$$

Hence Θ_f is an isomorphism commuting with both injection and projection. This proves necessity, sufficiency, and reconstruction separately. $\square$

The finite construction has input

$$\mathcal{I}_{\text{cen}} = (A_{\text{tab}}, G_{\text{tab}}, f_{\text{tab}}),\tag{22.19}$$

with canonical row orders. It executes the following deterministic procedure.

1. Verify that A_{tab} is an abelian group table and that G_{tab} is a group table.
2. Check both normalization rows and all $|G|^3$ instances of the cocycle equality (22.11). On failure, return the lexicographically first triple (g, h, k) and the two unequal values.
3. Order $A \times G$ lexicographically by the stored row indices and fill every product using (22.12).
4. Fill inverses with (22.16), construct i_f, π_f, s_f , and independently verify the five exactness tests (22.6).
5. Verify (22.17) for every a, b, g and verify that the factor set recovered from s_f equals the input table entry by entry.

Every loop is finite. Steps 1–2 prove that the construction is defined; the theorem proves that steps 3–5 must succeed. Conversely, if the output table is associative with the stated canonical section, the comparison (22.15) forces every instance of (22.11), so the cocycle test is also necessary.

The fixed output is

$$\begin{aligned}(\mathbf{h}_A, \mathbf{h}_G, f_{\text{tab}}, \text{normalization_rows}, \text{cocycle_rows}, \\ E_{f_{\text{tab}}}, i_{f_{\text{tab}}}, \pi_{f_{\text{tab}}}, s_{f_{\text{tab}}}, \text{exactness_witness}, \text{centrality_witness}, \mathbf{h}_{\text{independentverification}}).\end{aligned}\tag{22.20}$$

For $A = G = C_2$, let $f(\bar{1}, \bar{1}) = \bar{1}$ and let all normalized entries be zero. Formula (22.12) gives $(0, \bar{1})^2 = (\bar{1}, \bar{0})$, so $E_f \cong C_4$; this recovers (22.8). The zero cocycle instead gives $C_2 \times C_2$, the split boundary. As a negative test, for $G = C_3 = \langle x \rangle$ and $A = C_2$, set $f(x, x) = 1$ and every other normalized entry to zero. At $(g, h, k) = (x, x, x^2)$, the two sides of (22.11) are respectively 1 and 0, so step 2 rejects the table with an explicit witness. If $A = 1$ or $G = 1$, (22.12) reduces respectively to G or A , covering both degenerate boundaries.

22.3 Semidirect products and split reconstruction

Let A, G be groups, now written multiplicatively, and let

$$\theta : G \longrightarrow \text{Aut}(A), \quad g \longmapsto \theta_g\tag{22.21}$$

be a left action. With the global functional-composition convention, the action law is

$$\theta_1 = \text{id}_A, \quad \theta_{gh} = \theta_g \theta_h, \quad (\theta_g \theta_h)(a) = \theta_g(\theta_h(a)).\tag{22.22}$$

Define on $A \times G$

$$(a, g)(b, h) = (a\theta_g(b), gh).\tag{22.23}$$

The position of θ_g fixes the left-action convention; replacing it silently by θ_h would define a different formula and generally destroy associativity.

Theorem 22.3.1 (Semidirect product and split extensions). *Formula (22.23) defines a group $A \rtimes_\theta G$. The maps*

$$i_\theta(a) = (a, 1), \quad \pi_\theta(a, g) = g, \quad s_\theta(g) = (1, g) \quad (22.24)$$

form a split short exact sequence, and

$$s_\theta(g)i_\theta(a)s_\theta(g)^{-1} = i_\theta(\theta_g(a)). \quad (22.25)$$

Conversely, every split extension (22.1) with a fixed homomorphic section s determines the unique action

$$\theta_g(a) = i^{-1}(s(g)i(a)s(g)^{-1}), \quad (22.26)$$

and the map

$$\Phi : A \rtimes_\theta G \longrightarrow E, \quad \Phi(a, g) = i(a)s(g) \quad (22.27)$$

is an extension isomorphism preserving the chosen section.

Proof. The two associations of three pairs are

$$\begin{aligned} ((a, g)(b, h))(c, k) &= (a\theta_g(b)\theta_{gh}(c), ghk), \\ (a, g)((b, h)(c, k)) &= (a\theta_g(b\theta_h(c)), ghk) \\ &= (a\theta_g(b)(\theta_g\theta_h)(c), ghk). \end{aligned} \quad (22.28)$$

They agree by (22.22), so multiplication is associative. Equation (22.22) makes $(1, 1)$ a two-sided identity. The inverse is

$$(a, g)^{-1} = (\theta_{g^{-1}}(a^{-1}), g^{-1}), \quad (22.29)$$

because the product in the displayed order has first coordinate

$$a\theta_g\theta_{g^{-1}}(a^{-1}) = aa^{-1} = 1,$$

while the reverse product has first coordinate

$$\theta_{g^{-1}}(a^{-1})\theta_{g^{-1}}(a) = \theta_{g^{-1}}(a^{-1}a) = 1.$$

Thus (22.23) is a group law.

Both maps i_θ, π_θ are homomorphisms, i_θ is injective, π_θ is onto, and

$$\ker \pi_\theta = A \times \{1\} = \operatorname{im} i_\theta.$$

The map s_θ is a homomorphism with $\pi_\theta s_\theta = \operatorname{id}_G$, proving split exactness. Substitution of (22.29) in the left side of (22.25) yields

$$(1, g)(a, 1)(1, g^{-1}) = (\theta_g(a), 1),$$

which proves the action formula.

Now start with a split extension and fixed homomorphic section s . Normality of $i(A) = \ker \pi$ shows that the conjugate in (22.26) lies in $i(A)$, and injectivity of i makes its inverse image unique; hence the formula is well defined. Conjugation is an automorphism of $i(A)$, so every θ_g is an automorphism of A . Since $s(gh) = s(g)s(h)$,

$$\begin{aligned} i(\theta_{gh}(a)) &= s(g)s(h)i(a)s(h)^{-1}s(g)^{-1} \\ &= i(\theta_g(\theta_h(a))). \end{aligned}$$

Injectivity of i proves $\theta_{gh} = \theta_g\theta_h$, and $\theta_1 = \operatorname{id}$ follows from $s(1) = 1$.

The normal form (22.5) makes Φ bijective. From (22.26),

$$s(g)i(b) = i(\theta_g(b))s(g),$$

and therefore

$$\begin{aligned}\Phi(a, g)\Phi(b, h) &= i(a)s(g)i(b)s(h) \\ &= i(a\theta_g(b))s(gh) = \Phi(a\theta_g(b), gh).\end{aligned}\tag{22.30}$$

Thus Φ is a homomorphism and hence an isomorphism. Formula (22.25) shows that any action compatible with the fixed i and s must equal (22.26), proving uniqueness. $\square$

Corollary 22.3.2 (Direct-product boundary). *The split extension is an internal direct product $E = i(A) \times s(G)$ if and only if $\theta_g = \text{id}_A$ for every $g \in G$.*

Proof. The two factors form an internal direct product exactly when $[s(g), i(a)] = 1$ for all a, g . By (22.26), this is equivalent to $i(\theta_g(a)) = i(a)$. Injectivity of i makes that condition equivalent to $\theta_g(a) = a$ for every a, g . $\square$

For finite tables the input is

$$\mathcal{I}_{\rtimes} = (A_{\text{tab}}, G_{\text{tab}}, (\theta_{g\text{tab}})_{g \in G}).\tag{22.31}$$

The deterministic verifier first checks the two group tables. For each g , it checks

$$\theta_g(aa') = \theta_g(a)\theta_g(a') \quad (a, a' \in A)$$

and verifies that the image row set is all of A . It then checks both equalities in (22.22) on every $(g, h, a) \in G^2 \times A$. Only after these obligations pass does it fill (22.23), compute (22.29), and independent verification exactness, the section, and (22.25). The first failed equality, if any, is retained as a rejection witness.

The canonical output is

$$\begin{aligned}(\mathbf{h}_A, \mathbf{h}_G, (\theta_{g\text{tab}}), \text{automorphism_witnesses}, \text{action_witness}, \\ (A \rtimes_{\theta} G)_{\text{tab}}, i_{\theta\text{tab}}, \pi_{\theta\text{tab}}, s_{\theta\text{tab}}, \text{conjugation_rows}, \mathbf{h}_{\text{independentverification}}).\end{aligned}\tag{22.32}$$

All ordering is inherited from the pair of canonical row indices, so identical inputs serialize identically.

For the positive nontrivial-action test take

$$A = C_3 = \langle r \rangle, \quad G = C_2 = \langle t \rangle, \quad \theta_t(r) = r^{-1}.\tag{22.33}$$

The presentation obtained from (22.23) is

$$\langle r, t \mid r^3 = t^2 = 1, trt^{-1} = r^{-1} \rangle \cong S_3.$$

The action is nontrivial, so the corollary excludes a direct product. The trivial action gives $A \times G$. If $A = 1$ or $G = 1$, the construction reduces to the other factor. As a negative action table, let $G = C_3 = \langle x \rangle$, $A = C_3$, set θ_x to inversion and θ_{x^2} to the identity. Then

$$\theta_{x^2}\theta_x = \theta_x \neq \theta_1 = \theta_{x^3},$$

so the verifier rejects the pair (x^2, x) before constructing a product table.

22.4 Nonsplit extensions and exhaustive certificates

An extension is *nonsplit* when it has no homomorphic section. This is a universal negative statement:

$$\nexists s : G \rightarrow E \quad (\pi s = \text{id}_G \text{ and } s(gh) = s(g)s(h) \text{ for all } g, h). \quad (22.34)$$

The absence of a preferred section, or the nonzero factor set of one chosen section, does not by itself prove (22.34).

Proposition 22.4.1 (Finite section enumeration). *Let (22.1) be an extension of finite groups. Choose canonical row orders and put*

$$\mathcal{S}_\pi = \{s : G \rightarrow E : s(1) = 1, \pi(s(g)) = g \text{ for every } g \in G\}. \quad (22.35)$$

Then the extension is split if and only if some $s \in \mathcal{S}_\pi$ passes every equality $s(gh) = s(g)s(h)$. It is nonsplit if and only if every $s \in \mathcal{S}_\pi$ has a witness pair

$$(g_s, h_s) \quad \text{with} \quad s(g_s h_s) \neq s(g_s)s(h_s). \quad (22.36)$$

The list which assigns to each s its lexicographically first witness is a finite, independently verifiable nonsplitting certificate.

Proof. Surjectivity of π makes every fiber nonempty, so $\mathcal{S}_\pi$ is finite and nonempty. A member s is already a normalized right inverse of π ; it is a splitting exactly when it preserves all products. This proves the first equivalence. Negating the finite existential statement gives

$$\forall s \in \mathcal{S}_\pi \exists (g, h) \in G^2 : s(gh) \neq s(g)s(h),$$

which is the second equivalence. Choosing the first failed pair is deterministic. Replay reconstructs the finite candidate set from the fibers and checks the displayed inequality for every listed row, so no unlisted section is being assumed away. $\square$

For a central extension, the negative certificate has a smaller cochain form. Let s be a normalized section with cocycle f as in (22.10). Every other normalized section q has a unique expression

$$q_t(g) = i(t(g))s(g), \quad t : G \rightarrow A, \quad t(1) = 0. \quad (22.37)$$

Existence follows because $q(g)s(g)^{-1} \in \ker \pi = i(A)$; injectivity of i gives uniqueness. Define

$$(\delta t)(g, h) = t(g) + t(h) - t(gh). \quad (22.38)$$

Centrality permits the $i(t(h))$ factor to pass through $s(g)$, so

$$\begin{aligned} q_t(g)q_t(h) &= i(t(g) + t(h) + f(g, h))s(gh) \\ &= i(f(g, h) + (\delta t)(g, h))q_t(gh). \end{aligned} \quad (22.39)$$

Theorem 22.4.2 (Exact central nonsplitting criterion). *For the central extension represented by f , the following are equivalent:*

- (i) *the extension splits;*
- (ii) *there is a normalized $t : G \rightarrow A$ such that $f + \delta t = 0$ on G^2 ;*
- (iii) *the canonical model E_f contains a subgroup H for which $\pi_f|_H : H \rightarrow G$ is an isomorphism.*

Consequently it is nonsplit exactly when the finite equation system

$$t(g) + t(h) - t(gh) = -f(g, h) \quad (g, h \in G), \quad t(1) = 0 \quad (22.40)$$

has no solution.

Proof. By (22.39), q_t is a homomorphism exactly when $f + \delta t = 0$, proving (i) $\Leftrightarrow$ (ii), because (22.37) parametrizes all normalized sections. The equivalence of (i) and (iii) is the complement criterion of Statement 22.1, whose hypotheses hold by the central construction theorem Statement 22.2. Negating (ii) gives (22.40) with no solution. These arguments prove both directions; no inference from merely $f \neq 0$ has been used. $\square$

There is also a useful one-sided obstruction. If $g \in G$ has order m and a splitting s exists, then

$$s(g)^m = s(g^m) = s(1) = 1. \quad (22.41)$$

Therefore, if every element of the fiber $\pi^{-1}(g)$ has nontrivial m -th power, the extension is nonsplit. The converse is false in general: the existence of suitable individual lifts does not force those lifts to satisfy all multiplication relations. Accordingly (22.41) is recorded only as a sufficient rejection test, never as an equivalence.

The deterministic algorithm accepts either a certified finite extension or central cocycle data. In general mode it constructs $\mathcal{S}_\pi$ in lexicographic order and returns the first homomorphic section, or the complete witness list (22.36). In central mode it enumerates the $|A|^{|G|-1}$ normalized cochains t , checks (22.40) in lexicographic order on G^2 , and returns either the first solution or, for every t , its first failed equation. The order test (22.41) may terminate rejection early only when its exhaustive fiber premise is itself stored.

The fixed record is

$$\begin{aligned} &(\text{mode, source_hashes, fiber_rows, order_obstructions,} \\ &\text{candidate_count, } s_{\text{split}} \text{ or NONE, } t_{\text{split}} \text{ or NONE,} \\ &\text{first_failure_table, SPLIT or NONSPLIT, h_{independent verification}}). \end{aligned} \quad (22.42)$$

The candidate count and canonical enumeration rule allow independent verification to verify that the negative table is complete.

For the extension $C_2 \rightarrow C_4 \rightarrow C_2$, the nonidentity quotient element has order two, whereas its two lifts are the odd residue classes and both square to $\bar{2} \neq \bar{0}$. Equation (22.41) therefore certifies nonsplitting. Equivalently, its cocycle has $f(\bar{1}, \bar{1}) = \bar{1}$; every normalized cochain satisfies

$$(\delta t)(\bar{1}, \bar{1}) = 2t(\bar{1}) = 0 \quad \text{in } C_2,$$

so (22.40) is unsatisfiable.

To show why $f \neq 0$ is not a nonsplitting certificate, take $A = C_3$, $G = C_2 = \langle x \rangle$, and $f(x, x) = 1$, with normalized entries zero. Put $t(x) = 1$. Then $(\delta t)(x, x) = 2$, so $f + \delta t = 0$ in C_3 . The section q_t splits E_f , even though the original canonical section has nonzero factor set. The zero cocycle and every extension with $G = 1$ give the split boundary.

22.5 Extension equivalence and the cohomology class

Consider two extensions of the same identified groups G by A :

$$\begin{aligned} 1 &\rightarrow A \xrightarrow{i} E \xrightarrow{\pi} G \rightarrow 1, \\ 1 &\rightarrow A \xrightarrow{i'} E' \xrightarrow{\pi'} G \rightarrow 1. \end{aligned} \quad (22.43)$$

They are *equivalent extensions* when there is a group isomorphism $\Phi : E \rightarrow E'$ for which

$$\Phi i = i', \quad \pi' \Phi = \pi. \quad (22.44)$$

Thus an equivalence acts as the identity on the named kernel and quotient. This is an equivalence relation: the identity map satisfies (22.44), an inverse isomorphism satisfies the reversed squares, and composition preserves both squares. An abstract isomorphism $E \cong E'$ which does not satisfy (22.44) is not sufficient.

For central extensions, write A additively and give it the trivial G -action. Let $C^1(G, A)$ be the abelian group of normalized maps $t : G \rightarrow A$, $t(1) = 0$, under pointwise addition. Let $Z^2(G, A)$ be the normalized maps $f : G^2 \rightarrow A$ satisfying (22.11), and define

$$\begin{aligned} (\delta t)(g, h) &= t(g) + t(h) - t(gh), \\ B^2(G, A) &= \{\delta t : t \in C^1(G, A)\}, \\ H^2(G, A) &= Z^2(G, A)/B^2(G, A). \end{aligned} \tag{22.45}$$

The formulas are well defined. Pointwise sums and negatives preserve the cocycle equation, so $Z^2(G, A)$ is an abelian group. Moreover

$$\begin{aligned} (\delta t)(g, h) + (\delta t)(gh, k) \\ &= t(g) + t(h) + t(k) - t(ghk) \\ &= (\delta t)(h, k) + (\delta t)(g, hk). \end{aligned} \tag{22.46}$$

and normalization is immediate; hence $B^2(G, A) \leq Z^2(G, A)$. The quotient in (22.45) therefore exists.

If s is a normalized section with cocycle f and $s_t(g) = i(t(g))s(g)$, equation (22.39) gives

$$f_{s_t} = f_s + \delta t. \tag{22.47}$$

Every normalized section is uniquely of this form. Consequently the class $[f_s] \in H^2(G, A)$ is independent of the chosen section.

Theorem 22.5.1 (Equivalence classification of central extensions). *The assignment*

$$[1 \rightarrow A \rightarrow E \rightarrow G \rightarrow 1] \longmapsto [f_s] \in H^2(G, A) \tag{22.48}$$

is a bijection from equivalence classes of central extensions with fixed kernel and quotient identifications to $H^2(G, A)$. More explicitly, two normalized cocycles f, f' define equivalent extensions if and only if

$$f' = f + \delta t \tag{22.49}$$

for a normalized $t : G \rightarrow A$. In that case an equivalence and its inverse are

$$\begin{aligned} \Phi_t : E_f &\longrightarrow E_{f'}, & \Phi_t(a, g) &= (a - t(g), g), \\ \Phi_t^{-1} : E_{f'} &\longrightarrow E_f, & \Phi_t^{-1}(a, g) &= (a + t(g), g). \end{aligned} \tag{22.50}$$

Proof. Assume (22.49). Using the two multiplication laws and $\delta t(g, h) = t(g) + t(h) - t(gh)$,

$$\begin{aligned} \Phi_t((a, g)(b, h)) &= (a + b + f(g, h) - t(gh), gh), \\ \Phi_t(a, g)\Phi_t(b, h) &= (a + b - t(g) - t(h) + f'(g, h), gh) \\ &= (a + b + f(g, h) - t(gh), gh). \end{aligned}$$

Thus Φ_t is a homomorphism. The second line of (22.50) is visibly its inverse. Since $t(1) = 0$,

$$\Phi_t(a, 1) = (a, 1), \quad \pi_{f'}\Phi_t(a, g) = g,$$

so both squares (22.44) commute. Hence cohomologous cocycles yield equivalent extensions.

Conversely, let $\Phi : E_f \rightarrow E_{f'}$ satisfy (22.44). For each g , the quotient condition puts $\Phi(0, g)$ in the fiber above g , so there is a unique $u(g) \in A$ with

$$\Phi(0, g) = (u(g), g), \quad u(1) = 0.$$

Because $(a, g) = (a, 1)(0, g)$ and Φ fixes the kernel,

$$\Phi(a, g) = (a + u(g), g). \quad (22.51)$$

Apply multiplicativity to $(0, g)(0, h)$. Its two sides have first coordinates

$$f(g, h) + u(gh) \quad \text{and} \quad u(g) + u(h) + f'(g, h).$$

Therefore $f' = f - \delta u = f + \delta(-u)$, proving necessity in (22.49). Statement 22.2 shows that every cocycle produces a central extension, so (22.48) is surjective; the necessity just proved gives injectivity. $\square$

Corollary 22.5.2 (Zero class and splitting). *A central extension splits if and only if its class in $H^2(G, A)$ is zero.*

Proof. The class is zero exactly when $f = \delta u$ for some normalized u . Taking $t = -u$ gives $f + \delta t = 0$, which is equivalent to splitting by Statement 22.4. The reverse implication is the same calculation read backwards. $\square$

For later symmetry questions it is essential to distinguish *weak isomorphism*. Allow $R \in \text{Aut}(A)$ and $P \in \text{Aut}(G)$, and require an isomorphism inducing R and P rather than identities. A map of the form

$$\Psi(a, g) = (R(a) + u(g), P(g)) \quad (22.52)$$

is a homomorphism $E_f \rightarrow E_{f'}$ if and only if

$$f'(P(g), P(h)) = R(f(g, h)) - \delta u(g, h). \quad (22.53)$$

Indeed, comparing first coordinates of $\Psi((a, g)(b, h))$ and $\Psi(a, g)\Psi(b, h)$ gives exactly (22.53). Bijectivity follows from those of R, P , with inverse

$$(a', g') \mapsto (R^{-1}(a' - u(P^{-1}g')), P^{-1}g'). \quad (22.54)$$

Ordinary equivalence is the special case $R = P = \text{id}$.

For finite A, G , an equivalence verifier checks both inputs by Statement 22.2, enumerates all $|A|^{|G|-1}$ normalized cochains in canonical order, and tests (22.49) on all pairs. A success returns the first t , the full table (22.50), its inverse, and all commuting-square rows. A failure returns, for every t , the first pair on which (22.49) fails. This is complete for exactly the same finite-quantifier reason as Statement 22.4. The fixed record is

$$\begin{aligned} &(\mathbf{h}_A, \mathbf{h}_G, \mathbf{h}_f, \mathbf{h}_{f'}, \text{candidate_count}, \\ &t \text{ or NONE}, \Phi_{t\text{tab}} \text{ or NONE}, \text{inverse_rows}, \\ &\text{square_rows}, \text{failure_table}, \text{EQUIVALENT or INEQUIVALENT}, \\ &\mathbf{h}_{\text{independentverification}}). \end{aligned} \quad (22.55)$$

For $A = G = C_2$, the zero cocycle and the cocycle $f(\bar{1}, \bar{1}) = \bar{1}$ are inequivalent: every normalized t satisfies $(\delta t)(\bar{1}, \bar{1}) = 2t(\bar{1}) = 0$. Their middle groups are respectively $C_2 \times C_2$ and C_4 . In contrast, for $A = C_3, G = C_2$, the cocycle $f(x, x) = 1$ is cohomologous to zero because $t(x) = 1$ gives $f + \delta t = 0$, as computed in Statement 22.4. When $A = 1$ or $G = 1$, every normalized cocycle is zero and the equivalence class is unique.

22.6 Lifting automorphisms through a central extension

Let E_f be the central extension of G by the finite abelian group A constructed in Statement 22.2. Fix

$$R \in \text{Aut}(A), \quad P \in \text{Aut}(G). \quad (22.56)$$

A lift of (R, P) is an automorphism $T \in \text{Aut}(E_f)$ which makes both extension squares commute:

$$Ti_f = i_f R, \quad \pi_f T = P\pi_f. \quad (22.57)$$

These equations specify separately the action on the kernel and quotient.

For a normalized cochain $t : G \rightarrow A$, define

$$T_{R,P,t}(a, g) = (R(a) + t(g), P(g)). \quad (22.58)$$

It is bijective as a set map, with inverse

$$T_{R,P,t}^{-1}(a, g) = (R^{-1}(a - t(P^{-1}g)), P^{-1}g). \quad (22.59)$$

Normalization is necessary and sufficient for (22.58) to send the identity to the identity.

Theorem 22.6.1 (Cohomological lift criterion). *The following statements are equivalent:*

- (i) *a lift of (R, P) exists;*
- (ii) *there is a normalized $t : G \rightarrow A$ satisfying*

$$Rf(g, h) + t(gh) = t(g) + t(h) + f(Pg, Ph) \quad (g, h \in G); \quad (22.60)$$

- (iii) *the two cohomology classes satisfy*

$$R_*[f] = P^*[f] \quad \text{in } H^2(G, A), \quad (22.61)$$

where $(R_*f)(g, h) = R(f(g, h))$ and $(P^*f)(g, h) = f(Pg, Ph)$.

When (ii) holds, $T_{R,P,t}$ is a lift. Conversely every lift has this form for a unique normalized t .

Proof. Compare the images of a product with the product of the images:

$$\begin{aligned} T_{R,P,t}((a, g)(b, h)) &= (Ra + Rb + Rf(g, h) + t(gh), P(gh)), \\ T_{R,P,t}(a, g)T_{R,P,t}(b, h) &= (Ra + t(g) + Rb + t(h) + f(Pg, Ph), P(g)P(h)). \end{aligned} \quad (22.62)$$

Because P is a homomorphism, the second coordinates agree. The first coordinates agree for every a, b, g, h exactly when (22.60) holds. Hence (ii) makes the bijection (22.58) a group automorphism; direct substitution proves both equations (22.57).

Conversely, let T satisfy (22.57). The quotient square puts $T(0, g)$ over $P(g)$, so uniquely

$$T(0, g) = (t(g), P(g)).$$

The kernel square and $(a, g) = (a, 1)(0, g)$ then force

$$T(a, g) = (R(a), 1)(t(g), P(g)) = (R(a) + t(g), P(g)).$$

Since an automorphism fixes the identity, $t(1) = 0$. Its multiplicativity and (22.62) force (22.60). This proves (i) $\Leftrightarrow$ (ii) and uniqueness.

Rewrite (22.60) as

$$R_*f - P^*f = \delta t. \quad (22.63)$$

Thus (ii) implies (iii). Conversely, equality of the classes means that $R_*f - P^*f = \delta t$ for some normalized t , which is (22.60). The induced operations on cohomology are well defined because

$$R_*(\delta q) = \delta(Rq), \quad P^*(\delta q) = \delta(q \circ P), \quad (22.64)$$

as follows by substituting (22.38) on both sides. This verifies the quotient-level assertion rather than assuming representatives are fixed. $\square$

For fixed R, P , the set of lifts is either empty or a torsor for

$$Z^1(G, A) = \{z : G \rightarrow A : z(1) = 0, z(gh) = z(g) + z(h)\} = \text{Hom}(G, A). \quad (22.65)$$

Indeed, if t, t' solve (22.60), subtraction gives $\delta(t' - t) = 0$, which is exactly the homomorphism equation in (22.65). Conversely $t + z$ is another solution whenever $\delta z = 0$. Addition is free and transitive, proving both existence of the action and uniqueness of the translating z .

The lift composition law is also explicit. If the displayed maps are defined on the same cocycle model and are automorphisms, then

$$T_{R,P,t} T_{S,Q,u} = T_{RS,PQ, Ru+t \circ Q}, \quad (22.66)$$

because the first coordinate of the composite is $RSa + R(u(g)) + t(Qg)$. Iteration gives, by induction,

$$T_{R,P,t}^m(a, g) = \left(R^m a + \sum_{j=0}^{m-1} R^{m-1-j} t(P^j g), P^m g \right). \quad (22.67)$$

The induction step applies R to the previous sum and appends $t(P^m g)$, producing the same formula with $m + 1$. Consequently $T^m = 1$ if and only if

$$R^m = 1, \quad P^m = 1, \quad \sum_{j=0}^{m-1} R^{m-1-j} t(P^j g) = 0 \quad \text{for every } g \in G. \quad (22.68)$$

Existence of a lift and order m are therefore separate conditions. For $m = 8$, exact order eight additionally requires $T^4 \neq 1$; once $T^8 = 1$, every proper divisor of its order divides 4.

The finite algorithm receives canonical tables for A, G, f, R, P, t . It first replays the cocycle certificate of Statement 22.2 and verifies that R, P are automorphisms. It checks (22.60) on all $|G|^2$ pairs, constructs the full table (22.58) and inverse (22.59), and checks the two squares (22.57). It then computes the least positive m for which (22.68) holds; termination follows because the permutation T of the finite set E_f has finite order. If t is not supplied, it enumerates all normalized cochains and either returns the first solution or a complete first-failure table as in Statement 22.4.

The canonical record is

$$\begin{aligned} &(\mathbf{h}_A, \mathbf{h}_G, \mathbf{h}_f, R_{\text{tab}}, P_{\text{tab}}, t_{\text{tab}} \text{ or NONE,} \\ &\text{obstruction_rows, } T_{\text{tab}} \text{ or NONE, } T_{\text{tab}}^{-1} \text{ or NONE,} \\ &\text{kernel_square, quotient_square, ord}(T) \text{ or NONE, power_rows, h_independentverification}). \end{aligned} \quad (22.69)$$

Generator images are the rows $T(i_f(a)) = i_f(Ra)$ together with $T(0, g) = (t(g), Pg)$; because these elements generate E_f , this is also a compact, unambiguous representation.

For a split positive test $f = 0, t = 0$, every pair R, P lifts as $(a, g) \mapsto (Ra, Pg)$. Taking $A = C_{17}$, $G = 1$, and $R(a) = 2a$ gives exact order eight because

$$2^4 \equiv -1 \pmod{17}, \quad 2^8 \equiv 1 \pmod{17}. \quad (22.70)$$

For a genuine obstruction, let p be odd and take the Heisenberg extension of $G = \mathbb{F}_p^2$ by $A = \mathbb{F}_p$, whose commutator form is

$$\beta((x, y), (x', y')) = xy' - x'y.$$

Let $P(x, y) = (y, x)$ and $R = \text{id}_A$. Any lift would preserve commutators and hence require

$$R\beta(e_1, e_2) = \beta(Pe_1, Pe_2).$$

The two sides are 1 and -1 , unequal for odd p . Thus no lift exists; this is a witness to failure of (22.61). When $A = 1$, the unique lift of P is P itself, giving the kernel-free boundary.

22.7 Parity across an extension

Write $C_2 = \{0, 1\}$ additively. A parity on a group H is a homomorphism $H \rightarrow C_2$; it is nontrivial precisely when it is surjective. Let $E_f = A \times G$ be the central extension constructed from the normalized cocycle f in Statement 22.2. For a homomorphism $\lambda : A \rightarrow C_2$ and a normalized function $\eta : G \rightarrow C_2$, put

$$\widehat{\epsilon}_{\lambda, \eta}(a, g) = \lambda(a) + \eta(g). \quad (22.71)$$

Theorem 22.7.1 (Parity-extension criterion). *Every homomorphism $E_f \rightarrow C_2$ has the form (22.71), for a unique pair (λ, η) . The displayed function is a homomorphism if and only if*

$$(\delta\eta)(g, h) := \eta(g) + \eta(h) - \eta(gh) = \lambda(f(g, h)) \quad (g, h \in G). \quad (22.72)$$

Consequently, a prescribed kernel parity λ extends to E_f if and only if

$$[\lambda \circ f] = 0 \quad \text{in } H^2(G, C_2). \quad (22.73)$$

When extensions exist, their set is a torsor for $\text{Hom}(G, C_2)$.

Proof. If $\widehat{\epsilon} : E_f \rightarrow C_2$ is a homomorphism, define $\lambda(a) = \widehat{\epsilon}(a, 1)$ and $\eta(g) = \widehat{\epsilon}(0, g)$. Since $(a, g) = (a, 1)(0, g)$, formula (22.71) follows and proves uniqueness. Conversely, compare the two sides of the homomorphism equation:

$$\begin{aligned} \widehat{\epsilon}_{\lambda, \eta}((a, g)(b, h)) &= \lambda(a) + \lambda(b) + \lambda(f(g, h)) + \eta(gh), \\ \widehat{\epsilon}_{\lambda, \eta}(a, g) + \widehat{\epsilon}_{\lambda, \eta}(b, h) &= \lambda(a) + \lambda(b) + \eta(g) + \eta(h). \end{aligned} \quad (22.74)$$

They agree for all a, b, g, h exactly when (22.72) holds. The composition $\lambda \circ f$ is a normalized C_2 -valued two-cocycle because λ is a homomorphism, so (22.72) has a solution precisely when its class vanishes. Finally, the difference of two solutions has zero coboundary and hence belongs to $\text{Hom}(G, C_2)$; adding such a homomorphism gives every other solution uniquely. $\square$

The criterion is independent of the section used to encode the extension. Indeed, replacing f by $f + \delta t$ changes its image by

$$\lambda \circ (f + \delta t) = \lambda \circ f + \delta(\lambda \circ t), \quad (22.75)$$

so the cohomology class in (22.73) is unchanged. In particular, if $\epsilon : G \rightarrow C_2$ is already a quotient parity, then

$$\widetilde{\epsilon} = \epsilon \circ \pi_f, \quad \widetilde{\epsilon}(a, g) = \epsilon(g), \quad (22.76)$$

always supplies a parity on E_f : it is the case $\lambda = 0, \eta = \epsilon$ of the theorem. No splitting hypothesis is involved.

We now specialize the descent statement to the surface group used in the construction. Let

$$q : \Gamma_B \twoheadrightarrow Q, \quad K = \ker q, \quad \chi : \Gamma_B \rightarrow C_2 \quad (22.77)$$

be a finite quotient and the fixed Bolza parity. Then there is a unique parity $\epsilon_Q : Q \rightarrow C_2$ satisfying $\epsilon_Q q = \chi$ if and only if

$$K \subseteq \ker \chi. \quad (22.78)$$

Necessity follows by applying $\epsilon_Q q = \chi$ to K . For sufficiency define $\epsilon_Q(q(\gamma)) = \chi(\gamma)$. If $q(\gamma) = q(\gamma')$, then $\gamma'^{-1}\gamma \in K$, so (22.78) gives $\chi(\gamma) = \chi(\gamma')$; the definition is therefore well defined. It is visibly a homomorphism, and surjectivity of q proves uniqueness. This proves descent rather than merely assigning labels to selected generators.

Let $g_0, \dots, g_7$ be the eight physical side-pairing generators and set $q_j = q(g_j)$. The required physical-generator test is

$$\epsilon_Q(q_j) = 1 \quad (0 \leq j < 8). \quad (22.79)$$

It agrees with the fixed source parity because $\chi(g_j) = 1$. The genus-two surface relator is also consistent: every homomorphism into the abelian group C_2 kills commutators, and hence

$$\epsilon_Q(q([a_1, b_1][a_2, b_2])) = 0 + 0 = 0. \quad (22.80)$$

This parity check supplements, but does not replace, the group-valued check that the relator itself evaluates to 1_Q .

Suppose moreover that $Q = E_f$ and the order-eight symmetry is a lift $T = T_{R,P,t}$ from Statement 22.6. The parity is symmetry-compatible when $\hat{\epsilon}_{\lambda,\eta}T = \hat{\epsilon}_{\lambda,\eta}$. Substitution in (22.71) gives the exact finite criterion

$$\begin{aligned} \lambda R &= \lambda, \\ \lambda(t(g)) + \eta(Pg) &= \eta(g) \quad (g \in G). \end{aligned} \quad (22.81)$$

If $q_0, \dots, q_7$ generate Q , $T(q_j) = q_{j+1 \bmod 8}$, and (22.79) holds, then compatibility follows as well: the two homomorphisms $\epsilon_Q T$ and ϵ_Q agree on a generating set. Thus cyclic symmetry cannot silently change the assigned physical parity.

The finite construction algorithm takes canonical tables for A, G, f, λ , an optional candidate η , and optional tables for T, q , and the images of the source presentation generators. It has two explicitly different modes. In validator mode it decides only whether the supplied η works. In decision mode η is omitted and the algorithm enumerates all $2^{|G|-1}$ normalized maps $G \rightarrow C_2$ in lexicographic truth-table order. It returns the first solution, or, only after exhausting that whole finite list, certifies that the prescribed λ is nonextendable. It performs the following exhaustive checks.

1. Replay the group and cocycle certificates, verify that λ is a homomorphism. For each candidate η under consideration, test (22.72) on every $(g, h) \in G^2$. Store either the successful table or that candidate's lexicographically first failing pair. A validator failure means only **CANDIDATE_FAIL**; a decision-mode failure after all $2^{|G|-1}$ candidates means **NONEXTENDABLE**.
2. For a successful candidate, construct all $|A||G|$ values in (22.71) and check the homomorphism equation on every ordered pair of elements of E_f . This redundant pass detects encoding errors.
3. In surface-quotient mode, evaluate every defining relator under the supplied generator images, verify surjectivity of q , test (22.79), and compare $\epsilon_Q q$ with χ on the presentation generators. Equality there proves equality on every word and therefore certifies (22.78).
4. If T is supplied, independent verification its lift certificate, test (22.81) on all of A and G , and independently compare $\epsilon_Q(Tx)$ with $\epsilon_Q(x)$ for every $x \in Q$.

In decision mode, termination and completeness follow from exhaustive enumeration of the finite candidate space. By the parity-extension theorem, finding a candidate is equivalent to **EXTENDS**, while the complete list of candidate failure rows proves **NONEXTENDABLE**; no single failed candidate is used as a nonexistence certificate. Ordinary validation failure returns the lexicographically first violated row, while success returns the complete truth tables and hashes. Extension existence, surface descent, symmetry compatibility, and overall validation have separate classification fields, so failure of an optional check cannot be mistaken for a

cohomological obstruction. A canonical record is

$$\begin{aligned}
& (\mathbf{h}_A, \mathbf{h}_G, \mathbf{h}_f, \lambda_{\text{tab}}, \text{mode}, \text{candidate_count}, \eta_{\text{tab}} \text{ or NONE}, \\
& (\lambda f)_{\text{tab}}, (\delta \eta)_{\text{tab}} \text{ or NONE}, \text{candidate_failure_rows}, \\
& \hat{\epsilon}_{\text{tab}} \text{ or NONE}, \text{hom_rows}, \text{presentation_images}, \\
& \text{relator_rows}, \text{physical_parities}, \text{descent_rows}, \text{symmetry_rows}, \\
& \text{extension_status} \in \{\text{CANDIDATE_FAIL}, \text{EXTENDS}, \text{NONEXTENDABLE}\}, \\
& \text{surface_status} \in \{\text{NONE}, \text{SATISFIED}, \text{SURFACE_FAIL}\}, \\
& \text{symmetry_status} \in \{\text{NONE}, \text{SATISFIED}, \text{SYMMETRY_FAIL}\}, \\
& \text{overall_status} \in \{\text{SATISFIED}, \text{REJECT}\}, \text{first_failure or NONE}, \mathbf{h}_{\text{independentverification}}).
\end{aligned} \tag{22.82}$$

For a positive source test, take $Q = C_2$ and $q = \chi$. Then every physical image is 1, (22.80) holds, and the quotient parity is the identity. For any central extension above this Q , (22.76) pulls that parity back. In contrast, let $A = G = C_2$ and use the cocycle

$$f(x, y) = \left\lfloor \frac{x+y}{2} \right\rfloor \pmod{2}, \tag{22.83}$$

which gives $E_f \cong C_4$. The quotient parity still pulls back, but the nonzero kernel parity $\lambda = \text{id}_{C_2}$ does not extend: at $g = h = 1$, every normalized η has $(\delta \eta)(1, 1) = 0$, whereas $\lambda(f(1, 1)) = 1$. This is a concrete obstruction for (22.73). Finally, the trivial quotient of Γ_B cannot carry the fixed nontrivial parity, since its kernel is all of Γ_B , violating (22.78).

22.8 Orders of quotients and extensions

Order computations are meaningful only after the maps defining a quotient or extension have been certified. We therefore treat an order as the last field of a certificate, not as a guess inferred from a presentation.

Theorem 22.8.1 (Order formula for a finite extension). *Let*

$$1 \longrightarrow A \xrightarrow{i} E \xrightarrow{\pi} G \longrightarrow 1 \tag{22.84}$$

be exact. If A and G are finite, then E is finite and

$$|E| = |A| |G|. \tag{22.85}$$

Conversely, if E is finite, then A and G are finite and (22.85) holds.

Proof. For each $g \in G$, choose one $e_g \in \pi^{-1}(g)$. Exactness gives $\ker \pi = i(A)$. The map

$$A \longrightarrow \pi^{-1}(g), \quad a \longmapsto i(a)e_g \tag{22.86}$$

is injective because i is injective. If e lies in the target fiber, then $\pi(ee_g^{-1}) = 1$, so $ee_g^{-1} = i(a)$ for a unique a ; hence (22.86) is surjective. Every one of the $|G|$ disjoint fibers therefore has $|A|$ elements, proving (22.85). If E is finite, its subgroup $i(A)$ and its image G are finite, and the same fiber argument applies. $\square$

The proof neither assumes that the sequence splits nor that the kernel is central. In particular, every normalized central cocycle model and every semidirect product from this chapter satisfies

$$|E_f| = |A \times G| = |A| |G|, \quad |A \rtimes_{\theta} G| = |A| |G|. \tag{22.87}$$

Thus changing the cohomology class can change the isomorphism type but not the order. For a finite group E and a normal subgroup $N \trianglelefteq E$, (22.84) applied to the natural sequence gives

$$|E/N| = \frac{|E|}{|N|}. \quad (22.88)$$

For a surjective quotient map $q : \Gamma \twoheadrightarrow Q$ with $K = \ker q$, the first isomorphism theorem supplies

$$Q \cong \Gamma/K, \quad |Q| = [\Gamma : K] \quad \text{when } Q \text{ is finite.} \quad (22.89)$$

If $L \trianglelefteq \Gamma$, $L \leq K$, and the displayed quotients are finite, there is a short exact sequence

$$1 \longrightarrow K/L \longrightarrow \Gamma/L \longrightarrow \Gamma/K \longrightarrow 1, \quad (22.90)$$

where the middle map sends γL to γK . Well-definedness uses $L \leq K$, its kernel is K/L , and it is surjective. Hence

$$[\Gamma : L] = [K : L] [\Gamma : K]. \quad (22.91)$$

The normality of L in Γ is essential for (22.90) as a sequence of groups; it must not be omitted merely because the numerical index formula may hold for subgroups.

More generally, suppose $r < \infty$, the groups $E_0, A_1, \dots, A_r$ are finite, and every row in the finite tower

$$1 \rightarrow A_j \rightarrow E_j \rightarrow E_{j-1} \rightarrow 1 \quad (1 \leq j \leq r),$$

is short exact. Then every E_j is finite, and induction on (22.85) gives

$$|E_r| = |E_0| \prod_{j=1}^r |A_j|. \quad (22.92)$$

If $B \leq A$ is central in E , then $i(B) \trianglelefteq E$ and quotienting (22.84) produces

$$1 \rightarrow A/B \rightarrow E/i(B) \rightarrow G \rightarrow 1, \quad |E/i(B)| = |A/B| |G| = \frac{|E|}{|B|}. \quad (22.93)$$

The maps in this new sequence are induced from i and π ; checking their well-definedness prevents an invalid division by the order of a subgroup which is not normal.

Algorithm specification. The required module is

$$\text{OrderCertify}(\mathcal{T}_A, \mathcal{T}_E, \mathcal{T}_G, i_{\text{tab}}, \pi_{\text{tab}}; \mathcal{P}, \mathcal{X}), \quad (22.94)$$

where the first three inputs are finite multiplication tables in fixed canonical element orders, the next two are complete map tables, $\mathcal{P}$ is an optional finite presentation with proposed generator images in E , and $\mathcal{X}$ is an optional list of certified normal subgroups or an extension tower. Integers are stored with arbitrary precision. The output is either **Reject**(s, w), containing a stage name and the lexicographically first witness, or an immutable certificate containing the exact orders, fiber data, index data, and an independent verification hash.

Here is deterministic pseudocode; every loop follows the supplied canonical order.

STEP 1. For each of $\mathcal{T}_A, \mathcal{T}_E, \mathcal{T}_G$, test closure, the named identity, every inverse, and associativity on every ordered triple. Reject at the first failing row.

STEP 2. Test $i(1) = 1$, $\pi(1) = 1$, and both homomorphism equations on every ordered pair. Test injectivity of i , surjectivity of π , and the elementwise equality

$$\{e \in E : \pi(e) = 1\} = \{i(a) : a \in A\}. \quad (22.95)$$

- STEP 3. For each $g \in G$, choose the first element e_g in its fiber and enumerate the pairs $(a, i(a)e_g)$. Test that these pairs give a bijection onto $\pi^{-1}(g)$. Reject if a duplicate, omission, or wrong fiber occurs.
- STEP 4. If $\mathcal{P} = \langle X \mid R \rangle$ is present, evaluate every $r \in R$ under the supplied images and require $r = 1_E$. Close those images under products and inverses and require the generated subgroup to equal E . The presentation's universal property then gives a surjective map q , so report $[\Gamma : \ker q] = |E|$ by (22.89).
- STEP 5. For every subgroup or tower entry in $\mathcal{X}$, first test membership, subgroup closure, normality, and the stated maps. Construct the relevant cosets in canonical representative order and independent verification (22.88), (22.91), (22.92), or (22.93).
- STEP 6. Compute $|A||G|$ independently from the input lengths and require equality with $|E|$. Serialize each order both as a decimal integer and as its sorted prime-exponent vector. Hash the input tables, all witness rows, and the output using the prescribed byte encoding.

Theorem 22.8.2 (Correctness and reproducibility of OrderCertify). *The algorithm terminates. It accepts if and only if its mandatory tables define the short exact sequence (22.84) and every supplied optional presentation, subgroup, quotient, or tower datum passes its stated certification checks. On verification every reported order and index identity is correct. Equal canonical inputs produce byte-identical outputs.*

Proof. All sets and all loops are finite, including the triple associativity tests and subgroup closures, so the procedure terminates. If it accepts, Steps 1 and 2 prove that the inputs are groups and maps forming an exact sequence. Step 3 constructs the bijections used in the proof of (22.85), so its count is correct independently of the mere table lengths. Step 4 proves every defining relation and proves surjectivity, which justifies (22.89). Step 5 verifies the hypotheses before invoking each corresponding theorem. Thus every output invariant is valid.

Conversely, valid canonical tables for (22.84), together with valid values for every optional datum that is actually supplied, pass Steps 1 and 2. The proof of the order theorem shows that every Step 3 fiber test passes. The valid optional presentation and tower certificates pass their defining tests, and (22.85) proves the final equality in Step 6. Thus the algorithm accepts. Fixed traversal orders, first-witness rules, canonical representatives, unique integer encodings, and a specified serialization order remove every nondeterministic choice, proving reproducibility. $\square$

The canonical successful output is

$$\begin{aligned}
& (h_A, h_E, h_G, i_{\text{tab}}, \pi_{\text{tab}}, \text{exactness_rows}, \text{fiber_bijections}, \\
& |A|, |G|, |E|, \text{prime_vectors}, \text{presentation_relators}, \text{generation_rows}, \\
& \text{coset_rows}, \text{tower_products}, \text{SATISFIED}, h_{\text{independentverification}}).
\end{aligned} \tag{22.96}$$

The minimal positive test family includes the nonsplit extension $1 \rightarrow C_2 \rightarrow C_4 \rightarrow C_2 \rightarrow 1$, for which $4 = 2 \cdot 2$, and the split extension $1 \rightarrow C_3 \rightarrow S_3 \rightarrow C_2 \rightarrow 1$, for which $6 = 3 \cdot 2$. It also includes the boundary rows $1 \rightarrow 1 \rightarrow G \rightarrow G \rightarrow 1$ and $1 \rightarrow A \rightarrow A \rightarrow 1 \rightarrow 1$. A negative table obtained by changing one value of the map $C_4 \rightarrow C_2$ fails the homomorphism or kernel test before any order is certified.

Two useful non-implications are recorded explicitly. For a finite group Q , a nontrivial parity $Q \twoheadrightarrow C_2$ does imply

$$2 \mid |Q|, \tag{22.97}$$

by (22.85). However, an automorphism of exact order eight does not imply $8 \mid |Q|$: the group C_{17} has the order-eight automorphism $a \mapsto 2a$ from (22.70). Only after forming the semidirect

product $Q \rtimes C_8$ does (22.87) give order $8|Q|$. These tests prevent symmetry order from being confused with group order.

Part VIII

Second cohomology and equivariant extension design

22.9 The computable group $H^2(G, M)$

Let G be a group and let M be an abelian group equipped with a specified left action $g \cdot m$ by automorphisms. Thus

$$1 \cdot m = m, \quad (gh) \cdot m = g \cdot (h \cdot m), \quad g \cdot (m + n) = g \cdot m + g \cdot n. \quad (23.1)$$

All cochains below are inhomogeneous and normalized. Put $C^0(G, M) = M$; for $n \geq 1$, let $C^n(G, M)$ be the abelian group of functions $G^n \rightarrow M$ which vanish whenever any argument is 1_G . In degrees zero, one, and two the coboundary is

$$\begin{aligned} (\delta m)(g) &= g \cdot m - m, \\ (\delta t)(g, h) &= g \cdot t(h) - t(gh) + t(g), \\ (\delta f)(g, h, k) &= g \cdot f(h, k) - f(gh, k) + f(g, hk) - f(g, h). \end{aligned} \quad (23.2)$$

Normalization and (23.1) show directly that these formulas again give normalized cochains.

The two-cocycles, two-coboundaries, and second cohomology group are

$$\begin{aligned} Z^2(G, M) &= \ker(\delta : C^2 \rightarrow C^3), \\ B^2(G, M) &= \text{im}(\delta : C^1 \rightarrow C^2), \\ H^2(G, M) &= Z^2(G, M)/B^2(G, M). \end{aligned} \quad (23.3)$$

In particular, f is a normalized two-cocycle exactly when

$$g \cdot f(h, k) + f(g, hk) = f(g, h) + f(gh, k) \quad (g, h, k \in G). \quad (23.4)$$

Lemma 22.9.1 (The quotient in (23.3) is well defined). *For every normalized $t \in C^1(G, M)$, $\delta(\delta t) = 0$. Hence $B^2(G, M) \leq Z^2(G, M)$.*

Proof. Substitute the degree-one formula from (23.2) into the degree-two formula. For $g, h, k \in G$, the result is

$$\begin{aligned} &g \cdot \{h \cdot t(k) - t(hk) + t(h)\} \\ &\quad - \{gh \cdot t(k) - t(ghk) + t(gh)\} \\ &\quad + \{g \cdot t(hk) - t(ghk) + t(g)\} \\ &\quad - \{g \cdot t(h) - t(gh) + t(g)\} = 0. \end{aligned} \quad (23.5)$$

Every term cancels, using $g \cdot (h \cdot t(k)) = (gh) \cdot t(k)$. Thus every coboundary is a cocycle. Since δ is additive, both displayed sets are subgroups of C^2 , proving the assertion. $\square$

The normalized complex computes the usual group cohomology. In the degrees used here this can also be seen constructively: inserting an identity argument defines the standard degeneracy homotopy, which removes all identity-coordinate values from a cochain without changing its class. Consequently no cohomology class is lost by storing only the $(|G| - 1)^n$ potentially nonzero rows of a finite normalized n -cochain.

When the action is trivial, (23.2) becomes

$$(\delta t)(g, h) = t(g) + t(h) - t(gh), \quad (23.6)$$

and (23.4) becomes the central cocycle identity (22.11). Thus the notation agrees exactly with C22 rather than differing by a hidden sign. For a general action, $H^2(G, M)$ classifies extensions which induce that prescribed action of G on M ; central extensions are the trivial-action case. This classification will be constructed explicitly in Statement 22.12.

Finite computable expression. Suppose G and M are finite and their tables and action maps are given in canonical orders. Enumerate the normalized cochain groups, apply (23.2) row by row, and form

$$\mathcal{Z}_2 = \{f : \delta f = 0\}, \quad \mathcal{B}_2 = \{\delta t : t \in C^1(G, M)\}. \quad (23.7)$$

Membership, addition, and equality are all decidable from the finite tables. Sort $\mathcal{Z}_2$, choose the first unused cocycle as the next coset representative, and mark its translate by every element of $\mathcal{B}_2$. This terminates and produces a canonical quotient table for $H^2(G, M)$.

If $M = \mathbb{F}_p^d$ and the action is given by matrices $\rho(g) \in \text{GL}_d(\mathbb{F}_p)$, the same computation is linear algebra. Lexicographically order the nonidentity tuples and let D_1, D_2 be the matrices of the two maps in (23.2). Then

$$D_2 D_1 = 0, \quad H^2(G, M) = \ker D_2 / \text{im } D_1, \quad \dim H^2 = \text{nullity}(D_2) - \text{rank}(D_1). \quad (23.8)$$

Reduced row echelon form with a prescribed pivot rule selects a unique ordered basis of cocycle representatives. For general finite abelian M , exhaustive table enumeration remains the reference algorithm; primary decomposition and Smith normal form may accelerate it but do not change its output specification.

The fixed serialization is

$$(\mathbf{h}_G, \mathbf{h}_M, \rho_{\text{tab}}, \text{tuple_orders}, D_1 \text{ or } \delta_1^{\text{tab}}, D_2 \text{ or } \delta_2^{\text{tab}}, \\ \text{Z2_rows}, \text{B2_rows}, \text{coset_representatives}, \text{H2_table}, \text{dimension_or_order}, \mathbf{h}_{\text{independentverification}}). \quad (23.9)$$

The identities $D_2 D_1 = 0$, $B^2 \leq Z^2$, and the coset partition are independently verified before the record is accepted. Canonical tuple order, pivots, representatives, and byte encoding make repeated execution on the same input reproducible.

For the smallest positive test, let $G = C_2 = \langle s \rangle$, $M = C_2$, with trivial action. A normalized two-cochain is determined by $x = f(s, s)$. Both choices of x satisfy (23.4), while every normalized t has

$$(\delta t)(s, s) = t(s) + t(s) - t(1) = 0. \quad (23.10)$$

Hence $H^2(C_2, C_2) \cong C_2$; the zero and nonzero classes yield $C_2 \times C_2$ and C_4 , respectively. At the boundary $G = 1$, every positive-degree normalized cochain is zero, so $H^2(1, M) = 0$. As a nontrivial-action zero test, let C_2 act on C_3 by inversion. Direct evaluation of (23.4) shows a cocycle is determined by $x = f(s, s)$ with $x = -x$; hence $x = 0$ and

$$H^2(C_2, C_3^{\text{inv}}) = 0. \quad (23.11)$$

These examples test a nonzero class, a degenerate domain, and an action whose second cohomology vanishes.

22.10 Constructing normalized two-cocycles

Fix finite canonical tables for a group G , a finite abelian group M , and a left action $\rho : G \rightarrow \text{Aut}(M)$. A construction must distinguish validating one proposed table from deciding the entire cocycle system. The module

$$\text{Cocycle2}(\mathcal{T}_G, \mathcal{T}_M, \rho_{\text{tab}}; \text{mode}, \mathcal{C}) \quad (23.12)$$

has three modes. In **VALIDATE**, $\mathcal{C}$ is one candidate $f : G^2 \rightarrow M$. In **FIRST** or **ALL**, $\mathcal{C}$ is an optional list of prescribed coordinate values, and the module respectively returns the first solution or the complete set of solutions. Failure of **VALIDATE** rejects only that candidate; nonexistence is reported only after the finite solution space has been exhausted. Malformed group, module, action, mode, table, or constraint encodings return **INVALID_INPUT** with a stage name, the first structural failure, and the raw-input hash. This is distinct from a complete, well-formed candidate which fails normalization or the cocycle equation; the latter returns **CANDIDATE_FAIL**.

The construction uses the ordered coordinate set

$$\Omega_2 = (G \setminus \{1\})^2. \quad (23.13)$$

Coordinates outside Ω_2 are fixed to zero. For an assignment $x \in M^{\Omega_2}$, let f_x be the resulting normalized table and define its defect

$$\Delta_x(g, h, k) = g \cdot f_x(h, k) - f_x(gh, k) + f_x(g, hk) - f_x(g, h). \quad (23.14)$$

Then f_x is a two-cocycle if and only if every one of the $|G|^3$ defect rows is zero.

Deterministic pseudocode. Every loop below uses the supplied canonical order.

STEP 1. Verify that **mode** $\in \{\text{VALIDATE}, \text{FIRST}, \text{ALL}\}$ and that $\mathcal{C}$ has the mode-dependent shape specified above; otherwise return **INVALID_INPUT**. Then verify the group table of G , the abelian group table of M , and every action law

$$\rho(1) = 1, \quad \rho(gh) = \rho(g)\rho(h), \quad \rho(g)(m + n) = \rho(g)m + \rho(g)n. \quad (23.15)$$

Also verify that each $\rho(g)$ is bijective.

STEP 2. Construct Ω_2 . Check that every prescribed coordinate belongs to it and that no coordinate receives two unequal values; a failure is structural **INVALID_INPUT**. In validator mode require a well-typed value in every G^2 row; a missing, duplicated, or foreign-valued row is likewise structural. If that complete table has a nonzero identity-coordinate row, return **CANDIDATE_FAIL** with the first normalization witness.

STEP 3. For a normalized validator candidate compute (23.14) over G^3 and return **VALID**, or **CANDIDATE_FAIL** with the first nonzero defect.

STEP 4. In construction mode enumerate M^{Ω_2} in lexicographic order, skipping assignments that violate $\mathcal{C}$. Test (23.14) for every remaining assignment. In **FIRST** mode stop at the first cocycle; in **ALL** mode store every cocycle.

STEP 5. For each accepted f , construct the twisted multiplication

$$(m, g)(n, h) = (m + g \cdot n + f(g, h), gh) \quad (23.16)$$

on $M \times G$. Check its identity, inverse, and associativity tables independently of the defect test.

STEP 6. Serialize the sorted solutions, defect certificates, multiplication tables, coordinate order, and hashes. In **FIRST** mode return **NONE** only after every admissible assignment has failed; in **ALL** mode an empty solution list is the complete nonexistence certificate.

The reference enumeration is deliberately finite and unconditional. When $M = \mathbb{F}_p^d$, the implementation may replace Step 4 by linear algebra. With coordinates ordered as in (23.13), equation (23.14) is the homogeneous system

$$D_2x = 0. \quad (23.17)$$

Prescribed coordinates append equations $Lx = b$. Deterministic reduced row echelon form either returns an inconsistency row or an affine solution

$$x = x_0 + \sum_{j=1}^r u_j v_j, \quad u_j \in \mathbb{F}_p, \quad (23.18)$$

where free columns and basis vectors are in increasing pivot order. Enumerating the u_j lexicographically gives exactly the same ordered solution set as the reference procedure after sorting full truth tables.

Theorem 22.10.1 (Correctness of Cocycle2). *The algorithm terminates. It returns the classification*

INVALID_INPUT

exactly when a structural check in Steps 1 or 2 fails. On structurally valid input, validator mode returns VALID exactly for normalized two-cocycles. Construction mode returns exactly all normalized two-cocycles satisfying the prescribed coordinates, or the canonical first one in FIRST mode. For every returned f , (23.16) is a group law with identity $(0, 1)$.

Proof. All table checks and candidate spaces are finite, so termination is immediate. Steps 2 and 3 test normalization and every instance of (23.4), proving the validator assertion. In construction mode every possible normalized table occurs once in M^{Ω_2} ; filtering by $\mathcal{C}$ and (23.14) is therefore sound and complete. The $\mathbb{F}_p$ branch is equivalent because (23.17) is simply the coordinate matrix of those same defect equations.

It remains to justify the redundant group table. The two associations of three factors in (23.16) have first coordinates

$$\begin{aligned} ((m, g)(n, h))(r, k) &= m + g \cdot n + f(g, h) + (gh) \cdot r + f(gh, k), \\ (m, g)((n, h)(r, k)) &= m + g \cdot n + (gh) \cdot r + g \cdot f(h, k) + f(g, hk). \end{aligned} \quad (23.19)$$

They agree exactly by (23.4). Normalization makes $(0, 1)$ a two-sided identity. A right inverse of (m, g) is

$$(g^{-1} \cdot (-m - f(g, g^{-1})), g^{-1}). \quad (23.20)$$

Associativity and the right-inverse equation imply it is also the left inverse; alternatively the latter follows by substituting (g^{-1}, g) in (23.4). Hence the returned table is a group. $\square$

The fixed output record is

$$\begin{aligned} &(\mathbf{h}_G, \mathbf{h}_M, \rho_{\text{tab}}, \text{mode}, \Omega_2, \mathcal{C}, \text{candidate_count}, \\ &\text{solution_tables}, \text{defect_rows}, \text{linear_certificate or NA}, \\ &\text{twisted_tables}, \text{group_law_rows}, \text{classification}, \\ &\text{stage or NONE}, \text{first_failure or NONE}, \\ &\mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification or NONE}}). \end{aligned} \quad (23.21)$$

Fields unavailable after structural rejection are serialized as NONE, so invalid input also has an unambiguous record. The classification field takes one of the fixed values

INVALID_INPUT, VALID, CANDIDATE_FAIL,
FOUND, ALL_FOUND, NO_SOLUTION.

Fixed traversal and serialization make identical inputs produce byte-identical records.

For $G = M = C_2$ with trivial action, the two tables determined by $f(s, s) = 0$ and $f(s, s) = 1$ both pass; their twisted groups are C_2^2 and C_4 . If $G = C_3 = \langle s \rangle$, $M = C_2$ has trivial action, and a normalized candidate has $f(s, s) = 1$ with every other row zero, then

$$\Delta(s, s, s^2) = -f(s^2, s^2) - f(s, s) = -1 \neq 0, \quad (23.22)$$

so validator mode returns that explicit failure. Finally, when $G = 1$, Ω_2 is empty and the unique zero table is returned. These tests cover two valid cohomology classes, an invalid candidate, and the empty-coordinate boundary.

22.11 Coboundaries and changes of section

Continue with the left G -module M from Statement 22.9. For a normalized one-cochain $t : G \rightarrow M$, its two-coboundary is

$$b_t(g, h) = (\delta t)(g, h) = g \cdot t(h) - t(gh) + t(g). \quad (23.23)$$

The identity-coordinate rows vanish:

$$b_t(1, h) = b_t(g, 1) = 0. \quad (23.24)$$

The calculation (23.5) proves that b_t is a two-cocycle, so adding a coboundary preserves the cocycle condition.

The kernel of $\delta : C^1 \rightarrow C^2$ is the group of normalized crossed homomorphisms

$$\begin{aligned} Z^1(G, M) = \{t : G \rightarrow M : t(1) = 0, \\ t(gh) = t(g) + g \cdot t(h) \text{ for all } g, h\}. \end{aligned} \quad (23.25)$$

Therefore there is a short exact sequence of abelian groups

$$0 \longrightarrow Z^1(G, M) \longrightarrow C^1(G, M) \xrightarrow{\delta} B^2(G, M) \longrightarrow 0, \quad (23.26)$$

and the first isomorphism theorem gives

$$B^2(G, M) \cong C^1(G, M) / Z^1(G, M). \quad (23.27)$$

If G and M are finite, this also yields the independently checkable count

$$|B^2(G, M)| = \frac{|C^1(G, M)|}{|Z^1(G, M)|} = \frac{|M|^{|G|-1}}{|Z^1(G, M)|}. \quad (23.28)$$

Theorem 22.11.1 (Coboundaries are precisely section changes). *Let $f, f' \in Z^2(G, M)$, and give $M \times G$ the two twisted group laws (23.16). The corresponding extensions with their fixed identifications of kernel M and quotient G are equivalent if and only if*

$$f' = f + \delta t \quad (23.29)$$

for a normalized one-cochain t . When (23.29) holds, the equivalence and its inverse are

$$\begin{aligned} \Phi_t : E_f &\longrightarrow E_{f'}, & \Phi_t(m, g) &= (m - t(g), g), \\ \Phi_t^{-1} : E_{f'} &\longrightarrow E_f, & \Phi_t^{-1}(m, g) &= (m + t(g), g). \end{aligned} \quad (23.30)$$

Proof. Assume (23.29). In the f' -law, the product of the two images is

$$\begin{aligned} (m - t(g), g)(n - t(h), h) \\ = (m + g \cdot n - t(g) - g \cdot t(h) + f(g, h) + (\delta t)(g, h), gh) \\ = (m + g \cdot n + f(g, h) - t(gh), gh), \end{aligned} \quad (23.31)$$

which is $\Phi_t((m, g)(n, h))$. Thus Φ_t is a homomorphism; (23.30) gives its inverse, and it fixes the named kernel and quotient.

Conversely, let $\Phi : E_f \rightarrow E_{f'}$ be such an equivalence. Since it induces the identity on G , write $\Phi(0, g) = (-t(g), g)$ uniquely. It fixes the identity, so $t(1) = 0$. Because $(m, g) = (m, 1)(0, g)$ and Φ fixes the kernel, $\Phi(m, g) = (m - t(g), g)$. Comparing the image of $(0, g)(0, h)$ with the product of the images gives

$$f(g, h) - t(gh) = -t(g) - g \cdot t(h) + f'(g, h),$$

which is exactly (23.29). This proves both directions. $\square$

Two witnesses t, t' give the same coboundary precisely when

$$t' - t \in Z^1(G, M), \quad (23.32)$$

by (23.25). Thus the set of witnesses for a fixed coboundary is either empty or a coset of $Z^1(G, M)$. In particular, the witness is not generally unique; a certificate which silently selects one without a canonical rule is not reproducible.

Computable membership. For finite canonical tables, a candidate $b : G^2 \rightarrow M$ is a coboundary exactly when the finite system

$$g \cdot t(h) - t(gh) + t(g) = b(g, h), \quad t(1) = 0 \quad (23.33)$$

has a solution. Solver selection is deterministic. If the canonical input contains a verified encoding of $M = \mathbb{F}_p^d$, together with verified action matrices, the mandatory solver is **LINEAR**; otherwise it is **ENUMERATE**. A claimed vector-space encoding which fails its field, basis, table, or action checks returns **INVALID_INPUT** with the first structural witness and does not silently fall back to enumeration.

The **ENUMERATE** solver examines all $|M|^{|G|-1}$ normalized t 's, tests every ordered pair, and returns the first solution in truth-table order. If none works, its failure certificate is the ordered list containing every candidate truth table and that candidate's first failing pair. This is a complete nonmembership certificate; a failed single candidate is not.

For the **LINEAR** solver, equation (23.33) is

$$D_1 x = b. \quad (23.34)$$

Deterministic row reduction of the augmented matrix $[D_1 \mid b]$ returns either a contradiction row or a canonical particular solution plus the ordered nullspace basis Z^1 . Its failure certificate is the reduced augmented matrix, pivot log, and the first contradiction row. The fixed output is

$$\begin{aligned} &(\mathbf{h}_G, \mathbf{h}_M, \rho_{\text{tab}}, b_{\text{tab}}, \text{normalized}, \text{cocycle_rows}, \\ &\text{solver} \in \{\text{ENUMERATE}, \text{LINEAR}\}, \text{candidate_count}, \\ &t_{\text{first}} \text{ or NONE}, Z^1_{\text{rows}}, \text{failure_kind}, \\ &\text{failure_certificate}, \text{stage or NONE}, \\ &\text{classification} \in \{\text{COBOUNDARY}, \text{NOT_COBOUNDARY}, \text{INVALID_INPUT}\}, \\ &\mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}} \text{ or NONE}). \end{aligned} \quad (23.35)$$

The failure kind is **ENUMERATION_LIST**, **RREF_CONTRADICTION**, **STRUCTURAL**, or **NONE**, and the corresponding certificate has the format specified above. Fields unavailable after structural rejection are serialized as **NONE**. Replaying (23.33), (23.25), and the count (23.28) verifies all stored invariants. The deterministic solver dispatch, fixed table and pivot orders, and typed failure format make identical inputs produce byte-identical records.

For a nonzero positive example, let $G = M = C_3 = \langle s \rangle$ with trivial action and choose

$$t(1) = 0, \quad t(s) = 1, \quad t(s^2) = 0.$$

Then the nonidentity block of b_t , with rows and columns ordered (s, s^2) , is

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \pmod{3}. \quad (23.36)$$

It is nonzero but represents the zero class, and (23.30) explicitly splits the associated change of cocycle. By contrast, for trivial C_2 -action on C_2 , every normalized t has $\delta t = 0$, whereas the cocycle $f(s, s) = 1$ is nonzero. Thus it is a cocycle but not a coboundary. At the boundary $G = 1$, both C^1 and B^2 are zero. These cases distinguish zero cochains, nonzero coboundaries, and genuinely nonzero cohomology classes.

22.12 Constructing the extension class

Let

$$1 \longrightarrow M \xrightarrow{i} E \xrightarrow{\pi} G \longrightarrow 1 \quad (23.37)$$

be an extension with M abelian. A normalized section is a set map $s : G \rightarrow E$ satisfying $\pi s = \text{id}_G$ and $s(1) = 1$. It need not be a homomorphism. Conjugation by a lift defines

$$i(g \cdot m) = s(g)i(m)s(g)^{-1}. \quad (23.38)$$

This action is independent of the section: replacing $s(g)$ by $i(a)s(g)$ adds conjugation by $i(a)$, which is trivial on the abelian subgroup $i(M)$. Multiplication and conjugation in E then prove all module laws (23.1).

The factor set of s is the unique normalized function $f_s : G^2 \rightarrow M$ satisfying

$$i(f_s(g, h)) = s(g)s(h)s(gh)^{-1}. \quad (23.39)$$

Existence follows because the right side lies in $\ker \pi = i(M)$; uniqueness follows from injectivity of i .

Theorem 22.12.1 (Extension class and classification). *The factor set f_s is a normalized two-cocycle for the action (23.38), and its class*

$$\text{cl}(E) = [f_s] \in H^2(G, M) \quad (23.40)$$

is independent of s . With M , G , and their action fixed, cl is a bijection from strict equivalence classes of extensions (23.37) to $H^2(G, M)$.

Proof. From (23.39),

$$s(g)s(h) = i(f_s(g, h))s(gh). \quad (23.41)$$

Associate $s(g)s(h)s(k)$ first on the left and then on the right. Using (23.38), the respective kernel coordinates are

$$f_s(g, h) + f_s(gh, k), \quad g \cdot f_s(h, k) + f_s(g, hk). \quad (23.42)$$

Their equality is precisely (23.4), so $f_s \in Z^2(G, M)$. Normalization follows from $s(1) = 1$.

Every other normalized section has the unique form

$$s_t(g) = i(t(g))s(g), \quad t(1) = 0.$$

Substitution into (23.39), with $s(g)i(t(h))s(g)^{-1} = i(g \cdot t(h))$, gives

$$f_{s_t}(g, h) = f_s(g, h) + t(g) + g \cdot t(h) - t(gh) = f_s(g, h) + (\delta t)(g, h). \quad (23.43)$$

Thus (23.40) is independent of the section. A strict extension equivalence carries normalized sections to normalized sections and preserves (23.38)–(23.39), so it preserves the class.

For surjectivity, any $f \in Z^2(G, M)$ defines the twisted group $E_f = M \times G$ by (23.16), with kernel $m \mapsto (m, 1)$, quotient $(m, g) \mapsto g$, and section $g \mapsto (0, g)$. Its factor set is exactly f . For injectivity, equal classes mean $f' = f + \delta t$, and the strict equivalence (23.30) applies. Therefore the induced map on equivalence classes is bijective. $\square$

The original extension is explicitly recovered from its factor set. The map

$$\Psi_s : E_{f_s} \longrightarrow E, \quad \Psi_s(m, g) = i(m)s(g) \quad (23.44)$$

is bijective because every fiber of π is a unique $i(M)$ -coset of $s(g)$. Moreover,

$$\begin{aligned} \Psi_s(m, g)\Psi_s(n, h) &= i(m)s(g)i(n)s(h) \\ &= i(m + g \cdot n + f_s(g, h))s(gh) \\ &= \Psi_s((m, g)(n, h)), \end{aligned} \quad (23.45)$$

so it is an extension isomorphism, not merely a bijection of sets.

Algorithm specification. The finite module

$$\text{ExtensionClass}(\mathcal{T}_M, \mathcal{T}_E, \mathcal{T}_G, i_{\text{tab}}, \pi_{\text{tab}}, \rho_{\text{tab}}) \quad (23.46)$$

receives complete canonical tables. It outputs either `INVALID_INPUT` with a stage, the first witness, and the input hash, or a class certificate. Its deterministic pseudocode is:

- STEP 1. Verify all three group tables, commutativity of M , and every homomorphism, injectivity, surjectivity, and kernel equality in (23.37).
- STEP 2. For every $g \in G$, choose $s(g)$ as the first element of $\pi^{-1}(g)$, except that $s(1) = 1_E$. This canonical rule makes the section reproducible.
- STEP 3. Compute (23.38) for every (g, m) , using the unique preimage under i . Verify the module laws and require equality with the supplied ρ_{tab} .
- STEP 4. Compute (23.39) for every (g, h) . Replay normalization and every cocycle row (23.4), then invoke the fixed H^2 quotient procedure of Statement 22.9 to locate the canonical coset representative r of $[f_s]$. Apply the deterministic coboundary solver of the deterministic coboundary solver of Statement 22.11 to $b = r - f_s$, and require the signed reduction equation

$$r = f_s + \delta t_{\text{reduction}}. \quad (23.46a)$$

Store that solver's canonical first or particular witness and its verification rows. If $r = f_s$, bypass the solver and fix $t_{\text{reduction}} = 0$, `reduction_solver` = `NONE`, and `reduction_witness_certificate` = `ZERO`.

- STEP 5. Construct the full twisted table (23.16) and all values of Ψ_s . Check bijectivity, both extension squares, and the homomorphism identity (23.45) on every ordered pair.
- STEP 6. Return the factor table, canonical class representative, signed reduction equation, canonical witness certificate, reconstruction tables, and independent verification hash.

Theorem 22.12.2 (Correctness of `ExtensionClass`). *The algorithm terminates and accepts exactly when the inputs define a finite exact extension with abelian kernel whose induced action is the specified ρ . On verification it returns the section-independent class (23.40) and a certified isomorphism (23.44). Equal canonical inputs give byte-identical output.*

Proof. All loops range over finite tables. verification in Steps 1–3 proves the exact-extension and action hypotheses. Equations (23.41)–(23.43) prove that Step 4 returns the correct, section-independent class. Equations (23.44)–(23.45) prove Step 5. Conversely, a valid input extension passes Steps 1–3; the unique factor set passes Step 4 by the theorem, and the proved isomorphism passes Step 5. The quotient procedure fixes r , while the deterministic coboundary solver of Statement 22.11 fixes the witness of $r - f_s = \delta t_{\text{reduction}}$; in the zero case the explicit rule fixes the zero witness. Canonical element, section, quotient-representative, and serialization orders therefore prove reproducibility. $\square$

The fixed successful record is

$$\begin{aligned} &(\mathbf{h}_M, \mathbf{h}_E, \mathbf{h}_G, i_{\text{tab}}, \pi_{\text{tab}}, \rho_{\text{tab}}, s_{\text{tab}}, f_s^{\text{tab}}, \\ &\text{cocycle_rows, class_representative, } t_{\text{reduction}}, \text{reduction_equation_rows,} \\ &\text{reduction_solver or NONE, reduction_witness_certificate or ZERO,} \\ &\text{twisted_table, } \Psi_s^{\text{tab}}, \text{isomorphism_rows, SATISFIED, } h_{\text{independentverification}}). \end{aligned} \quad (23.47)$$

For $1 \rightarrow C_2 \rightarrow C_4 \rightarrow C_2 \rightarrow 1$, the canonical section $s(0) = 0, s(1) = 1$ gives $f_s(1, 1) = 1$, the nonzero class in $H^2(C_2, C_2)$. For C_2^2 with the evident section, the factor set and class are zero. The split noncentral example

$$1 \rightarrow C_3 \rightarrow S_3 \rightarrow C_2 \rightarrow 1 \quad (23.48)$$

induces inversion on C_3 ; its homomorphic section gives $f_s = 0$, in agreement with (23.11). With $M = 0$, the unique class reconstructs $E \cong G$. Finally, changing one value of $\pi : C_4 \rightarrow C_2$ causes a homomorphism or kernel failure in Step 1, so no extension class is issued from an invalid quotient table.

22.13 C_8 -invariant cohomology classes

Let $P \in \text{Aut}(G)$ and $R \in \text{Aut}(M)$ satisfy the semilinear compatibility

$$R(g \cdot m) = P(g) \cdot R(m) \quad (g \in G, m \in M) \quad (23.49)$$

and the C_8 -relations

$$P^8 = 1_G, \quad R^8 = 1_M. \quad (23.50)$$

Neither P nor R is required to have exact order eight. Let ζ denote the distinguished generator of C_8 . On normalized n -cochains, define

$$(\mathcal{S}_n c)(g_1, \dots, g_n) = R\left(c(P^{-1}g_1, \dots, P^{-1}g_n)\right), \quad (23.51)$$

and put $\mathcal{S}_0 = R$.

Theorem 22.13.1 (Cohomological C_8 -representation). *The maps $\mathcal{S}_n$ are cochain automorphisms and*

$$\delta \mathcal{S}_n = \mathcal{S}_{n+1} \delta. \quad (23.52)$$

Consequently

$$\overline{\mathcal{S}}[f] = [\mathcal{S}_2 f] \quad (23.53)$$

is a well-defined automorphism of $H^2(G, M)$, and $\zeta \mapsto \overline{\mathcal{S}}$ defines a representation of C_8 . A class $[f]$ is C_8 -invariant exactly when

$$\mathcal{S}_2 f - f = \delta u \quad (23.54)$$

for some normalized $u : G \rightarrow M$.

Proof. Formula (23.51) preserves normalization and has inverse

$$(\mathcal{S}_n^{-1}c)(g_1, \dots, g_n) = R^{-1}(c(Pg_1, \dots, Pg_n)). \quad (23.55)$$

For a one-cochain t , use (23.49) with $P^{-1}g$ to compute

$$\begin{aligned} (\mathcal{S}_2\delta t)(g, h) &= R\{P^{-1}g \cdot t(P^{-1}h) - t(P^{-1}(gh)) + t(P^{-1}g)\} \\ &= g \cdot (\mathcal{S}_1t)(h) - (\mathcal{S}_1t)(gh) + (\mathcal{S}_1t)(g) \\ &= (\delta\mathcal{S}_1t)(g, h). \end{aligned} \quad (23.56)$$

The same substitution in the alternating cochain formula proves (23.52) in every degree. Hence cocycles and coboundaries are preserved, which proves (23.53) is well defined. Iteration gives

$$(\mathcal{S}_n^k c)(g_1, \dots, g_n) = R^k c(P^{-k}g_1, \dots, P^{-k}g_n), \quad (23.57)$$

so (23.50) implies $\overline{\mathcal{S}}^8 = 1$, the defining relation of C_8 . Finally, $\overline{\mathcal{S}}[f] = [f]$ is equivalent to their difference being a coboundary, which is (23.54). $\square$

The invariant criterion is the cohomological form of the lift equation. Substituting $g = Px, h = Py$ in (23.54), and putting $t(x) = u(Px)$, gives

$$Rf(x, y) - f(Px, Py) = t(x) + P(x) \cdot t(y) - t(xy). \quad (23.58)$$

Thus the pullback convention in (23.51) agrees with the direct lift calculation; no inverse of P is lost when one changes viewpoints.

Explicit finite representation. Use Statement 22.9 to list canonical representatives

$$\mathcal{R} = (r_0, \dots, r_{N-1}) \quad \text{for } H^2(G, M), \quad r_0 = 0. \quad (23.59)$$

For every i , compute $s_i = \mathcal{S}_2r_i$, locate the unique representative $r_{j(i)}$ in its cohomology coset, and use the deterministic Statement 22.11 solver to certify

$$r_{j(i)} = s_i + \delta t_i. \quad (23.60)$$

If $r_{j(i)} = s_i$, fix $t_i = 0$, solver **NONE**, and certificate **ZERO**. Otherwise store the solver, its canonical witness, and all verification rows. The generator-image table is

i	r_i	$\mathcal{S}_2r_i$	$j(i)$	$r_{j(i)} = \mathcal{S}_2r_i + \delta t_i$
0	0	0	0	$t_0 = 0$
1	r_1	s_1	$j(1)$	t_1
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$N-1$	r_{N-1}	s_{N-1}	$j(N-1)$	t_{N-1}

(23.61)

It is a complete, explicit representation of the generator ζ .

The finite verification algorithm first validates the group, module, action, P, R , (23.49), and (23.50). It constructs the H^2 table and (23.61), then performs these independent checks:

1. test (23.52) on every normalized one- and two-cochain table;
2. verify every reduction equation (23.60) and that j is a permutation preserving the addition table of H^2 ;
3. compute j^8 and require it to be the identity; compute

$$d = \min\{e \in \{1, 2, 4, 8\} : j^e = 1\}; \quad (23.62)$$

4. return the invariant subgroup as the sorted fixed-point set

$$H^2(G, M)^{C_8} = \{[r_i] : j(i) = i\}. \quad (23.63)$$

If exact order eight is requested, the module accepts the order condition only when $d = 8$; otherwise $d \mid 8$ is the correct representation order. Structural failures have classification `INVALID_INPUT`; failed covariance, relation, or exact-order requirements have their own stage and first witness.

The canonical serialization is

$$\begin{aligned} &(\mathbf{h}_G, \mathbf{h}_M, \rho_{\text{tab}}, P_{\text{tab}}, R_{\text{tab}}, \text{cochain_action_rows}, \\ &\mathcal{R}, \text{generator_image_table}, \text{reduction_witnesses}, j_{\text{tab}}, j_{\text{tab}}^{-1}, \\ &d, \text{power_rows}, \text{invariant_classes}, \text{require_exact8}, \text{classification}, \\ &\text{first_failure or NONE}, \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}}). \end{aligned} \quad (23.64)$$

Canonical cohomology representatives, solver dispatch, permutation order, and byte encoding make repeated output identical.

For a nonfaithful positive test, take $P = R = 1$. Then $\overline{\mathcal{S}} = 1$, every class is invariant, and $d = 1$; this is a valid C_8 -representation. At the boundary $H^2 = 0$, the unique zero class is invariant and again $d = 1$.

For an exact-order test, write $G = C_{17}$ additively, take $M = \mathbb{F}_{17}$ with trivial action, let $P(a) = 2a$, and $R = 1$. The carry cocycles

$$f_c(a, b) = c \left\lfloor \frac{\bar{a} + \bar{b}}{17} \right\rfloor \pmod{17} \quad (c \in \mathbb{F}_{17}) \quad (23.65)$$

represent all classes in $H^2(C_{17}, \mathbb{F}_{17})$. Since $P^{-1}(a) = 9a$, pullback replaces the extension relation $x^{17} = c$ by $(x^9)^{17} = 9c$. Hence

$$\overline{\mathcal{S}}[f_c] = [f_{9c}]. \quad (23.66)$$

The scalar 9 has exact order eight modulo 17, so the representation has $d = 8$, and its only invariant class is $c = 0$. This example checks a genuine eight-cycle and rejects every proposed nonzero invariant class.

22.14 Automorphism lifting conditions

Let $E_f = M \times G$ have the twisted multiplication (23.16), and let (P, R) satisfy both (23.49) and (23.50). A lift of this pair is an automorphism $T : E_f \rightarrow E_f$ for which

$$T(m, 1) = (R(m), 1), \quad \pi T = P\pi. \quad (23.67)$$

For a normalized cochain $t : G \rightarrow M$, define

$$T_{R,P,t}(m, g) = (R(m) + t(g), P(g)). \quad (23.68)$$

As a set map it is bijective, with

$$T_{R,P,t}^{-1}(m, g) = (R^{-1}(m - t(P^{-1}g)), P^{-1}g). \quad (23.69)$$

Theorem 22.14.1 (Exact lift criterion). *Every lift satisfying (23.67) has the form (23.68) for a unique normalized t . Conversely, (23.68) is a lift if and only if*

$$Rf(g, h) + t(gh) = t(g) + P(g) \cdot t(h) + f(Pg, Ph) \quad (g, h \in G). \quad (23.70)$$

Such a lift exists if and only if the cohomology class $[f]$ is fixed by the C_8 -action of Statement 22.13.

Proof. The kernel and quotient conditions force $T(0, g) = (t(g), P(g))$ and

$$T(m, g) = T((m, 1)(0, g)) = (R(m) + t(g), P(g)).$$

An automorphism fixes the identity, so $t(1) = 0$, proving necessity and uniqueness of the form. Compare the images of a product:

$$\begin{aligned} T((m, g)(n, h)) &= (Rm + R(g \cdot n) + Rf(g, h) + t(gh), P(gh)), \\ T(m, g)T(n, h) &= (Rm + t(g) + P(g) \cdot (Rn + t(h)) + f(Pg, Ph), P(g)P(h)). \end{aligned} \quad (23.71)$$

Compatibility (23.49) cancels the terms containing n ; the remaining equality is exactly (23.70). Hence that equation makes the bijection (23.68) a homomorphism and therefore an automorphism satisfying (23.67).

Equation (23.70) is the direct form of (23.58). More explicitly, if $\mathcal{S}_2 f - f = \delta u$, set $t(x) = u(Px)$; then (23.58) is (23.70). Conversely, from a solution t , set $u(g) = t(P^{-1}g)$ and reverse the substitution to obtain (23.54). Thus a solution exists exactly when $[f]$ is invariant. $\square$

If t, t' are solutions for the same R, P , their difference $z = t' - t$ satisfies

$$z(gh) = z(g) + P(g) \cdot z(h). \quad (23.72)$$

This is not generally the ordinary $Z^1(G, M)$ equation. Rather, $w = z \circ P^{-1}$ belongs to $Z^1(G, M)$, and $z = w \circ P$. Hence the set of lifts, when nonempty, is a torsor for $Z^1(G, M)$ through this reparameterization.

Composition is explicit. Whenever both maps are automorphisms of the same twisted model,

$$T_{R,P,t} T_{S,Q,u} = T_{RS,PQ, R \circ u + t \circ Q}. \quad (23.73)$$

Induction gives

$$T_{R,P,t}^k(m, g) = \left(R^k m + \sum_{\ell=0}^{k-1} R^{k-1-\ell} t(P^\ell g), P^k g \right). \quad (23.74)$$

Therefore $T^8 = 1$ if and only if

$$R^8 = 1, \quad P^8 = 1, \quad \sum_{\ell=0}^7 R^{7-\ell} t(P^\ell g) = 0 \quad \text{for every } g \in G. \quad (23.75)$$

If (23.75) holds, exact order eight is equivalent to $T^4 \neq 1$. Existence of a lift and existence of a lift of exact order eight are different assertions.

Finite decision and representation algorithm. The input consists of canonical tables for

$$(G, M, \rho, f, P, R; \text{mode}, \text{require_exact8}), \quad (23.76)$$

where **mode** is **VALIDATE** with a supplied t , or **SEARCH** without one. The deterministic procedure is:

- STEP 1. Validate the mode and input shapes, the group and module tables, the action, cocycle, automorphisms, (23.49), and (23.50). Structural failure returns **INVALID_INPUT** with stage, first witness, and input hash.
- STEP 2. Recompute the induced C_8 -action on $[f]$. If it is not invariant, return **NO_LIFT** together with the non-coboundary certificate for $\mathcal{S}_2 f - f$.
- STEP 3. In validator mode, test (23.70) on all G^2 . Failure returns **CANDIDATE_FAIL**, not **NO_LIFT**.

STEP 4. In search mode enumerate all $|M|^{|G|-1}$ normalized cochains in truth-table order and retain exactly the solutions of (23.70). If exact order eight is required, retain only solutions passing (23.75) and $T^4 \neq 1$. Return the first retained cochain, or `NO_EXACT8_LIFT` only after the complete lift set has been exhausted. In that negative branch, store for every lift its full power rows and first failed exact-order condition; the ordered list is the complete nonexistence certificate.

STEP 5. Construct T, T^{-1} from (23.68)–(23.69), check both extension squares, the homomorphism table, both inverse identities, and all power rows. Compute the exact permutation order.

In validator mode a valid lift which fails a requested exact-order condition receives `ORDER_FAIL`; it does not invalidate the cohomological existence result.

Let $m_1, \dots, m_a$ generate M , and let $g_1, \dots, g_b$ generate G . The lift is completely represented by the generator images

$$\begin{array}{c|c} \text{generator} & T\text{-image} \\ \hline (m_i, 1) & (R(m_i), 1) \\ (0, g_j) & (t(g_j), P(g_j)). \end{array} \quad (23.77)$$

The algorithm also verifies the full element table, so this compact display cannot conceal a failure on relations.

Theorem 22.14.2 (Correctness and reproducibility). *The algorithm terminates. On structurally valid input its verification conditions are exactly the following:*

- (i) in `VALIDATE` mode without the exact-order request, the supplied t satisfies (23.70);
- (ii) in `VALIDATE` mode with that request, the supplied t satisfies (23.70), (23.75), and $T^4 \neq 1$;
- (iii) in `SEARCH` mode without that request, at least one normalized t satisfies (23.70);
- (iv) in `SEARCH` mode with that request, at least one normalized t satisfies (23.70), (23.75), and $T^4 \neq 1$.

The classification `NO_LIFT` means that no t satisfies (23.70), whereas `NO_EXACT8_LIFT` means that lifts exist but none has exact order eight. Every accepted map, inverse, extension square, and reported order is correct. Equal canonical inputs give byte-identical outputs.

Proof. Finiteness gives termination and exhaustive search. The lift theorem proves soundness and completeness for (23.70); (23.74)–(23.75) prove the order filter. Direct full-table checks prove every serialized mapping assertion independently. Fixed candidate, generator, failure, and serialization orders prove reproducibility. $\square$

The fixed record is

$$\begin{aligned} &(\mathbf{h}_G, \mathbf{h}_M, \mathbf{h}_f, \rho_{\text{tab}}, P_{\text{tab}}, R_{\text{tab}}, \text{mode}, \text{require_exact8}, \\ &\text{invariance_certificate}, \text{candidate_count}, t_{\text{tab}} \text{ or } \text{NONE}, \\ &T_{\text{tab}} \text{ or } \text{NONE}, T_{\text{tab}}^{-1} \text{ or } \text{NONE}, \text{generator_images}, \\ &\text{square_rows}, \text{homomorphism_rows}, \text{inverse_rows}, \text{power_rows}, \text{ord}(T) \text{ or } \text{NONE}, \\ &\text{search_failure_rows}, \text{classification}, \text{first_failure} \text{ or } \text{NONE}, \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}} \text{ or } \text{NONE}). \end{aligned} \quad (23.78)$$

The classification field takes exactly one value from

$$\begin{aligned} &\text{INVALID_INPUT}, \quad \text{NO_LIFT}, \quad \text{CANDIDATE_FAIL}, \quad \text{ORDER_FAIL}, \\ &\text{NO_EXACT8_LIFT}, \quad \text{LIFT_FOUND}, \quad \text{EXACT8_LIFT_FOUND}. \end{aligned}$$

The two success classifications have no failure witness. The two search nonexistence classifications carry the complete cohomological or exhaustive order certificate described above.

For an exact-order positive test, take the split extension with $M = C_2$, $G = C_{17}$, trivial action, $f = 0$, $R = 1$, $P(g) = 2g$, and $t = 0$. Then $T(m, g) = (m, 2g)$ has exact order eight. For the nonzero carry class in (23.65), (23.66) shows that the class is not invariant, so no lift exists. Finally, with $G = M = C_2$, $P = R = 1$, and $f = 0$, the class is invariant and lifts exist, but every solution of (23.70) is a homomorphism $G \rightarrow M$; the resulting lift has order one or two, never eight. This is the minimal test separating invariant-class lifting from the exact-order requirement.

22.15 The Bockstein correction

Let

$$0 \longrightarrow M' \xrightarrow{\iota} \widetilde{M} \xrightarrow{q} M'' \longrightarrow 0 \quad (23.79)$$

be a short exact sequence of left G -modules. The maps are G -equivariant. If $c \in Z^n(G, M'')$, choose a normalized cochain lift $\tilde{c} \in C^n(G, \widetilde{M})$ with $q\tilde{c} = c$. Since coboundaries commute with coefficient maps,

$$q(\delta\tilde{c}) = \delta(q\tilde{c}) = \delta c = 0. \quad (23.80)$$

Exactness gives a unique normalized $b_c \in C^{n+1}(G, M')$ such that

$$\iota(b_c) = \delta\tilde{c}. \quad (23.81)$$

The cochain b_c is the Bockstein correction attached to the chosen lift.

Theorem 22.15.1 (Connecting homomorphism and lift obstruction). *The cochain b_c is a cocycle, and*

$$\beta : H^n(G, M'') \longrightarrow H^{n+1}(G, M'), \quad \beta[c] = [b_c] \quad (23.82)$$

is a well-defined homomorphism. Moreover, $\beta[c] = 0$ if and only if c has a normalized cocycle lift $\hat{c} \in Z^n(G, \widetilde{M})$.

Proof. Apply δ to (23.81):

$$\iota(\delta b_c) = \delta(\iota b_c) = \delta^2\tilde{c} = 0.$$

Injectivity of ι gives $\delta b_c = 0$. If another lift is $\tilde{c}' = \tilde{c} + \iota a$, then

$$b'_c = b_c + \delta a, \quad (23.83)$$

so its class is unchanged. If $c' = c + \delta d$, choose a lift $\tilde{d}$ of d . The cochain $\tilde{c} + \delta\tilde{d}$ lifts c' and has the same coboundary as $\tilde{c}$; any other lift changes the result only by (23.83). Thus (23.82) is independent of both choices. Additivity follows by choosing the sum of two lifts.

If $\beta[c] = 0$, write $b_c = \delta a$. Then

$$\hat{c} = \tilde{c} - \iota a, \quad \delta\hat{c} = \iota(b_c - \delta a) = 0, \quad (23.84)$$

and $q\hat{c} = c$. Conversely, a cocycle lift has zero defect in (23.81), so its Bockstein class is zero. $\square$

The construction is natural. A commutative morphism of short exact sequences of G -modules carries a chosen lift and its defect to the corresponding lift and defect in the target sequence; hence the induced cohomology maps commute with β . If (23.79) splits G -equivariantly, every cocycle can be lifted through the splitting, so

$$\beta = 0. \quad (23.85)$$

Thus a nonzero Bockstein measures failure of an equivariant coefficient splitting at the cochain level.

For the standard mod- p sequence with trivial action,

$$0 \longrightarrow \mathbb{F}_p \xrightarrow{a \mapsto pa} \mathbb{Z}/p^2 \xrightarrow{q} \mathbb{F}_p \longrightarrow 0, \quad (23.86)$$

take $n = 1$ and lift a homomorphism $c : G \rightarrow \mathbb{F}_p$ by least residues $\tilde{c}(g) \in \{0, \dots, p-1\}$. Formula (23.81) becomes the fully computable carry

$$b_c(g, h) = \frac{\tilde{c}(g) + \tilde{c}(h) - \tilde{c}(gh)}{p} \pmod{p}. \quad (23.87)$$

The numerator is divisible by p because c is a homomorphism. Changing the integer representatives changes (23.87) by a coboundary, as predicted by (23.83). Ignoring this correction would incorrectly treat $\tilde{c}$ as a cocycle over $\mathbb{Z}/p^2$.

Finite computation and fixed output. The finite procedure below computes the degree- $n = 1$ Bockstein $H^1(G, M'') \rightarrow H^2(G, M')$; the theorem above remains valid in every degree. For finite canonical tables in (23.79), first verify all module tables, maps, exactness, and equivariance. Then verify that the input has one well-typed value $c(g)$ for every $g \in G$, that $c(1) = 0$, and that every row

$$c(gh) = c(g) + g \cdot c(h) \quad (23.87a)$$

holds. Failure returns `INVALID_INPUT` with the first normalization or cocycle witness. Only after this check, choose the lift of every value of c to be the first element of its q -fiber, with zero lifted to zero. Compute every defect row $\delta\tilde{c}$, invert ι uniquely to obtain b_c , and independently verify its cocycle equation. The cohomology engine of Statement 22.9 reduces b_c to the canonical class representative. If it is zero, the deterministic solver of Statement 22.11 finds the correction a in (23.84); the zero defect fixes $a = 0$, solver `NONE`, and certificate `ZERO`. If the class is nonzero, its reduction certificate is the obstruction witness.

The canonical record is

$$\begin{aligned} &(\mathbf{h}_{M'}, \mathbf{h}_{\tilde{M}}, \mathbf{h}_{M''}, \iota_{\text{tab}}, q_{\text{tab}}, \rho_{\text{tabs}}, n = 1, c_{\text{tab}}, \\ &\tilde{c}_{\text{tab}}, (\delta\tilde{c})_{\text{tab}}, b_c^{\text{tab}}, \text{cocycle_rows}, \text{beta_representative}, \\ &a_{\text{tab}} \text{ or } \text{NONE}, \text{correction_certificate}, \text{LIFTS or OBSTRUCTED}, \\ &\text{first_failure or NONE}, \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}}). \end{aligned} \quad (23.88)$$

Malformed input instead returns `INVALID_INPUT` with stage, first witness, and input hash. All loops are finite. Canonical fiber choices, cohomology reduction, solver dispatch, and serialization make repeated output identical. Equations (23.80)–(23.84) prove the correctness of every accepted record.

For the smallest nonzero test, let $G = C_p$ additively and $c(a) = a \in \mathbb{F}_p$. Formula (23.87) is

$$b_c(a, b) = \left\lfloor \frac{\bar{a} + \bar{b}}{p} \right\rfloor \pmod{p}. \quad (23.89)$$

This class is nonzero. Indeed, if $b_c = \delta t$, summing the rows $(a, 1)$ over $a \in C_p$ would give 1 on the left, while the right side telescopes to $p t(1) = 0$ in $\mathbb{F}_p$. It encodes the nonsplit extension C_{p^2} . The zero cocycle has zero Bockstein and lifts by zero. Likewise every class has zero Bockstein for a split coefficient sequence, verifying the boundary (23.85).

22.16 Power functions and commutator cocycles

Let p be prime, let V and A be finite $\mathbb{F}_p$ -vector spaces, and let $f : V^2 \rightarrow A$ be a normalized two-cocycle for the trivial action. The central extension $E_f = A \times V$ has multiplication

$$(a, x)(b, y) = (a + b + f(x, y), x + y). \quad (23.90)$$

We use the commutator convention $[u, v] = uvu^{-1}v^{-1}$. Define

$$\kappa_f(x, y) = f(x, y) - f(y, x). \quad (23.91)$$

Since V is abelian and A is central,

$$[(0, x), (0, y)] = (\kappa_f(x, y), 0). \quad (23.92)$$

Theorem 22.16.1 (Commutator and power invariants). *The map $\kappa_f : V \times V \rightarrow A$ is alternating and $\mathbb{F}_p$ -bilinear. It depends only on the cohomology class of f .*

For $n \geq 0$, put

$$q_{f,n}(x) = \sum_{r=0}^{n-1} f(rx, x), \quad q_{f,0}(x) = 0. \quad (23.93)$$

Here the $n = 0$ sum is empty. Then

$$(0, x)^n = (q_{f,n}(x), nx). \quad (23.94)$$

In particular, the p -power function

$$q_f(x) = q_{f,p}(x), \quad (0, x)^p = (q_f(x), 0) \quad (23.95)$$

is independent of the normalized section and of the cocycle representative. It satisfies

$$q_f(x + y) = q_f(x) + q_f(y) - \binom{p}{2} \kappa_f(x, y). \quad (23.96)$$

Thus q_f is linear for odd p , while for $p = 2$

$$q_f(x + y) = q_f(x) + q_f(y) + \kappa_f(x, y), \quad (23.97)$$

so q_f is a quadratic refinement of the commutator form.

Proof. Every commutator of E_f lies in the central subgroup A , so E_f has nilpotency class at most two. The class-two identities

$$[uv, w] = [u, w][v, w], \quad [u, vw] = [u, v][u, w] \quad (23.98)$$

prove additivity of (23.92) in both variables. Alternation is $[u, u] = 1$, and scalar linearity follows by repeated addition in the elementary abelian groups. If $f' = f + \delta t$, then $\delta t(x, y) = t(x) + t(y) - t(x + y)$ is symmetric, so its contribution to (23.91) cancels.

Formula (23.94) follows by induction: multiplying the n -th power by $(0, x)$ appends $f(nx, x)$ to the first coordinate. To show representative independence at $n = p$, telescope:

$$\begin{aligned} q_{f+\delta t,p}(x) - q_{f,p}(x) &= \sum_{r=0}^{p-1} \{t(rx) + t(x) - t((r+1)x)\} \\ &= p t(x) + t(0) - t(px) = 0, \end{aligned} \quad (23.99)$$

because A and V have exponent p . The same argument says that changing a section by t does not change p -th powers.

In a class-two group,

$$(uv)^p = u^p v^p [v, u]^{\binom{p}{2}}. \quad (23.100)$$

The section elements satisfy $(0, x)(0, y) = (f(x, y), 0)(0, x + y)$. The central factor has p -th power zero, so apply (23.100) and use $[(0, y), (0, x)] = (-\kappa_f(x, y), 0)$ to obtain (23.96). If p is odd, $p \mid \binom{p}{2}$, whereas for $p = 2$ the minus sign equals the plus sign. This proves (23.97). $\square$

The pair (κ_f, q_f) separates two kinds of extension data. The alternating form records noncommutativity. The power function records the carry, or Bockstein, component; it may be nonzero even when $\kappa_f = 0$. In particular, retaining only commutator rows would lose nonsplit abelian extensions.

Construction algorithm and I/O specification. The module

$$\text{PowerCommutator}(\mathcal{T}_V, \mathcal{T}_A, p, f_{\text{tab}}) \quad (23.101)$$

uses canonical vector and cocycle tables. Its pseudocode is:

- STEP 1. Verify that p is prime and that the supplied tables, bases, scalar multiplications, and additions define finite $\mathbb{F}_p$ -vector spaces. Structural failure returns `INVALID_INPUT` with stage and first witness.
- STEP 2. Require one well-typed value in every row of f . Test normalization and every cocycle identity. A well-formed failed candidate returns `CANDIDATE_FAIL` with its first normalization or cocycle defect.
- STEP 3. Construct the full group table (23.90), its inverses, and every commutator. Compute (23.91) independently and compare it with every row of (23.92).
- STEP 4. Test alternation, both additivity laws, and scalar linearity of κ_f exhaustively on V .
- STEP 5. Compute every partial sum $q_{f,n}(x)$ for $0 \leq n \leq p$, compare (23.94) with repeated multiplication, and test (23.96) on every ordered pair.
- STEP 6. If $p = 2$, serialize all quadratic-polarization rows (23.97); if p is odd, serialize all addition and scalar rows proving linearity of q_f .

Both early exits use the separate fixed record

$$\text{Reject}(\text{classification}, \text{stage}, \text{first_failure}, h_{\text{input}}),$$

where the classification is `INVALID_INPUT` or `CANDIDATE_FAIL`. Thus an early rejection never requires uncomputed group, commutator, or power fields.

Theorem 22.16.2 (Correctness and reproducibility). *The algorithm terminates. On structurally valid input it returns `SATISFIED` exactly when f is a normalized two-cocycle; every reported commutator, power, bilinearity, and polarization row is then correct. Equal canonical inputs produce byte-identical outputs.*

Proof. Every loop is finite. Step 2 is exactly the cocycle criterion of Statement 22.10. The theorem above proves the formulas used in Steps 3–6, while the redundant full group and repeated-power tables independently check their implementation. Fixed basis, element, pair, triple, partial-sum, failure, and byte orders prove reproducibility. $\square$

The fixed successful output is

$$\begin{aligned} &(p, h_V, h_A, h_f, \text{group_table}, \text{inverse_rows}, \\ &\kappa_f^{\text{tab}}, \text{commutator_rows}, \text{bilinearity_rows}, q_{f,n}^{\text{tab}} \ (0 \leq n \leq p), \\ &\text{power_rows}, \text{polarization_rows}, \text{classification}, \text{first_failure or NONE}, \\ &h_{\text{input}}, h_{\text{independentverification}} \text{ or NONE}). \end{aligned} \quad (23.102)$$

Its classification is `SATISFIED`; the other two classifications occur only in the separate rejection record above.

For the carry test, let $V = A = \mathbb{F}_p$ and

$$f(a, b) = \left\lfloor \frac{\bar{a} + \bar{b}}{p} \right\rfloor \pmod{p}. \quad (23.103)$$

Then $\kappa_f = 0$, while $q_f(x) = x$; the extension is C_{p^2} , and the power function is exactly the Bockstein correction (23.89). For odd p , the Heisenberg cocycle on $V = \mathbb{F}_p^2$,

$$f((x, y), (x', y')) = xy', \quad (23.104)$$

has

$$\kappa_f((x, y), (x', y')) = xy' - x'y, \quad q_f = 0. \quad (23.105)$$

For $p = 2$, the same formula gives $q_f(x, y) = xy$, whose polarization is the alternating form in (23.105). These tests respectively isolate the power component, the commutator component, and their quadratic interaction. A table violating one row of (23.4) supplies the negative candidate test.

22.17 Nonsplit C_8 -equivariant extensions

Let M be a finite left G -module, let $f \in Z^2(G, M)$, and let E_f be the twisted extension (23.16). Fix a compatible pair (P, R) satisfying (23.49)–(23.50). A C_8 -equivariant extension consists of a homomorphism

$$\alpha : C_8 \longrightarrow \text{Aut}(E_f) \quad (23.106)$$

whose distinguished generator acts on the kernel by R and on the quotient by P . If $T = \alpha(\zeta)$, this is equivalent to

$$T = T_{R,P,t} \text{ satisfying (23.70),} \quad T^8 = 1. \quad (23.107)$$

The action is faithful exactly when T has exact order eight, or equivalently, under (23.107), when $T^4 \neq 1$.

Theorem 22.17.1 (Existence of a nonsplit equivariant extension). *The data (M, G, ρ, f, P, R) admit a nonsplit C_8 -equivariant extension if and only if*

$$\begin{aligned} [f] &\neq 0 \quad \text{in } H^2(G, M), \\ \text{there is a normalized } t : G &\rightarrow M \text{ satisfying (23.70),} \\ \sum_{\ell=0}^7 R^{7-\ell} t(P^\ell g) &= 0 \quad \text{for every } g \in G. \end{aligned} \quad (23.108)$$

It admits a faithful such action if and only if t can additionally be chosen so that $T_{R,P,t}^4 \neq 1$.

Proof. By Statement 22.12, the extension splits exactly when $[f] = 0$, proving the first condition. By Statement 22.14, the second condition is exactly the existence of the required lift. Since (23.50) is already assumed, formula (23.75) says that its eighth power is the identity exactly when the third condition holds. Then $\zeta^k \mapsto T^k$ is a well-defined homomorphism from C_8 . Conversely, any equivariant extension supplies $T = \alpha(\zeta)$, hence all three conditions. The kernel of a homomorphism from C_8 is trivial precisely when the image of its generator has order eight, which gives the faithful statement. $\square$

Once $T^8 = 1$, the action may also be internalized as the finite semidirect product

$$\widehat{E} = E_f \rtimes_T C_8, \quad (e, k)(e', \ell) = (eT^k(e'), k + \ell \bmod 8). \quad (23.109)$$

It fits into

$$1 \longrightarrow E_f \longrightarrow \widehat{E} \longrightarrow C_8 \longrightarrow 1, \quad |\widehat{E}| = 8|M||G|. \quad (23.110)$$

Writing $z = (1_{E_f}, 1)$, the defining covariance relations are

$$z^8 = 1, \quad z(e, 0)z^{-1} = (T(e), 0) \quad (e \in E_f). \quad (23.111)$$

These hold whether or not the action is faithful; faithfulness concerns the order of the conjugation automorphism, not the order of z in the semidirect product.

Construction algorithm. The input is

$$(\mathcal{T}_G, \mathcal{T}_M, \rho_{\text{tab}}, f_{\text{tab}}, P_{\text{tab}}, R_{\text{tab}}; \text{require_faithful}). \quad (23.112)$$

The flag is Boolean and belongs to the canonical input. The deterministic pseudocode is:

- STEP 1. Verify every finite table, the module action, the candidate cocycle, the automorphisms, semilinear compatibility, the eighth-power relations, and the Boolean flag. Structural failure returns `INVALID_INPUT`; a well-formed noncocycle returns `CANDIDATE_FAIL`.
- STEP 2. Apply the deterministic Statement 22.11 solver to f . If $f = \delta a$, return `SPLIT_REJECT` with the canonical splitting witness a and all verification rows $\delta a(g, h) = f(g, h)$. If not, retain the complete non-coboundary certificate.
- STEP 3. Enumerate every normalized $t : G \rightarrow M$ in truth-table order and retain exactly those satisfying (23.70). If none exist, return `NO_LIFT` with the Statement 22.13 noninvariance certificate.
- STEP 4. For every retained lift, compute its power rows from (23.74). Retain those satisfying $T^8 = 1$; if the faithful flag is true, also require $T^4 \neq 1$. Choose the first survivor.
- STEP 5. If no lift satisfies $T^8 = 1$, return `NO_C8_LIFT`. If some satisfy that relation but none is faithful when faithfulness is required, return `NO_FAITHFUL_C8_LIFT`. In either case store every lift's power table and first failed relation, proving exhaustive nonexistence.
- STEP 6. Construct E_f , T, T^{-1} , the generator-image table (23.77), and $\hat{E}$. Check all group tables, extension maps, inverse and covariance rows, (23.111), exact orders, and (23.110).
- STEP 7. When G and M satisfy the elementary central hypotheses of Statement 22.16, also compute and attach the power and commutator invariant certificate; otherwise serialize that optional field as `NA`.

Theorem 22.17.2 (Algorithm correctness). *The algorithm terminates and returns `SATISFIED` exactly when the input admits the nonsplit C_8 -equivariant extension described in (23.108), with faithfulness additionally imposed exactly when the flag is true. Every accepted relation, order, and covariance identity is correct. Equal canonical inputs give byte-identical output.*

Proof. The cochain and element searches are finite. Step 2 is equivalent to nonsplitting, Steps 3–5 exhaust exactly the witnesses in (23.108), and the faithful filter is the last assertion of the existence theorem. Thus verification is necessary and sufficient. Step 6 evaluates every operation and relation in the constructed finite tables; (23.109) and the semidirect-product theorem prove associativity and (23.110). Fixed cochain, witness, power, generator, and serialization orders prove reproducibility. $\square$

The generator-level representation is

generator of E_f	T -image	relation in $\hat{E}$	verified image
$(m_i, 1)$	$(R(m_i), 1)$	z^8	1
$(0, g_j)$	$(t(g_j), P(g_j))$	zez^{-1}	$T(e)$.

(23.113)

Full-table verification supplements these compact generator rows.

The canonical record is

($h_G, h_M, h_f, \rho_{\text{tab}}, P_{\text{tab}}, R_{\text{tab}}, \text{require_faithful},$
 $\text{nonsplit_certificate}, \text{lift_candidate_count}, \text{lift_failure_rows}, t_{\text{tab}}$ or **NONE**,
 $a_{\text{split}}^{\text{tab}}$ or **NONE**, $\text{split_verification_rows}$ or **NONE**,
 T_{tab} or **NONE**, $\text{ord}(T)$ or **NONE**, generator_images , covariance_rows ,
 $\widehat{T}_E$ or **NONE**, $|\widehat{E}|$ or **NONE**, $\text{power_commutator_certificate}$ or **NA** or **NONE**,
 $\text{classification}, \text{stage}$ or **COMPLETE**, first_failure or **NONE**, $h_{\text{input}}, h_{\text{independentverification}}$ or **NONE**).
(23.114)

The mutually exclusive classifications are

INVALID_INPUT, CANDIDATE_FAIL, SPLIT_REJECT, NO_LIFT,
NO_C8_LIFT, NO_FAITHFUL_C8_LIFT, SATISFIED.

On **SATISFIED**, the stage is **COMPLETE**. On rejection it is the first failing stage. The two split fields are populated exactly for **SPLIT_REJECT**; the nonsplit certificate is populated only after the non-coboundary branch of Step 2. The optional power-commutator field is **NA** only on a successful input outside the hypotheses of Statement 22.16, and is **NONE** on an exit before Step 7. All other fields unavailable after an early exit are fixed to **NONE**.

For a nonfaithful positive test, take the nonsplit C_2 -by- C_2 extension C_4 and $P = R = T = 1$. It gives the valid action $C_8 \rightarrow \text{Aut}(C_4)$ with trivial image. For a faithful test, set

$$M = C_2, \quad G = C_2 \times C_{17}, \quad f((a, x), (b, y)) = \left\lfloor \frac{a+b}{2} \right\rfloor \pmod{2}, \quad (23.115)$$

take $P(a, x) = (a, 2x)$, $R = 1$, and $t = 0$. The extension is $E_f \cong C_4 \times C_{17}$, nonsplit because its restriction over the C_2 -factor is C_4 ; T has exact order eight through the C_{17} -factor, and $|\widehat{E}| = 544$. The nonzero carry class in (23.65) is a negative equivariance test because (23.66) makes it noninvariant.

To force the distinct **NO_C8_LIFT** branch, take

$$M = C_2, \quad G = C_{32}, \quad f(a, b) = \left\lfloor \frac{\bar{a} + \bar{b}}{32} \right\rfloor \pmod{2}, \quad P(a) = 3a, \quad R = 1. \quad (23.116)$$

Then $E_f \cong C_{64}$, and it is nonsplit. An automorphism of C_{64} lifting P and R is multiplication by a unit $u \equiv 3 \pmod{32}$; hence the complete lift list is $u = 3, 35 \pmod{64}$. Both lifts exist, but direct repeated squaring gives

$$3^8 \equiv 35^8 \equiv 33 \not\equiv 1 \pmod{64}.$$

Consequently every lift fails $T^8 = 1$, exactly producing **NO_C8_LIFT**. The earlier C_4 example with $\text{require_faithful} = \text{true}$ produces **NO_FAITHFUL_C8_LIFT**. Finally, $f = 0$ is rejected as **SPLIT_REJECT** with the explicit witness $a = 0$ and all rows $\delta a = f = 0$.

22.18 Exact finite-extension construction algorithm

This construction completes the cohomological analysis by turning a class of $H^2(G, M)$, an optional cocycle table, and optional automorphism-lift requirements into finite multiplication tables and independently verifiable certificates. Throughout, G is a finite group, M is a finite abelian group written additively, and $\rho : G \rightarrow \text{Aut}(M)$ is a left action. The identities of G and M occur first in their canonical element orders.

Canonical class catalogue. Order normalized cochains lexicographically by their value tables. Enumerate $C_N^1(G, M)$ and $C_N^2(G, M)$, and form

$$\begin{aligned} Z^2 &= \{f \in C_N^2 : g \cdot f(h, k) - f(gh, k) + f(g, hk) - f(g, h) = 0 \text{ for all } g, h, k\}, \\ B^2 &= \{\delta a : a \in C_N^1\}, \quad f \sim f' \iff f' - f \in B^2. \end{aligned} \quad (23.117)$$

For every coset of B^2 in Z^2 , select its least table, then sort the selected tables. This gives

$$\mathcal{R} = (r_0, \dots, r_{q-1}), \quad r_0 = 0, \quad H^2(G, M) = \{[r_0], \dots, [r_{q-1}]\}. \quad (23.118)$$

The zero table is least because zero is first in the order on M . Thus (23.118), including the index of every class, is determined by the input tables rather than by a choice of basis or solver.

Lemma 22.18.1 (Canonicalization and transport). *Suppose that a normalized candidate f_{in} represents $[r_j]$. The finite search for the least normalized $a : G \rightarrow M$ satisfying*

$$f_{\text{in}} = r_j + \delta a \quad (23.119)$$

terminates. Moreover

$$\Phi_a : E_{f_{\text{in}}} \longrightarrow E_{r_j}, \quad \Phi_a(m, g) = (m + a(g), g) \quad (23.120)$$

is an extension isomorphism. If $T_{R,P,t}$ is a lift on E_{r_j} , its transport to the supplied model is

$$\Phi_a^{-1} T_{R,P,t} \Phi_a = T_{R,P,t_{\text{in}}}, \quad t_{\text{in}}(g) = R(a(g)) + t(g) - a(Pg). \quad (23.121)$$

Proof. Membership in $[r_j]$ is exactly the existence assertion in (23.119); the cochain set is finite, so the least witness exists. For $(m, g), (n, h) \in E_{f_{\text{in}}}$, the first coordinate of the image of their product is

$$m + g \cdot n + r_j(g, h) + \delta a(g, h) + a(gh) = m + a(g) + g \cdot (n + a(h)) + r_j(g, h).$$

This is the first coordinate of $\Phi_a(m, g)\Phi_a(n, h)$ in E_{r_j} ; the second coordinates are both gh . The map is bijective with inverse $(m, g) \mapsto (m - a(g), g)$, and it fixes M and G . Conjugating $T_{R,P,t}$ by these two formulas gives (23.121) term by term. $\square$

Input specification. The canonical input is

$$\begin{aligned} \mathcal{I} = (\mathcal{T}_G, \mathcal{T}_M, \rho_{\text{tab}}; \text{source}, j \text{ or NONE}, f_{\text{in}} \text{ or NONE}; \\ \text{require_nonsplit}, \text{equivariance}, P_{\text{tab}} \text{ or NONE}, R_{\text{tab}} \text{ or NONE}). \end{aligned} \quad (23.122)$$

where

$$\begin{aligned} \text{source} &\in \{\text{CLASS}, \text{COCYCLE}, \text{BOTH}\}, \\ \text{equivariance} &\in \{\text{NONE}, \text{LIFT}, \text{C8}, \text{FAITHFUL_C8}\}. \end{aligned} \quad (23.123)$$

In **CLASS** mode only j is present; in **COCYCLE** mode only f_{in} is present; in **BOTH** mode both are present and must describe the same class. The nonsplit flag is Boolean. The tables P, R are absent exactly in **NONE** mode. Otherwise they must be compatible automorphisms as in (23.49); the two C_8 modes additionally require $P^8 = R^8 = 1$. These shape requirements are part of input validity, not mathematical rejection branches.

Deterministic pseudocode.

- STEP 1. Validate the two finite algebra tables, the action axioms, all modes and presence rules, and the Boolean flag. When present, validate P, R , semilinear compatibility, and the required eighth-power relations. A structural defect returns `INVALID_INPUT` with its stage and first failed row.
- STEP 2. Construct (23.117)–(23.118) by exhaustive canonical enumeration. Verify independently that B^2 is a subgroup of Z^2 , that the listed cosets are disjoint and exhaustive, and that every representative is the least member of its coset.
- STEP 3. Resolve the source. A supplied j must satisfy $0 \leq j < q$; an out-of-range or ill-typed index returns `INVALID_INPUT`. A supplied f_{in} must be well typed, normalized, and satisfy every cocycle row; a well-formed failure returns `CANDIDATE_FAIL`. Locate its unique class and compute the least witness (23.119). In `BOTH` mode a different located index returns `CLASS_MISMATCH` together with both indices and the complete coset certificate. Set the working cocycle to $f = r_j$.
- STEP 4. If `require_nonsplit` is true and $j = 0$, return `SPLIT_REJECT`. Store $a = 0$ for the canonical model and, when a candidate was supplied, its witness (23.119), which then proves $f_{\text{in}} = \delta a$.
- STEP 5. Construct $E_f = M \times G$ using

$$(m, g)(n, h) = (m + g \cdot n + f(g, h), gh), \quad (23.124)$$

identity $(0, 1)$, and inverse

$$(m, g)^{-1} = (-g^{-1} \cdot (m + f(g, g^{-1})), g^{-1}). \quad (23.125)$$

Build the full multiplication and inverse tables and check every associativity, identity, and inverse row.

- STEP 6. Serialize

$$i(m) = (m, 1), \quad \pi(m, g) = g, \quad s(g) = (0, g), \quad (23.126)$$

and exhaustively verify that i is injective, π is a surjective homomorphism, and

$$\text{im } i = \ker \pi, \quad s(g)i(m)s(g)^{-1} = i(g \cdot m), \quad s(g)s(h) = i(f(g, h))s(gh). \quad (23.127)$$

Also verify $|E_f| = |M||G|$.

- STEP 7. If the equivariance mode is `NONE`, go to Step 10. Otherwise enumerate all normalized $t : G \rightarrow M$, retaining exactly the solutions of

$$Rf(g, h) + t(gh) = t(g) + P(g) \cdot t(h) + f(Pg, Ph). \quad (23.128)$$

If there is no solution, return `NO_LIFT` with every candidate's first failed row and a non-coboundary certificate for the semilinear transform of f minus f . In either `C8` mode this is exactly the Statement 22.13 noninvariance certificate. In `LIFT` mode choose the first solution.

- STEP 8. In either `C8` mode compute, for every retained t ,

$$N_8(t)(g) = \sum_{\ell=0}^7 R^{7-\ell} t(P^\ell g). \quad (23.129)$$

Retain exactly the cochains with $N_8(t) = 0$. If there are none, return `NO_C8_LIFT` with the complete ordered list of norm failures. In `FAITHFUL_C8` mode retain further only the lifts with $T_{R,P,t}^4 \neq 1$. If none remain, return `NO_FAITHFUL_C8_LIFT` with all fourth-power rows. Choose the first surviving t .

STEP 9. Construct T, T^{-1} by (23.68)–(23.69), verify its complete homomorphism, inverse, kernel, quotient, and power tables, and compute its exact permutation order. In a C_8 mode also construct $E_f \rtimes_T C_8$ by (23.109) and verify every covariance row and the order $8|M||G|$. If a candidate model was supplied, attach Φ_a ; when a lift was requested, also attach the transported lift (23.121), checking every available table independently.

STEP 10. When the hypotheses of Statement 22.16 hold, attach (κ_f, q_f) and all polarization rows. Otherwise record NA. Recompute every hash from the serialized bytes and independent verification Steps 3–9 from the record before returning SATISFIED with stage COMPLETE.

Theorem 22.18.2 (Exactness and defining relations). *If Steps 1–6 are reached, (23.124) defines a group and*

$$1 \longrightarrow M \xrightarrow{i} E_f \xrightarrow{\pi} G \longrightarrow 1 \quad (23.130)$$

is exact. Its extension class is $[r_j]$, its induced action on M is ρ , and (23.127), together with the multiplication tables of M and G , is a complete set of normal-form relations.

Proof. The two possible first coordinates of a triple product are

$$\begin{aligned} ((m, g)(n, h))(u, k) &= m + g \cdot n + f(g, h) + (gh) \cdot u + f(gh, k), \\ (m, g)((n, h)(u, k)) &= m + g \cdot n + (gh) \cdot u + g \cdot f(h, k) + f(g, hk). \end{aligned} \quad (23.131)$$

They agree exactly by the cocycle identity in (23.117). Normalization gives the identity, and substituting (23.125) on the right gives zero in the first coordinate. The left-inverse calculation agrees because the cocycle row at (g^{-1}, g, g^{-1}) gives $f(g^{-1}, g) = g^{-1} \cdot f(g, g^{-1})$.

The formulas in (23.126) show directly that i is a homomorphism and injective, while π is a homomorphism and is surjective because $\pi s = 1_G$. Its kernel is precisely $M \times \{1\}$, which is $\text{im } i$. Direct multiplication proves both remaining relations in (23.127). They move every kernel letter to the left and reduce every word uniquely to $(m, g) = i(m)s(g)$; hence no additional relation is hidden. The factor set of s is $f = r_j$, so the class is $[r_j]$, and conjugation gives the prescribed action. $\square$

Theorem 22.18.3 (Algorithm soundness, completeness, and reproducibility). *The algorithm terminates. On SATISFIED it returns an exact finite extension representing precisely the requested class, together with a valid lift, C_8 -action, or faithful C_8 -action exactly as requested. Conversely, every object satisfying the input specification and the requested nonsplit and equivariance conditions causes SATISFIED. Equal canonical inputs produce byte-identical outputs.*

Proof. Every cochain set, coset, lift set, and group table is finite, so all searches terminate. Equations (23.117)–(23.119) prove that Step 3 finds the unique requested class and loses no candidate model; the transport lemma proves the latter assertion also for lifts. Statement 22.12 gives the split criterion used in Step 4. The preceding theorem proves the group and exact-sequence assertions.

For completeness in the LIFT mode, no eighth-power hypothesis is needed. Put $T(m, g) = (R(m) + t(g), P(g))$. Directly,

$$\begin{aligned} T((m, g)(n, h)) &= (Rm + R(g \cdot n) + Rf(g, h) + t(gh), P(gh)), \\ T(m, g)T(n, h) &= (Rm + t(g) + P(g) \cdot (Rn + t(h)) + f(Pg, Ph), P(g)P(h)). \end{aligned}$$

Bijectivity of P, R and semilinear compatibility (23.49) show that these rows agree exactly when (23.128) holds. Hence that equation is necessary and sufficient for a lift even without (23.50). In the two C_8 modes, the iterate formula (23.74) proves that, under the separately validated equations $P^8 = R^8 = 1$, (23.129) vanishes exactly when $T^8 = 1$. Among such lifts, $T^4 \neq 1$

is equivalent to exact order eight. Thus the successive exhaustive filters are both sound and complete, including all three nonexistence branches. Fixed element, cochain, coset, witness, failure, and byte orders make every choice canonical; the final independent verification check proves reproducibility. $\square$

Output and failure specification. The success record is

$$\begin{aligned}
& (h_{\text{input}}, h_G, h_M, h_\rho, \text{source}, \text{equivariance}, \text{require_nonsplit}, \\
& P_{\text{tab}} \text{ or NONE}, R_{\text{tab}} \text{ or NONE}, \mathcal{R}, j, f_{\text{can}}, f_{\text{in}} \text{ or NONE}, \\
& a_{\text{transfer}} \text{ or NONE}, \text{transfer_rows} \text{ or NONE}, \mathcal{T}_{E_f}, i_{\text{tab}}, \pi_{\text{tab}}, s_{\text{tab}}, \\
& \text{associativity_rows}, \text{inverse_rows}, \text{exactness_rows}, \text{relation_rows}, |E_f|, \\
& \text{lift_candidate_count}, t_{\text{tab}} \text{ or NONE}, T_{\text{tab}} \text{ or NONE}, \text{ord}(T) \text{ or NONE}, \\
& \widehat{\mathcal{T}}_E \text{ or NONE}, \text{power_commutator_certificate} \text{ or NA or NONE}, \\
& \text{classification}, \text{COMPLETE}, \text{NONE}, h_{\text{independentverification}}).
\end{aligned} \tag{23.132}$$

A rejection record retains the common input and class-resolution fields, then the stage, first failure, and every certificate computed before that stage; all unavailable later fields are **NONE**. The mutually exclusive classifications are

$$\begin{aligned}
& \text{INVALID_INPUT}, \text{CANDIDATE_FAIL}, \text{CLASS_MISMATCH}, \text{SPLIT_REJECT}, \\
& \text{NO_LIFT}, \text{NO_C8_LIFT}, \text{NO_FAITHFUL_C8_LIFT}, \text{SATISFIED}.
\end{aligned} \tag{23.133}$$

In particular, a malformed table is never confused with a mathematical nonexistence result, and a failed supplied cocycle is never reported as evidence that its alleged class has no lift.

Minimum test family. The trivial quotient checks $1 \rightarrow M \rightarrow M \rightarrow 1$. With $M = G = C_2$, the zero cocycle produces $C_2 \times C_2$ and triggers **SPLIT_REJECT** only when nonsplitting is required; the carry cocycle produces C_4 . For a genuinely nonzero transfer test, take $G = M = C_3 = \langle s \rangle$ with trivial action, ordered as $1 < s < s^2$ and $0 < 1 < 2$, and use the least normalized witness

$$a(1) = 0, \quad a(s) = 0, \quad a(s^2) = 1, \quad (\delta a)_{\{s, s^2\}^2} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \neq 0.$$

Run **BOTH** with $j = 0$, $f_{\text{in}} = \delta a$, $\text{require_nonsplit} = \text{false}$, $P(g) = -g$, and $R = 1$. On the canonical model $r_0 = 0$, choose $t = 0$. Then (23.121) gives

$$t_{\text{in}}(s) = a(s) - a(s^2) = 2, \quad t_{\text{in}}(s^2) = a(s^2) - a(s) = 1,$$

so the test exercises the nonzero witness in (23.119), the isomorphism (23.120), and the sign of the transported lift in (23.121). Separately, pairing the C_2 carry candidate with the zero-class index checks **CLASS_MISMATCH**. A single corrupted cocycle row checks **CANDIDATE_FAIL**, and a malformed operation row checks **INVALID_INPUT**.

For equivariance branches, the nonzero C_{17} carry class in (23.65)–(23.66) checks **NO_LIFT**. The $C_2 \rightarrow C_{64} \rightarrow C_{32}$ construction (23.116) checks **NO_C8_LIFT**. The nonsplit C_4 extension with trivial action checks **NO_FAITHFUL_C8_LIFT** when faithfulness is required. Finally,

$$M = C_2, \quad G = C_2 \times C_{17}, \quad f((a, x), (b, y)) = \left\lfloor \frac{a+b}{2} \right\rfloor \bmod 2, \quad P(a, x) = (a, 2x), \quad R = 1 \tag{23.134}$$

produces the faithful positive output $E_f \cong C_4 \times C_{17}$, $\text{ord}(T) = 8$, and $|E_f \rtimes_T C_8| = 544$. Thus every classification in (23.133), every source mode, and every equivariance mode has a finite deterministic test.

Part IX

Finite-field module extension constructions

22.19 Finite-field modules

The module constructor starts from exact finite data rather than an unspecified field object. Let p be an integer, let $e \geq 1$, and let $\mu(X) \in \mathbb{F}_p[X]$ be monic of degree e . When p is prime and μ is irreducible, put

$$K = \mathbb{F}_p[X]/(\mu), \quad q = |K| = p^e. \quad (24.1)$$

Residue classes are represented by coefficient vectors $(c_0, \dots, c_{e-1})$, ordered lexicographically with the zero vector first. Addition is coefficientwise modulo p ; multiplication is polynomial multiplication followed by the unique remainder modulo μ .

Lemma 22.19.1 (Certified field construction). *Let $\Lambda(e)$ be the set of prime divisors of e . After primality of p and the degree and leading coefficient of μ have been checked, the quotient in (24.1) is a field if and only if*

$$\begin{aligned} X^{p^\ell} &\equiv X \pmod{\mu}, \\ \gcd(\mu, X^{p^{e/\ell}} - X) &= 1 \quad \text{for every } \ell \in \Lambda(e). \end{aligned} \quad (24.2)$$

When these tests pass, every nonzero residue u has a unique inverse, computable by the extended Euclidean algorithm.

Proof. The first congruence says that every irreducible factor degree divides e ; it also excludes repeated factors because $X^{p^\ell} - X$ has derivative -1 . For a prime divisor $\ell \mid e$, the gcd condition excludes every factor whose degree divides e/ℓ . If a monic degree- e polynomial passing the tests were reducible, each irreducible factor would have some proper degree $d \mid e$. Choose a prime $\ell \mid (e/d)$. Then $d \mid e/\ell$, so that factor would divide the corresponding gcd, a contradiction. The converse follows because an irreducible degree- e polynomial divides $X^{p^\ell} - X$ and divides none of the proper-subfield polynomials in (24.2). Bezout's identity gives $a(X)u(X) + b(X)\mu(X) = 1$ for $u \neq 0$, so $a \bmod \mu$ is the inverse; uniqueness follows by cancellation in a field. $\square$

Let Q be a finite group with a verified multiplication table and an ordered tuple of proposed generators $\mathcal{X} = (x_1, \dots, x_r)$. Fix $d \geq 0$ and matrices

$$A_i \in \text{Mat}_d(K) \quad (1 \leq i \leq r). \quad (24.3)$$

The proposed module is the additive group

$$V = K^d, \quad |V| = q^d = p^{ed}, \quad (24.4)$$

with generator action $x_i \cdot v = A_i v$. The case $d = 0$ is allowed: $V = 0$, its unique 0-by-0 matrix is the identity, and $|V| = 1$.

Canonical word and matrix construction. Use breadth-first search in the Cayley graph with letters ordered

$$x_1, x_1^{-1}, x_2, x_2^{-1}, \dots, x_r, x_r^{-1} \quad (24.5)$$

to assign every reached $g \in Q$ its shortest, then lexicographically least, word w_g . Put

$$A_g = w_g(A_1, \dots, A_r), \quad A_{1_Q} = I_d, \quad (24.6)$$

where an inverse letter is evaluated using A_i^{-1} . Here A_{1_Q} is the matrix of the identity element of Q ; A_1 remains the image of the first listed generator x_1 .

Theorem 22.19.2 (Finite-field module criterion). *The input data determine, by the canonical word construction, a well-defined left $K[Q]$ -module on V with the prescribed generator matrices if and only if:*

$$\langle x_1, \dots, x_r \rangle = Q, \quad A_i \in \text{GL}_d(K), \quad A_g A_h = A_{gh} \quad (g, h \in Q). \quad (24.7)$$

Under (24.7), the complete representation and group-algebra action are

$$\rho : Q \rightarrow \text{GL}_d(K), \quad \rho(g) = A_g, \quad \left(\sum_{g \in Q} c_g g \right) v = \sum_{g \in Q} c_g A_g v. \quad (24.8)$$

Proof. Necessity follows from a representation: the listed generators must reach every group element for (24.6) to construct the requested full table; generator images are invertible; and the homomorphism law is the last identity in (24.7).

Conversely, the first condition makes A_g defined for every g . The empty word gives $A_{1_Q} = I_d$, and every A_g is invertible because it is a word in invertible generator matrices. The full table identity also gives

$$A_{g^{-1}} A_g = A_{1_Q} = A_g A_{g^{-1}}, \quad (24.9)$$

so all A_g are invertible and $\rho(gh) = \rho(g)\rho(h)$. Thus

$$1 \cdot v = v, \quad (gh) \cdot v = g \cdot (h \cdot v), \quad g \cdot (av + bw) = a(g \cdot v) + b(g \cdot w), \quad (24.10)$$

which are precisely the left module axioms. Induction on word length then shows that any word representing g , not only the selected w_g , evaluates to A_g ; this separately proves word independence. Bilinear extension yields (24.8), and associativity of the group algebra follows from the homomorphism law. $\square$

Executable construction and I/O specification. The input is

$$\text{FiniteFieldModule}(p, e, \mu_{\text{coeff}}; \mathcal{T}_Q, \mathcal{X}, d, A_1^{\text{in}}, \dots, A_r^{\text{in}}). \quad (24.11)$$

Here all integers, field coefficients, group elements, generators, matrix rows, and bytes use their declared canonical orders. The deterministic pseudocode is:

- STEP 1. Check types, bounds $p \geq 2, e \geq 1, d \geq 0$, the polynomial shape, the group table, the ordered generator tuple, and every matrix shape. An ill-typed or malformed object returns `INVALID_INPUT` with the first failed row.
- STEP 2. Test p by trial division through $\lfloor \sqrt{p} \rfloor$, factor e in increasing order, and run (24.2) using repeated squaring and monic Euclidean remainders. Composite p or a failed irreducibility row returns `FIELD_FAIL` with its divisor, gcd, or congruence witness.
- STEP 3. Enumerate the q residue vectors, construct the complete addition, multiplication, additive-inverse, and nonzero multiplicative-inverse tables, and independently check all field axioms and $|K| = p^e$.
- STEP 4. Run the ordered search (24.5). If some element of Q is unreachable, return the classification

`GENERATION_FAIL`

with the reached subgroup and least missing element.

- STEP 5. Row-reduce every A_i^{in} over the constructed field. A singular generator image returns `MATRIX_FAIL` with its rank, pivot rows, and nonzero kernel vector. Otherwise compute all inverse matrices.

- STEP 6. Evaluate (24.6) along the canonical word tree, then test every ordered pair in (24.7). The first failed pair returns `REPRESENTATION_FAIL` with $g, h, gh, A_g A_h, A_{gh}$.
- STEP 7. Enumerate $V = K^d$, construct all addition and scalar tables and every action row $g \cdot v = A_g v$, and independently test (24.10). Serialize bases, coordinate maps, generator images, full matrices, canonical words, and the group-algebra interface.
- STEP 8. Recompute $|V| = q^d = p^{ed}$, hash every canonical input and output table, independent verification Steps 3–7, and return `SATISFIED` only if the independent verification bytes agree.

Theorem 22.19.3 (Algorithm correctness and reproducibility). *The procedure terminates. It returns `SATISFIED` if and only if the well-typed input has prime p , irreducible μ , and satisfies all three conditions in (24.7), including $\langle \mathcal{X} \rangle = Q$. Equivalently, the canonical word data determine a representation $\rho : Q \rightarrow \text{GL}_d(K)$ with precisely the prescribed generator images. On success all field, vector-space, action, and group-algebra tables are correct, and equal canonical inputs produce byte-identical outputs.*

Proof. Trial division, polynomial powering, finite table checks, breadth-first search, Gaussian elimination, and all action loops terminate. The field lemma proves Step 2 in both directions. The search is exhaustive, so Step 4 succeeds exactly when $\mathcal{X}$ generates Q . Gaussian elimination is equivalent to invertibility, and the module criterion proves that Step 6 succeeds exactly for the requested representation. Formula (24.10) proves the independently generated action table. Every tie is resolved by fixed element, letter, pivot, pair, and byte orders; the independent verification check therefore proves reproducibility. $\square$

The successful record is

$$(p, e, \mu_{\text{coeff}}, q, \mathcal{T}_K, \mathcal{T}_Q, \mathcal{X}, \text{word_table}, d, \text{basis}, \\ A_i^{\text{in}}, A_i^{-1}, (A_g)_{g \in Q}, \mathcal{T}_V, \text{coordinate_maps}, \text{action_table}, \\ \text{field_axiom_rows}, \text{representation_rows}, \text{module_axiom_rows}, |V|, \\ \text{classification}, \text{COMPLETE}, \text{NONE}, h_{\text{input}}, h_{\text{independentverification}}).$$
(24.12)

The mutually exclusive classifications are

$$\text{INVALID_INPUT}, \quad \text{FIELD_FAIL}, \quad \text{GENERATION_FAIL}, \quad \text{MATRIX_FAIL}, \\ \text{REPRESENTATION_FAIL}, \quad \text{SATISFIED}.$$
(24.13)

Every rejection record contains its stage, first failure, all witnesses named above, and hashes of all fields available at that stage; later fields are `NONE`.

Minimum test family. Take $p = 2, e = 1, \mu = X, Q = 1, d = 0$ for the zero module boundary. For a nontrivial irreducible test, take

$$K = \mathbb{F}_2[\alpha]/(\alpha^2 + \alpha + 1), \quad Q = C_3 = \langle x \rangle, \quad d = 1, \quad A_x = (\alpha).$$
(24.14)

Since $\alpha^3 = 1 \neq \alpha$, this gives a faithful one-dimensional module of order 4. Over $\mathbb{F}_3$, the matrix

$$Q = C_2 = \langle y \rangle, \quad A_y = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$
(24.15)

gives a valid two-dimensional module of order 9 and checks that reducibility is not confused with invalidity. The polynomial $X^2 + 1 = (X + 1)^2$ over $\mathbb{F}_2$ checks `FIELD_FAIL`; an empty generator list for nontrivial C_2 checks `GENERATION_FAIL`; the zero 1-by-1 generator matrix checks `MATRIX_FAIL`; and mapping a generator of C_3 to $2 \in \mathbb{F}_5^\times$ checks `REPRESENTATION_FAIL`, since $2^3 \neq 1$.

22.20 Linear C_8 -actions and covariance

Let (K, Q, V, ρ) be a certified Statement 22.19 module, and write $A_g = \rho(g)$. Let

$$C_8 = \langle \zeta \mid \zeta^8 = 1 \rangle \quad (24.16)$$

and let $P \in \text{Aut}(Q)$ satisfy $P^8 = 1$. A P -covariant K -linear C_8 -action on V is determined by a matrix $J \in \text{GL}_d(K)$ satisfying

$$J^8 = I_d, \quad JA_gJ^{-1} = A_{P(g)} \quad \text{for every } g \in Q. \quad (24.17)$$

The special case $P = 1_Q$ says that J centralizes the Q -action. No faithfulness is implicit in the word *action*.

Theorem 22.20.1 (Linear action and exact order). *A matrix J defines a homomorphism*

$$\sigma_J : C_8 \longrightarrow \text{GL}_d(K), \quad \sigma_J(\zeta^k) = J^k \quad (24.18)$$

if and only if $J^8 = I_d$. Its exact order is

$$o(J) = \min\{m \in \{1, 2, 4, 8\} : J^m = I_d\}, \quad (24.19)$$

and the action is faithful if and only if $o(J) = 8$, equivalently $J^4 \neq I_d$.

Proof. If (24.18) is a homomorphism, the defining relation of C_8 gives $J^8 = I_d$. Conversely, if $J^8 = I_d$, then for residues $k, \ell \in \{0, \dots, 7\}$,

$$J^k J^\ell = J^{k+\ell} = J^{k+\ell \bmod 8}, \quad (24.20)$$

so (24.18) is well defined and multiplicative. The order of an element whose eighth power is one divides 8, giving (24.19). A homomorphism from a cyclic group is injective exactly when the image of its generator has the same order. Among the divisors of 8, the proper ones divide 4, which proves the final equivalence. $\square$

The covariance equation has a second exact interpretation. Form

$$H = Q \rtimes_P C_8, \quad (g, k)(h, \ell) = (gP^k(h), k + \ell \bmod 8). \quad (24.21)$$

Theorem 22.20.2 (Covariant extension of the representation). *Assume $J^8 = I_d$. The formula*

$$\tilde{\rho}(g, k) = A_g J^k \quad (24.22)$$

defines a representation $H \rightarrow \text{GL}_d(K)$ extending ρ and sending $(1_Q, 1)$ to J if and only if the covariance rows in (24.17) hold.

Proof. If $\tilde{\rho}$ is a representation, the relation $(1_Q, 1)(g, 0)(1_Q, 1)^{-1} = (P(g), 0)$ in H gives $JA_gJ^{-1} = A_{P(g)}$. Conversely, induction on k gives

$$J^k A_h J^{-k} = A_{P^k(h)}. \quad (24.23)$$

Therefore

$$\begin{aligned} \tilde{\rho}(g, k)\tilde{\rho}(h, \ell) &= A_g J^k A_h J^\ell \\ &= A_g A_{P^k(h)} J^{k+\ell} = \tilde{\rho}(gP^k(h), k + \ell \bmod 8), \end{aligned} \quad (24.24)$$

where $J^8 = I_d$ makes the exponent independent of the chosen residue. This proves both well-definedness and multiplicativity. $\square$

Representation and search algorithm. The canonical input is

$$\text{C8LinearAction}(\mathcal{R}_V, P_{\text{tab}}; \text{mode}, J_{\text{in}} \text{ or NONE}, \text{require_faithful}), \quad (24.25)$$

where $\mathcal{R}_V$ is the successful Statement 22.19 record, $\text{mode} \in \{\text{VALIDATE}, \text{SEARCH}\}$, and the last field is Boolean. A validator input contains exactly one matrix J_{in} ; a search input contains none. For fixed serialization, $\text{candidate_count} = 1$ in VALIDATE mode and $\text{candidate_count} = q^{d^2}$ in SEARCH mode.

- STEP 1. Rehash and independent verification the Statement 22.19 record. Validate the mode, field presence, Boolean flag, the full table of P , bijectivity, $P(gh) = P(g)P(h)$, and $P^8 = 1_Q$. Structural failure returns INVALID_INPUT with the first row and stage.
- STEP 2. In VALIDATE mode, verify the shape and entries of J_{in} , row-reduce it, and then check every row of (24.17) in the order J invertible, covariance, $J^8 = I_d$. A well-formed failed candidate returns CANDIDATE_FAIL with the first failed matrix row.
- STEP 3. In SEARCH mode, enumerate all q^{d^2} matrices by row-major field coordinates. Retain in order exactly the invertible matrices satisfying every covariance row and $J^8 = I_d$. If this complete list is empty, return NO_C8_ACTION with its candidate count and each matrix's first failed stage.
- STEP 4. Compute J, J^2, J^4, J^8 for every surviving matrix and its order from (24.19). If faithfulness is required, retain exactly those with $J^4 \neq I_d$. In validator mode a rejected supplied action returns ORDER_FAIL. In search mode, if C_8 -actions exist but none is faithful, return NO_FAITHFUL_C8_ACTION with the complete ordered fourth-power table. Choose the first remaining matrix.
- STEP 5. Construct all eight generator-image rows $\zeta^k \mapsto J^k$, and verify (24.20) for all 64 ordered pairs. Construct the full table (24.21), its inverses, and every matrix in (24.22), and verify $|H| = 8|Q|$.
- STEP 6. Verify the original Q -generator images, $J^8 = I_d$, every covariance relation

$$JA_{x_i}J^{-1} = A_{P(x_i)} \quad (1 \leq i \leq r), \quad (24.26)$$

and then independently verify (24.17) on all of Q and the homomorphism law for every pair of elements of H .

- STEP 7. Serialize the exact order, kernel

$$\ker \sigma_J = \langle \zeta^{o(J)} \rangle, \quad |\ker \sigma_J| = \frac{8}{o(J)}, \quad (24.27)$$

all generator images, covariance and relation rows, canonical input/output hashes, and a independent verification hash. Return SATISFIED only after byte-identical independent verification.

Theorem 22.20.3 (Algorithm correctness and reproducibility). *On inputs passing Step 1, the validator terminates and returns SATISFIED exactly when its supplied J satisfies (24.17) and, when requested, $J^4 \neq I_d$. On inputs passing Step 1, the search terminates and returns SATISFIED exactly when such a matrix exists. Every accepted output is a representation of $Q \rtimes_P C_8$, its reported order and kernel are exact, and equal canonical inputs produce byte-identical records.*

Proof. After Step 1, all matrix and group tables are finite. In validator mode Steps 2 and 4 are precisely the stated equations. In search mode Step 3 visits every matrix exactly once; hence its empty-list certificate is necessary and sufficient for nonexistence. Step 4 first distinguishes existence of an action from existence of a faithful one and then uses the exact-order theorem. The covariant-extension theorem proves Steps 5–6, while Lagrange’s theorem in the cyclic group gives (24.27). Fixed field, matrix, element, pair, failure, and byte orders prove determinism; the final independent verification proves reproducibility. $\square$

The success record is

$$\begin{aligned} &(\mathbf{h}_{\mathcal{R}_V}, P_{\text{tab}}, \text{mode}, \text{require_faithful}, q, d, \text{candidate_count}, \\ &J_{\text{tab}}, J_{\text{tab}}^{-1}, o(J), \ker \sigma_J, \text{C8_generator_images}, \text{C8_relation_rows}, \\ &\text{covariance_rows}, \mathcal{T}_H, (\tilde{\rho}(h))_{h \in H}, \text{semidirect_relation_rows}, |H|, \\ &\text{classification}, \text{COMPLETE}, \text{NONE}, \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}}). \end{aligned} \quad (24.28)$$

The mutually exclusive classifications are

$$\begin{aligned} &\text{INVALID_INPUT}, \quad \text{CANDIDATE_FAIL}, \quad \text{ORDER_FAIL}, \\ &\text{NO_C8_ACTION}, \quad \text{NO_FAITHFUL_C8_ACTION}, \quad \text{SATISFIED}. \end{aligned} \quad (24.29)$$

Rejection records keep the common input, mode, candidate counts, first failure, and all matrices or exhaustive failure rows available at the exit stage; every later field is **NONE**.

Minimum test family. For a faithful scalar action take

$$Q = 1, \quad K = \mathbb{F}_{17}, \quad d = 1, \quad J = (2). \quad (24.30)$$

Since $2^4 = -1$ and $2^8 = 1$, this has exact order eight. The identity matrix with $P = 1_Q$ is a valid nonfaithful action on every Statement 22.19 module. For nontrivial covariance, take $K = \mathbb{F}_2$, $Q = C_3 = \langle x \rangle$, $P(x) = x^{-1}$, and

$$A_x = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (24.31)$$

Then $A_x^3 = I$, $J^2 = I$, and $JA_xJ^{-1} = A_x^{-1}$, giving a positive twisted test.

For **NO_C8_ACTION**, use the one-dimensional $\mathbb{F}_4[C_3]$ -module (24.14) with $P(x) = x^{-1}$. Every invertible 1-by-1 matrix is scalar and hence commutes with A_x , whereas

$$A_x = (\alpha) \neq (\alpha^{-1}) = A_{x^{-1}}.$$

Thus covariance is impossible. For **NO_FAITHFUL_C8_ACTION**, take $Q = 1, K = \mathbb{F}_2, d = 1$; the only invertible matrix is I . A singular validator matrix, a covariance defect in (24.31), and the same identity matrix with the faithful flag give three validator tests: the first two return **CANDIDATE_FAIL**, and the last returns **ORDER_FAIL**. A malformed row in P_{tab} tests **INVALID_INPUT**.

22.21 Irreducible factors and semisimple decomposition

Let $K = \mathbb{F}_q$, let B be a finite group, and let $V = K^d$ carry a certified representation $\rho : B \rightarrow \text{GL}(V)$. The acting group B is selected from the C24 interfaces:

$$B = Q \quad \text{with matrices } A_g, \quad \text{or} \quad B = Q \rtimes_P C_8 \quad \text{with matrices } \tilde{\rho}(g, k). \quad (24.32)$$

Thus a decomposition relative to the second choice is automatically C_8 -stable, whereas a decomposition relative only to Q need not be fixed term by term by J .

A subspace $W \leq V$ is a $K[B]$ -submodule when

$$\rho(b)W \subseteq W \quad (b \in B). \quad (24.33)$$

It is enough to test (24.33) on the fixed generators of B . A nonzero module is irreducible if its only submodules are 0 and itself. It is semisimple if it is an internal direct sum of irreducible submodules.

Canonical subspace representation. Represent every r -dimensional subspace by the unique d -by- r reduced column-echelon matrix whose columns form its ordered basis. Enumerate ranks $0, \dots, d$, pivot sets lexicographically, and free entries in field order. Membership, intersection, sum, and invariance are decided by Gaussian reduction:

$$W \text{ is invariant} \iff \text{rank}[B_W \ \rho(s)B_W] = \text{rank } B_W \quad \text{for every acting generator } s. \quad (24.34)$$

This is an exact finite enumeration of the submodule lattice.

If $W \leq U$ are invariant, reduce vectors modulo the canonical basis of W and choose the least nonpivot coordinate representatives for U/W . Define

$$\rho_{U/W}(b)(u + W) = \rho(b)u + W. \quad (24.35)$$

Lemma 22.21.1 (Well-defined quotient and irreducibility test). *Equation (24.35) is well defined and gives a $K[B]$ -module. The quotient U/W is irreducible exactly when its canonical submodule enumeration contains only 0 and U/W .*

Proof. If $u + W = u' + W$, then $u - u' \in W$. Invariance gives $\rho(b)(u - u') \in W$, so the two images in (24.35) agree. Linearity, the identity, and multiplication follow from the same properties of ρ . Submodules of U/W are, by inverse image under the quotient map, in inclusion-preserving bijection with submodules of U containing W . Hence absence of a proper nonzero enumerated submodule is exactly irreducibility. $\square$

Theorem 22.21.2 (Canonical composition series). *Every finite-dimensional $K[B]$ -module has a canonically constructed chain*

$$0 = V_0 < V_1 < \dots < V_t = V \quad (24.36)$$

whose factors V_i/V_{i-1} are irreducible.

Proof. If $V/V_i \neq 0$, enumerate its nonzero submodules and choose one of smallest positive dimension, breaking ties by its reduced basis. Such a submodule is irreducible: a proper nonzero submodule would have smaller positive dimension. Let V_{i+1} be its full inverse image in V . Then V_{i+1}/V_i is irreducible by the quotient lemma. Its dimension is positive, so $\dim V_{i+1} > \dim V_i$; after at most d iterations the chain reaches V . Every enumeration and tie break uses fixed orders, proving canonicity. $\square$

Composition factors do not by themselves imply a direct-sum decomposition. The exact criterion used by the constructor is:

Theorem 22.21.3 (Complement criterion for semisimplicity). *For a finite-dimensional $K[B]$ -module U , the following are equivalent:*

- (i) U is a direct sum of irreducible submodules;
- (ii) every submodule $W \leq U$ has an invariant complement $C \leq U$, so $U = W \oplus C$;

- (iii) *the least nonzero submodule S of U selected by the canonical enumeration has an invariant complement, and the recursively selected complement has the same property until zero is reached.*

When these conditions hold, recursively taking the least complement produces a canonical irreducible direct decomposition.

Proof. Suppose $U = S_1 \oplus \cdots \oplus S_n$ with each S_i irreducible. We prove (ii) by induction on n . Put $U' = S_1 \oplus \cdots \oplus S_{n-1}$ and let $\pi : U \rightarrow S_n$ be projection. If $\pi(W) = 0$, apply induction inside U' and add S_n to the complement. Otherwise $\pi(W) = S_n$ by irreducibility. Induction gives

$$U' = (W \cap U') \oplus C'.$$

For $u' + s \in U$, choose $w \in W$ with $\pi(w) = s$; then $u' + s - w \in U'$, and its $W \cap U'$ component may be absorbed into w . Hence $U = W + C'$, while $W \cap C' = 0$, so C' is the required invariant complement. Thus (i) implies (ii), and (ii) implies (iii).

Assume (iii). A least positive-dimensional submodule S is irreducible. Write $U = S \oplus C$ and apply the recursive hypothesis to C . Induction on dimension expresses C , and hence U , as a direct sum of irreducibles. This proves (iii) implies (i). Choosing the least available basis at every finite search gives the canonical last assertion. $\square$

When $\text{char } K \nmid |B|$, complements also have a direct construction. Given any linear projection $p : U \rightarrow W$ with $p|_W = 1_W$, set

$$\bar{p} = \frac{1}{|B|} \sum_{b \in B} \rho(b) p \rho(b)^{-1}. \quad (24.37)$$

Then $\bar{p}$ commutes with every $\rho(c)$, restricts to the identity on W , and $\ker \bar{p}$ is an invariant complement. Indeed, reindexing $b \mapsto cb$ proves commutation, and invariance of W makes every summand act as the identity on W . Thus $U = W \oplus \ker \bar{p}$. Formula (24.37) is not used when the characteristic divides $|B|$; the exhaustive complement search remains valid in that modular case.

For canonical serialization, process every invariant $W \leq V$ in the stored lattice order. Starting from its canonical reduced column-echelon basis, scan the standard coordinate vectors $e_1, \dots, e_d$ and append a vector exactly when it increases rank. This gives the lexicographically first such extension to a basis of V . Define p_W to be the identity on the basis vectors of W and zero on the appended vectors, and use p_W in (24.37). Thus the initial projection, the averaged projection, and its kernel are all functions of the canonical input.

Decomposition algorithm and I/O. The canonical input is

$$\text{IrreducibleDecompose}(\mathcal{R}_V, \mathcal{R}_J \text{ or NONE}; \text{acting_source}, \text{mode}), \quad (24.38)$$

where

$$\text{acting_source} \in \{\text{Q}, \text{Q_SEMIDIRECT_C8}\}, \quad \text{mode} \in \{\text{COMPOSITION}, \text{DIRECT}\}. \quad (24.39)$$

The field $\mathcal{R}_J$ is NONE exactly for source Q. For source Q_SEMIDIRECT_C8, it is present exactly as a successful, hash-matching Statement 22.20 record.

STEP 1. Rehash and independent verification all supplied records; validate source and mode presence rules. Select B , its canonical generators, and its complete action matrices from (24.32). A mismatch or malformed record returns INVALID_INPUT.

STEP 2. Enumerate every reduced column-echelon subspace and test (24.34). Store the complete submodule lattice, its inclusions, sums, intersections, dimensions, and generator invariance rows.

STEP 3. Starting from $V_0 = 0$, apply the construction in (24.36). For every factor build quotient coordinates and action matrices from (24.35), and independently enumerate its submodules to certify irreducibility.

STEP 4. If the mode is **COMPOSITION**, serialize the series and factors and proceed to Step 7. Direct-decomposition fields are fixed to **NONE**.

STEP 5. In **DIRECT** mode start with $U = V$. Choose the least nonzero submodule of minimum positive dimension. Enumerate all invariant $C \leq U$, retaining those for which

$$S \cap C = 0, \quad \dim S + \dim C = \dim U. \quad (24.40)$$

If none exists, return **NONSEMISIMPLE** with S , the composition series, and the ordered rank/intersection failure for every candidate C . Otherwise choose the least complement, append S , replace U by C , and repeat.

STEP 6. Verify that every returned summand is nonzero and irreducible and that the block-column map

$$\Psi : S_1 \oplus \cdots \oplus S_m \longrightarrow V, \quad (v_1, \dots, v_m) \longmapsto \sum_i v_i \quad (24.41)$$

is a $K[B]$ -linear isomorphism by full rank and action rows.

STEP 7. If $\text{char } K \nmid |B|$, construct p_W by the fixed coordinate-vector scan for every invariant $W \leq V$ in lattice order, then independently construct and verify all averaged projections (24.37). Otherwise record that field as **NA**. Hash and independent verification the lattice, quotient, factor, and direct-sum tables before returning **SATISFIED**.

Theorem 22.21.4 (Algorithm correctness and reproducibility). *On structurally valid input, composition mode always returns **SATISFIED** with irreducible factors. Direct mode returns **SATISFIED** exactly when V is semisimple as a module for the selected acting group; otherwise it returns **NONSEMISIMPLE** with a complete complement obstruction. Equal canonical inputs produce byte-identical records.*

Proof. The finite subspace enumeration proves termination and makes every invariance and complement claim decidable. The composition-series theorem proves Step 3. The complement criterion proves both directions of Steps 5–6: success constructs an irreducible direct sum, whereas failure to complement the selected irreducible submodule is incompatible with semisimplicity. Formula (24.37) supplies an independent check in the nonmodular case but is not needed for completeness. Fixed rank, pivot, field, subspace, complement, and byte orders prove canonicity and reproducibility. $\square$

The canonical record is

$$\begin{aligned} &(\mathbf{h}_{\mathcal{R}_V}, \mathbf{h}_{\mathcal{R}_J} \text{ or } \mathbf{NONE}, \text{acting_source}, \text{mode}, B, \text{action_matrices}, \\ &\text{submodule_lattice}, \text{inclusion_rows}, \text{sum_intersection_rows}, (V_i)_{i=0}^t, \text{factor_records}, \\ &(S_j)_{j=1}^m \text{ or } \mathbf{NONE}, \text{complement_rows}, \Psi_{\text{tab}} \text{ or } \mathbf{NONE}, \text{averaged_projection_rows or NA}, \\ &\text{classification}, \text{stage}, \text{first_failure} \text{ or } \mathbf{NONE}, \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}} \text{ or } \mathbf{NONE}). \end{aligned} \quad (24.42)$$

The classifications are

$$\mathbf{INVALID_INPUT}, \quad \mathbf{NONSEMISIMPLE}, \quad \mathbf{SATISFIED}. \quad (24.43)$$

On **SATISFIED** the stage is **COMPLETE**; unavailable fields after rejection are **NONE**.

Minimum test family. The zero module has the empty composition and direct decompositions. For $B = 1$ and $V = \mathbb{F}_2^2$, every line is irreducible and direct mode returns a canonical sum of two lines. The module (24.15) over $\mathbb{F}_3[C_2]$ decomposes into the $+1$ and -1 eigenlines. The matrix A_x in (24.31) makes $\mathbb{F}_2^2$ an irreducible C_3 -module because it has no eigenvector over $\mathbb{F}_2$.

For the modular obstruction take $K = \mathbb{F}_2$, $B = C_2 = \langle y \rangle$, and

$$\rho(y) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}. \quad (24.44)$$

The line Ke_1 is the unique proper nonzero submodule. Any complement is a different line, but none is invariant; hence the two composition factors are trivial while direct mode returns NONSEMISIMPLE. Finally, use the same matrix as a nonfaithful C_8 -generator with $Q = 1$. Relative to Q the module is a direct sum of trivial lines, while relative to $Q \rtimes C_8$ it has the obstruction (24.44). This checks that the acting-source flag changes the mathematical question rather than only the serialization.

22.22 Indecomposable modules and Krull–Schmidt construction

Continue with the finite $K[B]$ -module selected in Statement 22.21. The zero module has the empty indecomposable decomposition. A nonzero module U is indecomposable when

$$U = U_1 \oplus U_2 \implies U_1 = 0 \text{ or } U_2 = 0. \quad (24.45)$$

This is weaker than irreducibility: the modular Jordan module (24.44) has a proper nonzero submodule but no invariant complement.

The computable endomorphism algebra is

$$\text{End}_{K[B]}(U) = \{X \in \text{Mat}_{\dim U}(K) : X\rho_U(s) = \rho_U(s)X \text{ for every acting generator } s\}. \quad (24.46)$$

It is the solution space of a finite homogeneous linear system.

Theorem 22.22.1 (Idempotent splitting criterion). *A nonzero finite-dimensional module U is indecomposable if and only if the only idempotents in $\text{End}_{K[B]}(U)$ are 0 and I_U . For every idempotent e ,*

$$U = \text{im } e \oplus \ker e, \quad (24.47)$$

and both summands are $K[B]$ -submodules.

Proof. If $e^2 = e$, then e commutes with the B -action, so its image and kernel are invariant. Every $u \in U$ has the decomposition

$$u = e(u) + (u - e(u)), \quad e(u - e(u)) = 0, \quad (24.48)$$

and an element in both summands has the form $e(v)$ and is killed by e , hence equals $e^2(v) = 0$. This proves (24.47). A nontrivial idempotent therefore decomposes U . Conversely, if $U = U_1 \oplus U_2$ with both summands nonzero, projection onto U_1 along U_2 is a nontrivial $K[B]$ -linear idempotent. $\square$

Lemma 22.22.2 (Fitting decomposition). *For every $\phi \in \text{End}_{K[B]}(U)$, there is $n \leq \dim U$ such that*

$$U = \ker \phi^n \oplus \text{im } \phi^n. \quad (24.49)$$

Consequently, if U is indecomposable, every endomorphism is either invertible or nilpotent, and $\text{End}_{K[B]}(U)$ is a local ring.

Proof. The increasing kernels and decreasing images stabilize by dimension; choose n after both have stabilized. If $y = \phi^n(x) \in \ker \phi^n$, then $x \in \ker \phi^{2n} = \ker \phi^n$, so $y = 0$. Rank-nullity now proves (24.49). For indecomposable U , one of the two summands is zero. If the kernel is zero, ϕ is invertible; if the image is zero, $\phi^n = 0$. Thus every nonunit ϕ is nilpotent and

$$(1 - \phi)^{-1} = 1 + \phi + \cdots + \phi^{n-1}. \quad (24.50)$$

The criterion that, for every ϕ , either ϕ or $1 - \phi$ is a unit is equivalent to locality; hence the endomorphism ring is local. $\square$

Theorem 22.22.3 (Finite Krull–Schmidt theorem). *Every finite-dimensional $K[B]$ -module has a decomposition*

$$V = U_1 \oplus \cdots \oplus U_m \quad (24.51)$$

into nonzero indecomposable submodules. If $V = W_1 \oplus \cdots \oplus W_n$ is another such decomposition, then $m = n$ and, after a permutation, $U_i \cong W_i$ for every i .

Proof. Existence follows by induction on dimension. If V is indecomposable, stop. Otherwise (24.45) gives two nonzero proper direct summands; apply induction to each.

For uniqueness, fix inclusions and projections for both decompositions. On U_i , the identity is a finite sum

$$1_{U_i} = \sum_{j=1}^n (p_i \iota'_j)(p'_j \iota_i), \quad (24.52)$$

where primes refer to the W -decomposition. The endomorphism ring of U_i is local by the Fitting lemma. Nonunits in a local ring form an ideal, so at least one summand in (24.52) is a unit. For that j , the map $p'_j \iota_i : U_i \rightarrow W_j$ is a split injection. Its image is a nonzero direct summand of indecomposable W_j , hence it is all of W_j , and $U_i \cong W_j$. Moreover this restriction is an isomorphism, so U_i is a graph over W_j relative to $V = W_j \oplus \bigoplus_{k \neq j} W_k$. Therefore

$$V = U_i \oplus \bigoplus_{k \neq j} W_k.$$

Quotienting by U_i cancels the matched summands. Induction on the number of summands proves the asserted permutation and equality $m = n$. $\square$

Canonical construction algorithm. The input is a closed Statement 22.21 record:

$$\text{IndecomposableDecompose}(\mathcal{R}_{\text{irr}}), \quad (24.53)$$

where $\mathcal{R}_{\text{irr}}$ may have classification **SATISFIED** or **NONSEMISIMPLE**; both contain the verified acting source, module, complete submodule lattice, and composition series.

- STEP 1. Rehash and independent verification $\mathcal{R}_{\text{irr}}$, its Statement 22.19 record, and, when present, its Statement 22.20 record. Require consistent field, module, acting-source, lattice, and action hashes. A mismatch or malformed record returns **INVALID_INPUT**.
- STEP 2. Initialize an ordered queue with V . The zero module skips to Step 6 with an empty leaf list.
- STEP 3. Remove the first nonzero module U . In its canonical basis, solve (24.46) by fixed-pivot row reduction; enumerate every element of the resulting endomorphism space in coefficient order, compute $e^2 - e$, and retain all idempotents.
- STEP 4. If the only idempotents are 0, I_U , store U as a certified indecomposable leaf. Otherwise choose the least nontrivial idempotent, construct canonical reduced bases for $\text{im } e$ and $\ker e$, verify (24.47), and insert the two nonzero summands into the queue in basis order.

STEP 5. Repeat Steps 3–4 until the queue is empty. Dimension strictly decreases at every split, so this process terminates.

STEP 6. Form the block-column sum map

$$\Theta : \bigoplus_{i=1}^m U_i \longrightarrow V, \quad (u_i)_i \longmapsto \sum_i u_i, \quad (24.54)$$

and verify by full rank and action rows that it is a $K[B]$ -linear isomorphism. Independently check each leaf against the complement data in the Statement 22.21 submodule lattice.

STEP 7. For every pair U_i, U_j of equal dimension, solve the intertwining equations

$$X \rho_{U_i}(s) = \rho_{U_j}(s) X \quad \text{for all acting generators } s, \quad (24.55)$$

enumerate the solution space, and declare the pair isomorphic exactly when an invertible X exists. Choose the least such X , verify equivalence and transitivity of the resulting relation, and serialize the sorted isomorphism types and their multiplicities.

STEP 8. Reconstruct the full action matrices by conjugating the block diagonal leaf action through Θ . Hash and independent verification all endomorphism, idempotent, split, intertwiner, and reconstruction rows before returning SATISFIED.

Theorem 22.22.4 (Algorithm correctness and reproducibility). *On structurally valid input the algorithm terminates and returns SATISFIED with an indecomposable direct decomposition of V . Its sorted multiset of isomorphism types is independent of the chosen idempotent splits, while the actual returned subspaces are uniquely determined by the declared canonical orders. Equal canonical inputs produce byte-identical records.*

Proof. The endomorphism spaces and their matrix elements are finite. The idempotent criterion proves every stop and split decision, and both children of a nontrivial split have dimension strictly between zero and that of their parent. Thus the algorithm terminates with indecomposable leaves, and (24.47) inductively proves that (24.54) is an isomorphism. Equation (24.55) is exactly the condition for a linear map to be B -linear; finite enumeration makes invertibility an exact isomorphism test. Krull–Schmidt proves independence of the final isomorphism multiset. Fixed pivot, coefficient, idempotent, basis, queue, intertwiner, type, and byte orders prove deterministic serialization and independent verification. $\square$

The canonical output is

$$\begin{aligned} &(\mathbf{h}_{\mathcal{R}_{\text{irr}}}, K, B, V, \text{split_tree}, \text{endomorphism_bases}, \text{idempotent_tables}, \\ & (U_i)_{i=1}^m, \text{leaf_action_matrices}, \text{indecomposability_certificates}, \Theta_{\text{tab}}, \\ & \text{intertwiner_spaces}, \text{least_isomorphisms}, \text{isomorphism_types}, \text{multiplicities}, \text{reconstruction_rows}, \\ & \text{classification}, \text{COMPLETE}, \text{NONE}, \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}}). \end{aligned} \quad (24.56)$$

The only classifications are

$$\text{INVALID_INPUT} \quad \text{and} \quad \text{SATISFIED}. \quad (24.57)$$

An invalid record stores its first failed stage and witness; all later fields are NONE.

Minimum test family. The zero module returns no leaves. The irreducible $\mathbb{F}_2[C_3]$ -module from (24.31) returns one two-dimensional leaf. The semisimple module (24.15) returns its two non-isomorphic eigenlines. The Jordan module (24.44) returns one two-dimensional indecomposable leaf even though its composition length is two.

For multiplicity and uniqueness, take $K = \mathbb{F}_2$, $B = C_2 = \langle y \rangle$, and

$$\rho(y) = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (24.58)$$

The algorithm splits this module into two isomorphic copies of the indecomposable block (24.44), records one isomorphism type with multiplicity two, and reconstructs the displayed matrix. Altering one parent action hash gives the `INVALID_INPUT` test.

22.23 Construction and classification of extension modules

Let $K = \mathbb{F}_q$, let B be the acting group selected in Statement 22.21, and let U, W be certified finite-dimensional $K[B]$ -modules with action matrices

$$A_g = \rho_U(g) \in \mathrm{GL}_u(K), \quad D_g = \rho_W(g) \in \mathrm{GL}_w(K). \quad (24.59)$$

They may be indecomposable leaves produced by Statement 22.22, but the construction applies to arbitrary compatible certified modules.

Put $\mathcal{H} = \mathrm{Hom}_K(W, U)$. It is a left $K[B]$ -module under

$$g \cdot X = A_g X D_g^{-1}. \quad (24.60)$$

Indeed, $1 \cdot X = X$, and

$$(gh) \cdot X = A_g A_h X D_h^{-1} D_g^{-1} = g \cdot (h \cdot X). \quad (24.61)$$

A normalized crossed homomorphism is a function $c : B \rightarrow \mathcal{H}$ with

$$c_1 = 0, \quad c_{gh} = c_g + g \cdot c_h. \quad (24.62)$$

For $L \in \mathcal{H}$, its coboundary is

$$(\delta L)_g = g \cdot L - L = A_g L D_g^{-1} - L. \quad (24.63)$$

Theorem 22.23.1 (Block construction and exactness). *For a normalized table c , put*

$$\eta_g = c_g D_g, \quad R_g = \begin{pmatrix} A_g & \eta_g \\ 0 & D_g \end{pmatrix}. \quad (24.64)$$

Then $g \mapsto R_g$ is a representation of B if and only if c satisfies (24.62). In that case $E_c = U \oplus W$, with action R_g , fits into the exact sequence

$$0 \longrightarrow U \xrightarrow{i} E_c \xrightarrow{\pi} W \longrightarrow 0, \quad i(x) = (x, 0), \quad \pi(x, y) = y. \quad (24.65)$$

The induced actions on the submodule and quotient are precisely ρ_U and ρ_W , and

$$\dim E_c = u + w, \quad |E_c| = q^{u+w} = |U||W|. \quad (24.66)$$

Proof. Block multiplication gives

$$R_g R_h = \begin{pmatrix} A_g A_h & A_g \eta_h + \eta_g D_h \\ 0 & D_g D_h \end{pmatrix}. \quad (24.67)$$

Thus $R_g R_h = R_{gh}$ is equivalent to

$$\eta_{gh} = A_g \eta_h + \eta_g D_h. \quad (24.68)$$

Multiplying (24.68) on the right by $D_{gh}^{-1} = D_h^{-1} D_g^{-1}$ and using (24.64) produces

$$c_{gh} = A_g c_h D_g^{-1} + c_g = c_g + g \cdot c_h,$$

which is (24.62); every step is reversible. Normalization gives $R_1 = I$. The map i is injective and B -linear, π is surjective and B -linear, and $\ker \pi = U \oplus 0 = \operatorname{im} i$, proving exactness. The dimension and order formulas follow from the fixed vector-space direct sum. $\square$

Theorem 22.23.2 (Equivalence, splitting, and H^1). *Let c, c' satisfy (24.62). The extensions $E_c, E_{c'}$ are equivalent by a $K[B]$ -isomorphism inducing the identity on U and W if and only if*

$$c' = c + \delta L \quad \text{for some } L \in \mathcal{H}. \quad (24.69)$$

For this equation the isomorphism from $E_{c'}$ to E_c is

$$F_L(x, y) = (x + Ly, y). \quad (24.70)$$

The extension splits as a $K[B]$ -module if and only if $[c] = 0$ in

$$H^1(B, \mathcal{H}) = Z^1(B, \mathcal{H}) / B^1(B, \mathcal{H}). \quad (24.71)$$

Consequently

$$\operatorname{Ext}_{K[B]}^1(W, U) \cong H^1(B, \operatorname{Hom}_K(W, U)). \quad (24.72)$$

Proof. Every linear map preserving the injection and quotient has the unique form (24.70). Compare its two intertwining products:

$$\begin{aligned} F_L R_{c'}(g)(x, y) &= (A_g x + \eta'_g y + L D_g y, D_g y), \\ R_c(g) F_L(x, y) &= (A_g x + A_g L y + \eta_g y, D_g y). \end{aligned} \quad (24.73)$$

Equality is equivalent to $\eta'_g = \eta_g + A_g L - L D_g$. Right multiplication by D_g^{-1} turns this into (24.69), proving both directions and the sign in (24.70).

If $c = \delta L$, then $\eta_g = A_g L - L D_g$, and

$$s_L : W \longrightarrow E_c, \quad s_L(y) = (-Ly, y) \quad (24.74)$$

is B -linear and satisfies $\pi s_L = 1_W$. Conversely, any B -linear section has form $y \mapsto (-Ly, y)$, and its intertwining equation gives $c = \delta L$. Hence splitting is equivalent to the zero cohomology class. Finally, every short exact sequence of K -vector spaces has a linear section, whose coordinates give (24.64); changing the section changes c by (24.69). Thus equivalence classes of all module extensions are exactly the quotient (24.71). This bijection respects addition: the pullback–pushout definition of the Baer sum of E_c and E_d has off-diagonal block $\eta_g(c) + \eta_g(d) = (c_g + d_g) D_g$, and its zero and inverse classes correspond respectively to 0 and $-c$. It is therefore an isomorphism of abelian groups, proving (24.72). $\square$

Canonical class and construction algorithm. Enumerate normalized functions $B \rightarrow \mathcal{H}$ in matrix-table order, retain the solutions of (24.62), enumerate all coboundaries (24.63), and sort the cosets by their least representatives:

$$\mathcal{C} = (c^{(0)}, \dots, c^{(N-1)}), \quad c^{(0)} = 0. \quad (24.75)$$

The input is

$$\begin{aligned} &\text{ModuleExtension}(\mathcal{R}_U, \mathcal{R}_W; \text{source}, j \text{ or NONE}, c_{\text{in}} \text{ or NONE}; \\ &\quad \text{require_nonsplit}), \end{aligned} \quad (24.76)$$

where **source** is **CLASS**, **COCYCLE**, or **BOTH**. Its presence specification is exhaustive and exclusive:

source	j	c_{in}
CLASS	$0 \leq j < N$	NONE
COCYCLE	NONE	a full table $(c_g \in \mathcal{H})_{g \in B}$
BOTH	$0 \leq j < N$	a full table $(c_g \in \mathcal{H})_{g \in B}$.

No omitted or additional source field is permitted; table keys are exactly the canonical elements of B , every entry has shape $u \times w$ over K , and **require_nonsplit** is Boolean.

STEP 1. Rehash and independent verification both module records. Require the same field, acting group, generator order, and action-source hash; validate all source fields and the Boolean flag. A mismatch or malformed field returns **INVALID_INPUT**.

STEP 2. Construct the full matrix space $\mathcal{H}$, verify (24.60), enumerate Z^1, B^1 , and construct the disjoint exhaustive catalogue (24.75), including all coset and leastness rows.

STEP 3. Validate any supplied cocycle table against (24.62). A well-formed failure returns **CANDIDATE_FAIL**. Locate its unique class; in **COCYCLE** mode assign its catalogue index to j . In **BOTH** mode, a class different from the supplied index j returns **CLASS_MISMATCH** with both coset certificates.

STEP 4. Set $c = c^{(j)}$. When a candidate is present, choose the least L satisfying

$$c_{\text{in}} = c + \delta L \quad (24.77)$$

and store every verification row. If nonsplitting is required and $j = 0$, return **SPLIT_REJECT** with a typed splitting certificate. In **CLASS** mode the certified model is E_0 , the witness is $L_{\text{split}} = 0$, and the section is $s_0(y) = (0, y)$. When c_{in} is present, the certified model is $E_{c_{\text{in}}}$, the witness is the least $L_{\text{split}} = L_{\text{transfer}}$ from (24.77), and the section is $s_{L_{\text{split}}}(y) = (-L_{\text{split}}y, y)$ as in (24.74). Serialize every B -linearity row and every row of $\pi s_{L_{\text{split}}} = 1_W$ before exit.

STEP 5. Construct every matrix R_g , its inverse, and the full action table on E_c . Verify $R_g R_h = R_{gh}$, (24.62), (24.68), and all identity and inverse rows independently.

STEP 6. Construct i, π , the canonical vector-space section $s_0(y) = (0, y)$, and all exactness and induced-action rows in (24.65). Verify (24.66). When a candidate is present, construct F_L, F_L^{-1} and verify every row of (24.70)–(24.73).

STEP 7. Serialize generator images

$$x_k \mapsto \begin{pmatrix} A_{x_k} & \eta_{x_k} \\ 0 & D_{x_k} \end{pmatrix}, \quad (24.78)$$

the class, extension, exactness, transport, hashes, and independent verification rows. Return **SATISFIED** only after byte-identical independent verification.

Theorem 22.23.3 (Algorithm correctness and reproducibility). *On structurally valid input, the algorithm returns SATISFIED exactly when the source fields determine one class in (24.75) and that class is nonzero whenever nonsplitting is required. Its output is an exact $K[B]$ -module extension in the requested class; any supplied cohomologous model is connected to it by the verified map F_L . Equal canonical inputs produce byte-identical records.*

Proof. All matrix-function and cochain spaces are finite. The exhaustive coset construction proves unique class resolution; (24.77) is exactly membership in the selected coset. The two theorems above prove representation, exactness, equivalence, and the split rejection in both directions. Fixed matrix, function, coset, witness, generator, failure, and byte orders prove deterministic selection and serialization. $\square$

The successful record is

$$\begin{aligned} &(\mathbf{h}_{\mathcal{R}_U}, \mathbf{h}_{\mathcal{R}_W}, K, B, u, w, \text{source}, \text{require_nonsplit}, \mathcal{H}, \text{Hom_action_rows}, \mathcal{C}, j, \\ & c_{\text{in}} \text{ or NONE}, c, L_{\text{transfer}} \text{ or NONE}, \text{coboundary_rows}, (\eta_g)_{g \in B}, \\ & \mathcal{T}_{E_c}, (R_g)_{g \in B}, i_{\text{tab}}, \pi_{\text{tab}}, s_{0, \text{tab}}, \text{exactness_rows}, |E_c|, \\ & F_{L, \text{tab}} \text{ or NONE}, \text{transport_rows} \text{ or NONE}, \text{generator_images}, \\ & \text{classification}, \text{COMPLETE}, \text{NONE}, \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}}). \end{aligned} \quad (24.79)$$

The mutually exclusive classifications are

$$\text{INVALID_INPUT}, \quad \text{CANDIDATE_FAIL}, \quad \text{CLASS_MISMATCH}, \quad \text{SPLIT_REJECT}, \quad \text{SATISFIED}. \quad (24.80)$$

Each rejection record contains all class, witness, stage, and first failure fields available before exit; later fields are NONE. In particular its fixed tail contains

$$\begin{aligned} &(\text{split_model} \text{ or NONE}, L_{\text{split}} \text{ or NONE}, \\ & s_{L_{\text{split}}, \text{tab}} \text{ or NONE}, \text{splitting_rows} \text{ or NONE}, \\ & \text{stage}, \text{first_failure}, \text{classification}). \end{aligned}$$

Minimum test family. For $B = 1$, the catalogue has only the zero class and every module extension splits. For a nonsplit modular test, take

$$K = \mathbb{F}_2, \quad B = C_2 = \langle y \rangle, \quad U = W = K_{\text{triv}}, \quad c_y = 1. \quad (24.81)$$

Then every coboundary is zero, (24.62) holds, and

$$R_y = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad (24.82)$$

recovering the indecomposable module (24.44).

For a nonzero split cocycle, take $K = \mathbb{F}_3$, $B = C_2$, $A_y = -1$, $D_y = 1$, and $L = 1$. Then

$$(\delta L)_y = (-1)(1)(1)^{-1} - 1 = 1 \pmod{3}, \quad (24.83)$$

so a nonzero candidate canonicalizes to the zero class and (24.74) gives an explicit splitting. Pairing (24.81) with the zero-class index tests CLASS_MISMATCH; changing its sole nonidentity cocycle value to a new formal symbol tests INVALID_INPUT. For a well-typed cocycle failure take $K = \mathbb{F}_3$, let $B = C_2 = \langle y \rangle$, let $U = W$ be trivial one-dimensional modules, and set $c_1 = 0, c_y = 1$. The $y^2 = 1$ instance of (24.62) would require $0 = c_y + y \cdot c_y = 1 + 1 = 2$, so this input returns CANDIDATE_FAIL.

22.24 Fixed submodules and quotient modules

Let $K = \mathbb{F}_q$, let B be a finite acting group with fixed canonical element order, and let $V = K^n$ be a certified $K[B]$ -module with

$$M_g = \rho_V(g) \in \mathrm{GL}_n(K), \quad M_g M_k = M_{gk}. \quad (24.84)$$

The record may be a module from Statement 22.19–Statement 22.22 or an extension module from Statement 22.23. No semisimplicity assumption is made.

Fixed submodule. For a certified subgroup $H \leq B$, form the vertically stacked matrix

$$\mathcal{D}_H = \mathrm{stack}_{h \in H}(M_h - I_n), \quad V^H = \{v : M_h v = v \ \forall h \in H\} = \ker \mathcal{D}_H. \quad (24.85)$$

It is enough to stack a certified generating set of H , but the algorithm uses every element so that its certificate is directly checkable. In the principal case $H = C_8 = \langle r \mid r^8 = 1 \rangle$, with $T = M_r$,

$$V^{C_8} = \ker(T - I_n), \quad (24.86)$$

because a vector fixed by T is fixed by every power of T . Rank–nullity gives the exact dimension and size

$$d_H = \dim_K V^H = n - \mathrm{rank} \mathcal{D}_H, \quad |V^H| = q^{d_H}. \quad (24.87)$$

Theorem 22.24.1 (Normalizer scope of fixed points). *The space V^H is a $K[N_B(H)]$ -submodule. In particular it is a $K[B]$ -submodule when $H \trianglelefteq B$. Without normalization, no B -stability conclusion is valid.*

Proof. If $b \in N_B(H)$, $v \in V^H$, and $h \in H$, then

$$M_h M_b v = M_b M_{b^{-1}hb} v = M_b v. \quad (24.88)$$

Thus $M_b v \in V^H$. Taking all $b \in B$ proves the normal-subgroup case. The calculation also identifies exactly where normalization is needed. $\square$

Invariant submodule and quotient coordinates. Let $J : K^s \hookrightarrow K^n$ be a full-column-rank canonical basis matrix for a subspace $U = \mathrm{im} J$. The subspace is B -invariant exactly when, for every $g \in B$, there is a unique matrix L_g such that

$$M_g J = J L_g. \quad (24.89)$$

In that event $L_g \in \mathrm{GL}_s(K)$, and the quotient action

$$\overline{M}_g(v + U) = M_g v + U \quad (24.90)$$

is well defined and invertible. Choose the lexicographically first list C of standard basis columns for which

$$P = [J \ C] \in \mathrm{GL}_n(K), \quad Q = [0 \ I_{n-s}] P^{-1}, \quad S = C. \quad (24.91)$$

Thus Q gives canonical quotient coordinates and S is a vector-space section. Direct multiplication yields

$$QJ = 0, \quad QS = I_{n-s}, \quad \ker Q = \mathrm{im} J, \quad v = J[I_s \ 0] P^{-1} v + SQv. \quad (24.92)$$

The quotient matrices are

$$\overline{M}_g = Q M_g S \in \mathrm{GL}_{n-s}(K), \quad (24.93)$$

and satisfy

$$Q M_g = \overline{M}_g Q, \quad \overline{M}_g \overline{M}_k = \overline{M}_{gk}. \quad (24.94)$$

Equivalently,

$$P^{-1}M_gP = \begin{pmatrix} L_g & X_g \\ 0 & \overline{M}_g \end{pmatrix} \quad (24.95)$$

for a uniquely determined X_g . Consequently

$$\dim_K(V/U) = n - s, \quad |V/U| = q^{n-s} = |V|/|U|. \quad (24.96)$$

Theorem 22.24.2 (Quotient construction and fixed-point comparison). *Equations (24.89)–(24.95) construct the exact sequence of $K[B]$ -modules*

$$0 \longrightarrow U \xrightarrow{J} V \xrightarrow{Q} V/U \longrightarrow 0. \quad (24.97)$$

For any $H \leq B$, restriction to fixed points has the natural exact sequence

$$0 \longrightarrow U^H \longrightarrow V^H \longrightarrow (V/U)^H \xrightarrow{\partial} H^1(H, U). \quad (24.98)$$

In particular, fixed points need not commute with quotient formation.

Proof. Existence in (24.89) is invariance; uniqueness follows from injectivity of J . Since $M_{g^{-1}}U \subseteq U$, the restrictions L_g are invertible. If $v - v' \in U$, then $M_g(v - v') \in U$, proving (24.90) is well defined. Equations (24.92)–(24.94) then prove representation and exactness.

For $z \in (V/U)^H$, choose a lift v with $Qv = z$. There is a unique $a_h \in K^s$ satisfying

$$Ja_h = M_hv - v. \quad (24.99)$$

The representation law and (24.89) give

$$a_{hk} = a_h + L_h a_k, \quad v \mapsto v + Ju \implies a_h \mapsto a_h + (L_h u - u). \quad (24.100)$$

Hence $\partial z = [a]$ is independent of the lift. Its class is zero exactly when some lift $v - Ju$ is H -fixed. This identifies $\ker \partial$ with the image of V^H , while the kernel of $V^H \rightarrow (V/U)^H$ is U^H , proving (24.98). $\square$

Thus

$$(V/U)^H = \text{im}(V^H \rightarrow V/U) \iff \partial = 0. \quad (24.101)$$

If $\text{char } K \nmid |H|$, the right side always holds. Indeed, for a cocycle a , put $A = \sum_{k \in H} a_k$; then $L_h A = A - |H|a_h$, so $a_h = L_h(-|H|^{-1}A) - (-|H|^{-1}A)$. In modular characteristic the connecting map can be nonzero.

Canonical construction algorithm. When fixed points are requested, the subgroup specification H_{spec} is typed as either (GENERAL, H , NONE) or (C8, H , r). The first accepts any certified subgroup; the second asserts that the supplied r generates H and has exact order eight in B . Thus an arbitrary noncyclic subgroup of order eight is never subjected to the cyclic test. The typed input is

$$\text{FixedQuotient}(\mathcal{R}_V; \text{scope}, H_{\text{spec}} \text{ or NONE}, J_{\text{in}} \text{ or NONE}), \quad (24.102)$$

with the exhaustive presence rules

scope	H_{spec}	J_{in}
FIXED	one typed subgroup specification	NONE
QUOTIENT	NONE	an $n \times s$ matrix
BOTH	one typed subgroup specification	an $n \times s$ matrix.

(24.103)

The boundary values $s = 0, n$ are permitted. Any other presence pattern is structurally invalid.

- STEP 1. Rehash and independent verification $\mathcal{R}_V$, including every row of (24.84), and validate K, B, n, scope and the exact presence pattern. A malformed field, stale hash, or failed parent independent verification returns `INVALID_INPUT`.
- STEP 2. When H_{spec} is present, validate its discriminator and exact field presence, then verify identity, inverses, closure, the canonical enumeration of H , and its generators. Any type, presence, identity, inverse, closure, enumeration, or generator failure returns `INVALID_INPUT` with the least failing field or group-table row. Only in C8 mode verify $H = (1, r, \dots, r^7)$, $r^8 = 1$, and that no smaller positive power of r is 1; failure has the same classification.
- STEP 3. Build $\mathcal{D}_H$, compute its fixed-pivot reduced row-echelon form and canonical kernel basis F . Verify every row of $M_h F = F$, rank-nullity, (24.87), the normalizer, and every induced normalizer-action matrix on F .
- STEP 4. When J_{in} is present, require a declared integer $0 \leq s \leq n$, shape $n \times s$, entries in K , and rank s . A field, integer, bound, shape, or rank failure returns `INVALID_INPUT` with its shape row or fixed-pivot rank witness. Replace a valid matrix by the fixed-pivot canonical basis J of the same column space and solve (24.89) in group order. Only when this structurally valid full-rank subspace fails (24.89), return `NONINVARIANT_REJECT` with the least g , least offending column, its reduced membership remainder, and all preceding successful rows.
- STEP 5. Construct C, P, Q, S by (24.91), and verify (24.92) before constructing $\overline{M}_g$. Verify all rows of (24.93)–(24.97), including identities, inverses, dimensions, and cardinalities.
- STEP 6. In BOTH mode, compute fixed-pivot bases of $U^H, V^H, (V/U)^H$. Enumerate $Z^1(H, U)$ and $B^1(H, U)$, construct $H^1(H, U)$ by least coset representatives, and serialize (24.99)–(24.101) for every quotient-fixed basis vector.
- STEP 7. Serialize the canonical bases, matrices, action tables, exactness rows, normalizer scope, connecting classes, first-failure data, input hash, and independent verification hash. Return `SATISFIED` only after a byte-identical independent verification.

Theorem 22.24.3 (Algorithm correctness and termination). *The algorithm terminates on every finite typed input. It returns `SATISFIED` exactly when the parent is a certified module, every requested subgroup is valid, and every requested quotient subspace is B -invariant. On passage it constructs precisely the fixed and quotient modules in (24.85) and (24.90), with exact dimensions, orders, and comparison map (24.98). Equal canonical inputs produce byte-identical records.*

Proof. Every group, vector space, matrix space, and cochain space enumerated is finite. Gaussian elimination and the fixed standard-column completion terminate. Equations (24.85)–(24.100) prove soundness of every accepted object; the failed membership remainder proves the only mathematical rejection. Fixed pivot, element, basis, coset, failure, and serialization orders prove determinism. Exhaustive independent verification proves completeness against the stated typed specification. $\square$

The successful record is

$$\begin{aligned}
&(\mathbf{h}_{\mathcal{R}_V}, K, B, n, \text{scope}, H_{\text{spec}} \text{ or NONE}, J_{\text{in}} \text{ or NONE}, \\
&\mathcal{D}_H \text{ or NONE}, F \text{ or NONE}, d_H \text{ or NONE}, N_B(H) \text{ or NONE}, \\
&J, L_g, C, P, Q, S, \overline{M}_g \text{ or NONE}, \text{exactness_rows}, \\
&\text{fixed_quotient_rows or NONE}, \text{connecting_classes or NONE}, \\
&\text{classification, COMPLETE, NONE}, \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}}).
\end{aligned} \tag{24.104}$$

Absent scope-dependent fields are **NONE**. The mutually exclusive classifications are

$$\text{INVALID_INPUT}, \quad \text{NONINVARIANT_REJECT}, \quad \text{SATISFIED}. \quad (24.105)$$

Every rejection record has the same schema, retains all fields available before exit, names the failing stage and first failed predicate, and sets all later fields to **NONE**.

Minimum test family. For $H = 1$, (24.85) returns V . For $U = 0$ and $U = V$, the quotients are respectively V and 0 , with the dimensions and orders in (24.96). A modular strict-inclusion test is

$$K = \mathbb{F}_2, \quad H = C_8 = \langle r \rangle, \quad T = M_r = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad U = Ke_1, \quad (24.106)$$

where the action factors through C_2 , so $T^8 = I$. Here

$$V^H = U, \quad (V/U)^H = V/U, \quad \text{im}(V^H \rightarrow V/U) = 0, \quad (24.107)$$

and $\partial \neq 0$, proving that fixed points and quotients cannot be silently interchanged in characteristic 2. With the same T , $J_{\text{in}} = e_2$ is not invariant and tests **NONINVARIANT_REJECT**. A wrong-width J_{in} , a nonclosed H -table, and a stale parent hash independently test **INVALID_INPUT**.

22.25 Parity modules and certified descent

Write $C_2 = \{0, 1\}$ additively. Let $K = \mathbb{F}_q$, let $V = K^n$, and let G act through certified matrices M_g . A kernel-parity functional is an additive homomorphism

$$\lambda : (V, +) \longrightarrow C_2, \quad \lambda(v + w) = \lambda(v) + \lambda(w). \quad (24.108)$$

This is an $\mathbb{F}_2$ -linear functional on the $\mathbb{F}_2$ -space underlying V when $\text{char } K = 2$. When $\text{char } K \neq 2$, it is necessarily zero: every nonzero v has odd prime order while C_2 has exponent two. It is not silently treated as a K -linear functional when $K \neq \mathbb{F}_2$.

Definition 22.25.1 (Parity module). The pair (V, λ) is a G -parity module if

$$\lambda(M_g v) = \lambda(v) \quad (g \in G, v \in V). \quad (24.109)$$

If a certified order-eight module operator T is present, it is a G - and C_8 -parity module when also $\lambda(Tv) = \lambda(v)$ for every v .

verification homomorphism on an extension. Let $f : G \times G \rightarrow V$ be a normalized two-cocycle for the given action and let the certified group extension be

$$E_f = V \times G, \quad (v, g)(w, h) = (v + M_g w + f(g, h), gh). \quad (24.110)$$

For an additive λ and a normalized function $\eta : G \rightarrow C_2$, define

$$\epsilon_{\lambda, \eta}(v, g) = \lambda(v) + \eta(g). \quad (24.111)$$

Theorem 22.25.2 (Exact parity-verification criterion). *Every homomorphism $E_f \rightarrow C_2$ has the unique form (24.111). The displayed function is a homomorphism if and only if*

$$\begin{aligned} \lambda(M_g w) &= \lambda(w) & (g \in G, w \in V), \\ (\delta\eta)(g, h) &:= \eta(g) + \eta(h) - \eta(gh) = \lambda(f(g, h)) & (g, h \in G). \end{aligned} \quad (24.112)$$

Thus a prescribed parity module (V, λ) extends to E_f if and only if

$$[\lambda \circ f] = 0 \quad \text{in } H^2(G, C_2). \quad (24.113)$$

The condition is not the stronger pointwise demand $\lambda \circ f = 0$; a nonzero coboundary is allowed.

Proof. For a homomorphism ϵ , set $\lambda(v) = \epsilon(v, 1)$ and $\eta(g) = \epsilon(0, g)$. Since $(v, g) = (v, 1)(0, g)$, (24.111) and uniqueness follow. Comparing the two sides of the homomorphism equation gives

$$\begin{aligned}\epsilon((v, g)(w, h)) &= \lambda(v) + \lambda(M_g w) + \lambda(f(g, h)) + \eta(gh), \\ \epsilon(v, g) + \epsilon(w, h) &= \lambda(v) + \lambda(w) + \eta(g) + \eta(h).\end{aligned}\tag{24.114}$$

Setting $h = 1$ and varying w forces the first line of (24.112); the remaining equality is its second line. Conversely those two identities make (24.114) equal for every pair. Finally, $\lambda \circ f$ is a normalized C_2 -valued cocycle because λ is additive and G -invariant, so solvability is exactly the vanishing of its cohomology class. $\square$

Changing the section by a cochain $t : G \rightarrow V$ replaces f by its cohomologous factor set and changes

$$\lambda \circ f \longmapsto \lambda \circ f + \delta(\lambda \circ t),\tag{24.115}$$

so (24.113) is independent of coordinates. Once one η exists, all others are obtained by adding $\text{Hom}(G, C_2)$.

Eightfold compatibility. Suppose a certified lift of the order-eight symmetry has the form

$$\Theta(v, g) = (Tv + t(g), Pg),\tag{24.116}$$

where T is additive and invertible, $P \in \text{Aut}(G)$, and the lift equations have already been independently verified. Substitution into (24.111) shows that $\epsilon_{\lambda, \eta} \Theta = \epsilon_{\lambda, \eta}$ if and only if

$$\lambda T = \lambda, \quad \lambda(t(g)) + \eta(Pg) = \eta(g) \quad (g \in G).\tag{24.117}$$

Both rows are required. Invariance of λ on the module alone does not imply invariance of the verification homomorphism when the translation cochain t is nonzero.

Surface relation, physical parity, and descent. Use the fixed genus-two presentation

$$\Gamma_B = \langle a_1, b_1, a_2, b_2 \mid \mathcal{R} = [a_1, b_1][a_2, b_2] = 1 \rangle, \quad \chi(a_i) = \chi(b_i) = 1.\tag{24.118}$$

The exponent-sum homomorphism from the free group sends each commutator to zero, hence

$$\chi_F(\mathcal{R}) = \chi_F([a_1, b_1]) + \chi_F([a_2, b_2]) = 0 + 0 = 0 \quad \text{in } C_2.\tag{24.119}$$

Thus χ_F factors to the fixed surjective parity $\chi : \Gamma_B \rightarrow C_2$.

Let

$$\pi : \Gamma_B \twoheadrightarrow Q, \quad N = \ker \pi\tag{24.120}$$

be a certified finite quotient; in the present construction Q may be E_f . There is a unique homomorphism $\epsilon_Q : Q \rightarrow C_2$ satisfying $\epsilon_Q \pi = \chi$ if and only if

$$N \subseteq \ker \chi.\tag{24.121}$$

Necessity follows on N . Conversely, define $\epsilon_Q(\pi(\gamma)) = \chi(\gamma)$; (24.121) makes this well defined, and surjectivity gives uniqueness. This is descent to the quotient, rather than a bit assignment on selected elements.

Let $g_0, \dots, g_7$ be the certified physical side-pairing words and write $q_j = \pi(g_j)$. The required physical test is

$$\epsilon_Q(q_j) = 1 \quad (0 \leq j < 8).\tag{24.122}$$

The certificate also evaluates each g_j as a word in a_1, b_1, a_2, b_2 , proves $\chi(g_j) = 1$, and checks that its group evaluation is q_j . Separately, it records both

$$\pi(\mathcal{R}) = 1_Q, \quad \epsilon_Q(\pi(\mathcal{R})) = \epsilon_Q([\pi(a_1), \pi(b_1)]) + \epsilon_Q([\pi(a_2), \pi(b_2)]) = 0.\tag{24.123}$$

The second equality holds because every homomorphism to the abelian group C_2 kills commutators; it does not replace the first, group-valued relator check.

Canonical construction algorithm. The input is

$$\text{ParityModule}(\mathcal{R}_{E_f}, \mathcal{R}_\pi; \text{mode}, \text{kernel_policy}, \lambda_{\text{in}} \text{ or NONE}, \eta_{\text{in}} \text{ or NONE}, \mathcal{R}_\Theta \text{ or NONE}), \quad (24.124)$$

where $\text{mode} \in \{\text{VALIDATE}, \text{DECIDE}\}$ and $\text{kernel_policy} \in \{\text{ANY}, \text{TRIVIAL}, \text{NONTRIVIAL}\}$. Here ANY imposes no condition on λ , TRIVIAL requires $\lambda = 0$, and NONTRIVIAL requires $\lambda \neq 0$; the record stores the tested equality or a least vector v with $\lambda(v) = 1$. The exhaustive presence specification is

mode	λ_{in}	η_{in}
VALIDATE	full table $V \rightarrow C_2$	full normalized table $G \rightarrow C_2$
DECIDE	NONE	NONE.

(24.125)

The optional symmetry record is either a complete closed record or NONE; partial symmetry fields are invalid. The quotient record is required to have codomain exactly the same encoded group $Q = E_f$: its canonical element order, multiplication table, identity, and hash must equal those in $\mathcal{R}_{E_f}$. If supplied, $\mathcal{R}_\Theta$ must act on that same encoded E_f .

- STEP 1. Rehash and independent verification the finite field, action, normalized cocycle, extension multiplication table, and quotient map. Verify the exact codomain table/hash match $Q = E_f$, surjectivity, all presentation relators in Q , all physical word evaluations, modes, policies, and exact field presence. Also require any symmetry record to name the same E_f . Structural failure returns INVALID_INPUT.
- STEP 2. Enumerate additive maps $V \rightarrow C_2$ in truth-table order, or validate the supplied full table. Test (24.108), (24.109), and the explicitly defined kernel-policy predicate, storing either $\lambda = 0$ or its least nonzero-value witness. In odd characteristic the enumeration contains only zero. If $\mathcal{R}_\Theta$ is present, independent verification it and also test $\lambda T = \lambda$.
- STEP 3. For each retained λ , enumerate all normalized $\eta : G \rightarrow C_2$ in lexicographic truth-table order, or use the supplied table. Test every row of (24.112), retaining the least failed predicate for every rejected pair.
- STEP 4. For a surviving pair, construct the full table (24.111) and independently test $\epsilon(xy) = \epsilon(x) + \epsilon(y)$ on every ordered pair in E_f^2 . This full multiplication-table test is the verification homomorphism certificate.
- STEP 5. Test $\epsilon\pi = \chi$ on a_1, b_1, a_2, b_2 , all eight rows of (24.122), and both rows of (24.123). Equality on the presentation generators, together with the homomorphism and relator certificates, proves equality on every element and hence (24.121).
- STEP 6. If $\mathcal{R}_\Theta$ is present, test the second row of (24.117) on every g , and independently compare $\epsilon(\Theta x)$ with $\epsilon(x)$ for every $x \in E_f$.
- STEP 7. In VALIDATE mode, any well-typed mathematical failure returns CANDIDATE_FAIL with the first failing row. In DECIDE mode, return the first accepted pair; only exhaustion of the entire finite pair list returns NO_ACCEPTING_PARITY with all candidate failure rows.
- STEP 8. Serialize all tables, cohomology rows, physical and relator rows, symmetry rows, failure data, hashes, and the chosen pair. Return SATISFIED only after byte-identical independent verification.

Theorem 22.25.3 (Algorithm correctness, completeness, and termination). *On structurally valid input, the algorithm returns SATISFIED exactly when there exists a parity satisfying the selected kernel policy, the extension-homomorphism criterion, physical-generator parity, surface descent, and any requested order-eight compatibility. A validator failure rejects only its supplied pair; a decision-mode nonexistence result is issued only after exhaustive search. Equal canonical inputs produce byte-identical records.*

Proof. The additive-map and normalized-function spaces have respectively at most $2^{|V|}$ and $2^{|G|-1}$ tables, so the ordered search is finite. The verification theorem proves Steps 2–4 in both directions. The descent argument proves Step 5, and direct substitution proves Step 6. Thus every passing pair has all requested properties, while every possible pair appears in decision mode. Fixed table, element, pair, predicate, and serialization orders prove reproducibility. $\square$

The successful record is

$$\begin{aligned}
&(\mathbf{h}_{E_f}, \mathbf{h}_\pi, \mathbf{h}_\Theta \text{ or NONE, mode, kernel_policy,} \\
&\lambda_{\text{tab}}, \eta_{\text{tab}}, (\lambda f)_{\text{tab}}, (\delta\eta)_{\text{tab}}, \epsilon_{\text{tab}}, \\
&\text{additivity_rows, invariance_rows, homomorphism_rows,} \\
&\text{presentation_rows, physical_rows, relator}_Q\text{_rows, relator}_{C_2}\text{_rows,} \\
&\text{descent_rows, symmetry_rows or NONE, candidate_failure_rows,} \\
&\text{classification, COMPLETE, NONE, h}_{\text{input}}, \mathbf{h}_{\text{independentverification}}).
\end{aligned} \tag{24.126}$$

The mutually exclusive terminal classifications are

$$\text{INVALID_INPUT, CANDIDATE_FAIL, NO_ACCEPTING_PARITY, SATISFIED.} \tag{24.127}$$

Each rejection record has the same schema, retains every field available before exit, identifies stage and first failure, and sets later fields to NONE.

Minimum test family. For the positive descent test take $V = 0$, $G = C_2$, $f = 0$, hence $E_f = Q = C_2$, take $\pi = \chi$, and set $\epsilon_Q = \text{id}_{C_2}$. All physical images have value one and (24.123) holds. Pullback from any quotient parity is recovered by $\lambda = 0$. For a genuine modular kernel parity take $K = \mathbb{F}_2$, $G = 1$, $V = C_2$, $f = 0$, and $\lambda = \text{id}$. Add the abstract certified C_8 action whose generator has $T = I_V$, with $P = 1, t = 0$. The action on V and E_f is nonfaithful, but both rows of (24.117) pass.

For an obstruction test take $V = G = C_2$ with trivial action, $\lambda = \text{id}$, and the normalized carry cocycle

$$f(x, y) = \left\lfloor \frac{x + y}{2} \right\rfloor \pmod{2}. \tag{24.128}$$

Then $E_f \cong C_4$, and at $(1, 1)$ every normalized η has $(\delta\eta)(1, 1) = 0$, whereas $\lambda(f(1, 1)) = 1$, so the prescribed kernel parity cannot extend. The trivial quotient of Γ_B has $N = \Gamma_B \not\subseteq \ker \chi$, and therefore tests

$$\text{NO_ACCEPTING_PARITY.}$$

For a concrete validator failure in the positive $V = 0, G = C_2, E_f = C_2, \pi = \chi$ setup, supply $\lambda = 0, \eta = 0$. The resulting zero map is a homomorphism but $\epsilon\pi \neq \chi$, so the classification is CANDIDATE_FAIL. A partial η -table, broken relator image, mismatched quotient codomain, or stale hash tests INVALID_INPUT.

22.26 Exact dimension and group-order control

Let $K = \mathbb{F}_q$ with $q = p^e$, and let $V = K^d$ be a certified module. Its additive group has the exact order

$$|V| = q^d = p^{ed}. \quad (24.129)$$

This identity depends only on the field and dimension, not on irreducibility, indecomposability, faithfulness, or a chosen cohomology class.

Theorem 22.26.1 (Module operations and exact size). *For every short exact sequence of finite-dimensional K -modules*

$$0 \longrightarrow U \longrightarrow V \longrightarrow W \longrightarrow 0, \quad (24.130)$$

one has

$$\dim_K V = \dim_K U + \dim_K W, \quad |V| = |U||W|. \quad (24.131)$$

Consequently,

$$\begin{aligned} \dim(V_1 \oplus V_2) &= d_1 + d_2, & |V_1 \oplus V_2| &= q^{d_1+d_2}, \\ \dim(V/U) &= d - s, & |V/U| &= q^{d-s} \quad (\dim U = s). \end{aligned} \quad (24.132)$$

These formulas include $d = 0$, $U = 0$, and $U = V$.

Proof. Choose a basis of U , extend its image to a basis of V , and map the added vectors to W . Exactness makes their images a basis of W , proving the dimension identity. Equation (24.129) converts addition of dimensions into multiplication of cardinalities. The two specializations give (24.132). $\square$

From a module kernel to a group order. Suppose a certified group extension is exact:

$$1 \longrightarrow (V, +) \longrightarrow E \longrightarrow G \longrightarrow 1. \quad (24.133)$$

Every fiber over G is a coset of V , so

$$|E| = q^d |G| = p^{ed} |G|. \quad (24.134)$$

No splitting or centrality assumption is used. Distinct factor sets may change the isomorphism type of E , but not (24.134). Conversely, a bare G -module does not itself multiply a group order: the factor q^d is justified only after an exact group-extension record, or after an explicitly constructed semidirect product, has passed.

If $U \leq V$ is G -invariant of dimension s , then it is normal in E : conjugation by an element above g restricts on V to the certified action of g . Quotienting (24.133) gives

$$1 \longrightarrow V/U \longrightarrow E/U \longrightarrow G \longrightarrow 1, \quad |E/U| = q^{d-s} |G| = \frac{|E|}{q^s}. \quad (24.135)$$

The invariance and normality certificates are indispensable; dimension subtraction alone does not define a group quotient.

For a tower of certified exact extensions

$$1 \rightarrow W_j \rightarrow Q_j \rightarrow Q_{j-1} \rightarrow 1, \quad |W_j| = q_j^{\Delta_j} = p_j^{e_j \Delta_j} \quad (1 \leq j \leq r), \quad (24.136)$$

induction gives

$$|Q_r| = |Q_0| \prod_{j=1}^r q_j^{\Delta_j}. \quad (24.137)$$

The overflow-free prime-exponent form is

$$\nu_\ell(|Q_r|) = \nu_\ell(|Q_0|) + \sum_{\substack{1 \leq j \leq r \\ p_j = \ell}} e_j \Delta_j \quad (\ell \text{ prime}). \quad (24.138)$$

Arbitrary-precision integers and (24.138), rather than floating logarithms, are the authoritative representation.

Order-eight bookkeeping. An order-eight automorphism of Q_r does not imply $8 \mid |Q_r|$. If a new external C_8 factor is actually adjoined by a certified semidirect product, then and only then

$$|Q_r \rtimes C_8| = 8|Q_r|. \quad (24.139)$$

If the order-eight action is already internal to the recorded group, there is no additional factor. The order-control input therefore uses the exclusive modes `NONE`, `ALREADY_INCLUDED`, and `EXTERNAL_SEMIDIRECT`.

Exact budget headroom. For current exact order N , field size $q_\star \geq 2$, and positive integer cap M , define

$$d_{\max}(N, q_\star, M) = \max\{a \in \mathbb{Z}_{\geq 0} : Nq_\star^a \leq M\}, \quad (24.140)$$

when $N \leq M$, and `NONE` otherwise. It is computed by integer multiplication or monotone binary search: find the first a with $Nq_\star^a > M$, then return $a - 1$. Thus

$$Nq_\star^{d_{\max}} \leq M < Nq_\star^{d_{\max}+1}, \quad (24.141)$$

which is a complete integer witness and involves no rounding.

Separation is an independent obligation. Let $\pi : \Gamma_B \twoheadrightarrow Q_r$ be a certified quotient and let $\mathcal{D}$ be a finite ordered list of dangerous nonidentity source words. The required separation predicate is

$$\text{Sep}(\pi, \mathcal{D}) \iff \pi(w) \neq 1_{Q_r} \text{ for every } w \in \mathcal{D}. \quad (24.142)$$

Neither a larger d , a larger $|Q_r|$, nor unused budget implies (24.142). Each word image must be evaluated and certified.

Canonical control algorithm. The input is

$$\begin{aligned} &\text{ModuleOrderControl}(\mathcal{R}_{Q_0}; (\mathcal{R}_{V_j}, \mathcal{R}_{U_j} \text{ or } \text{NONE}, \mathcal{R}_{Q_j})_{j=1}^r; \\ &\quad \text{c8_mode}, \mathcal{R}_8 \text{ or } \text{NONE}, \mathcal{R}_\rtimes \text{ or } \text{NONE}; M \text{ or } \text{NONE}, q_\star \text{ or } \text{NONE}; \\ &\quad \mathcal{R}_\pi \text{ or } \text{NONE}, \mathcal{D} \text{ or } \text{NONE}), \end{aligned} \quad (24.143)$$

Here $r \geq 0$, and $r = 0$ is the permitted empty tower with final group Q_0 . The record $\mathcal{R}_{Q_j}$ certifies the exact row in (24.136). If $\mathcal{R}_{U_j}$ is absent, $W_j = V_j$ and $s_j = 0, \Delta_j = d_j$; otherwise it must certify a Q_{j-1} -invariant submodule of dimension s_j , the kernel is the certified quotient $W_j = V_j/U_j$, and $\Delta_j = d_j - s_j$. In the latter case $\mathcal{R}_{Q_j}$ contains an explicit action-preserving isomorphism $\kappa_j : V_j/U_j \rightarrow \ker(Q_j \rightarrow Q_{j-1})$; it does not identify U_j itself with a subgroup of Q_j .

The presence rules are

c8_mode	$\mathcal{R}_8$	$\mathcal{R}_\rtimes$
NONE	NONE	NONE
ALREADY_INCLUDED	internal certificate	NONE
EXTERNAL_SEMIDIRECT	action certificate	semidirect record

(24.144)

M	$q_\star$
NONE	NONE
positive integer	NONE or a prime power ≥ 2 .

An internal certificate names the exact hash of Q_r , exhibits the already-counted C_8 subgroup or action data, and proves that no new group is being formed. An external action certificate also names Q_r , while the semidirect record constructs and projects the new group. All other combinations are invalid. The separation record and danger list are either both present or both `NONE`.

- STEP 1. Rehash and independent verification the base group, all module records, and every exact-extension row. Validate r , list lengths, finite fields, dimensions, modes, arbitrary-precision integer encodings, and all presence rules. Factor $|Q_0|$ by fixed increasing-prime trial division, store each division row and the final primality witness, and reconstruct the integer from its exponent vector. When q_* is present, factor it by the same procedure and require exactly one prime factor. Structural failure returns `INVALID_INPUT`.
- STEP 2. For each layer, recompute $q_j = p_j^{e_j}$ from its field certificate and $q_j^{d_j}$ from repeated squaring. If a submodule is present, independent verification its invariance, rank, quotient map, and $\Delta_j = d_j - s_j$. If absent, record and verify $s_j = 0, \Delta_j = d_j$.
- STEP 3. Replay injection, projection, homomorphism, surjectivity, and kernel-equality rows for (24.136). Construct a fiber bijection for every element of Q_{j-1} , and independently require $|Q_j| = q_j^{\Delta_j} |Q_{j-1}|$. When U_j is present, independent verification every bijectivity, action-intertwining, and kernel-image row of $\kappa_j : V_j/U_j \rightarrow \ker(Q_j \rightarrow Q_{j-1})$.
- STEP 4. Accumulate the decimal integer and prime vector (24.138), and compare them with the canonical table length of every Q_j . Replay the $|Q_0|$, q_j , and any q_* factorization witnesses and reconstruct every integer. Any mismatch is `INVALID_INPUT`, not a budget result.
- STEP 5. Process `c8_mode`. In external mode independent verification the matching $\mathcal{R}_8$ action, semidirect multiplication table, exact projection to C_8 , and (24.139), then replace N by $8|Q_r|$. In already-included mode independent verification the matching internal certificate and its Q_r hash but do not change $N = |Q_r|$.
- STEP 6. If M is present, compare $N \leq M$. If false, return `BUDGET_EXCEEDED` with $N, M, N - M$. If q_* is present, compute (24.140) and store both inequalities (24.141).
- STEP 7. If $(\mathcal{R}_\pi, \mathcal{D})$ is present, independent verification the quotient map, its exact target-table/hash match, the nonidentity source certificates for all $w \in \mathcal{D}$, and every word evaluation. The least w with $\pi(w) = 1$ returns `SEPARATION_FAIL` with its evaluation trace.
- STEP 8. Serialize all dimensions, quotient ranks, powers, exact rows, fiber bijections, prime vectors, C_8 rows, budget inequalities, separation traces, failure data, and hashes. Return `SATISFIED` only after byte-identical independent verification.

Theorem 22.26.2 (Correctness, completeness, and reproducibility). *The algorithm terminates and returns `SATISFIED` exactly when all structural and exactness certificates are valid, the resulting exact order satisfies any cap, and every requested dangerous word is separated. Every reported dimension, order, prime vector, and headroom bound is exact. Equal canonical inputs produce byte-identical records.*

Proof. All groups, modules, lists, and tables are finite. Repeated squaring and integer binary search terminate. Equations (24.129)–(24.138) prove every layer count, while the independently verified semidirect record proves the only possible factor eight. Equations (24.140)–(24.141) prove the budget answer in both directions, and exhaustive word evaluation is equivalent to (24.142). Hence verification is sound and all stated rejections are complete for their stage. Fixed layer, prime, element, word, failure, integer, and byte orders prove reproducibility. $\square$

For compact serialization notation, define the complete j -th layer record by

$$\mathcal{L}_j := (p_j, e_j, q_j, d_j, s_j, \Delta_j, \\ \mathbf{h}_{V_j}, \mathbf{h}_{U_j} \text{ or } \text{NONE}, \\ \mathbf{h}_{Q_j}).$$

The successful record is

$$\begin{aligned}
& (\mathbf{h}_{Q_0}, r, (\mathcal{L}_j)_{j=1}^r, \\
& (|Q_j|)_{j=0}^r, \text{fiber_rows}, \\
& \text{kernel_isomorphism_rows}, \text{prime_vectors}, \\
& \text{c8_mode}, \mathbf{h}_{\mathcal{R}_8} \text{ or NONE}, \text{c8_rows or NONE}, \\
& N, M \text{ or NONE}, q_\star \text{ or NONE}, d_{\max} \text{ or NONE}, \text{budget_rows}, \\
& \text{danger_words or NONE}, \text{separation_traces or NONE}, \\
& \text{classification, COMPLETE, NONE, } \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}}).
\end{aligned} \tag{24.145}$$

The fixed precedence and mutually exclusive terminal classifications are

$$\text{INVALID_INPUT} \prec \text{BUDGET_EXCEEDED} \prec \text{SEPARATION_FAIL} \prec \text{SATISFIED}. \tag{24.146}$$

Every rejection record retains all fields available before its exit, stores stage and first witness, and sets later fields to NONE.

Minimum test family. Take $Q_0 = C_3$, $K = \mathbb{F}_2$, and the trivial C_3 -action on a certified two-dimensional kernel layer. Then $|Q_1| = 3 \cdot 2^2 = 12$, regardless of whether the extension splits. Quotienting the kernel by either invariant line gives order $3 \cdot 2 = 6$. With $N = 12$, $M = 24$, and $q_\star = 2$, the exact headroom is

$$d_{\max} = 1, \quad 12 \cdot 2 \leq 24 < 12 \cdot 2^2. \tag{24.147}$$

External semidirect mode changes 12 to 96; already-included mode leaves it 12. A cap $M = 11$ tests BUDGET_EXCEEDED, while $d = 0$, $U = 0$, and $U = V$ test dimension boundaries; $r = 0$ tests the empty tower.

Finally take the declared source Γ_B and $Q = C_2$. Define $\pi_1(a_1) = 1$ and all other presentation generators to be zero; define $\pi_2(b_1) = 1$ and all others to be zero. Both maps are surjective and kill the surface relator, and they have the same target order, but for the dangerous word b_1 , only π_2 separates it. This explicit equal-order pair proves that no order or dimension certificate can replace (24.142). A stale exactness hash, a noninvariant proposed submodule, a false semidirect table, and a target-table mismatch test INVALID_INPUT.

Part X

Arithmetic and congruence quotients

23 Arithmetic representation of the Bolza group

23.1 The common generator number field

Fix the chosen linear lifts from C02 and C05. Put

$$\xi = \zeta_{16} = e^{\pi i/8}, \quad \zeta = \xi^2, \quad \beta = \sqrt{2 + 2\sqrt{2}} > 0, \quad E = \mathbb{Q}(\xi, \beta). \tag{25.1}$$

The inequality specifies the distinguished complex embedding; it is not part of the abstract field presentation.

Theorem 23.1.1 (Degree and a certified basis). *The cyclotomic field $F = \mathbb{Q}(\xi)$ has degree eight, $\beta \notin F$, and*

$$[E : \mathbb{Q}] = 16, \quad \mathcal{B}_E = (1, \xi, \dots, \xi^7, \beta, \beta\xi, \dots, \beta\xi^7) \tag{25.2}$$

is a $\mathbb{Q}$ -basis of E .

Proof. The element ξ has exact order sixteen and is a root of $\Phi_{16}(X) = X^8 + 1$. This polynomial is irreducible: after the translation $X \mapsto X + 1$,

$$\Phi_{16}(X + 1) = (X + 1)^8 + 1 = X^8 + 8X^7 + 28X^6 + 56X^5 + 70X^4 + 56X^3 + 28X^2 + 8X + 2.$$

Every nonleading coefficient is divisible by 2, while the constant term 2 is not divisible by 4. Eisenstein's criterion at 2 makes the translated polynomial irreducible over $\mathbb{Q}$. Translation is an automorphism of $\mathbb{Q}[X]$, so Φ_{16} is irreducible as well. It is therefore the minimal polynomial of ξ , and $[F : \mathbb{Q}] = 8$. Moreover

$$\sqrt{2} = \xi^2 + \xi^{-2} = \xi^2 - \xi^6, \quad \beta^2 = 2 + 2\sqrt{2}, \quad (25.3)$$

because $\xi^8 = -1$. Suppose $\beta \in F$. Under the distinguished embedding β is real, so complex conjugation fixes it and hence $\beta \in F^+ = F \cap \mathbb{R}$. The real embedding of F^+ induced by $\xi \mapsto \xi^3$ sends $\sqrt{2} = \xi^2 + \xi^{-2}$ to $-\sqrt{2}$. Applying it to (25.3) would give

$$\sigma(\beta)^2 = 2 - 2\sqrt{2} < 0, \quad (25.4)$$

impossible because $\sigma(\beta) \in \mathbb{R}$. Thus $X^2 - (2 + 2\sqrt{2})$ is irreducible over F , $[E : F] = 2$, and the tower basis theorem gives (25.2). $\square$

For completeness, β has rational minimal polynomial

$$m_\beta(X) = X^4 - 4X^2 - 4. \quad (25.5)$$

Indeed, (25.3) gives $m_\beta(\beta) = 0$, while $\mathbb{Q}(\beta^2) = \mathbb{Q}(\sqrt{2})$ has degree two. The same negative-conjugate argument shows that $2 + 2\sqrt{2}$ is not a square in $\mathbb{Q}(\sqrt{2})$; hence adjoining β gives another factor two. Thus m_β has degree four and is minimal.

All specified coefficients, and no accidental enlargement. Set $A = 1 + \sqrt{2}$ and $B = \beta$. The physical matrices and the linear rotation are

$$G_j = \begin{pmatrix} A & B\zeta^j \\ B\zeta^{-j} & A \end{pmatrix} \quad (0 \leq j < 8), \quad C = \begin{pmatrix} \xi & 0 \\ 0 & \xi^{-1} \end{pmatrix}. \quad (25.6)$$

Equations (25.1)–(25.3) place every entry in E . Products, inverses of determinant-one matrices, and the presentation words A_1, B_1, A_2, B_2 therefore also have entries in E . Conversely, the upper-left entry of C is ξ and the upper-right entry of G_0 is β , so the field generated by all entries of the chosen lifts is exactly

$$\mathbb{Q}(\text{entries of } C, G_0, \dots, G_7) = E. \quad (25.7)$$

This equality concerns the fixed coherent linear lifts; projective classes alone do not canonically select their signs.

Exact coordinate model. Let x, y denote the coordinate images of ξ, β . Relative to $\mathcal{B}_E$, multiplication reduces uniquely by

$$x^8 = -1, \quad y^2 = 2 + 2(x^2 - x^6), \quad xy = yx. \quad (25.8)$$

Thus every element has a unique form

$$z = \sum_{a=0}^7 u_a x^a + y \sum_{a=0}^7 v_a x^a, \quad u_a, v_a \in \mathbb{Q}. \quad (25.9)$$

For nonzero z , inversion is obtained by solving the exact 16×16 rational linear system for multiplication by z . The system is nonsingular because (25.8) presents the field E , not merely a quotient algebra.

The distinguished roots are also isolated without floating ambiguity. The polynomial m_β is negative at 2, positive at $9/4$, and has derivative $4X(X^2 - 2) > 0$ on that interval; hence

$$2 < \beta < \frac{9}{4} \quad (25.10)$$

selects its unique positive root there. The root-of-unity datum $\xi = e^{\pi i/8}$, equivalently its primitive-sixteenth-root index, selects the required root of $X^8 + 1$.

Canonical construction algorithm. The input

$$\text{GeneratorField}(\mathcal{R}_{\text{geom}}; \Phi_{16}, m_\beta, \text{root}_\xi, (2, 9/4)_\beta) \quad (25.11)$$

contains the closed exact records for (25.6), including the physical and presentation matrices and their source hashes.

- STEP 1. Rehash and independent verification the geometric matrix record. Validate the rational polynomial coefficients, degrees, primitive-root index, rational isolating endpoints, matrix dimensions, and exact field presence. A malformed field or stale parent record returns `INVALID_INPUT`.
- STEP 2. Verify $\Phi_{16} = X^8 + 1$, independent verification the displayed translation expansion, and record the seven nonconstant binomial rows $8, 28, 56, 70, 56, 28, 8 \equiv 0 \pmod{2}$, together with $2 \not\equiv 0 \pmod{4}$ and the invertible translation row. These independent verification the Eisenstein proof. Verify $\xi^8 = -1$, hence exact order sixteen, and construct $F = \mathbb{Q}[X]/(\Phi_{16})$ in the power basis.
- STEP 3. Verify (25.3), (25.5), the signs and derivative at the endpoints of (25.10), and the negative real conjugate (25.4). Record the degree tower $1, 8, 16$ and the proof that the sixteen vectors in (25.2) are independent and spanning.
- STEP 4. Construct the multiplication matrices of x and y from (25.8), then all sums, products, equality tests, and inverses in the unique coordinates (25.9). Check associativity on the basis triples and verify both defining relations independently.
- STEP 5. Encode A, B, ζ, ξ^{-1} , every entry of (25.6), every inverse generator, and every presentation generator in $\mathcal{B}_E$ -coordinates. Reconstruct the source matrices exactly and verify the two inclusions in (25.7).
- STEP 6. Construct complex conjugation by $x \mapsto x^{-1} = -x^7$, $y \mapsto y$, verify that it preserves (25.8), and serialize its full 16×16 rational matrix.
- STEP 7. Serialize polynomials, root identifiers, tower degrees, basis, structure constants, matrix-coordinate tables, equality rows, conjugation, source hashes, and independent verification hash. Return `SATISFIED` only after byte-identical independent verification.

Theorem 23.1.2 (Algorithm correctness and reproducibility). *The algorithm terminates and returns `SATISFIED` exactly when the supplied data define the distinguished field E in (25.1) and all specified generator and rotation entries reconstruct in that field. On passage (25.2), (25.7), and the exact arithmetic model (25.8)–(25.9) are certified. Equal canonical inputs produce byte-identical records.*

Proof. Polynomial division, rational Gaussian elimination, and the finite basis and matrix loops terminate. The degree theorem proves that the reduction rules yield unique coordinates and that inversion systems for nonzero elements are nonsingular. The exact reconstructions prove coefficient containment, while the explicit entries ξ, β prove the reverse inclusion. Fixed polynomial, monomial, rational, matrix-entry, root, failure, and byte orders prove reproducibility. $\square$

The successful record is

$$\begin{aligned} &(\mathbf{h}_{\text{geom}}, \Phi_{16}, m_\beta, \text{root}_\xi, (2, 9/4)_\beta, [E : \mathbb{Q}], \\ &\text{irreducibility_rows}, \mathcal{B}_E, \text{structure_constants}, \text{conjugation_matrix}, \\ &\text{generator_coordinates}, \text{presentation_coordinates}, \text{containment_rows}, \\ &\text{classification}, \text{COMPLETE}, \text{NONE}, \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}}). \end{aligned} \quad (25.12)$$

The mutually exclusive classifications are

$$\text{INVALID_INPUT}, \quad \text{FIELD_MISMATCH}, \quad \text{SATISFIED}. \quad (25.13)$$

A structurally valid but incorrectly reconstructed coefficient returns **FIELD_MISMATCH** with its least matrix-entry equality row. Every rejection preserves all earlier fields and sets later ones to **NONE**.

Minimum tests. The positive record uses (25.1), (25.6), and the sixteen distinct coordinate basis vectors. Replacing the $(1, 2)$ -entry of G_0 by $\beta + 1$ is well typed but returns **FIELD_MISMATCH**. Supplying a nonprimitive root ξ^2 , reversing the β isolating interval, deleting one basis coordinate, or changing a source hash returns **INVALID_INPUT**. The conjugate embedding $\xi \mapsto \xi^3$ supplies the required negative-sign test (25.4).

23.2 The full integer ring and certified integral bases

Let E and the rational basis $\mathcal{B}_E$ be the certified degree-sixteen field from Statement 23.1. Its full ring of integers is

$$\mathcal{O}_E = \{\alpha \in E : \alpha \text{ satisfies a monic polynomial in } \mathbb{Z}[X]\}. \quad (25.14)$$

Because ξ and β satisfy monic integer polynomials, the subring

$$\mathcal{A}_0 = \mathbb{Z}[\xi, \beta] = \bigoplus_{a=0}^7 \mathbb{Z}\xi^a \oplus \bigoplus_{a=0}^7 \mathbb{Z}\beta\xi^a \subseteq \mathcal{O}_E \quad (25.15)$$

is an order. The equality follows from the integral reduction rules $\xi^8 = -1$ and $\beta^2 = 2 + 2(\xi^2 - \xi^6)$. Thus $\mathcal{B}_E$ is an integral basis of the order $\mathcal{A}_0$; it is not asserted, without a maximality certificate, to be an integral basis of $\mathcal{O}_E$.

Trace, discriminant, and index. For an order $\mathcal{A} \subset E$ with $\mathbb{Z}$ -basis $\omega = (\omega_1, \dots, \omega_{16})$, define

$$\text{Tr}_{E/\mathbb{Q}}(\alpha) = \text{tr}(m_\alpha), \quad \text{disc}(\omega) = \det(\text{Tr}_{E/\mathbb{Q}}(\omega_i \omega_j))_{i,j}. \quad (25.16)$$

Here m_α is the rational multiplication matrix in the Statement 23.1 basis. The trace form is nondegenerate because $E/\mathbb{Q}$ is separable, so every order discriminant is a nonzero integer.

Theorem 23.2.1 (Discriminant-index identity). *If $\mathcal{A} \subseteq \mathcal{A}'$ are two orders and $I = [\mathcal{A}' : \mathcal{A}]$, then*

$$\text{disc}(\mathcal{A}) = I^2 \text{disc}(\mathcal{A}'). \quad (25.17)$$

In particular,

$$\text{disc}(\mathcal{A}) = [\mathcal{O}_E : \mathcal{A}]^2 \text{disc}(\mathcal{O}_E). \quad (25.18)$$

Proof. Write the basis of $\mathcal{A}$ as the basis of $\mathcal{A}'$ times an integer change-of-basis matrix S . The trace Gram matrices satisfy $T_{\mathcal{A}} = S^T T_{\mathcal{A}'} S$. Taking determinants gives $\det T_{\mathcal{A}} = (\det S)^2 \det T_{\mathcal{A}'}$, while $|\det S| = I$. This proves both formulas. $\square$

A finite local maximality certificate. For a rational prime p , call $\mathcal{A}$ p -maximal if $p \nmid [\mathcal{O}_E : \mathcal{A}]$.

Theorem 23.2.2 (Integral-basis criterion). *An order $\mathcal{A}$ equals $\mathcal{O}_E$ if and only if, for every prime $p \mid \text{disc}(\mathcal{A})$, no nonzero class $\bar{x} \in \mathcal{A}/p\mathcal{A}$ has an integral lift x/p . Equivalently, every such x/p has a multiplication characteristic polynomial with at least one noninteger coefficient:*

$$\chi_{x/p}(T) = \det(TI - m_{x/p}) \notin \mathbb{Z}[T]. \quad (25.19)$$

Proof. An element $\alpha \in E$ is integral if and only if the characteristic polynomial of m_α is monic with integer coefficients. The forward direction follows because all conjugates of an algebraic integer are integral and the characteristic polynomial is a power of its minimal polynomial; the reverse direction follows because α is a root of that monic integer polynomial.

If a nonzero $\bar{x}$ in the theorem has x/p integral, then $\mathcal{A}[x/p]$ is a strictly larger order and its index over $\mathcal{A}$ is divisible by p , so $\mathcal{A}$ is not p -maximal. Conversely, if $p \mid [\mathcal{O}_E : \mathcal{A}]$, the finite abelian group $\mathcal{O}_E/\mathcal{A}$ has an element of order p . Represent it by $\alpha \in \mathcal{O}_E$. Then $x = p\alpha \in \mathcal{A}$, $\bar{x} \neq 0$, and $x/p = \alpha$ is integral. Finally, every prime dividing the index divides the order discriminant by (25.18). Testing only the prime divisors of that discriminant is therefore exhaustive. $\square$

Certified construction modes. The typed input is

$$\text{IntegerRing}(\mathcal{R}_E; \text{mode}, \Omega_{\text{in}} \text{ or NONE}), \quad (25.20)$$

with the exclusive rules

mode	Ω_{in}	
ORDER_ONLY	NONE	(25.21)
CERTIFY	sixteen full-rank vectors in E	
CONSTRUCT	NONE.	

ORDER_ONLY certifies only $\mathcal{A}_0$ and carries no assertion that it is maximal. CERTIFY tests a proposed full integral basis. CONSTRUCT enlarges $\mathcal{A}_0$ until the criterion above certifies $\mathcal{O}_E$.

- STEP 1. Rehash and independent verification $\mathcal{R}_E$, its field relations, basis, structure constants, and degree. Validate the mode and exact candidate presence. Structural failure returns INVALID_INPUT.
- STEP 2. Construct $\mathcal{A}_0$ from (25.15), verify integer multiplication structure constants, compute its trace Gram matrix and nonzero discriminant, and factor the absolute discriminant by increasing-prime trial division with primality and reconstruction rows. If **mode** = ORDER_ONLY, set $\mathcal{A}_{\text{final}} = \mathcal{A}_0$, record **final_order_kind** = A0, set every local maximality, overorder, and full-ring index field to NONE, and return ORDER_ESTABLISHED only after byte-identical independent verification.
- STEP 3. In CERTIFY mode, express the proposed Ω_{in} in $\mathcal{B}_E$ -coordinates. Require rational full rank, integrality of every element via its multiplication characteristic polynomial, containment of 1 and $\mathcal{A}_0$, and integer multiplication structure constants. Any failure returns INVALID_BASIS with the least failed row. On success set $\mathcal{A} = \bigoplus_i \mathbb{Z}\omega_i$, and store the integer change-of-basis matrix proving $\mathcal{A}_0 \subseteq \mathcal{A}$. In CONSTRUCT mode instead initialize $\mathcal{A} := \mathcal{A}_0$.
- STEP 4. For the current order $\mathcal{A}$, compute its exact trace discriminant and factor it. For every prime divisor p , list all nonzero vectors of $\mathcal{A}/p\mathcal{A}$ in lexicographic coordinate order. For each representative x , compute $m_{x/p}$ and (25.19).

- STEP 5. In **CERTIFY** mode, the first integral x/p returns **NOT_MAXIMAL** with $p, \bar{x}, \chi_{x/p}$ and the strict overorder witness. If all lists are exhausted, return the maximality certificate $\mathcal{A} = \mathcal{O}_E$.
- STEP 6. In **CONSTRUCT** mode, upon finding an integral x/p , form the ring generated by $\mathcal{A}$ and x/p : repeatedly adjoin all basis products, compute the containing lattice by rational Hermite normal form, and stop when multiplication closure stabilizes. Every intermediate lattice lies in $\mathcal{O}_E$ and contains $\mathcal{A}$; because $\mathcal{O}_E/\mathcal{A}$ is finite, it has only finitely many subgroups, so this inner loop terminates at $\mathcal{A}[x/p]$. Replace $\mathcal{A}$ by this strict overorder and repeat Step 4.
- STEP 7. For every enlargement, verify the lattice containment, determinant index, and (25.17). The absolute nonzero discriminant strictly decreases by a square factor, so construction terminates.
- STEP 8. Serialize the final basis, change-of-basis matrices, structure constants, traces, discriminants, factorizations, every local residue row, overorder witnesses, indices, failure fields, and input/independent verification hashes.

Theorem 23.2.3 (Correctness, termination, and conditional use). *CONSTRUCT terminates with a certified integral basis of $\mathcal{O}_E$. CERTIFY returns MAXIMAL exactly for such a basis, and a NOT_MAXIMAL witness is conclusive. ORDER_ONLY proves all specified matrices integral in $\mathcal{O}_E$ through the inclusion $\mathcal{A}_0 \subseteq \mathcal{O}_E$, but makes no unsupported claim about prime ideals or an integral basis of the full ring. Equal canonical inputs produce byte-identical records.*

Proof. Steps 1–3 construct genuine orders. The integral-basis criterion proves both outcomes of Steps 4–5. Every construction enlargement is a proper order inclusion, and (25.17) makes the positive absolute discriminant decrease, proving termination. At termination no local witness exists, so the resulting order is $\mathcal{O}_E$. Fixed factor, residue-vector, polynomial, lattice, pivot, failure, and byte orders prove reproducibility. $\square$

If a certified order $\mathcal{A} \subseteq \mathcal{O}_E$ has known index I , then for every rational prime $p \nmid I$,

$$\mathcal{A} \otimes \mathbb{Z}_{(p)} = \mathcal{O}_E \otimes \mathbb{Z}_{(p)}. \quad (25.22)$$

Indeed, the finite quotient $\mathcal{O}_E/\mathcal{A}$ has order I and is annihilated by I . When $p \nmid I$, the integer I is a unit in $\mathbb{Z}_{(p)}$, so localization annihilates that quotient. Exactness of localization then gives (25.22). Thus order-based arithmetic may be transferred locally only after the index primes have been excluded. With an unknown index, this localization claim is unavailable.

The successful final-order record is

$$\begin{aligned} &(\mathbf{h}_E, \text{mode}, \text{final_order_kind}, \Omega_{\mathcal{A}_{\text{final}}}, \text{basis_change}, \text{structure_constants}, \\ &\text{trace_matrix}_{\mathcal{A}_{\text{final}}}, \text{disc}(\mathcal{A}_{\text{final}}), \text{factorization_rows}, \\ &\text{local_maximality_rows}, \text{overorder_chain}, \text{index_rows}, \\ &\text{classification}, \text{COMPLETE}, \text{NONE}, \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}}). \end{aligned} \quad (25.23)$$

The mutually exclusive classifications are

$$\begin{aligned} &\text{INVALID_INPUT}, \text{INVALID_BASIS}, \text{NOT_MAXIMAL}, \\ &\text{ORDER_ESTABLISHED}, \text{MAXIMAL}. \end{aligned} \quad (25.24)$$

The trace matrix, discriminant, and factorization rows always belong to $\mathcal{A}_{\text{final}}$. The order-only record has the same schema with only the local-maximality rows, overorder chain, and full-ring index rows set to **NONE**, $\text{final_order_kind} = \mathbf{A0}$, $\Omega_{\mathcal{A}_{\text{final}}} = \mathcal{B}_E$, and classification **ORDER_ESTABLISHED**. A maximal result has $\text{final_order_kind} = \mathbf{OE}$ and $\mathcal{A}_{\text{final}} = \mathcal{O}_E$.

Minimum tests. For $\mathbb{Q}(\sqrt{5})$, the order $\mathbb{Z}[\sqrt{5}]$ with basis $(1, \sqrt{5})$ has discriminant 20. At $p = 2$, the nonzero residue class of $1 + \sqrt{5}$ yields

$$\alpha = \frac{1 + \sqrt{5}}{2}, \quad \alpha^2 - \alpha - 1 = 0, \quad (25.25)$$

so it is a certified NOT_MAXIMAL witness. In contrast, $(1, \sqrt{2})$ is the full basis of $\mathbb{Z}[\sqrt{2}] = \mathcal{O}_{\mathbb{Q}(\sqrt{2})}$; exhaustive local tests certify maximality even though its discriminant 8 is divisible by 2. These two tests show that divisibility of an order discriminant alone neither proves nor disproves maximality. A dependent candidate basis, a nonintegral vector $1/2$, or a noninteger multiplication coefficient tests INVALID_BASIS.

23.3 Exact denominator clearing and local reduction

Let $\Omega = (\omega_1, \dots, \omega_{16})$ be a certified integral basis of the full ring $\mathcal{O}_E$ from a maximal Statement 23.2 record. For $\alpha \in E$, define its denominator ideal

$$\mathfrak{d}(\alpha) = \{a \in \mathcal{O}_E : a\alpha \in \mathcal{O}_E\}. \quad (25.26)$$

For a finite set $S \subset E$, put

$$\mathfrak{d}(S) = \bigcap_{\alpha \in S} \mathfrak{d}(\alpha) = \{a \in \mathcal{O}_E : aS \subseteq \mathcal{O}_E\}. \quad (25.27)$$

These are nonzero integral ideals: write all elements of S in the integral basis and let $m > 0$ be the least common multiple of their rational coordinate denominators; then

$$mS \subseteq \mathcal{O}_E, \quad (m) \subseteq \mathfrak{d}(S). \quad (25.28)$$

Thus a common integer always clears a finite family. The ideal $\mathfrak{d}(S)$ is sharper and need not be principal.

Theorem 23.3.1 (Exact local denominator criterion). *Let $\mathfrak{p}$ be a nonzero prime ideal of $\mathcal{O}_E$. Every $\alpha \in S$ belongs to the local ring $(\mathcal{O}_E)_{\mathfrak{p}}$ if and only if*

$$\mathfrak{d}(S) \not\subseteq \mathfrak{p}. \quad (25.29)$$

Equivalently, the family has a common denominator outside $\mathfrak{p}$.

Proof. If $a \in \mathfrak{d}(S) \setminus \mathfrak{p}$, then $\alpha = (a\alpha)/a \in (\mathcal{O}_E)_{\mathfrak{p}}$ for all $\alpha \in S$. Conversely, write each of the finitely many elements as $\alpha = b_{\alpha}/c_{\alpha}$ with $b_{\alpha}, c_{\alpha} \in \mathcal{O}_E$ and $c_{\alpha} \notin \mathfrak{p}$. The product $c = \prod_{\alpha \in S} c_{\alpha}$ lies outside $\mathfrak{p}$, because $\mathfrak{p}$ is prime, and satisfies $cS \subseteq \mathcal{O}_E$. Hence $c \in \mathfrak{d}(S) \setminus \mathfrak{p}$, proving (25.29). $\square$

The traditional notation $\mathfrak{p} \nmid \mathfrak{d}(S)$ will mean precisely $\mathfrak{d}(S) \not\subseteq \mathfrak{p}$. Equation (25.28) shows that $\mathfrak{p} \nmid (m)$ is sufficient for good localization, but it may exclude extra primes if the chosen scalar contains cancelled factors.

Matrices and presentation relations. Let $\mathcal{M} = (M_1, \dots, M_t) \subset \mathrm{SL}_n(E)$ and let $S_{\mathcal{M}}$ contain every entry of every M_i and M_i^{-1} . At a prime satisfying

$$\mathfrak{d}_{\mathcal{M}} := \mathfrak{d}(S_{\mathcal{M}}) \not\subseteq \mathfrak{p}, \quad (25.30)$$

entrywise reduction through

$$(\mathcal{O}_E)_{\mathfrak{p}} \longrightarrow \kappa(\mathfrak{p}) = \mathcal{O}_E/\mathfrak{p} \quad (25.31)$$

defines matrices $\overline{M}_i \in \mathrm{SL}_n(\kappa(\mathfrak{p}))$. Every word identity and presentation relator among the M_i remains valid after reduction, because (25.31) is a ring homomorphism. Generator distinctness, faithfulness, and exact element orders may collapse and must be tested after reduction.

If determinant one is independently certified, including inverse entries in $S_{\mathcal{M}}$ is redundant for existence: the adjugate formula makes them integral wherever the original entries are. They are retained by the algorithm as a consistency check. Multiplying a matrix by a clearing scalar and then reducing at a prime containing that scalar is not a reduction of the original matrix:

$$(mM_i)(mM_j) = m^2 M_i M_j \neq m(M_i M_j) \quad \text{unless the scalar effects are separately normalized.} \quad (25.32)$$

In particular, scalar clearing cannot rescue a prime forbidden by (25.29).

Conjugate models. If $M'_i = PM_i P^{-1}$ for $P \in \text{GL}_n(E)$, the exact denominator ideal for the displayed model is computed from the entries of M'_i and their inverses. To assert that reduction preserves the conjugacy itself, one must additionally require

$$P, P^{-1} \in M_n((\mathcal{O}_E)_{\mathfrak{p}}); \quad (25.33)$$

otherwise both matrix families might reduce while the conjugating matrix does not.

Exact lattice computation. Let $R_{\alpha} \in M_{16}(\mathbb{Q})$ be multiplication by α in the integral basis. Coordinate vectors identify (25.26) with

$$\mathfrak{d}(\alpha) \cong \{z \in \mathbb{Z}^{16} : R_{\alpha} z \in \mathbb{Z}^{16}\}. \quad (25.34)$$

After clearing rational matrix denominators, Smith or Hermite normal form computes a canonical basis for this full-rank lattice. Canonical lattice intersection applied to (25.27) computes $\mathfrak{d}(S)$; multiplication by each ω_i verifies that the resulting lattice is an ideal.

Canonical construction algorithm. The input is

$$\text{DenominatorClear}(\mathcal{R}_{\mathcal{O}_E}, \mathcal{R}_{\mathcal{M}}; \text{model_kind}, P \text{ or NONE}), \quad (25.35)$$

where `model_kind` is `DIRECT` or `CONJUGATE`. Direct mode requires $P = \text{NONE}$; conjugate mode requires a full invertible matrix P and the exact equality $M'_i = PM_i P^{-1}$ for every specified generator.

- STEP 1. Rehash and independent verification the Statement 23.2 parent. Require `final_order_kind = OE` and classification `MAXIMAL`; an order-only or nonmaximal parent returns `NONMAXIMAL_PARENT` not a denominator conclusion.
- STEP 2. Validate the matrix dimensions, rational field coordinates, determinant-one rows, inverses, model discriminator, and exact presence of P . Replay every defining group and presentation identity. Structural failure returns `INVALID_INPUT`.
- STEP 3. Enumerate $S_{\mathcal{M}}$ in generator, inverse, row, column order. Compute every lattice (25.34), verify its index and membership equivalence in both directions, and intersect them to obtain $\mathfrak{d}_{\mathcal{M}}$.
- STEP 4. Compute the least common multiple m of all reduced rational coordinate denominators. Verify every row of (25.28), the ideal containment $(m) \subseteq \mathfrak{d}_{\mathcal{M}}$, and whether equality holds; equality is not presumed.
- STEP 5. In conjugate mode, compute separately the denominator ideals of the entries of P and P^{-1} , and record the additional local exclusion required by (25.33).
- STEP 6. Verify ideal closure, nonzeroness, canonical Hermite basis, lattice indices, and multiplication rows. Reconstruct all cleared entries and all original rational entries exactly.

STEP 7. Serialize per-entry denominator ideals, their intersection, scalar clearing data, conjugacy data, presentation rows, first failures, source hashes, and independent verification hash. Return SATISFIED only after byte-identical independent verification.

Theorem 23.3.2 (Algorithm correctness and exact exclusion). *The algorithm terminates. On passage it returns the exact nonzero ideal $\mathfrak{d}_{\mathcal{M}}$, and a nonzero prime $\mathfrak{p}$ permits entrywise reduction of the displayed matrix model exactly when (25.30) holds. If conjugacy preservation is requested, the additional condition (25.33) is exact. Equal canonical inputs produce byte-identical records.*

Proof. All rational lattice and finite matrix operations terminate. Equations (25.26)–(25.34) prove that each computed lattice has the required membership property and that their intersection is the exact common denominator ideal. The local criterion proves necessity and sufficiency at every prime. Fixed element, matrix-entry, lattice, pivot, denominator, failure, and byte orders prove reproducibility. $\square$

The successful record is

$$\begin{aligned} &(\mathbf{h}_{\mathcal{O}_E}, \mathbf{h}_{\mathcal{M}}, \text{model_kind}, P \text{ or NONE}, \\ &(\mathfrak{d}(\alpha))_{\alpha \in S_{\mathcal{M}}}, \mathfrak{d}_{\mathcal{M}}, \text{ideal_rows}, m, (m), \\ &\text{scalar_containment_rows}, \text{conjugacy_denominators or NONE}, \\ &\text{presentation_rows}, \text{classification}, \text{COMPLETE, NONE}, \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}}). \end{aligned} \quad (25.36)$$

The classifications are

$$\text{INVALID_INPUT}, \quad \text{NONMAXIMAL_PARENT}, \quad \text{SATISFIED}. \quad (25.37)$$

The canonical lattice computation has no separate mathematical rejection: every membership, HNF, and intersection row is recomputed from the exact input and must agree on byte independent verification.

Minimum tests. For the specified model (25.6), every entry is already integral, so

$$\mathfrak{d}_{\mathcal{M}} = \mathcal{O}_E, \quad m = 1. \quad (25.38)$$

For the scalar family $S = \{1/2\}$, $\mathfrak{d}(S) = 2\mathcal{O}_E$; precisely the prime ideals above 2 fail (25.29). Presenting the same element as $2/4$ still gives the exact ideal $2\mathcal{O}_E$, while the unreduced scalar 4 would overexclude, testing representation independence. The rational conjugator $P = \text{diag}(2, 1)$ and its inverse test the separate condition (25.33). An order-only parent, a singular P , a false inverse, and a stale source hash test the two structural rejection classifications.

23.4 Algebraic-integer traces and exact trace obstructions

Fix the maximal Statement 23.2 record for $\mathcal{O}_E$, and let $\mathcal{M} = (M_1, \dots, M_s)$ be the specified determinant-one matrix family. The input required in this subsection is the global integrality branch of Statement 23.3:

$$M_i \in \text{SL}_2(\mathcal{O}_E) \quad (1 \leq i \leq s), \quad \mathfrak{d}_{\mathcal{M}} = \mathcal{O}_E. \quad (25.39)$$

Merely having a denominator ideal which is invertible at a selected prime is not enough for the conclusions below. Such a model is locally reducible at that prime, but its traces need not be algebraic integers globally.

Lemma 23.4.1 (Integral word and trace closure). *Let $w = x_{i_1}^{\epsilon_1} \cdots x_{i_\ell}^{\epsilon_\ell}$, where $\epsilon_j \in \{1, -1\}$, and set*

$$M(w) = M_{i_1}^{\epsilon_1} \cdots M_{i_\ell}^{\epsilon_\ell}, \quad t_w = \text{tr } M(w). \quad (25.40)$$

Then

$$M(w) \in \text{SL}_2(\mathcal{O}_E), \quad t_w \in \mathcal{O}_E. \quad (25.41)$$

If the verified presentation relations identify two words w, w' , then $M(w) = M(w')$, so their traces and all ideals defined from them are independent of the chosen representative.

Proof. The ring $M_2(\mathcal{O}_E)$ is closed under multiplication. If

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathcal{O}_E), \quad \text{then} \quad A^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \in \text{SL}_2(\mathcal{O}_E). \quad (25.42)$$

Thus every factor and inverse factor in (25.40) is integral, and induction on ℓ proves the first assertion in (25.41). The trace is the sum of the two diagonal entries, hence is integral. Finally, independent verification of a defining relation replaces a subword by another subword with the same matrix; a finite sequence of such replacements therefore preserves the entire product, not merely its trace. $\square$

Principal trace data and their norms. For every specified word define

$$a_w = t_w - 2, \quad \Delta_w = t_w^2 - 4, \quad \mathfrak{a}_w = (a_w), \quad \mathfrak{D}_w = (\Delta_w). \quad (25.43)$$

Both elements lie in $\mathcal{O}_E$, so both displayed principal ideals are legitimate integral ideals, including the zero ideal. Multiplication by $\alpha \in \mathcal{O}_E$ is a $\mathbb{Z}$ -linear endomorphism m_α of the free lattice $\mathcal{O}_E$. In a certified integral basis Ω ,

$$N_{E/\mathbb{Q}}(\alpha) = \det[m_\alpha]_\Omega \in \mathbb{Z}. \quad (25.44)$$

This integer is independent of Ω , and it vanishes exactly when $\alpha = 0$. For $\alpha \neq 0$, the absolute ideal norm obeys

$$N((\alpha)) = [\mathcal{O}_E : \alpha \mathcal{O}_E] = |N_{E/\mathbb{Q}}(\alpha)|. \quad (25.45)$$

No finite ideal norm is assigned to (0) : the quotient $\mathcal{O}_E/(0)$ is infinite. Accordingly, a record with $a_w = 0$ or $\Delta_w = 0$ stores the field norm 0, the zero principal ideal, and the ideal-norm field **NONE**. It must not silently store ideal norm zero.

The trace of an inverse gives a useful independent verification check. From the Cayley–Hamilton identity for a determinant-one matrix,

$$M(w)^{-1} = t_w I - M(w), \quad t_{w^{-1}} = t_w, \quad a_{w^{-1}} = a_w, \quad \Delta_{w^{-1}} = \Delta_w. \quad (25.46)$$

Likewise, for every $P \in \text{GL}_2(E)$,

$$\text{tr}(PM(w)P^{-1}) = t_w. \quad (25.47)$$

Thus trace data do not depend on the specified conjugate model. Equation (25.47) does not make a nonintegral conjugate model reducible: the separate denominator conditions of Statement 23.3 still govern its entries and its conjugator.

What trace divisibility does and does not certify. Let $\mathfrak{p}$ be a nonzero prime ideal and write $\kappa(\mathfrak{p}) = \mathcal{O}_E/\mathfrak{p}$. Define the two entry ideals

$$\begin{aligned}\mathfrak{J}_w^+ &= ((M(w) - I)_{11}, (M(w) - I)_{12}, (M(w) - I)_{21}, (M(w) - I)_{22}), \\ \mathfrak{J}_w^- &= ((M(w) + I)_{11}, (M(w) + I)_{12}, (M(w) + I)_{21}, (M(w) + I)_{22}).\end{aligned}\tag{25.48}$$

For later projective use, also form the lift-sign-invariant ideal

$$\mathfrak{J}_w^{\text{proj}} := \mathfrak{J}_w^+ \mathfrak{J}_w^-.$$

Replacing $M(w)$ by $-M(w)$ interchanges the two factors and therefore leaves $\mathfrak{J}_w^{\text{proj}}$ unchanged. Here parentheses around several elements denote the ideal they generate. Entrywise reduction gives the exact linear criterion

$$\overline{M(w)} = I \iff \mathfrak{J}_w^+ \subseteq \mathfrak{p}.\tag{25.49}$$

Taking traces shows only the one-way implication

$$\mathfrak{J}_w^+ \subseteq \mathfrak{p} \implies a_w \in \mathfrak{p}.\tag{25.50}$$

Its converse is false. Hence $a_w \notin \mathfrak{p}$ certifies that the linear reduction is nonidentity, while $a_w \in \mathfrak{p}$ requires the full four-entry test (25.49).

The sign of an SL_2 -lift is immaterial in the projective group. Since the only roots of $X^2 - 1$ in the field $\kappa(\mathfrak{p})$ are $1, -1$, with the two coinciding in characteristic 2, the exact projective criterion is

$$[\overline{M(w)}] = 1 \text{ in } \text{PSL}_2(\kappa(\mathfrak{p})) \iff \mathfrak{J}_w^+ \subseteq \mathfrak{p} \text{ or } \mathfrak{J}_w^- \subseteq \mathfrak{p} \iff \mathfrak{J}_w^{\text{proj}} \subseteq \mathfrak{p}.\tag{25.51}$$

The last equivalence uses primality of $\mathfrak{p}$: a prime ideal contains a product of ideals exactly when it contains at least one factor. The lift-invariant discriminant supplies only the obstruction

$$[\overline{M(w)}] = 1 \implies \Delta_w \in \mathfrak{p}.\tag{25.52}$$

Thus $\Delta_w \notin \mathfrak{p}$ certifies projective nonidentity, whereas divisibility of Δ_w is not sufficient for projective identity. This distinction prevents either $(t_w - 2)$ or $(t_w^2 - 4)$ from being used as a replacement for matrix comparison.

Theorem 23.4.2 (Finite trace obstruction and exact fallback). *If $a_w \neq 0$, only finitely many nonzero prime ideals can defeat the linear trace obstruction $a_w \notin \mathfrak{p}$; they are exactly the prime-ideal divisors of $\mathfrak{a}_w$. If $\Delta_w \neq 0$, only finitely many can defeat the projective trace obstruction; they are exactly the divisors of $\mathfrak{D}_w$. At every remaining prime the appropriate reduced image is nonidentity. At an exceptional prime, (25.49) or (25.51), rather than trace divisibility, is necessary and sufficient.*

Proof. A nonzero principal ideal in the Dedekind domain $\mathcal{O}_E$ has a unique factorization into finitely many nonzero prime ideals. Membership $\alpha \in \mathfrak{p}$ is equivalent to $(\alpha) \subseteq \mathfrak{p}$, hence to occurrence of $\mathfrak{p}$ in the factorization of (α) . Apply this first to a_w and (25.50), and then to Δ_w and (25.52). The exact statements at the exceptional primes are the entrywise equivalences (25.49) and (25.51). $\square$

Construction and certificate. The typed input consists of a maximal $\mathcal{O}_E$ -record, the global Statement 23.3 matrix record, verified determinant and inverse rows, verified presentation rows, a finite ordered word list W , an ordered sublist $\mathcal{D} \subseteq W$ whose projective images must survive, and a finite ordered list $\mathcal{P} = (\mathfrak{p}_1, \dots, \mathfrak{p}_r)$ of requested nonzero prime ideals. Every member of $\mathcal{P}$ carries a canonical ideal basis, an $\mathcal{O}_E$ -parent hash, and a certified residue-field presentation. Either list may be empty. Execute:

1. Replay the field, integral-basis, multiplication-table, matrix, determinant, inverse, relation, source-hash, and parent-hash rows. Verify that $\mathcal{D}$ is an ordered sublist of W and that every member of $\mathcal{P}$ is a nonzero prime ideal of the same certified $\mathcal{O}_E$, with the asserted residue field. Any malformed structure, false arithmetic row, or stale hash returns `INVALID_INPUT`. Any structurally valid Statement 23.2 parent whose classification is not `MAXIMAL`, or whose final-order kind is not `OE`, returns `NONMAXIMAL_PARENT`. This includes both `ORDER_ESTABLISHED` and `NOT_MAXIMAL` records. Only for a parent with classification `MAXIMAL` and final-order kind `OE` is the matrix denominator tested; such a parent with $\mathfrak{d}_{\mathcal{M}} \neq \mathcal{O}_E$ returns `NONINTEGRAL_MODEL`.
2. For every $w \in W$, multiply exact coordinate matrices in left-to-right word order. Verify determinant one, integral coordinates, the inverse identities (25.46), and all available representative-change rows.
3. Compute t_w, a_w, Δ_w , their coordinate vectors, principal-ideal Hermite bases, multiplication determinants, and the nonzero ideal factorizations. Treat the zero cases exactly as specified after (25.45).
4. Compute both four-entry ideals in (25.48) and their product $\mathfrak{J}_w^{\text{proj}}$. For every requested prime, independent verification the residue field and compare the ideal-containment tests with direct entrywise matrix reduction. Record trace obstruction, exact linear result, and exact projective result as separate fields. In particular, a failed trace filter means `TRACE_EST_INSUFFICIENT`, followed by the exact matrix test.
5. Compute $\mathfrak{J}_{\mathcal{D}}$ from (25.56), its canonical ideal basis, and, when nonzero, its prime-ideal factorization and all simultaneous-containment rows (25.57). If it is zero, return the terminal classification

`UNSEPARABLE_WORD`,

together with the least occurrence w in $\mathcal{D}$ satisfying $M(w) = I$ or $M(w) = -I$ and the corresponding sign. The empty list uses the empty-product convention stated below and passes this step.

6. Serialize in fixed word, letter, matrix-entry, basis, prime, factor, pivot, and byte order. Re-compute from serialized input and return `SATISFIED` only after byte-identical independent verification.

Theorem 23.4.3 (Algorithm correctness and termination). *The construction terminates for every finite $W, \mathcal{D}, \mathcal{P}$. On `SATISFIED` it returns algebraic integers t_w, a_w, Δ_w , exact principal ideals and norms with the zero convention above, exact linear and projective reduction tests at every requested prime, and the exact simultaneous-separation ideal. On `UNSEPARABLE_WORD` it returns a directly independently verifiable global $\pm I$ witness. Equal canonical inputs produce byte-identical records.*

Proof. The word, danger, coordinate, and prime lists are finite. Matrix multiplication, integral-basis arithmetic, Hermite normal form, determinants, and ideal factorization over a fixed number field are terminating exact algorithms. The closure lemma proves integrality; (25.44)–(25.45) prove the norm fields; and (25.49)–(25.52) and (25.56)–(25.58) prove the separation and failure fields. The fixed traversal and serialization orders establish determinism and byte-identical independent verification. $\square$

The completed record is

$$\begin{aligned}
&(\mathbf{h}_{\mathcal{O}_E}, \mathbf{h}_{\mathcal{M}}, W, \mathcal{D}, \mathcal{P}, \\
&(M(w), t_w, a_w, \Delta_w, \mathbf{a}_w, \mathfrak{D}_w, N_{E/\mathbb{Q}}(a_w), N_{E/\mathbb{Q}}(\Delta_w), \\
&\quad N(\mathbf{a}_w) \text{ or NONE}, N(\mathfrak{D}_w) \text{ or NONE}, \mathfrak{J}_w^+, \mathfrak{J}_w^-, \mathfrak{J}_w^{\text{proj}}, \text{reduction_rows}_w)_{w \in W}, \\
&\mathfrak{J}_{\mathcal{D}}, \text{factorization}_{\mathcal{D}} \text{ or NONE}, \text{simultaneous_rows}, \\
&\text{failure_word} \text{ or NONE}, \text{failure_sign} \text{ or NONE}, \text{presentation_rows}, \\
&\text{stage}, \text{first_failure}, \text{failure_kind}, \text{classification}, \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}} \text{ or NONE}).
\end{aligned} \tag{25.53}$$

The mutually exclusive classifications are

$$\begin{aligned}
&\text{INVALID_INPUT}, \quad \text{NONMAXIMAL_PARENT}, \quad \text{NONINTEGRAL_MODEL}, \\
&\quad \text{UNSEPARABLE_WORD}, \quad \text{SATISFIED}.
\end{aligned} \tag{25.54}$$

For SATISFIED, the failure fields are NONE. For UNSEPARABLE_WORD, the failure kind is

$$\text{GLOBAL_PROJECTIVE_TRIVIAL}.$$

The independent verification hash then covers the least-word and sign witness.

Minimum tests. The empty word gives $M = I$, $t = 2$, $a = 0$, and $\Delta = 0$; both ideal-norm fields must be NONE. The integral unipotent matrix

$$U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \tag{25.55}$$

has determinant one and trace 2, but $U \neq I$; modulo every prime its upper-right entry remains 1. It therefore tests both failed converses in (25.50) and (25.52). The matrices I and $-I$ test the distinction between linear and projective identity, including residue characteristic 2; placing either word in $\mathcal{D}$ returns UNSEPARABLE_WORD, failure kind GLOBAL_PROJECTIVE_TRIVIAL, and its signed witness. A word and its formal inverse test (25.46), and conjugation by $\text{diag}(2, 1)$ tests trace invariance without licensing entrywise reduction of a nonintegral conjugate. Finally, an order-only ring record, a denominator ideal different from $\mathcal{O}_E$, a false inverse row, and a stale source hash test the other three rejection classifications and their deterministic routing.

Simultaneous projective separation. Let $\mathcal{D}$ be a finite ordered list of words whose projective images are required to survive. Define

$$\mathfrak{J}_{\mathcal{D}} := \prod_{w \in \mathcal{D}} \mathfrak{J}_w^{\text{proj}}. \tag{25.56}$$

The empty product is, by definition, $\mathcal{O}_E$. Thus an empty $\mathcal{D}$ passes vacuously, has ideal norm 1, and has no bad prime ideal. For every nonzero prime ideal,

$$\begin{aligned}
&[\overline{M(w)}] \neq 1 \text{ for every } w \in \mathcal{D} \\
&\iff \mathfrak{J}_{\mathcal{D}} \not\subseteq \mathfrak{p}.
\end{aligned} \tag{25.57}$$

Indeed, repeated primality says that $\mathfrak{p}$ contains the product in (25.56) exactly when it contains at least one factor. Moreover,

$$\mathfrak{J}_w^{\text{proj}} = 0 \iff M(w) = I \text{ or } M(w) = -I. \tag{25.58}$$

The forward implication follows because $\mathcal{O}_E$ is a domain: one factor is zero, so all entries of $M(w) - I$, or all entries of $M(w) + I$, vanish. Consequently $\mathfrak{J}_{\mathcal{D}} = 0$ is the UNSEPARABLE_WORD

branch, with failure kind `GLOBAL_PROJECTIVE_TRIVIAL`; no finite-bad-prime assertion is then allowed. If every listed matrix differs from both signs of the identity, $\mathfrak{J}_{\mathcal{D}} \neq 0$, its prime-ideal factorization is finite, and (25.57) is the exact simultaneous certificate. Pairwise injectivity on a finite list is obtained by applying the same construction to every difference word uv^{-1} .

Finally, rational norm factors are only a coarse sieve. If $\mathfrak{p} \cap \mathbb{Z} = (\ell)$ and $0 \neq \alpha \in \mathfrak{p}$, then

$$\ell \mid \left| N_{E/\mathbb{Q}}(\alpha) \right|. \quad (25.59)$$

Conversely, divisibility of the field norm by ℓ guarantees that at least one prime ideal above ℓ contains α , not that every prime above ℓ does. Exact bad prime ideals must therefore come from ideal factorization or containment rows, never from a rational integer norm alone.

23.5 The faithful integral representation

Let

$$\Pi = \langle a_1, b_1, a_2, b_2 \mid \mathcal{R} = [a_1, b_1][a_2, b_2] = 1 \rangle. \quad (25.60)$$

Statement 4.7 identifies this particular presentation with the concrete Bolza side-pairing group Γ_B , not merely with an abstract group of the same isomorphism type. Fix the four coherent lift matrices from Statement 5.1,

$$\mathcal{L} = (A_1, B_1, A_2, B_2) \in \mathrm{SU}(1, 1)^4, \quad (25.61)$$

with their specified order, signs, exact entries, and source hash. Their projective classes are the four geometric side pairings and their lifted relator is exactly

$$[A_1, B_1][A_2, B_2] = I, \quad (25.62)$$

not merely a central sign.

Lemma 23.5.1 (Integral closure of the specified lift). *Every matrix in $\mathcal{L}$, every inverse letter, and every finite word in those letters belongs to $\mathrm{SL}_2(\mathcal{O}_E)$.*

Proof. Statement 5.2 places the entries of every physical side-pairing lift G_j in $\mathcal{O}_E$. The formulas defining (25.61) use finite products and inverses of the G_j . Products remain integral, and for

$$X = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix} \in \mathrm{SL}_2(\mathcal{O}_E) \quad \text{one has} \quad X^{-1} = \begin{pmatrix} x_{22} & -x_{12} \\ -x_{21} & x_{11} \end{pmatrix}. \quad (25.63)$$

Thus inverse letters are integral as well. Induction on word length proves the claim. Equivalently, this replays the global-integrality branch $\mathfrak{d}_{\mathcal{M}} = \mathcal{O}_E$ of Statement 23.3 and the closure statement of Statement 23.4; no power basis is being asserted to be an integral basis of $\mathcal{O}_E$. $\square$

Theorem 23.5.2 (Existence and integrality). *There is a unique homomorphism with the specified generator images,*

$$\rho_E : \Gamma_B \longrightarrow \mathrm{SL}_2(\mathcal{O}_E), \quad (a_1, b_1, a_2, b_2) \longmapsto (A_1, B_1, A_2, B_2). \quad (25.64)$$

Its composition with projectivization is exactly the geometric side-pairing representation ρ_{geom} .

Proof. The assignment of the four free generators to (25.61) extends uniquely to the free group. Equation (25.62) says that the defining relator, hence its normal closure, lies in the kernel. The universal property of the quotient presentation therefore yields a unique homomorphism from $\Pi = \Gamma_B$. The lemma places its entire image in $\mathrm{SL}_2(\mathcal{O}_E)$. On each presentation generator its projective class is, by the specified lift table, the corresponding geometric side pairing. Two homomorphisms agreeing on generators agree everywhere, so $\mathrm{proj} \circ \rho_E = \rho_{\mathrm{geom}}$. $\square$

Relation verification proves descent from the free group, but does not by itself prove injectivity. That second assertion comes from the specific polygon representation.

Theorem 23.5.3 (Faithfulness in characteristic zero). *The homomorphism ρ_E in (25.64) is injective. Its image contains no matrix $-I$.*

Proof. The Poincaré polygon certificate of Statement 4.7, together with the invertible Nielsen substitution between the boundary and surface generators, proves that the concrete projective map is faithful. Consequently

$$\begin{aligned} \ker \rho_E &\subseteq \ker(\text{proj} \circ \rho_E) \\ &= \ker \rho_{\text{geom}} = \{1\}. \end{aligned} \quad (25.65)$$

The reverse containment is automatic, so the kernel is trivial. If $\rho_E(\gamma) = -I$, then the projective image of γ is the identity; projective faithfulness gives $\gamma = 1$, whereas

$$\rho_E(1) = I \neq -I. \quad (25.66)$$

This contradiction excludes $-I$ from the image. $\square$

The image can therefore be denoted without ambiguity by

$$\Gamma_E := \rho_E(\Gamma_B) \leq \text{SL}_2(\mathcal{O}_E), \quad \Gamma_B \xrightarrow[\sim]{\rho_E} \Gamma_E. \quad (25.67)$$

For words u, v this gives the exact characteristic-zero equality tests

$$\begin{aligned} u = v \text{ in } \Gamma_B &\iff \rho_E(uv^{-1}) = I \iff \rho_E(u) = \rho_E(v), \\ [\rho_E(u)] &= [\rho_E(v)] \iff u = v \text{ in } \Gamma_B. \end{aligned} \quad (25.68)$$

The last equivalence uses exclusion of a hidden $-I$, not merely linear injectivity.

Lift and conjugacy discipline. Multiplying selected generators by central signs can preserve the projective representation and can even preserve the relator. Such a choice is a different linear lift, generally a twist by a character

$$\chi : \Gamma_B \longrightarrow \{\pm 1\}, \quad \rho_E^\chi(\gamma) = \chi(\gamma)\rho_E(\gamma). \quad (25.69)$$

It is not silently substituted for the specified map (25.64). Therefore the ordered matrices, their signs, the exact lifted relator, and the source hash are part of the representation identity.

For $P \in \text{GL}_2(E)$, the conjugate map

$$\rho_E^P(\gamma) = P\rho_E(\gamma)P^{-1} \quad (25.70)$$

is faithful because conjugation is an automorphism of $\text{GL}_2(E)$. It is an integral representation only if all displayed conjugate generator matrices are proved to lie in $\text{SL}_2(\mathcal{O}_E)$. A rational conjugator and a local Statement 23.3 denominator certificate do not imply global integrality.

Characteristic-zero faithfulness is not modular faithfulness. At an allowed prime ideal $\mathfrak{p}$, reduction gives a homomorphism

$$\bar{\rho}_{\mathfrak{p}} : \Gamma_B \longrightarrow \text{SL}_2(\kappa(\mathfrak{p})). \quad (25.71)$$

The target is finite while Γ_B is infinite, so this map cannot be injective. In particular, (25.65) cannot be reduced modulo $\mathfrak{p}$. Statement 23.4 proves only that a prescribed finite list of nontrivial elements or difference words survives after excluding the prime divisors of its exact entry-ideal product:

$$\mathfrak{J}_{\mathcal{D}} \not\subseteq \mathfrak{p} \iff [\bar{\rho}_{\mathfrak{p}}(w)] \neq 1 \quad \text{for all } w \in \mathcal{D}. \quad (25.72)$$

Neither a trace filter nor passage of finitely many tests upgrades (25.72) to faithfulness on the infinite group.

Construction and certificate. The typed input comprises the exact surface presentation; the Statement 4.7 polygon-faithfulness and Nielsen records; the Statement 5.1 coherent-lift table and lifted-boundary-cycle row; the Statement 5.2–Statement 5.3 integral-entry, determinant, multiplication, and inverse rows; the maximal Statement 23.2 $\mathcal{O}_E$ -record; the global Statement 23.3 denominator record; and the Statement 23.4 trace/entry-ideal parent hash. Execute:

1. Replay every supplied record's internal hash, field embeddings, integral-basis coordinates, multiplication rows, generator words, matrix entries, determinants, inverses, and fixed generator order. This step verifies internal consistency but compares the candidate lift separately from the specified coherent table. Malformed data or a stale internal hash return `INVALID_INPUT`. A structurally valid Statement 23.2 parent which is not `MAXIMAL/OE` returns `NONMAXIMAL_PARENT`; a global denominator ideal different from $\mathcal{O}_E$ returns `NONINTEGRAL_MODEL`.
2. Multiply the surface relator and the cyclically conjugate boundary word in fixed order. Require the exact matrix I . A result $-I$, or any other matrix, returns `RELATOR_FAILURE`.
3. Projectivize every candidate generator and compare it with the corresponding geometric side-pairing row. A discrepancy returns `PROJECTIVE_MISMATCH`.
4. After both semantic checks pass, compare the four candidate matrices, signs, exact coordinate tuples, and specified source hash with the coherent-lift table. Any difference returns `LIFT_MISMATCH`, even if the relator and projectivization are unchanged.
5. Replay the finite Poincaré polygon hypotheses, boundary cycles, side pairings, inverse Nielsen substitutions, and the induced presentation isomorphism. Failure returns the classification

`POLYGON_CERTIFICATE_FAILURE`.
6. Reconstruct the universal-property descent, the equality of projectivized maps on generators, and the kernel calculation (25.65). Serialize every dependency and proof row in fixed source, generator, word, matrix-entry, basis, and byte order. Return `SATISFIED` only after byte-identical independent verification.

Theorem 23.5.4 (Algorithm correctness). *The construction terminates and returns `SATISFIED` exactly when the supplied specified lift defines the faithful integral representation (25.64). Every rejection carries its first failing stage and a finite independent verification witness. Equal canonical inputs produce byte-identical records.*

Proof. All input lists and coordinate operations are finite. The first two steps prove that the candidate assignment is integral and descends to the presented group. The third step identifies its projectivization with the geometric generator map; the fourth binds it to the specified linear lift; and the fifth certifies that the geometric map is the specific faithful side-pairing action, rather than an abstractly isomorphic copy. Equation (25.65) then proves linear injectivity. Fixed traversal, normalization, failure, and byte orders prove determinism. $\square$

The completed record is

$$\begin{aligned}
 &(\mathbf{h}_{\text{presentation}}, \mathbf{h}_{\text{polygon}}, \mathbf{h}_{\mathcal{O}_E}, \mathbf{h}_{\mathcal{M}}, \mathbf{h}_{\text{trace}}, \\
 &\mathcal{L}, \text{integrality_rows}, \text{determinant_rows}, \text{inverse_rows}, \text{relator_rows}, \\
 &\text{projective_matching_rows}, \text{Nielsen_rows}, \text{polygon_rows}, \text{kernel_rows}, \\
 &\text{stage}, \text{first_failure}, \text{classification}, \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}} \text{ or } \text{NONE}).
 \end{aligned} \tag{25.73}$$

The mutually exclusive classifications are

$$\begin{aligned} & \text{INVALID_INPUT, NONMAXIMAL_PARENT,} \\ & \text{NONINTEGRAL_MODEL, LIFT_MISMATCH,} \\ & \text{RELATOR_FAILURE, PROJECTIVE_MISMATCH,} \\ & \text{POLYGON_CERTIFICATE_FAILURE,} \\ & \text{SATISFIED.} \end{aligned} \tag{25.74}$$

Minimum tests. The specified Bolza data must return SATISFIED. An internally consistent determinant-one candidate table whose recomputed surface product is not I tests the relator-failure route before any exact identity comparison. Conjugating the entire specified table by a noncentral integral determinant-one matrix preserves the relator but changes the geometric projective generator rows, thereby testing the projective-mismatch route. Replacing one lift by its negative leaves every commutator and every projective class unchanged but tests the lift-mismatch route at the later exact-identity stage. A false vertex-cycle row or a noninvertible Nielsen table tests the polygon classification. Order-only, not-maximal, nonintegral-denominator, false-inverse, and stale-hash inputs test the remaining structural routes. Finally, a finite reduction that separates every member of a supplied test list but has finite image must still be labelled nonfaithful on Γ_B , testing the warning following (25.71).

23.6 Arithmetic realization of the eightfold symmetry

Retain the faithful integral image $\Gamma_E = \rho_E(\Gamma_B)$ from Statement 23.5. With the distinguished primitive sixteenth root from Statement 23.1, define

$$C = \begin{pmatrix} \xi & 0 \\ 0 & \xi^{-1} \end{pmatrix}, \quad \xi = \zeta_{16}. \tag{25.75}$$

Because ξ is a root of unity,

$$\xi, \xi^{-1} \in \mathcal{O}_E^\times, \quad \det C = 1, \quad C, C^{-1} \in \text{SL}_2(\mathcal{O}_E). \tag{25.76}$$

Thus conjugation by C is defined over the same global integral model at every nonzero prime ideal; no new denominator is introduced.

Lemma 23.6.1 (Exact integral covariance). *For the physical side-pairing lifts, with indices modulo 8,*

$$CG_jC^{-1} = G_{j+1} \quad (0 \leq j < 8). \tag{25.77}$$

Consequently

$$C\Gamma_EC^{-1} = \Gamma_E, \quad C \in N_{\text{SL}_2(\mathcal{O}_E)}(\Gamma_E). \tag{25.78}$$

Proof. The diagonal entries of G_j are unchanged by diagonal conjugation. The upper-right entry is multiplied by $\xi^2 = \zeta_8$, and the lower-left entry by $\xi^{-2} = \zeta_8^{-1}$. In the exact formula for G_j , this increments the side index and proves (25.77) entry by entry in $\mathcal{O}_E$. The matrices $G_0, \dots, G_7$ generate Γ_E . Hence (25.77) gives the inclusion $C\Gamma_EC^{-1} \subseteq \Gamma_E$; applying the same argument to C^{-1} gives the reverse inclusion. $\square$

Let ϕ_8 be the geometric automorphism certified in Statement 7.5. Faithfulness of ρ_E transports it to

$$\Phi_8 : \Gamma_E \longrightarrow \Gamma_E, \quad \Phi_8(\rho_E(\gamma)) = \rho_E(\phi_8(\gamma)) = C\rho_E(\gamma)C^{-1}. \tag{25.79}$$

The definition is independent of the chosen word for γ because ρ_E and ϕ_8 are homomorphisms; the last equality follows first from (25.77) and then from products and inverses.

Theorem 23.6.2 (Arithmetic C_8 -action). *The subgroup generated by $\Phi_8 = \text{Ad}_C|_{\Gamma_E}$ in $\text{Aut}(\Gamma_E)$ is cyclic of exact order 8. It is the arithmetic realization of the geometric C_8 -action.*

Proof. Since ξ has exact order 16,

$$C^8 = -I, \quad C^{16} = I, \quad \text{ord}_{\text{SL}_2(E)}(C) = 16. \quad (25.80)$$

The central matrix $-I$ acts trivially by conjugation, so $\Phi_8^8 = 1$. For $1 \leq k < 8$, repeated use of (25.77) gives

$$\Phi_8^k(G_0) = G_k \neq G_0. \quad (25.81)$$

The inequality is certified by the Statement 7.5 exact key comparison, or geometrically by the distinct points $G_k o$. Thus no smaller positive power is the identity. $\square$

Equation (25.80) is essential: C itself is not an order-eight element of the linear group. Rather, its projective class $c_8 = [C]$ has exact order 8. If $\bar{\Gamma}_E$ is the projective image, then

$$\langle \bar{\Gamma}_E, c_8 \rangle = \bar{\Gamma}_E \rtimes_{\Phi_8} \langle c_8 \rangle, \quad \langle c_8 \rangle \cong C_8. \quad (25.82)$$

Indeed, normalization follows from (25.78). If $c_8^k \in \bar{\Gamma}_E$ for $0 < k < 8$, the corresponding deck transformation fixes the center of the disk, contradicting freeness of the Γ_B -action. Therefore the intersection is trivial. At the linear level, if $\hat{\Gamma}_E = \langle \Gamma_E, C \rangle$, projectivization gives the central extension

$$1 \longrightarrow \{\pm I\} \longrightarrow \hat{\Gamma}_E \longrightarrow \bar{\Gamma}_E \rtimes_{\Phi_8} C_8 \longrightarrow 1. \quad (25.83)$$

Here $-I = C^8$ belongs to $\hat{\Gamma}_E$ but, by Statement 23.5, not to Γ_E . Formula (25.83), rather than an unsupported linear semidirect-product claim with an order-eight lift, records the exact relationship.

Reduction at prime ideals. For every nonzero prime ideal $\mathfrak{p}$, (25.76)–(25.77) reduce entrywise to

$$\bar{C} \in \text{SL}_2(\kappa(\mathfrak{p})), \quad \bar{C} \bar{G}_j \bar{C}^{-1} = \bar{G}_{j+1}. \quad (25.84)$$

Hence conjugation induces an automorphism on the finite image

$$Q_{\mathfrak{p}} := \bar{\rho}_{\mathfrak{p}}(\Gamma_B), \quad \Phi_{8,\mathfrak{p}}(q) = \bar{C} q \bar{C}^{-1}, \quad \Phi_{8,\mathfrak{p}}^8 = 1. \quad (25.85)$$

The order may be a proper divisor of 8. Exact order survives if the seven characteristic-zero witnesses in (25.81) remain distinct. Using the lift-sign-invariant entry ideals of Statement 23.4, set

$$\mathfrak{K}_k = \mathfrak{J}^{\text{proj}}(G_k G_0^{-1}), \quad \mathfrak{K}_{C_8} = \prod_{k=1}^7 \mathfrak{K}_k. \quad (25.86)$$

Each factor is nonzero by Statement 23.5 and (25.81). Therefore

$$\mathfrak{K}_{C_8} \not\subseteq \mathfrak{p} \implies \text{ord}(\Phi_{8,\mathfrak{p}}) = 8. \quad (25.87)$$

This condition is sufficient for the exact order: it preserves all seven inequalities $\bar{G}_k \neq \bar{G}_0$. If it fails, the order is *not* declared smaller; the reduced generator permutation must be tested directly. In residue characteristic 2, $-I = I$ and the linear order of $\bar{C}$ can collapse, while the covariance relation (25.84) still remains valid.

Construction and certificate. The typed input consists of the Statement 23.1 primitive-root and field record, a maximal Statement 23.2 ring record, the Statement 23.5 faithful integral representation record, the eight exact physical generator matrices, the Statement 7.3 generator-action tables, and the Statement 7.5 exact-order certificate. It also contains a finite, canonically ordered request list $\mathcal{P} = (\mathfrak{p}_1, \dots, \mathfrak{p}_r)$, possibly empty. Each request carries a canonical ideal basis, the common $\mathcal{O}_E$ -parent hash, a nonzero-prime certificate, and a certified presentation of $\kappa(\mathfrak{p})$. Execute:

1. Replay all internal hashes, field relations, the isolating choice of ξ , its exact order 16, the maximal parent, and every matrix coordinate, determinant, inverse, and generator key. Verify the canonical order and absence of duplicates in $\mathcal{P}$, as well as every parent, primality, and residue-field row. Invalid data return `INVALID_INPUT`. A valid nonmaximal parent returns `NONMAXIMAL_PARENT`. A representation which is not globally integral returns

`NONINTEGRAL_REPRESENTATION`.

2. Construct C from the certified ξ , verify (25.76), and multiply all eight covariance rows (25.77) in index order. Reconstruct the normalizer equality by both inclusions.
3. Transport conjugation through the physical, boundary, and presentation-generator dictionaries. Compare every word and matrix row with the Statement 7.3 table and verify the surface relator.
4. Verify $C^8 = -I$, $C^{16} = I$, all seven inequalities $G_k \neq G_0$, and the eight generator rows for $\Phi_8^8 = 1$. Reconstruct (25.82)–(25.83), including the trivial-intersection and $-I \notin \Gamma_E$ witnesses.
5. For each $\mathfrak{p} \in \mathcal{P}$, in canonical order, reduce (25.84) and compute (25.86). Keep an order-divides-eight result distinct from an exact-order-eight result. Emit exactly one reduction row per request; a duplicate or foreign-prime row is invalid.
6. Serialize in fixed field, generator, word, matrix-entry, prime, ideal, factor, and byte order. Return `SATISFIED` only after byte-identical independent verification; otherwise return the first failing stage and exact witness.

Theorem 23.6.3 (Algorithm correctness). *The construction terminates and returns `SATISFIED` exactly when the supplied records realize the specified eightfold symmetry by the integral normalizer C . On passage, the characteristic-zero action has exact order 8, every requested reduction has a valid induced action of order dividing 8, and an exact-order claim is made only when its reduced witness rows prove it.*

Proof. All lists and matrix operations are finite. Equations (25.75)–(25.78) prove integral normalization, (25.79)–(25.81) prove the exact automorphism and its order, and (25.82)–(25.83) prove the projective and linear extension interfaces. Ring homomorphisms preserve the covariance identities, proving (25.84)–(25.85). Statement 23.4 applied to the seven difference words proves (25.87). Fixed traversal and serialization orders give deterministic independent verification. $\square$

The completed record is

$$\begin{aligned}
& (h_E, h_{\mathcal{O}_E}, h_{\rho_E}, h_{\phi_8}, \mathcal{P}, C, C^{-1}, \\
& \text{unit_rows, covariance_rows, normalizer_rows, action_tables,} \\
& \text{linear_order_rows, automorphism_order_rows, extension_rows,} \\
& (\mathfrak{K}_k)_{k=1}^7, \mathfrak{K}_{C_8}, (\text{reduction_row}_{\mathfrak{p}})_{\mathfrak{p} \in \mathcal{P}}, \\
& \text{stage, first_failure, classification, } h_{\text{input}}, h_{\text{independentverification}} \text{ or NONE}).
\end{aligned} \tag{25.88}$$

The mutually exclusive terminal classifications are

$$\begin{aligned} & \text{INVALID_INPUT,} & \text{NONMAXIMAL_PARENT,} \\ & \text{NONINTEGRAL_REPRESENTATION,} \\ & \text{SATISFIED.} \end{aligned} \tag{25.89}$$

Semantic row failures use `INVALID_INPUT`. The first-failure field identifies the failed row; no overlapping terminal classification is introduced.

Minimum tests. The specified Bolza data must pass and produce linear order 16 for C , projective order 8, and automorphism order 8. Replacing ξ by ξ^2 , changing one covariance index, or changing one generator-action word must be rejected at its first semantic row. Deleting the witness $G_k \neq G_0$ must forbid an exact-order conclusion. An order-only or not-maximal parent and a locally but not globally integral representation test the three structural classifications. At a prime above 2, the implementation must accept covariance while independently recomputing the possibly collapsed order. Finally, a prime containing one factor $\mathfrak{K}_k$ must disable the sufficient certificate (25.87) without being misreported as proof that the reduced order is smaller.

23.7 Arithmetic quotients and the independent parity factor

The parity character is the presentation-theoretic homomorphism from Statement 9.1,

$$\chi : \Gamma_B \twoheadrightarrow C_2, \quad \chi(g_j) = 1 \quad (0 \leq j < 8), \quad K_\chi = \ker \chi. \tag{25.90}$$

It is independent of the matrix coefficient field. In particular, determinant, trace, residue characteristic, and the central sign of an SL_2 -lift do not encode χ .

For a nonzero prime ideal $\mathfrak{p}$, the Statement 23.6 reduction has two distinct quotient kinds:

$$\begin{aligned} Q_{\mathfrak{p}}^{\text{lin}} &= \text{im}(\bar{\rho}_{\mathfrak{p}}), & \pi_{\mathfrak{p}}^{\text{lin}} : \Gamma_B &\twoheadrightarrow Q_{\mathfrak{p}}^{\text{lin}}, \\ Q_{\mathfrak{p}}^{\text{proj}} &= \text{im}(\text{proj} \circ \bar{\rho}_{\mathfrak{p}}), & \pi_{\mathfrak{p}}^{\text{proj}} : \Gamma_B &\twoheadrightarrow Q_{\mathfrak{p}}^{\text{proj}}. \end{aligned} \tag{25.91}$$

The target is always the corestricted image, so both displayed maps are surjective. A parity realization on either quotient is a homomorphism

$$\epsilon_{\mathfrak{p}}^{\square} : Q_{\mathfrak{p}}^{\square} \longrightarrow C_2, \quad \epsilon_{\mathfrak{p}}^{\square} \pi_{\mathfrak{p}}^{\square} = \chi, \quad \square \in \{\text{lin}, \text{proj}\}. \tag{25.92}$$

Theorem 23.7.1 (Exact parity-descent criterion). *Let $\pi : \Gamma_B \twoheadrightarrow Q$ be any finite quotient. The following are equivalent:*

- (i) *there is $\epsilon : Q \rightarrow C_2$ with $\epsilon\pi = \chi$;*
 - (ii) *$\ker \pi \subseteq K_\chi$;*
 - (iii) *$\pi(u) = \pi(v) \implies \chi(u) = \chi(v)$ for all words u, v ;*
 - (iv) *$\ker \pi$ contains no odd-parity element.*
- (25.93)

When it exists, ϵ is unique and surjective.

Proof. If (i) holds and $\gamma \in \ker \pi$, then $\chi(\gamma) = \epsilon(1) = 0$, proving (ii). If (ii) holds, define $\epsilon(\pi(\gamma)) = \chi(\gamma)$. If $\pi(\gamma) = \pi(\delta)$, then $\gamma\delta^{-1} \in \ker \pi \subseteq K_\chi$, so the definition is independent of the lift. It is a homomorphism and satisfies (i). Surjectivity of π makes the factor unique, and surjectivity of χ makes it onto. Conditions (ii), (iii), and (iv) are the same kernel statement written respectively for elements, fibres, and the nonzero bit of C_2 . $\square$

Linear and projective identity fibres. For a word w , criterion (25.93) specializes to

$$\begin{aligned} \text{linear obstruction: } & \chi(w) = 1, \quad \overline{\rho_E(w)} = I, \\ \text{projective obstruction: } & \chi(w) = 1, \quad \overline{\rho_E(w)} = I \text{ or } -I. \end{aligned} \quad (25.94)$$

In residue characteristic 2, the two signs coincide, but the parity bit does not disappear. Conversely, in odd characteristic an odd word reducing to $-I$ obstructs projective parity descent even though it is not in the kernel of the linear quotient. This is why the quotient-kind field in (25.91) is mathematically essential.

The value of a signed spelling $w = s_1^{e_1} \cdots s_m^{e_m}$ is $\chi(w) = \sum_i e_i = m \bmod 2$. This is invariant under all group relations and cancellations, but it is not a claim about the integer value of minimal word length. Likewise, $\det \rho_E(w) = 1$ for both parities, and the unipotent trace example of Statement 23.4 shows why no trace test can recover the bit.

Finite construction or finite obstruction. Let $q_j = \pi(g_j)$, and enumerate Q by deterministic breadth-first search from 1_Q using the ordered right edges $x \mapsto xq_j$. Store $\epsilon(1_Q) = 0$. Whenever a processed edge has endpoint $y = xq_j$, propose

$$\epsilon(y) = \epsilon(x) + 1 \pmod{2}. \quad (25.95)$$

If y is new, store the bit and predecessor word; if it already has the opposite bit, let u be the new path word and v the stored path word. Then

$$\pi(u) = \pi(v), \quad \kappa = uv^{-1} \in \ker \pi, \quad \chi(\kappa) = 1. \quad (25.96)$$

This is an independently verifiable obstruction to (25.93).

Theorem 23.7.2 (Arithmetic parity algorithm). *The breadth-first construction terminates. It returns a complete one-bit table precisely when parity descends, and otherwise returns the first conflict (25.96). In the successful case,*

$$\epsilon(\pi(w)) = \chi(w) = |w| \pmod{2} \quad (25.97)$$

for every physical-shell spelling w , and the table is the unique factor in (25.92).

Proof. The exact matrix image Q is finite and generated by $(q_0, \dots, q_7)$, so every element is enqueued exactly once. Induction along predecessor paths proves (25.97). A conflicting update gives (25.96), so parity cannot descend. Conversely, any odd-parity kernel word is an odd closed walk at 1_Q ; repeated applications of (25.95) would return to the stored zero with proposed bit one, so the finite traversal must encounter a conflict. If there is none, (25.97) is independent of spelling and the proof of (25.93) constructs the required homomorphism. $\square$

If parity descends, every q_j has bit one, the quotient has even order, and its physical-shell Cayley multigraph is bipartite with

$$Q_0 = \epsilon^{-1}(0), \quad Q_1 = \epsilon^{-1}(1), \quad |Q_0| = |Q_1|. \quad (25.98)$$

These are consequences, not alternative matrix criteria. An odd order image, a trivial image, or any identity shell generator is an immediate negative test.

Compatibility with the reduced C_8 -action. Statement 23.6 supplies an induced automorphism $\Phi_{8,p}^\square$ on either quotient kind, even when its reduced order is smaller than eight. If parity descends, then

$$\epsilon_p^\square \Phi_{8,p}^\square \pi_p^\square(\gamma) = \chi(\phi_8(\gamma)) = \chi(\gamma) = \epsilon_p^\square \pi_p^\square(\gamma). \quad (25.99)$$

Surjectivity of $\pi_p^\square$ therefore gives

$$\epsilon_p^\square \Phi_{8,p}^\square = \epsilon_p^\square, \quad \Phi_{8,p}^\square(Q_i) = Q_i. \quad (25.100)$$

Thus rotation preserves the two parity parts even if its own order collapses. The group $Q_p^\square$ here is the image of Γ_B , not the larger group obtained by adjoining $\overline{C}$. A parity assigned to that external rotation generator is additional data and is not determined by χ .

Construction and certificate. The typed input contains the Statement 9.1 parity record and Statement 9.4 invariance rows; the maximal Statement 23.2 parent; the Statement 23.5 faithful integral representation; the Statement 23.6 finite canonical prime list and reduction/action rows; and a finite, canonically ordered request list

$$\mathcal{R}_{\text{par}} = ((\mathbf{p}_i, \square_i))_{i=1}^m, \quad m \geq 0, \quad \mathcal{R}_{\text{par}} \subseteq \mathcal{P} \times \{\text{lin}, \text{proj}\}.$$

The list may be empty and has no duplicate pair. For every member it contains exactly one exact finite-image multiplication table, identity, and ordered physical-shell tuple. Execute:

1. Replay hashes, field and maximal-parent rows, global integrality, prime and residue-field certificates, quotient kind, every matrix element, multiplication row, identity, inverse, and shell generator. Verify the order and uniqueness of $\mathcal{R}_{\text{par}}$, membership $\mathbf{p} \in \mathcal{P}$, and presence of exactly one table for every request. Verify that the shell generates exactly the declared corestricted image. A missing, duplicate, foreign, or malformed request or table returns `INVALID_INPUT`; valid nonmaximal and nonintegral parents retain their separate structural classifications.
2. Replay the surface-relator parity accumulator and all eight rows $\chi(g_j) = 1$. Compare direct word parity with every stored predecessor word.
3. Run (25.95) in fixed queue and shell order. On the first conflict, verify both matrix evaluations and serialize the odd kernel witness (25.96). Otherwise verify the full bit table, all multiplication rows, surjectivity, and (25.97).
4. Replay reduced covariance and verify (25.99)–(25.100) whenever a parity table exists. If parity is obstructed, set the symmetry-row field to `NONE`. Do not require the reduced rotation to have exact order eight.
5. Emit exactly one result per member of $\mathcal{R}_{\text{par}}$, reject duplicate and foreign result rows, and serialize in prime, quotient-kind, queue, shell, element, matrix-entry, bit, witness, and byte order.

Every well-formed request completes with one of the mutually exclusive per-quotient results

$$\text{PARITY_FACTORS}, \quad \text{PARITY_OBSTRUCTED}. \quad (25.101)$$

The second is a certified mathematical answer, not an item-level failure. On item-level `SATISFIED`, the completed record is

$$\begin{aligned} &(\mathbf{h}_\chi, \mathbf{h}_{\rho_E}, \mathbf{h}_{\Phi_8}, \mathcal{R}_{\text{par}}, \\ & (Q_p^\square, (q_j)_{j=0}^7, \text{BFS_rows}, \epsilon_p^\square \text{ or } \text{NONE}, \\ & \text{symmetry_rows or NONE}, (u, v, \kappa, \text{evaluation_rows}) \text{ or NONE}, \\ & \text{parity_result})_{(\mathbf{p}, \square) \in \mathcal{R}_{\text{par}}}, \\ & \text{COMPLETE, NONE, SATISFIED, } \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}}). \end{aligned} \quad (25.102)$$

For `PARITY_FACTORS`, the bit table and symmetry rows are present and the conflict tuple is `NONE`. For `PARITY_OBSTRUCTED`, the bit table and symmetry rows are `NONE`, while the full tuple $(u, v, \kappa, \text{evaluation_rows})$ is present. No other presence pattern is valid. If $\mathcal{R}_{\text{par}}$ is empty, the row tuple is empty and the item returns `SATISFIED` vacuously after validating its parents. The item-level terminal classifications are

$$\begin{aligned} & \text{INVALID_INPUT}, \quad \text{NONMAXIMAL_PARENT}, \\ & \text{NONINTEGRAL_REPRESENTATION}, \quad \text{SATISFIED}. \end{aligned} \quad (25.103)$$

An early structural rejection has the separate record

$$\text{REJECT}(\text{classification}, \text{stage}, \text{first_failure}, \text{h}_{\text{input}}), \quad \text{classification} \neq \text{SATISFIED}. \quad (25.104)$$

It contains no quotient, BFS, parity, or symmetry field.

Theorem 23.7.3 (Certificate correctness). *The construction terminates and, on item-level SATISFIED, gives for every requested arithmetic quotient either its unique parity factor or a canonical odd kernel witness. Equal canonical inputs produce byte-identical records.*

Proof. The request list and each finite image are finite. The arithmetic parity algorithm proves the per-quotient dichotomy. Equations (25.99)–(25.100) prove the symmetry rows on factorable requests. The presence rules cover obstructed and early-rejection branches. Fixed validation, first-conflict, traversal, and serialization orders prove termination and reproducibility. $\square$

Minimum tests. The quotient $\Gamma_B/K_\chi \cong C_2$ returns the identity parity table, whereas the trivial quotient returns an odd closed-edge witness. A quotient with an identity shell image, or with odd order, must be obstructed. A projective test in which an odd word reduces to $-I$ must fail even if the corresponding linear quotient has no such kernel word. Characteristic 2 must trigger no special positive answer: the same BFS criterion is still run. Altering one multiplication row, omitting a generated element, changing the quotient kind, duplicating a request, or inserting a foreign-prime row tests structural rejection. Finally, a valid parity table together with a reduced rotation of order 1, 2, or 4 still has to satisfy (25.100), testing independence from the rotation order.

23.8 Reduction-ready equivariant parity representation

Fix a nonzero prime ideal $\mathfrak{p}$ from the certified Statement 23.6 list and a quotient kind $\square \in \{\text{lin}, \text{proj}\}$. Write

$$\pi = \pi_{\mathfrak{p}}^{\square} : \Gamma_B \twoheadrightarrow Q_{\mathfrak{p}}^{\square} = Q, \quad K = \ker \pi. \quad (25.105)$$

The base reduction is C_8 -equivariant, but Statement 23.7 shows that its matrix coordinates need not carry parity. Define the augmented map

$$\begin{aligned} \hat{\pi} : \Gamma_B &\longrightarrow Q \times C_2, \\ \gamma &\longmapsto (\pi(\gamma), \chi(\gamma)), \quad \tilde{Q} := \text{im } \hat{\pi}. \end{aligned} \quad (25.106)$$

The codomain is corestricted to $\tilde{Q}$; the direct product need not be the image.

Theorem 23.8.1 (Kernel-intersection construction). *The augmented map is a surjection onto the finite group $\tilde{Q}$, and*

$$\tilde{K} := \ker \hat{\pi} = K \cap K_\chi. \quad (25.107)$$

The second projection restricts to the unique parity factor

$$\tilde{\epsilon} : \tilde{Q} \twoheadrightarrow C_2, \quad \tilde{\epsilon} \hat{\pi} = \chi. \quad (25.108)$$

Thus every augmented quotient carries parity, regardless of whether the base arithmetic quotient does.

Proof. Both components in (25.106) are homomorphisms, so their product is a homomorphism. Corestriction makes it surjective. An element maps to $(1, 0)$ exactly when it lies in both component kernels, proving (25.107). Projection to the second coordinate gives (25.108), and it is onto because χ is onto. Finally $\widetilde{K} \subseteq K_\chi$, so the uniqueness part of the Statement 23.7 criterion applies. $\square$

The first-coordinate projection

$$r : \widetilde{Q} \twoheadrightarrow Q, \quad r(q, b) = q \quad (25.109)$$

has kernel canonically isomorphic to $K/(K \cap K_\chi)$. Restriction of χ to K therefore yields the exact dichotomy

$K \subseteq K_\chi$	$\widetilde{Q}$	$\ker r$	$ \widetilde{Q} $
yes	$\{(q, \epsilon(q)) : q \in Q\}$	1	$ Q $
no	$Q \times C_2$	C_2	$2 Q $.

(25.110)

For the second row, choose $k \in K$ with $\chi(k) = 1$. Multiplication by $\widehat{\pi}(k) = (1, 1)$ flips the second coordinate over every q , so both bits occur and the image is the full direct product. Hence the construction introduces no unnecessary doubling when parity already factors.

C_8 -equivariance. Let $\Phi = \Phi_{8,p}^\square$ be the reduced automorphism. Since $\pi\phi_8 = \Phi\pi$ and $\chi\phi_8 = \chi$, define

$$\widetilde{\Phi} : \widetilde{Q} \longrightarrow \widetilde{Q}, \quad \widetilde{\Phi}(q, b) = (\Phi(q), b). \quad (25.111)$$

For every $\gamma \in \Gamma_B$,

$$\widetilde{\Phi} \widehat{\pi}(\gamma) = (\Phi\pi(\gamma), \chi(\gamma)) = \widehat{\pi}(\phi_8(\gamma)). \quad (25.112)$$

This proves both well-definedness on the image and equivariance. Moreover

$$\widetilde{\epsilon}\widetilde{\Phi} = \widetilde{\epsilon}, \quad \text{ord}(\widetilde{\Phi}) = \text{ord}(\Phi) \mid 8. \quad (25.113)$$

Equality of the orders follows because $r\widetilde{\Phi} = \Phi r$ gives one divisibility, while (25.111) gives the other. Parity augmentation does not repair a collapsed rotation order; exact order eight still requires the Statement 23.6 witnesses.

Exact survival of a finite danger list. Let $\mathcal{D} = (w_1, \dots, w_t)$ be an ordered list whose images are required to be nonidentity. For each word,

$$\widehat{\pi}(w) = 1_{\widetilde{Q}} \iff \chi(w) = 0 \text{ and } \pi(w) = 1_Q. \quad (25.114)$$

Thus every odd word survives automatically. For an even word define

$$\mathfrak{S}_w^{\text{lin}} = \mathfrak{J}_w^+, \quad \mathfrak{S}_w^{\text{proj}} = \mathfrak{J}_w^{\text{proj}}, \quad \mathfrak{S}_{\mathcal{D}}^\square = \prod_{\substack{w \in \mathcal{D} \\ \chi(w)=0}} \mathfrak{S}_w^\square. \quad (25.115)$$

The empty product is $\mathcal{O}_E$. If every $w \in \mathcal{D}$ is nonidentity in the characteristic-zero projective representation, then every even factor is nonzero and

$$\mathfrak{S}_{\mathcal{D}}^\square \not\subseteq \mathfrak{p} \iff \widehat{\pi}(w) \neq 1 \text{ for all } w \in \mathcal{D}. \quad (25.116)$$

Proof. For odd w , the second coordinate in (25.106) is nonzero. For even w , (25.114) reduces survival to the exact entry-ideal criterion: $\mathfrak{J}_w^+ \subseteq \mathfrak{p}$ in the linear case and $\mathfrak{J}_w^{\text{proj}} \subseteq \mathfrak{p}$ in the projective case. Primality says that $\mathfrak{p}$ contains the product in (25.115) exactly when it contains at least one even factor, proving (25.116). Characteristic-zero nonidentity and Statement 23.5 faithfulness exclude $M(w) = \pm I$, so no factor is zero. $\square$

If a supplied danger word is already the identity in Γ_B , its parity is even and its factor in (25.115) is zero. The request is then globally unseparable, not merely bad at the selected prime. For pairwise injectivity on a finite word list, use all difference words uv^{-1} ; pairs of opposite parity separate automatically.

Reduction-ready construction. The typed input consists of the maximal Statement 23.2 parent; the fixed Statement 23.5 integral representation; the Statement 23.6 canonical prime and equivariance rows; the Statement 23.7 parity record and per-base-quotient factor/obstruction rows; and a finite canonical request list

$$\mathcal{R}_{\text{red}} = ((\mathbf{p}_i, \square_i, \mathcal{D}_i))_{i=1}^m, \quad m \geq 0. \quad (25.117)$$

Each $\mathbf{p}_i$ belongs to the certified prime list, $\square_i$ is linear or projective, and $\mathcal{D}_i$ is a finite ordered word list with exact source hashes. In addition,

$$((\mathbf{p}_i, \square_i))_{i=1}^m \subseteq \mathcal{R}_{\text{par}}.$$

For each projected key, the input contains exactly one hash-matching Statement 23.7 group table and parity-result row, and one hash-matching Statement 23.6 action row.

The accepted request normal form is executable. Group raw triples by the canonical ideal-Hermite bytes of $\mathbf{p}$ and then by kind, with $\text{lin} < \text{proj}$. Within a group, replace all danger entries by the set of pairs $(h_{\text{source}}, \text{canonical_word_bytes})$; multiplicity is discarded. Sort this set first by source-hash bytes and then by word bytes. Sort the resulting groups by the same prime-kind order. The canonical $\mathcal{R}_{\text{red}}$ has exactly one triple per group. Any input not already in this normal form is rejected. Execute:

1. Replay every parent, field, prime, residue, representation, covariance, quotient-kind, parity, word, and source-hash row. Verify canonical request order, normalized danger sets, and absence of duplicate or foreign requests. Require every projected key to occur in $\mathcal{R}_{\text{par}}$, and compare the exact group-table, parity-result, and action-row hashes. Invalid data return `INVALID_INPUT`; structurally valid nonmaximal and nonintegral parents retain their distinct terminal classifications.
2. Enumerate Q and $\tilde{Q}$ by fixed breadth-first traversal of the ordered physical shell. Verify every pair multiplication, inverse, identity, both coordinate projections, surjectivity, (25.107), and the order dichotomy (25.110).
3. Construct $\tilde{\epsilon}$, verify its complete table and every physical generator bit, then construct $\tilde{\Phi}$ and independent verification (25.111)–(25.113). Preserve the reduced rotation’s proved order; do not promote a divisor to eight.
4. Evaluate every danger word in characteristic zero, in Q , and in $\tilde{Q}$. Partition by its certified parity, compute the appropriate ideals (25.115), and compare direct reduction with (25.114)–(25.116).
5. If a least danger word is globally identity, record `UNSEPARABLE_WORD`. Otherwise record `ALL_SEPARATED` or `COLLISION_AT_PRIME`, with the least colliding even word in the latter case. An empty danger list is all-separated with product $\mathcal{O}_E$.
6. Emit exactly one row per request and serialize in prime, kind, queue, group-element, shell, word, bit, matrix-entry, ideal, failure, and byte order. Reject missing, duplicate, and foreign result rows.

Theorem 23.8.2 (Construction correctness and termination). *For every well-formed finite request list the algorithm terminates. On item-level passage it returns a finite C_8 -equivariant quotient $\tilde{Q}$ carrying parity, the exact base-to-augmented order dichotomy, and a necessary-and-sufficient separation result for every finite danger list. Equal canonical inputs produce byte-identical records.*

Proof. Every residue matrix group and every enumerated image is finite, so the breadth-first computations terminate. Equations (25.106)–(25.110) prove the group, kernel, parity, and cardinality rows. Equations (25.111)–(25.113) prove equivariance and order. Equations (25.114)–(25.116) prove the danger-list result, including the automatic odd-parity branch. Fixed traversal, first-witness, and serialization orders prove reproducibility. $\square$

On SATISFIED, the completed record is

$$\begin{aligned}
& (h_{\rho_E}, h_{\Phi_8}, h_{\chi}, \mathcal{R}_{\text{red}}, \\
& (Q, \tilde{Q}, \hat{\pi}, r, \tilde{\epsilon}, \tilde{\Phi}, \text{kernel_rows}, \text{order_rows}, \\
& \quad h_{Q,\text{par}}, h_{\text{action}}, \mathcal{D}, \text{parity_partition}, \\
& \quad \mathfrak{S}_{\mathcal{D}}^{\square}, \text{danger_rows}, \text{separation_result}, \\
& \quad \text{witness or NONE})_{\mathcal{R}_{\text{red}}}, \\
& \text{COMPLETE, NONE, SATISFIED, } h_{\text{input}}, h_{\text{independentverification}}).
\end{aligned} \tag{25.118}$$

The per-request separation results are

$$\begin{aligned}
& \text{ALL_SEPARATED,} \quad \text{COLLISION_AT_PRIME,} \\
& \text{UNSEPARABLE_WORD.}
\end{aligned} \tag{25.119}$$

All three are certified per-request answers compatible with the item-level classification SATISFIED. For ALL_SEPARATED, the witness field is NONE. For COLLISION_AT_PRIME, it contains the least colliding even word in canonical danger order, its two coordinate evaluations, and the exact ideal-containment row. For UNSEPARABLE_WORD, it contains the least characteristic-zero identity word and its exact matrix, parity, and word-equality rows. No other presence pattern is valid. The item-level classifications are

$$\begin{aligned}
& \text{INVALID_INPUT,} \quad \text{NONMAXIMAL_PARENT,} \\
& \text{NONINTEGRAL_REPRESENTATION,} \quad \text{SATISFIED.}
\end{aligned} \tag{25.120}$$

An item-level rejection uses

$$\text{REJECT}(\text{classification}, \text{stage}, \text{first_failure}, h_{\text{input}}) \tag{25.121}$$

and contains no quotient fields. An empty request list returns a vacuous SATISFIED after all common parents have been validated.

Minimum tests. If the base quotient already carries parity, (25.110) must return its graph with unchanged order. The trivial base quotient must augment to C_2 , not remain trivial. A base quotient obstructed by an odd kernel word must double, and the stored obstruction word must map to $(1, 1)$. An all-odd danger list is automatically separated at every prime and has ideal product $\mathcal{O}_E$. A nontrivial even word whose entry ideal is contained in $\mathfrak{p}$ tests the collision result; the identity word tests the globally unseparable result. Linear and projective requests must use respectively $\mathfrak{J}^+$ and $\mathfrak{J}^{\text{proj}}$. A characteristic two request must retain the parity bit even though matrix signs merge. Finally, missing, duplicate, foreign, nonmaximal, nonintegral, and stale-hash inputs test every structural rejection route.

24 Prime-ideal reduction and finite matrix images

24.1 Nonzero prime ideals and absolute norm

Let E and its certified full ring of integers $\mathcal{O}_E$ be as in C25. An integral ideal is an additive subgroup $\mathfrak{a} \subseteq \mathcal{O}_E$ satisfying $\mathcal{O}_E \mathfrak{a} \subseteq \mathfrak{a}$. A prime ideal for finite reduction means throughout

$$(0) \subsetneq \mathfrak{p} \subsetneq \mathcal{O}_E, \quad xy \in \mathfrak{p} \implies x \in \mathfrak{p} \text{ or } y \in \mathfrak{p}. \tag{26.1}$$

The qualifier nonzero is essential and is never suppressed in a machine record.

Lemma 24.1.1 (Finite index of nonzero ideals). *Every nonzero integral ideal $\mathfrak{a}$ contains a nonzero rational integer. Consequently $\mathcal{O}_E/\mathfrak{a}$ is finite.*

Proof. Choose $0 \neq \alpha \in \mathfrak{a}$. Its monic minimal polynomial over $\mathbb{Q}$ has integer coefficients because α is an algebraic integer:

$$m_\alpha(X) = X^r + c_{r-1}X^{r-1} + \cdots + c_1X + c_0, \quad c_0 \neq 0. \quad (26.2)$$

Substitution gives

$$c_0 = -\alpha(\alpha^{r-1} + c_{r-1}\alpha^{r-2} + \cdots + c_1) \in (\alpha) \subseteq \mathfrak{a}. \quad (26.3)$$

Thus $(|c_0|) \subseteq \mathfrak{a}$. The additive group $\mathcal{O}_E$ is free of rank $d = [E : \mathbb{Q}] = 16$, so

$$|\mathcal{O}_E/(|c_0|)| = |c_0|^d < \infty. \quad (26.4)$$

The quotient by $\mathfrak{a}$ is a quotient of this finite group and is finite. $\square$

Theorem 24.1.2 (Prime, maximal, and residue field). *For a nonzero proper integral ideal $\mathfrak{p}$, the following are equivalent:*

- (i) $\mathfrak{p}$ is prime;
 - (ii) $\mathcal{O}_E/\mathfrak{p}$ is an integral domain;
 - (iii) $\mathcal{O}_E/\mathfrak{p}$ is a field;
 - (iv) $\mathfrak{p}$ is maximal.
- (26.5)

In particular every nonzero prime ideal of $\mathcal{O}_E$ is maximal and has a finite residue field

$$\kappa(\mathfrak{p}) := \mathcal{O}_E/\mathfrak{p}. \quad (26.6)$$

Proof. The equivalence of (i) and (ii), and of (iii) and (iv), is the standard ideal–quotient correspondence, proved directly by pulling products and ideals back through the quotient map. The lemma makes the quotient in (ii) finite. If $0 \neq \bar{x}$ lies in a finite integral domain, multiplication by $\bar{x}$ is injective and hence surjective; therefore $\bar{x}\bar{y} = 1$ for some $\bar{y}$. Thus (ii) implies (iii). Every field is an integral domain, giving the reverse implication. $\square$

For every nonzero integral ideal define its absolute norm by

$$N\mathfrak{a} := [\mathcal{O}_E : \mathfrak{a}] = |\mathcal{O}_E/\mathfrak{a}| \in \mathbb{Z}_{>0}. \quad (26.7)$$

Thus

$$N\mathfrak{p} = |\kappa(\mathfrak{p})|, \quad N\mathcal{O}_E = 1. \quad (26.8)$$

No finite value is assigned to $N(0)$. If $(0) \neq \mathfrak{a} \subseteq \mathfrak{b}$, then

$$N\mathfrak{a} = [\mathfrak{b} : \mathfrak{a}] N\mathfrak{b}, \quad (26.9)$$

so inclusion reverses numerical size. The notation $\mathfrak{p} \mid \mathfrak{a}$ means $\mathfrak{a} \subseteq \mathfrak{p}$, not the opposite containment. This convention agrees with unique ideal factorization and the denominator conventions of Statement 23.3.

The zero and unit boundaries. Because $\mathcal{O}_E$ is a domain, (0) is prime in the purely ring-theoretic sense. It is excluded from (26.1): $\mathcal{O}_E/(0) = \mathcal{O}_E$ is infinite and is not a finite residue field. The unit ideal has finite norm 1, but it is not proper, its quotient is the zero ring, and it is not prime. Neither ideal is a permitted reduction modulus.

Canonical ideal and residue certificate. Let $\Omega = (\omega_1, \dots, \omega_d)$ be the certified integral basis. A nonzero ideal is a full-rank sublattice, represented by the unique column Hermite normal form matrix $H_{\mathfrak{a}}$ whose columns are its basis coordinates. Then

$$N\mathfrak{a} = |\det H_{\mathfrak{a}}|. \quad (26.10)$$

Ideal closure is checked by multiplying every HNF basis vector by every ω_i and solving the resulting integer membership systems. Canonical residue representatives are obtained by HNF reduction; addition and multiplication are performed in $\mathcal{O}_E$ and reduced back to those representatives.

For a candidate nonzero proper ideal, primality can therefore be certified without probabilistic factoring: construct its complete finite quotient table and verify that each nonzero residue has a multiplicative inverse. Failure returns the lexicographically least nonzero residue with no inverse and the least zero-divisor pair. Indeed, if $0 \neq x$ is a nonunit in a finite commutative ring, the map $m_x : y \mapsto xy$ is not surjective, since 1 is not in its image. A nonsurjective self-map of a finite set is noninjective, so $xy = xz$ for some $y \neq z$. Then

$$x(y - z) = 0, \quad x \neq 0, \quad y - z \neq 0.$$

Thus the exhaustive fixed product scan always finds a zero-divisor witness. Passage returns the full inverse table. By (26.5), these are respectively negative and positive primality certificates.

Construction and certificate. The typed input contains a maximal Statement 23.2 $\mathcal{O}_E$ -record and a finite ordered list of candidate integral lattices, each supplied in canonical HNF or as the distinguished zero lattice. Execute:

1. Replay the field, integral basis, multiplication table, maximal parent, HNF, and source hashes. A structurally valid parent whose classification is not **MAXIMAL**, or whose final kind is not **OE**, returns **NONMAXIMAL_PARENT**. Other malformed common data return **INVALID_INPUT**.
2. In candidate order, distinguish the zero lattice, the full unit lattice, and every other nonzero lattice. A nonzero rank-deficient lattice is **NOT_AN_IDEAL**, with its exact rank witness: a nonzero ideal contains $n\mathcal{O}_E$ by (26.3), hence has rank d . For a full-rank candidate, verify ideal closure and compute (26.10). A failed closure row has the same class with its least multiplication witness.
3. For every surviving proper ideal, build the complete quotient table in canonical representative order. Verify additive and multiplicative identities, distributivity, and the parent reduction map.
4. For every surviving proper ideal, scan nonzero residues and candidate inverses in fixed order. If every residue has an inverse, classify the ideal as **NONZERO_PRIME**; otherwise classify it as **NONPRIME_PROPER** and store the first inverse failure and least zero-divisor witness.
5. Emit exactly one row per candidate and serialize in candidate, HNF-column, basis, residue, operation, inverse, witness, and byte order. Reject missing, duplicate, or foreign rows.

Theorem 24.1.3 (Certification correctness and termination). *The construction terminates. Every well-formed candidate receives exactly one correct classification among*

$$\begin{aligned} &\text{ZERO_IDEAL, UNIT_IDEAL, NOT_AN_IDEAL,} \\ &\text{NONPRIME_PROPER, NONZERO_PRIME.} \end{aligned} \quad (26.11)$$

For the last class, the output residue table is a finite field of cardinality $N\mathfrak{p}$. Equal canonical inputs produce byte-identical records.

Proof. The candidate list, quotient representatives, multiplication rows, and inverse searches are finite. HNF membership proves ideal closure, (26.10) proves the quotient cardinality, and (26.5) proves the two proper-ideal classifications. The zero, unit, and nonideal branches are disjoint by definition. The rank argument and the finite multiplication-map argument prove existence of both nonideal and nonprime witnesses. Fixed scan and serialization orders give deterministic witnesses and independent verification. $\square$

On item-level passage, the completed record is

$$\begin{aligned} &(\mathbf{h}_{\mathcal{O}_E}, ((H_{\mathbf{a}}, \text{ideal_rows}, N\mathbf{a} \text{ or NONE,} \\ &\quad \text{quotient_table or NONE, inverse_rows or NONE, witness or NONE, class)}_{\mathbf{a}}, \quad (26.12) \\ &\quad \text{COMPLETE, NONE, SATISFIED, } \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}})). \end{aligned}$$

All five classes are certified per-candidate answers compatible with item-level **SATISFIED**. Their presence rules are:

- **ZERO_IDEAL**: the zero-lattice and rank rows are present; norm, quotient, inverse, and witness fields are **NONE**.
- **UNIT_IDEAL**: ideal rows, norm 1, and the zero-ring table are present; the inverse list is empty and the witness is **NONE**.
- **NOT_AN_IDEAL**: the least rank or closure witness is present; norm, quotient, and inverse fields are **NONE**.
- **NONPRIME_PROPER**: ideal rows, positive norm, quotient table, inverse-scan rows, first nonunit, and least zero-divisor pair are present.
- **NONZERO_PRIME**: ideal rows, positive norm, quotient field table, and complete inverse rows are present; the witness is **NONE**.

No other presence pattern is valid. Only the last class may be exported as a reduction modulus. The item-level terminal classifications are

$$\text{INVALID_INPUT,} \quad \text{NONMAXIMAL_PARENT,} \quad \text{SATISFIED.} \quad (26.13)$$

An early rejection uses

$$\text{REJECT}(\text{classification, stage, first_failure, } \mathbf{h}_{\text{input}}) \quad (26.14)$$

and contains no candidate-classification fields.

Minimum tests. In $\mathbb{Z}[i]$, the ideal $(1+i)$ is prime with norm 2, while (2) is not prime because $(1+i)^2 = 2i \in (2)$ but $1+i \notin (2)$. These test the two proper branches. The zero and unit ideals test both forbidden boundaries. A reproducible nonideal test is

$$L = \mathbb{Z} \cdot 1 \oplus \mathbb{Z} \cdot 2i \subset \mathbb{Z}[i], \quad i \cdot 1 = i \notin L.$$

A nonzero rank-one lattice tests the rank branch. A false HNF, wrong determinant norm, missing residue, nonmaximal parent, duplicate candidate row, and stale hash test the structural rejections.

24.2 Finite residue fields and certified residue degree

Let $\mathfrak{p}$ be a row exported by Statement 24.1 with class `NONZERO_PRIME`, and put $\kappa(\mathfrak{p}) = \mathcal{O}_E/\mathfrak{p}$. The rational prime below $\mathfrak{p}$ and the residue degree are part of the data; they are not inferred from a bare field cardinality.

Theorem 24.2.1 (Finite residue-field structure). *There is a unique rational prime p such that*

$$\mathfrak{p} \cap \mathbb{Z} = (p). \quad (26.15)$$

The quotient map identifies the prime field of $\kappa(\mathfrak{p})$ with $\mathbb{F}_p$. If

$$f = f(\mathfrak{p}/p) := [\kappa(\mathfrak{p}) : \mathbb{F}_p], \quad q = |\kappa(\mathfrak{p})|, \quad (26.16)$$

then

$$\kappa(\mathfrak{p}) \cong \mathbb{F}_{p^f}, \quad q = p^f. \quad (26.17)$$

In particular

$$N\mathfrak{p} = p^f. \quad (26.18)$$

For the degree-16 field fixed in C25,

$$1 \leq f \leq [E : \mathbb{Q}] = 16. \quad (26.19)$$

The isomorphism in (26.17) is not canonical unless a presentation of the residue field is included.

Proof. Contraction along $\mathbb{Z} \hookrightarrow \mathcal{O}_E$ sends a prime ideal to a prime ideal. The contraction is proper because $1 \notin \mathfrak{p}$, and it is nonzero because the finite-index lemma of Statement 24.1 supplies $0 \neq n \in \mathfrak{p} \cap \mathbb{Z}$. The nonzero prime ideals of $\mathbb{Z}$ are exactly (p) for unique positive rational primes p , proving (26.15).

The kernel of $\mathbb{Z} \rightarrow \kappa(\mathfrak{p})$ is (p) , so the image is $\mathbb{F}_p$ and the residue field has characteristic p . Its additive group is a finite vector space over $\mathbb{F}_p$. If its dimension is f , choosing a basis gives exactly p^f vectors and proves (26.17). Equation (26.18) follows from the norm identity of Statement 24.1. Since $p\mathcal{O}_E \subseteq \mathfrak{p}$, reduction gives a surjection

$$\mathcal{O}_E/p\mathcal{O}_E \twoheadrightarrow \mathcal{O}_E/\mathfrak{p}. \quad (26.20)$$

The source has $\mathbb{F}_p$ -dimension 16, which proves (26.19). $\square$

Intrinsic checks. The Frobenius map

$$F_p : \kappa(\mathfrak{p}) \longrightarrow \kappa(\mathfrak{p}), \quad x \longmapsto x^p \quad (26.21)$$

is an automorphism. Its exact order is f . Indeed every element of a field with $q = p^f$ elements satisfies $x^q = x$, whereas $F_p^e = \text{id}$ for $0 < e < f$ would put all p^f elements among the at most p^e roots of $T^{p^e} - T$. Consequently the field is the splitting field over $\mathbb{F}_p$ of the separable polynomial

$$T^{p^f} - T = \prod_{a \in \kappa(\mathfrak{p})} (T - a). \quad (26.22)$$

This also proves that two finite fields of cardinality p^f are isomorphic, but it does not distinguish prime ideals lying over the same p , nor does it choose an isomorphism.

Canonical presentation relative to the input record. Use the Statement 24.1 canonical residue order. First find the least positive integer n for which $n\bar{1} = 0$ in the complete addition table. Record this integer as p , verify that it is prime, and verify that $m\bar{1} = 0$ holds exactly when $p \mid m$. Starting with $b_1 = \bar{1}$, repeatedly append the least residue outside the $\mathbb{F}_p$ -span of the previous rows. This terminates with a canonical basis

$$B_{\mathfrak{p}} = (b_1, \dots, b_f), \quad \kappa(\mathfrak{p}) = \bigoplus_{j=1}^f \mathbb{F}_p b_j. \quad (26.23)$$

The enumeration of coefficient tuples verifies both $q = p^f$ and every basis coordinate.

For an explicit polynomial model, scan residues in canonical order and choose the first θ for which

$$\text{span}_{\mathbb{F}_p} \{1, \theta, \dots, \theta^{f-1}\} = \kappa(\mathfrak{p}). \quad (26.24)$$

Such an element exists. One direct proof is that the finite subgroup $\kappa(\mathfrak{p})^\times$ of a field is cyclic: if m is its exponent, the root bound applied to $X^m - 1$, together with the primary decomposition of the finite abelian group, forces an element of order $|\kappa(\mathfrak{p})^\times|$. A multiplicative generator generates the whole field over its prime field.

There are then unique coefficients $c_i \in \mathbb{F}_p$ satisfying

$$g_{\mathfrak{p}}(T) = T^f + c_{f-1}T^{f-1} + \dots + c_1T + c_0, \quad g_{\mathfrak{p}}(\theta) = 0. \quad (26.25)$$

The spanning condition makes this the minimal polynomial of θ , so it is irreducible, and evaluation induces the fully specified isomorphism

$$\mathbb{F}_p[T]/(g_{\mathfrak{p}}) \xrightarrow{\sim} \kappa(\mathfrak{p}), \quad T \mapsto \theta. \quad (26.26)$$

For $f = 1$, the least choice may be $\theta = 0$ and $g_{\mathfrak{p}} = T$; this is a valid degree-one presentation.

Construction and certificate. The typed input consists of the completed Statement 24.1 record, one selected candidate row, and the exact C25 parent hashes inherited by that row. Execute the following finite computation:

1. Replay the C25 parent, the prime-ideal HNF, ideal-closure rows, norm, quotient table, inverse table, class, and all hashes.
2. Require the selected row to have class `NONZERO_PRIME`. Recompute the characteristic by the additive-order scan and certify (26.15).
3. Construct (26.23), set $f = |B_{\mathfrak{p}}|$, and verify $N\mathfrak{p} = q = p^f$ and $1 \leq f \leq 16$.
4. Compute θ , $g_{\mathfrak{p}}$, all evaluation rows in (26.26), and all f Frobenius rows. Verify irreducibility both from the degree of θ and by the exact finite-field test

$$T^{p^f} - T \equiv 0 \pmod{g_{\mathfrak{p}}}, \quad \gcd(g_{\mathfrak{p}}, T^{p^{f/\ell}} - T) = 1 \quad (\ell \text{ prime}, \ell \mid f).$$

5. Serialize integers, HNF columns, residues, basis coordinates, polynomial coefficients, evaluation rows, and Frobenius rows in their stated orders; reject missing, duplicate, or foreign rows.

Theorem 24.2.2 (Certification correctness and termination). *The construction terminates and passes exactly when the selected row defines the finite residue field of its recorded nonzero prime ideal. On passage it returns a correct rational prime, residue degree, cardinality, polynomial presentation, evaluation isomorphism, and Frobenius action. Equal canonical inputs produce byte-identical records.*

Proof. Every scan is over the finite quotient table or a finite coefficient set. Statement 24.1 proves the input quotient is a field. The structure theorem proves the characteristic and dimension identities; the fixed Gaussian elimination proves (26.23); the primitive-element argument proves that the scan in (26.24) succeeds; and the minimal-polynomial argument proves (26.26). Fixed orders give deterministic independent verification. $\square$

On passage the completed record is

$$\begin{aligned} &(\mathbf{h}_{\mathcal{O}_E}, H_{\mathbf{p}}, p, f, q, B_{\mathbf{p}}, \theta, g_{\mathbf{p}}, \\ &\text{addition_rows, multiplication_rows, evaluation_rows, frobenius_rows,} \\ &\text{COMPLETE, NONE, SATISFIED, } \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}}). \end{aligned} \quad (26.27)$$

All displayed payload fields are present on passage. The terminal classifications are

$$\text{INVALID_INPUT, NONPRIME_MODULUS, SATISFIED.} \quad (26.28)$$

Malformed common data, failed parent independent verification, or inconsistent tables give INVALID_INPUT. If a well-formed selected Statement 24.1 row is not in the prime class, its terminal classification is

$$\text{NONPRIME_MODULUS.}$$

Either rejection returns only

$$\text{REJECT(classification, stage, first_failure, } \mathbf{h}_{\text{input}}) \quad (26.29)$$

and contains none of the passage payload fields.

Minimum tests. In $\mathbb{Z}[i]$, the ramified prime $(1 + i)$ gives $p = 2, f = 1, q = 2$. The split prime $(2 + i)$ gives $p = 5, f = 1, q = 5$. The inert prime (3) gives $p = 3, f = 2, q = 9$ and admits the model $\mathbb{F}_3[T]/(T^2 + 1)$. The nonprime ideal (2) , a false characteristic, wrong dimension, reducible polynomial, incomplete evaluation table, short Frobenius orbit, nonmaximal parent, and stale hash exercise every rejection branch. Characteristic 2 and degree 1 are valid boundaries; a cardinality without the prime HNF or polynomial/evaluation presentation is never an exact output.

24.3 Good reduction relative to a finite certificate

Good reduction is always relative to a finite list of equations and inequalities. It is not a claim that reduction is faithful on the infinite group, and it is not a claim that the reduced image equals an ambient matrix group.

Denominator and nonvanishing ideals. For a finite set $S \subset E$, recall the nonzero denominator ideal

$$\mathfrak{d}(S) = \{a \in \mathcal{O}_E : aS \subseteq \mathcal{O}_E\}. \quad (26.30)$$

Define its integral numerator ideal by

$$\mathfrak{n}(S) := \sum_{s \in S} s \mathfrak{d}(S) \subseteq \mathcal{O}_E. \quad (26.31)$$

If S contains a nonzero element, then $\mathfrak{n}(S) \neq 0$. For a nonzero prime $\mathfrak{p}$ satisfying $\mathfrak{d}(S) \not\subseteq \mathfrak{p}$, all members of S are integral at $\mathfrak{p}$, and

$$\bar{s} = 0 \text{ for every } s \in S \iff \mathfrak{n}(S) \subseteq \mathfrak{p}. \quad (26.32)$$

Indeed $\mathfrak{d}(S)\mathcal{O}_{E,\mathfrak{p}} = \mathcal{O}_{E,\mathfrak{p}}$, so after localization (26.31) is the ideal generated by S ; reduction kills every generator exactly when that ideal lies in the localized maximal ideal. Contraction gives (26.32).

Let $\mathcal{A}$ be the finite list of all matrix entries which must be reducible, including entries of inverses or a conjugator when the chosen model uses them, and put

$$\mathfrak{D} := \mathfrak{d}(\mathcal{A}). \quad (26.33)$$

For the globally integral Statement 23.5 model, $\mathfrak{D} = \mathcal{O}_E$. Let $\mathcal{V} = (S_1, \dots, S_t)$ be the ordered list of nonvanishing packets:

- a singleton $\{\det A\}$ for each matrix whose reduction must remain invertible, unless its certified determinant is the unit 1;
- the four entries of $A - B$ for every required linear distinction, including $A \neq I$ by taking $B = I$;
- both four-entry packets from $A - B$ and $A + B$ for every required projective distinction $[A] \neq [B]$;
- the seven Statement 23.6 projective orbit packets defining $\mathfrak{K}_{C_8}$;
- every selected Statement 23.8 even-word packet required to survive.

Every packet must contain a characteristic-zero nonzero entry. Its zero/nonzero check and source hash are part of the input certificate. For the integral packets of Statement 23.4, $\mathfrak{n}(S_j)$ is exactly the corresponding entry ideal.

Definition 24.3.1 (Good-reduction obstruction). The obstruction ideal attached to the finite datum is

$$\mathfrak{B}_{\text{red}} := \mathfrak{D} \prod_{j=1}^t \mathfrak{n}(S_j). \quad (26.34)$$

A nonzero prime ideal $\mathfrak{p}$ is *certifiably good* for that datum when

$$\mathfrak{B}_{\text{red}} \not\subseteq \mathfrak{p}. \quad (26.35)$$

The empty packet product is $\mathcal{O}_E$. A zero packet is not silently deleted: it means the requested characteristic-zero inequality is false and the datum is globally unsatisfiable.

Because $\mathfrak{p}$ is prime, (26.35) is equivalent to

$$\mathfrak{D} \not\subseteq \mathfrak{p} \quad \text{and} \quad \mathfrak{n}(S_j) \not\subseteq \mathfrak{p} \text{ for every } j. \quad (26.36)$$

Thus the definition is independent of a chosen scalar denominator.

Theorem 24.3.2 (Preservation at good primes). *At every prime satisfying (26.35), all requested entries reduce unambiguously, every requested determinant remains nonzero, every listed linear or projective shell distinction survives, and the Statement 23.6 conjugation orbit has length eight. Hence the induced automorphism on the reduced image has exact order eight. Every selected Statement 23.8 danger word also has its certified survival property.*

Proof. The factor $\mathfrak{D}$ and the local denominator theorem of Statement 23.3 define entrywise reduction. Formula (26.32) applied to determinant singletons proves invertibility. Applied to the entries of $A - B$, it proves linear distinction. In the projective case both $A - B$ and $A + B$ retain a nonzero entry, so the reductions are neither equal nor opposite; this is precisely the Statement 23.4 projective entry-ideal test, including characteristic 2.

The seven orbit factors multiply to $\mathfrak{K}_{C_8}$. Condition (26.36) therefore preserves all seven inequalities between the base point and its nontrivial iterates, so Statement 23.6 gives exact order eight. The same argument with the Statement 23.8 linear or projective packet proves the danger-word claims. $\square$

Theorem 24.3.3 (Only finitely many excluded primes). *If every requested characteristic-zero inequality is true, then $\mathfrak{B}_{\text{red}} \neq (0)$, and the set*

$$\text{Bad}(\mathfrak{B}_{\text{red}}) := \{\mathfrak{p} \neq (0) : \mathfrak{B}_{\text{red}} \subseteq \mathfrak{p}\} \quad (26.37)$$

is finite. It is exactly the set of prime-ideal divisors occurring in the factorization

$$\mathfrak{B}_{\text{red}} = \prod_{\nu=1}^u \mathfrak{q}_{\nu}^{e_{\nu}}, \quad e_{\nu} > 0. \quad (26.38)$$

Consequently infinitely many prime ideals are certifiably good.

Proof. All factors of (26.34) are nonzero ideals in the domain $\mathcal{O}_E$, so their product is nonzero. The ring of integers is Dedekind, and every nonzero ideal has the finite unique factorization (26.38). A prime contains the product exactly when it contains one factor, which gives (26.37). Finally, every rational prime has a prime-ideal divisor of $p\mathcal{O}_E$, while the finite set in (26.37) lies over only finitely many rational primes. Rational primes outside that finite set supply infinitely many good prime ideals. $\square$

Membership in (26.37) is an exclusion from this uniform certificate, not proof that an advertised property actually fails. In particular, if $\mathfrak{p}$ contains one orbit factor, the reduced action must be tested directly; it may still have order eight for a different witness. Conversely, a good prime need not make the image equal to $\text{SL}_2(\kappa(\mathfrak{p}))$ or $\text{PSL}_2(\kappa(\mathfrak{p}))$. Ambient generation is a separate finite group computation.

Construction and certificate. The typed input consists of the maximal C25 parent, the Statement 23.3 denominator record, the faithful integral Statement 23.5 representation, the Statement 23.6 action/orbit ideals, optional Statement 23.8 survival requests, and a finite canonical list of Statement 24.2 prime-field records. Execute:

1. Replay all field, ring, matrix, determinant, inverse, covariance, word, parity, prime, residue, and source hashes. Reconstruct $\mathcal{A}$, its exact denominator ideal, and the ordered packet list. Reject duplicates or foreign packets.
2. For each packet, certify its characteristic-zero nonvanishing and compute its HNF numerator ideal by (26.31). A least zero packet returns `UNSATISFIABLE_REQUIREMENT`.
3. Form (26.34), factor it into canonical prime HNFs, and verify the product, exponents, norms, and increasing factor order.
4. For every requested prime, compute the complete factor containment bit vector. Classify it as `GOOD` when all bits are zero and otherwise as `EXCLUDED`, storing the least failing factor and its denominator, determinant, shell, orbit, or danger kind.
5. Serialize in parent, packet, matrix-entry, word, ideal, prime, factor, request, witness, and byte order. Reject missing, duplicate, or foreign rows.

Theorem 24.3.4 (Certification correctness and termination). *The construction terminates. On passage, its ideal factorization is a complete finite exclusion list for the stated sufficient good-reduction certificate, and every requested prime has the correct `GOOD` or `EXCLUDED` class. Equal canonical inputs produce byte-identical records.*

Proof. Denominator, HNF, ideal product, containment, and number-field ideal factorization algorithms terminate on finite exact data. Equations (26.32) and (26.36) prove each condition bit; unique ideal factorization proves completeness of the exclusion list. Fixed orders prove deterministic independent verification. $\square$

On item-level passage the completed record is

$$\begin{aligned} &(\mathbf{h}_{\mathcal{O}_E}, \mathbf{h}_{\rho_E}, \mathbf{h}_{\Phi_S}, \mathfrak{D}, \mathcal{V}, (\mathbf{n}(S_j))_{j=1}^t, \\ &\mathfrak{B}_{\text{red}}, ((H_{\mathbf{q}_\nu}, e_\nu, N\mathbf{q}_\nu))_{\nu=1}^u, (\text{containment_bits, class, witness or NONE})_{\mathfrak{p}}, \\ &\text{COMPLETE, NONE, SATISFIED, } \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}}). \end{aligned} \quad (26.39)$$

Both per-prime classes are compatible with item-level SATISFIED. A good row has no witness. An excluded row has the complete bit vector and least failing factor, but makes no claim that the corresponding reduced property is false. An empty request list still returns the obstruction ideal and its complete factorization.

The item-level terminal classifications are

$$\begin{aligned} &\text{INVALID_INPUT,} \quad \text{NONMAXIMAL_PARENT,} \\ &\text{UNSATISFIABLE_REQUIREMENT,} \quad \text{SATISFIED.} \end{aligned} \quad (26.40)$$

An early rejection returns only

$$\text{REJECT}(\text{classification, stage, first_failure, } \mathbf{h}_{\text{input}}) \quad (26.41)$$

and contains none of the obstruction or per-prime fields.

Minimum tests. For the integral determinant-one Bolza matrices, the common denominator and determinant factors are the unit ideal. The scalar set $\{1/2\}$ excludes exactly primes above 2 from local integrality; the singleton $\{3\}$ excludes primes above 3 from a required nonvanishing test. A shell difference divisible by $(1+i) \subset \mathbb{Z}[i]$ tests exact containment. Identical matrices in a required shell distinction produce a zero packet and the global-unsatisfiable classification. A characteristic-2 orbit collapse tests the order factor. Finally, a prime dividing the recorded obstruction but passing an alternative direct orbit test must remain EXCLUDED, not be mislabeled as actual bad reduction, and a proper reduced image must not be mislabeled as an ambient-generation failure of this item.

24.4 Entrywise reduction of the specified generators

Fix a Statement 24.2 residue-field record for a nonzero prime $\mathfrak{p}$, and require its Statement 24.3 row to be GOOD for the finite matrix datum. Although the preferred Statement 23.5 matrices are globally integral, the correct common domain for an allowed conjugate model is the local ring $\mathcal{O}_{E,\mathfrak{p}}$.

The local residue map. For $a, s \in \mathcal{O}_E$ with $s \notin \mathfrak{p}$, define

$$r_{\mathfrak{p}} : \mathcal{O}_{E,\mathfrak{p}} \longrightarrow \kappa(\mathfrak{p}), \quad r_{\mathfrak{p}}(a/s) = \bar{a} \bar{s}^{-1}. \quad (26.42)$$

This is a well-defined surjective ring homomorphism and

$$\ker r_{\mathfrak{p}} = \mathfrak{p}\mathcal{O}_{E,\mathfrak{p}}. \quad (26.43)$$

Indeed, equality $a/s = b/t$ gives $u(at - bs) = 0$ for some $u \notin \mathfrak{p}$. Since $\mathcal{O}_E$ is a domain, $at = bs$, and reduction followed by multiplication by $(\bar{s}\bar{t})^{-1}$ proves well-definedness. The ring laws are immediate from fraction arithmetic, constants prove surjectivity, and $\bar{a}\bar{s}^{-1} = 0$ exactly when $a \in \mathfrak{p}$, proving the kernel statement.

Apply $r_{\mathfrak{p}}$ entrywise. The resulting map satisfies

$$\begin{aligned} R_{\mathfrak{p}}(A + B) &= R_{\mathfrak{p}}(A) + R_{\mathfrak{p}}(B), \\ R_{\mathfrak{p}}(AB) &= R_{\mathfrak{p}}(A)R_{\mathfrak{p}}(B), \quad R_{\mathfrak{p}}(I) = I \end{aligned} \quad (26.44)$$

for matrices of compatible sizes over $\mathcal{O}_{E,\mathfrak{p}}$. Each identity follows entry by entry because every matrix entry is a finite sum of products.

Let

$$\mathcal{G} = (G_0, \dots, G_7), \quad G_j \in \mathrm{SL}_2(\mathcal{O}_{E,\mathfrak{p}}), \quad \overline{G}_j := R_{\mathfrak{p}}(G_j). \quad (26.45)$$

Since the determinant is an integral polynomial in the entries,

$$\det \overline{G}_j = r_{\mathfrak{p}}(\det G_j) = 1, \quad \overline{G}_j \in \mathrm{SL}_2(\kappa(\mathfrak{p})). \quad (26.46)$$

Reducing the two identities $G_j G_j^{-1} = G_j^{-1} G_j = I$ gives

$$R_{\mathfrak{p}}(G_j^{-1}) = \overline{G}_j^{-1}. \quad (26.47)$$

Thus no division by a possibly vanishing reduced determinant is introduced after reduction.

Theorem 24.4.1 (Compatibility with free-word evaluation). *Let $F(\mathbf{x})$ be the free group on the ordered physical letters and let $\mathrm{ev}_{\mathcal{G}} : F(\mathbf{x}) \rightarrow \mathrm{SL}_2(\mathcal{O}_{E,\mathfrak{p}})$ be the generator assignment $x_j \mapsto G_j$. For every word w ,*

$$R_{\mathfrak{p}}(\mathrm{ev}_{\mathcal{G}}(w)) = \mathrm{ev}_{\overline{\mathcal{G}}}(w). \quad (26.48)$$

Hence entrywise reduction respects products, inverse letters, and free cancellation.

Proof. Induct on word length, using (26.44) for a positive final letter and (26.47) for a negative final letter. The empty word maps to the identity on both sides. Cancellation is preserved because a reduced matrix and its recorded inverse multiply to I . $\square$

If the Statement 23.6 normalizing matrix C is included among the certified local matrices, it may also be reduced. Every already verified matrix equality, such as $CG_j C^{-1} = G_{j+1}$, then remains an equality after applying $R_{\mathfrak{p}}$. This observation does not prove that the reduced conjugation action has exact order eight; that conclusion needs the orbit certificate and belongs to Statement 24.7. Likewise, (26.48) is still a statement about a free-group assignment. Its descent through the surface relator is the separate theorem Statement 24.5.

Canonical local computation. The Statement 23.3 record supplies the common denominator ideal $\mathfrak{D} \not\subseteq \mathfrak{p}$. In its ordered canonical HNF basis, select the least-indexed basis element outside $\mathfrak{p}$ and call it $c_{\mathfrak{p}}$. Such an element exists, since otherwise the whole basis, and hence $\mathfrak{D}$, would lie in $\mathfrak{p}$. Write every required entry α as $a_{\alpha}/c_{\mathfrak{p}}$ with $a_{\alpha} = c_{\mathfrak{p}}\alpha \in \mathcal{O}_E$, and compute

$$\overline{\alpha} = \overline{a_{\alpha}} \overline{c_{\mathfrak{p}}}^{-1}. \quad (26.49)$$

The definition (26.42) proves independence from this computational choice. For the integral model take $c_{\mathfrak{p}} = 1$.

Construction and certificate. The typed input consists of the maximal C25 parent, the Statement 23.3 denominator and local-integrality rows, the Statement 23.5 ordered determinant-one generator matrices and inverses, a completed Statement 24.1 record with one selected candidate row, a Statement 24.2 residue-field presentation or **NONE**, a Statement 24.3 prime-classification row or **NONE**, and an optional finite canonical test-word list. The two conditional payloads must be present exactly when the selected Statement 24.1 row has class **NONZERO_PRIME**. Execute:

1. Replay only the unconditional parent, integral-basis, multiplication, denominator, matrix, determinant, inverse, selected Statement 24.1 candidate, word-source, and hash rows. Do not read either conditional payload at this stage. Require exactly eight physical generators in their specified order.

2. Validate the selected Statement 24.1 row before reading either conditional payload. A well-formed row outside `NONZERO_PRIME` returns `NONPRIME_MODULUS`. On the prime branch, require and independent verification the Statement 24.2 prime, field, and polynomial payload, matching its HNF, and then require and independent verification the Statement 24.3 row for this exact matrix packet. A valid `EXCLUDED` row returns `PRIME_NOT_ESTABLISHED_GOOD`; a valid `GOOD` row continues. Missing, premature, or foreign conditional payloads are invalid.
3. Construct c_p , its residue inverse, every numerator a_α , and every reduced entry by (26.49). Compare direct reductions made with every stored alternative local denominator.
4. Assemble the eight matrices $\overline{G}_j$; verify all determinant, inverse, and pairwise product rows using the complete residue-field tables.
5. Evaluate every optional test word on both sides of (26.48) in fixed left-to-right order and compare all intermediate rows.
6. Serialize in field, prime, denominator, generator, matrix-entry, inverse, product, word, failure, and byte order; reject missing, duplicate, or foreign rows.

Theorem 24.4.2 (Construction correctness and termination). *The construction terminates. On passage it returns exactly the entrywise reductions of the specified generators in $\text{SL}_2(\kappa(\mathfrak{p}))$, their correct inverses and product table, and the correct reduction of every requested free word. Equal canonical inputs produce byte-identical records.*

Proof. All HNF, residue, matrix, and word computations range over finite stored data. The good-prime row guarantees the required local denominator. Equations (26.42)–(26.49) prove every returned algebraic row, and the fixed scan and serialization orders prove deterministic independent verification. $\square$

On passage the completed record is

$$\begin{aligned} & (h_{\mathcal{O}_E}, H_p, p, f, h_{\kappa(\mathfrak{p})}, c_p, \overline{c_p}^{-1}, \\ & ((G_j, \text{entry_numerators}, \overline{G}_j, \det \overline{G}_j, \overline{G}_j^{-1})_{j=0}^7, \text{product_rows}, \text{word_rows}, \\ & \text{COMPLETE, NONE, SATISFIED}, h_{\text{input}}, h_{\text{independentverification}}). \end{aligned} \quad (26.50)$$

Every displayed payload field is present on passage. The item-level terminal classifications are

$$\begin{aligned} & \text{INVALID_INPUT}, \quad \text{NONMAXIMAL_PARENT}, \quad \text{NONPRIME_MODULUS}, \\ & \text{PRIME_NOT_ESTABLISHED_GOOD}, \quad \text{SATISFIED}. \end{aligned} \quad (26.51)$$

An early rejection returns only

$$\text{REJECT}(\text{classification}, \text{stage}, \text{first_failure}, h_{\text{input}}) \quad (26.52)$$

and contains no reduced-matrix fields.

Minimum tests. An integral determinant-one generator tests $c_p = 1$. The scalar entry $1/2$ at a prime above 3 tests $\bar{1} \bar{2}^{-1}$, while a prime above 2 must be rejected by the good-prime condition. A nontrivial product and an inverse letter test (26.44), (26.47), and (26.48). A false determinant, mismatched inverse, a well-formed nonprime Statement 24.1 row, wrong field table, excluded prime, duplicate generator, nonmaximal parent, and stale hash exercise all structural failures. In characteristic 2, opposite matrices may coalesce, but each determinant-one reduction remains valid; no projective distinction, surface-relation descent, exact action order, or ambient-generation claim is produced by this item.

24.5 Descent of the surface relation

Retain the concrete presentation fixed in Statement 23.5,

$$\Gamma_B = \langle a_1, b_1, a_2, b_2 \mid [a_1, b_1][a_2, b_2] = 1 \rangle, \quad (26.53)$$

with commutator convention $[x, y] = xyx^{-1}y^{-1}$. Fix a successful Statement 24.4 generator-reduction record at $\mathfrak{p}$.

The reduced standard marking. The Statement 5.1 dictionary expresses the four presentation lifts in terms of the physical shell. Apply the same words to the reduced shell:

$$\begin{aligned} \overline{A}_1 &= \overline{G}_0, & \overline{B}_1 &= \overline{G}_1^{-1}, \\ \overline{A}_2 &= \overline{G}_1^{-1}\overline{G}_0\overline{G}_2, & \overline{B}_2 &= \overline{G}_3^{-1}\overline{G}_0^{-1}\overline{G}_1. \end{aligned} \quad (26.54)$$

The free-word compatibility of Statement 24.4 gives, entry by entry,

$$\overline{A}_i = R_{\mathfrak{p}}(A_i), \quad \overline{B}_i = R_{\mathfrak{p}}(B_i) \quad (i = 1, 2). \quad (26.55)$$

Thus the direct reduction of the standard tuple and its reconstruction from the physical tuple are required to agree, not merely to be conjugate or projectively equal.

Theorem 24.5.1 (Reduction of the surface relation). *The four matrices in (26.54) satisfy*

$$[\overline{A}_1, \overline{B}_1][\overline{A}_2, \overline{B}_2] = I \quad \text{in } \mathrm{SL}_2(\kappa(\mathfrak{p})). \quad (26.56)$$

Consequently there is a unique homomorphism

$$\rho_{\mathfrak{p}} : \Gamma_B \longrightarrow \mathrm{SL}_2(\kappa(\mathfrak{p})), \quad (a_1, b_1, a_2, b_2) \longmapsto (\overline{A}_1, \overline{B}_1, \overline{A}_2, \overline{B}_2). \quad (26.57)$$

Proof. Statement 23.5 proves the exact linear identity $[A_1, B_1][A_2, B_2] = I$. Statement 24.4 commutes with multiplication and inversion, so

$$\begin{aligned} [\overline{A}_1, \overline{B}_1][\overline{A}_2, \overline{B}_2] &= R_{\mathfrak{p}}([A_1, B_1][A_2, B_2]) \\ &= R_{\mathfrak{p}}(I) = I, \end{aligned} \quad (26.58)$$

which proves (26.56). The assignment on the four generators first defines a unique homomorphism from their free group. Equation (26.56) puts the defining relator, and hence its normal closure, in the kernel. The universal property of the quotient presentation (26.53) gives (26.57), with uniqueness because the four generator images are fixed. $\square$

Equivalently, for the free-group evaluation map $\tilde{\rho}_{\mathfrak{p}}$ and quotient map $\eta : F(a_1, b_1, a_2, b_2) \twoheadrightarrow \Gamma_B$, one has

$$\tilde{\rho}_{\mathfrak{p}} = \rho_{\mathfrak{p}} \circ \eta, \quad \langle\langle [a_1, b_1][a_2, b_2] \rangle\rangle \subseteq \ker \tilde{\rho}_{\mathfrak{p}}. \quad (26.59)$$

This is the precise well-definedness certificate for evaluating any representative word of an element of Γ_B .

Exact scope. Killing the one defining relator proves existence of the homomorphism; it proves neither injectivity nor surjectivity onto the ambient matrix group. The all-identity four-tuple already satisfies (26.56). The image subgroup, its enumeration, and the index of the kernel are Statement 24.6. The reduced C_8 -action is Statement 24.7.

The exact identity in Statement 23.5 is stronger than a projective relation equal only to $\pm I$. In odd residue characteristic, a characteristic-zero value $-I$ would remain nonidentity and would not support (26.57). In characteristic 2, its reduction would equal I , so an arbitrary tuple could happen to descend at that one prime. The specified construction never uses that accidental collapse as a replacement for the exact parent relation. For any arbitrary four-tuple over a residue field, the necessary and sufficient descent test is still direct equality in (26.56).

Construction and certificate. The typed input consists of the fixed Statement 23.5 presentation and standard-lift record, the Statement 5.1 physical-to-standard word dictionary, and a successful Statement 24.4 record with its parent, prime, field, denominator, and generator hashes. An optional finite canonical list of surface-group words may be supplied for evaluation. Execute:

1. Replay all field, ring, presentation, lift-sign, word-dictionary, matrix, determinant, inverse, prime, residue, local-denominator, good-prime, Statement 24.4, source, and hash rows. Require the exact specified genus-two relator and generator order.
2. Construct the four matrices by (26.54), and independently reduce all four Statement 23.5 standard matrices. Compare the eight matrices entry by entry as in (26.55).
3. Compute both commutators in the fixed order $A_i B_i A_i^{-1} B_i^{-1}$, recording every intermediate product, then compute their product and compare it with the identity matrix.
4. Require both the direct characteristic-zero reduction (26.58) and the residue-field multiplication route to give the same identity. A disagreement is invalid input with the first differing matrix-entry row.
5. Store (26.57) by its marked generator tuple, presentation relator, and deterministic word-evaluator. Evaluate every optional test word by the Statement 24.4 left-to-right convention.
6. Serialize in presentation, generator, dictionary, matrix-entry, commutator, intermediate-product, word, failure, and byte order; reject missing, duplicate, or foreign rows.

Theorem 24.5.2 (Certification correctness and termination). *The construction terminates. On passage its marked tuple defines exactly the unique homomorphism (26.57), and every requested word row is its correct value. Equal canonical inputs produce byte-identical records.*

Proof. All matrix and word lists are finite. Statement 24.4 proves the two construction routes agree; (26.58) proves the relator row; and the presentation universal property proves existence, uniqueness, and well-definedness. Fixed multiplication, first-failure, and serialization orders prove deterministic independent verification. $\square$

On passage the completed record is

$$(\mathbf{h}_{\text{presentation}}, H_{\mathbf{p}}, \mathbf{h}_{\kappa(\mathbf{p})}, (\overline{A}_1, \overline{B}_1, \overline{A}_2, \overline{B}_2), \\ \text{dictionary_comparison_rows}, \text{commutator_rows}, \text{relator_row}, \rho_{\mathbf{p}}, \text{word_rows}, \\ \text{COMPLETE}, \text{NONE}, \text{SATISFIED}, \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}}).$$
(26.60)

All payload fields are present on passage. Since Statement 24.4 has already closed the prime and good-reduction branches, the terminal classifications here are

$$\text{INVALID_INPUT}, \quad \text{SATISFIED}.$$
(26.61)

An early rejection returns only

$$\text{REJECT}(\text{classification}, \text{stage}, \text{first_failure}, \mathbf{h}_{\text{input}})$$
(26.62)

and contains no reduced-relator or homomorphism fields.

Minimum tests. The specified Bolza tuple must reproduce both sides of (26.58). The all-identity tuple satisfies the relator and demonstrates that descent does not prove generation or faithfulness. A tuple with relator $-I$ must fail in odd characteristic and may pass only after direct equality testing in characteristic 2. Reversing the commutator convention, changing one inverse, changing one physical-to-standard dictionary word, duplicating an intermediate row, supplying a foreign field, or changing a parent hash must return the first precise invalid-input witness.

24.6 The finite matrix image and congruence kernel

Let $\rho_{\mathfrak{p}}$ be the homomorphism certified in Statement 24.5 and write $k = \kappa(\mathfrak{p})$, $q = |k| = p^f$. Define the actual linear image

$$Q_{\mathfrak{p}} := \text{im } \rho_{\mathfrak{p}} = \langle \overline{A}_1, \overline{B}_1, \overline{A}_2, \overline{B}_2 \rangle \leq \text{SL}_2(k). \quad (26.63)$$

The equality with the generated subgroup follows because the four marked elements generate $\Gamma_{\mathfrak{B}}$. It does not say that $Q_{\mathfrak{p}}$ is the full ambient group.

Lemma 24.6.1 (Ambient cardinality). *For a field with q elements,*

$$|\text{SL}_2(k)| = q(q^2 - 1). \quad (26.64)$$

Proof. The first column of an invertible 2-by-2 matrix has $q^2 - 1$ choices and its second column has $q^2 - q$ choices. Thus $|\text{GL}_2(k)| = (q^2 - 1)(q^2 - q)$. The determinant map onto $k^\times$ has fibers of equal size and $|k^\times| = q - 1$. Its identity fiber is $\text{SL}_2(k)$, giving (26.64). $\square$

Deterministic image enumeration. Let the signed alphabet be

$$\mathcal{X} = (a_1, b_1, a_2, b_2, a_1^{-1}, b_1^{-1}, a_2^{-1}, b_2^{-1}) \quad (26.65)$$

in that order. Start a queue with I , attach the empty word, and right-multiply each dequeued matrix by the eight marked images in order. A matrix not previously present in canonical entry bytes is appended to the queue together with its first discovery word. When the queue empties, its carrier is exactly $Q_{\mathfrak{p}}$. Indeed every discovered matrix is a word in the marked images, while induction on word length shows that every such word is eventually discovered. Termination follows from (26.64).

After enumeration, compute every pair product and inverse in the carrier and require closure, associativity by direct table rows, a two-sided identity, and two-sided inverses. Matrix multiplication already proves these laws; the exhaustive table independently detects serialization or lookup errors. Lagrange's theorem gives the useful check

$$|Q_{\mathfrak{p}}| \mid q(q^2 - 1). \quad (26.66)$$

Theorem 24.6.2 (Finite quotient and kernel index). *Put*

$$K_{\mathfrak{p}} := \ker \rho_{\mathfrak{p}} = \{\gamma \in \Gamma_{\mathfrak{B}} : \rho_{\mathfrak{p}}(\gamma) = I\}. \quad (26.67)$$

Corestriction gives a surjection

$$\rho_{\mathfrak{p}}^Q : \Gamma_{\mathfrak{B}} \twoheadrightarrow Q_{\mathfrak{p}} \quad (26.68)$$

with kernel $K_{\mathfrak{p}}$. Hence

$$\Gamma_{\mathfrak{B}}/K_{\mathfrak{p}} \cong Q_{\mathfrak{p}}, \quad [\Gamma_{\mathfrak{B}} : K_{\mathfrak{p}}] = |Q_{\mathfrak{p}}| < \infty. \quad (26.69)$$

In particular $K_{\mathfrak{p}}$ is normal and has finite index.

Proof. The image is finite because it is a subgroup of the finite group in (26.64). Corestriction is surjective by the definition of image and does not change the kernel. The first isomorphism theorem proves the group isomorphism in (26.69), whose coset cardinality proves the index identity. Every homomorphism kernel is normal. $\square$

The quotient table gives an exact membership procedure for the infinite kernel:

$$w \in K_{\mathfrak{p}} \iff \text{ev}_{Q_{\mathfrak{p}}}(w) = I. \quad (26.70)$$

This tests a supplied word; it does not enumerate every kernel element.

A finite kernel generating certificate. For each $q_0 \in Q_p$, let t_{q_0} be the first BFS word representing q_0 . First-discovery words and all their prefixes form a Schreier transversal. For every q_0 and every positive presentation generator x , record

$$s(q_0, x) = t_{q_0} x t_{q_0 \rho_p(x)}^{-1} \in K_p. \quad (26.71)$$

Schreier rewriting proves

$$K_p = \langle s(q_0, x) : q_0 \in Q_p, x \in \{a_1, b_1, a_2, b_2\} \rangle. \quad (26.72)$$

Identity generators may be deleted only after their word values have been recorded. Thus the kernel is represented both by its exact membership evaluator and by a finite generating list.

Projective companion. If a projective image is requested, set

$$Z_p := Q_p \cap \{\lambda I : \lambda \in k^\times, \lambda^2 = 1\}. \quad (26.73)$$

This is $\{I\}$ in characteristic 2, and otherwise is either $\{I\}$ or $\{I, -I\}$. The projective image is the actual quotient Q_p/Z_p , with

$$|Q_p^{\text{proj}}| = |Q_p|/|Z_p|. \quad (26.74)$$

It is not replaced by $\text{PSL}_2(k)$ without a separate generation check.

Construction and certificate. The typed input is a successful Statement 24.5 marked homomorphism record, including its Statement 24.2 field and Statement 24.4 matrix parents, plus a Boolean request for the projective companion. Execute:

1. Replay all parent, prime, field, polynomial, matrix, determinant, inverse, presentation, relator, word-dictionary, good-prime, source, and hash rows.
2. Run the BFS in the exact alphabet order (26.65), storing every queue transition, discovery word, duplicate lookup, and final carrier entry.
3. Build the complete multiplication and inverse tables; compare each product with direct matrix multiplication and verify (26.66).
4. Corestrict the marked map, construct the kernel evaluator (26.70), and build every Schreier row (26.71). Verify each listed generator evaluates to I and independent verification Schreier rewriting on the defining relator and all optional test words.
5. If requested, enumerate Z_p , construct the projective cosets and their table, and verify (26.74).
6. Compare the linear carrier with the complete ambient carrier. Record `FULL_AMBIENT` on equality and `PROPER_IMAGE` otherwise.
7. Serialize in field, matrix, queue, discovery-word, group-table, kernel, Schreier, projective-coset, class, failure, and byte order; reject missing, duplicate, or foreign rows.

Theorem 24.6.3 (Certification correctness and termination). *The construction terminates. On passage it returns the exact finite image, its marked quotient table, its normal finite-index kernel, the exact index, and, when requested, the exact projective image. The ambient-generation class is necessary and sufficient. Equal canonical inputs produce byte-identical records.*

Proof. The BFS explores at most $q(q^2 - 1)$ matrices and the remaining tables are finite. Its two containments prove that the carrier is exactly (26.63). The first isomorphism theorem proves the kernel and index rows, Schreier's theorem proves (26.72), and quotienting by the explicit scalar intersection proves (26.74). Carrier equality is equivalent to full ambient generation. Fixed orders prove independent verification. $\square$

On passage the completed record is

$$\begin{aligned} & (H_{\mathfrak{p}}, p, f, q, h_{\rho_{\mathfrak{p}}}, Q_{\mathfrak{p}}, \text{discovery_words}, \text{transition_rows}, \\ & \text{multiplication_table}, \text{inverse_table}, K_{\mathfrak{p}}, \text{kernel_evaluator}, \text{schreier_rows}, \\ & Z_{\mathfrak{p}} \text{ or NONE}, Q_{\mathfrak{p}}^{\text{proj}} \text{ or NONE}, \text{ambient_class}, \\ & \text{COMPLETE}, \text{NONE}, \text{SATISFIED}, h_{\text{input}}, h_{\text{independentverification}}). \end{aligned} \quad (26.75)$$

The two ambient classes are

$$\text{FULL_AMBIENT}, \quad \text{PROPER_IMAGE}, \quad (26.76)$$

and both are certified answers compatible with item-level passage. Projective fields are all present exactly when requested. The terminal classifications are

$$\text{INVALID_INPUT}, \quad \text{SATISFIED}. \quad (26.77)$$

An early rejection returns only

$$\text{REJECT}(\text{classification}, \text{stage}, \text{first_failure}, h_{\text{input}}) \quad (26.78)$$

and contains no image or kernel fields.

Minimum tests. The all-identity marked tuple gives $Q_{\mathfrak{p}} = 1$, $K_{\mathfrak{p}} = \Gamma_{\mathbf{B}}$, and index 1, while remaining a proper ambient image. Standard generators of $\text{SL}_2(\mathbb{F}_2)$ must enumerate all six elements. A single finite-order generator tests a proper cyclic image and its Schreier rows. An odd-characteristic image containing $-I$ halves its projective order, while characteristic 2 does not. A missing queue transition, non-first discovery word, false duplicate, incorrect product or inverse, nonkernel Schreier word, improper ambient promotion, omitted optional field, foreign parent, and stale hash test every structural rejection.

24.7 The eightfold action after reduction

Fix the Statement 24.6 image $Q_{\mathfrak{p}}$ together with the same good prime, residue field, reduced physical shell, and Statement 23.6 normalizing matrix

$$C = \begin{pmatrix} \xi & 0 \\ 0 & \xi^{-1} \end{pmatrix}, \quad \overline{C} = R_{\mathfrak{p}}(C) \in \text{SL}_2(\kappa(\mathfrak{p})). \quad (26.79)$$

No denominator is needed for C , since ξ is a unit of $\mathcal{O}_E$.

Reducing the eight integral covariance identities gives, with indices modulo 8,

$$\overline{C} \overline{G}_j \overline{C}^{-1} = \overline{G}_{j+1} \quad (0 \leq j < 8). \quad (26.80)$$

The physical shell generates the same image as the four standard generators. Hence (26.80), and its inverse version, prove

$$\overline{C} Q_{\mathfrak{p}} \overline{C}^{-1} = Q_{\mathfrak{p}}. \quad (26.81)$$

Theorem 24.7.1 (Reduced C_8 -action). *Conjugation by $\overline{C}$ defines an automorphism*

$$\alpha_{\mathfrak{p}} : Q_{\mathfrak{p}} \longrightarrow Q_{\mathfrak{p}}, \quad \alpha_{\mathfrak{p}}(q) = \overline{C} q \overline{C}^{-1}. \quad (26.82)$$

It is equivariant with the geometric automorphism:

$$\rho_{\mathfrak{p}} \circ \phi_8 = \alpha_{\mathfrak{p}} \circ \rho_{\mathfrak{p}}. \quad (26.83)$$

For a Statement 24.3 good prime, its order is exactly eight.

Proof. Normalization proves that (26.82) maps Q_p to itself, and conjugation by $\overline{C}^{-1}$ is its inverse. Equation (26.83) holds on every physical generator by (26.80), hence on all of Γ_B .

The exact integral identities from Statement 23.6 reduce to

$$\overline{C}^8 = -I, \quad \overline{C}^{16} = I. \quad (26.84)$$

The first right-hand side is central, including when it equals I in characteristic 2, so $\alpha_p^8 = \text{id}$. The good-prime orbit factors give

$$\alpha_p^k(\overline{G}_0) = \overline{G}_k \neq \overline{G}_0 \quad (1 \leq k < 8). \quad (26.85)$$

No smaller positive power is therefore the identity, and

$$\text{ord}(\alpha_p) = 8. \quad (26.86)$$

□

Failure of the Statement 24.3 obstruction test would establish only that this uniform witness is unavailable. It would not prove that the reduced action has smaller order; the action table would then have to be tested directly. The present construction accepts the certified good branch and therefore may assert (26.86).

Kernel stability and quotient interpretation. Equivariance gives

$$\phi_8(K_p) = K_p. \quad (26.87)$$

Indeed, if $\rho_p(\gamma) = I$, then $\rho_p(\phi_8(\gamma)) = \alpha_p(I) = I$; applying the same argument to ϕ_8^{-1} gives equality. Under the first-isomorphism identification from Statement 24.6, the induced quotient automorphism is exactly

$$\overline{\phi}_{8,p} : \Gamma_B/K_p \longrightarrow \Gamma_B/K_p, \quad \gamma K_p \longmapsto \phi_8(\gamma) K_p, \quad (26.88)$$

and corresponds to α_p .

If the projective companion of Statement 24.6 was requested, its scalar kernel Z_p is preserved by conjugation. Therefore α_p descends to the projective image. The Statement 24.3 projective orbit packets preserve eight distinct projective orbit points, so that descended automorphism also has exact order eight.

The matrix C is not the action order. Every accepted good prime has odd residue characteristic. Indeed, in characteristic 2 the group $\kappa(\mathfrak{p})^\times$ has odd order, while $\overline{\xi}^{16} = 1$; hence $\overline{\xi} = 1$, $\overline{C} = I$, and (26.80) collapses all eight shell matrices. Such a prime is necessarily excluded by the good-orbit certificate. At an accepted odd-characteristic prime, (26.84) and $-I \neq I$ force $\overline{C}$ to have order 16, while $\text{Ad}_{\overline{C}}|_{Q_p}$ has order 8. The latter, rather than a guessed order of $\overline{C}$, is the recorded action order.

Nor is the concrete matrix subgroup $\langle Q_p, \overline{C} \rangle$ automatically the semidirect product $Q_p \rtimes C_8$: its intersection with $\langle \overline{C} \rangle$ must be computed, and central scalars may occur. The abstract semidirect product defined by the certified action is always legitimate, but no unsupported trivial-intersection claim is made here.

Construction and certificate. The typed input consists of the Statement 23.6 integral covariance and exact-order record containing C , the Statement 24.3 GOOD orbit row, the Statement 24.4 reduced shell and local residue-map data, the Statement 24.2 field table, the Statement 24.5 equivariant marked homomorphism data, and the complete Statement 24.6 image and optional projective tables. The matrix $\overline{C}$ is computed in this item, not read from the Statement 24.4 output. Execute:

1. Replay every parent, unit, matrix, determinant, inverse, covariance, prime, field, good-orbit, presentation, image, discovery-word, projective, source, and hash row. Require exact agreement of the prime and matrix packets at every level.
2. Reduce C entrywise and verify its determinant, inverse, all powers through 16, and both identities in (26.84).
3. Verify all eight rows (26.80), then conjugate every enumerated element of Q_p . Require a permutation of the carrier preserving every multiplication and inverse row.
4. Compare the conjugated value of every BFS discovery word with evaluation of its exact ϕ_8 -image. This gives a complete table check of (26.83).
5. Compose the action table in increasing positive powers. Verify the first identity power is 8, and separately independent verification all seven good-orbit inequalities (26.85).
6. Apply ϕ_8 to every Statement 24.6 Schreier generator and verify its value is I ; if projective data are present, construct and verify the induced projective action and its order.
7. Serialize in prime, matrix, covariance, image-carrier, action, power, orbit, kernel, projective, failure, and byte order; reject missing, duplicate, or foreign rows.

Theorem 24.7.2 (Certification correctness and termination). *The construction terminates. On passage it returns the exact order-eight automorphism of the finite image, complete equivariance and action tables, and an exact kernel-stability certificate. The optional projective action is present exactly with its parent table. Equal canonical inputs produce byte-identical records.*

Proof. All matrix, carrier, action, power, orbit, and Schreier lists are finite. Equations (26.80)–(26.83) prove well-definedness and equivariance; (26.84)–(26.86) prove exact order; and (26.87) proves kernel stability. Exhaustive table comparisons and fixed orders prove deterministic independent verification. $\square$

On passage the completed record is

$$(H_p, h_{Q_p}, \overline{C}, \overline{C}^{-1}, \text{power_rows}, \text{covariance_rows}, \\ \alpha_p, \text{action_table}, \text{equivariance_rows}, \text{orbit_rows}, \text{kernel_stability_rows}, \\ \text{projective_action or NONE, COMPLETE, NONE, SATISFIED}, h_{\text{input}}, h_{\text{independentverification}}).$$

(26.89)

The projective-action field is present exactly when the Statement 24.6 projective table is present. The terminal classifications are

$$\text{INVALID_INPUT}, \quad \text{SATISFIED}.$$

(26.90)

An early rejection returns only

$$\text{REJECT}(\text{classification}, \text{stage}, \text{first_failure}, h_{\text{input}})$$

(26.91)

and contains no action fields.

Minimum tests. A certified good prime must have odd characteristic and give $\text{ord}(\overline{C}) = 16$ but $\text{ord}(\alpha_p) = 8$. A characteristic-2 row must be excluded: direct reduction gives $\overline{C} = I$ and a collapsed shell, so a forged GOOD row is rejected rather than promoted to order eight. Changing one covariance entry, action-table value, power row, discovery-word image, Schreier value, optional projective row, prime parent, or hash exercises every structural rejection. Finally, a nontrivial intersection $Q_p \cap \langle \overline{C} \rangle$ must not be reported as a concrete semidirect product.

24.8 Parity descent and the canonical augmented quotient

Let $\square \in \{\text{lin}, \text{proj}\}$. For the linear kind take the Statement 24.6 corestricted map; for the projective kind take its optional projective companion:

$$\pi_{\mathfrak{p}}^{\square} : \Gamma_{\text{B}} \twoheadrightarrow Q_{\mathfrak{p}}^{\square}, \quad K_{\mathfrak{p}}^{\square} := \ker \pi_{\mathfrak{p}}^{\square}. \quad (26.92)$$

Every assertion below applies separately to each requested kind. For the unadorned notation below, take $\square = \text{lin}$.

Theorem 24.8.1 (Exact parity-descent criterion). *The fixed parity character $\chi : \Gamma_{\text{B}} \twoheadrightarrow C_2$ factors through $\pi_{\mathfrak{p}}^{\square}$ if and only if*

$$K_{\mathfrak{p}}^{\square} \subseteq K_{\chi} := \ker \chi. \quad (26.93)$$

When it exists, the factor $\epsilon_{\mathfrak{p}}^{\square} : Q_{\mathfrak{p}}^{\square} \rightarrow C_2$ is unique and surjective.

Proof. If $\chi = \epsilon_{\mathfrak{p}}^{\square} \pi_{\mathfrak{p}}^{\square}$, then every element of $K_{\mathfrak{p}}^{\square}$ has parity zero, proving necessity. Conversely define $\epsilon_{\mathfrak{p}}^{\square}(\pi_{\mathfrak{p}}^{\square}(\gamma)) = \chi(\gamma)$. If two representatives have the same image, their quotient lies in $K_{\mathfrak{p}}^{\square}$, so (26.93) proves that their parities agree. The resulting function is a homomorphism. Surjectivity of $\pi_{\mathfrak{p}}^{\square}$ proves uniqueness, and surjectivity of χ proves surjectivity of the factor. $\square$

Finite decision and obstruction. Use the ordered physical-shell Cayley graph of $Q_{\mathfrak{p}}^{\square}$. Give the identity bit zero and propagate a bit flip across every shell edge, because $\chi(g_j) = 1$. If a previously labelled endpoint receives the opposite proposed bit, the new and stored path words u, v give

$$\pi_{\mathfrak{p}}^{\square}(u) = \pi_{\mathfrak{p}}^{\square}(v), \quad \kappa = uv^{-1} \in K_{\mathfrak{p}}^{\square}, \quad \chi(\kappa) = 1. \quad (26.94)$$

This is a finite independently verifiable witness that (26.93) fails. If no conflict occurs, the complete bit table is the unique $\epsilon_{\mathfrak{p}}^{\square}$. Conversely every odd kernel word is an odd closed walk at the identity and forces such a conflict, so the procedure is necessary and sufficient.

The canonical repair when parity does not descend. Whether or not (26.93) holds, define

$$\begin{aligned} \widehat{\pi}_{\mathfrak{p}}^{\square} : \Gamma_{\text{B}} &\longrightarrow Q_{\mathfrak{p}}^{\square} \times C_2, \\ \gamma &\longmapsto (\pi_{\mathfrak{p}}^{\square}(\gamma), \chi(\gamma)), \quad \widetilde{Q}_{\mathfrak{p}}^{\square} := \text{im } \widehat{\pi}_{\mathfrak{p}}^{\square}. \end{aligned} \quad (26.95)$$

Corestriction makes this map surjective, and

$$\widehat{K}_{\mathfrak{p}}^{\square} := \ker \widehat{\pi}_{\mathfrak{p}}^{\square} = K_{\mathfrak{p}}^{\square} \cap K_{\chi}. \quad (26.96)$$

Projection to the first coordinate is onto. Its kernel is the image $\chi(K_{\mathfrak{p}}^{\square})$, which is either zero or all of C_2 . Therefore the exact dichotomy is

$$\widetilde{Q}_{\mathfrak{p}}^{\square} = \begin{cases} \{(q, \epsilon_{\mathfrak{p}}^{\square}(q)) : q \in Q_{\mathfrak{p}}^{\square}\}, & K_{\mathfrak{p}}^{\square} \subseteq K_{\chi}, \\ Q_{\mathfrak{p}}^{\square} \times C_2, & K_{\mathfrak{p}}^{\square} \not\subseteq K_{\chi}. \end{cases} \quad (26.97)$$

In particular

$$\begin{aligned} [\Gamma_{\text{B}} : \widehat{K}_{\mathfrak{p}}^{\square}] &= |\widetilde{Q}_{\mathfrak{p}}^{\square}| \\ &= \begin{cases} |Q_{\mathfrak{p}}^{\square}|, & K_{\mathfrak{p}}^{\square} \subseteq K_{\chi}, \\ 2|Q_{\mathfrak{p}}^{\square}|, & K_{\mathfrak{p}}^{\square} \not\subseteq K_{\chi}. \end{cases} \end{aligned} \quad (26.98)$$

Thus intersection preserves finite index and increases it by at most two; the factor two occurs exactly when base parity fails.

The second projection restricts to the canonical quotient parity

$$\widehat{\epsilon}_{\mathfrak{p}}^{\square} : \widetilde{Q}_{\mathfrak{p}}^{\square} \twoheadrightarrow C_2, \quad \widehat{\epsilon}_{\mathfrak{p}}^{\square} \widehat{\pi}_{\mathfrak{p}}^{\square} = \chi. \quad (26.99)$$

It is always present, including when no parity exists on the base image.

C_8 -compatibility. Let $\alpha_p^\square$ be the Statement 24.7 action on the selected image. Since Statement 9.4 proves $\chi\phi_8 = \chi$, define

$$\tilde{\alpha}_p^\square(q, b) = (\alpha_p^\square(q), b) \quad \text{on } \tilde{Q}_p^\square. \quad (26.100)$$

For every γ ,

$$\tilde{\alpha}_p^\square \hat{\pi}_p^\square(\gamma) = \hat{\pi}_p^\square(\phi_8(\gamma)), \quad \text{ord}(\tilde{\alpha}_p^\square) = 8. \quad (26.101)$$

The equality proves that the image in (26.95) is preserved. The formula makes the order divide eight, while its surjective projection onto the base intertwines with the exact-order-eight action $\alpha_p^\square$, giving the reverse divisibility. The augmented action preserves the parity projection.

Linear versus projective parity. The quotient kind cannot be discarded. In odd characteristic, an odd word reducing to $-I$ is not in the linear kernel but is in the projective kernel, so it can obstruct projective parity without obstructing linear parity. In characteristic 2 the two matrix signs coincide, but the independent C_2 bit never disappears. Although Statement 24.7 good eight-orbit primes are necessarily odd, the criterion and augmentation themselves remain valid for any finite quotient.

Construction and certificate. The typed input contains the Statement 9.1 parity character and Statement 9.4 invariance rows, the complete Statement 24.6 linear image and optional projective image, the Statement 24.7 action tables, and a duplicate-free canonical kind list. Execute:

1. Replay every presentation, physical-word, parity, invariance, prime, field, image, group-table, kernel, projective, action, source, and hash row. A projective request requires its exact optional parent; an unrequested projective payload is invalid.
2. For each kind, run the deterministic bit propagation in the already fixed carrier and shell-edge order. Store either the complete bit table or the first conflict (26.94).
3. On success, verify the proposed parity on every product-table row and every marked generator. On conflict, verify both path values, the kernel identity, and the odd bit.
4. Enumerate the actual pair image in (26.95), build its complete table and both projections, and compare it with the appropriate graph or direct product in (26.97). Verify (26.98)–(26.99).
5. Construct (26.100), verify its complete table, equivariance, exact order, and parity invariance.
6. Serialize in kind, carrier, edge, bit, conflict, pair, group-table, kernel, action, failure, and byte order; reject missing, duplicate, or foreign rows.

Theorem 24.8.2 (Certification correctness and termination). *The construction terminates. It decides parity descent exactly and returns either the unique base parity or a least odd-kernel obstruction. In both cases it returns the canonical augmented quotient with kernel intersection, exact order, parity, and order-eight action. Equal canonical inputs produce byte-identical records.*

Proof. Every carrier and table is finite. The conflict argument proves the criterion algorithm; (26.95)–(26.99) prove the kernel, image, order, and parity statements; and (26.100)–(26.101) prove symmetry compatibility. Fixed traversal and serialization orders prove independent verification. $\square$

For each requested kind, let $\mathcal{R}^\square$ denote the record

$$(\square, Q_p^\square, K_p^\square, \text{bit_table or NONE}, \text{conflict or NONE}, \\ \tilde{Q}_p^\square, \tilde{K}_p^\square, \hat{\pi}_p^\square, \hat{\epsilon}_p^\square, \tilde{\alpha}_p^\square, \\ \text{group_rows}, \text{order_rows}, \text{action_rows}, \text{parity_class}, h_{\text{independentverification}}^\square).$$

The item-level passage envelope, which is always present, is

$$(\text{kind_list}, (\mathcal{R}^\square)_{\square \in \text{kind_list}}, \\ \text{COMPLETE}, \text{NONE}, \text{SATISFIED}, h_{\text{input}}, h_{\text{independentverification}}).$$
 (26.102)

The two per-kind classes are

$$\text{PARITY_DESCENDS}, \quad \text{PARITY_AUGMENTED}.$$
 (26.103)

Both are certified answers compatible with item-level passage. The first has the complete base bit table and no conflict; the second has no base bit table and has the least conflict. All augmented fields are present in both. The item-level classifications are

$$\text{INVALID_INPUT}, \quad \text{SATISFIED}.$$
 (26.104)

An empty kind list passes after common-parent independent verification: its per-kind sequence is empty, but the complete item-level envelope is still returned. An early rejection returns only

$$\text{REJECT}(\text{classification}, \text{stage}, \text{first_failure}, h_{\text{input}})$$
 (26.105)

and contains no parity or augmented-quotient fields.

Minimum tests. For $Q = C_2$ and $\pi = \chi$, parity descends and the augmented image is its graph with unchanged order. For the trivial base quotient, parity does not descend and augmentation gives C_2 , not the trivial group. A first odd closed walk tests (26.94) and exact doubling. In odd characteristic, an odd word with linear value $-I$ tests the difference between linear and projective requests. Characteristic 2 tests that sign collapse does not erase the external parity bit. A false bit, nonleast conflict, false graph, unnecessary doubling, wrong action order, missing projective parent, duplicate kind, foreign group table, and stale hash exercise every structural rejection.

24.9 Exact quotient orders and ambient bounds

Let $q = p^f$ be the Statement 24.2 residue-field order and let the complete Statement 24.6 carrier have

$$n_p := |Q_p| = [\Gamma_B : K_p].$$
 (26.106)

This carrier count is the exact quotient order. A formula involving only q is, in general, only an ambient upper bound.

Linear order. The Statement 24.6 ambient calculation gives

$$N_{\text{lin}}(q) := |\text{SL}_2(\mathbb{F}_q)| = q(q^2 - 1) = q(q - 1)(q + 1).$$
 (26.107)

Since $Q_p \leq \text{SL}_2(\mathbb{F}_q)$, Lagrange's theorem gives

$$n_p \mid N_{\text{lin}}(q), \quad 1 \leq n_p \leq q(q^2 - 1).$$
 (26.108)

The ambient coindex is the exact positive integer

$$c_{\text{lin}} := [\text{SL}_2(\mathbb{F}_q) : Q_{\mathfrak{p}}] = \frac{q(q^2 - 1)}{n_{\mathfrak{p}}}. \quad (26.109)$$

Therefore

$$Q_{\mathfrak{p}} = \text{SL}_2(\mathbb{F}_q) \iff c_{\text{lin}} = 1. \quad (26.110)$$

The forward implication is immediate; the reverse follows because a subgroup of a finite group having the same cardinality is the whole group. Here the independently verified BFS carrier is the generation proof. Merely observing that a proposed order is a large divisor of $N_{\text{lin}}(q)$, or seeing many random matrices, proves neither the order nor equality.

For the specified good-reduction datum, the identity and the eight physical shell matrices are pairwise distinct. Hence the additional certified lower bound is

$$9 \leq n_{\mathfrak{p}} \leq q(q^2 - 1). \quad (26.111)$$

This lower bound is conditional on the full shell-nonidentity record; it is not applied to an arbitrary reduced tuple. Similarly, the existence of an automorphism of order eight implies $8 \mid |\text{Aut}(Q_{\mathfrak{p}})|$, not $8 \mid |Q_{\mathfrak{p}}|$.

Projective order. Let

$$z_{\mathfrak{p}} = |Z_{\mathfrak{p}}| \in \{1, 2\}, \quad g_q = \gcd(2, q - 1). \quad (26.112)$$

Statement 24.6 proves

$$\begin{aligned} n_{\mathfrak{p}}^{\text{proj}} &= |Q_{\mathfrak{p}}^{\text{proj}}| = n_{\mathfrak{p}}/z_{\mathfrak{p}}, \\ N_{\text{proj}}(q) &= |\text{PSL}_2(\mathbb{F}_q)| = q(q^2 - 1)/g_q, \\ n_{\mathfrak{p}}^{\text{proj}} &\mid N_{\text{proj}}(q). \end{aligned} \quad (26.113)$$

In characteristic 2, $g_q = z_{\mathfrak{p}} = 1$. In odd characteristic, projectivization divides the image order by two exactly when $-I \in Q_{\mathfrak{p}}$. The exact projective coindex is

$$c_{\text{proj}} = \frac{N_{\text{proj}}(q)}{n_{\mathfrak{p}}^{\text{proj}}}, \quad Q_{\mathfrak{p}}^{\text{proj}} = \text{PSL}_2(\mathbb{F}_q) \iff c_{\text{proj}} = 1. \quad (26.114)$$

Parity-augmented order. For either kind $\square$, let $n^{\square} = |Q_{\mathfrak{p}}^{\square}|$. The Statement 24.8 class determines the exact multiplier

$$d^{\square} = \begin{cases} 1, & \text{PARITY_DESCENDS,} \\ 2, & \text{PARITY_AUGMENTED,} \end{cases} \quad |\tilde{Q}_{\mathfrak{p}}^{\square}| = d^{\square} n^{\square}. \quad (26.115)$$

Thus

$$\begin{aligned} |\tilde{Q}_{\mathfrak{p}}^{\text{lin}}| &\mid 2q(q^2 - 1), \\ |\tilde{Q}_{\mathfrak{p}}^{\text{proj}}| &\mid \frac{2q(q^2 - 1)}{g_q}, \end{aligned} \quad (26.116)$$

but the multiplier is never guessed from residue characteristic or matrix signs. If parity descends then $Q_{\mathfrak{p}}^{\square}$ has a quotient C_2 , so its order is even. Failure to descend places no such evenness conclusion on the base image; the augmented image is nevertheless even.

If the abstract symmetry extension is explicitly requested, Statement 24.8 supplies an action of exact order eight and hence

$$\left| \tilde{Q}_{\mathfrak{p}}^{\square} \rtimes_{\tilde{\alpha}^{\square}} C_8 \right| = 8d^{\square} n^{\square}. \quad (26.117)$$

This formula concerns the abstract semidirect product. It does not identify it with the concrete subgroup generated by matrices $Q_{\mathfrak{p}}$ and $\overline{C}$, whose intersection data must be checked separately.

Exact computational certificate. The input consists of Statement 24.2 field data, the Statement 24.6 linear and optional projective carriers, Statement 24.7 action data, and the requested Statement 24.8 parity-augmented records. Execute:

1. Replay every prime, field, carrier, product, inverse, kernel, projective-center, action, parity, augmentation, source, and hash row.
2. Recompute $q = p^f$, $N_{\text{lin}}(q)$, and $N_{\text{proj}}(q)$ by exact integer arithmetic. Count the canonical carriers rather than accepting supplied order fields.
3. Divide the ambient orders by the carrier counts with quotient and remainder. Require zero remainders, verify (26.108)–(26.114), and compare full-ambient classes with coindex one.
4. Recompute $d^\square$ from the exact Statement 24.8 parity class, count the augmented carrier independently, and verify (26.115)–(26.116).
5. If the shell lower bound is asserted, independent verification all eight nonidentity and pairwise-distinction rows. If the optional symmetry extension is requested, enumerate its pair carrier and verify (26.117).
6. Factor every reported positive integer by deterministic increasing-prime trial division and verify the product. Store both the decimal integer and its canonical prime-power list.
7. Serialize in kind, integer, prime factor, carrier, coindex, parity multiplier, optional extension, failure, and byte order; reject missing, duplicate, or foreign rows.

Theorem 24.9.1 (Order-certificate correctness and termination). *The construction terminates and returns the exact linear, optional projective, parity-augmented, and optional symmetry-extension orders, together with sharp ambient divisibility and equality tests. Equal canonical inputs produce byte-identical records.*

Proof. All carriers are finite and all integer operations terminate; trial division stops at the shrinking square-root bound. Lagrange proves the base divisibilities, explicit scalar quotients prove the projective formula, and Statement 24.8 proves the parity multiplier. Direct carrier counts make every reported order exact. Fixed arithmetic and serialization orders prove independent verification. $\square$

On passage the completed record is

$$\begin{aligned}
 &(p, f, q, N_{\text{lin}}, n_p, c_{\text{lin}}, \text{linear_class}, \\
 &N_{\text{proj}} \text{ or NONE}, n_p^{\text{proj}} \text{ or NONE}, c_{\text{proj}} \text{ or NONE}, \\
 &((\square, d^\square, |\tilde{Q}_p^\square|, \text{extension_order or NONE}))_{\text{kind_list}}, \\
 &\text{factorization_rows, bound_rows, COMPLETE, NONE, SATISFIED, } h_{\text{input}}, h_{\text{independentverification}}).
 \end{aligned}
 \tag{26.118}$$

Projective fields are present exactly with the projective parent, and extension orders exactly with their request. An empty kind list is allowed; the top-level linear order and success envelope remain present. The terminal classifications are

$$\text{INVALID_INPUT, SATISFIED.} \tag{26.119}$$

An early rejection returns only

$$\text{REJECT(classification, stage, first_failure, } h_{\text{input}}) \tag{26.120}$$

and contains no order fields.

Minimum tests. The trivial image has order 1 and is proper. A full $\mathrm{SL}_2(\mathbb{F}_2)$ carrier has order 6 and coindex 1. A proper unipotent subgroup of order q tests a strict divisor. In odd characteristic, presence and absence of $-I$ test both projective divisors. Direct parity tests unchanged order and an odd-kernel conflict tests exact doubling. A proposed large divisor without its carrier, a false shell lower bound, an inference $8 \mid |Q|$ from the action alone, nonzero division remainder, wrong prime factorization, unsupported full-ambient class, unrequested optional field, foreign parent, and stale hash exercise every rejection route.

24.10 Linear and projective congruence kernels

Fix the faithful integral representation $\rho_E : \Gamma_B \rightarrow \mathrm{SL}_2(\mathcal{O}_E)$ and write $M(\gamma) = \rho_E(\gamma)$. For a nonzero prime ideal $\mathfrak{p}$, the ambient principal congruence subgroup is

$$\mathrm{SL}_2(\mathcal{O}_E; \mathfrak{p}) := \ker(\mathrm{SL}_2(\mathcal{O}_E) \longrightarrow \mathrm{SL}_2(\kappa(\mathfrak{p}))). \quad (26.121)$$

Definition 24.10.1 (Linear congruence kernel). The linear kernel relative to the fixed representation is

$$\begin{aligned} K_{\mathfrak{p}}^{\mathrm{lin}} &:= \rho_E^{-1}(\mathrm{SL}_2(\mathcal{O}_E; \mathfrak{p})) \\ &= \{\gamma \in \Gamma_B : M(\gamma) \equiv I \pmod{\mathfrak{p}}\} = \ker \rho_{\mathfrak{p}}. \end{aligned} \quad (26.122)$$

Let $\mathfrak{J}_{\gamma}^+$ and $\mathfrak{J}_{\gamma}^-$ be the four-entry ideals generated by the entries of $M(\gamma) - I$ and $M(\gamma) + I$, respectively, and set $\mathfrak{J}_{\gamma}^{\mathrm{proj}} = \mathfrak{J}_{\gamma}^+ \mathfrak{J}_{\gamma}^-$. The exact membership tests are

$$\begin{aligned} \gamma \in K_{\mathfrak{p}}^{\mathrm{lin}} &\iff \mathfrak{J}_{\gamma}^+ \subseteq \mathfrak{p}, \\ \gamma \in K_{\mathfrak{p}}^{\mathrm{proj}} &\iff \mathfrak{J}_{\gamma}^{\mathrm{proj}} \subseteq \mathfrak{p}. \end{aligned} \quad (26.123)$$

The second line refers to the following distinct kernel.

Definition 24.10.2 (Projective congruence kernel). Let $\rho_{\mathfrak{p}}^{\mathrm{proj}}$ be the composition of $\rho_{\mathfrak{p}}$ with projectivization. Then

$$\begin{aligned} K_{\mathfrak{p}}^{\mathrm{proj}} &:= \ker \rho_{\mathfrak{p}}^{\mathrm{proj}} \\ &= \{\gamma \in \Gamma_B : \overline{M(\gamma)} = \lambda I \text{ for some } \lambda \in \kappa(\mathfrak{p})^{\times}, \lambda^2 = 1\}. \end{aligned} \quad (26.124)$$

The determinant-one condition forces $\lambda^2 = 1$. In odd characteristic the only choices are 1 and -1 ; in characteristic 2 they coincide. Primality of $\mathfrak{p}$ then proves the second equivalence in (26.123): a product of the two entry ideals lies in $\mathfrak{p}$ exactly when one factor does. No arbitrary scalar is accepted.

Theorem 24.10.3 (Relation between the two kernels). *One has*

$$K_{\mathfrak{p}}^{\mathrm{lin}} \trianglelefteq K_{\mathfrak{p}}^{\mathrm{proj}} \trianglelefteq \Gamma_B. \quad (26.125)$$

With

$$Z_{\mathfrak{p}} = Q_{\mathfrak{p}} \cap \{\lambda I : \lambda^2 = 1\},$$

restriction of $\rho_{\mathfrak{p}}$ gives the short exact sequence

$$1 \longrightarrow K_{\mathfrak{p}}^{\mathrm{lin}} \longrightarrow K_{\mathfrak{p}}^{\mathrm{proj}} \longrightarrow Z_{\mathfrak{p}} \longrightarrow 1. \quad (26.126)$$

Consequently

$$\begin{aligned} [\Gamma_B : K_{\mathfrak{p}}^{\mathrm{lin}}] &= |Q_{\mathfrak{p}}|, \\ [K_{\mathfrak{p}}^{\mathrm{proj}} : K_{\mathfrak{p}}^{\mathrm{lin}}] &= |Z_{\mathfrak{p}}|, \\ [\Gamma_B : K_{\mathfrak{p}}^{\mathrm{proj}}] &= |Q_{\mathfrak{p}}|/|Z_{\mathfrak{p}}|. \end{aligned} \quad (26.127)$$

Proof. Both groups are kernels of homomorphisms from Γ_B , so they are normal, and projectivization gives the inclusion. On K_p^{proj} , the linear reduction lands in Z_p ; its kernel is exactly the linear congruence kernel. Every element of Z_p has a preimage in Γ_B because Q_p is the actual image, and any such preimage lies in the projective kernel. This proves exactness. The index formulas follow from Statement 24.6 and the finite short exact sequence. $\square$

In particular,

$$K_p^{\text{lin}} = K_p^{\text{proj}} \iff |Z_p| = 1. \quad (26.128)$$

In odd characteristic the inclusion is strict exactly when $-I \in Q_p$; in characteristic 2 the two kernels are equal. Their quotient orders, word membership, and later systolic certificates must therefore retain the kind field.

Higher linear levels. For $m \geq 1$, define

$$K_{p^m}^{\text{lin}} := \{\gamma \in \Gamma_B : M(\gamma) - I \in M_2(\mathfrak{p}^m)\}. \quad (26.129)$$

This is the kernel of reduction into $\text{SL}_2(\mathcal{O}_E/\mathfrak{p}^m)$, hence is normal and finite index, and

$$K_p^{\text{lin}} \supseteq K_{p^2}^{\text{lin}} \supseteq K_{p^3}^{\text{lin}} \supseteq \cdots. \quad (26.130)$$

The level- m projective analogue must use the actual scalar kernel over $\mathcal{O}_E/\mathfrak{p}^m$; near primes above 2, it is not defined here by blindly writing only $\pm I$.

Parity and symmetry refinements. Statement 24.8 uses the distinct intersections

$$\widetilde{K}_p^\square = K_p^\square \cap K_\chi, \quad \square \in \{\text{lin}, \text{proj}\}. \quad (26.131)$$

These are not renamed congruence kernels: they are parity-refined congruence kernels and may have twice the base index. Statement 24.7 equivariance proves that both base kernels are ϕ_8 -stable, since the linear identity and the projective scalar center are preserved by the corresponding action. Statement 9.4 then makes the two parity-refined kernels stable as well.

Construction and certificate. The typed input contains the Statement 23.5 integral representation and matrix evaluator, the Statement 24.2 prime and residue table, the complete Statement 24.6 linear image and optional projective companion, the Statement 24.7 action, the Statement 24.8 parity records, a finite canonical list of levels m , and a finite canonical test-word list. Execute:

1. Replay every representation, presentation, matrix, prime, field, image, center, kernel, action, parity, word-source, and hash row. Require projective and parity fields exactly when requested.
2. For each word, compute $M(\gamma)$ in characteristic zero, its two entry ideals, its residue matrix, and the linear and optional projective membership bits. Compare ideal containment, direct matrix equality, and Statement 24.6 table lookup.
3. If the projective parent was requested, enumerate Z_p , construct the map in (26.126), and verify its complete kernel, image, coset, exact-sequence, and projective-index rows. Otherwise require every projective field to be NONE.
4. For each requested m , test all four entries of $M(\gamma) - I$ for membership in $\mathfrak{p}^m$, verify the ideal power HNF, and compare all nested-level bits.
5. Verify ϕ_8 -stability on the Statement 24.6 Schreier generators and verify every requested parity-intersection bit.

6. Serialize in kind, level, word, matrix-entry, ideal, center, exact-sequence, index, parity, action, failure, and byte order; reject missing, duplicate, or foreign rows.

Theorem 24.10.4 (Certification correctness and termination). *The construction terminates and returns exact linear, optional projective, higher-level linear, and parity-refined kernel certificates with their distinct membership and index data. Equal canonical inputs produce byte-identical records.*

Proof. All word, level, field, center, and carrier lists are finite; HNF ideal arithmetic terminates. Equations (26.122)–(26.124) prove membership, the exact sequence proves all index rows, and kernel functoriality proves the level and stability rows. Fixed orders prove deterministic independent verification. $\square$

On passage the completed record is

$$(H_{\mathfrak{p}}, h_{\rho_E}, K_{\mathfrak{p}}^{\text{lin}}, K_{\mathfrak{p}}^{\text{proj}} \text{ or NONE}, Z_{\mathfrak{p}} \text{ or NONE}, \\ \text{linear_index_rows, exact_sequence_rows or NONE, projective_index_rows or NONE,} \\ (K_{\mathfrak{p}^m}^{\text{lin}})_{\text{level_list}}, \text{membership_rows,} \\ \text{parity_intersection_rows, stability_rows, COMPLETE, NONE, SATISFIED, } h_{\text{input}}, h_{\text{independentverification}}).$$

(26.132)

The projective kernel, center, exact-sequence rows, and projective index rows are present exactly with their parent request and are **NONE** otherwise; the linear index rows are always present. An empty word or level list is allowed, but the top-level linear kernel, linear index data, and success envelope remain present, while the projective branch still obeys its parent request. The terminal classifications are

$$\text{INVALID_INPUT, SATISFIED.} \quad (26.133)$$

An early rejection returns only

$$\text{REJECT(classification, stage, first_failure, } h_{\text{input}}) \quad (26.134)$$

and contains no kernel fields.

Minimum tests. The identity word lies in both kernels. In odd characteristic a word reducing to $-I$ lies projectively but not linearly in the kernel and makes the inclusion strict; characteristic 2 makes the two first-level kernels equal. A noncentral determinant-one residue must fail projective membership even if its trace mimics a scalar. A word congruent to I modulo $\mathfrak{p}$ but not $\mathfrak{p}^2$ tests the higher-level filtration. An odd kernel word tests the parity intersection. A false scalar, swapped kernel kind, wrong center size, broken exact-sequence map, nonnested level bit, changed lift-sign/representation hash, duplicate word, foreign field, and stale parent exercise every structural rejection.

25 Systolic bounds for congruence kernels

25.1 Congruence conditions for kernel elements

Scope and inherited data. Let E be the number field, $\mathcal{O}_E$ its certified full ring of integers, and $\rho_E : \Gamma_B \rightarrow \text{SL}_2(\mathcal{O}_E)$ the faithful integral representation fixed in C25. Let $\mathfrak{p} \subsetneq \mathcal{O}_E$ be a nonzero prime ideal accepted by C26, and let $m \in \mathbb{Z}_{\geq 1}$. This item concerns the *linear* principal congruence kernel of Statement 24.10. It does not identify that kernel with the optional projective kernel.

For any ideal $\mathfrak{a} \subseteq \mathcal{O}_E$, set

$$M_2(\mathfrak{a}) := \{(b_{ij}) \in M_2(\mathcal{O}_E) : b_{ij} \in \mathfrak{a} \text{ for all } i, j\}. \quad (27.1)$$

Entrywise reduction modulo $\mathfrak{p}^m$ gives

$$\begin{aligned}\rho_{\mathfrak{p}^m}^{\text{lin}} : \Gamma_B &\longrightarrow \text{SL}_2(\mathcal{O}_E/\mathfrak{p}^m), \\ K_{\mathfrak{p}^m}^{\text{lin}} &:= \ker \rho_{\mathfrak{p}^m}^{\text{lin}} = \{\gamma \in \Gamma_B : \rho_E(\gamma) - I \in M_2(\mathfrak{p}^m)\}.\end{aligned}\tag{27.2}$$

Thus, for $M(\gamma) = \rho_E(\gamma) = (m_{ij})$,

$$\gamma \in K_{\mathfrak{p}^m}^{\text{lin}} \iff \begin{cases} m_{11} - 1 \in \mathfrak{p}^m, \\ m_{12} \in \mathfrak{p}^m, \\ m_{21} \in \mathfrak{p}^m, \\ m_{22} - 1 \in \mathfrak{p}^m. \end{cases}\tag{27.3}$$

Every membership assertion here is exact ideal membership in the fixed integral basis; a floating-point residue is not a witness.

Defect variables. For a matrix satisfying the right side of (27.3), define

$$B_\gamma := M(\gamma) - I = (b_{ij}) \in M_2(\mathfrak{p}^m), \quad \tau_+(M(\gamma)) := \text{tr } M(\gamma) - 2.\tag{27.4}$$

The subscript $+$ records that the identity lift, rather than a negative or arbitrary scalar lift, has been selected.

Lemma 25.1.1 (Universal two-by-two determinant identity). *For every commutative ring A and every $B = (b_{ij}) \in M_2(A)$,*

$$\begin{aligned}\det(I + B) &= (1 + b_{11})(1 + b_{22}) - b_{12}b_{21} \\ &= 1 + \text{tr } B + \det B.\end{aligned}\tag{27.5}$$

Proof. The first equality is the definition of the determinant. Expanding its right side gives $1 + b_{11} + b_{22} + b_{11}b_{22} - b_{12}b_{21}$, which is the second line. No division, valuation, completion, or characteristic assumption is used. $\square$

Lemma 25.1.2 (Quadratic trace defect). *If $M = I + B \in \text{SL}_2(\mathcal{O}_E)$, then*

$$\text{tr } B = -\det B, \quad \text{tr } M - 2 = -\det B = b_{12}b_{21} - b_{11}b_{22}.\tag{27.6}$$

If in addition $B \in M_2(\mathfrak{p}^m)$, then

$$b_{11}b_{22}, b_{12}b_{21} \in \mathfrak{p}^m \mathfrak{p}^m = \mathfrak{p}^{2m}, \quad \det B \in \mathfrak{p}^{2m}.\tag{27.7}$$

Proof. Substitute $\det M = 1$ into (27.5) and cancel 1. This proves (27.6). The definition of the ideal product puts each product of two entries of $\mathfrak{p}^m$ in $\mathfrak{p}^{2m}$; closure under subtraction proves (27.7). $\square$

Theorem 25.1.3 (Kernel congruence condition). *For every $\gamma \in K_{\mathfrak{p}^m}^{\text{lin}}$,*

$$\text{tr } \rho_E(\gamma) - 2 \in \mathfrak{p}^{2m}.\tag{27.8}$$

Equivalently, the trace-defect image satisfies

$$\tau_+(K_{\mathfrak{p}^m}^{\text{lin}}) \subseteq \mathfrak{p}^{2m}, \quad \text{tr } \rho_E(\gamma) \equiv 2 \pmod{\mathfrak{p}^{2m}}.\tag{27.9}$$

Proof. Membership in the kernel gives (27.3), hence the defect matrix in (27.4) belongs to $M_2(\mathfrak{p}^m)$. The representation has determinant one. Applying (27.6) and (27.7) proves both formulations. $\square$

Scope of the bound. Equation (27.8) includes the identity element, for which the trace defect is zero. It does not say that the trace defect of a nonidentity kernel element is nonzero; that requires the torsion-free cocompact Fuchsian argument isolated in Statement 25.2. Nor does (27.8) by itself give an archimedean trace bound or a systole bound. Those require the norm and embedding hypotheses of the subsequent tasks.

Projective normalization rule. Suppose an optional projective certificate supplies an *exact* sign $\varepsilon \in \{1, -1\}$ such that $M(\gamma) - \varepsilon I \in M_2(\mathfrak{p}^m)$. Then $\varepsilon M(\gamma) \in \mathrm{SL}_2(\mathcal{O}_E)$ and the linear lemma gives

$$\varepsilon \operatorname{tr} M(\gamma) - 2 \in \mathfrak{p}^{2m}. \quad (27.10)$$

Without that exact sign certificate, a scalar class modulo $\mathfrak{p}^m$ is insufficient. Indeed, if $M = \lambda I + B$ with $B \in M_2(\mathfrak{p}^m)$, then

$$\lambda(\operatorname{tr} M - 2\lambda) = 1 - \lambda^2 - \det B. \quad (27.11)$$

The term $1 - \lambda^2$ need not lie in $\mathfrak{p}^{2m}$, especially in the higher-level situations over primes above 2 already separated in Statement 24.10. Consequently the projective branch may not reuse (27.8) by deleting the sign.

Sharpness and failed converses. The exponent $2m$ is optimal at the ambient matrix level. Choose $x \in \mathfrak{p}^m \setminus \mathfrak{p}^{m+1}$ and put

$$M_x = \begin{pmatrix} 1+x^2 & x \\ x & 1 \end{pmatrix}, \quad \det M_x = 1, \quad \operatorname{tr} M_x - 2 = x^2. \quad (27.12)$$

Here $M_x - I \in M_2(\mathfrak{p}^m)$, while $x^2 \in \mathfrak{p}^{2m} \setminus \mathfrak{p}^{2m+1}$. This example asserts sharpness for $\mathrm{SL}_2(\mathcal{O}_E)$, not that M_x lies in the particular surface-group image.

The converse to (27.8) is false even with determinant one:

$$U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in \mathrm{SL}_2(\mathcal{O}_E), \quad \operatorname{tr} U - 2 = 0, \quad U - I \notin M_2(\mathfrak{p}^m). \quad (27.13)$$

The determinant-one hypothesis is also essential. Over $\mathbb{Z}$, for an odd prime p , $x = p^m$, and $A_x = (1+x)I$, one has

$$A_x - I \in M_2((p)^m), \quad \operatorname{tr} A_x - 2 = 2p^m \notin (p)^{2m}, \quad \det A_x \neq 1. \quad (27.14)$$

Exact construction and certificate. The typed input is the Statement 24.10 representation, ideal, level, and linear-kernel record, a finite canonical test-word list, and the optional projective-sign request

$$\mathbf{R}_{\text{proj}} = \text{NONE} \quad \text{or} \quad \mathbf{R}_{\text{proj}} = ((w_i, \varepsilon_i))_{i \in I}, \quad w_i \in \text{test_word_list}, \quad \varepsilon_i \in \{1, -1\}.$$

When present, this list is in canonical word-byte order and contains each requested word exactly once. Its presence bit, including the distinction between **NONE** and a present empty list, and its source hash belong to the typed input.

Replay the representation, ideal, word-list, and request hashes; reject $m < 1$, a zero or nonprime $\mathfrak{p}$, a nonintegral matrix, or a failed determinant check. Also reject a missing, duplicate, or foreign requested word, an illegal sign, noncanonical order, or a sign for which $M(w_i) - \varepsilon_i I \notin M_2(\mathfrak{p}^m)$. For each word, compute M , the four entries of $B = M - I$, their exact $\mathfrak{p}^m$ -membership bits, $\operatorname{tr} B$, $\det B$, and the two sides of (27.6). When all four bits pass, verify ideal membership of both product terms and of the trace defect in $\mathfrak{p}^{2m}$. For every requested projective row, compute the normalized matrix $\varepsilon_i M(w_i)$ and verify (27.10).

On passage the completed record is

$$\begin{aligned}
&(\mathfrak{p}, h_{\mathfrak{p}}, m, h_{\rho_E}, K_{\mathfrak{p}^m}^{\text{lin}}, \text{entry_membership_rows}, \\
&\text{determinant_identity_rows}, \text{trace_defect_rows}, \text{projective_sign_rows or NONE}, \\
&R_{\text{proj}} \text{ or NONE}, h_{R_{\text{proj}}} \text{ or NONE}, \\
&\text{COMPLETE, NONE, SATISFIED}, h_{\text{input}}, h_{\text{independentverification}}).
\end{aligned} \tag{27.15}$$

If $R_{\text{proj}} = \text{NONE}$, the projective sign rows and request hash are both **NONE**. If the request is present but empty, those rows are the present empty sequence and its hash is present. Otherwise there is exactly one output row for each request. The input and independent verification hashes distinguish all three cases. An empty test-word list is allowed, but the symbolic kernel theorem, ideal-power record, and success envelope remain present; then a present projective request must also be empty. Any failed precondition returns only

$$\text{REJECT}(\text{INVALID_INPUT}, \text{stage}, \text{first_failure}, h_{\text{input}}). \tag{27.16}$$

Theorem 25.1.4 (Certificate correctness and termination). *The procedure terminates. Every passing linear row satisfies (27.8), every optional projective row satisfies exactly the normalized claim (27.10), and equal canonical inputs produce byte-identical records.*

Proof. The word list and all matrix-entry lists are finite. Exact number field arithmetic, ideal multiplication, HNF membership, determinants, and traces terminate. The two lemmas prove soundness; fixed generator, word, matrix-entry, ideal-basis, and byte orders prove determinism. $\square$

Mandatory tests. The identity tests the zero defect without asserting nonzeroness. The matrices M_x , U , and A_x test sharpness, failure of the converse, and necessity of determinant one. Additional tests cover $m = 1$, $m > 1$, one failing entry, an exact negative projective lift, an absent projective request, a present empty request, missing, duplicate, foreign, illegal, and false sign rows, a scalar class without a sign certificate, an empty word list, a bad prime, level zero, a determinant failure, a changed source hash, and byte-identical independent verification.

Interface closure. Statement 25.2 uses the containment (27.8) to prove nonzeroness for nonidentity surface-group elements. Linear-kernel membership is determined by the exact matrix criterion, with $\mathfrak{p}^{2m}$ retained exactly and the projective lift normalized as above.

25.2 Trace divisibility and nonvanishing

Inherited hypotheses and notation. Retain the number field, prime ideal, level $m \geq 1$, and faithful integral lift $\rho_E : \Gamma_B \rightarrow \text{SL}_2(\mathcal{O}_E)$ of Statement 25.1. Under the distinguished complex embedding $\iota : E \hookrightarrow \mathbb{C}$ of Statement 23.1, the matrices lie in $\text{SU}(1, 1)$, and their projective image is the torsion-free cocompact Fuchsian group defining the Bolza surface. For $\gamma \in K_{\mathfrak{p}^m}^{\text{lin}}$, define the positive-lift trace defect

$$t_+(\gamma) := \text{tr } \rho_E(\gamma) - 2 \in \mathcal{O}_E. \tag{27.17}$$

Statement 25.1 proves, without any geometric hypothesis,

$$t_+(\gamma) \in \mathfrak{p}^{2m}. \tag{27.18}$$

The purpose of the present item is to certify the additional nonvanishing needed before taking a principal-ideal norm.

Lemma 25.2.1 (Classification in the Bolza group). *Every $1 \neq \gamma \in \Gamma_B$ acts hyperbolically on $\mathbb{H}_R^2$. For either geometric lift $M = \iota(\rho_E(\gamma)) \in \text{SU}(1, 1)$, its trace is real and*

$$|\text{tr } M| > 2. \tag{27.19}$$

Proof. An orientation-preserving isometry of $\mathbb{H}_R^2$ is elliptic, parabolic, hyperbolic, or the identity. Discreteness makes every elliptic stabilizer finite, so torsion-freeness excludes a nonidentity elliptic element. A parabolic element would determine a cusp in the quotient, which is impossible because the Bolza quotient is compact. The identity projective isometry is excluded by faithfulness of the surface-group action. Hence the element is hyperbolic. The standard $\mathrm{SU}(1, 1)$ form has diagonal entries $a, \bar{a}$, so its trace $a + \bar{a}$ is real. Cayley conjugation carries it to $\mathrm{SL}_2(\mathbb{R})$ without changing the trace. The real trace classification then gives (27.19), independently of the lift sign. $\square$

Theorem 25.2.2 (Nonzero divisible trace defect). *For every $1 \neq \gamma \in K_{\mathfrak{p}^m}^{\mathrm{lin}}$,*

$$0 \neq t_+(\gamma) = \mathrm{tr} \rho_E(\gamma) - 2 \in \mathfrak{p}^{2m}. \quad (27.20)$$

Equivalently, its nonzero principal ideal satisfies

$$(0) \subsetneq (t_+(\gamma)) \subseteq \mathfrak{p}^{2m}. \quad (27.21)$$

Proof. Containment is (27.18). If $t_+(\gamma) = 0$, injectivity of ι gives $\mathrm{tr} \iota(\rho_E(\gamma)) = 2$, contradicting (27.19). For a ring element t , membership $t \in \mathfrak{p}^{2m}$ is equivalent to $(t) \subseteq \mathfrak{p}^{2m}$. Nonvanishing makes (t) a nonzero ideal, proving (27.21). $\square$

Valuation form. For $0 \neq a \in E$, normalize the prime-ideal valuation by

$$(a) = \mathfrak{p}^{v_{\mathfrak{p}}(a)} \mathfrak{a}, \quad \mathfrak{p} \nmid \mathfrak{a}. \quad (27.22)$$

Unique ideal factorization and (27.21) give the computable statement

$$v_{\mathfrak{p}}(t_+(\gamma)) \geq 2m \quad (1 \neq \gamma \in K_{\mathfrak{p}^m}^{\mathrm{lin}}). \quad (27.23)$$

No value $v_{\mathfrak{p}}(0)$ is needed. A software convention that writes $v_{\mathfrak{p}}(0) = +\infty$ must still exclude the identity before emitting the nonzero principal-ideal record.

Lift and sign discipline. The algebraic element in (27.20) is $\mathrm{tr} M - 2$, not $|\mathrm{tr} M| - 2$. Replacing M by $-M$ can destroy the congruence $M \equiv I \pmod{\mathfrak{p}^m}$, so a later archimedean argument may not silently change its sign. For an optional projective row carrying the exact sign $\varepsilon \in \{1, -1\}$ certified in Statement 25.1, define

$$t_{\varepsilon}(\gamma) := \varepsilon \mathrm{tr} \rho_E(\gamma) - 2. \quad (27.24)$$

Then the geometric matrix $\iota(\varepsilon \rho_E(\gamma))$ represents the same projective isometry. For $1 \neq \gamma$, (27.19), injectivity of ι , and the Statement 25.1 congruence give

$$0 \neq t_{\varepsilon}(\gamma) \in \mathfrak{p}^{2m}. \quad (27.25)$$

This projective conclusion is emitted only for an exact requested sign; a residue scalar without that sign remains insufficient.

Necessity of the hypotheses. Compactness cannot be removed. Over $\mathbb{Z}$, for a prime integer p and $x = p^m$,

$$P_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}, \quad P_x \equiv I \pmod{p^m}, \quad \mathrm{tr} P_x - 2 = 0. \quad (27.26)$$

The nonidentity matrix P_x is parabolic and can occur in a nonuniform congruence group. Thus divisibility alone does not prove nonvanishing. Faithfulness is also indispensable: any nonidentity element in the kernel of a nonfaithful representation maps to I and has zero trace defect. The identity of the faithful Bolza group is the boundary case excluded explicitly in (27.20).

Exact evaluation specification. The typed input consists of a passing Statement 25.1 record, the fixed geometric-action and faithfulness hashes, a finite canonical list of word identifiers claimed to represent nonidentity elements of the linear kernel, and the inherited three-state optional projective-sign request. For every word, recompute M , require $M \neq I$, verify the four linear kernel entry tests, compute t_+ in the integral basis, and require both a nonzero coordinate vector and exact $\mathfrak{p}^{2m}$ -membership. Factor the nonzero principal ideal only after those conditions and verify (27.22)–(27.23). For each requested projective row, independent verification its exact sign, recompute t_ε , and verify (27.25). Missing, duplicate, foreign, identity, scalar-only, or hash-inconsistent rows are invalid input.

The completed record is

$$\begin{aligned} &(\mathfrak{p}, m, h_{\rho_E}, h_{\text{geometry}}, \text{linear_trace_rows}, \text{principal_ideal_rows}, \\ &\text{valuation_rows}, \text{projective_trace_rows or NONE}, R_{\text{proj}} \text{ or NONE}, \\ &\text{COMPLETE, NONE, SATISFIED}, h_{\text{input}}, h_{\text{independentverification}}). \end{aligned} \quad (27.27)$$

The absent, present-empty, and present-nonempty projective states are preserved exactly as in Statement 25.1. An empty linear test list is allowed and returns empty row sequences while retaining the universal theorem and success envelope. It is not evidence that the whole kernel was enumerated. A failed condition returns only

$$\text{REJECT}(\text{INVALID_INPUT}, \text{stage}, \text{first_failure}, h_{\text{input}}). \quad (27.28)$$

Theorem 25.2.3 (Correctness and termination). *The evaluation terminates, every passing nonidentity linear row satisfies (27.20)–(27.23), every passing projective row satisfies (27.25), and equal canonical inputs produce byte-identical outputs.*

Proof. All input lists and matrices are finite. Exact field arithmetic, HNF ideal membership, principal-ideal factorization, comparison with zero, and fixed-order serialization terminate. The classification lemma proves nonvanishing, Statement 25.1 proves divisibility, and unique ideal factorization proves the valuation row. $\square$

Mandatory tests and interface closure. A certified nonidentity word in a Statement 24.10 linear kernel is the positive test; the identity is the rejected boundary. The parabolic matrix P_x tests that compactness is genuinely used, and a deliberately nonfaithful representation tests the faithfulness condition. Further tests cover positive and negative real geometric traces, $m = 1$, $m > 1$, exact valuation $2m$, valuation greater than $2m$, absent and empty projective requests, a correct negative sign, a false sign, duplicate and foreign words, stale hashes, and deterministic independent verification. Statement 25.3 may consume only the nonzero principal ideals (27.21); it assigns a finite ideal norm only after an established nonzero defect.

25.3 Algebraic norm lower bound

Input and norm conventions. Let $0 \neq t \in \mathcal{O}_E$ be a linear or exactly sign-normalized projective trace defect certified by Statement 25.2, so that $(t) \subseteq \mathfrak{p}^{2m}$. Write

$$q := N(\mathfrak{p}) := |\mathcal{O}_E/\mathfrak{p}|, \quad N(I) := |\mathcal{O}_E/I| \quad (27.29)$$

for a nonzero integral ideal I . Thus $q = p^f \geq 2$ is the exact residue-field order from Statement 24.2. The absolute value of a field norm is used throughout; the sign of a real norm is not discarded until after the exact determinant is computed.

Lemma 25.3.1 (Ideal norm reverses containment). *For nonzero integral ideals $I \subseteq J \subseteq \mathcal{O}_E$, the canonical map is a surjection and*

$$\mathcal{O}_E/I \twoheadrightarrow \mathcal{O}_E/J, \quad N(I) = N(J)[J : I] \geq N(J). \quad (27.30)$$

Equality holds exactly when $I = J$.

Proof. Reduction of a residue class modulo I further modulo J is well defined because $I \subseteq J$, and every class modulo J has a preimage. Its kernel is J/I , whose order is $[J : I]$. The finite-index formula proves the assertion and its equality case. $\square$

Lemma 25.3.2 (Principal ideal norm). *If $0 \neq a \in \mathcal{O}_E$, then*

$$N((a)) = |N_{E/\mathbb{Q}}(a)|. \quad (27.31)$$

Proof. Multiplication by a is an injective endomorphism of the $\mathbb{Z}$ -lattice $\mathcal{O}_E$. In an integral basis its cokernel is $\mathcal{O}_E/a\mathcal{O}_E$, while its determinant is the field norm. The lattice index is the absolute determinant. $\square$

Theorem 25.3.3 (Exact algebraic norm lower bound). *Every trace defect supplied by Statement 25.2 satisfies*

$$|N_{E/\mathbb{Q}}(t)| = N((t)) \geq N(\mathfrak{p}^{2m}) = q^{2m}. \quad (27.32)$$

The inequality is exact and uses no archimedean approximation.

Proof. Statement 25.2 gives a nonzero principal ideal $(t) \subseteq \mathfrak{p}^{2m}$. Apply (27.30), then (27.31), and use multiplicativity of ideal norm. $\square$

Valuation and equality certificate. Factor the nonzero principal ideal uniquely as

$$(t) = \mathfrak{p}^e \mathfrak{a}, \quad e = v_{\mathfrak{p}}(t) \geq 2m, \quad \mathfrak{p} \nmid \mathfrak{a}. \quad (27.33)$$

Then

$$|N_{E/\mathbb{Q}}(t)| = q^e N(\mathfrak{a}) = q^{2m} c_t, \quad c_t := q^{e-2m} N(\mathfrak{a}) \in \mathbb{Z}_{\geq 1}. \quad (27.34)$$

Consequently,

$$|N_{E/\mathbb{Q}}(t)| = q^{2m} \iff (t) = \mathfrak{p}^{2m}. \quad (27.35)$$

Thus an equality claim requires equality of ideal HNFs, not merely a decimal agreement of logarithms.

Optimization formulation. The associated exact optimization problem is

$$\begin{aligned} \mathcal{F}_{\mathfrak{p},m} &:= \mathfrak{p}^{2m} \setminus \{0\}, \\ \mu_{\mathfrak{p},m} &:= \min_{a \in \mathcal{F}_{\mathfrak{p},m}} |N_{E/\mathbb{Q}}(a)|. \end{aligned} \quad (27.36)$$

The feasible set is nonempty and all objective values are positive integers, so the minimum exists by well ordering. If a_{HNF} is the first nonzero vector in the canonical HNF basis of $\mathfrak{p}^{2m}$, then

$$q^{2m} \leq \mu_{\mathfrak{p},m} \leq |N_{E/\mathbb{Q}}(a_{\text{HNF}})|. \quad (27.37)$$

Moreover, the lower bound is attained exactly when $\mathfrak{p}^{2m}$ is principal. The theorem (27.32) needs only the lower bound; it does not claim that an optimizer has been found.

Exact numerical realization. Fix the C25 integral basis $\boldsymbol{\omega} = (\omega_1, \dots, \omega_d)$. For $a \in \mathcal{O}_E$, let $A(a) \in M_d(\mathbb{Z})$ be the matrix of multiplication by a . Then

$$N_{E/\mathbb{Q}}(a) = \det A(a), \quad |N_{E/\mathbb{Q}}(a)| = |\det A(a)|. \quad (27.38)$$

If the columns of the integer HNF matrix H_I are the coordinates of a basis of I , then

$$N(I) = |\det H_I|. \quad (27.39)$$

Equations (27.38)–(27.39) independently recompute both sides of (27.32). The exact divisibility check is

$$|\det A(t)| = q^{2m} c_t, \quad c_t \in \mathbb{Z}_{\geq 1}, \quad (27.40)$$

performed by integer division with zero remainder.

Floating point is permitted only for display. With directed rounding, compute enclosures

$$\begin{aligned} [\ell_q^-, \ell_q^+] &\ni \log q, \\ L_N^- &:= \text{round}_\downarrow(2m\ell_q^-), & L_N^- &\leq \log \mu_{\mathfrak{p},m} \leq U_{\text{HNF}}^+, \\ U_{\text{HNF}}^+ &:= \text{round}_\uparrow \log |N_{E/\mathbb{Q}}(a_{\text{HNF}})|. \end{aligned} \quad (27.41)$$

The precision is increased until both outward-rounding directions and the displayed decimal endpoints are stable. These intervals report the exact integer certificate; they never replace it.

Independent small examples. For $E = \mathbb{Q}$, $\mathfrak{p} = (3)$, $m = 2$, and $t = 3^4$,

$$(t) = \mathfrak{p}^{2m}, \quad |N_{\mathbb{Q}/\mathbb{Q}}(t)| = 81 = q^{2m}. \quad (27.42)$$

The choice $t = 3^5$ gives the strict case $243 = 81 \cdot 3$. For $E = \mathbb{Q}(i)$, with integral basis $(1, i)$, $\mathfrak{p} = (1 + i)$, $q = 2$, $m = 1$, and $t = 2i$,

$$A(2i) = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}, \quad |\det A(2i)| = 4 = q^2. \quad (27.43)$$

The ideal and multiplication-matrix computations therefore cross validate the equality case in degrees one and two.

Construction and I/O specification. The typed input contains a passing Statement 25.2 record, the C25 integral basis and multiplication table, the Statement 24.2 prime/norm/HNF record, a finite canonical list of nonzero trace-defect rows, and the inherited three-state projective request. Replay every hash. For each row, reject zero, reconstruct (t) , verify its containment in $\mathfrak{p}^{2m}$, compute both determinants in (27.38)–(27.39), factor the principal ideal, and verify the exponent, cofactor, equality flag, and outward log interval. Missing, duplicate, foreign, sign-inconsistent, or noncanonically ordered rows are invalid input.

The completed record is

$$\begin{aligned} (E, \omega, \mathfrak{p}, q, m, \text{principal_ideal_rows}, \text{valuation_factor_rows}, \\ \text{exact_norm_rows}, \text{cofactor_rows}, (\mu_{\text{lower}}, \mu_{\text{candidate_upper}}), \text{interval_rows}, \\ \text{projective_norm_rows or NONE, COMPLETE, NONE, SATISFIED}, h_{\text{input}}, h_{\text{independentverification}}). \end{aligned} \quad (27.44)$$

Absent, present-empty, and present-nonempty projective rows retain their Statement 25.2 presence states. Empty trace-row lists are allowed, but the ideal-power optimization bounds and success envelope remain present. Any failed condition returns only

$$\text{REJECT}(\text{INVALID_INPUT}, \text{stage}, \text{first_failure}, h_{\text{input}}). \quad (27.45)$$

Theorem 25.3.4 (Correctness and termination). *The construction terminates, every passing row satisfies the exact bound (27.32), all displayed intervals enclose their exact quantities, and equal canonical inputs produce byte-identical records.*

Proof. All row and basis lists are finite. HNF arithmetic, determinant, ideal multiplication, ideal factorization, exact division, and adaptive directed-rounding computations terminate under the fixed precision-escalation policy. The preceding lemmas prove the bound; fixed basis, row, factor, interval, and byte orders prove independent verification determinism. $\square$

Mandatory tests and subsequent boundary. The two equality examples (27.42)–(27.43) independently test the norm identities; 3^5 tests strictness. Zero, a defect in $\mathfrak{p}^{2m-1} \setminus \mathfrak{p}^{2m}$, reversed ideal containment, wrong q , a nonzero division remainder, a false equality flag, insufficient interval precision, optional presence, empty lists, and stale hashes are rejection or boundary tests. Statement 25.4 may use (27.32) only together with certified bounds for every nondistinguished embedding; the algebraic norm alone is not a lower bound for the distinguished real absolute value.

25.4 Trace lower bounds from archimedean caps

Embedding convention. The field $E = \mathbb{Q}(\zeta_{16}, \beta)$ has no real embeddings: the image of a primitive sixteenth root of unity cannot lie in $\mathbb{R}$. Its sixteen embeddings form eight conjugate pairs. Choose representatives $\tau_0, \tau_1, \dots, \tau_7$, with $\tau_0 = \iota : E \hookrightarrow \mathbb{C}$ the geometric embedding fixed in Statement 23.1. Every place is therefore counted twice in the norm:

$$|N_{E/\mathbb{Q}}(t)| = |\iota(t)|^2 \prod_{k=1}^7 |\tau_k(t)|^2. \quad (27.46)$$

Let $t \neq 0$ be a trace defect certified in Statement 25.3. For each nondistinguished complex place, assume a certified positive upper bound

$$|\tau_k(t)| \leq D_k \quad (1 \leq k \leq 7), \quad (27.47)$$

where every $D_k > 0$. Define the paired-place denominator

$$\mathcal{C} := \prod_{k=1}^7 D_k^2, \quad (27.48)$$

with an empty product equal to 1.

Theorem 25.4.1 (Distinguished trace-defect lower bound). *If the caps (27.47) hold, then*

$$|\iota(t)| \geq B_{\mathfrak{p},m}(\mathcal{C}) := \frac{q^m}{\sqrt{\mathcal{C}}} > 0. \quad (27.49)$$

Proof. Every field embedding is injective, so no factor $\tau(t)$ vanishes. Isolate the geometric conjugate pair in (27.46), use the exact norm lower bound $|N_{E/\mathbb{Q}}(t)| \geq q^{2m}$ from Statement 25.3, and replace each remaining factor by its certified upper bound. Because all denominator factors are positive, this gives $|\iota(t)|^2 \geq q^{2m}/\mathcal{C}$. Taking the positive square root proves (27.49). $\square$

From defect to matrix trace. For the geometric lift $M = \iota(\rho_E(\gamma))$, Statement 25.2 proves that its trace $2 + \iota(t_+(\gamma))$ is real. The reverse triangle inequality and (27.49) give

$$|\operatorname{tr} M| = |2 + \iota(t_+)| \geq |\iota(t_+)| - 2 \geq B_{\mathfrak{p},m}(\mathcal{C}) - 2. \quad (27.50)$$

Statement 25.2 separately gives $|\operatorname{tr} M| > 2$. Hence a nonnegative certified trace floor is

$$T_{\text{abs}} := \max\{2, B_{\mathfrak{p},m}(\mathcal{C}) - 2\}, \quad |\operatorname{tr} M| \geq T_{\text{abs}}. \quad (27.51)$$

The value 2 in this weak floor does not replace the strict hyperbolicity statement. A nontrivial quantitative improvement for translation length from the sign-free branch requires $B_{\mathfrak{p},m}(\mathcal{C}) > 4$.

If an exact real-sign certificate proves $\iota(t) > 0$, then the stronger positive-trace branch is

$$\operatorname{tr} M = 2 + \iota(t) \geq 2 + B_{\mathfrak{p},m}(\mathcal{C}). \quad (27.52)$$

For an exactly normalized projective row, the same formula applies to εM and t_ε only when the sign of $\iota(t_\varepsilon)$ is certified. Congruence normalization and positive-real-trace normalization are separate input bits. The latter is represented by the independent request

$$R_+ = \text{NONE} \quad \text{or} \quad R_+ = (\text{trace_key}_i)_{i \in I}.$$

When present, R_+ is in canonical trace-key order and has no duplicates. Every requested key must prove $\text{Im } \iota(t) = 0$ and carry a certified isolating interval for $\text{Re } \iota(t)$ whose lower endpoint is strictly positive. Absence, a present empty request, and a present nonempty request are distinct states and have distinct hashes. This request is never inferred from, or merged with, the projective congruence sign ε .

Uniform and rowwise forms. If the caps depend on γ , (27.49) is only a rowwise bound. For a family $\mathcal{T}$ of nonzero trace defects, a uniform profile consists of constants satisfying

$$\sup_{t \in \mathcal{T}} |\tau_k(t)| \leq D_k \quad (1 \leq k \leq 7). \quad (27.53)$$

Only such a profile produces a common number $B_{\mathbf{p},m}(\mathcal{C})$ for all rows in the family. Uniformity in γ is required by Statement 25.6; dependence on m is allowed when recorded explicitly. In rowwise mode the serialized bound data are the keyed sequence

$$(\text{trace_key}(t), \mathcal{C}_t^+, B_t^-)_t.$$

In uniform mode there is one common pair $(\mathcal{C}_{\text{unif}}^+, B_{\text{unif}}^-)$, and every keyed row references the same profile hash. A rowwise collection is never collapsed to one scalar pair.

Optimization problem. For a fixed cap profile, set

$$\mathcal{F}(\mathbf{p}, m, \mathbf{D}) := \{ 0 \neq a \in \mathbf{p}^{2m} : |\tau_k(a)| \leq D_k \text{ for } 1 \leq k \leq 7 \}, \quad (27.54)$$

and, when this set is nonempty,

$$\lambda_\iota := \min_{a \in \mathcal{F}} |\iota(a)|. \quad (27.55)$$

The minimum exists. Given one feasible a_0 , points with $|\iota(a)| \leq |\iota(a_0)|$ lie in a bounded region of the Minkowski embedding, whose intersection with the ideal lattice is finite. The theorem gives the rigorous optimization bound

$$B_{\mathbf{p},m}(\mathcal{C}) \leq \lambda_\iota \leq |\iota(a_0)|. \quad (27.56)$$

This does not assert that a supplied trace row realizes the minimum.

Directed-rounding evaluation. Caps may be exact positive algebraic or rational numbers. When they are evaluated numerically, each stored endpoint $D_{k,t}^+$ must be a proved upper bound for its trace key t . Compute

$$\begin{aligned} \mathcal{C}_t^+ &:= \text{round}_\uparrow \left(\prod_{k=1}^7 (D_{k,t}^+)^2 \right), \\ S_t^+ &:= \text{round}_\uparrow \left(\sqrt{\mathcal{C}_t^+} \right), \\ B_t^- &:= \text{round}_\downarrow \left(\frac{q^m}{S_t^+} \right). \end{aligned} \quad (27.57)$$

Then the machine-checkable direction is

$$0 < B_t^- \leq B_{\mathfrak{p},m}(\mathcal{C}_t) \leq |\iota(t)|. \quad (27.58)$$

The precision escalator rejects zero, negative, infinite, NaN, or unproved cap endpoints and continues until the displayed lower endpoint is stable. A downward-rounded $\max\{2, B_t^- - 2\}$, or $2 + B_t^-$ in the certified positive branch, is the only decimal trace floor exported. In uniform mode the same calculation is made once from the common cap endpoints and its pair is referenced by all rows.

Small cross-checks and a necessary warning. For the imaginary quadratic fixture $E_0 = \mathbb{Q}(i)$, take $\mathfrak{p} = (2 + i)$, so $q = 5$, $m = 1$, and $t = (2 + i)^2 \in \mathfrak{p}^2$. There is only the distinguished conjugate pair, hence $\mathcal{C} = 1$, and

$$|N_{E_0/\mathbb{Q}}(t)| = 25, \quad B_{\mathfrak{p},1} = 5 = |\iota(t)|. \quad (27.59)$$

This independently checks the square and square-root directions. A synthetic eight-pair fixture separately checks all seven nondistinguished squared cap factors.

The nondistinguished caps are essential. In $E_1 = \mathbb{Q}(i, \sqrt{2})$, choose the complex place with $\sqrt{2} > 0$ and let $u = 1 - \sqrt{2}$. Then $|\iota(u)| < 1$, while the other complex pair has modulus greater than 1. For any fixed $0 \neq a \in \mathfrak{p}^{2m}$,

$$t_n = au^n \in \mathfrak{p}^{2m}, \quad |N(t_n)| = |N(a)|, \quad |\iota(t_n)| \longrightarrow 0. \quad (27.60)$$

Thus the norm bound alone cannot control the distinguished embedding.

Construction and certificate. The typed input contains a passing Statement 25.3 norm record, the complete ordered list of eight conjugate pairs with ι marked, exact complex isolating data, either one complete cap profile per trace key or one explicitly uniform cap profile, a finite canonical trace-row list, the independent three-state positive-real-sign request R_+ with its source hash, and the inherited optional projective request. Verify the signature $r_1 = 0, r_2 = 8$, complex-pair multiplicities, positivity and coverage of every cap, the field-norm product, and each embedding enclosure. In rowwise mode require exactly one cap profile and one bound row for every trace key; in uniform mode require the declared family-coverage proof and common profile hash. Validate R_+ 's canonical order, uniqueness, key membership, exact zero imaginary parts, and strictly positive real isolating intervals. Compute exact symbolic bounds when possible and the directed intervals (27.57) in all cases. A missing place, duplicated place, unproved cap, row/family scope mismatch, missing, duplicate, foreign, or false positive-sign row, or any sign substitution is rejected.

The completed record is

$$\begin{aligned} & (E, \iota, \text{embedding_pair_rows}, \text{cap_scope}, \text{cap_rows}, \text{keyed_bound_rows}, \\ & (\mathcal{C}_{\text{unif}}^+, B_{\text{unif}}^-) \text{ or NONE}, \text{trace_defect_lower_rows}, \\ & \text{absolute_trace_floor_rows}, R_+ \text{ or NONE}, h_{R_+} \text{ or NONE}, \\ & \text{positive_trace_rows} \text{ or NONE}, \\ & \text{projective_trace_rows} \text{ or NONE}, \\ & \text{COMPLETE, NONE, SATISFIED}, h_{\text{input}}, h_{\text{independentverification}}). \end{aligned} \quad (27.61)$$

The positive-trace rows are NONE exactly when $R_+ = \text{NONE}$, are present-empty exactly when the request is present-empty, and otherwise contain exactly one row per requested key. Optional projective and empty lists preserve their own parent presence bits. A present uniform cap profile with an empty trace list certifies only the stated cap-domain implication, not coverage of the

full kernel. The case $B_t^- \leq 4$ is a valid weak bound and returns SATISFIED; it is not an input failure. Failed preconditions return

$$\text{REJECT}(\text{INVALID_INPUT}, \text{stage}, \text{first_failure}, \mathbf{h}_{\text{input}}). \quad (27.62)$$

Theorem 25.4.2 (Correctness and termination). *The construction terminates. Every passing row satisfies (27.49)–(27.52) in its declared scope, every numeric endpoint is outward rounded, and equal canonical inputs produce byte-identical records.*

Proof. All embedding, cap, and trace lists are finite. Certified algebraic evaluation and adaptive interval arithmetic terminate under the fixed precision policy. The product formula proves the lower bound, the reverse triangle inequality proves the trace floor, and fixed place, row, interval, and byte orders prove deterministic independent verification. $\square$

Mandatory tests and subsequent boundary. The Gaussian equality fixture tests the empty nondistinguished product and geometric square; an eight-pair fixture tests all exponents two. A strict cap, a loose cap, a rowwise profile mislabeled uniform, a missing conjugate, zero and negative caps, an inward-rounded endpoint, rowwise keys with distinct bounds, shared uniform-profile references, $B_t^- \leq 4$ passing with only the weak floor, all three positive-sign request states, missing, duplicate, foreign, and false positive-sign rows, positive and negative traces, projective presence states, empty rows, stale hashes, and independent verification are mandatory. Statement 25.5 may use the sign-free floor only with its $B > 4$ condition; the stronger $2 + B$ floor additionally requires a certified positive real sign.

25.5 Translation-length lower bounds

Geometric input. The hyperbolic plane has curvature $-R^{-2}$ with $R > 0$. For a nonidentity Bolza element γ , let $M = \iota(\rho_E(\gamma)) \in \text{SU}(1, 1)$ under the distinguished complex embedding. By Statement 25.2 its trace is real and the projective isometry is hyperbolic. Its translation length is

$$\ell_h(\gamma) := \inf_{z \in \mathbb{H}_R^2} d_{\mathbb{H}_R^2}(z, \gamma z) = 2R \operatorname{arcosh} \left(\frac{|\operatorname{tr} M|}{2} \right). \quad (27.63)$$

This is a global conjugacy invariant; it is not the based displacement $d(o, \gamma o)$.

Lemma 25.5.1 (Monotonicity). *For $x \geq 2$, define*

$$F_R(x) := 2R \operatorname{arcosh}(x/2). \quad (27.64)$$

Then F_R is continuous and increasing, and for $x > 2$,

$$F'_R(x) = \frac{R}{\sqrt{x^2/4 - 1}} > 0. \quad (27.65)$$

Proof. The derivative follows from $(\operatorname{arcosh} y)' = (y^2 - 1)^{-1/2}$ and the chain rule. Continuity at $x = 2$ follows from $\operatorname{arcosh}(1) = 0$. $\square$

Sign-free lower bound. For a trace key t , let $B_t^- > 0$ be the directed lower endpoint certified by Statement 25.4. Its sign-free trace floor and length floor are

$$\begin{aligned} T_{\text{sf},t}^- &:= \max\{2, B_t^- - 2\}, \\ L_{\text{sf},t}^- &:= F_R(T_{\text{sf},t}^-) = 2R \operatorname{arcosh} \left(\max \left\{ 1, \frac{B_t^- - 2}{2} \right\} \right). \end{aligned} \quad (27.66)$$

Theorem 25.5.2 (Sign-free translation-length bound). *Every linear or exactly normalized projective row to which the Statement 25.4 cap profile applies satisfies*

$$\ell_h(\gamma) \geq L_{\text{sf},t}^- \quad (27.67)$$

This bound is positive from the arithmetic floor precisely when $B_t^- > 4$. If $B_t^- \leq 4$, (27.66) returns the valid weak bound zero and does not manufacture a positive systolic statement.

Proof. Statement 25.4 gives $|\text{tr } M| \geq \max\{2, B_t^- - 2\}$. Apply the increasing function F_R and use (27.63). $\square$

Certified positive-trace branch. For a key requested by the independent positive-real-sign input of Statement 25.4, one has $\text{tr } M \geq 2 + B_t^-$. Therefore

$$L_{+,t}^- := 2R \text{arcosh} \left(1 + \frac{B_t^-}{2} \right), \quad \ell_h(\gamma) \geq L_{+,t}^- > 0. \quad (27.68)$$

For a projective row this is evaluated on the exact normalized lift εM ; its translation length equals that of M . The positive branch is absent unless the separate R_+ request and strict positive isolating interval pass.

Logarithmic forms. For $y \geq 1$,

$$\text{arcosh } y = \log \left(y + \sqrt{y^2 - 1} \right) \geq \log y. \quad (27.69)$$

Thus $B_t^- > 4$ gives the simpler sign-free estimate

$$L_{\text{sf},t}^- \geq 2R \log \left(\frac{B_t^- - 2}{2} \right), \quad (27.70)$$

while every positive branch gives

$$L_{+,t}^- \geq 2R \log \left(1 + \frac{B_t^-}{2} \right). \quad (27.71)$$

The exact arcosh expressions, not these weaker logarithmic forms, are the primary certificate.

Upper bounds for individual rows. If exact algebraic evaluation supplies a certified interval

$$2 < T_t^- \leq |\text{tr } M| \leq T_t^+, \quad 0 < R^- \leq R \leq R^+, \quad (27.72)$$

then monotonicity gives the two-sided rowwise enclosure

$$2R^- \text{arcosh}(T_t^-/2) \leq \ell_h(\gamma) \leq 2R^+ \text{arcosh}(T_t^+/2). \quad (27.73)$$

An upper endpoint for one row is not an upper bound for the kernel systole and is never serialized as such.

Optimization specification. First fix a scope S , either one trace key or one explicitly named family with a fixed membership/domain hash. Let $\mathcal{P}_{\text{adm}}(S)$ be the cap profiles whose place coverage, scope identifier, domain hash, and outward endpoints pass Statement 25.4 for that same S . For the positive branch, admissibility additionally requires positive-real-sign coverage of all of S : membership in R_+ for a singleton, or a universal positive-sign certificate for a family. For either branch $\beta \in \{\text{sf}, +\}$, define the formal objective

$$L_{\beta,S}^{\text{opt}} := \sup_{\mathcal{P} \in \mathcal{P}_{\text{adm}}(S)} L_{\beta} \left(\frac{q^m}{\sqrt{\mathcal{C}(\mathcal{P})}} \right). \quad (27.74)$$

Tightening a valid upper cap decreases $\mathcal{C}$, increases the argument, and cannot decrease the certified length. A chosen profile is a feasible certificate; no attainment of the supremum is claimed. Profiles with different scope identifiers or domain hashes are never combined in one supremum.

Directed numerical certificate. For lower bounds, use the already downward endpoint B_t^- , the tagged lower scale $R_{\downarrow} > 0$, and downward rounding at every monotone operation. Near 1, evaluate

$$\text{arcosh } y = \log \left(y + \sqrt{y-1} \sqrt{y+1} \right) \quad (27.75)$$

with directed square roots and logarithm. For a rowwise upper bound, use T_t^+ , $R_{\uparrow}$, and upward rounding throughout. Increase precision until the displayed endpoints are stable and ordered.

As a cross-check, with $R = 1$, the sign-free value $B^- = 6$ and the positive value $B^- = 2$ both give

$$2 \text{arcosh}(2) = 2 \log(2 + \sqrt{3}) \in [2.633915793849633, 2.633915793849634]. \quad (27.76)$$

The interval contains the independently computed high-precision value. The boundary $B^- = 4$ gives the sign-free lower bound $2 \text{arcosh}(1) = 0$.

Construction and output. The curvature-scale input is the tagged union

scale_kind	R_{exact}	(R^-, R^+)
EXACT	$R_{\text{exact}} > 0$	NONE
INTERVAL	NONE	$0 < R^- \leq R^+$

with exactly one branch present. Set $R_{\downarrow} = R_{\uparrow} = R_{\text{exact}}$ in the exact branch and $R_{\downarrow} = R^-$, $R_{\uparrow} = R^+$ in the interval branch. The branch tag and payload are included in the input and independent verification hashes.

The remaining typed input contains a passing Statement 25.4 record, the keyed B_t^- rows, the cap-scope identifiers and hashes, the three-state positive-sign request, the three-state projective request, and an optional finite canonical list of exact trace intervals for rowwise upper bounds. Verify all parent keys, presence bits, interval directions, $R_{\downarrow} > 0$, hyperbolic trace conditions, and branch assignments. Reject a missing, duplicated, or mixed scale branch. Compute every lower endpoint using $R_{\downarrow}$ downward and every requested upper endpoint upward. Missing, duplicate, foreign, inward, nonfinite, sign-mismatched, or stale rows are invalid.

The completed record is

$$\begin{aligned} &(\text{scale_kind}, R_{\text{exact}} \text{ or NONE}, (R^-, R^+) \text{ or NONE}, \\ &\text{cap_scope}, \text{scope_hash}, \text{keyed_B_rows}, \text{sign_free_length_rows}, \\ &R_+ \text{ or NONE}, \text{positive_length_rows} \text{ or NONE}, \\ &\text{projective_length_rows} \text{ or NONE}, \text{rowwise_upper_rows} \text{ or NONE}, \\ &\text{COMPLETE}, \text{NONE}, \text{SATISFIED}, h_{\text{input}}, h_{\text{independentverification}}). \end{aligned} \quad (27.77)$$

The exact and interval scale payloads are mutually exclusive and follow `scale_kind` exactly. Absent, present-empty, and present-nonempty optional inputs retain their exact states and hashes. Empty trace rows return empty sign-free rows and retain the symbolic formulas and success envelope; they prove no kernel-wide bound. A failed condition returns

$$\text{REJECT}(\text{INVALID_INPUT}, \text{stage}, \text{first_failure}, \text{h}_{\text{input}}). \quad (27.78)$$

Theorem 25.5.3 (Correctness and termination). *The construction terminates. Every passing lower endpoint is at most the exact translation length, every requested upper endpoint is at least it, and equal canonical inputs give byte-identical records.*

Proof. The input lists are finite. Certified real evaluation and adaptive directed rounding terminate under the fixed precision policy. Equations (27.63)–(27.68) and monotonicity prove the lower bounds; (27.72)–(27.73) prove the optional upper bounds. Fixed key, branch, interval, and byte orders prove independent verification determinism. $\square$

Mandatory tests and subsequent boundary. Equation (27.76) is the independent numerical cross-check. Tests also include $B^- < 4$, $B^- = 4$, $B^- > 4$, a positive-sign request, negative trace without that request, exact projective normalization, both optional-empty states, singleton and family optimization scopes, an attempted mixed-scope supremum, exact and interval scale branches, a double-present and a missing scale branch, $T^- = 2$, an invalid $T^- > T^+$, $R^- \leq 0$, inward rounding, precision escalation, stale hashes, and independent verification. Statement 25.6 may take an infimum only over rows covered by one kernel-wide uniform cap theorem; a finite test list or rowwise collection is not such a proof.

25.6 Global systole lower bound

Kernel scope and geometric quantity. Fix a passing linear congruence record from Statement 25.1 and put

$$K_m := K_{\mathfrak{p}^m}^{\text{lin}}, \quad S_m := K_m \setminus \{1\}, \quad X_m := K_m \backslash \mathbb{H}_R^2. \quad (27.79)$$

The parent Bolza lattice is torsion-free and cocompact, hence so is its finite-index subgroup K_m . Thus X_m is a compact hyperbolic surface. Its systole is

$$\text{sys}(X_m) := \inf_{\gamma \in S_m} \ell_h(\gamma) > 0. \quad (27.80)$$

Compactness also makes the infimum a minimum, but the lower-bound argument below does not depend on choosing a minimizing element.

Universal family certificate. A Statement 25.5 record is *kernel-wide* only if its scope fields satisfy

$$\begin{aligned} \text{scope_kind} &= \text{FAMILY}, & \text{scope_id} &= \text{linear_kernel}(\mathfrak{p}, m), \\ \text{scope_set} &= S_m, & \text{scope_hash} &= H(\text{kernel_record}, \mathfrak{p}, m, S_m). \end{aligned} \quad (27.81)$$

The symbolic set S_m is committed by its defining predicate and parent hashes; no enumeration of this infinite set is required. In addition the arithmetic cap theorem, sign branch, and scale branch must prove the universally quantified statement

$$\text{Coverage}_m : \quad \forall \gamma \in S_m, \quad \ell_h(\gamma) \geq L_m^-, \quad L_m^- \geq 0. \quad (27.82)$$

Every auxiliary hypothesis used to obtain L_m^- must itself have the same family scope and domain hash. A collection of rowwise bounds, even one containing many tested elements, is not Coverage_m .

Theorem 25.6.1 (Kernel-wide systole bound). *If (27.81)–(27.82) pass, then*

$$\text{sys}(X_m) \geq L_m^-. \quad (27.83)$$

Proof. The set in the infimum (27.80) is exactly the quantified domain in (27.82). Taking the infimum of that universal inequality proves (27.83). No passage from a finite sample to the full kernel occurs. $\square$

Dangerous elements and a strict target. For a requested target $L_\star \geq 0$, define

$$\mathcal{D}_{\leq L_\star}(K_m) := \{\gamma \in S_m : \ell_h(\gamma) \leq L_\star\}. \quad (27.84)$$

The universal certificate gives the exact logical implication

$$L_m^- > L_\star \implies \mathcal{D}_{\leq L_\star}(K_m) = \emptyset \implies \text{sys}(X_m) > L_\star. \quad (27.85)$$

If $L_m^- = L_\star$, only $\text{sys}(X_m) \geq L_\star$ follows. Equality is therefore a boundary case and is never serialized as a strict success.

For directed intervals $L_m^- \in [\underline{L}_m, \bar{L}_m]$ and $L_\star \in [\underline{L}_\star, \bar{L}_\star]$, the comparison verdict is

$$\text{verdict} := \begin{cases} \text{PROVED_STRICT}, & \underline{L}_m > \bar{L}_\star, \\ \text{BOUNDARY_NOT_STRICT}, & \underline{L}_m = \bar{L}_\star, \\ \text{NOT_PROVED}, & \bar{L}_m \leq \underline{L}_\star \text{ and } \underline{L}_m \neq \bar{L}_\star, \\ \text{INTERVAL_INDETERMINATE}, & \text{otherwise.} \end{cases} \quad (27.86)$$

Only the first state proves (27.85). Precision may be increased for the last state, but overlapping exact values must not be decided by floating-point tie breaking.

Optimization is domain-fixed. Let $\mathcal{U}_m$ be the admissible uniform Statement 25.5 profiles that all prove (27.82) for the same S_m , kernel record, scale branch, and domain hash. The strongest available certificate is formally

$$L_m^{\text{opt}} := \sup_{\mathcal{P} \in \mathcal{U}_m} L_m^-(\mathcal{P}). \quad (27.87)$$

A supplied profile gives a feasible lower bound; attainment of the supremum is not assumed. Profiles for different levels, kernels, projective normalizations, or scale records cannot be mixed.

Subgroup and projective boundaries. If a certified character χ defines $H_m := K_m \cap \ker \chi$, then

$$H_m \setminus \{1\} \subseteq S_m, \quad \text{sys}(H_m \setminus \mathbb{H}_R^2) \geq \text{sys}(X_m) \geq L_m^-. \quad (27.88)$$

This inheritance is valid because the infimum is taken over a smaller set. At level one, Statement 24.10 separately defines the projective kernel and gives

$$K_{\mathfrak{p}}^{\text{lin}} \subseteq K_{\mathfrak{p}}^{\text{proj}} \quad (m = 1 \text{ only}). \quad (27.89)$$

The projective kernel is larger, so a lower bound for the linear kernel does not transfer to it. No general higher-level projective kernel is inferred from Statement 25.1 sign-normalized rows, especially at primes above 2. A level- m projective claim requires its own separately defined kernel record, universal scope predicate, exact normalized-lift rule, and domain hash; otherwise its classification is PROJECTIVE_CERTIFICATE_ABSENT.

Typed construction. Input consists of the passing Statement 25.1 kernel record, one passing Statement 25.5 family record, the exact family predicate and lineage hashes, an optional character record, and an optional target in the tagged form

$$\text{target_kind} \in \{\text{ABSENT}, \text{EXACT}, \text{INTERVAL}\}, \quad \begin{cases} \text{ABSENT} : & \text{NONE}, \\ \text{EXACT} : & L_\star \geq 0, \\ \text{INTERVAL} : & 0 \leq \underline{L}_\star \leq \overline{L}_\star. \end{cases} \quad (27.90)$$

The Statement 25.5 parent must descend from the corrected distinguished complex-pair certificate of Statement 25.4, whose bound is $q^m/\sqrt{C}$. Any record descended from the obsolete single real-place bound is stale and rejected before comparison. Exactly one payload branch is present. In the exact branch normalize

$$\underline{L}_\star = \overline{L}_\star = L_\star.$$

In the interval branch the displayed payload endpoints are $\underline{L}_\star, \overline{L}_\star$. In the absent branch no target endpoints exist and no comparison is evaluated. Verify parent hashes, kernel level, set predicates, uniform coverage, curvature scale, outward rounding, and every optional presence bit. Then output

$$\begin{aligned} &(\text{linear_kernel_id}, \text{kernel_hash}, \text{scope_hash}, \text{coverage_proof_hash}, \\ &[\underline{L}_m, \overline{L}_m], \text{dangerous_set_status}, \text{target_kind}, \text{verdict or NONE}, \\ &\text{parity_inheritance or NONE}, \text{projective_status}, \text{COMPLETE}, \text{SATISFIED}, h_{\text{input}}, h_{\text{independent verification}}). \end{aligned} \quad (27.91)$$

Without a target, the dangerous-set classification is

$$\text{dangerous_set_status} = \text{NOT_REQUESTED}.$$

With a target it is `EMPTY_PROVED` exactly for a strict proof; otherwise it is

$$\text{dangerous_set_status} = \text{EMPTINESS_NOT_PROVED}.$$

A failed structural condition returns

$$\text{REJECT}(\text{INVALID_INPUT}, \text{stage}, \text{first_failure}, h_{\text{input}}). \quad (27.92)$$

Weak bounds, boundary verdicts, and unresolved target comparisons remain valid passing records; they are not silently promoted to computational admissibility.

Theorem 25.6.2 (Correctness, completeness, and termination). *Every passing record proves (27.83). When $\text{target_kind} \in \{\text{EXACT}, \text{INTERVAL}\}$, its dangerous-set classification is sound and complete relative to the supplied universal certificate and the strict comparison rule:*

$$\text{EMPTY_PROVED} \iff \text{Coverage}_m \text{ passes and } \underline{L}_m > \overline{L}_\star. \quad (27.93)$$

For the absent branch,

$$\text{target_kind} = \text{ABSENT}, \quad \text{dangerous_set_status} = \text{NOT_REQUESTED}.$$

No target endpoint is evaluated in that branch. Its verdict field is `NONE`, and `NONE` is forbidden in both target-present branches. The construction terminates and equal canonical inputs give byte-identical outputs.

Proof. Soundness of the systole row is the kernel-wide theorem. Equation (27.85) and directed comparison prove dangerous-set soundness. The reverse implication in (27.93) is the construction's exhaustive branch rule, not a claim that failure of this certificate implies the existence of a short geodesic. All validation lists and hashes are finite; the family coverage object is a finite proof certificate for a quantified theorem. Fixed field, branch, and byte orders prove termination and independent verification determinism. $\square$

Mandatory boundary tests. Tests include a uniform profile with $\underline{L}_m = 6.01$ and target 6, which is `PROVED_STRICT`; equality $6 = 6$, which is `BOUNDARY_NOT_STRICT`; overlapping intervals; zero weak bounds; an empty and a nonempty finite test list, neither accepted as family coverage; a stale or foreign domain hash; a profile omitting one quantified branch; valid parity-subgroup inheritance; an attempted linear-to-projective transfer; all target presence states; inward rounding; and independent verification. Statement 25.8 may accept a requested strict threshold only from `PROVED_STRICT`.

25.7 Asymptotic order–systole relation

One fixed congruence tower. Fix an admissible prime ideal $\mathfrak{p}$, write $q = N\mathfrak{p} > 1$, and let

$$Q_m := \rho_m(\Gamma_B) \cong \Gamma_B/K_m, \quad n_m := |Q_m| = [\Gamma_B : K_m] \quad (27.94)$$

for every m in a cofinal level set $\mathcal{I} = \{m \geq m_0 : \text{level_gate}(m)\}$. The kernel and image equalities must be certified at every level. Changing $\mathfrak{p}$, q , representation, or kernel convention starts a different tower.

Order envelopes. An admissible order certificate has constants

$$C_Q \geq 1, \quad d > 0, \quad 1 \leq n_m \leq C_Q q^{dm} \quad (m \in \mathcal{I}), \quad (27.95)$$

where C_Q and d are independent of m . It bounds the actual image order, not merely a guessed presentation order. There is always the coarse matrix-environment certificate

$$|\mathcal{O}_E/\mathfrak{p}^m| = q^m, \quad n_m \leq |M_2(\mathcal{O}_E/\mathfrak{p}^m)| = q^{4m}. \quad (27.96)$$

Thus $C_Q = 1, d = 4$ is valid whenever the Statement 25.1 integral reduction map is defined. A sharper determinant-one or local group formula may replace it only with a proof covering the same level set.

Lemma 25.7.1 (Logarithmic order consequence). *Every passing order envelope gives*

$$m \log q \geq \frac{\log n_m - \log C_Q}{d}. \quad (27.97)$$

Proof. Take natural logarithms in (27.95). Since $d > 0$, rearrangement and division preserve the inequality direction. $\square$

Uniform affine systolic envelope. A single-level Statement 25.6 record is insufficient for an asymptotic statement. The required family certificate has constants

$$a > 0, \quad C_0 \geq 0, \quad \text{sys}(X_m) \geq a m \log q - C_0 \quad (m \in \mathcal{I}), \quad (27.98)$$

independent of m , and its universal scope covers every nonidentity element of every K_m in $\mathcal{I}$.

For example, suppose one cap constant $\mathcal{C} > 0$ is valid for the whole tower and the curvature scale $R > 0$ is fixed. The positive-real-sign branch of Statement 25.5 gives

$$\begin{aligned} \text{sys}(X_m) &\geq 2R \log \left(1 + \frac{q^m}{2\sqrt{\mathcal{C}}} \right) \\ &\geq 2Rm \log q - 2R \log(2\sqrt{\mathcal{C}}). \end{aligned} \quad (27.99)$$

It therefore yields (27.98) with

$$a = 2R, \quad C_0 = 2R \max\{0, \log(2\sqrt{\mathcal{C}})\}. \quad (27.100)$$

The larger nonnegative constant only weakens the lower bound.

For the sign-free branch, require the uniform threshold

$$q^m / \sqrt{\mathcal{C}} \geq 4 \quad (m \in \mathcal{I}). \quad (27.101)$$

Then $B - 2 \geq B/2$ for $B = q^m / \sqrt{\mathcal{C}}$, so Statement 25.5 gives

$$\text{sys}(X_m) \geq 2Rm \log q - 2R \log(4\sqrt{\mathcal{C}}), \quad (27.102)$$

and one may use $C_0 = 2R \max\{0, \log(4\sqrt{\mathcal{C}})\}$. A cap depending on m , or coverage of only finitely many elements at each level, does not establish (27.98).

Theorem 25.7.2 (Order-systole elimination). *If (27.95) and (27.98) hold on the same tower, then*

$$\text{sys}(X_m) \geq A \log n_m - B \quad (m \in \mathcal{I}), \quad A := \frac{a}{d}, \quad B := C_0 + \frac{a}{d} \log C_Q. \quad (27.103)$$

Proof. Multiply (27.97) by $a > 0$ and substitute the result into (27.98). This proves (27.103), including levels for which its right-hand side is negative. No lower order bound can replace the upper order bound (27.95) in this argument. $\square$

Asymptotic meaning and converse form. The constants $A > 0$ and $B \geq 0$ are fixed on the tower. Hence

$$n_m \longrightarrow \infty \implies \text{sys}(X_m) \longrightarrow \infty. \quad (27.104)$$

Order divergence is a separate fact; it is not inferred from an upper order envelope. Equivalently, (27.95) and (27.98) imply

$$\log n_m \leq \log C_Q + \frac{d}{a} (\text{sys}(X_m) + C_0). \quad (27.105)$$

Neither display asserts optimality of A .

Directed numerical certificate. For interval inputs use certified endpoints

$$a \geq a^- > 0, \quad d \leq d^+, \quad C_0 \leq C_0^+, \quad C_Q \leq C_Q^+. \quad (27.106)$$

With outward rounding define

$$\begin{aligned} A^- &:= \text{round}_\downarrow(a^-/d^+), \\ B^+ &:= \text{round}_\uparrow(C_0^+ + A^- \log C_Q^+). \end{aligned} \quad (27.107)$$

The certified numerical output is

$$\text{sys}(X_m) \geq A^- \log n_m - B^+. \quad (27.108)$$

To justify the coefficient replacement, put $Y_m = \log n_m - \log C_Q^+$. If $Y_m < 0$, then $A^- Y_m \leq 0 < a^- m \log q$. If $Y_m \geq 0$, the order envelope gives $Y_m \leq d^+ m \log q$, and $A^- \leq a^-/d^+$ gives

$$a^- m \log q \geq A^- Y_m.$$

In both cases, subtracting C_0^+ and rounding the logarithm and sum upward proves (27.108).

The requested output precision is finite. Since $a^-/d^+ > 0$, successively increasing working precision eventually produces $A^- > 0$; certified division and logarithm terminate at each precision. Thus this numerical branch always terminates for valid finite endpoints and has no unresolved state.

Typed construction and independent verification. Input contains a fixed tower record, a cofinality proof for $\mathcal{I}$, a passing order-envelope proof, a passing uniform Statement 25.6 family proof, branch-specific cap and threshold data, and a directed arithmetic policy. Scope identifiers and domain hashes of the two envelopes must coincide. Verify $q > 1$, $C_Q \geq 1$, $d, a > 0$, $C_0 \geq 0$, independence from m , universal level and element coverage, and every parent hash.

The canonical output is

$$\begin{aligned} &(\text{tower_id}, \text{level_predicate}, \text{cofinality_proof_hash}, q, \\ &C_Q, d, a, C_0, A, B, A^-, B^+, \\ &\text{order_proof_hash}, \text{systole_proof_hash}, \text{ASYMPTOTIC_PROVED}, \\ &\text{COMPLETE}, \text{SATISFIED}, h_{\text{input}}, h_{\text{independentverification}}). \end{aligned} \quad (27.109)$$

Exact symbolic constants are mandatory; A^- is a certified one-sided lower coefficient and B^+ a certified one-sided upper offset, exactly as required by (27.108). Missing cofinality, an m -dependent constant, finite sample coverage, mismatched towers, or an inward endpoint returns

$$\text{REJECT}(\text{INVALID_ASYMPTOTIC_INPUT}, \text{stage}, \text{first_failure}, h_{\text{input}}). \quad (27.110)$$

Theorem 25.7.3 (Correctness and termination). *Every passing output proves (27.103) for all levels in the committed cofinal set. The construction terminates, and equal canonical inputs give byte-identical outputs.*

Proof. The order and systolic family proofs discharge (27.95) and (27.98); the elimination theorem gives the exact output. Monotonic directed operations in (27.106)–(27.108) give the numerical enclosure. All proof objects have finite encodings, and validation uses fixed field, branch, level-predicate, and byte orders. $\square$

Mandatory tests. Tests include the coarse choice $C_Q = 1$, $d = 4$; a sharper valid order envelope; $q = 2$; the positive and sign-free cap branches; the exact sign-free threshold; an m -dependent cap; a finite level list; a noncofinal level predicate; mismatched primes or kernel conventions; a lower order bound incorrectly substituted for (27.95); $d = 0$; $a = 0$; inward rounding; precision escalation; stale hashes; and independent verification. With $a = 2R$, $d = 4$, $C_Q = 1$, the coarse relation reduces to $\text{sys}(X_m) \geq (R/2) \log n_m - C_0$.

25.8 Congruence-quotient computational admissibility

Candidate coherence. A computational candidate is based on one passing Statement 25.7 tower record and one selected level $m \in \mathcal{I}$, where membership is checked against the exact committed level predicate. Its finite image, marked quotient map, and kernel are

$$\begin{aligned} \mathfrak{C}_m := &(\text{tower_id}, h_{\text{tower}}, h_{\mathcal{I}}, \mathfrak{p}, q, m, \\ &\text{LINEAR}, Q_m, \pi_m, K_m, h_Q, h_\pi, h_K). \end{aligned} \quad (27.111)$$

Before any computational predicate is evaluated, every child record must carry the same tower identifier, tower hash, level-predicate hash, prime, level, kernel convention, and candidate hashes. In particular

$$Q_m = \pi_m(\Gamma_B), \quad K_m = \ker \pi_m, \quad Q_m \cong \Gamma_B / K_m. \quad (27.112)$$

A failed membership test $m \notin \mathcal{I}$, stale hash, different level, relabelled carrier, projective normalization, or foreign kernel is invalid input, not a negative computational result. The present construction follows the certified linear-kernel path; it does not infer a projective certificate.

The six independent conditions. Let $G_i(\mathfrak{C}_m) \in \{0, 1\}$ be the following exact predicates.

1. *Marked shell.* The complete group and inverse tables pass Statement 11.1; the four marked images satisfy the surface relator and generate Q_m . The eight physical images q_j are nonidentity and pairwise distinct, and satisfy

$$q_{j+4} = q_j^{-1} \quad (j \in \mathbb{Z}/8). \quad (27.113)$$

Thus both the quotient map and the eight-direction shell, rather than only an abstract finite group, are certified.

2. *Exact C_8 -equivariance.* A complete permutation α of the carrier is a multiplication-preserving bijection and

$$\alpha\pi_m = \pi_m\phi_8, \quad \alpha(q_j) = q_{j+1}, \quad \alpha^8 = 1, \quad \alpha^4 \neq 1. \quad (27.114)$$

The last two tests make the order exactly eight. Merely knowing that the order divides eight fails this condition.

3. *Parity factorization.* A complete bit table defines a homomorphism $\epsilon : Q_m \rightarrow C_2$ satisfying

$$\chi = \epsilon\pi_m, \quad \epsilon(q_j) = 1 \quad (0 \leq j < 8). \quad (27.115)$$

Equivalently $K_m \subseteq K_\chi$. Arithmetic origin and C_8 -equivariance do not imply this independent condition.

4. *Global systole.* The Statement 25.6 record has universal scope $K_m \setminus \{1\}$, the identical kernel hash, and the configured computational target interval $[\underline{L}_{\text{prod}}, \overline{L}_{\text{prod}}]$. It must report

$$\begin{aligned} \underline{L}_m &> \overline{L}_{\text{prod}}, \\ \text{verdict} &= \text{PROVED_STRICT}, \\ \text{dangerous_set_status} &= \text{EMPTY_PROVED}. \end{aligned} \quad (27.116)$$

Equality is a failure of the computational threshold, not a strict pass. A finite dangerous list cannot replace the universal record.

5. *Representation inventory.* The exact list $\mathcal{R}_m$ passes Statement 11.9: each matrix family is a representation, each commutant has dimension one, the characters are pairwise orthonormal, and

$$\sum_{\rho \in \mathcal{R}_m} (\dim \rho)^2 = |Q_m|. \quad (27.117)$$

Approximate matrices without exact reconstruction and a partial list do not pass.

6. *Resource envelope.* One declared algorithm, hardware profile, scalar format, execution schedule, and error policy pass all six Statement 11.10 comparisons:

$$\begin{aligned} \widehat{M} &\leq M_*, & \widehat{W} &\leq W_*, & \widehat{T} &\leq T_*, \\ E_{\text{asm}} &\leq \varepsilon_{\text{asm}}, & E_{\text{solve}} &\leq \varepsilon_{\text{solve}}, & E_{\text{obs}} &\leq \varepsilon_{\text{obs}}. \end{aligned} \quad (27.118)$$

Budget equality passes. Failure of an upper estimate means not certified under that declared implementation; it does not by itself prove absolute computational impossibility.

Criterion 25.8.1 (Computational admissibility). *After candidate coherence passes,*

$$\text{Admissible}(\mathfrak{C}_m) \iff \bigwedge_{i=1}^6 G_i(\mathfrak{C}_m) = 1. \quad (27.119)$$

Proof. Necessity is the declared computational specification: each condition represents a separate property needed by the subsequent construction. Conversely, the six exact certificates respectively provide a valid marked quotient shell, faithful order-eight symmetry, parity descent, strict global geometric separation, complete Fourier sector inventory, and a certified feasible execution envelope. Their shared candidate hashes make the data composable, so the conjunction is sufficient. $\square$

Deterministic decision algorithm. Input consists of $\mathfrak{C}_m$, the passing Statement 25.7 parent and level-predicate hashes, an exact proof that $m \in \mathcal{I}$, six mandatory nonempty certificate records, the computational target, the representation field description, and the resource record. The Statement 25.7 parent must descend from the corrected complex-pair norm bound and its $a = 2R$ systolic slope; an obsolete $a = 4R$ lineage is rejected as stale input. First validate schemas and cross-record identities in canonical field order. If coherent, evaluate all six conditions independently in the fixed order above, retaining every classification and witness rather than short-circuiting after the first failure. For condition i , define its complete result record

$$\begin{aligned} r_i &:= (\text{classification}_i, h_{\text{cert},i}, h_{\text{result rows},i}, \text{witness}_i \text{ or NONE}), \\ \text{condition_result_vector} &:= (r_{\text{shell}}, r_{C_8}, r_{\text{parity}}, r_{\text{systole}}, r_{\text{reps}}, r_{\text{resource}}). \end{aligned} \quad (27.120)$$

Each classification is SATISFIED or a condition-specific failure. The witness is NONE exactly on success and otherwise is the minimal exact failure witness. Result-row and certificate hashes are always present, so both decision branches retain all diagnostics. No condition record may be absent or not requested.

Define the candidate identifier canonically by

$$\text{candidate_id} := H(\text{tower_id}, h_{\text{tower}}, h_{\mathcal{I}}, m, h_Q, h_{\pi}, h_K, h_{\text{target}}, h_{\text{cert},1}, \dots, h_{\text{cert},6}).$$

If all entries pass, return

$$\begin{aligned} &(\text{candidate_id}, h_Q, h_{\pi}, h_K, h_{\text{tower}}, \\ &\text{condition_result_vector}, \text{computational_target}, \text{PRODUCTION_ADMISSIBLE}, \text{COMPLETE}, \text{SATISFIED}, \\ &h_{\text{input}}, h_{\text{independentverification}}). \end{aligned} \quad (27.121)$$

If one or more conditions fail, the same envelope instead contains

$$\begin{aligned} &(\text{condition_result_vector}, \text{first_failed_gate}, \text{all_failure_witnesses}, \\ &\text{PRODUCTION_REJECTED}, \text{COMPLETE}, \text{VIOLATED}, \\ &h_{\text{input}}, h_{\text{independentverification}}). \end{aligned} \quad (27.122)$$

The first failure is selected only for deterministic indexing; all failure evidence remains serialized. Schema or identity failure prevents meaningful condition composition and returns

$$\text{REJECT}(\text{INVALID_INPUT}, \text{stage}, \text{first_mismatch}, h_{\text{input}}). \quad (27.123)$$

Logical boundary. Computational rejection does not retract valid arithmetic conclusions. For example, an exact congruence homomorphism may remain a theorem even when its reduced rotation has smaller order, parity does not descend, the strict length target is unresolved, an irreducible is missing, or the selected algorithm exceeds its resource estimate. Conversely, large order, arithmetic provenance, or favorable asymptotics cannot stand in for any missing condition.

Theorem 25.8.2 (Soundness, completeness, and termination). *The algorithm returns*

PRODUCTION_ADMISSIBLE

exactly for the candidates satisfying (27.119). It terminates, and equal canonical inputs produce byte-identical outputs.

Proof. Candidate coherence is a finite exact comparison. The shell, automorphism, parity, and representation tests range over finite tables; the systole condition checks a completed proof record; and the resource condition evaluates finitely many certified inequalities. Therefore all six decisions terminate. The success branch is exactly their conjunction and the failure branch its negation, proving soundness and completeness relative to the declared specification. Fixed carrier, condition, witness, field, and byte orders prove independent verification determinism. $\square$

Mandatory tests. The suite contains one coherent all-pass fixture and six single-fault fixtures, one for each condition. Boundary tests include a surface relation without surjectivity; a shell collision; an action of order one or four; a missing parity factor; systole equality at the target; a finite rather than universal dangerous list; the C_2 inventory with its sign representation omitted; exact resource-budget equality; an exceeded upper estimate; a mixed level or carrier hash; a projective record substituted for the linear kernel; reordered but canonically equivalent input; and independent verification. Every noncoherence test is `INVALID_INPUT`; every coherent single-fault test is `PRODUCTION_REJECTED` with exactly the intended first failed condition.

26 Congruence covering towers

26.1 Prime-ideal-power tower

Fixed arithmetic data. Retain the faithful integral representation

$$\rho_E : \Gamma_B \longrightarrow \mathrm{SL}_2(\mathcal{O}_E) \quad (28.1)$$

certified in C25 and one nonzero proper prime ideal $\mathfrak{p} \subset \mathcal{O}_E$, equipped with passing Statement 24.1 prime-ideal and Statement 24.2 residue-field records. A Statement 24.3 `GOOD` record is optional additional orbit-survival evidence; it is not required to define the linear prime-power tower. A Statement 24.3 `EXCLUDED` record blocks only the conclusions whose hypotheses require that orbit-survival condition. It does not say that the residue ring, reduction homomorphism, or tower fails to exist. Put $q = N\mathfrak{p}$. For every $m \geq 1$, the quotient map of rings induces

$$\rho_m : \Gamma_B \longrightarrow \mathrm{SL}_2(\mathcal{O}_E/\mathfrak{p}^m), \quad \rho_m(\gamma) = \rho_E(\gamma) \bmod \mathfrak{p}^m. \quad (28.2)$$

Reduction respects sums, products, determinants, and inverses, so ρ_m is a group homomorphism.

Definition 26.1.1 (Linear prime-power tower).

$$\begin{aligned} K_m(\mathfrak{p}) &:= \ker \rho_m \\ &= \{\gamma \in \Gamma_B : \rho_E(\gamma) - I \in M_2(\mathfrak{p}^m)\}, \\ Q_m(\mathfrak{p}) &:= \rho_m(\Gamma_B). \end{aligned} \quad (28.3)$$

The four entry-membership tests in (28.3) are exact ideal-membership computations. The definition is linear; a projective scalar congruence does not silently replace I .

Theorem 26.1.2 (Finite normal covering levels). *For every $m \geq 1$,*

$$K_m(\mathfrak{p}) \triangleleft \Gamma_B, \quad [\Gamma_B : K_m(\mathfrak{p})] = |Q_m(\mathfrak{p})| < \infty. \quad (28.4)$$

Consequently

$$X_m(\mathfrak{p}) := K_m(\mathfrak{p}) \backslash \mathbb{H}_R^2 \longrightarrow X := \Gamma_B \backslash \mathbb{H}_R^2 \quad (28.5)$$

is a finite regular hyperbolic covering with deck group Q_m .

Proof. A homomorphism kernel is normal, and the first isomorphism theorem identifies the quotient with the image. Since

$$|\mathcal{O}_E/\mathfrak{p}^m| = q^m, \quad (28.6)$$

the ambient matrix set is finite, hence so is Q_m . A finite-index normal subgroup of the torsion-free cocompact Fuchsian group defines the stated regular cover. $\square$

What is and is not certified. The construction gives the exact image subgroup

$$Q_m \leq \mathrm{SL}_2(\mathcal{O}_E/\mathfrak{p}^m). \quad (28.7)$$

Equality requires an independent generation or strong-approximation certificate at that level. Neither good reduction nor the cardinality q^m proves surjectivity onto the ambient special linear group. Likewise, one computed level is not evidence for all levels: the tower is committed by the universal predicate (28.3).

Computable construction. Input consists of the exact field/order record, the integral matrix images of a fixed ordered generating tuple, passing Statement 24.1 and Statement 24.2 records for the prime ideal and residue field, the linear-kernel convention, the fixed symbolic level predicate $\mathbb{Z}_{\geq 1}$, and one requested finite level $m \geq 1$. Both level fields are mandatory: the predicate defines the universal tower, while m selects the finite record constructed in this run. A Statement 24.3 record is an optional tagged field. If supplied, verify its classification and parent hash, but neither GOOD nor EXCLUDED controls verification of this tower constructor. Verify primality, properness, the residue norm, denominator avoidance, determinant one, the surface relator, faithfulness lineage, and all mandatory parent hashes. For a requested level, compute the finite residue ring in canonical basis order, reduce generator matrices, close their image under multiplication, and record the kernel by the exact congruence predicate rather than by an attempted enumeration of Γ_B .

The level record is

$$\begin{aligned} &(\text{tower_id}, \mathfrak{p}, q, m, \text{LINEAR}, h_{\rho_E}, \text{residue_ring_hash}, \\ &\text{generator_image_table}, |Q_m|, \text{image_group_hash}, \text{kernel_predicate_hash}, \\ &\text{Statement 24.3_evidence or NONE}, \\ &\text{NORMAL, FINITE_INDEX, COMPLETE, SATISFIED}, h_{\text{input}}, h_{\text{independent verification}}). \end{aligned} \quad (28.8)$$

The tower identifier is the hash

$$\text{tower_id} := H(E, \mathcal{O}_E, \rho_E, \mathfrak{p}, \text{LINEAR}, \text{level_predicate}). \quad (28.9)$$

It is independent of a selected level; each level record binds both this identifier and its own m . No symbolic infinity of tables is materialized. Further levels are obtained by replaying the same terminating level constructor with a new m , while retaining the identical tower identifier.

Invalid input returns

$$\text{REJECT}(\text{INVALID_TOWER_INPUT}, \text{stage}, \text{first_failure}, h_{\text{input}}). \quad (28.10)$$

In particular $m = 0$, $\mathfrak{p} = (0)$, $\mathfrak{p} = \mathcal{O}_E$, a composite ideal, a denominator prime, or a projective/linear schema collision is rejected.

Theorem 26.1.3 (Correctness and termination). *Every passing level record represents exactly (28.2)–(28.5). Finite-level construction terminates, and equal canonical inputs give byte-identical outputs.*

Proof. Ideal arithmetic and reduction in the finite ring terminate. The image closure visits at most q^{4m} matrices. Exact multiplication and equality give a terminating breadth-first closure. Equations (28.2)–(28.4) prove correctness; fixed basis, carrier, generator, field, and byte orders prove deterministic independent verification. $\square$

Minimal tests. As a small independent fixture, let $\mathbb{Z} \rightarrow \mathrm{SL}_2(\mathbb{Z})$ send

$$n \mapsto \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}. \quad (28.11)$$

For $\mathfrak{p} = (3)$, reduction modulo 3^m has image of order 3^m and kernel $3^m\mathbb{Z}$, directly checking (28.3)–(28.4). The boundary suite includes $m = 1$, $m = 2$, $m = 0$, the zero and unit ideals, a nonprime ideal, a bad denominator prime, a relator failure, determinant failure, a proper image subgroup, stale hashes, carrier relabelling, a valid prime carrying optional Statement 24.3 EXCLUDED evidence, all optional-field presence cases, and independent verification.

26.2 Nested congruence kernels

Ideal and kernel filtrations. Fix the passing Statement 26.1 tower. For integers $n \geq m \geq 1$,

$$\mathfrak{p}^n \subseteq \mathfrak{p}^m \implies K_n(\mathfrak{p}) \subseteq K_m(\mathfrak{p}). \quad (28.12)$$

Indeed, every entry of $\rho_E(\gamma) - I$ that lies in $\mathfrak{p}^n$ also lies in $\mathfrak{p}^m$. Thus

$$\Gamma_B \geq K_1 \geq K_2 \geq \cdots \quad (28.13)$$

is a descending normal finite-index filtration.

Theorem 26.2.1 (Canonical transition morphism). *For $n \geq m \geq 1$, the formula*

$$p_{n,m} : Q_n \longrightarrow Q_m, \quad \gamma K_n \longmapsto \gamma K_m \quad (28.14)$$

defines a unique surjective homomorphism satisfying

$$p_{n,m}\rho_n = \rho_m, \quad \ker p_{n,m} = K_m/K_n. \quad (28.15)$$

Proof. If $\gamma K_n = \delta K_n$, then $\delta^{-1}\gamma \in K_n \subseteq K_m$, proving well-definedness. Multiplication is inherited from cosets. Every γK_m is the image of γK_n , proving surjectivity. The kernel identity follows directly. Since ρ_n is surjective onto its image, the commuting identity determines $p_{n,m}$ uniquely. $\square$

Coherence. For $\ell \geq n \geq m \geq 1$,

$$p_{m,m} = \mathrm{id}_{Q_m}, \quad p_{\ell,m} = p_{n,m} \circ p_{\ell,n}. \quad (28.16)$$

Both sides send γK_ℓ to γK_m . Consequently the finite images form an inverse system. Orders obey

$$|Q_m| \mid |Q_n|, \quad |Q_n| = |K_m/K_n| |Q_m|. \quad (28.17)$$

Equality $K_n = K_m$ is allowed: then $p_{n,m}$ is an isomorphism, so strict growth is not inferred from ideal containment alone.

Matrix realization. The residue-ring reduction

$$r_{n,m} : \mathcal{O}_E/\mathfrak{p}^n \twoheadrightarrow \mathcal{O}_E/\mathfrak{p}^m \quad (28.18)$$

applied entrywise satisfies

$$p_{n,m}(\rho_n(\gamma)) = \rho_m(\gamma). \quad (28.19)$$

This computes the transition on the actual image subgroup; it does not require either image to equal its ambient special linear group.

Typed construction. The mandatory input contains one Statement 26.1 tower ID and two passing level records with $n \geq m$, their carrier tables, generator images, kernel-predicate hashes, and the residue reduction map. It also contains the tag

$$\text{coherence_kind} \in \{\text{NONE}, \text{TRIPLE}\}.$$

If the tag is **NONE**, no third-level datum or additional transition datum is present. If it is **TRIPLE**, the input contains a tagged triple (ℓ, n', m') , requires $(n', m') = (n, m)$ and $\ell \geq n \geq m \geq 1$, contains the third passing level record, its carrier and hashes, and all data needed to construct independently $p_{\ell,n}, p_{n,m}, p_{\ell,m}$. The tag, its canonical payload (empty for **NONE**), and the hash of their serialization are mandatory input fields. These alternatives are exhaustive: the tag is **NONE** if and only if every optional coherence field is the distinguished token **NONE**, and it is **TRIPLE** if and only if every such field is present.

Verify all parent identities, the common tower, prime, residue and canonical-key conventions, the level order, entrywise reduction, and the images of every marked generator. Enumerate the finite carrier of every transition source, reduce each matrix, locate the unique canonical key in the target, and check every ordered multiplication pair. In the **TRIPLE** branch, separately certify that all three tables are total, single-valued, surjective homomorphisms and agree with direct entrywise residue reduction. The baseline $\text{transition_table}_{n,m}$ must be byte-for-byte identical to the $p_{n,m}$ table used by the triple certificate. Then compute on every key $x \in Q_\ell$ the two serialized rows

$$x \mapsto p_{\ell,m}(x), \quad x \mapsto p_{n,m}(p_{\ell,n}(x)),$$

in the fixed Q_ℓ -carrier order, and compare them entry by entry.

The fixed-arity output is

$$\begin{aligned} &(\text{tower_id}, n, m, h_{Q_n}, h_{Q_m}, \text{transition_table}_{n,m}, h_{p_{n,m}}, \\ &\text{kernel_fiber_keys}, |K_m/K_n|, \text{SURJECTIVE}, \text{HOMOMORPHISM}, \text{coherence_kind}, \\ &h_{\text{coherence request}}, h_{Q_\ell}, \text{transition_table}_{\ell,n}, h_{p_{\ell,n}}, \text{transition_table}_{\ell,m}, h_{p_{\ell,m}}, \\ &h_{\text{direct rows}}, h_{\text{composite rows}}, h_{\text{rowwise comparison}}, \text{composition classification}, \\ &\text{COMPLETE}, \text{SATISFIED}, h_{\text{input}}, h_{\text{independent verification}}). \end{aligned} \quad (28.20)$$

The field $h_{\text{coherence request}}$ is always present: in the **NONE** branch it hashes the canonical serialization of that tag with empty payload. For **NONE**, the nine fields from h_{Q_ℓ} through $\text{composition classification}$ are all **NONE**. For **TRIPLE**, none of those nine fields is **NONE**, and

$$\text{composition classification} = \text{SATISFIED}$$

exactly when the two row lists agree at every Q_ℓ -key. The comparison hash binds the request hash, all three level hashes, all three transition-table hashes, both row-list hashes, and the entrywise verdicts. Hence a purported composition certificate cannot be detached from the three maps that it compares.

An unknown tag, a foreign tower, mixed prime or convention, $n < m$ or $\ell < n$, triple/primary pair mismatch, tag/presence mismatch, missing or nonunique image key, a nontotal, nonsurjective, nonhomomorphic, or reduction-inconsistent table, baseline/triple table mismatch, failed composition, malformed optional level, or stale request, parent, table, row, or independent verification hash returns

$$\text{REJECT}(\text{INVALID_TRANSITION}, \text{stage}, \text{first_failure}, h_{\text{input}}). \quad (28.21)$$

Theorem 26.2.2 (Correctness and termination). *Every passing table is exactly the canonical map (28.14); all requested coherence identities hold, and construction terminates.*

Proof. The theorem above proves uniqueness and algebraic correctness. Both carriers and all table checks are finite. Fixed level, carrier, generator, pair, and byte orders prove deterministic independent verification. $\square$

Tests. For the unipotent fixture of Statement 26.1, $Q_m \cong \mathbb{Z}/3^m\mathbb{Z}$, and $p_{n,m}$ is ordinary reduction modulo 3^m . Tests include $n = m$, adjacent and skipped levels, a NONE request, a passing three-level composite, an altered composite row, a missing third-level field under TRIPLE, a forbidden third-level field under NONE, $n < m$, equal consecutive kernels, foreign primes, carrier relabelling, a missing fiber, an altered generator image, stale hashes, and independent verification.

26.3 Growth and exact certification of quotient orders

Setting and quantities. Fix the faithful integral representation and the prime-power tower of Statement 26.1, and retain the canonical transitions of Statement 26.2. Write

$$R_m := \mathcal{O}_E/\mathfrak{p}^m, \quad k := \mathcal{O}_E/\mathfrak{p} \cong \mathbb{F}_q, \quad A_m := \mathrm{SL}_2(R_m). \quad (28.22)$$

There are three different numbers: the ambient order $|A_m|$, the actual image order $|Q_m|$, and the index $[\Gamma_B : K_m]$. The first is determined by q, m ; the last two are equal, but need not equal the first.

Lemma 26.3.1 (Cardinality of the prime-power residue ring). *For $m \geq 1$, the ring R_m is finite local with maximal ideal $\mathfrak{p}/\mathfrak{p}^m$, and*

$$|R_m| = q^m, \quad |\mathfrak{p}/\mathfrak{p}^m| = q^{m-1}. \quad (28.23)$$

Proof. For $0 \leq j < m$, multiplication after localization at $\mathfrak{p}$ identifies $\mathfrak{p}^j/\mathfrak{p}^{j+1}$ with a one-dimensional vector space over k . Thus every factor in

$$R_m \supset \mathfrak{p}/\mathfrak{p}^m \supset \cdots \supset \mathfrak{p}^{m-1}/\mathfrak{p}^m \supset 0 \quad (28.24)$$

has q elements. Multiplying the factor cardinalities gives $|R_m| = q^m$; omitting the first factor gives the second formula. An element outside $\mathfrak{p}/\mathfrak{p}^m$ has nonzero residue in k , hence is a unit, proving locality and uniqueness of the maximal ideal. $\square$

Lemma 26.3.2 (Successive special-linear reduction). *For $j \geq 1$, reduction induces a surjection*

$$\mathrm{red}_{j+1,j} : A_{j+1} \twoheadrightarrow A_j \quad (28.25)$$

whose kernel has order q^3 .

Proof. Take $A \in A_j$ and lift its four entries arbitrarily to a matrix B over R_{j+1} . Its determinant is $1 + u$, where $u \in I_j := \mathfrak{p}^j/\mathfrak{p}^{j+1}$. The reduction of A modulo $\mathfrak{p}$ is invertible, so at least one entry of B , and therefore the complementary determinant cofactor, is a unit. If that cofactor is c , changing the corresponding entry by $-uc^{-1} \in I_j$ changes the determinant by exactly $-u$: the determinant is affine in each individual entry. The corrected lift has determinant one. This proves surjectivity.

An element of the kernel is uniquely $I + X$ with $X \in M_2(I_j)$. Since $I_j^2 = 0$ for $j \geq 1$,

$$\det(I + X) = 1 + \mathrm{tr} X. \quad (28.26)$$

Hence the determinant-one condition is the single linear equation $\mathrm{tr} X = 0$. The k -space I_j has dimension one, so the solution space of traceless 2-by-2 matrices has dimension three and cardinality q^3 . $\square$

Theorem 26.3.3 (Exact ambient order and actual-image bound). *For every $m \geq 1$,*

$$|A_m| = q^{3(m-1)} |\mathrm{SL}_2(k)| = q^{3m-2}(q^2 - 1), \quad (28.27)$$

and

$$|Q_m| = [\Gamma_B : K_m], \quad |Q_m| \mid q^{3m-2}(q^2 - 1). \quad (28.28)$$

In particular $1 \leq |Q_m| < q^{3m}$.

Proof. A first column of a matrix in $\text{SL}_2(k)$ may be any nonzero vector, giving $q^2 - 1$ choices. For a fixed nonzero first column, the determinant-one equation for the second column is a nonzero affine linear equation in two variables and has q solutions. Therefore

$$|\text{SL}_2(k)| = q(q^2 - 1). \quad (28.29)$$

Iterating (28.25) from level m to level one and using its kernel size gives (28.27). The first isomorphism theorem applied to ρ_m gives the equality in (28.28). Since $Q_m \leq A_m$, Lagrange's theorem gives divisibility. Finally $q^{3m-2}(q^2 - 1) < q^{3m}$. $\square$

Monotonicity, plateaux, and divergence. The transition theorem of Statement 26.2 gives, for $n \geq m$,

$$|Q_n| = |\ker p_{n,m}| |Q_m| = |K_m/K_n| |Q_m|. \quad (28.30)$$

Thus the orders form a nondecreasing divisibility sequence. The inequality is strict exactly when $K_n \subsetneq K_m$; ideal containment by itself does not prove that strictness.

Theorem 26.3.4 (Unbounded actual-image orders). *For the faithful integral model fixed in Statement 26.1,*

$$\bigcap_{m \geq 1} K_m = \{1\}, \quad \sup_{m \geq 1} |Q_m| = \infty. \quad (28.31)$$

Proof. If γ belongs to every K_m , then each entry of $\rho_E(\gamma) - I$ belongs to $\bigcap_m \mathfrak{p}^m$. A nonzero algebraic integer has finite $\mathfrak{p}$ -adic valuation, so this intersection is zero. Hence $\rho_E(\gamma) = I$, and faithfulness gives $\gamma = 1$.

Suppose instead that $(|Q_m|)$ were bounded. Being a nondecreasing sequence of integers, it would be constant from some level m_0 . For $m \geq m_0$, the surjection $p_{m+1,m}$ would then be an isomorphism. Formula (28.15) would give $K_{m+1} = K_m$, so $\bigcap_m K_m = K_{m_0}$. But K_{m_0} has finite index in the infinite group Γ_B , and is therefore infinite, contradicting (28.31)'s first equality. $\square$

Logical boundary. The theorem proves unboundedness, not a levelwise exponential lower bound and not eventual strict growth. Equality

$$|Q_m| = q^{3m-2}(q^2 - 1) \quad (28.32)$$

is equivalent to $\rho_m(\Gamma_B) = A_m$, because a subgroup of a finite group with the same order is the whole group. The forward direction therefore requires an exact generation certificate at the specified level. A strong-approximation result may provide such surjectivity under additional hypotheses, but no such theorem is assumed here.

Finite exact certificate. Input is a passing Statement 26.1 level record at m , including the complete canonical carrier of Q_m . It also contains

$$\text{comparison_kind} \in \{\text{NONE}, \text{PAIR}\}.$$

The request tag and its serialization hash are always present. The **NONE** branch forbids comparison payload. The **PAIR** branch additionally contains a passing Statement 26.2 transition record $p_{n,m}$ with $n \geq m$, both complete carriers, and all parent hashes from the same tower.

Count the Q_m -carrier, compute (28.27) by exact integer arithmetic, test divisibility, and compare the two integers before deciding ambient equality. In the pair branch, count Q_n and the kernel fiber and verify (28.30) by multiplication. The fixed-arity output is

$$\begin{aligned} &(\text{tower_id}, q, m, h_{Q_m}, |Q_m|, [\Gamma_B : K_m], |A_m|, \text{DIVIDES}, \\ &\text{ambient equality classification}, \text{comparison_kind}, h_{\text{request}}, \\ &(n, h_{Q_n}, |Q_n|, |K_m/K_n|, \text{order identity classification}) \text{ or NONE}, \\ &\text{COMPLETE}, \text{SATISFIED}, h_{\text{input}}, h_{\text{independentverification}}). \end{aligned} \quad (28.33)$$

Presence is all-or-none for the parenthesized comparison block. The ambient equality classification is **SURJECTIVE** exactly when the two finite orders are equal and is otherwise **PROPER_IMAGE**; it is never inferred from the upper bound. Unknown tags, partial pair payloads, mismatched tower or levels, duplicate carrier keys, incomplete image closure, a false divisibility or order identity, an incorrect ambient formula, or a stale hash return

$$\text{REJECT}(\text{INVALID_ORDER_CERTIFICATE}, \text{stage}, \text{first_failure}, h_{\text{input}}). \quad (28.34)$$

Theorem 26.3.5 (Correctness and termination of the order certificate). *Every passing record reports the exact order of the actual congruence image, the exact ambient order, and, when requested, the exact transition multiplier. Construction terminates.*

Proof. Completeness of the canonical carriers makes their counts exact. Equations (28.27), (28.28), and (28.30) prove every reported identity. All counts, integer operations, carrier uniqueness checks, and finite table checks terminate. Fixed tower, level, carrier, integer, and byte orders give deterministic independent verification. $\square$

Tests and subsequent use. For the unipotent fixture of Statement 26.1, $|Q_m| = 3^m$, whereas

$$|\text{SL}_2(\mathbb{Z}/3^m\mathbb{Z})| = 3^{3m-2}(3^2 - 1),$$

so the image is proper and the adjacent multiplier is 3. Tests include $m = 1$, $q = 2$, a proper image, an artificially certified full image, an equal-kernel plateau, adjacent and skipped pairs, both request tags, partial payloads, a wrong carrier count, a wrong ambient factor, nondivisibility, a false multiplier, foreign towers, and stale hashes. Statement 26.4 may use the exact indices to compute covering degrees and genera; metric systole growth, however, requires the independent trace hypotheses stated there.

26.4 Global systole growth in the congruence tower

Global quantity and monotonicity. For the linear congruence kernels of Statement 26.1, put

$$X_m := K_m \setminus \mathbb{H}_R^2, \quad s_m := \text{sys}(X_m) = \inf_{1 \neq \gamma \in K_m} \ell_h(\gamma). \quad (28.35)$$

The infimum uses translation length, not displacement from a fixed base point. If $n \geq m$, Statement 26.2 gives $K_n \subseteq K_m$, and hence

$$s_n \geq s_m. \quad (28.36)$$

This is non-strict: equal consecutive kernels, or merely equal shortest lengths, may produce a plateau.

The complete closed dangerous set. Fix $L \geq 0$, and let D_P be the diameter constant of the fixed Bolza fundamental polygon. Retain the complete C15 element-covering list

$$\mathcal{D}_L := \{\delta \in B_{\text{geo}}(L + 2D_P) : \delta \neq 1, \ell_h(\delta) \leq L\}. \quad (28.37)$$

The cutoff is closed. Thus an element of length exactly L is dangerous. C15 proves that (28.37) is finite and that every ambient conjugacy class of translation length at most L has a representative in it. The larger radius $L + 2D_P$ is essential; $B_{\text{geo}}(L)$, a based-displacement sample, or an arbitrary finite trace list is not a complete replacement.

Lemma 26.4.1 (Normal-kernel dangerous-set criterion). *For every $m \geq 1$,*

$$s_m > L \iff K_m \cap \mathcal{D}_L = \emptyset. \quad (28.38)$$

Proof. If $\delta \in K_m \cap \mathcal{D}_L$, then $\delta \neq 1$ and $\ell_h(\delta) \leq L$, so (28.35) gives $s_m \leq L$.

Conversely, if $s_m \leq L$, compactness of X_m realizes its systole by some $1 \neq \gamma \in K_m$ with $\ell_h(\gamma) \leq L$. Completeness of (28.37) supplies $\eta \in \Gamma_B$ such that $\delta = \eta\gamma\eta^{-1} \in \mathcal{D}_L$. Since K_m is normal, $\delta \in K_m$. This proves the reverse implication. $\square$

Exact exit level of a dangerous element. For a nonzero integral matrix $A = (a_{ij})$, define

$$v_{\mathfrak{p}}(A) := \min_{a_{ij} \neq 0} v_{\mathfrak{p}}(a_{ij}). \quad (28.39)$$

Faithfulness implies $\rho_E(\delta) - I \neq 0$ for every $\delta \in \mathcal{D}_L$, so

$$e_{\mathfrak{p}}(\delta) := v_{\mathfrak{p}}(\rho_E(\delta) - I) < \infty. \quad (28.40)$$

By the linear-kernel definition,

$$\delta \in K_m \iff m \leq e_{\mathfrak{p}}(\delta). \quad (28.41)$$

All zero entries are ignored in the minimum; at least one nonzero entry exists, so no infinite valuation is exported.

Define the strict tail threshold

$$M_L := \begin{cases} 1, & \mathcal{D}_L = \emptyset, \\ 1 + \max_{\delta \in \mathcal{D}_L} e_{\mathfrak{p}}(\delta), & \mathcal{D}_L \neq \emptyset. \end{cases} \quad (28.42)$$

Theorem 26.4.2 (Effective systole divergence). *For every $L \geq 0$,*

$$m \geq M_L \implies s_m > L. \quad (28.43)$$

Consequently

$$\lim_{m \rightarrow \infty} \text{sys}(X_m) = +\infty. \quad (28.44)$$

Proof. For $m \geq M_L$, equations (28.40)–(28.42) exclude every $\delta \in \mathcal{D}_L$ from K_m . The intersection in (28.38) is empty, proving the strict inequality (28.43). Since this holds for each real $L \geq 0$, it is exactly the definition of divergence to $+\infty$. $\square$

Why quotient-order growth is insufficient. Statement 26.3 proves both $|Q_m| \rightarrow \infty$ and $\bigcap_m K_m = \{1\}$. The order statement alone does not imply (28.44): order contains no information about whether a particular short element remains in the kernel. The proof above uses the stronger residual statement together with the complete finite dangerous set. It also explains why nested ideals alone, without faithfulness or a separation theorem, do not prove metric growth.

A quantitative arithmetic rate under uniform hypotheses. There is a second, independent route. Assume that for all $m \geq m_0$ and every $1 \neq \gamma \in K_m$, one common nondistinguished-place cap profile from corrected Statement 25.4 applies, with

$$\mathcal{C} = \prod_{k=1}^7 D_k^2 > 0, \quad B_m = \frac{q^m}{\sqrt{\mathcal{C}}}. \quad (28.45)$$

The constants and their family-domain hash must be independent of m and γ . In the sign-free branch also require

$$B_m \geq 4 \quad (m \geq m_0). \quad (28.46)$$

Then the corrected trace and length chain gives

$$\begin{aligned} s_m &\geq 2R \operatorname{arcosh} \left(\frac{B_m - 2}{2} \right) \\ &\geq 2R \log \left(\frac{B_m}{4} \right) = 2Rm \log q - 2R \log(4\sqrt{\mathcal{C}}) \\ &\geq 2Rm \log q - C_{\text{sf}}, \end{aligned} \quad (28.47)$$

where

$$C_{\text{sf}} := 2R \max\{0, \log(4\sqrt{C})\}. \quad (28.48)$$

The first inequality takes an infimum only after the cap theorem has covered every nonidentity kernel element. The second uses $\text{arcosh } y \geq \log y$ and $B_m - 2 \geq B_m/2$.

If instead an exact family-wide positive-real-trace certificate is present, corrected Statement 25.5 yields

$$s_m \geq 2R \log \left(1 + \frac{B_m}{2}\right) \geq 2Rm \log q - C_+, \quad C_+ := 2R \max\{0, \log(2\sqrt{C})\}. \quad (28.49)$$

Congruence sign normalization does not supply this positive-sign certificate. In either branch the coefficient of $m \log q$ is $2R$, not $4R$.

For $C_\beta \in \{C_{\text{sf}}, C_+\}$, the explicit rate threshold

$$M_{L,\beta}^{\text{rate}} := \max \left\{ m_0, \left\lfloor \frac{L + C_\beta}{2R \log q} \right\rfloor + 1 \right\} \quad (28.50)$$

satisfies $m \geq M_{L,\beta}^{\text{rate}} \Rightarrow s_m > L$. The added one enforces strictness when the quotient in (28.50) is an integer.

Typed finite construction. The mandatory input contains the fixed linear tower, one exact threshold $L \geq 0$, a complete passing Statement 15.8 element-covering record through radius $L + 2D_P$, exact matrices for every retained key, and all lineage hashes. Class mode is allowed only with a passing complete conjugacy-partition certificate; equal traces alone never deduplicate elements. For every dangerous key, independently recompute its word, matrix, nonidentity classification, closed length comparison, and valuation (28.40), then compute (28.42).

An optional single-level query has the tagged form

$$\text{level_check_kind} \in \{\text{ABSENT}, \text{PRESENT}\}, \quad \text{PRESENT} \implies m_{\text{test}} \geq 1.$$

Its tag and request hash are mandatory. The absent branch forbids a level payload. The present branch requires exactly one integer m_{test} in the same tower and checks every valuation row.

An optional rate request has the tag

$$\text{rate_kind} \in \{\text{ABSENT}, \text{SIGN_FREE}, \text{POSITIVE}\}. \quad (28.51)$$

Its tag and request hash are always present. The absent branch forbids rate payload. Either present branch requires a kernel-wide, all-level family record with the same tower, prime, scale, kernel convention, and scope hash; the sign-free branch verifies (28.46), and the positive branch verifies its independent sign certificate.

The fixed-arity completed output is

$$\begin{aligned} &(\text{tower_id}, L, h_{\mathcal{D}_L}, |\mathcal{D}_L|, \text{valuation_rows}, M_L, \\ &\text{COMPLETE_CLOSED_DANGEROUS_SET}, \text{STRICT_TAIL_PROVED}, \\ &\text{level_check_kind}, h_{\text{level_request}}, m_{\text{test}} \text{ or NONE}, \\ &\text{level_status}, \text{level_witness} \text{ or NONE}, \\ &\text{rate_kind}, h_{\text{rate_request}}, \\ &(m_0, q, R, C, C_\beta, 2R, M_{L,\beta}^{\text{rate}}, h_{\text{family_proof}}) \text{ or NONE}, \\ &\text{COMPLETE}, \text{terminal_status} \in \{\text{SATISFIED}, \text{VIOLATED}\}, h_{\text{input}}, h_{\text{independentverification}}). \end{aligned} \quad (28.52)$$

For an absent level query, $m_{\text{test}} = \text{NONE}$, $\text{level_status} = \text{NOT_REQUESTED}$, the witness is NONE , and the terminal classification is SATISFIED . For a present query, exhaustively test

$$m_{\text{test}} \leq e_p(\delta) \quad (\delta \in \mathcal{D}_L).$$

If no row passes this membership test, the level classification is `PROVED_STRICT`, the witness is `NONE`, and the terminal classification is `SATISFIED`. Otherwise the level classification is `COUNTEREXAMPLE`, the terminal classification is `VIOLATED`, and the witness is the least offending canonical key together with its word, matrix hash, certified closed length comparison, valuation, reduced identity matrix, and the inequality $m_{\text{test}} \leq e_p(\delta)$. Thus a failure is mathematical evidence that $s_{m_{\text{test}}} \leq L$, not a structural error.

The rate block is wholly absent exactly for `rate_kind = ABSENT` and wholly present otherwise. Unknown level or rate tags, partial payloads, a radius smaller than $L + 2D_P$, open instead of closed cutoff, incomplete enumeration, based/global substitution, unproved class deduplication, identity leakage, nonfaithful lineage, mixed primes or kernel conventions, a missing valuation, a nonuniform cap, absent positive-sign proof, obsolete q^{2m}/C lineage, or stale hashes returns

$$\text{REJECT}(\text{INVALID_SYSTOLE_GROWTH_INPUT}, \text{stage}, \text{first_failure}, h_{\text{input}}). \quad (28.53)$$

If an exact cutoff comparison cannot decide equality, no completed dangerous list or tail threshold is emitted. The fixed hold record is

$$\text{HOLD}(\text{UNDECIDED_CUTOFF}, \text{stage}, \text{first_undecidable}, h_{\text{input}}, h_{\text{verified prefix}}). \quad (28.54)$$

It is neither `SATISFIED` nor `VIOLATED` and proves no systolic assertion. The four terminal classes are therefore disjoint: structural or independent verification failure gives (28.53), undecidable exact comparison gives (28.54), and only a complete independent verification gives the `SATISFIED` or witness-bearing `VIOLATED` branch of (28.52).

Theorem 26.4.3 (Certificate correctness and termination). *Every completed output of (28.52), regardless of its terminal classification, proves the tail statement (28.43). If `level_check_kind = PRESENT`, then terminal `SATISFIED` proves $s_{m_{\text{test}}} > L$, while terminal `VIOLATED` proves $s_{m_{\text{test}}} \leq L$ by its witness. If the level query is absent, no single-level assertion is made. Every present rate block proves (28.47) or (28.49) on its declared all-level domain. The finite construction terminates and replays deterministically.*

Proof. C15 makes $\mathcal{D}_L$ finite and complete. Exact number-field arithmetic computes all valuations, so (28.41)–(28.43) prove the mandatory output. Corrected Statement 25.4–Statement 25.6 prove the optional kernel-wide length floors, and the displayed inequalities derive the rate. All word, carrier, interval, valuation, and hash lists are finite. Fixed key, place, level, interval, and byte orders prove termination and deterministic independent verification. $\square$

Boundary tests and subsequent use. Tests include an empty dangerous list, a one-element list of valuation zero, several unequal valuations, length exactly L , an insufficient L -radius scan, an insufficient based scan, normal and deliberately nonnormal kernels, equal-trace distinct elements, absent and present level checks, a strict level pass, the least-key failure witness, malformed level payloads, all rate tags, $B_{m_0} = 4$, $B_{m_0} < 4$, a level-dependent cap, missing positive sign, the obsolete squared bound, strict-threshold equality in (28.50), undecidable cutoff, foreign parents, stale hashes, and independent verification. Statement 26.5 may use the resulting tail only after separately proving that the C_8 action preserves every kernel; Statement 26.6 treats parity by a separate subgroup argument.

26.5 Eightfold equivariance at every prime-power level

Integral covariance at level m . Retain the arithmetic symmetry of Statement 23.6:

$$C = \begin{pmatrix} \xi & 0 \\ 0 & \xi^{-1} \end{pmatrix} \in \text{SL}_2(\mathcal{O}_E), \quad \rho_E(\phi_8(\gamma)) = C \rho_E(\gamma) C^{-1}. \quad (28.55)$$

Here C^{-1} is also integral, $C^8 = -I$, and ϕ_8 has exact order eight on Γ_B . Let C_m denote the reduction of C in $\mathrm{SL}_2(R_m)$. Reduction of (28.55) gives

$$\rho_m(\phi_8(\gamma)) = C_m \rho_m(\gamma) C_m^{-1}. \quad (28.56)$$

This identity holds at every nonzero prime ideal and every $m \geq 1$; no division by a new denominator is required.

Lemma 26.5.1 (Kernel invariance). *For every $m \geq 1$,*

$$\phi_8(K_m) = K_m. \quad (28.57)$$

Proof. If $\gamma \in K_m$, write $\rho_E(\gamma) = I + B$ with $B \in M_2(\mathfrak{p}^m)$. Equation (28.55) gives

$$\rho_E(\phi_8(\gamma)) - I = C B C^{-1}. \quad (28.58)$$

Every entry on the right is a finite sum of products of integral entries of C, C^{-1} with entries in $\mathfrak{p}^m$; the ideal property puts it in $\mathfrak{p}^m$. Thus $\phi_8(K_m) \subseteq K_m$. Applying the same argument to ϕ_8^{-1} , whose matrix normalizer is C^{-1} , gives the reverse inclusion. $\square$

Theorem 26.5.2 (Descended action and its well-definedness). *The formula*

$$\alpha_m : Q_m \longrightarrow Q_m, \quad \alpha_m(\gamma K_m) := \phi_8(\gamma) K_m \quad (28.59)$$

is a unique automorphism satisfying

$$\alpha_m(\rho_m(\gamma)) = C_m \rho_m(\gamma) C_m^{-1}, \quad \rho_m \phi_8 = \alpha_m \rho_m, \quad \alpha_m^8 = \mathrm{id}_{Q_m}. \quad (28.60)$$

Proof. If $\gamma K_m = \delta K_m$, then $\delta^{-1}\gamma \in K_m$. Kernel invariance gives $\phi_8(\delta)^{-1}\phi_8(\gamma) \in K_m$, so (28.59) is independent of the representative. It is a homomorphism because ϕ_8 is, and the same construction with ϕ_8^{-1} gives its inverse. Equation (28.56) proves the conjugation formula and the commuting square. Surjectivity of ρ_m onto Q_m makes α_m unique. Finally $\alpha_m^8 \rho_m = \rho_m \phi_8^8 = \rho_m$, and surjectivity again gives the last identity. $\square$

Action order versus normalizing-matrix order. Equation (28.60) only proves

$$d_m := \mathrm{ord}(\alpha_m) \mid 8. \quad (28.61)$$

The order of C_m is not d_m : central powers of C_m act trivially by conjugation. In odd residue characteristic C_m may have order 16; in characteristic two its central signs may coalesce. Neither fact alone decides the order of α_m . Likewise, $d_m = 8$ does not imply $8 \mid |Q_m|$: an automorphism's order is an order in $\mathrm{Aut}(Q_m)$, not the order of an element of Q_m . Since $d_m \mid 8$, exact order eight is equivalent to

$$\alpha_m^4 \neq \mathrm{id}_{Q_m}, \quad (28.62)$$

and can be proved by one moved carrier key. The stronger seven-row shell certificate

$$\alpha_m^k(\rho_m(G_0)) \neq \rho_m(G_0) \quad (1 \leq k < 8)$$

is sufficient but not necessary.

Equivariance of the inverse system. For $n \geq m$, the canonical transition map of Statement 26.2 satisfies

$$p_{n,m}\alpha_n = \alpha_m p_{n,m}. \quad (28.63)$$

Indeed, both sides send γK_n to $\phi_8(\gamma)K_m$. Consequently

$$d_m \mid d_n \mid 8. \quad (28.64)$$

To prove the first divisibility, write d_n for the order upstairs; then surjectivity of $p_{n,m}$ and (28.63) give $\alpha_m^{d_n} p_{n,m} = p_{n,m}$, hence $\alpha_m^{d_n} = 1$. Thus action order can rise along the tower but cannot fall. In particular,

$$d_{m_0} = 8 \implies d_n = 8 \quad (n \geq m_0). \quad (28.65)$$

A passing first-level good-orbit certificate from Statement 24.3 therefore forces exact order eight at every higher prime-power level. If that certificate is absent, each finite action table still determines d_m exactly; failure of the sufficient orbit test is not evidence of collapse.

Linear and projective scopes. This subsection concerns the linear kernel $\rho_m^{-1}(I)$. A projective quotient instead uses the preimage of a scalar subgroup and needs its own scalar-kernel, normalization, carrier, and orbit records. No projective action or exact-order assertion is inferred from the linear table.

Geometric lift on the covers. The disk rotation represented projectively by C normalizes K_m , so

$$\hat{r}_m : K_m \backslash \mathbb{H}_R^2 \longrightarrow K_m \backslash \mathbb{H}_R^2, \quad K_m z \longmapsto K_m(Cz) \quad (28.66)$$

is well defined. If $K_m z = K_m z'$, then $z' = kz$ for some $k \in K_m$, and $Cz' = (CkC^{-1})Cz$ with $CkC^{-1} \in K_m$. Its inverse is induced by C^{-1} , and it covers the order-eight rotation of the Bolza surface. This cover isometry and the automorphism α_m are related by conjugation on deck transformations, but their order statements are distinct: the finite deck-group action may have order $d_m < 8$.

Exact finite construction. Input contains the Statement 23.6 integral normalizer record, the exact ϕ_8 generator table, one passing Statement 26.1 level m with the complete Q_m carrier and multiplication table, and all parent hashes. Reduce C, C^{-1} , verify determinant and inverse rows, independent verification (28.56) on every marked generator, and conjugate every carrier matrix. Require a total single-valued carrier permutation preserving all multiplication and inverse rows. Compare each discovery word with its exact ϕ_8 -image, then compose the action table until the first identity power in $\{1, 2, 4, 8\}$ is found.

An optional transition request has the tag

$$\text{transition_kind} \in \{\text{NONE}, \text{PAIR}\}. \quad (28.67)$$

The tag and request hash are mandatory. The PAIR branch contains a passing Statement 26.2 record $p_{n,m}$, the complete action table at n , and common tower hashes. It checks (28.63) on every canonical key of Q_n and verifies $d_m \mid d_n$. The NONE branch forbids all transition payload.

The fixed output is

$$\begin{aligned} &(\text{tower_id}, m, C_m, C_m^{-1}, \text{covariance_rows}, \text{action_table}_m, \\ &\text{equivariance_rows}, \text{kernel_stability_rows}, \text{power_rows}, d_m, \\ &\text{order_status} \in \{\text{DIVISOR_OF_8}, \text{EXACTLY_8}\}, \\ &\text{transition_kind}, h_{\text{transition request}}, \\ &(n, h_{p_{n,m}}, h_{\alpha_n}, d_n, \text{commuting_rows}, \text{ORDER_DIVISIBILITY_PASS}) \text{ or NONE,} \\ &\text{cover_lift_record}, \text{COMPLETE}, \text{SATISFIED}, h_{\text{input}}, h_{\text{independent verification}}). \end{aligned} \quad (28.68)$$

The order classification is `EXACTLY_8` exactly when $d_m = 8$; otherwise it is `DIVISOR_OF_8` and the exact value $d_m \in \{1, 2, 4\}$ remains explicit. The transition block is present wholly and only for `PAIR`.

An unknown tag, partial pair, nonintegral normalizer, mixed tower or level, failed covariance, missing or duplicate action image, a nonbijective or nonhomomorphic table, a false identity power, unsupported exact-eight claim, failed commuting row, incorrect order divisibility, a matrix-order/action-order substitution, or stale hash returns

$$\text{REJECT}(\text{INVALID_EQUIVARIANCE_INPUT}, \text{stage}, \text{first_failure}, \text{h}_{\text{input}}). \quad (28.69)$$

Theorem 26.5.3 (Construction correctness and termination). *Every passing record contains the unique descended automorphism, its exact finite order, a complete equivariance certificate, and a well-defined cover lift. Every requested transition square commutes. The construction terminates and replays deterministically.*

Proof. Equations (28.57)–(28.60) prove kernel stability, well-definedness, and uniqueness. Exhaustive carrier permutation and product checks construct the exact automorphism. Since $\alpha_m^8 = 1$, at most eight finite table compositions determine its exact order. Equation (28.63) proves each pair record. All carriers, tables, power lists, and hash lists are finite; fixed generator, carrier, power, level, and byte orders prove termination and independent verification. $\square$

Tests and subsequent boundary. Tests include odd and characteristic-two residue rings, matrix order 16 with action order 8, synthetic action orders 1, 2, 4, 8, a moved α_m^4 -key, a collapsed shell witness followed by complete table recovery, a first-level exact-eight certificate and its higher-level propagation, both transition tags, a changed commuting row, an action-order decrease, failed representative independence, a forged implication $8 \mid |Q_m|$, an attempted linear-to-projective transfer, foreign towers, stale hashes, and independent verification. No concrete matrix extension is called a semidirect product without a separate intersection computation. Statement 26.6 must still verify parity factorization independently.

26.6 Parity-compatible congruence levels

Parity character and two kernels. Let

$$\chi : \Gamma_B \twoheadrightarrow C_2 = \{0, 1\} \quad (28.70)$$

be the presentation-compatible parity homomorphism of C09, written additively, and retain its exact symmetry identity

$$\chi \circ \phi_8 = \chi. \quad (28.71)$$

For every linear congruence kernel K_m , define

$$\widetilde{K}_m := K_m \cap \ker \chi, \quad \widetilde{Q}_m := \Gamma_B / \widetilde{K}_m. \quad (28.72)$$

Both factors in the intersection are normal and of finite index. Therefore $\widetilde{K}_m \triangleleft \Gamma_B$ has finite index.

Theorem 26.6.1 (Exact parity-factorization criterion). *There exists a homomorphism $\chi_m : Q_m \rightarrow C_2$ satisfying*

$$\chi = \chi_m \circ \rho_m \quad (28.73)$$

if and only if

$$K_m \subseteq \ker \chi. \quad (28.74)$$

When it exists, χ_m is unique and surjective.

Proof. If (28.73) holds and $k \in K_m$, then $\chi(k) = \chi_m(e) = 0$, proving necessity. Conversely, assume (28.74) and define $\chi_m(\gamma K_m) := \chi(\gamma)$. If $\gamma K_m = \delta K_m$, then $\delta^{-1}\gamma \in K_m \subseteq \ker \chi$, so $\chi(\gamma) = \chi(\delta)$; this separately proves well-definedness. The homomorphism identity follows from that of χ , and (28.73) is the definition. Surjectivity of ρ_m gives uniqueness, while surjectivity of χ gives surjectivity of χ_m . $\square$

The augmented quotient always carries parity. Regardless of (28.74), define

$$\tilde{\chi}_m : \tilde{Q}_m \rightarrow C_2, \quad \tilde{\chi}_m(\gamma \tilde{K}_m) := \chi(\gamma). \quad (28.75)$$

This is well defined because $\tilde{K}_m \subseteq \ker \chi$, and it is the unique surjective map with $\chi = \tilde{\chi}_m \tilde{\rho}_m$.

The product homomorphism

$$\Psi_m = (\rho_m, \chi) : \Gamma_B \longrightarrow Q_m \times C_2 \quad (28.76)$$

has kernel exactly $\tilde{K}_m$. Hence the first isomorphism theorem identifies

$$\tilde{Q}_m \cong \text{im } \Psi_m. \quad (28.77)$$

Proposition 26.6.2 (Graph/product dichotomy). *Exactly one of the following occurs.*

(i) *If $K_m \subseteq \ker \chi$, then*

$$\text{im } \Psi_m = \{(q, \chi_m(q)) : q \in Q_m\}, \quad \tilde{K}_m = K_m, \quad |\tilde{Q}_m| = |Q_m|. \quad (28.78)$$

(ii) *If $K_m \not\subseteq \ker \chi$, then*

$$\text{im } \Psi_m = Q_m \times C_2, \quad [K_m : \tilde{K}_m] = 2, \quad |\tilde{Q}_m| = 2|Q_m|. \quad (28.79)$$

Proof. Under (28.74), equation (28.73) makes every pair in (28.76) equal to $(q, \chi_m(q))$; surjectivity onto Q_m gives the whole graph. The kernel equality then gives $\tilde{K}_m = K_m$.

If (28.74) fails, choose $k \in K_m$ with $\chi(k) = 1$. Then $\Psi_m(k) = (e, 1)$. For any $q \in Q_m$, choose γ with $\rho_m(\gamma) = q$; multiplying $\Psi_m(\gamma)$ by either $(e, 0)$ or $(e, 1)$ produces both $(q, 0)$ and $(q, 1)$. Thus the image is the full product. Restriction of χ to K_m is surjective with kernel $\tilde{K}_m$, so the remaining index and order formulas follow. $\square$

Nested tower and bonding maps. For $n \geq m$,

$$\tilde{K}_n \subseteq \tilde{K}_m, \quad \tilde{\rho}_{n,m} : \tilde{Q}_n \twoheadrightarrow \tilde{Q}_m, \quad \gamma \tilde{K}_n \mapsto \gamma \tilde{K}_m. \quad (28.80)$$

The inclusion proves representative independence; the displayed formula proves homomorphism and surjectivity, and

$$\ker \tilde{\rho}_{n,m} = \tilde{K}_m / \tilde{K}_n, \quad \tilde{\chi}_m \tilde{\rho}_{n,m} = \tilde{\chi}_n. \quad (28.81)$$

The graph classification is upward persistent: if (28.74) holds at m , it holds at every $n \geq m$. Equivalently, the product classification at level n forces product classification at every lower level $m \leq n$. No assertion is made that a switch to the graph branch must occur.

Eightfold action and exact order. Equations (28.57) and (28.71) give

$$\phi_8(\tilde{K}_m) = \phi_8(K_m) \cap \phi_8(\ker \chi) = K_m \cap \ker \chi = \tilde{K}_m. \quad (28.82)$$

Thus

$$\tilde{\alpha}_m(\gamma \tilde{K}_m) := \phi_8(\gamma) \tilde{K}_m \quad (28.83)$$

is well defined by the same representative argument as Statement 26.5, and

$$\tilde{\chi}_m \tilde{\alpha}_m = \tilde{\chi}_m, \quad \tilde{p}_{n,m} \tilde{\alpha}_n = \tilde{\alpha}_m \tilde{p}_{n,m}. \quad (28.84)$$

Under the graph identification (28.78), $\tilde{\alpha}_m$ is α_m transported by the graph isomorphism. Under the product identification (28.79), it is

$$\tilde{\alpha}_m(q, b) = (\alpha_m(q), b), \quad (28.85)$$

because parity is invariant. Therefore in both branches

$$\text{ord}(\tilde{\alpha}_m) = \text{ord}(\alpha_m). \quad (28.86)$$

In particular an exact-eight conclusion is inherited precisely when it was actually certified for the base action; parity augmentation does not manufacture or destroy action order.

Covering and systolic consequences. The subgroup inclusion $\tilde{K}_m \subseteq K_m$ produces a finite regular cover

$$\tilde{X}_m := \tilde{K}_m \backslash \mathbb{H}_R^2 \longrightarrow X_m := K_m \backslash \mathbb{H}_R^2 \quad (28.87)$$

of degree 1 in the graph case and 2 in the product case. Taking an infimum over the smaller nonidentity set gives

$$\text{sys}(\tilde{X}_m) \geq \text{sys}(X_m). \quad (28.88)$$

Hence the divergence and every valid lower bound of Statement 26.4 transfer to the parity-compatible tower. This conclusion uses subgroup inclusion, not parity factorization through Q_m .

Finite decision and construction. Input contains the C09 character and invariance records, a passing Statement 26.1 level with complete carrier and marked generators, and the Statement 26.5 action table. Propagate a bit from the identity through the ordered Cayley graph by the marked generator parities. If a key receives two different bits, independent verification the two discovery words; their quotient is an explicit odd element of K_m . If no conflict occurs, the bit table is the unique χ_m of (28.73). This finite procedure decides exactly between (28.78) and (28.79).

Construct the complete pair image of (28.76), its multiplication and inverse tables, the projection to Q_m , $\tilde{\chi}_m$, and the action (28.83). An optional transition request has tag

$$\text{transition_kind} \in \{\text{NONE}, \text{PAIR}\}. \quad (28.89)$$

The pair branch contains complete augmented records at $n \geq m$ and verifies (28.80)–(28.81), (28.84), order divisibility, and graph/product monotonicity on every carrier key. Presence is all-or-none; the tag and request hash always exist.

The fixed output is

$$\begin{aligned} &(\text{tower_id}, m, \text{parity_case} \in \{\text{GRAPH}, \text{PRODUCT}\}, \\ &\text{bit_table or NONE, odd_kernel_witness or NONE,} \\ &\text{pair_carrier, pair_table, } |\tilde{Q}_m|, \text{projection_table, parity_table,} \\ &\text{augmented_action_table, ord}(\tilde{\alpha}_m), \text{order_equality_row,} \\ &\text{transition_kind, } h_{\text{transition request}}, \text{transition_block or NONE,} \\ &\text{NORMAL, FINITE_INDEX, PARITY_SURJECTIVE, COMPLETE, SATISFIED, } h_{\text{input}}, h_{\text{independent verification}}). \end{aligned} \quad (28.90)$$

In the graph branch the complete bit table is present and the odd-kernel witness is **NONE**. In the product branch the bit table is **NONE** and the least canonical conflict, including both words and their odd kernel quotient, is present.

An unknown tag, partial optional payload, mismatched character, tower, prime, level, or action, inconsistent bit propagation, an invalid conflict word, incomplete pair image, wrong graph/product size, failed parity homomorphism, altered action order, failed bonding row, forbidden branch field, or stale hash returns

$$\text{REJECT}(\text{INVALID_PARITY_TOWER_INPUT}, \text{stage}, \text{first_failure}, \text{h}_{\text{input}}). \quad (28.91)$$

Theorem 26.6.3 (Construction correctness and termination). *Every passing record constructs exactly $\Gamma_B/(K_m \cap \ker \chi)$, decides parity descent to Q_m , and certifies the unique parity and eightfold action on the augmented quotient. Every requested bonding square commutes. The construction terminates and replays deterministically.*

Proof. Finite Cayley-graph propagation either defines a consistent function on every Q_m -key or produces two words for the same key with opposite parity. The factorization theorem proves the two outcomes complete and mutually exclusive. Equations (28.76)–(28.86) prove the group, kernel, parity, action, and transition claims. All carriers and tables are finite; fixed key, edge, bit, level, conflict, and byte orders prove termination and independent verification. $\square$

Tests and subsequent boundary. Tests include graph and product fixtures, a least odd-kernel conflict, an altered parity bit, both possible order multipliers 1, 2, exact action orders 1, 2, 4, 8, invariant and deliberately noninvariant characters, upward graph persistence, a forbidden reverse implication, both transition tags, a failed parity bonding row, strict systole inheritance, foreign parents, stale hashes, and independent verification. Statement 26.7 changes the prime and must reconstruct these parity records separately rather than transferring a bit table between residue characteristics.

26.7 Congruence towers at two distinct prime ideals

Two genuinely separate local records. Let $\mathfrak{p} \neq \mathfrak{r}$ be nonzero prime ideals of $\mathcal{O}_E$, with

$$q_{\mathfrak{p}} = N\mathfrak{p}, \quad q_{\mathfrak{r}} = N\mathfrak{r}.$$

Apply Statement 26.1 separately to obtain

$$K_m^{\mathfrak{p}} := \ker \rho_{\mathfrak{p},m}, \quad K_n^{\mathfrak{r}} := \ker \rho_{\mathfrak{r},n}. \quad (28.92)$$

The two tower identifiers include the canonical ideal bases and are therefore different. Distinct prime ideals can nevertheless lie over the same rational prime; neither distinct residue characteristics nor nonisomorphic finite images follows from $\mathfrak{p} \neq \mathfrak{r}$.

Chinese remainder theorem and the actual image. Distinct prime ideals in a Dedekind domain are comaximal, so for $m, n \geq 1$,

$$\mathfrak{p}^m + \mathfrak{r}^n = \mathcal{O}_E, \quad \mathcal{O}_E/(\mathfrak{p}^m \mathfrak{r}^n) \cong \mathcal{O}_E/\mathfrak{p}^m \times \mathcal{O}_E/\mathfrak{r}^n. \quad (28.93)$$

The isomorphism sends a residue class to its two reductions. Its kernel is $\mathfrak{p}^m \cap \mathfrak{r}^n = \mathfrak{p}^m \mathfrak{r}^n$, and surjectivity follows from a Bézout relation for the two powers.

Define the joint homomorphism and kernel

$$\begin{aligned} \rho_{m,n}^{\mathfrak{p},\mathfrak{r}} &:= (\rho_{\mathfrak{p},m}, \rho_{\mathfrak{r},n}), \\ L_{m,n} &:= K_m^{\mathfrak{p}} \cap K_n^{\mathfrak{r}}, \\ H_{m,n} &:= \text{im } \rho_{m,n}^{\mathfrak{p},\mathfrak{r}}. \end{aligned} \quad (28.94)$$

Then

$$\ker \rho_{m,n}^{\mathfrak{p},\mathfrak{r}} = L_{m,n}, \quad H_{m,n} \cong \Gamma_B/L_{m,n}. \quad (28.95)$$

Both coordinate projections from $H_{m,n}$ are surjective, so it is a subdirect product of $Q_m^{\mathfrak{p}}$ and $Q_n^{\mathfrak{r}}$.

Theorem 26.7.1 (Exact direct-product criterion). *For all $m, n \geq 1$,*

$$|H_{m,n}| = \frac{|Q_m^{\mathfrak{p}}||Q_n^{\mathfrak{r}}|}{[\Gamma_B : K_m^{\mathfrak{p}}K_n^{\mathfrak{r}}]}. \quad (28.96)$$

Consequently the following are equivalent:

$$\begin{aligned} H_{m,n} &= Q_m^{\mathfrak{p}} \times Q_n^{\mathfrak{r}}, \\ |H_{m,n}| &= |Q_m^{\mathfrak{p}}||Q_n^{\mathfrak{r}}|, \\ K_m^{\mathfrak{p}}K_n^{\mathfrak{r}} &= \Gamma_B. \end{aligned} \quad (28.97)$$

Proof. The two kernels are normal, so their product is a subgroup. The second isomorphism theorem gives

$$K_m^{\mathfrak{p}}/L_{m,n} \cong K_m^{\mathfrak{p}}K_n^{\mathfrak{r}}/K_n^{\mathfrak{r}}.$$

Multiplying the resulting index identities yields

$$[\Gamma_B : L_{m,n}] = \frac{[\Gamma_B : K_m^{\mathfrak{p}}][\Gamma_B : K_n^{\mathfrak{r}}]}{[\Gamma_B : K_m^{\mathfrak{p}}K_n^{\mathfrak{r}}]}.$$

Use (28.95) and Statement 26.1 to obtain (28.96). A subgroup of the finite direct product equals it exactly when their orders agree, proving all three equivalences. $\square$

Thus the ring decomposition (28.93) does not by itself prove group independence: one must determine the actual joint image. Abstract isomorphism or nonisomorphism of the two coordinate quotients also does not decide (28.97).

The two-parameter inverse system. If $m' \geq m$ and $n' \geq n$, then

$$L_{m',n'} \subseteq L_{m,n},$$

and the formula

$$P_{m',n';m,n} : H_{m',n'} \twoheadrightarrow H_{m,n}, \quad \gamma L_{m',n'} \longmapsto \gamma L_{m,n} \quad (28.98)$$

is well defined and surjective. Its kernel is $L_{m,n}/L_{m',n'}$, and componentwise reduction makes all rectangular composition diagrams commute. There is no canonical map $Q_m^{\mathfrak{p}} \rightarrow Q_n^{\mathfrak{r}}$ merely because the prime ideals are distinct; (28.98) belongs to the joint bi-tower.

Eightfold action. Statement 26.5 proves invariance of both coordinate kernels, hence $\phi_8(L_{m,n}) = L_{m,n}$. The induced action

$$\alpha_{m,n}^{\mathfrak{p},\mathfrak{r}}(\gamma L_{m,n}) := \phi_8(\gamma) L_{m,n} \quad (28.99)$$

corresponds inside the product to $(\alpha_m^{\mathfrak{p}}, \alpha_n^{\mathfrak{r}})$. Let their exact orders be $d_{\mathfrak{p},m}$ and $d_{\mathfrak{r},n}$. Since the coordinate projections of $H_{m,n}$ are surjective,

$$\text{ord}(\alpha_{m,n}^{\mathfrak{p},\mathfrak{r}}) = \text{lcm}(d_{\mathfrak{p},m}, d_{\mathfrak{r},n}). \quad (28.100)$$

Indeed, a common multiple acts trivially componentwise. Conversely, if a power acts trivially on $H_{m,n}$, surjectivity of both projections makes that power trivial on both coordinate images.

Parity on the joint tower. Define

$$\tilde{L}_{m,n} := L_{m,n} \cap \ker \chi. \quad (28.101)$$

By Statement 26.6, parity factors through $H_{m,n}$ exactly when $L_{m,n} \subseteq \ker \chi$. Otherwise the image of $(\rho_{m,n}^{\mathfrak{p},\mathfrak{r}}, \chi)$ is $H_{m,n} \times C_2$; in the factorizing case it is the graph of the unique descended parity. Thus the parity-compatible quotient has order $|H_{m,n}|$ or $2|H_{m,n}|$, respectively. Equation (28.71) preserves $\tilde{L}_{m,n}$, and the exact order of its descended C_8 -action remains the value in (28.100). These parity records must be reconstructed for the joint image; a bit table from either coordinate quotient is not a substitute.

Metric consequence. Since $L_{m,n}$ is contained in both coordinate kernels,

$$\text{sys}(L_{m,n} \setminus \mathbb{H}_R^2) \geq \max\{\text{sys}(K_m^p \setminus \mathbb{H}_R^2), \text{sys}(K_n^t \setminus \mathbb{H}_R^2)\}. \quad (28.102)$$

The same inequality with an additional weak increase holds after intersecting with $\ker \chi$. Along the diagonal $m = n = t$, either coordinate's Statement 26.4 theorem proves systole divergence. No comparison of residue characteristics is needed for this metric conclusion.

Exact finite construction. Input contains two distinct canonical prime records, their Statement 26.1 level records at m, n , complete coordinate carriers and tables, Statement 26.5 action records, the C09 parity record, and all hashes. Verify primality, inequality of ideal bases, common coefficient ring and representation, the rational primes below each ideal, and a Bézout certificate for (28.93). Enumerate $H_{m,n}$ by closing the paired marked generators inside the coordinate product. Build both projection tables, the exact kernel-intersection predicate, and the complete multiplication table. Compare its size with the coordinate product to decide (28.97), rather than assuming independence.

An optional rectangle request has tag

$$\text{rectangle_kind} \in \{\text{NONE}, \text{TRANSITION}\}. \quad (28.103)$$

The transition branch supplies $m' \geq m, n' \geq n$, a complete upper joint image, and the four coordinate bonding records. Check (28.98), parity preservation, and C_8 -equivariance on every upper carrier key. The branch tag and hash are mandatory and the payload is all-or-none.

The fixed output is

$$\begin{aligned} &(\text{tower_id}_p, \text{tower_id}_t, m, n, \text{residue_characteristic_relation}, \\ &\text{CRT_record}, \text{joint_carrier}, \text{joint_table}, |H_{m,n}|, \text{projection_tables}, \\ &\text{product_status} \in \{\text{FULL}, \text{PROPER_SUBDIRECT}\}, \text{index_identity_row}, \\ &\text{joint_action_table}, \text{ord}(\alpha_{m,n}^{p,t}), \text{lcm_row}, \\ &\text{parity_case}, \text{parity_augmented_record}, \text{systole_inheritance_row}, \\ &\text{rectangle_kind}, h_{\text{rectangle_request}}, \text{rectangle_block or NONE}, \\ &\text{COMPLETE, SATISFIED}, h_{\text{input}}, h_{\text{independentverification}}). \end{aligned} \quad (28.104)$$

The residue-characteristic relation is computed as SAME or DISTINCT, never inferred from ideal inequality. Unknown tags, equal prime ideals, a false CRT witness, foreign representations, incomplete coordinate images, a non-surjective projection, assumed full product, wrong index or least-common-multiple row, reused parity data, an asserted cross-prime bonding map, partial rectangle payload, failed commuting row, or stale hash returns

$$\text{REJECT}(\text{INVALID_TWO_PRIME_INPUT}, \text{stage}, \text{first_failure}, h_{\text{input}}). \quad (28.105)$$

Theorem 26.7.2 (Construction correctness and termination). *Every passing record is exactly the joint congruence quotient $\Gamma_B/L_{m,n}$, with correct projections, product classification, eight-fold action, and parity augmentation. Every requested rectangle commutes. The construction terminates and replays deterministically.*

Proof. The Chinese remainder and first isomorphism theorems prove (28.93)–(28.95); the index calculation proves the product decision. Kernel invariance and parity invariance prove the induced structures. All coordinate products are finite, so paired-generator closure and all exhaustive table checks terminate. Fixed prime, level, carrier, pair, parity, rectangle, and byte orders prove deterministic independent verification. $\square$

Tests and subsequent boundary. Tests include distinct ideals above the same rational prime, ideals above different rational primes, a full product, a proper diagonal subdirect product, correct and altered CRT witnesses, both parity branches, action orders with each possible least common multiple dividing eight, diagonal systole inheritance, both rectangle tags, a failed rectangular composite, a forged cross-prime map, foreign parents, stale hashes, and independent verification. Statement 26.8 may impose additional independence criteria, but none is retroactively assumed here.

26.8 Independence, non-equivalence, and the common local limit

Three notions that must not be conflated. Let $\mathcal{T}_p$ and $\mathcal{T}_r$ be the distinct prime-power towers of Statement 26.7. We distinguish:

- (i) *data separation*: their prime and tower identifiers are different;
- (ii) *product independence* at (m, n) : $H_{m,n} = Q_m^p \times Q_n^r$;
- (iii) *non-equivalence*: specified finite levels, or all levels in specified tails, are not isomorphic as abstract groups.

By (28.97), product independence is equivalent to

$$K_m^p K_n^r = \Gamma_B. \quad (28.106)$$

Neither data separation nor non-equivalence implies (28.106), and the Chinese remainder theorem alone proves none of these group-level statements.

Composition-factor invariant. For a finite group G , write $\text{CF}(G)$ for the full multiset of isomorphism keys of its simple composition factors, with multiplicity, and write $\text{CF}_{\text{na}}(G)$ for the submultiset of its nonabelian factors.

Lemma 26.8.1 (Composition factors in an extension). *If*

$$1 \longrightarrow N \longrightarrow G \longrightarrow H \longrightarrow 1 \quad (28.107)$$

is exact, then the composition-factor multiset of G is the multiset union of those of N and H .

Proof. Take a composition series for N and one for H . Pull every subgroup in the latter series back to G , and place the former series below N . The isomorphism theorem identifies the successive quotients above N with those of H , giving a composition series of G . The Jordan–Hölder theorem says that the multiset is independent of the chosen series: its proof refines any two subnormal series by the butterfly lemma, and composition series admit no proper refinement. Thus the displayed union is an invariant, not an artifact of the constructed series. $\square$

Nonabelian factors are stable within one prime tower. Let p_0 be the rational prime below p . The kernel of

$$\text{SL}_2(\mathcal{O}_E/\mathfrak{p}^m) \longrightarrow \text{SL}_2(\mathcal{O}_E/\mathfrak{p}) \quad (28.108)$$

is a p_0 -group: Statement 26.3 decomposes it into $m - 1$ successive kernels, each of order $q_p^3 = p_0^{3f_p}$. The actual transition kernel $\ker(Q_m^p \rightarrow Q_1^p)$ is a subgroup of that ambient kernel and is therefore also a p_0 -group. Its composition factors are all cyclic of order p_0 . Applying (28.107) gives

$$\text{CF}_{\text{na}}(Q_m^p) = \text{CF}_{\text{na}}(Q_1^p) \quad (m \geq 1). \quad (28.109)$$

The identical argument for the rational prime r_0 below r gives

$$\text{CF}_{\text{na}}(Q_n^r) = \text{CF}_{\text{na}}(Q_1^r) \quad (n \geq 1). \quad (28.110)$$

Theorem 26.8.2 (Tailwise non-equivalence criterion). *If exact finite-group certificates prove*

$$\text{CF}_{\text{na}}(Q_1^{\mathfrak{p}}) \neq \text{CF}_{\text{na}}(Q_1^{\mathfrak{r}}), \quad (28.111)$$

then

$$Q_m^{\mathfrak{p}} \not\cong Q_n^{\mathfrak{r}} \quad \text{for every } m, n \geq 1. \quad (28.112)$$

Consequently no cofinal levelwise isomorphism between the two inverse systems exists.

Proof. An abstract group isomorphism carries a composition series to a composition series and therefore preserves its factor multiset. Equations (28.109)–(28.111) make the two multisets unequal at every pair of levels, proving (28.112). A cofinal levelwise isomorphism would in particular provide an isomorphism between some such pair, which is impossible. $\square$

Different rational residue characteristics can help produce (28.111), but do not replace it. Small Lie-type groups have exceptional cross-characteristic isomorphisms, and a proper image may have no expected Lie-type factor. Every simple factor must therefore be identified by an exact multiplication table, normal-subgroup certificate, and isomorphism key. If (28.111) fails, the result is `NON_EQUIVALENCE_NOT_PROVED`, not an isomorphism claim.

A weaker scheduled certificate. Let $(m_j, n_j)_{j \geq 1}$ be a committed infinite schedule, cofinal in both coordinates. If exact records establish unequal full composition-factor multisets for each j , then every scheduled pair is nonisomorphic. This proves scheduled non-equivalence only. It does not exclude an isomorphism $Q_m^{\mathfrak{p}} \cong Q_n^{\mathfrak{r}}$ outside the schedule and is never promoted to (28.112).

Tower-wide product independence is a separate hypothesis. For cofinal predicates $I_{\mathfrak{p}}, I_{\mathfrak{r}} \subseteq \mathbb{Z}_{\geq 1}$, call the towers product-independent on the cofinal grid when

$$\forall (m, n) \in I_{\mathfrak{p}} \times I_{\mathfrak{r}}, \quad K_m^{\mathfrak{p}} K_n^{\mathfrak{r}} = \Gamma_B. \quad (28.113)$$

By Statement 26.7 this is necessary and sufficient for every joint image on the grid to be the full direct product. A finite list of passing pairs is not a proof of the universal quantifier. Product independence, composition-factor non-equivalence, and local convergence below are logically independent certificates.

Common local geometric limit. Assume both towers pass the Statement 26.4 systole-divergence theorem. Given $r > 0$, choose levels satisfying

$$\text{sys}(X_m^{\mathfrak{p}}) > 2r, \quad \text{sys}(X_n^{\mathfrak{r}}) > 2r. \quad (28.114)$$

Then every closed radius- r ball in either surface is isometric to a radius- r ball in the curvature- $-R^{-2}$ hyperbolic plane. Indeed, if two points of one lifted ball were identified by a nonidentity deck transformation γ , their distance would be at most $2r$, and $\ell_h(\gamma) \leq d(z, \gamma z) \leq 2r$, contradicting the strict inequality in (28.114). Thus both towers converge uniformly locally to

$$(\mathbb{H}_R^2, g_{\text{hyp}}), \quad (28.115)$$

in the sense that every fixed-radius ball embeds isometrically at all sufficiently deep levels. Equality $\text{sys} = 2r$ is a boundary and does not certify a closed radius- r embedding.

Theorem 26.8.3 (Independent conditional conclusions). *For the two prime towers:*

- (a) *Statement 26.4 for each tower implies the common local limit (28.115).*
- (b) *The exact mismatch (28.111) implies tailwise non-equivalence (28.112).*

(c) *The family hypothesis (28.113) implies full joint products on its cofinal grid.*

No conclusion in this list is used as a proof of another.

Proof. Part (a) is the ball-lifting argument above, part (b) is the tailwise non-equivalence theorem, and part (c) applies the two directions of (28.97) at every quantified pair. Their hypotheses are stored in different proof objects and domain hashes. $\square$

Exact construction and classifications. Input contains two complete Statement 26.1 tower records, their exact level-one group tables and required Statement 26.2 transition records, and separate Statement 26.4 divergence proofs. Every joint-image leaf additionally carries a hash-matching Statement 26.7 record for the exact pair of levels declared by that leaf; a Statement 26.7 pair record is never typed as a tower record. The input also contains a requested radius $r > 0$ and two tagged optional claims:

$$\begin{aligned} \text{noniso_kind} &\in \{\text{NONE}, \text{BASE_SIGNATURE}, \text{SCHEDULE}\}, \\ \text{independence_kind} &\in \{\text{NONE}, \text{COFINAL_GRID}\}. \end{aligned} \tag{28.116}$$

Tags and request hashes are mandatory. Optional payloads are all-or-none and bind their exact level predicates or schedule.

Enumerate normal subgroups only for the finitely many carriers that occur in certificate leaves; no universal branch enumerates schedule or grid instances. Verify each resulting composition series, certify simplicity of every quotient by exhaustive normal-subgroup testing, and canonicalize factor isomorphism keys. In base-signature mode compute (28.111).

Both universal modes accept only a finite certificate in the fixed, versioned proof-schema language IPT-28.08. A certificate is a finite typed directed acyclic graph. Its leaves are exact finite-group calculations, hash-bound earlier conclusions, or normalized integer equalities and inequalities. A leaf asserting a joint-image property is accepted only with the hash-matching Statement 26.7 record for its declared level pair. Its internal nodes are substitution, conjunction, modus ponens, and natural-number induction, each with explicit premises. The checker visits the graph in topological order; an induction node has one base subgraph and one symbolic successor subgraph, never an infinite list of instances.

A **SCHEDULE** payload contains an integer $j_0 \geq 1$, a finite expression for $j \mapsto (m_j, n_j)$, finite expressions for two tail-witness functions, checked proofs that each coordinate eventually exceeds every input bound, and a finite proof graph whose root is

$$\forall j \geq j_0, \quad \text{CF}(Q_{m_j}^p) \neq \text{CF}(Q_{n_j}^r).$$

A schedule certificate also contains exact finite full-factor mismatch records for every $1 \leq j < j_0$. The checker emits the scheduled positive classification only after those prefix records and the displayed inductive root together certify every scheduled pair $j \geq 1$. A **COFINAL_GRID** payload contains finite descriptions of the two level predicates, witness functions producing arbitrarily large admitted levels, and a finite proof graph whose root is the universally quantified assertion (28.113); every joint-image leaf in that graph obeys the Statement 26.7 pair-record rule above. Thus the verifier checks only finitely many typed nodes. An infinite list, sampled levels, or an untyped external theorem is not a certificate. Use the Statement 26.4 strict tail functions at threshold $2r$ to return sufficient local levels.

The fixed output is

$$\begin{aligned} &(\text{tower_ids}, r, M_{2r}^p, M_{2r}^r, \text{COMMON_LOCAL_LIMIT_PROVED}, \\ &\text{noniso_kind}, h_{\text{noniso_request}}, \text{noniso_certificate}, \text{noniso_status}, \\ &\text{independence_kind}, h_{\text{independence_request}}, \text{grid_proof or NONE}, \text{independence_status}, \\ &\text{C8_scope_hashes}, \text{parity_scope_hashes}, \text{COMPLETE}, \text{SATISFIED}, h_{\text{input}}, h_{\text{independentverification}}). \end{aligned} \tag{28.117}$$

The non-equivalence classification has exactly the four values

NOT_REQUESTED,
TAILWISE_NON_EQUIVALENCE_PROVED,
SCHEDULED_NON_EQUIVALENCE_PROVED,
NON_EQUIVALENCE_NOT_PROVED.

The independence classification has exactly the three values

NOT_REQUESTED,
PRODUCT_INDEPENDENCE_PROVED,
PRODUCT_INDEPENDENCE_NOT_PROVED.

The tagged-union presence rules are exact. For `noniso_kind = NONE`, the nonisomorphism certificate is `NONE`. For `BASE_SIGNATURE`, it consists of precisely the two level-one composition-series and canonical factor-key records. For `SCHEDULE`, it is precisely one finite IPT-28.08 schedule certificate and never an infinite sequence of records. Similarly, `independence_kind = NONE` forces the grid proof to be `NONE`, whereas `COFINAL_GRID` permits exactly one finite grid certificate.

With no nonisomorphism request, the corresponding classification is

NOT_REQUESTED.

A valid request whose exact factors do not prove the requested mismatch remains a structurally valid record but returns

NON_EQUIVALENCE_NOT_PROVED.

With no independence request, the grid-proof field is

NONE,

and the independence classification is

NOT_REQUESTED.

A structurally valid grid request whose root assertion is not proved returns

PRODUCT_INDEPENDENCE
_NOT_PROVED.

The line break is typographic only and denotes the single classification key listed above. Only a passing universal proof emits the positive product-independence classification.

An unknown tag, a payload violating its presence rule, malformed finite syntax, an invalid witness or proof-graph inference, an infinite list, finite samples substituted for a family proof, invalid composition series, unproved simple quotient, noncanonical factor key, residue characteristic used as an isomorphism key, mixed tower or kernel conventions, non-strict radius threshold, missing C_8 or parity scope, unsupported full-product claim, or stale hash returns

REJECT(INVALID_INDEPENDENCE_INPUT, stage, first_failure, h_{input}). (28.118)

Theorem 26.8.4 (Construction correctness and termination). *Every passing record proves the common local limit. Each optional positive classification proves exactly its declared non-equivalence or product-independence statement and no stronger one. All finite branches terminate and equal canonical inputs independent verification identically.*

Proof. Finite group tables have finitely many subgroups, so exhaustive normality and simplicity checks terminate. A proof certificate is a finite acyclic graph, and every node rule has a terminating exact checker; induction is checked from its finite base and successor subgraphs, not by enumerating instances. The extension lemma proves base-signature propagation, while the accepted root sequents prove exactly the scheduled or cofinal-grid claims displayed above. Statement 26.4 provides finite strict tail levels for $2r$. Fixed tower, series, factor, expression, graph, radius, and byte orders prove deterministic independent verification. $\square$

Tests and final boundary. Tests include distinct prime ideals over the same rational prime, distinct rational characteristics with equal small simple factors, different exact base signatures, equal nonabelian but different full factor multisets, a proper image with no Lie-type factor, a scheduled but not tailwise result, a sampled grid falsely labeled universal, passing and failing product pairs, $r > 0$, the exact boundary $\text{sys} = 2r$, all optional tags and empty states, mixed C_8 or parity scopes, stale hashes, and independent verification. The two towers may share the same local hyperbolic limit while every finite level is algebraically nonisomorphic; neither fact is a contradiction.

Part XI

Finite classical groups and semilinear constructions

27 Finite classical groups

27.1 The general linear group

Finite-field and dimension specification. Fix a prime p , an integer $f \geq 1$, and a monic irreducible polynomial $F(T) \in \mathbb{F}_p[T]$ of degree f . The concrete field used by every calculation is

$$\mathbb{F}_q := \mathbb{F}_p[T]/(F), \quad q := p^f. \quad (29.1)$$

Elements are coefficient tuples $(a_0, \dots, a_{f-1}) \in \{0, \dots, p-1\}^f$, ordered lexicographically, with arithmetic reduced modulo F . Distinct irreducible polynomials of degree f give isomorphic abstract fields, but not identical serialized carriers. Consequently the polynomial and its field-table hash are part of every computational record. We fix $n \in \mathbb{Z}_{\geq 1}$ and use column vectors; matrices act on the left, so $(AB)v = A(Bv)$.

Definition 27.1.1 (General linear group). Let $V = \mathbb{F}_q^n$. Then

$$\begin{aligned} \text{GL}(n, q) &:= \{A \in \text{M}_n(\mathbb{F}_q) : \det A \neq 0\} \\ &= \text{M}_n(\mathbb{F}_q)^\times = \text{Aut}_{\mathbb{F}_q}(V). \end{aligned} \quad (29.2)$$

The notation includes the concrete field record (29.1); it never means matrices over an unspecified ring of cardinality q .

Theorem 27.1.2 (Equivalent invertibility tests). *For $A \in \text{M}_n(\mathbb{F}_q)$, the following are equivalent:*

- (i) $\det A \neq 0$,
 - (ii) $\text{rank } A = n$,
 - (iii) $\ker A = \{0\}$,
 - (iv) the columns of A are an ordered basis of V ,
 - (v) there is B with $AB = BA = I_n$.
- (29.3)

Proof. Gaussian elimination over a field converts A by invertible row operations to a unique reduced row-echelon form. It has n pivots exactly when the kernel is zero, exactly when the columns are a basis. In that case elimination applied to $[A \mid I_n]$ constructs $B = A^{-1}$. Conversely $AB = I_n$ forces zero kernel. Finally the determinant changes by a nonzero scalar under a row swap or nonzero row scaling and is unchanged by row addition, so it is nonzero exactly when elimination reaches I_n . □

Theorem 27.1.3 (Group structure and faithful natural action). *Matrix multiplication makes $\text{GL}(n, q)$ a finite group. Evaluation defines a faithful action*

$$\alpha : \text{GL}(n, q) \longrightarrow \text{Sym}(V), \quad \alpha(A)(v) = Av. \quad (29.4)$$

Proof. The identity is invertible, determinants multiply, and (29.3) supplies an inverse for every member; associativity is inherited from matrix multiplication. Thus the set is a group. Each invertible linear map permutes V , and $\alpha(AB) = \alpha(A)\alpha(B)$ under the fixed composition convention. If $\alpha(A)$ is the identity permutation, then $Ae_i = e_i$ for every standard basis vector, whence $A = I_n$. Since $M_n(\mathbb{F}_q)$ has q^{n^2} elements, the group is finite. $\square$

Order from ordered bases. After linearly independent columns $v_1, \dots, v_i$ have been chosen, their span has q^i elements. The next column therefore has $q^n - q^i$ choices. The correspondence between invertible matrices and ordered bases is bijective by (29.3), so

$$|\text{GL}(n, q)| = \prod_{i=0}^{n-1} (q^n - q^i). \quad (29.5)$$

Factoring q^i from the i -th factor gives the equivalent integer formula

$$|\text{GL}(n, q)| = q^{n(n-1)/2} \prod_{j=1}^n (q^j - 1). \quad (29.6)$$

Dividing (29.5) by the q^{n^2} matrices in the full matrix algebra also gives the exact density

$$\frac{|\text{GL}(n, q)|}{q^{n^2}} = \prod_{j=1}^n (1 - q^{-j}). \quad (29.7)$$

These are integer identities derived from the same independent column choices, not heuristic probabilities.

Exact finite construction. Input is

$$(p, f, F, n, \text{LEFT_COLUMN_ACTION}, \text{field_encoding_version}). \quad (29.8)$$

First verify $p \in \mathbb{Z}$ and $p \geq 2$, and then test divisibility by every integer $2 \leq d \leq \lfloor \sqrt{p} \rfloor$. Check $f, n \geq 1$ and that F is monic of degree f . Enumerate every monic polynomial D with $1 \leq \deg D \leq \lfloor f/2 \rfloor$, use exact polynomial division, and reject if the remainder of F upon division by D is zero. The range is empty when $f = 1$. Construct the addition, multiplication, zero, one, and inverse tables of (29.1); the no-divisor test proves that every nonzero residue has an inverse.

Enumerate vectors in lexicographic coefficient order and matrices in row-major order. For each of the q^{n^2} matrices, use the first available nonzero pivot in the fixed field order. Retain exactly the matrices with n pivots, store the elimination transcript and inverse, and then construct the multiplication table and the permutations (29.4). Independently compute (29.5) with exact integers and require equality with the carrier count. This is a complete construction of the full group, not merely the subgroup generated by a supplied list.

The fixed output record is

$$\begin{aligned} &(p, f, q, F, \text{field_table_hash}, n, \text{row_major_carrier}, \\ &\text{rank_and_inverse_certificates}, \text{multiplication_table}, \text{inverse_table}, \\ &\text{natural_action_table}, |\text{GL}(n, q)|, \text{ORDER_FORMULA_VERIFIED}, \\ &\text{GROUP_AXIOMS_VERIFIED}, \\ &\text{FAITHFUL_ACTION_VERIFIED}, \text{COMPLETE}, \text{SATISFIED}, \\ &\text{h}_{\text{input}}, \text{h}_{\text{independentverification}}). \end{aligned} \quad (29.9)$$

Every field element, vector, matrix, permutation, table row, and certificate has the fixed order above. An invalid type, nonprime p , inconsistent q , reducible or wrongly normalized F , $f < 1$, $n < 1$, arithmetic-table mismatch, singular retained matrix, missing invertible matrix, order mismatch, action-convention collision, or stale hash returns

$$\text{REJECT}(\text{INVALID_GL_INPUT}, \text{stage}, \text{first_failure}, \text{h}_{\text{input}}). \quad (29.10)$$

Theorem 27.1.4 (Construction correctness, termination, and independent verification). *Every passing record constructs exactly $\text{GL}(n, q)$ with its faithful natural action and order (29.5). The construction terminates, and equal canonical inputs produce byte-identical outputs.*

Proof. The irreducibility and table checks are finite. Matrix enumeration has q^{n^2} iterations, each elimination is finite, and all carrier tables have finite nested loops. The equivalences (29.3) prove exact membership, the group and action theorem proves the tables' meaning, and the ordered-basis argument proves the expected count. Fixed encodings and traversal orders prove deterministic independent verification. $\square$

Minimal tests and boundaries. For $p = 2, f = 1, F = T, n = 2$, formula (29.5) gives

$$|\text{GL}(2, 2)| = (4 - 1)(4 - 2) = 6. \quad (29.11)$$

The constructor finds exactly six matrices, and their faithful permutation action on the three nonzero vectors identifies the result with S_3 . At the admitted boundary $n = 1$,

$$\text{GL}(1, q) = \mathbb{F}_q^\times, \quad |\text{GL}(1, q)| = q - 1; \quad (29.12)$$

in particular $\text{GL}(1, 2)$ is the one-element group. The singular matrix

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \in \text{M}_2(\mathbb{F}_2) \quad (29.13)$$

has rank one and must be excluded. Rejection tests include $n = 0, f = 0$, composite p , a reducible polynomial such as $T^2 + T$ over $\mathbb{F}_2$, a declared $q \neq p^f$, altered field or matrix order, incorrect inverses, a carrier of size other than (29.5), and independent verification under a stale field-table hash. Although some authors define a trivial $\text{GL}(0, q)$, this construction specification deliberately requires $n \geq 1$. Statement 27.2 must independently filter this full carrier by $\det A = 1$ and certify the resulting subgroup; the present $\text{GL}(n, q)$ carrier is not silently retyped as an $\text{SL}(n, q)$ carrier.

27.2 Special and projective special linear groups

Determinant kernel. Continue with the concrete field and matrix conventions of Statement 27.1. The determinant is a surjective homomorphism

$$\det : \text{GL}(n, q) \longrightarrow \mathbb{F}_q^\times. \quad (29.14)$$

Indeed, multiplicativity gives the homomorphism property, and for every $u \in \mathbb{F}_q^\times$, $\text{diag}(u, 1, \dots, 1)$ has determinant u .

Definition 27.2.1 (Special linear group).

$$\text{SL}(n, q) := \ker(\det) = \{A \in \text{GL}(n, q) : \det A = 1\}. \quad (29.15)$$

It is the exact determinant-one subcarrier of the Statement 27.1 group, not the full general-linear carrier with a changed type label.

Theorem 27.2.2 (Order of the special linear group). *For every $n \geq 1$,*

$$\begin{aligned} |\mathrm{SL}(n, q)| &= \frac{|\mathrm{GL}(n, q)|}{q-1} \\ &= q^{n(n-1)/2} \prod_{j=2}^n (q^j - 1), \end{aligned} \quad (29.16)$$

where the product is empty when $n = 1$.

Proof. The multiplicative group of the field has $q-1$ elements. Surjectivity in (29.14) and the first isomorphism theorem give $\mathrm{GL}(n, q)/\mathrm{SL}(n, q) \cong \mathbb{F}_q^\times$. Division of (29.6) by $q-1$, its $j=1$ factor, gives the second expression. $\square$

The scalar center. The finite subgroup $\mathbb{F}_q^\times$ of a field is cyclic. For completeness, decompose the finite abelian group into its primary cyclic factors and choose one element attaining the exponent e . Every element then satisfies $x^e = 1$. A nonzero polynomial of degree e has at most e roots, so $q-1 \leq e$; Lagrange gives $e \leq q-1$. Hence $e = q-1$. It follows that

$$\#\{\lambda \in \mathbb{F}_q^\times : \lambda^n = 1\} = d, \quad d := \gcd(n, q-1). \quad (29.17)$$

Theorem 27.2.3 (Center of $\mathrm{SL}(n, q)$). *For $n \geq 1$,*

$$Z(\mathrm{SL}(n, q)) = Z_n(q) := \{\lambda I_n : \lambda \in \mathbb{F}_q^\times, \lambda^n = 1\}, \quad |Z_n(q)| = d. \quad (29.18)$$

Proof. For $n = 1$, both groups in (29.18) are trivial. Suppose $n \geq 2$. Every displayed scalar lies in $\mathrm{SL}(n, q)$ and is central. Conversely, if A is central, it commutes with every elementary transvection $E_{ij}(1) = I_n + e_i e_j^\top$, $i \neq j$, which has determinant one. Comparing $A e_i e_j^\top = e_i e_j^\top A$ first forces every off-diagonal entry of A to vanish and then forces all diagonal entries to be equal. Thus $A = \lambda I_n$, and determinant one is equivalent to $\lambda^n = 1$. Equation (29.17) gives the order. $\square$

Definition 27.2.4 (Projective special linear group).

$$\mathrm{PSL}(n, q) := \mathrm{SL}(n, q)/Z_n(q). \quad (29.19)$$

This quotient is defined for every $n \geq 1$; no simplicity claim is part of the notation.

Corollary 27.2.5 (Order of the projective quotient).

$$|\mathrm{PSL}(n, q)| = \frac{|\mathrm{SL}(n, q)|}{\gcd(n, q-1)} = \frac{q^{n(n-1)/2}}{d} \prod_{j=2}^n (q^j - 1). \quad (29.20)$$

Faithful projective action. Let

$$\mathbb{P}^{n-1}(\mathbb{F}_q) := (\mathbb{F}_q^n \setminus \{0\})/\mathbb{F}_q^\times, \quad |\mathbb{P}^{n-1}(\mathbb{F}_q)| = \frac{q^n - 1}{q - 1}. \quad (29.21)$$

The action of $\mathrm{SL}(n, q)$ on one-dimensional subspaces has kernel exactly $Z_n(q)$. For $n \geq 2$, an element fixing every line has each e_i as an eigenvector; fixing the line of $e_i + e_j$ forces the corresponding eigenvalues to agree. It is therefore scalar, and membership in SL places it in $Z_n(q)$. The reverse inclusion is immediate. For $n = 1$, both the acting quotient and its kernel calculation are trivial. Thus quotient descent gives a faithful homomorphism

$$\bar{\alpha} : \mathrm{PSL}(n, q) \hookrightarrow \mathrm{Sym}(\mathbb{P}^{n-1}(\mathbb{F}_q)). \quad (29.22)$$

Exact construction. Input is a passing Statement 27.1 record, including its field, carrier, multiplication, inverse, and elimination hashes. Recompute every determinant in canonical carrier order, verify the homomorphism table and the diagonal surjectivity witnesses, and retain precisely the determinant-one matrices. Restrict the parent multiplication and inverse tables and require closure.

Next enumerate all field scalars in field order, retain exactly those satisfying $\lambda^n = 1$, and compare their scalar matrices both with an exhaustive center computation in the SL table and with $d = \gcd(n, q - 1)$. Form right cosets of $Z_n(q)$; centrality makes right and left cosets identical. The canonical representative is the least row-major matrix in each coset. Multiplication of representatives followed by coset lookup constructs the PSL table.

Represent a projective point by the unique nonzero vector whose first nonzero coordinate is 1. Sort these normalized representatives lexicographically, construct the action table, verify its kernel is exactly $Z_n(q)$, and descend it to the quotient. The resulting fixed output is

$$\begin{aligned} &(\mathbf{h}_{\text{GL}}, n, q, \text{determinant_table}, \text{determinant_witnesses}, \\ &\text{SL_carrier}, \text{SL_tables}, |\text{SL}(n, q)|, \text{SL_ORDER_VERIFIED}, \\ &d, \text{scalar_center_table}, \text{CENTER_VERIFIED}, \\ &\text{PSL_cosets}, \text{PSL_tables}, |\text{PSL}(n, q)|, \text{PSL_ORDER_VERIFIED}, \\ &\text{projective_point_table}, \text{faithful_projective_action}, \\ &\text{COMPLETE}, \text{SATISFIED}, \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}}). \end{aligned} \quad (29.23)$$

An invalid or stale parent, determinant-table mismatch, non-surjective determinant, failure of determinant-one closure, a scalar-root count different from d , disagreement between the scalar list and the exhaustive center, a noncanonical or duplicate coset, quotient-table failure, an incorrect normalized projective point, action-kernel mismatch, an order mismatch, or an unsupported simplicity flag returns

$$\text{REJECT}(\text{INVALID_SL_PSL_INPUT}, \text{stage}, \text{first_failure}, \mathbf{h}_{\text{input}}). \quad (29.24)$$

Theorem 27.2.6 (Construction correctness and termination). *Every passing record constructs exactly $\text{SL}(n, q)$, its scalar center, and $\text{PSL}(n, q)$, with orders (29.16) and (29.20) and the faithful action (29.22). The algorithm terminates and canonical inputs independent verification byte-identically.*

Proof. All carriers inherited from Statement 27.1 are finite. Determinant filtering, center enumeration, exhaustive commutation tests, coset partitioning, quotient multiplication, and projective action construction therefore terminate. Equations (29.15), (29.18), and (29.19) prove that the filtered carrier, center, and quotient are the declared groups. The two order theorems and kernel calculation prove the numerical and action fields. Fixed parent, scalar, matrix, coset-representative, projective-point, and byte orders prove independent verification. $\square$

Tests and low-dimensional boundaries. For $n = 2, q = 2$, the determinant target has order one, so

$$\text{SL}(2, 2) = \text{GL}(2, 2), \quad d = 1, \quad |\text{PSL}(2, 2)| = 6. \quad (29.25)$$

Its faithful action on the three projective points is the six-element permutation group. For $n = 2, q = 3$,

$$|\text{GL}(2, 3)| = 48, \quad |\text{SL}(2, 3)| = 24, \quad d = 2, \quad |\text{PSL}(2, 3)| = 12. \quad (29.26)$$

This checks that quotienting by the entire four-element scalar group of $\text{GL}(2, 5)$, rather than only the two scalars with $\lambda^2 = 1$, would be invalid. At $n = 1$,

$$\text{SL}(1, q) = \text{PSL}(1, q) = 1. \quad (29.27)$$

The suite also tests $q = 2$, nontrivial and trivial scalar centers, a forged determinant-one row, a missing scalar root, left/right coset agreement, projective normalization, nonfaithful descended actions, stale parent hashes, and independent verification. In particular $\text{PSL}(2, 2)$ and $\text{PSL}(2, 3)$ are not labeled simple; any later use of simplicity requires its own dimension-and-field condition and proof.

27.3 The projective general linear group

Scalar quotient. Continue with the concrete Statement 27.1 field and general-linear carrier. The nonzero scalar matrices form the central subgroup

$$Z_{\text{GL}}(n, q) := \{\lambda I_n : \lambda \in \mathbb{F}_q^\times\} \leq \text{GL}(n, q), \quad |Z_{\text{GL}}(n, q)| = q - 1. \quad (29.28)$$

For $n \geq 2$, commuting with every elementary transvection forces a central matrix to be scalar, by the entry comparison used in Statement 27.2. For $n = 1$, every invertible matrix is scalar. Thus

$$Z(\text{GL}(n, q)) = Z_{\text{GL}}(n, q) \quad (n \geq 1). \quad (29.29)$$

Definition 27.3.1 (Projective general linear group).

$$\text{PGL}(n, q) := \text{GL}(n, q) / Z_{\text{GL}}(n, q). \quad (29.30)$$

The quotient identifies matrices precisely when they differ by a nonzero field scalar. It does not impose determinant one.

Theorem 27.3.2 (Order and projective action).

$$\begin{aligned} |\text{PGL}(n, q)| &= \frac{|\text{GL}(n, q)|}{q - 1} \\ &= q^{n(n-1)/2} \prod_{j=2}^n (q^j - 1). \end{aligned} \quad (29.31)$$

Moreover, the natural action on $\mathbb{P}^{n-1}(\mathbb{F}_q)$ descends to a faithful action

$$\beta : \text{PGL}(n, q) \hookrightarrow \text{Sym}(\mathbb{P}^{n-1}(\mathbb{F}_q)). \quad (29.32)$$

Proof. Centrality makes (29.30) a group quotient, and Lagrange gives (29.31). Every scalar fixes every one-dimensional subspace. If a general-linear matrix fixes every such subspace, the basis-line and $e_i + e_j$ -line argument of Statement 27.2 makes it scalar. Hence the kernel of the original action is exactly $Z_{\text{GL}}(n, q)$, and quotient descent proves faithfulness. When $n = 1$, both $\text{PGL}(1, q)$ and the induced permutation group are trivial, so the statement remains valid. $\square$

Exact relation to PSL. Put $d = \gcd(n, q - 1)$, and let

$$(\mathbb{F}_q^\times)^n := \{\lambda^n : \lambda \in \mathbb{F}_q^\times\}. \quad (29.33)$$

The determinant is not itself well defined on projective classes: replacing A by λA multiplies its determinant by λ^n . Its residue class modulo n -th powers is well-defined, however:

$$\begin{aligned} \delta : \text{PGL}(n, q) &\longrightarrow \mathbb{F}_q^\times / (\mathbb{F}_q^\times)^n, \\ [A] &\longmapsto \det(A) (\mathbb{F}_q^\times)^n. \end{aligned} \quad (29.34)$$

The first map in the sequence below is the explicitly typed map

$$\begin{aligned} \iota : \text{PSL}(n, q) &\longrightarrow \text{PGL}(n, q), \\ AZ_n(q) &\longmapsto AZ_{\text{GL}}(n, q). \end{aligned}$$

Theorem 27.3.3 (Projective determinant exact sequence). *The map δ is a surjective homomorphism and*

$$1 \longrightarrow \mathrm{PSL}(n, q) \xrightarrow{\iota} \mathrm{PGL}(n, q) \xrightarrow{\delta} \mathbb{F}_q^\times / (\mathbb{F}_q^\times)^n \longrightarrow 1 \quad (29.35)$$

is exact. Consequently

$$[\mathrm{PGL}(n, q) : \mathrm{PSL}(n, q)] = d, \quad (29.36)$$

and $\mathrm{PGL}(n, q) = \mathrm{PSL}(n, q)$ exactly when $d = 1$.

Proof. The preceding scalar calculation proves well-definedness, and determinant multiplicativity proves the homomorphism law. Diagonal matrices $\mathrm{diag}(u, 1, \dots, 1)$ prove surjectivity. If $AZ_n(q) = BZ_n(q)$, then $AZ_{\mathrm{GL}}(n, q) = BZ_{\mathrm{GL}}(n, q)$, so ι is well defined. If the latter cosets agree for $A, B \in \mathrm{SL}$, then

$$B^{-1}A \in Z_{\mathrm{GL}}(n, q) \cap \mathrm{SL}(n, q) = Z_n(q),$$

so the original $Z_n(q)$ -cosets agree and ι is injective. If $\delta([A]) = 1$, then $\det A = \mu^n$ for some $\mu \in \mathbb{F}_q^\times$, and $\mu^{-1}A \in \mathrm{SL}(n, q)$ represents the same projective class. Hence $\ker \delta = \mathrm{im} \iota$. Since the cyclic group $\mathbb{F}_q^\times$ has order $q - 1$, the n -th-power image has order $(q - 1)/d$, so the final quotient has order d . $\square$

Although (29.31) equals the numerical formula for $|\mathrm{SL}(n, q)|$, no equality or isomorphism of those two groups follows. One is a quotient of GL ; the other is a determinant kernel.

Exact construction. Input contains hash-matching passing Statement 27.1 and Statement 27.2 records on the same field, dimension, carrier, and action convention. Enumerate the $q - 1$ nonzero field scalars, construct their scalar matrices, and compare the list with an exhaustive center computation in the Statement 27.1 multiplication table. Partition $\mathrm{GL}(n, q)$ into scalar cosets, choose the least row-major member as representative, and build quotient multiplication and inverse tables by coset lookup.

Independently normalize every nonzero vector as in Statement 27.2 and build the projective permutation table. Verify that the kernel in GL is exactly the full scalar subgroup and that the descended PGL action is faithful. Enumerate the n -th-power subgroup of $\mathbb{F}_q^\times$, build its cosets, evaluate (29.34) on every projective class, and verify the kernel as a PGL subcarrier. Map every Statement 27.2 coset $AZ_n(q)$ to the PGL coset $AZ_{\mathrm{GL}}(n, q)$; store the resulting table and verify that it is total, single-valued, injective, homomorphic, and has image exactly $\ker \delta$.

The fixed output is

$$\begin{aligned} &(\mathrm{h}_{\mathrm{GL}}, \mathrm{h}_{\mathrm{SL}/\mathrm{PSL}}, n, q, \mathrm{full_scalar_center}, \\ &\mathrm{PGL_cosets}, \mathrm{PGL_tables}, |\mathrm{PGL}(n, q)|, \\ &\mathrm{projective_point_table}, \mathrm{faithful_PGL_action}, \\ &(\mathbb{F}_q^\times)^n, \mathrm{power_quotient_table}, \mathrm{projective_determinant_table}, \\ &\mathrm{PSL_embedding_table}, \mathrm{EMBEDDING_AND_KERNEL_VERIFIED}, \\ &d, \mathrm{PSL_INDEX_VERIFIED}, \\ &\mathrm{PGL_EQUALS_PSL} \text{ or } \mathrm{PGL_STRICTLY_CONTAINS_PSL}, \\ &\mathrm{COMPLETE}, \mathrm{SATISFIED}, \mathrm{h}_{\mathrm{input}}, \mathrm{h}_{\mathrm{independentverification}}). \end{aligned} \quad (29.37)$$

Mismatched or stale parents, a noncentral retained scalar, an incorrect center, a duplicate or incomplete coset partition, quotient-table failure, a projective action with the wrong kernel, an incorrect n -th-power subgroup, a projective determinant depending on the representative, a kernel different from the image of the typed PSL embedding, a partial, multivalued, noninjective, or nonhomomorphic embedding table, an index different from d , or an equality classification inconsistent with $d = 1$ returns

$$\mathrm{REJECT}(\mathrm{INVALID_PGL_INPUT}, \mathrm{stage}, \mathrm{first_failure}, \mathrm{h}_{\mathrm{input}}). \quad (29.38)$$

Theorem 27.3.4 (Construction correctness and termination). *Every passing record constructs exactly $\text{PGL}(n, q)$, its faithful projective action, and the exact sequence (29.35). The constructor terminates and canonical inputs independent verification byte-identically.*

Proof. All parent carriers are finite. Scalar enumeration, center comparison, coset partitioning, quotient tables, projective permutations, power-subgroup enumeration, and kernel comparison are therefore finite. Equations (29.28)–(29.32) prove the quotient and action fields, while (29.34)–(29.36) prove the determinant-class fields and equality classification. Fixed carrier, scalar, representative, field-coset, permutation, and byte orders prove independent verification. $\square$

Tests and boundaries. For $n = 2, q = 3$,

$$|\text{PGL}(2, 3)| = 24, \quad |\text{PSL}(2, 3)| = 12, \quad d = 2. \quad (29.39)$$

The faithful action on the four projective points has order 24, so its image is all of S_4 ; by contrast $\text{SL}(2, 3)$ also has order 24 but has a two-element center, demonstrating that equality of orders is not a group identification. For $n = 2, q = 4, d = 1$, so the constructor must report $\text{PGL}(2, 4) = \text{PSL}(2, 4)$, both of order 60. At $n = 1$, both projective groups are trivial. Rejection tests include quotienting by only $Z_n(q)$ instead of the full general-linear scalar center, treating raw determinant as projectively well-defined, a wrong equality flag, noncanonical projective representatives, stale parent hashes, and independent verification.

27.4 The finite symplectic group

The fixed alternating form. Let $n \geq 1$, let $V = \mathbb{F}_q^{2n}$, and order its standard basis as $(e_1, \dots, e_n, f_1, \dots, f_n)$. Fix

$$J_n := \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}, \quad \langle x, y \rangle := x^\top J_n y. \quad (29.40)$$

The form is nondegenerate because J_n is invertible, and it is alternating because $\langle x, x \rangle = 0$ for every x . When the characteristic is 2, the displayed $-I_n$ equals I_n ; alternation is still checked directly from the zero diagonal and paired cross terms. It is never inferred merely from the phrase *skew-symmetric*.

Definition 27.4.1 (Symplectic group).

$$\text{Sp}(2n, q) := \{A \in \text{GL}(2n, q) : A^\top J_n A = J_n\}. \quad (29.41)$$

Theorem 27.4.2 (Group, determinant, and natural action). *The set (29.41) is a subgroup of $\text{SL}(2n, q)$. Its natural left action on V is faithful.*

Proof. The identity preserves J_n , and

$$(AB)^\top J_n (AB) = B^\top (A^\top J_n A) B = J_n. \quad (29.42)$$

If $A^\top J_n A = J_n$, then $A^{-1} = J_n^{-1} A^\top J_n$, so inverses also preserve the form. For the determinant statement, use the integral polynomial identity

$$\text{Pf}(A^\top S A) = \det(A) \text{Pf}(S) \quad (29.43)$$

for an alternating matrix S . It holds over $\mathbb{Z}$ and hence after reduction in every characteristic. Taking $S = J_n$, whose Pfaffian is a unit, makes (29.43) and (29.41) imply $\det A = 1$. Faithfulness follows because a matrix fixing every vector fixes each standard basis vector and is the identity. $\square$

Symplectic bases. An ordered tuple

$$\mathcal{B} = (v_1, \dots, v_n, w_1, \dots, w_n) \quad (29.44)$$

is a symplectic basis when

$$\langle v_i, v_j \rangle = \langle w_i, w_j \rangle = 0, \quad \langle v_i, w_j \rangle = \delta_{ij}. \quad (29.45)$$

Its column matrix belongs to $\text{Sp}(2n, q)$, and every symplectic matrix gives one such basis. Thus the group acts simply transitively on the set of ordered symplectic bases: there is a unique form-preserving linear map sending the standard symplectic basis to any prescribed one.

Lemma 27.4.3 (First hyperbolic pair). *There are*

$$(q^{2n} - 1)q^{2n-1} \quad (29.46)$$

choices for a pair (v_1, w_1) satisfying $v_1 \neq 0$ and $\langle v_1, w_1 \rangle = 1$. For every such pair,

$$V = \langle v_1, w_1 \rangle \perp \langle v_1, w_1 \rangle^\perp, \quad (29.47)$$

and the orthogonal complement is a nondegenerate symplectic space of dimension $2n - 2$.

Proof. There are $q^{2n} - 1$ nonzero choices for v_1 . Nondegeneracy makes $w \mapsto \langle v_1, w \rangle$ a nonzero linear functional, so each fiber, including the fiber over 1, has q^{2n-1} elements. The restricted form on $\langle v_1, w_1 \rangle$ has matrix $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and is nondegenerate. A nondegenerate subspace meets its orthogonal complement trivially, dimensions add to $2n$, and restriction to the complement is again nondegenerate and alternating. $\square$

Theorem 27.4.4 (Order of the symplectic group). *With $|\text{Sp}(0, q)| := 1$,*

$$|\text{Sp}(2n, q)| = (q^{2n} - 1)q^{2n-1} |\text{Sp}(2n - 2, q)|, \quad (29.48)$$

and therefore

$$|\text{Sp}(2n, q)| = q^{n^2} \prod_{i=1}^n (q^{2i} - 1). \quad (29.49)$$

Proof. Choose the first hyperbolic pair by (29.46), then choose a symplectic basis of its canonical orthogonal complement. This construction and deletion of the first pair are inverse, proving (29.48). Iteration contributes the product of $q^{2i} - 1$ and the exponent $\sum_{i=1}^n (2i - 1) = n^2$, proving (29.49). $\square$

Exact construction. Input is a passing Statement 27.1 record with dimension $2n$, together with the tag `STANDARD_ALTERNATING_FORM` and the basis order fixed in (29.40). Reconstruct J_n from n , verify nondegeneracy and alternation entrywise, and filter the full $\text{GL}(2n, q)$ carrier by the exact equality $A^\top J_n A = J_n$. Restrict the multiplication, inverse, and natural-action tables. Verify closure, determinant one against a hash-matching Statement 27.2 determinant table, and faithful action.

Independently enumerate the symplectic bases (29.44)–(29.45) in lexicographic vector order. Match each basis to its unique column matrix, verify the bijection with the filtered carrier, check the recurrence (29.48) at every rank down to zero, and compare the carrier count with (29.49). The fixed output is

$$\begin{aligned} &(\mathbf{h}_{\text{GL}(2n, q)}, \mathbf{h}_{\text{SL}(2n, q)}, n, q, \text{basis_order}, J_n, \\ &\text{form_table}, \text{Sp_carrier}, \text{Sp_tables}, \\ &\text{symplectic_basis_table}, \text{basis_matrix_bijection}, \\ &\text{recurrence_records}, |\text{Sp}(2n, q)|, \text{ORDER_VERIFIED}, \\ &\text{DETERMINANT_ONE_VERIFIED}, \text{FAITHFUL_ACTION_VERIFIED}, \\ &\text{COMPLETE}, \text{SATISFIED}, \mathbf{h}_{\text{input}}, \mathbf{h}_{\text{independentverification}}). \end{aligned} \quad (29.50)$$

An invalid or stale parent, $n < 1$, a dimension other than $2n$, a basis-order collision, a singular or nonalternating form, characteristic 2 rejected solely because $-1 = 1$, a retained matrix failing the form equation, a missing preserving matrix, closure or inverse failure, a determinant not equal to one, an invalid symplectic basis, a nonbijective basis table, a recurrence or order mismatch, or a stale form hash returns

$$\text{REJECT}(\text{INVALID_SYMPLECTIC_INPUT}, \text{stage}, \text{first_failure}, \text{h}_{\text{input}}). \quad (29.51)$$

Theorem 27.4.5 (Construction correctness and termination). *Every passing record constructs exactly $\text{Sp}(2n, q)$, its faithful natural action, and order (29.49). The algorithm terminates and equal canonical inputs independent verification byte-identically.*

Proof. The Statement 27.1 carrier, vector set, and all tuple spaces are finite. Form filtering, table restriction, basis enumeration, bijection checking, and recurrence verification therefore terminate. Equations (29.40)–(29.43) prove the subgroup and determinant fields; the basis bijection and recurrence prove completeness and order. Fixed field, basis, matrix, tuple, and byte orders prove independent verification. $\square$

Tests and characteristic-two boundary. At rank one,

$$\text{Sp}(2, q) = \text{SL}(2, q), \quad |\text{Sp}(2, q)| = q(q^2 - 1), \quad (29.52)$$

because for every 2×2 matrix A , $A^\top J_1 A = (\det A) J_1$. Thus $|\text{Sp}(2, 2)| = 6$, matching Statement 27.2. For $n = 2, q = 2$,

$$|\text{Sp}(4, 2)| = 2^4(2^2 - 1)(2^4 - 1) = 720. \quad (29.53)$$

Both are positive characteristic-two fixtures. Negative tests include a symmetric form with nonzero diagonal in characteristic 2, a singular alternating matrix, swapped basis order, a matrix with determinant one that does not preserve J_n , a preserving matrix omitted from the carrier, an incorrect complement dimension, a wrong recurrence exponent, stale parent/form hashes, and independent verification.

27.5 Special and projective unitary groups

Quadratic field and conjugation. Fix $q = p^f$ and a concrete field $\mathbb{K} = \mathbb{F}_{q^2}$ supplied by a Statement 27.1 field record of degree $2f$ over $\mathbb{F}_p$. Define

$$\sigma : \mathbb{K} \longrightarrow \mathbb{K}, \quad \sigma(x) = x^q. \quad (29.54)$$

Since $x^{q^2} = x$ in $\mathbb{K}$, $\sigma^2 = 1$. Its fixed set is a subfield. For $x \neq 0$, the equation $x^q = x$ is equivalent to $x^{q-1} = 1$. Since $\mathbb{K}^\times$ is cyclic of order $q^2 - 1$, it has exactly $\gcd(q - 1, q^2 - 1) = q - 1$ such elements. Including zero, the fixed subfield has exactly q elements:

$$\mathbb{K}^\sigma = \{x : x^q = x\} \cong \mathbb{F}_q. \quad (29.55)$$

Because $q < q^2$, the fixed set is proper, so σ is not the identity. Together with $\sigma^2 = 1$, this proves that σ has exact order two. All computations use the actual fixed subcarrier and its inherited tables, not an unbound abstract copy of $\mathbb{F}_q$.

Hermitian form and unitary group. On $V = \mathbb{K}^n$, with $n \geq 1$, fix the form

$$h(x, y) := x^{\sigma^\top} y = \sum_{i=1}^n x_i^q y_i. \quad (29.56)$$

It is conjugate-linear in the first variable, linear in the second, satisfies $h(y, x) = h(x, y)^\sigma$, and is nondegenerate. For a matrix A , write $A^{\sigma^\top} := (A^\sigma)^\top$.

Definition 27.5.1 (Unitary and special unitary groups).

$$\begin{aligned} U(n, q) &:= \{A \in \text{GL}_n(\mathbb{K}) : A^{\sigma^T} A = I_n\}, \\ SU(n, q) &:= \{A \in U(n, q) : \det A = 1\}. \end{aligned} \quad (29.57)$$

The parameter of $U(n, q)$ is the fixed-field size q , while its matrix entries lie in the q^2 -element field.

Norm-one determinant. The field norm is

$$N_{\mathbb{K}/\mathbb{F}_q}(a) = a^{q+1}. \quad (29.58)$$

Because $\mathbb{K}^\times$ is cyclic of order $q^2 - 1$, its norm-one kernel

$$C_{q+1} := \{a \in \mathbb{K}^\times : a^{q+1} = 1\} \quad (29.59)$$

is cyclic of order $q+1$, and the norm image has order $q-1$, hence is all of $\mathbb{F}_q^\times$. Taking determinants in (29.57) gives

$$\det(A)^\sigma \det(A) = 1,$$

so determinant maps $U(n, q)$ into C_{q+1} . Conversely $\text{diag}(u, 1, \dots, 1) \in U(n, q)$ for every $u \in C_{q+1}$, proving the exact sequence

$$1 \longrightarrow SU(n, q) \longrightarrow U(n, q) \xrightarrow{\det} C_{q+1} \longrightarrow 1. \quad (29.60)$$

Counting unit vectors. Let s_m be the number of $x \in \mathbb{K}^m$ with $h(x, x) = 1$. All nonzero fibers of $x \mapsto h(x, x) \in \mathbb{F}_q$ have the same size: multiplication by an element of any prescribed nonzero norm gives a bijection between them. Appending one coordinate to a vector of length $m-1$ therefore gives

$$s_m = (q+1)q^{2m-2} - q s_{m-1}, \quad s_1 = q+1. \quad (29.61)$$

Indeed, a prefix of norm 1 permits only final coordinate zero; each other prefix permits $q+1$ final coordinates of the required nonzero norm. Induction in (29.61) yields

$$s_m = q^{m-1}(q^m - (-1)^m). \quad (29.62)$$

Theorem 27.5.2 (Orders of U and SU).

$$|U(n, q)| = q^{n(n-1)/2} \prod_{i=1}^n (q^i - (-1)^i), \quad (29.63)$$

and

$$|SU(n, q)| = q^{n(n-1)/2} \prod_{i=2}^n (q^i - (-1)^i). \quad (29.64)$$

Proof. The columns of a unitary matrix are an ordered orthonormal basis, and every ordered orthonormal basis gives one unitary matrix. Choose its first vector in s_n ways. Its orthogonal complement is nondegenerate of dimension $n-1$. Every nondegenerate Hermitian space over $\mathbb{K}/\mathbb{F}_q$ admits an orthonormal basis: diagonalize the form, then rescale each nonzero diagonal coefficient using norm surjectivity. Thus the complement is isometric to the standard space, and

$$|U(n, q)| = s_n |U(n-1, q)|, \quad |U(0, q)| = 1. \quad (29.65)$$

Substitution of (29.62) gives (29.63). Exactness of (29.60) divides by $q+1$, the $i=1$ factor, and proves (29.64). $\square$

Scalar center and projectivization. Put $c = \gcd(n, q + 1)$. The scalar matrices lying in $SU(n, q)$ are

$$Z_U(n, q) := \{\lambda I_n : \lambda^{q+1} = 1, \lambda^n = 1\}, \quad |Z_U(n, q)| = c. \quad (29.66)$$

They form the full center of $SU(n, q)$. For $n \geq 2$, this has the following explicit proof. Choose a generator $u \in C_{q+1}$. Since $q + 1 \geq 3$, one has $u \neq 1$ and $u \neq u^{-1}$. For $i \neq j$, let

$$D_{ij}(u) = \text{diag}(1, \dots, u, \dots, u^{-1}, \dots, 1),$$

with u in position i and u^{-1} in position j . Its diagonal entries have norm one and its determinant is one, so $D_{ij}(u) \in SU(n, q)$. Comparing entries in commutation with all these matrices forces every off-diagonal entry of a central matrix to vanish. Next embed

$$S_{ij} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

in coordinates i, j , with the identity on all other coordinates. It satisfies $S_{ij}^{\sigma^T} S_{ij} = I_n$ and has determinant one, including in characteristic 2. Commutation of a diagonal central matrix with every S_{ij} forces all its diagonal entries to agree. Hence it is λI_n , and the unitary and determinant conditions are exactly the two equations in (29.66). The $n = 1$ group is trivial and satisfies the same formula.

Definition 27.5.3 (Projective special unitary group).

$$PSU(n, q) := SU(n, q)/Z_U(n, q). \quad (29.67)$$

Consequently

$$|PSU(n, q)| = \frac{q^{n(n-1)/2}}{\gcd(n, q+1)} \prod_{i=2}^n (q^i - (-1)^i). \quad (29.68)$$

The action of $SU(n, q)$ on $\mathbb{P}^{n-1}(\mathbb{K})$ has scalar kernel exactly $Z_U(n, q)$, so it descends to a faithful projective action of $PSU(n, q)$. Indeed, a matrix fixing every projective point fixes the lines of each e_i , and the lines of $e_i + e_j$ force all resulting eigenvalues to agree; it is scalar. Intersecting the scalar group with $SU(n, q)$ gives (29.66). This assertion does not label the quotient simple.

Exact construction. Input contains p, f, q , a passing Statement 27.1 record for $GL(n, q^2)$, and the tags `FROBENIUS_Q` and `STANDARD_HERMITIAN_FORM`. Compute the q -power table, verify it is a field automorphism of exact order 2 unless the input is rejected as a collapsed extension, and enumerate its fixed subfield. Compute the norm table and verify its fibers, kernel, and image.

Filter the general-linear carrier first by $A^{\sigma^T} A = I_n$ to construct U , then by $\det A = 1$ to construct SU . Verify all inherited group tables, the determinant exact sequence, the unit-vector recurrence, and the two order formulas. Enumerate the scalar list (29.66), compare it with the exhaustive center of the finite SU table, partition into canonical scalar cosets, and construct the PSU tables and faithful projective action.

The fixed output is

$$\begin{aligned} &(p, f, q, h_{GL(n, q^2)}, \text{sigma_table}, \text{fixed_field_table}, \\ &\text{norm_table}, \text{Hermitian_form_table}, \text{U_carrier}, \text{U_tables}, \\ &\text{SU_carrier}, \text{SU_tables}, \text{determinant_sequence}, \\ &\text{unit_vector_counts}, |U(n, q)|, |SU(n, q)|, \text{ORDERS_VERIFIED}, \\ &c, \text{scalar_center}, \text{PSU_cosets}, \text{PSU_tables}, |PSU(n, q)|, \\ &\text{faithful_projective_action}, \text{COMPLETE}, \text{SATISFIED}, \\ &h_{\text{input}}, h_{\text{independent verification}}). \end{aligned} \quad (29.69)$$

An invalid or stale parent, $n < 1$, $q \neq p^f$, a Frobenius table that is not multiplicative or does not square to the identity, a fixed field of size other than q , a wrong norm fiber, a non-Hermitian or singular form, a retained matrix failing its form or determinant equation, determinant image smaller than C_{q+1} , a recurrence or order mismatch, a scalar-center disagreement, an invalid quotient or projective kernel, an unsupported simplicity flag, or stale hashes returns

$$\text{REJECT}(\text{INVALID_UNITARY_INPUT}, \text{stage}, \text{first_failure}, \text{h}_{\text{input}}). \quad (29.70)$$

Theorem 27.5.4 (Construction correctness and termination). *Every passing record constructs exactly $U(n, q)$, $SU(n, q)$, and $PSU(n, q)$, their stated orders, and the faithful projective action. The algorithm terminates and canonical inputs independent verification byte-identically.*

Proof. All field, matrix, vector, center, and coset carriers are finite. Exponentiation, fixed-point enumeration, form filtering, table restriction, center comparison, quotient construction, and action checking therefore terminate. Equations (29.54)–(29.60) prove the field, form, and determinant records; (29.61)–(29.65) prove the orders; and (29.66)–(29.68) prove projectivization. Fixed field, matrix, scalar, coset, projective-point, and byte orders prove independent verification. $\square$

Tests and boundaries. For $n = 1$,

$$|U(1, q)| = q + 1, \quad SU(1, q) = PSU(1, q) = 1. \quad (29.71)$$

For $q = 2, n = 2$, over $\mathbb{F}_4$,

$$|SU(2, 2)| = 6, \quad c = 1, \quad |PSU(2, 2)| = 6. \quad (29.72)$$

For $q = 2, n = 3$,

$$|SU(3, 2)| = 2^3(2^2 - 1)(2^3 + 1) = 216, \quad c = 3, \quad |PSU(3, 2)| = 72. \quad (29.73)$$

The latter quotient is not labeled simple. Rejection tests include using $x \mapsto x^p$ instead of $x \mapsto x^q$ when $f > 1$, confusing the q -element fixed field with the q^2 -element coefficient field, using all $q^2 - 1$ scalars as the determinant image, dropping the $i = 1$ factor twice, a center defined by only one of its two scalar equations, a forged projective kernel, stale parent hashes, and independent verification.

27.6 A certified ledger of classical-group orders

Scope and notation. This section consolidates the order calculations of Statement 27.1–Statement 27.5 without changing their carriers. Thus $q = p^f$ is a prime power, $n \geq 1$, linear and symplectic matrices have entries in $\mathbb{F}_q$, and unitary matrices have entries in $\mathbb{F}_{q^2}$ with fixed field $\mathbb{F}_q$. Put

$$d_- := \gcd(n, q - 1), \quad d_+ := \gcd(n, q + 1), \quad e := \gcd(2, q - 1). \quad (29.74)$$

Empty products below equal 1. The letter P always denotes an actual quotient by the full scalar center; it is never a decorative change of notation.

Definition 27.6.1 (Three adapted-basis counts). Let $B_{\text{lin}}(n, q)$, $B_{\text{sp}}(n, q)$, and $B_{\text{uni}}(n, q)$ denote respectively the numbers of ordered bases of $\mathbb{F}_q^n$, ordered symplectic bases of $\mathbb{F}_q^{2n}$ for the form of Statement 27.4, and ordered orthonormal bases of $\mathbb{F}_{q^2}^n$ for the Hermitian form of Statement 27.5. The preceding bijective counts give

$$\begin{aligned} B_{\text{lin}}(n, q) &= \prod_{r=0}^{n-1} (q^n - q^r) = q^{n(n-1)/2} \prod_{i=1}^n (q^i - 1), \\ B_{\text{sp}}(n, q) &= q^{n^2} \prod_{i=1}^n (q^{2i} - 1), \\ B_{\text{uni}}(n, q) &= q^{n(n-1)/2} \prod_{i=1}^n (q^i - (-1)^i). \end{aligned} \quad (29.75)$$

Lemma 27.6.2 (Why the basis counts are group orders). *The three groups $\text{GL}(n, q)$, $\text{Sp}(2n, q)$, and $U(n, q)$ act simply transitively on the corresponding sets of adapted ordered bases. Their orders are therefore B_{lin} , B_{Sp} , and B_{uni} .*

Proof. For the linear case, the columns of a matrix form an ordered basis exactly when the matrix is invertible. For either form-preserving case, the column matrix A of an ordered basis has Gram matrix $A^\top J_n A$, respectively $A^\sigma A$. It is adapted exactly when this Gram matrix is the fixed standard Gram matrix. Thus an adapted basis has one and only one group element as its column matrix. Equivalently, there is a unique isometry carrying the standard adapted basis to the prescribed one. This proves freeness and transitivity in all three cases. $\square$

Theorem 27.6.3 (Complete order ledger). *With the contracts above,*

$$|\text{GL}(n, q)| = q^{n(n-1)/2} \prod_{i=1}^n (q^i - 1), \quad (29.76)$$

$$|\text{SL}(n, q)| = q^{n(n-1)/2} \prod_{i=2}^n (q^i - 1), \quad (29.77)$$

$$|\text{PGL}(n, q)| = q^{n(n-1)/2} \prod_{i=2}^n (q^i - 1), \quad (29.78)$$

$$|\text{PSL}(n, q)| = \frac{q^{n(n-1)/2}}{d_-} \prod_{i=2}^n (q^i - 1), \quad (29.79)$$

$$|\text{Sp}(2n, q)| = q^{n^2} \prod_{i=1}^n (q^{2i} - 1), \quad (29.80)$$

$$|\text{PSp}(2n, q)| = \frac{q^{n^2}}{e} \prod_{i=1}^n (q^{2i} - 1), \quad (29.81)$$

$$|U(n, q)| = q^{n(n-1)/2} \prod_{i=1}^n (q^i - (-1)^i), \quad (29.82)$$

$$|SU(n, q)| = q^{n(n-1)/2} \prod_{i=2}^n (q^i - (-1)^i), \quad (29.83)$$

$$|PSU(n, q)| = \frac{q^{n(n-1)/2}}{d_+} \prod_{i=2}^n (q^i - (-1)^i). \quad (29.84)$$

Here $\text{PSp}(2n, q)$ means $\text{Sp}(2n, q)/Z(\text{Sp}(2n, q))$.

Proof. Equation (29.76) is the first count in (29.75). In its original product, factor $q^n - q^r = q^r(q^{n-r} - 1)$; then $\sum_{r=0}^{n-1} r = n(n-1)/2$, and reversing $n-r$ gives the displayed product.

The determinant map from $\text{GL}(n, q)$ onto $\mathbb{F}_q^\times$ has kernel $\text{SL}(n, q)$. Its image has order $q-1$, so division of (29.76) by the $i=1$ factor proves (29.77). The scalar center of $\text{GL}(n, q)$ also has order $q-1$; quotienting by that center proves (29.78). The scalar center of $\text{SL}(n, q)$ has order d_- ; dividing (29.77) by this center proves (29.79). Notice that (29.77) and (29.78) are equalities of integers, not an assertion that the two named groups are isomorphic.

The second adapted-basis count in (29.75) is (29.80). The symplectic-center lemma below identifies its center as a group of order e , proving (29.81).

The third adapted-basis count is (29.82). The determinant map from $U(n, q)$ onto the norm-one cyclic group of order $q+1$ has kernel $SU(n, q)$. Dividing by the $i=1$ factor $q - (-1) = q+1$ proves (29.83). Finally the scalar center of $SU(n, q)$ has order d_+ , so its projective quotient has order (29.84). Every division made here is therefore supplied by a displayed surjective homomorphism or by an identified finite center. $\square$

Lemma 27.6.4 (The symplectic center). *For every $n \geq 1$,*

$$Z(\mathrm{Sp}(2n, q)) = \{\lambda I_{2n} : \lambda \in \mathbb{F}_q^\times, \lambda^2 = 1\}, \quad |Z(\mathrm{Sp}(2n, q))| = e. \quad (29.85)$$

Proof. For each nonzero v in the symplectic space, define

$$\tau_v(x) := x + \langle x, v \rangle v.$$

Alternation gives $\langle \tau_v x, \tau_v y \rangle = \langle x, y \rangle$: the two cross terms cancel, also in characteristic two, and the last term contains $\langle v, v \rangle = 0$. Thus $\tau_v \in \mathrm{Sp}(2n, q)$. If A is central, form preservation gives

$$A\tau_v A^{-1} = \tau_{Av} = \tau_v.$$

The nonzero rank-one operators $x \mapsto \langle x, Av \rangle Av$ and $x \mapsto \langle x, v \rangle v$ are equal. Equality of their images gives $Av = \lambda_v v$, and substitution gives $\lambda_v^2 = 1$. If v, w are independent, comparison of $A(v + w)$ in the basis v, w yields $\lambda_v = \lambda_w = \lambda_{v+w}$. Dependent nonzero vectors can be linked through a vector independent of both because the space has dimension $2n \geq 2$. Hence one scalar λ works for every nonzero vector and $A = \lambda I_{2n}$. Conversely such a scalar preserves the form exactly when $\lambda^2 = 1$, and is then central. Since $\mathbb{F}_q^\times$ is cyclic of order $q - 1$, the equation $\lambda^2 = 1$ has $\gcd(2, q - 1) = e$ solutions. $\square$

Proposition 27.6.5 (Two stages and no hidden divisor). *The provenance of every factor is*

target	adapted-basis count	successive exact divisors
GL	B_{lin}	1
SL	B_{lin}	$q - 1$ (determinant image)
PGL	B_{lin}	$q - 1$ (scalar center)
PSL	B_{lin}	$(q - 1)d_-$
Sp	B_{sp}	1
PSp	B_{sp}	e
U	B_{uni}	1
SU	B_{uni}	$q + 1$ (determinant image)
PSU	B_{uni}	$(q + 1)d_+$

(29.86)

In particular, determinant-one restriction and projectivization are different operations even when they happen to divide by the same integer.

Proof. The adapted-basis bijections prove the middle column. The exact sequences and center computations used in the proof of the order ledger prove every entry in the last column. For example, SL is a kernel of index $q - 1$, whereas PGL is a quotient by a subgroup of order $q - 1$. Similarly SU is a determinant kernel of index $q + 1$, whereas PSU requires the additional and generally different divisor d_+ . There is no quotient divisor for a symbol that does not begin with P . $\square$

Corollary 27.6.6 (Exact index checks). *All adjacent ratios forced by the constructions are*

$$\begin{aligned} [\mathrm{GL} : \mathrm{SL}] &= q - 1, & |\mathrm{GL}|/|\mathrm{PGL}| &= q - 1, & |\mathrm{PGL}|/|\mathrm{PSL}| &= d_-, \\ |\mathrm{Sp}|/|\mathrm{PSp}| &= e, & [U : SU] &= q + 1, & |SU|/|PSU| &= d_+. \end{aligned} \quad (29.87)$$

The middle linear ratio follows numerically from the formulas and agrees with the exact sequence of Statement 27.3.

Proof. Cancel the common products in (29.76)–(29.84). The cancellation is legitimate because all group orders are positive integers; the same ratios also equal the already identified image or center orders. Thus the identities are both arithmetically and group-theoretically certified. $\square$

Corollary 27.6.7 (*p*-parts). *Writing v_p for the exponent of p in a positive integer,*

$$\begin{array}{c|c} \text{families} & v_p(|G|) \\ \hline \text{GL, SL, PGL, PSL, } U, SU, PSU & f n(n-1)/2 \\ \text{Sp, PSp} & f n^2 \end{array} \quad (29.88)$$

Proof. For every $i \geq 1$, both $q^i - 1$ and $q^i + 1$ are prime to p . The integers d_- , d_+ , and e divide $q - 1$ or $q + 1$, so they too are prime to p . The only p -power in each ledger entry is therefore its displayed power of $q = p^f$, which proves the table. $\square$

Theorem 27.6.8 (Rank-one and small-field boundary analysis). *The ledger specializes to*

$$\begin{aligned} |\text{GL}(1, q)| &= q - 1, & |\text{SL}(1, q)| &= |\text{PGL}(1, q)| = |\text{PSL}(1, q)| = 1, \\ |\text{Sp}(2, q)| &= q(q^2 - 1) = |\text{SL}(2, q)|, & |\text{PSp}(2, q)| &= |\text{PSL}(2, q)|, \\ |U(1, q)| &= q + 1, & |SU(1, q)| &= |PSU(1, q)| = 1. \end{aligned} \quad (29.89)$$

It also gives

$$\begin{array}{c|cccc} (n, q) & |\text{GL}| & |\text{SL}| & |\text{PGL}| & |\text{PSL}| \\ \hline (2, 2) & 6 & 6 & 6 & 6 \\ (2, 3) & 48 & 24 & 24 & 12 \end{array} \quad \begin{array}{c|ccc} (n, q) & |U| & |SU| & |PSU| \\ \hline (2, 3) & 96 & 24 & 12 \\ (3, 2) & 648 & 216 & 72 \end{array} . \quad (29.90)$$

Proof. For $n = 1$, substitute into (29.76)–(29.84), use the empty product convention, and note $d_- = 1$, $d_+ = 1$. For the symplectic comparison, (29.80) at $n = 1$ is $q(q^2 - 1)$, which is (29.77) at linear dimension two; their centers both have order $\gcd(2, q - 1)$. Direct substitution gives the two finite tables. These tests include characteristic two, nontrivial linear centers, and nontrivial unitary centers. $\square$

Exact order-certificate algorithm. An input record consists of a family symbol

$$\mathcal{F} \in \{\text{GL, SL, PGL, PSL, Sp, PSp, } U, SU, PSU\}, \quad (29.91)$$

the integer $n \geq 1$, a prime-power certificate $q = p^f$, and the concrete field-table hashes required in Statement 27.1 and Statement 27.5. For Sp and PSp, n denotes the half-dimension; for all other symbols it denotes the matrix dimension. The algorithm performs the following finite operations.

1. Verify the prime-power and field records; for a unitary family, verify the q^2 -element carrier, the exact order-two Frobenius, and its q -element fixed subcarrier.
2. Build the applicable adapted-basis factor list in (29.75) using exact integers. Record each factor together with its family and index i .
3. Apply only the successive divisors assigned to the requested symbol in (29.86). Compute each greatest common divisor by the Euclidean algorithm and require exact zero remainder at every division.
4. Recompute the appropriate adjacent ratios in (29.87), the p -part in (29.88), and every applicable boundary value in (29.89)–(29.90). Reject the record if any equality fails.
5. Serialize the symbol, dimension convention, field hashes, factor list, divisor provenance, quotient order, and all check values; hash the serialization.

Theorem 27.6.9 (Termination, soundness, and independent verification). *The order-certificate algorithm terminates. It accepts exactly records whose returned integer is the order of the requested concrete classical group, and its serialized record deterministically replays that conclusion.*

Proof. Primality tests, repeated powering in finite tables, the Euclidean algorithm, and the loops of length n all terminate. Thus every step is finite. If the record is accepted, Step 1 binds all symbols to the carriers used in the proofs above; Step 2 computes the appropriate simply transitive adapted-basis count; and Step 3 applies precisely the determinant-image and scalar-center divisors proved in (29.86). The complete order ledger therefore proves that the returned integer is $|\mathcal{F}(n, q)|$. Conversely a valid concrete input, processed with the prescribed exact formulas, satisfies the divisibility, ratio, p -part, and boundary checks, so it is accepted. Serialization fixes every branch and every integer operation; recomputation from it is deterministic, and hash equality binds the independent verification to the same certificate. $\square$

Failure certificates. An invalid prime-power record, a mismatched fixed field, an unrecognized symbol, a dimension below one, a nonexact alleged division, or a failed ratio produces a typed rejection containing the first failed equality and the input hashes. In particular the procedure never repairs a missing $q - 1$, $q + 1$, d_- , d_+ , or e factor by silently relabeling the target group.

27.7 Exact automorphism groups and a guarded classification interface

Concrete input. Let Q be one of the concrete finite groups constructed in Statement 27.1–Statement 27.6. Here it is stored as

$$\mathcal{Q} = (E, \mu, \iota, e_Q; X, \text{enc}, H_Q), \quad (29.92)$$

where $E = \{0, \dots, N - 1\}$, $e_Q \in E$ is the identity, $\mu(a, b) = ab$ is the complete multiplication table, $\iota(a) = a^{-1}$, $X = (x_1, \dots, x_s)$ is an ordered generating tuple, enc is the canonical matrix or projective-class encoding, and H_Q hashes all these data. A projective element is represented by the lexicographically least nonzero scalar multiple of a matrix. Thus equality in Q is decidable before any automorphism is discussed.

Definition 27.7.1 (Permutation model of the automorphism group). Write permutations as functions and compose functionally: $(ab)(x) = a(b(x))$. Define

$$\text{Aut}_{\text{tab}}(\mathcal{Q}) := \{a \in \text{Sym}(E) : a(e_Q) = e_Q, a(\mu(u, v)) = \mu(a(u), a(v)) \text{ for all } u, v \in E\}. \quad (29.93)$$

An element a is serialized by the vector $(a(0), \dots, a(N - 1))$.

Theorem 27.7.2 (The table model is exact). $\text{Aut}_{\text{tab}}(\mathcal{Q})$ is a group and the identity-on-carrier identification gives a canonical isomorphism

$$\text{Aut}_{\text{tab}}(\mathcal{Q}) \cong \text{Aut}(Q). \quad (29.94)$$

For $a \in \text{Aut}(Q)$, its exact order is the least common multiple of the lengths of the cycles of its permutation vector.

Proof. The condition in (29.93) says precisely that the bijection a preserves multiplication and the identity. Hence it is a group automorphism. Conversely every abstract automorphism permutes the finite carrier and satisfies those equations. Identity, composition, and inverse permutations preserve the same equations, so the displayed set is a group and the correspondence is an isomorphism. A permutation power is the identity exactly when its exponent is divisible by every cycle length; their least common multiple is therefore its exact order. $\square$

Lemma 27.7.3 (Generator-image extension certificate). Fix images $Y = (y_1, \dots, y_s) \in Q^s$. Choose, by breadth-first search in the ordered alphabet $x_1, x_1^{-1}, \dots, x_s, x_s^{-1}$, one canonical word w_g for every $g \in Q$, and put

$$a_Y(g) := w_g(Y). \quad (29.95)$$

Then Y is the image tuple of an automorphism if and only if

$$\begin{aligned} a_Y(x_i) = y_i \quad (1 \leq i \leq s), \quad a_Y(uv) = a_Y(u)a_Y(v) \quad (u, v \in Q), \\ \{a_Y(g) : g \in Q\} = Q. \end{aligned} \quad (29.96)$$

If it exists, the automorphism with image tuple Y is unique.

Proof. If an automorphism sends X to Y , applying it to $g = w_g(X)$ forces its value to be (29.95); it consequently satisfies (29.96), proving necessity and uniqueness. Conversely the middle equality in (29.96) makes a_Y a homomorphism, and the last equality makes it surjective. A surjection of a finite set to itself is bijective, so a_Y is an automorphism with the declared generator images. $\square$

Unconditional enumeration. Enumerate the N^s tuples Y in lexicographic order, apply (29.95)–(29.96), and retain the accepted permutation vectors. The generator-image lemma proves that every automorphism appears once and only once. Thus this finite procedure is an unconditional completeness certificate for $\text{Aut}(Q)$; it may be large, but finiteness is not confused with tractability.

Theorem 27.7.4 (Inner automorphisms). *The map*

$$\begin{aligned} \text{inn} : Q &\longrightarrow \text{Aut}(Q), \\ g &\longmapsto (x \mapsto gxg^{-1}) \end{aligned} \quad (29.97)$$

is a homomorphism with kernel $Z(Q)$. Consequently $\text{Inn}(Q) \cong Q/Z(Q)$, and the permutation vector computed from (29.97) is an exact generator-image certificate.

Proof. Direct multiplication gives $\text{inn}_{gh} = \text{inn}_g \text{inn}_h$ under the fixed functional composition convention. The image of g is the identity exactly when $gx = xg$ for every $x \in Q$, which is exactly $g \in Z(Q)$. The first isomorphism theorem proves the quotient statement. Evaluating on every encoded carrier element yields the permutation vector, whose equations are then checked by (29.93). $\square$

Theorem 27.7.5 (Normalizer and diagonal interface). *Suppose the chosen faithful matrix realization places $\hat{Q}$ in $\text{GL}(V)$. Let*

$$N := N_{\text{GL}(V)}(\hat{Q}), \quad C := C_{\text{GL}(V)}(\hat{Q}).$$

Conjugation defines an exact sequence

$$1 \longrightarrow C \longrightarrow N \xrightarrow{c} \text{Aut}(\hat{Q}), \quad c(d)(x) = dxd^{-1}, \quad (29.98)$$

whose image is isomorphic to N/C . The order of $c(d)$ is the order of dC in N/C . Its outer class has the order of $d(\hat{Q}C)$ in $N/(\hat{Q}C)$.

Proof. Normalization makes conjugation preserve $\hat{Q}$, and conjugation is multiplicative in d . Its kernel consists exactly of matrices commuting with every element of $\hat{Q}$, namely C , so the first isomorphism theorem proves the image claim and the first order statement. The inner automorphisms in this image are conjugations by elements of $\hat{Q}$. Therefore their full preimage is $\hat{Q}C$, which proves the outer-class statement. Replacing C by the scalar matrices is permitted only after a separate absolute-irreducibility and centralizer certificate. $\square$

Diagonal automorphisms. For a classical group, the normalizer search may be restricted to the declared general-linear or similitude overgroup, but every returned d must still pass $d\hat{Q}d^{-1} = \hat{Q}$ on the full table or on certified generators. The two cosets in (29.98) are both recorded: dC controls the order of the actual automorphism, while $d(\hat{Q}C)$ controls only its outer class. They must not be interchanged.

Theorem 27.7.6 (Field automorphisms). *Let the coefficient field be $\mathbb{F}_{p^m}$, where $m = f$ for the linear and symplectic carriers and $m = 2f$ for the unitary carrier. If entrywise*

$$F_r(A) := A^{p^r} \quad (29.99)$$

preserves the defining matrix equations and the scalar equivalence used for projectivization, it induces an automorphism of Q . The coefficient map has order

$$h = \frac{m}{\gcd(m, r)}, \quad (29.100)$$

and the induced group automorphism has order dividing h . Its exact order is the cycle least common multiple in (29.94).

Proof. Frobenius preserves sums, products, inverses, determinants, and matrix multiplication. Applying it to a defining equation whose coefficients and form class are stable gives the same equation, so the group carrier is preserved. It also sends scalar multiples to scalar multiples, proving well-definedness on projective classes. The Galois group of $\mathbb{F}_{p^m}/\mathbb{F}_p$ is cyclic of order m , and the order of its r -th power is $m/\gcd(m, r)$. Hence F_r^h is the identity on all entries, so the induced order divides h . A collapse can occur on a small or subfield-defined group; the permutation-vector order, rather than h alone, decides exactness. $\square$

Theorem 27.7.7 (A concrete graph map for the linear family). *For $Q = \text{PSL}(n, q)$, the formula*

$$\gamma([A]) := [(A^{-1})^\top] \quad (29.101)$$

is a well-defined automorphism of order at most two. When $n = 2$ it is inner and must not be counted as a new outer factor.

Proof. Replacing A by λA replaces the displayed matrix by $\lambda^{-1}(A^{-1})^\top$, so its projective class is unchanged. Also

$$\gamma([AB]) = [(B^{-1}A^{-1})^\top] = [(A^{-1})^\top(B^{-1})^\top] = \gamma([A])\gamma([B]),$$

and applying γ twice returns $[A]$. If $n = 2$, then for $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and every $A \in \text{SL}(2, q)$, $(A^{-1})^\top = JAJ^{-1}$. Thus γ is conjugation by the projective class of J , hence inner. For unitary, symplectic, twisted, and exceptional low-rank realizations, a family-specific graph or graph-field formula is treated as a candidate and accepted only after the table test (29.93); no uniform order-two factor is presumed. $\square$

Definition 27.7.8 (Standard generated subgroup). Let I, D, F, Γ be the accepted permutation-vector sets arising respectively from inner, diagonal-normalizer, field, and graph or graph-field candidates. Their exact semilinear closure is

$$A_{\text{std}} := \langle I \cup D \cup F \cup \Gamma \rangle \leq \text{Aut}_{\text{tab}}(Q). \quad (29.102)$$

It is computed by closure under permutation composition and inverse; overlaps between the four sources are retained as equal vectors, not counted twice.

Theorem 27.7.9 (Guarded classification interface). *There are exactly two sound ways to certify that $A_{\text{std}} = \text{Aut}(Q)$.*

1. Enumeration branch: *compute $\text{Aut}_{\text{tab}}(Q)$ by (29.95)–(29.96) and compare the two sorted permutation-vector sets.*
2. Classification branch: *attach a theorem certificate whose family, rank, field, simplicity, and exception hypotheses match the concrete carrier and whose conclusion is precisely that every automorphism is a product of the accepted inner, diagonal, field, and graph candidates.*

In either branch, equality of the two groups follows. Without one of these certificates the only valid conclusion is $A_{\text{std}} \leq \text{Aut}(Q)$.

Proof. Every generator of A_{std} has passed (29.93), so the inclusion always holds. In the enumeration branch, equality is a literal equality of canonical finite sets. In the classification branch, the attached theorem gives the reverse inclusion because its conclusion expresses every automorphism as a word in the accepted generators. If neither reverse-inclusion argument is present, generation of a subgroup supplies no logical route to surjectivity, so equality cannot be asserted. $\square$

Mandatory exception router. The simple-projective classification branch is disabled, unless a more specific certified theorem is supplied, for every nonprojective target and for the following low-rank records:

record	identified behavior	required route	
PSL(1, q)	1	enumerate	
PSL(2, 2)	S_3	enumerate	
PSL(2, 3)	A_4	enumerate	
PSU(1, q)	1	enumerate	(29.103)
PSU(2, q)	type A_1	reroute or enumerate	
PSU(3, 2)	order 72	enumerate	
PSp(2, q)	PSL(2, q)	reroute or enumerate	
PSp(4, 2)	S_6	enumerate	

The S_6 row is especially important because its exceptional outer automorphism need not be recovered by a naive type- C_2 factor list. Other accidental isomorphisms are stored in an exception registry keyed by the carrier hash; they may alter the names and overlaps of outer generators even when the group is simple. The enumeration branch remains valid for every row.

Generator-image output table. For every accepted generator a of A_{std} , the record contains

source	source datum	$a(x_1), \dots, a(x_s)$	$ a $	test hash	
I	g	$gx_1g^{-1}, \dots, gx_sg^{-1}$	cycle lcm	H_a	
D	d	$dx_1d^{-1}, \dots, dx_sd^{-1}$	cycle lcm	H_a	(29.104)
F	r	$F_r(x_1), \dots, F_r(x_s)$	cycle lcm	H_a	
Γ	formula id	$\gamma(x_1), \dots, \gamma(x_s)$	cycle lcm	H_a	

Every row expands to the full permutation vector through (29.95) and passes (29.96); the abbreviated table is never the sole well-definedness evidence.

Theorem 27.7.10 (Canonical serialization and consistency). *Sort automorphisms by their full permutation vectors, choose the lexicographically first minimal generating tuple of the resulting permutation group, and serialize*

$$\mathcal{A}_Q = (H_Q, \text{branch}, \text{exception id}, I, D, F, \Gamma, A_{\text{std}}, \text{Aut}_{\text{tab}}(Q), \text{generator table}). \quad (29.105)$$

Repeated execution on the same input produces byte-identical output. The record certifies closure, homomorphism equations, identity and inverse consistency, exact element orders, and, when the final equality flag is set, completeness of the automorphism group.

Proof. All carrier, tuple, breadth-first, and permutation orders are fixed lexicographically. Multiplication, inverse, cycle decomposition, and subgroup closure are deterministic finite operations. Hence the selected vectors and their serialization are unique. Equations (29.93) and (29.96) certify the homomorphism and group laws; cycle decomposition certifies orders. The guarded-classification theorem proves that the equality flag is set only with a reverse-inclusion certificate. $\square$

Order-eight extraction. The subsequent Statement 27.8 search reads the sorted subset

$$\text{Aut}_8(Q) := \{a \in \text{Aut}_{\text{tab}}(Q) : |a| = 8\}. \quad (29.106)$$

Membership is exact because it is decided from permutation cycles. A nonempty $\text{Aut}_8(Q)$ supplies candidates only; surface-relator, generation, parity, equivariance, and dangerous-set conditions remain separate conditions.

27.8 Classical groups suitable for the C_8 -action

Typed marked record. Let Q be a concrete classical group whose carrier, order, and automorphism table have passed Statement 27.1–Statement 27.7. Let

$$\mathcal{D} = (d_1, \dots, d_t) \subset \Gamma_B \setminus \{1\} \quad (29.107)$$

be a finite ordered dangerous list, with every d_ν supplied as a canonical word and exact group key. The list is part of the input: no geometric conclusion may use elements not actually included in it. A marked candidate is

$$\mathfrak{M} = (Q; A_1, B_1, A_2, B_2; \alpha; \chi_Q; \mathcal{D}), \quad (29.108)$$

where $A_i, B_i \in Q$, α is a certified member of $\text{Aut}(Q)$, and $\chi_Q : Q \rightarrow C_2$ is a proposed parity table.

Definition 27.8.1 (The seven verification conditions). The candidate $\mathfrak{M}$ is C_8 -admissible relative to $\mathcal{D}$ when all of the following hold.

R, relator. $[A_1, B_1][A_2, B_2] = e_Q$, with $[x, y] = xyx^{-1}y^{-1}$.

G, generation. $\langle A_1, B_1, A_2, B_2 \rangle = Q$.

A, automorphism. α passes the full multiplication-table test (29.93).

O, exact order. $\text{ord}(\alpha) = 8$, computed from its permutation cycles.

E, equivariance. On the four presentation generators, $\alpha \circ \pi_{\mathfrak{M}} = \pi_{\mathfrak{M}} \circ \phi_8$.

P, parity. χ_Q is a homomorphism and $\chi_Q(A_1) = \chi_Q(B_1) = \chi_Q(A_2) = \chi_Q(B_2) = 1$.

D, separation. $\pi_{\mathfrak{M}}(d_\nu) \neq e_Q$ for every $1 \leq \nu \leq t$.

Here $\pi_{\mathfrak{M}}$ is initially the generator assignment

$$a_1 \mapsto A_1, \quad b_1 \mapsto B_1, \quad a_2 \mapsto A_2, \quad b_2 \mapsto B_2. \quad (29.109)$$

Condition R is what makes this assignment a map from Γ_B ; this fact is proved below rather than assumed in the definition.

Executable equivariance table. Using the exact Statement 7.3 words for ϕ_8 , condition E is the equality of the last two columns in

x	$\pi_{\mathfrak{M}}(x)$	$\pi_{\mathfrak{M}}(\phi_8(x))$	$\alpha(\pi_{\mathfrak{M}}(x))$
a_1	A_1	B_1^{-1}	$\alpha(A_1)$
b_1	B_1	$A_2^{-1}B_1A_1$	$\alpha(B_1)$
a_2	A_2	$A_2^{-1}B_1A_1B_1^{-1}A_1^{-1}B_1^{-1}B_2^{-1}$	$\alpha(A_2)$
b_2	B_2	$A_1B_1A_1^{-1}B_1^{-1}A_2$	$\alpha(B_2)$

(29.110)

All entries are evaluated in the supplied multiplication table of Q . Thus the covariance claim has four exact generator-image rows and no implicit word convention.

Theorem 27.8.2 (Relator and generation construct the quotient). *Conditions R and G hold if and only if (29.109) extends uniquely to a surjective homomorphism*

$$\pi_{\mathfrak{M}} : \Gamma_B \twoheadrightarrow Q. \quad (29.111)$$

Its kernel $K_{\mathfrak{M}}$ is normal and $\Gamma_B/K_{\mathfrak{M}} \cong Q$.

Proof. The Bolza surface group has presentation

$$\Gamma_B = \langle a_1, b_1, a_2, b_2 \mid [a_1, b_1][a_2, b_2] = 1 \rangle.$$

The universal property of a presented group says that the free generator assignment factors through Γ_B exactly when the relator evaluates to the identity. This is condition R, and uniqueness follows because the four classes generate Γ_B . The image of the resulting homomorphism is exactly $\langle A_1, B_1, A_2, B_2 \rangle$, so it is onto exactly when G holds. Every homomorphism kernel is normal, and the first isomorphism theorem gives the final isomorphism. $\square$

Theorem 27.8.3 (Generator covariance is full covariance). *Assume R, G, and A. Then the four equalities (29.110) hold if and only if*

$$\alpha \circ \pi_{\mathfrak{M}} = \pi_{\mathfrak{M}} \circ \phi_8 \quad \text{on all of } \Gamma_B. \quad (29.112)$$

When they hold, $\phi_8(K_{\mathfrak{M}}) = K_{\mathfrak{M}}$, and $\alpha^8 = \text{id}_Q$.

Proof. Both sides of (29.112) are homomorphisms $\Gamma_B \rightarrow Q$. They are equal exactly when they agree on the four presentation generators, which is the last-column comparison in (29.110). If $k \in K_{\mathfrak{M}}$, then

$$\pi_{\mathfrak{M}}(\phi_8(k)) = \alpha(\pi_{\mathfrak{M}}(k)) = \alpha(e_Q) = e_Q,$$

so $\phi_8(K_{\mathfrak{M}}) \subseteq K_{\mathfrak{M}}$. Iterating eight times and using $\phi_8^8 = 1$ upgrades inclusion to equality. Finally (29.112) and $\phi_8^8 = 1$ show that α^8 fixes the image of every element of Γ_B ; surjectivity of $\pi_{\mathfrak{M}}$ makes that image all of Q , hence $\alpha^8 = 1$. $\square$

Corollary 27.8.4 (The surface-relator seed orbit). *If conditions R, G, A, O, and E hold, then*

$$\mathbf{A}^{(k)} := (\alpha^k(A_1), \alpha^k(B_1), \alpha^k(A_2), \alpha^k(B_2)), \quad 0 \leq k < 8, \quad (29.113)$$

is an orbit of eight distinct generating surface-relator seeds, and

$$\mathbf{A}^{(k)} = (\pi_{\mathfrak{M}}(\phi_8^k(a_1)), \pi_{\mathfrak{M}}(\phi_8^k(b_1)), \pi_{\mathfrak{M}}(\phi_8^k(a_2)), \pi_{\mathfrak{M}}(\phi_8^k(b_2))). \quad (29.114)$$

Proof. An automorphism preserves the value of every group word, so applying α^k to the relator identity preserves the relation and applying it to a generating tuple preserves generation. Repeated use of (29.112) proves (29.114). If $\mathbf{A}^{(k)} = \mathbf{A}^{(0)}$ for $0 < k < 8$, then α^k fixes a generating set and hence all of Q , contradicting $\text{ord}(\alpha) = 8$. $\square$

Theorem 27.8.5 (Parity factor and invariance). *Under R and G, condition P is equivalent to the existence of a homomorphism $\chi_Q : Q \rightarrow C_2$ satisfying*

$$\chi_Q \circ \pi_{\mathfrak{M}} = \chi. \quad (29.115)$$

This factor is unique. If E also holds, then

$$\chi_Q \circ \alpha = \chi_Q. \quad (29.116)$$

Proof. If (29.115) holds, its values on the four presentation generators are one, proving the generator rows of P. Conversely a homomorphism χ_Q with those four values makes $\chi_Q \pi_{\mathfrak{M}}$ and χ agree on a generating set, hence everywhere. Surjectivity of $\pi_{\mathfrak{M}}$ gives uniqueness. Using (29.112), Statement 9.4, and (29.115),

$$\chi_Q \alpha \pi_{\mathfrak{M}} = \chi_Q \pi_{\mathfrak{M}} \phi_8 = \chi \phi_8 = \chi = \chi_Q \pi_{\mathfrak{M}}.$$

Canceling the surjective map $\pi_{\mathfrak{M}}$ proves (29.116). Since the four marked images generate and have parity one, χ_Q is onto and $|Q|$ is even. $\square$

Theorem 27.8.6 (Dangerous-set meaning). *Under R and G, condition D is equivalent to*

$$K_{\mathfrak{M}} \cap \mathcal{D} = \emptyset. \quad (29.117)$$

If the supplied list is the complete global list $\tilde{\mathcal{D}}_{\text{sys}}(L)$ of Statement 15.7, then D is further equivalent to

$$\text{sys}(\mathbb{H}/K_{\mathfrak{M}}) > L. \quad (29.118)$$

No such systolic conclusion follows from a merely partial list.

Proof. For each d_ν , membership in the kernel is equivalent to $\pi_{\mathfrak{M}}(d_\nu) = e_Q$. Applying this equivalence to all entries gives (29.117). When the list is the specified complete global list, the complete dangerous-set criterion (15.28) gives (29.118), including the closed boundary at length L . If the list is partial, an omitted kernel element can still have length at most L , so the stronger conclusion is not justified. $\square$

Corollary 27.8.7 (Orbit-compressed separation). *Assume R, G, A, and E, and suppose $\mathcal{D}$ is ϕ_8 -invariant. It suffices to test one canonical representative from each $\langle \phi_8 \rangle$ -orbit:*

$$\pi_{\mathfrak{M}}(d) \neq e_Q \implies \pi_{\mathfrak{M}}(\phi_8^k(d)) = \alpha^k(\pi_{\mathfrak{M}}(d)) \neq e_Q. \quad (29.119)$$

Proof. The equality is repeated covariance (29.112). An automorphism maps the identity only to the identity, so it maps a nonidentity element to a nonidentity element. Every list entry lies in the orbit of its representative, proving sufficiency; necessity is immediate. $\square$

Physical-shell refinement. For any marked candidate satisfying R and G, let $g_0, \dots, g_7$ be the fixed physical shell and set

$$q_j := \pi_{\mathfrak{M}}(g_j), \quad \mathcal{D}_{\text{shell}} := \{g_j : 0 \leq j < 8\} \cup \{g_i^{-1} g_j : 0 \leq i < j < 8\}. \quad (29.120)$$

Call this candidate *shell-suitable* when the eight q_j are distinct and nonidentity. This definition does not assume condition O.

Theorem 27.8.8 (Shell criterion and an alternative order witness). *Under R, G, A, and E,*

$$\begin{aligned} \mathfrak{M} \text{ is shell-suitable} &\iff K_{\mathfrak{M}} \cap \mathcal{D}_{\text{shell}} = \emptyset, \\ \alpha(q_j) &= q_{j+1} \quad (j \bmod 8). \end{aligned} \quad (29.121)$$

Moreover shell suitability makes condition O automatic.

Proof. An identity $q_j = e_Q$ is equivalent to $g_j \in K$, and a collision $q_i = q_j$ is equivalent to $g_i^{-1}g_j \in K$. This proves the first equivalence. Covariance and $\phi_8(g_j) = g_{j+1}$ prove the second line. The full-covariance theorem already gives $\alpha^8 = 1$. For $1 \leq k < 8$, $\alpha^k(q_0) = q_k \neq q_0$, so no smaller positive power is the identity. Hence $\text{ord}(\alpha) = 8$. $\square$

Theorem 27.8.9 (Necessary and sufficient suitability criterion). *A tuple (29.108) is an accepted C_8 -equivariant, parity-compatible, dangerous-set-separating classical quotient if and only if conditions*

$$R, G, A, O, E, P, D \quad (29.122)$$

all pass. Adding the shell condition is necessary and sufficient for the accepted quotient also to have the simple eight-neighbor physical shell. In the accepted case, the regular cover degree is

$$[\Gamma_B : K_{\mathfrak{M}}] = |Q|, \quad (29.123)$$

where $|Q|$ is independently recomputed from the applicable Statement 27.6 order formula.

Proof. Necessity follows by reading an accepted quotient: its marked images satisfy the presentation relation and generate its declared image; the declared action is an exact order-eight automorphism; descent is precisely (29.112); parity factorization is (29.115); and dangerous-set separation is (29.117). The physical-shell condition is equivalent to (29.121).

Conversely, the relator and generation theorem constructs a surjective $\pi_{\mathfrak{M}}$. Conditions A, O, and E supply the exact C_8 -action and prove kernel invariance. Condition P supplies the unique parity factor, and D supplies precisely the declared separation. Thus every required structure exists and has the stated compatibility. Finally the first isomorphism theorem identifies $\Gamma_B/K_{\mathfrak{M}}$ with Q , so their orders give (29.123). This equality is proved only after surjectivity; it is not a degree–order identification by notation. $\square$

Why no condition may be silently omitted. Each condition has its own independently verifiable failure witness:

R	nonidentity relator value	
G	least element outside the generated subgroup	
A	first failed multiplication or bijection row	
O	cycle lcm different from eight	
E	first unequal row of (29.110)	(29.124)
P	first parity multiplication conflict	
D	least dangerous word evaluating to e_Q	
shell	least identity or collision pair	

For example, the parity quotient $Q = C_2$, with all four presentation generators mapped to its nonzero element, passes the relator, generation, and parity conditions but has trivial automorphism group and therefore fails O. When the declared target is the trivial group, its all-identity tuple passes both R and G, but it fails P and O. These boundary records prevent relation, parity, and eightfold symmetry from being conflated.

Exhaustive finite search. Let $\text{Aut}_8^{\text{avail}}(Q)$ be the sorted set of order-eight automorphism vectors actually present in the supplied Statement 27.7 record. It is always a subset of $\text{Aut}_8(Q)$. A dedicated flag `AUT8_COMPLETE` certifies equality of these two sets; the stronger Statement 27.7 flag `AUT_COMPLETE` implies it. For a selected finite list of certified classical carriers, the constructor scans groups in increasing $(|Q|, \text{family}, n, q, H_Q)$ order. Inside a group it scans Q^4 lexicographically, tests R and G, scans the sorted $\text{Aut}_8^{\text{avail}}(Q)$, applies (29.110), constructs the unique parity factor by the Statement 9.5 table algorithm, and evaluates the dangerous list. Shell mode also applies (29.121). The first passing tuple is returned.

Theorem 27.8.10 (Search termination, soundness, and completeness). *The search over a finite carrier list terminates, and every SATISFIED output satisfies (29.122). If every input group has a certificate $\text{Aut}_8^{\text{avail}}(Q) = \text{Aut}_8(Q)$, namely AUT8_COMPLETE (for example, as implied by AUT_COMPLETE), then a completed search with no output proves that no marked candidate in the declared carriers satisfies the requested conditions. Without that certificate, a found candidate remains sound, but a negative result is only PARTIAL_NO_MATCH.*

Proof. There are finitely many groups, $|Q|^4$ marked tuples per group, finitely many automorphism vectors, and finitely many dangerous words, so all loops terminate. Each positive test is exactly a condition in the necessary-and-sufficient theorem, proving soundness. When the available order-eight sets are complete, the loops enumerate every possible tuple and every possible exact order-eight automorphism, so exhaustion proves nonexistence in the stated finite domain. If such a set is incomplete, an omitted order-eight automorphism could complete a tuple; this prevents a sound global negative conclusion but does not invalidate candidates already verified by the full table equations. $\square$

Canonical certificate. The accepted serialization is

$$\begin{aligned} \mathcal{C}_{29.08} := & (H_Q, \text{family}, n, q, |Q|, A_1, B_1, A_2, B_2, H_\alpha, \text{ord } \alpha, \\ & \text{four rows (29.110), } (\mathbf{A}^{(k)})_{k=0}^7, \chi_Q, (q_j)_{j=0}^7, \\ & H_{\mathcal{D}}, (\pi_{\mathfrak{M}}(d_\nu))_{\nu=1}^t, \text{condition flags, classification, } H_{\text{independent verification}}). \end{aligned} \quad (29.125)$$

All products, inverses, generated subgroups, automorphism cycles, parity bits, and word evaluations are recomputed from canonical tables. Fixed group, tuple, automorphism, word, and failure orders make repeated execution byte-identical. The possible terminal classifications are

$$\begin{aligned} \text{SATISFIED, NO_MATCH_ESTABLISHED, PARTIAL_NO_MATCH,} \\ \text{INVALID_INPUT, UNDECIDED} \end{aligned} \quad (29.126)$$

UNDECIDED is reserved for an unavailable exact dangerous-set comparison or an incomplete external threshold certificate; it is never promoted to separation.

28 Semilinear automorphisms

28.1 Inner automorphisms

Let G be a finite group with identity e_G , center $Z(G)$, and a verified multiplication table. For $g \in G$, define

$$\iota_g : G \longrightarrow G, \quad \iota_g(x) = gxg^{-1}. \quad (30.1)$$

Theorem 28.1.1 (Inner-order criterion). *The map*

$$\iota : G \longrightarrow \text{Aut}(G), \quad g \longmapsto \iota_g \quad (30.2)$$

is a homomorphism with kernel $Z(G)$. Consequently

$$\text{Inn}(G) \cong G/Z(G), \quad \text{ord}(\iota_g) = \text{ord}(gZ(G)). \quad (30.3)$$

In particular, ι_g has exact order eight if and only if

$$g^8 \in Z(G), \quad g^k \notin Z(G) \quad (1 \leq k < 8). \quad (30.4)$$

Proof. For $g, h, x \in G$, $\iota_{gh}(x) = g(hxh^{-1})g^{-1}$, so $\iota_{gh} = \iota_g \iota_h$ under functional composition. Moreover ι_g is the identity exactly when g commutes with every x , namely when $g \in Z(G)$. The first isomorphism theorem proves the first assertion in (30.3). Since $\iota_g^k = \iota_{g^k}$, its k -th power is the identity exactly when $g^k \in Z(G)$. The least such positive k is both the coset order and the automorphism order, proving (30.3)–(30.4). $\square$

Executable record and tests. The certificate stores g , the vector $(\iota_g(x))_{x \in G}$, its cycle decomposition, the list $(g^k)_{k=1}^8$, and center-membership bits. All are recomputed from the group table. For a positive order-eight test, take the dihedral group of 32 elements

$$G = \langle r, s \mid r^{16} = s^2 = e_G, \text{ } srs = r^{-1} \rangle. \quad (30.5)$$

Its center is $\{e_G, r^8\}$, so $rZ(G)$ has order eight and ι_r passes (30.4). If $g \in Z(G)$, the same constructor returns the identity permutation; this is the boundary test. Requiring $g^8 = e_G$ would wrongly reject the valid choice $g = r$, because $r^8 \neq e_G$ while $r^8 \in Z(G)$.

28.2 Diagonal automorphisms

Let $\hat{Q} \leq \text{GL}(V)$ be a faithful matrix realization of a finite group, and put

$$N = N_{\text{GL}(V)}(\hat{Q}), \quad C = C_{\text{GL}(V)}(\hat{Q}). \quad (30.6)$$

For $d \in N$, call $\delta_d(x) = dxd^{-1}$ a diagonal or similitude-induced automorphism when d belongs to the declared classical similitude overgroup.

Theorem 28.2.1 (Two diagonal coset orders). *Conjugation gives a homomorphism $N \rightarrow \text{Aut}(\hat{Q})$ with kernel C . Therefore*

$$\begin{aligned} \text{ord}(\delta_d) &= \text{ord}(dC \text{ in } N/C), \\ \text{ord}(\delta_d \text{ Inn}(\hat{Q})) &= \text{ord}(d(\hat{Q}C) \text{ in } N/(\hat{Q}C)). \end{aligned} \quad (30.7)$$

Proof. Normalization makes δ_d an automorphism, and $\delta_{d_1 d_2} = \delta_{d_1} \delta_{d_2}$. It is the identity precisely when d centralizes $\hat{Q}$, proving the first line by the first isomorphism theorem. Its image is inner precisely when some $q \in \hat{Q}$ has $\delta_d = \delta_q$, equivalently $q^{-1}d \in C$, or $d \in \hat{Q}C$. Passing to the outer quotient proves the second line. $\square$

The exact algorithm verifies $dxd^{-1} \in \hat{Q}$ on a certified generating set, expands to the full permutation vector, and computes both coset orders in (30.7). It may replace C by field scalars only after independently proving the centralizer identity. As a positive test, take $\hat{Q} = \text{SL}(2, 3)$ and

$$d = \text{diag}(-1, 1) \in \text{GL}(2, 3). \quad (30.8)$$

Normality of SL gives $d \in N$, while $\delta_d(E_{12}(1)) = E_{12}(-1) \neq E_{12}(1)$ and $d^2 = I$; hence δ_d has order two. Commuting with both $E_{12}(1)$ and $E_{21}(1)$ forces a 2-by-2 matrix to be scalar, verifying the centralizer used in this test. By contrast a nonidentity scalar $d = \lambda I$ has nontrivial matrix order but induces the identity automorphism, demonstrating why the matrix order is not the answer.

28.3 Field automorphisms

Let $K = \mathbb{F}_{p^m}$ be a concrete field record and define

$$F_r : K \longrightarrow K, \quad F_r(a) = a^{p^r}. \quad (30.9)$$

On matrices and vectors F_r is always applied entrywise.

Theorem 28.3.1 (Coefficient and induced orders). *The coefficient automorphism F_r has exact order*

$$h = \frac{m}{\gcd(m, r)}. \quad (30.10)$$

If the defining equations and scalar equivalence of a concrete matrix group Q are stable under F_r , entrywise application induces an automorphism $\overline{F}_r \in \text{Aut}(Q)$ whose order divides h . It has exact order eight if and only if its full permutation vector has cycle least common multiple eight.

Proof. The Galois group of $K/\mathbb{F}_p$ is cyclic of order m , generated by F_1 . The order of its r -th power is $m/\gcd(m, r)$, proving (30.10). Frobenius preserves field operations, determinants, products, inverses, and scalar proportionality. Stability of the defining equations therefore makes it a well-defined group automorphism, also on projective classes. Since F_r^h fixes every coefficient, the induced order divides h . The exact order of a finite permutation is the cycle least common multiple, giving the final criterion. $\square$

For a positive exact-order-eight test, let $K = \mathbb{F}_{2^8}$, choose a primitive element θ , and take $Q = \mathrm{GL}(1, 2^8) = K^\times$. The automorphism F_1 has order eight on Q , since $\theta^{2^k} \neq \theta$ for $1 \leq k < 8$ and $\theta^{2^8} = \theta$. On $\mathrm{PGL}(1, 2^8) = 1$, the same coefficient Frobenius induces the identity. This boundary test proves that (30.10) alone cannot certify the induced order after projectivization.

28.4 Graph automorphisms

For a general-linear carrier define

$$\gamma(A) := (A^{-1})^\top. \quad (30.11)$$

Theorem 28.4.1 (Inverse-transpose test). *The map γ is an involutory automorphism of $\mathrm{GL}(n, q)$ and preserves $\mathrm{SL}(n, q)$. It descends well definedly to $\mathrm{PGL}(n, q)$ and $\mathrm{PSL}(n, q)$. For the special-linear group in dimension two it is inner.*

Proof. For invertible A, B ,

$$\gamma(AB) = ((AB)^{-1})^\top = (A^{-1})^\top (B^{-1})^\top = \gamma(A)\gamma(B), \quad \gamma^2(A) = A. \quad (30.12)$$

Also $\det \gamma(A) = \det(A)^{-1}$, so determinant one is preserved. Replacing A by λA replaces $\gamma(A)$ by $\lambda^{-1}\gamma(A)$, proving projective well-definedness. In dimension two, with $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, direct multiplication gives $(A^{-1})^\top = JAJ^{-1}$ for every $A \in \mathrm{SL}(2, q)$, so the restriction is inner. $\square$

On $\mathrm{Sp}(2n, q)$, the defining equation gives $\gamma(A) = J_n A J_n^{-1}$, so this candidate again overlaps an inner automorphism. On the standard unitary carrier, $A^{\sigma^\top} A = I$ gives $\gamma(A) = A^\sigma$, so inverse-transpose overlaps the quadratic field involution. Thus graph, inner, and field sources are deduplicated by full permutation vectors. A proposed diagram map for another realization is accepted only after its generator images preserve every defining relation and its cycle order is computed. The rank-two identity above is the boundary test; for $\mathrm{SL}(3, 2)$, $\gamma(E_{12}(1)) = E_{21}(1) \neq E_{12}(1)$, providing a nontrivial involution test.

28.5 The semilinear group

Let K/k be a finite Galois extension and $V = K^d$ with $d \geq 1$. A σ -semilinear map satisfies

$$T(av + bw) = \sigma(a)T(v) + \sigma(b)T(w). \quad (30.13)$$

Every invertible such map has a unique pair representation

$$T_{A,\sigma}(v) = A\sigma(v), \quad A \in \mathrm{GL}_d(K), \quad \sigma \in \mathrm{Gal}(K/k), \quad (30.14)$$

where σ acts coordinatewise.

Theorem 28.5.1 (Semidirect-product laws). *The pairs in (30.14) form the semilinear group $\Gamma L_d(K/k)$, with*

$$\begin{aligned} (A, \sigma)(B, \tau) &= (A\sigma(B), \sigma\tau), \\ (A, \sigma)^{-1} &= (\sigma^{-1}(A^{-1}), \sigma^{-1}), \\ (A, \sigma)^k &= (A\sigma(A) \cdots \sigma^{k-1}(A), \sigma^k). \end{aligned} \quad (30.15)$$

Its action on V is faithful.

Proof. Composition on v gives

$$A\sigma(B\tau(v)) = A\sigma(B)(\sigma\tau)(v),$$

which proves multiplication. Applying it twice to the proposed inverse yields (I, id) on both sides. Induction proves the power formula. Associativity follows either from function composition or directly from

$$A\sigma(B)(\sigma\tau)(C) = A\sigma(B\tau(C)).$$

If $A\sigma(v) = v$ for all v , evaluation on standard basis vectors gives $A = I$; evaluation on ae_i then gives $\sigma(a) = a$ for every a , hence $\sigma = \text{id}$. This proves faithfulness. $\square$

The canonical record stores A , its inverse, the field automorphism exponent, and the action vector on all of V . For $d = 1$, $A = 1$, and $K = \mathbb{F}_{2^8}$, the pair $(1, F_1)$ has order eight. The identity pair is the boundary test. Matrix multiplication without the twist $\sigma(B)$ fails on any B having a coefficient moved by σ , providing a negative regression.

28.6 Constructing an automorphism of exact order eight

Let Q be a finite group, let $\beta \in \text{Aut}(Q)$, and let $g \in Q$. Put

$$\alpha := \iota_g \circ \beta, \quad c_k := g\beta(g) \cdots \beta^{k-1}(g). \quad (30.16)$$

Lemma 28.6.1 (Twisted-power formula). *For every $k \geq 1$,*

$$\alpha^k = \iota_{c_k} \circ \beta^k. \quad (30.17)$$

Proof. The case $k = 1$ is the definition. Since $\beta^k \iota_g = \iota_{\beta^k(g)} \beta^k$, multiplying the k -th expression by $\iota_g \beta$ gives

$$\iota_{c_k} \beta^k \iota_g \beta = \iota_{c_k \beta^k(g)} \beta^{k+1} = \iota_{c_{k+1}} \beta^{k+1}.$$

Induction proves the formula. $\square$

Theorem 28.6.2 (Exact order-eight criterion). *The automorphism α has exact order eight if and only if*

$$\begin{aligned} \iota_{c_8} \beta^8 &= \text{id}_Q, \\ \iota_{c_k} \beta^k &\neq \text{id}_Q \quad (1 \leq k < 8). \end{aligned} \quad (30.18)$$

If $\beta^8 = \text{id}_Q$, the first line is equivalent to $c_8 \in Z(Q)$.

Proof. Substitute $k = 1, \dots, 8$ in (30.17). Exact order eight means precisely that the eighth power is the identity and none of the first seven powers is. Under the additional hypothesis $\beta^8 = 1$, the eighth power is ι_{c_8} , which is the identity exactly when $c_8 \in Z(Q)$ by Statement 28.1. $\square$

The general first line cannot be split into $\beta^8 = 1$ and $c_8 \in Z(Q)$: a nontrivial inner β^8 can cancel ι_{c_8} . The constructor therefore forms the complete permutation vector of $\iota_g \beta$, computes its cycle least common multiple, and uses (30.17) as an independent independent verification. It scans the sorted candidate β -vectors and $g \in Q$, returns the first cycle-lcm eight record, or certifies no match only when the candidate automorphism set is complete.

Theorem 28.6.3 (Algorithm specification). *The exact-order-eight constructor terminates, every returned automorphism has order eight, and an exhaustive negative result is complete relative to its declared finite candidate set.*

Proof. Both scan domains and every permutation are finite. The cycle least common multiple proves soundness, while lexicographic exhaustion proves the relative negative statement. Fixed carrier, automorphism, element, and cycle orders make the result reproducible. $\square$

For a positive test, take $Q = \text{GL}(1, 2^8)$, $g = e_Q$, and $\beta = F_1$ from Statement 28.3; then $\alpha = \beta$ has exact order eight. Replacing Q by $\text{PGL}(1, 2^8) = 1$ yields order one and is the projective-collapse boundary test.

28.7 Automorphism-orbit generators

Assume $\alpha^8 = \text{id}_Q$. For a seed $s \in Q$, define

$$s_j := \alpha^j(s) \quad (j \in \mathbb{Z}/8\mathbb{Z}). \quad (30.19)$$

Theorem 28.7.1 (Orbit and inverse-direction criterion). *The ordered tuple satisfies*

$$\alpha(s_j) = s_{j+1}, \quad s_{j+4} = s_j^{-1} \quad (0 \leq j < 8) \quad (30.20)$$

if and only if the single seed equation

$$\alpha^4(s) = s^{-1} \quad (30.21)$$

holds. It has eight distinct nonidentity entries exactly when $s \neq e_Q$ and the $\langle \alpha \rangle$ -orbit of s has length eight.

Proof. The first identity in (30.20) is immediate from (30.19). If (30.21) holds, applying α^j gives

$$s_{j+4} = \alpha^j(\alpha^4(s)) = \alpha^j(s^{-1}) = \alpha^j(s)^{-1} = s_j^{-1}.$$

Conversely the $j = 0$ instance is (30.21). Orbit–stabilizer for the cyclic action says that the number of distinct entries is the orbit length. An automorphism fixes the identity, so an orbit containing it is the singleton identity orbit; this proves the last statement. $\square$

The subgroup

$$Q_s := \langle s_0, \dots, s_7 \rangle \quad (30.22)$$

is α -invariant, because α permutes its generators. Thus the orbit tuple generates the declared target precisely when $Q_s = Q$. The generator-image table stores

x	s_0	s_1	s_2	s_3	s_4	s_5	s_6	s_7
$\alpha(x)$	s_1	s_2	s_3	s_4	s_5	s_6	s_7	s_0
x^{-1}	s_4	s_5	s_6	s_7	s_0	s_1	s_2	s_3

(30.23)

and verifies it against the full group table.

For a positive test, let $Q = C_{17} = \langle s \rangle$ and $\alpha(s) = s^2$. The residue 2 has order eight modulo 17, and $2^4 = 16 \equiv -1 \pmod{17}$; hence the eight elements s^{2^j} are distinct, generate Q , and satisfy (30.21). The seed $s = e_Q$ satisfies (30.21) but its orbit has size one, which is the required negative boundary. Canonical powers, subgroup closure, table rows, and hashes make serialization unambiguous.

28.8 C_8 -equivariant finite images

Let Q be a verified finite group, let $\alpha \in \text{Aut}(Q)$, and choose $(A_1, B_1, A_2, B_2) \in Q^4$. The finite-image input table is

x	a_1	b_1	a_2	b_2
$\pi(x)$	A_1	B_1	A_2	B_2
$\pi(\phi_8(x))$	B_1^{-1}	$A_2^{-1}B_1A_1$	$A_2^{-1}B_1A_1B_1^{-1}A_1^{-1}B_1^{-1}B_2^{-1}$	$A_1B_1A_1^{-1}B_1^{-1}A_2$

(30.24)

Theorem 28.8.1 (Finite-image descent). *There is a C_8 -equivariant surjection $\pi : \Gamma_B \twoheadrightarrow Q$ with the displayed marking and quotient automorphism α if and only if*

$$\begin{aligned} [A_1, B_1][A_2, B_2] &= e_Q, & \langle A_1, B_1, A_2, B_2 \rangle &= Q, \\ \alpha(\pi(x)) &= \pi(\phi_8(x)) & (x = a_1, b_1, a_2, b_2). \end{aligned} \quad (30.25)$$

When these conditions hold,

$$\alpha\pi = \pi\phi_8, \quad \phi_8(\ker \pi) = \ker \pi, \quad \alpha^8 = \text{id}_Q. \quad (30.26)$$

The descended action has exact order eight only if its cycle least common multiple is eight.

Proof. The presentation universal property makes the relator condition equivalent to existence and uniqueness of the marked homomorphism; the generation condition is equivalent to surjectivity. Both $\alpha\pi$ and $\pi\phi_8$ are homomorphisms, so the last four equalities in (30.25) imply their equality everywhere. A kernel element is then sent by ϕ_8 to a kernel element. Applying the inclusion eight times and using $\phi_8^8 = 1$ gives equality. The same covariance and surjectivity give $\alpha^8 = 1$. None of these arguments rules out a smaller positive power, so the cycle check is necessary for exact order.

Conversely, any equivariant marked surjection necessarily evaluates the surface relator to the identity, has generating images, and satisfies the four generator covariance rows. This proves both directions. $\square$

Construction and serialization. The implementation scans marked tuples, rejects the first failed relator or generation row, expands each accepted automorphism candidate through (30.24), and compares the four covariance rows. It serializes the complete quotient table, marking, relator value, generated subgroup, automorphism vector and order, covariance table, kernel-invariance flag, and independent verification hash:

$$\mathcal{C}_{30.08} = (H_Q, A_1, B_1, A_2, B_2, H_\alpha, \text{ord } \alpha, \text{rows (30.24)–(30.26), classification, } H_{\text{independent verification}}). \quad (30.27)$$

All scans are finite and lexicographically fixed. The parity quotient $Q = C_2$, with all four images nonzero and $\alpha = \text{id}$, is a useful boundary: it satisfies the relator, generation, and covariance equations, but the descended action has order one. Thus equivariance of a finite image and exact C_8 -faithfulness remain distinct certified properties.

29 Generator-orbit construction

29.1 Seed elements

For a canonically ordered finite group Q , let $\mathcal{A}^{\text{avail}}(Q) \leq \text{Aut}(Q)$ be the automorphism set actually supplied by Statement 27.7. Define the available seed domain

$$\mathcal{S}^{\text{avail}}(Q) := \{(\alpha, s) : \alpha \in \mathcal{A}^{\text{avail}}(Q), \alpha^8 = \text{id}_Q, s \in Q\}. \quad (31.1)$$

A seed records a prospective rotation and one prospective directed neighbor. It becomes a surface-group quotient after the inverse-direction, relator, generation, parity, and separation conditions are imposed.

Theorem 29.1.1 (Complete, duplicate-free seed scan). *Order automorphisms by full permutation vector and group elements by their carrier indices. Lexicographic scanning of their Cartesian product, retaining precisely the pairs in (31.1), emits each available seed exactly once and misses none. If $\mathcal{A}^{\text{avail}}(Q) = \text{Aut}(Q)$, the scan is complete for Q .*

Proof. Every pair has a unique ordered index (i, j) ; testing $\alpha^8 = 1$ is an exact equality of finite permutation vectors. Thus a retained pair is a seed, no index is emitted twice, and every qualifying index is reached. Equality of the available and full automorphism sets upgrades relative coverage to absolute coverage. $\square$

After every block, the record stores the next index pair, counts of tested and retained pairs, the rolling hash, and the last emitted record. Resume first verifies the prefix hash and then starts at the stored successor, so no pair is repeated or skipped. For $Q = 1$ there is one boundary seed (id, e_Q) . For $Q = C_{17}$ the map $\alpha(s) = s^2$ and a generator s give the positive seed used below.

29.2 Exact C_8 -orbits

For $(\alpha, s) \in \mathcal{S}^{\text{avail}}(Q)$, put

$$\mathcal{O}_8(\alpha, s) = (s_0, \dots, s_7), \quad s_j = \alpha^j(s). \quad (31.2)$$

Theorem 29.2.1 (Orbit-length certificate). *The stabilizer*

$$H_s := \{k \in \mathbb{Z}/8\mathbb{Z} : \alpha^k(s) = s\} \quad (31.3)$$

is a subgroup and

$$|\{s_0, \dots, s_7\}| = \frac{8}{|H_s|}. \quad (31.4)$$

In particular the tuple has eight distinct entries exactly when

$$\alpha^k(s) \neq s \quad (1 \leq k < 8). \quad (31.5)$$

Proof. If $k, \ell \in H_s$, then $\alpha^{k-\ell}(s) = s$, so H_s is a subgroup of the cyclic group C_8 . The standard map $C_8/H_s \rightarrow \{s_j\}$, $kH_s \mapsto \alpha^k(s)$, is well defined and bijective, proving (31.4). Trivial stabilizer is equivalent to all seven inequalities in (31.5). Testing only $k = 1$ would not exclude orbit lengths two or four. $\square$

The exact action table is

$$\alpha : (s_0, s_1, \dots, s_7) \longmapsto (s_1, s_2, \dots, s_0). \quad (31.6)$$

The serialized record contains all eight carrier indices, the seven equality bits, H_s , the orbit length, and a independent verification hash. For C_{17} with $\alpha(s) = s^2$, the order of 2 modulo 17 is eight, so (31.5) passes. The identity seed has orbit length one and is the boundary.

29.3 Eight directional generators

Definition 29.3.1 (Directional shell condition). The orbit tuple is an eight-directional generating shell when

$$s_j \neq e_Q, \quad s_i \neq s_j \ (i \neq j), \quad \langle s_0, \dots, s_7 \rangle = Q. \quad (31.7)$$

Theorem 29.3.2 (Local and global meanings). *The first two conditions in (31.7) are necessary and sufficient for the labeled map from the eight Bolza directions at the base vertex to the eight outgoing Cayley darts at e_Q to be injective with no loop. The last condition is necessary and sufficient for the underlying Cayley graph to be connected.*

Proof. The dart labeled j ends at s_j . It is a loop exactly when $s_j = e_Q$, and two labels have the same endpoint exactly when $s_i = s_j$. This proves the local statement. Vertices reachable from e_Q by shell-labeled paths form the subgroup generated by the shell (in a finite group inverses are positive powers), so all vertices are reachable exactly when that subgroup is Q . $\square$

The C_{17} example passes all three tests. The identity seed fails both local tests and generation of a nontrivial target. The output fixes the eight endpoint indices, 28 collision bits, the generated-subgroup list, and the least missing element on failure.

29.4 Inverse-direction constraint

Theorem 29.4.1 (One seed equation is enough). *For the orbit (31.2), the four inverse-pair equations*

$$s_{j+4} = s_j^{-1} \quad (0 \leq j < 4) \quad (31.8)$$

hold if and only if

$$\alpha^4(s) = s^{-1}. \quad (31.9)$$

Proof. Equation (31.8) at $j = 0$ is (31.9). Conversely, applying the automorphism α^j to (31.9) gives

$$s_{j+4} = \alpha^j(s^{-1}) = \alpha^j(s)^{-1} = s_j^{-1}.$$

□

Under (31.8), the full shell subgroup equals $\langle s_0, s_1, s_2, s_3 \rangle$. The condition does not imply distinctness: if $\alpha = \text{id}$ and $s = s^{-1}$, then (31.9) holds but the orbit is constant. In the C_{17} test, $2^4 \equiv -1 \pmod{17}$, so (31.9) holds and the four inverse pairs are distinct.

29.5 The surface-relator constraint

From the corrected shell-to-presentation dictionary define

$$\begin{aligned} A_1 &:= s_0, & B_1 &:= s_1^{-1}, \\ A_2 &:= s_1^{-1}s_0s_2, & B_2 &:= s_3^{-1}s_0^{-1}s_1. \end{aligned} \quad (31.10)$$

Theorem 29.5.1 (Executable seed relator). *The assignment $(a_1, b_1, a_2, b_2) \mapsto (A_1, B_1, A_2, B_2)$ extends uniquely to a homomorphism $\pi_s : \Gamma_B \rightarrow Q$ if and only if*

$$[s_0, s_1^{-1}][s_1^{-1}s_0s_2, s_3^{-1}s_0^{-1}s_1] = e_Q. \quad (31.11)$$

Proof. Substitution of (31.10) into the Bolza defining relator $[a_1, b_1][a_2, b_2]$ gives exactly the left side of (31.11). The universal property of $\langle a_1, b_1, a_2, b_2 \mid [a_1, b_1][a_2, b_2] = 1 \rangle$ makes identity evaluation necessary and sufficient for the free assignment to factor through Γ_B , and makes the factor unique. □

Shell reconstruction. The C04/C08 Nielsen dictionary from which (31.10) was derived is an inverse pair of substitutions. Therefore the resulting homomorphism satisfies $\pi_s(g_j) = s_j$ for $0 \leq j < 4$. If (31.8) also holds, then for $0 \leq j < 4$,

$$\pi_s(g_{j+4}) = \pi_s(g_j^{-1}) = s_j^{-1} = s_{j+4},$$

so $\pi_s(g_j) = s_j$ for all eight shell indices.

The checker evaluates inverses and products in the displayed order, stores both commutators and their product, and emits the nonidentity value on failure. In an abelian target the equation is automatic, which is a boundary showing that relation satisfaction alone proves neither generation nor parity. Applying α^k to a passing identity proves that every iterated seed tuple also passes.

29.6 Surjective generation

Theorem 29.6.1 (Equivalent generation tests). *Assume (31.11), so that π_s exists. Then*

$$\begin{aligned} \pi_s \text{ is onto} &\iff \langle A_1, B_1, A_2, B_2 \rangle = Q \\ &\iff \langle s_0, s_1, s_2, s_3 \rangle = Q \\ &\iff \langle s_0, \dots, s_7 \rangle = Q \quad \text{under (31.8)}. \end{aligned} \quad (31.12)$$

Proof. The image of a homomorphism is generated by the images of the domain generators, proving the first equivalence. Formula (31.10) puts every A_i, B_i in the subgroup generated by $s_0, \dots, s_3$. Conversely,

$$s_0 = A_1, \quad s_1 = B_1^{-1}, \quad s_2 = s_0^{-1} s_1 A_2, \quad s_3 = s_0^{-1} s_1 B_2^{-1}, \quad (31.13)$$

so the two subgroups coincide. Under inverse closure, the last four shell entries are inverses of the first four, proving the final equivalence. $\square$

Breadth-first subgroup closure terminates and returns either the full ordered carrier or the least missing element. A named ambient group is never recorded as the quotient when this exact equality fails.

29.7 Parity constraint

Theorem 29.7.1 (Parity factor criterion). *Assume the inverse-direction equations (31.8). For a surjective seed map $\pi_s : \Gamma_B \twoheadrightarrow Q$, the following are equivalent:*

- (i) $\ker \pi_s \subseteq \ker \chi$,
 - (ii) there is a homomorphism $\chi_Q : Q \rightarrow C_2$ with $\chi_Q \pi_s = \chi$,
 - (iii) there is a homomorphism χ_Q with $\chi_Q(s_j) = 1$ for all j .
- (31.14)

The factor is unique; when the orbit generates and $\alpha(s_j) = s_{j+1}$, it satisfies $\chi_Q \alpha = \chi_Q$.

Proof. Factorization implies kernel inclusion. Conversely, define $\chi_Q(\pi_s(\gamma)) = \chi(\gamma)$; the inclusion makes this well defined, and it is plainly a homomorphism. Surjectivity gives uniqueness. The shell reconstruction above and $\chi(g_j) = 1$ show that (ii) implies (iii).

For the converse, (31.10) and additivity in C_2 give

$$\chi_Q(A_1) = \chi_Q(B_1) = \chi_Q(A_2) = \chi_Q(B_2) = 1, \quad (31.15)$$

since the last two words contain three odd shell factors. Thus $\chi_Q \pi_s$ agrees with χ on the four presentation generators, proving (ii). Finally the two homomorphisms $\chi_Q \alpha$ and χ_Q agree on every generating s_j , so they agree on Q . $\square$

The finite parity-table algorithm of Statement 9.5 supplies either the unique bit vector or two equal quotient words with different parities. The C_{17} orbit example fails this condition because an odd-order group has no surjection onto C_2 ; the C_2 parity quotient is the minimal positive factor but cannot carry an eight-element orbit.

29.8 Local injectivity constraint

Let $F \subset \Gamma_B$ be a finite deduplicated domain containing the identity, and define the deduplicated difference set

$$\Delta(F) := \{x^{-1}y : x, y \in F\}, \quad \Delta^\times(F) := \Delta(F) \setminus \{1\}. \quad (31.16)$$

Theorem 29.8.1 (Complete finite injectivity test). *For any homomorphism $\pi : \Gamma_B \rightarrow Q$ with kernel K ,*

$$\pi|_F \text{ is injective} \iff K \cap \Delta(F) = \{1\} \iff \pi(d) \neq e_Q \quad (d \in \Delta^\times(F)). \quad (31.17)$$

Proof. For $x, y \in F$, $\pi(x) = \pi(y)$ exactly when $\pi(x^{-1}y) = e_Q$, equivalently $x^{-1}y \in K$. Thus a collision exists exactly when a nonidentity difference belongs to the kernel, proving both equivalences. $\square$

The algorithm scans all $|F|^2$ ordered pairs, canonicalizes and deduplicates their differences, removes exactly the identity key, and evaluates every remaining word. The first failed entry stores its witness pair (x, y) . For $F = \{1\}$ the test is vacuous and passes; for $F = \{1, g\}$ it reduces to $\pi(g) \neq e_Q$. Checking merely the nonidentity elements of F can miss a collision between two distinct nonidentity members, so it is not a substitute for (31.17).

29.9 Seed equivalence and canonical representatives

For each subgroup $H \leq \text{Aut}(Q)$, define

$$\mathcal{S}_H(Q) := \{(\alpha, s) : \alpha \in H, \alpha^8 = \text{id}_Q, s \in Q\}.$$

The group H acts on its raw seed space by

$$u \cdot (\alpha, s) := (u\alpha u^{-1}, u(s)). \quad (31.18)$$

This is closed because $u, \alpha \in H$ implies $u\alpha u^{-1} \in H$. Define $(\alpha, s) \sim_H (\alpha', s')$ when they lie in the same H -orbit.

Theorem 29.9.1 (Seed equivalence and transported conditions). *Relation $\sim_H$ is an equivalence relation. If $(\alpha', s') = u \cdot (\alpha, s)$, then*

$$u(\alpha^j(s)) = (\alpha')^j(s') \quad (0 \leq j < 8), \quad (31.19)$$

and u preserves orbit length, inverse direction, relator satisfaction, generation, and nonidentity evaluations. A parity factor transports as $\chi'_Q = \chi_Q u^{-1}$.

Proof. The identity, inverse, and product automorphisms prove reflexivity, symmetry, and transitivity. Induction on j gives (31.19). An isomorphism preserves identity, inverses, products, equality, commutators, and generated subgroups, proving all listed conditions. Moreover $\chi'_Q(u(x)) = \chi_Q(x)$, so every parity row transports. $\square$

For full deduplication, take $H = \text{Aut}(Q)$; for available-subgroup deduplication, take $H = \mathcal{A}^{\text{avail}}(Q)$. In either case compute every H -orbit, choose its lexicographically least permutation-vector/element pair, and store orbit and stabilizer sizes with

$$|H \cdot (\alpha, s)| = \frac{|H|}{|\text{Stab}_H(\alpha, s)|}. \quad (31.20)$$

Sorted orbit keys give a coverage record whose counts sum to $|\mathcal{S}_H(Q)|$. The available instance is complete for $\sim_H$, but it must not claim full $\text{Aut}(Q)$ -deduplication unless the two groups are equal. Checkpoint resumption uses the next raw-seed index and the hash of all completed orbits.

29.10 Kernel equivalence

Let $\pi_i : \Gamma_B \twoheadrightarrow Q_i$ be surjections.

Theorem 29.10.1 (Kernel–target isomorphism equivalence). *The following are equivalent:*

- (i) $\ker \pi_1 = \ker \pi_2$,
 - (ii) there is a unique isomorphism $\theta : Q_1 \rightarrow Q_2$ satisfying $\theta\pi_1 = \pi_2$.
- (31.21)

If the quotients carry descended rotations $\alpha_i\pi_i = \pi_i\phi_8$, then the same θ satisfies

$$\theta\alpha_1 = \alpha_2\theta. \quad (31.22)$$

Proof. An intertwining isomorphism preserves exactly the identity preimages, so (ii) implies (i). Under (i), define

$$\theta(\pi_1(\gamma)) := \pi_2(\gamma). \quad (31.23)$$

If $\pi_1(\gamma) = \pi_1(\delta)$, then $\delta^{-1}\gamma$ lies in the common kernel, so the two π_2 -values agree; hence the map is well defined. It is a homomorphism and surjective. Its kernel is trivial because $\pi_2(\gamma) = e$ implies $\gamma \in \ker \pi_1$, so it is an isomorphism. Surjectivity of π_1 gives uniqueness. Finally, for every γ ,

$$\theta\alpha_1\pi_1(\gamma) = \theta\pi_1\phi_8(\gamma) = \pi_2\phi_8(\gamma) = \alpha_2\pi_2(\gamma) = \alpha_2\theta\pi_1(\gamma),$$

and canceling the surjection proves (31.22). $\square$

An exact finite certificate enumerates target isomorphisms by generator images and accepts precisely a θ matching all four marked rows; the theorem then certifies common kernel without enumerating the infinite subgroup. Composing a quotient with a target automorphism is the positive test. The identity quotient and the parity quotient have different kernels and different orders, providing a negative boundary. Raw seed equivalence is defined before a kernel exists; for fully accepted reconstructed seeds, (31.21)–(31.22) provide the corresponding marked-kernel comparison.

Part XII

Permutation and low-index constructions

30 Transitive permutation representations

30.1 Actions, stabilizers, and orbit certificates

Let G be a group acting on the left on a set X . Thus the action is a map satisfying

$$e_G x = x, \quad (gh)x = g(hx) \quad (g, h \in G, x \in X). \quad (32.1)$$

Equivalently, it is a homomorphism $\vartheta_X : G \rightarrow \text{Sym}(X)$. For $x \in X$, set

$$G_x := \{g \in G : gx = x\}, \quad Gx := \{gx : g \in G\}. \quad (32.2)$$

Theorem 30.1.1 (Stabilizer–orbit theorem). *The set G_x is a subgroup of G , and the map*

$$\beta_x : G/G_x \longrightarrow Gx, \quad gG_x \longmapsto gx \quad (32.3)$$

is a well-defined G -equivariant bijection. Consequently

$$|Gx| = [G : G_x] \quad (32.4)$$

as cardinal numbers, and in particular as ordinary integers when the orbit is finite.

Proof. The identity fixes x . If $g, h \in G_x$, then $(gh^{-1})x = g(h^{-1}x) = gx = x$, since $hx = x$ also gives $h^{-1}x = x$. Hence $G_x \leq G$.

For $g, h \in G$,

$$gG_x = hG_x \iff h^{-1}g \in G_x \iff h^{-1}gx = x \iff gx = hx. \quad (32.5)$$

This single equivalence proves both well-definedness and injectivity of β_x ; surjectivity follows from the definition of Gx . Finally, $\beta_x(kgG_x) = kgx = k\beta_x(gG_x)$, so the bijection is equivariant. Taking cardinalities proves (32.4). $\square$

For a finite input action, the exact certificate for a selected point stores the sorted stabilizer carrier, the sorted orbit carrier, and the two inverse tables

$$gG_x \longmapsto gx, \quad y \longmapsto \min\{g : gx = y\} G_x. \quad (32.6)$$

Closure of the stabilizer, constancy on left cosets, bijectivity, and equivariance are checked directly from the multiplication and action tables. This is stronger than checking only the numerical identity (32.4).

Small tests. For the natural action $S_3 \curvearrowright \{1, 2, 3\}$, the stabilizer of 1 has order two and the orbit has size three; (32.3) has three rows. For the trivial action on a nonempty set, $G_x = G$ and the orbit has one point. The empty action has no point stabilizer and is therefore excluded whenever a marked stabilizer is requested.

30.2 Classification of transitive actions

Theorem 30.2.1 (Coset classification). *A nonempty G -set X is transitive of finite degree d if and only if there is a subgroup $H \leq G$ of index d and a G -equivariant bijection*

$$G/H \cong X. \quad (32.7)$$

Pointed transitive G -sets (X, x) , up to pointed G -isomorphism, correspond to actual subgroups $H = G_x$. Unpointed transitive G -sets, up to G -isomorphism, correspond to conjugacy classes of subgroups.

Proof. If X is transitive, then $Gx = X$ for any $x \in X$, and Statement 30.1 supplies (32.7) with $H = G_x$ and $[G : H] = |X| = d$. Conversely, left multiplication on G/H is transitive because $gH = g \cdot H$.

Suppose $f : (G/H, H) \rightarrow (G/K, K)$ is a pointed equivariant isomorphism. Stabilizers are preserved, so

$$H = G_H = G_{f(H)} = G_K = K. \quad (32.8)$$

Without marked points, let $f(H) = gK$. Equivariance gives

$$H = G_H = G_{gK} = gKg^{-1}. \quad (32.9)$$

Conversely, if $H = gKg^{-1}$, then $aH \mapsto agK$ is a well-defined equivariant isomorphism. Thus equality classifies pointed models and conjugacy classifies unpointed models. $\square$

The distinction in (32.8)–(32.9) is part of the data specification. A change of basepoint conjugates the stabilizer and does not produce a new unpointed action. A degree alone does not determine the action: distinct conjugacy classes of index- d subgroups can give nonisomorphic transitive G -sets.

The canonical transitive-action record is

$$(d, H_{\text{can}}, (\sigma_{a_1}, \sigma_{b_1}, \sigma_{a_2}, \sigma_{b_2}), \text{based}), \quad (32.10)$$

where H_{can} is represented by the lexicographically least member of its conjugacy class when the basepoint is unmarked. The four permutations are retained only when $G = \Gamma_B$ or when a four-generator marking has been supplied; otherwise the complete ordered generator list is used.

30.3 Faithful degree and orbit decompositions

Every finite nonempty G -set decomposes uniquely into its orbits. Choosing one point from each orbit gives

$$X \cong \bigsqcup_{i=1}^r G/H_i, \quad |X| = \sum_{i=1}^r [G : H_i], \quad (32.11)$$

where changing the chosen point only conjugates H_i .

Theorem 30.3.1 (Kernel intersection formula). *For the action in (32.11),*

$$\ker \vartheta_X = \bigcap_{x \in X} G_x = \bigcap_{i=1}^r \text{core}_G(H_i), \quad \text{core}_G(H) := \bigcap_{g \in G} gHg^{-1}. \quad (32.12)$$

Proof. An element is in $\ker \vartheta_X$ exactly when it fixes every point, proving the first equality. In the orbit G/H_i , the stabilizer of gH_i is $gH_i g^{-1}$. Intersecting over that orbit gives $\text{core}_G(H_i)$, and intersecting over all orbits proves the second equality. $\square$

Define the finite faithful permutation degree by

$$\mu_f(G) := \min \left\{ \sum_{i=1}^r [G : H_i] : r \geq 1, [G : H_i] < \infty, \bigcap_{i=1}^r \text{core}_G(H_i) = 1 \right\}, \quad (32.13)$$

with value $+\infty$ if the displayed set is empty. We use the nonempty-action convention, so $\mu_f(1) = 1$.

Theorem 30.3.2 (Exact meaning of μ_f). *Formula (32.13) is the minimum size of a finite nonempty faithful G -set. If G is finite, the minimum exists and $\mu_f(G) \leq |G|$.*

Proof. By (32.11) every finite action has the displayed degree, and by (32.12) it is faithful exactly when the displayed intersection is trivial. Conversely, every finite family in (32.13) constructs the faithful disjoint-union action in (32.11). When G is finite, the regular action $G \curvearrowright G/1$ is a feasible action of degree $|G|$, and the positive integers are well-ordered. $\square$

One must not replace (32.13) by a minimum over a single core-free subgroup: a minimal faithful action can be intransitive and strictly smaller than every faithful transitive action. As an exact test, write $E = C_2^3$ as the vector space $\mathbb{F}_2^3$, and let H_i be the kernel of the i th coordinate functional. The three H_i are normal of index two and have trivial intersection, so the three degree-two coset actions form a faithful action of degree six. Every subgroup of E is normal; hence a core-free subgroup is trivial and every faithful transitive action has degree eight. No faithful action has degree below six: below that bound the nontrivial orbit sizes can only be one orbit of size two or four, or two orbits of size two; the intersection of their kernels has codimension at most two. Three nontrivial orbits already contribute at least $2 + 2 + 2$ points. Thus $\mu_f(E) = 6 < 8$, and all three orbit blocks are essential.

30.4 Coset-action construction and its kernel

Fix a subgroup $H \leq G$ and use left cosets. Define

$$\rho_H : G \longrightarrow \text{Sym}(G/H), \quad \rho_H(g)(xH) := gxH. \quad (32.14)$$

Theorem 30.4.1 (Coset action). *Formula (32.14) is a well-defined transitive left action and*

$$\ker \rho_H = \text{core}_G(H). \quad (32.15)$$

Proof. If $xH = yH$, then $x^{-1}y \in H$, while $(gx)^{-1}(gy) = x^{-1}y \in H$, so $gxH = gyH$. Associativity gives $\rho_H(g_1g_2) = \rho_H(g_1)\rho_H(g_2)$, and every coset xH is $\rho_H(x)(H)$, proving transitivity.

An element g fixes xH precisely when

$$gxH = xH \iff x^{-1}gx \in H \iff g \in xHx^{-1}. \quad (32.16)$$

It fixes every coset precisely when it belongs to the intersection of the right side over all $x \in G$, proving (32.15). $\square$

Finite table algorithm. Suppose $G = \{q_0 = e, q_1, \dots, q_{n-1}\}$ is canonically ordered and its exact multiplication table and a subgroup bit-vector for H are available. Scan $q_0, q_1, \dots$, choosing a new representative for each unmarked left coset. Sort the elements within each coset and order cosets by their least carrier index. If the resulting representatives are $t_1 = e, t_2, \dots, t_d$, define for every stored generator z

$$\sigma_z(i) = j \iff zt_iH = t_jH. \quad (32.17)$$

Membership is decided by $t_j^{-1}zt_i \in H$, so every row is exact.

Theorem 30.4.2 (Replay certificate). *The coset partition, the permutations σ_z , and the intersection of the conjugates of H determine an independently verifiable certificate for (32.14)–(32.15). If the stored generators generate G , checking their permutation rows and the multiplication table suffices to reconstruct all of ρ_H .*

Proof. The partition proves that the selected $t_i H$ are all distinct and exhaustive. Equation (32.17) gives a total bijection for each generator because left multiplication permutes cosets. Products of these rows define the action of every generator word; equality of group elements is resolved by the exact table, so the value is word-independent. Finally (32.16), evaluated for all representatives, proves the stored kernel. $\square$

Using right cosets with the multiplication rule in (32.14) would reverse conventions and generally fail well-definedness. The certificate therefore stores the literal flag `side = left` and rejects a convention mismatch.

30.5 Low-index subgroups and the represented quotient

Let $[G : H] = d < \infty$. After labeling the left cosets by $\{1, \dots, d\}$, (32.14) becomes a homomorphism $\rho_H : G \rightarrow S_d$. It satisfies

$$d \mid |\operatorname{im} \rho_H|, \quad d \leq |\operatorname{im} \rho_H| \leq d!. \quad (32.18)$$

The divisibility and lower bound refer to the transitive image action; the upper bound follows from $\operatorname{im} \rho_H \leq S_d$.

Theorem 30.5.1 (When index equals quotient order). *The following conditions are equivalent:*

- (i) $H \trianglelefteq G$,
 - (ii) $\operatorname{core}_G(H) = H$,
 - (iii) $|\operatorname{im} \rho_H| = d$,
 - (iv) $\operatorname{im} \rho_H$ acts regularly on G/H .
- (32.19)

Under these conditions the represented quotient is G/H . Without them it is $G/\operatorname{core}_G(H)$, not a group called G/H .

Proof. Normality is equivalent to equality of all conjugates of H , which is equivalent to (ii). By (32.15) and the first isomorphism theorem,

$$|\operatorname{im} \rho_H| = [G : \operatorname{core}_G(H)] = [G : H][H : \operatorname{core}_G(H)]. \quad (32.20)$$

Thus (ii) and (iii) are equivalent. A transitive finite action has group order equal to its degree exactly when a point stabilizer is trivial, which is regularity. The stabilizer in the image of the point H has order $[H : \operatorname{core}_G(H)]$, proving the equivalence with (iv). $\square$

Sharp positive and negative tests. For $G = S_d$ and the stabilizer $H = S_{d-1}$ in the natural action, $d \geq 2$, the image is all of S_d , so the upper bound $d!$ is attained. In particular, for S_3 and $H = \langle (12) \rangle$, the index is three, the core is trivial, and the represented quotient has order six rather than three. By contrast, $A_3 \leq S_3$ has index two and produces the quotient C_2 of order two. These tests reject the invalid inference *index d implies quotient order d* .

A low-index enumeration is complete only relative to a stated index range and a stated conjugacy convention. Each output row must include the subgroup generators, verified index, core generators, permutation image, and a proof that duplicate subgroups have been removed only up to conjugacy.

30.6 Normal cores and finite images

Theorem 30.6.1 (Finite quotient extracted from a finite orbit). *For every finite-index subgroup $H \leq G$,*

$$\text{core}_G(H) \trianglelefteq G, \quad G/\text{core}_G(H) \cong \text{im } \rho_H \leq S_d, \quad d = [G : H], \quad (32.21)$$

and therefore

$$[G : \text{core}_G(H)] = |\text{im } \rho_H| \mid d!, \quad d \leq [G : \text{core}_G(H)] \leq d!. \quad (32.22)$$

Proof. The core is the kernel of ρ_H by Statement 30.4, hence is normal. The first isomorphism theorem gives (32.21). Lagrange's theorem inside the finite group S_d gives the divisibility in (32.22), and transitivity gives the lower bound as in (32.18). $\square$

Although the definition of the core intersects all G -conjugates, only finitely many distinct conjugates occur when H has finite index. More precisely,

$$\text{core}_G(H) = \bigcap_{i=1}^d t_i H t_i^{-1} \quad (32.23)$$

for any left transversal $t_1, \dots, t_d$. Indeed, writing $g = t_i h$ with $h \in H$ gives $g H g^{-1} = t_i H t_i^{-1}$. This identity supplies a finite core certificate even when G itself is infinite.

Kernel-first extraction protocol. For a low-index subgroup record, the extractor performs the following exact conditions:

- | | | |
|--|---|-----------|
| <ol style="list-style-type: none"> 1 2 3 4 5 6 | <ol style="list-style-type: none"> verify the subgroup and its index d, construct the complete left-coset action, compute $K = \ker \rho_H$ and independently the intersection (32.23), require equality of the two kernel descriptions, construct $P = \langle \sigma_z : z \in Z \rangle \leq S_d$, verify $P = [G : K]$ whenever both sides are finite tables. | (32.24) |
|--|---|-----------|

The emitted finite quotient is the marked image P , together with the epimorphism $z \mapsto \sigma_z$; no quotient by a non-normal subgroup is emitted. A failure at any condition is retained as a diagnostic record, not silently promoted to a quotient.

30.7 Automorphisms in permutation representations

Let

$$\rho : \Gamma_B \longrightarrow S_d, \quad K := \ker \rho, \quad P := \text{im } \rho, \quad (32.25)$$

and let $\phi_8 \in \text{Aut}(\Gamma_B)$ be the order-eight Bolza rotation fixed in the earlier construction.

Theorem 30.7.1 (Descent to the permutation image). *The formula*

$$\bar{\phi}_8(\rho(\gamma)) := \rho(\phi_8(\gamma)) \quad (32.26)$$

defines an endomorphism of P exactly when $\phi_8(K) \subseteq K$. Because $\phi_8^8 = 1$, this inclusion is equivalent to

$$\phi_8(K) = K. \quad (32.27)$$

Under (32.27), $\bar{\phi}_8$ is an automorphism and $\bar{\phi}_8^8 = 1$; its order may be a proper divisor of eight.

Proof. If $\rho(\gamma) = \rho(\delta)$, then $\delta^{-1}\gamma \in K$. Formula (32.26) is independent of the representative exactly when $\phi_8(\delta^{-1}\gamma) \in K$, which for all representatives is precisely $\phi_8(K) \subseteq K$. Under this inclusion,

$$K = \phi_8^8(K) \subseteq \phi_8^7(K) \subseteq \dots \subseteq \phi_8(K) \subseteq K,$$

so every inclusion is equality. Applying (32.26) eight times gives the identity, and the inverse is induced by ϕ_8^{-1} . Passing to a quotient can collapse order, so only divisibility by eight follows. $\square$

Write $Z = (a_1, b_1, a_2, b_2)$ for the fixed generating tuple of Γ_B . A permutation $r \in S_d$ implements the descended automorphism in the labeled action precisely when

$$r\rho(z)r^{-1} = \rho(\phi_8(z)) \quad (z \in Z). \quad (32.28)$$

Theorem 30.7.2 (Four-equation implementation test). *The four equations (32.28) hold if and only if*

$$r\rho(\gamma)r^{-1} = \rho(\phi_8(\gamma)) \quad \text{for every } \gamma \in \Gamma_B. \quad (32.29)$$

If they hold, then $r \in N_{S_d}(P)$, condition (32.27) holds, and conjugation by r on P is $\bar{\phi}_8$.

Proof. The two sides of (32.29), viewed as maps from Γ_B to S_d , are homomorphisms. Homomorphisms agreeing on the generating set Z agree everywhere, proving the equivalence. Their common image is both rPr^{-1} and $\rho(\phi_8(\Gamma_B)) = P$, so r normalizes P . If $k \in K$, (32.29) gives $\rho(\phi_8(k)) = r1r^{-1} = 1$, proving kernel stability and the last assertion. $\square$

Implementation is an additional constraint: a descended automorphism of P need not be induced by conjugation with a permutation in the chosen degree- d embedding. When implementers exist, they form either the empty set or one coset of the centralizer

$$C_{S_d}(P) := \{c \in S_d : cp = pc \text{ for all } p \in P\}. \quad (32.30)$$

Indeed, two implementers r, r' induce the same automorphism exactly when $r'^{-1}r \in C_{S_d}(P)$.

The normalizer conjugation map

$$N_{S_d}(P) \longrightarrow \text{Aut}(P), \quad r \longmapsto (p \mapsto rpr^{-1}) \quad (32.31)$$

has kernel $C_{S_d}(P)$. Consequently,

$$\text{ord}(\bar{\phi}_8) = \text{ord}(rC_{S_d}(P)) \text{ in } N_{S_d}(P)/C_{S_d}(P), \quad (32.32)$$

not necessarily $\text{ord}(r)$. An exact-order-eight claim therefore requires the seven nonidentity power tests for $\bar{\phi}_8$, or equivalently for the coset in (32.32).

Finite implementation search. Enumerate $r \in S_d$ in lexicographic one-line order. For each r , test all four equations (32.28), with both sides stored as complete permutation vectors. On success, independently verify normalization, compute the centralizer coset, and record the eight power vectors of the induced automorphism. This scan is finite and exhaustive for the fixed labeled embedding. Relabeling the points by $u \in S_d$ conjugates simultaneously P and r ; canonicalization minimizes the concatenated vector over this simultaneous-conjugacy orbit.

For the regular degree-three copy $P = \langle(123)\rangle$, conjugation by (23) implements inversion, while elements of $P = C_{S_3}(P)$ implement the identity. This supplies respectively a positive implementation test and a boundary showing why the order of r alone does not certify the order of the induced automorphism.

30.8 Degree, group order, and quotient extraction

Let a finite group Q act transitively on a set Ω of degree d , choose $\omega \in \Omega$, and put $L = Q_\omega$. Then

$$d = [Q : L], \quad |Q| = d |L|. \quad (32.33)$$

Theorem 30.8.1 (Faithful, free, and regular are distinct conditions). *For the transitive action above,*

$$\begin{aligned} \text{faithful} &\iff \text{core}_Q(L) = 1, \\ \text{free} &\iff L = 1, \\ \text{regular} &\iff \text{transitive and } L = 1 \\ &\iff d = |Q|. \end{aligned} \quad (32.34)$$

Thus regular implies faithful, but faithful transitive need not imply regular.

Proof. The kernel is the core by Statement 30.4, proving the first line. All point stabilizers in a transitive action are conjugate, so one is trivial exactly when every stabilizer is trivial; this is freeness. A free transitive action is regular. Finally (32.33) gives $d = |Q|$ exactly when $|L| = 1$. $\square$

The natural action of S_3 on three points is faithful because the point stabilizer has trivial core, but it is not regular because $|L| = 2$. The left regular action of any finite Q has $L = 1$ and degree $|Q|$. These two exact tests prevent faithfulness from being conflated with freeness.

For a possibly infinite source group G , a transitive degree- d action extracts the finite marked quotient

$$Q := \text{im } \rho \cong G / \ker \rho, \quad d \mid |Q|, \quad d \leq |Q| \leq d!. \quad (32.35)$$

If $\rho = \rho_H$, then $\ker \rho = \text{core}_G(H)$. The action degree determines $|Q|$ only in the regular case.

Typed source and stabilizer data. The source is supplied by a canonical exact record `source_id`: either a finite presentation with its ordered generators and relators, or a finite multiplication table. The record contains the canonical bytes themselves and their hash; the hash is an integrity checksum, not a substitute for bitwise equality. A relation-value table certifies that the proposed generator permutations define ρ . Put

$$L := P_1, \quad H := \rho^{-1}(L), \quad K := \rho^{-1}(1_P). \quad (32.36)$$

Here $L \leq P$ is the image point stabilizer, whereas $H \leq G$ is the source point stabilizer. They are different typed objects, and

$$[P : L] = [G : H] = d, \quad K = \ker \rho = \text{core}_G(H), \quad [G : K] = |P|. \quad (32.37)$$

The certificates `H_cert` and `K_cert` are the respective exact predicates

$$\rho(g)(1) = 1, \quad \rho(g) = 1_P; \quad (32.38)$$

for a finite table they additionally contain the full subgroup bit-vectors. Thus the data do not pretend that P_1 is a subgroup of the source.

There are two scopes for faithfulness and regularity, and they obey different tests:

$$\begin{array}{llll} \text{source faithful} & \iff & K = 1_G, & \text{image faithful} \iff \text{true}, \\ \text{source regular} & \iff & H = 1_G, & \text{image regular} \iff L = 1_P \iff H = K. \end{array} \quad (32.39)$$

The image action is faithful because P is literally a subgroup of S_d . For example, the trivial one-point action of a nontrivial G has $P = L = 1$ and $H = K = G$: its image action is faithful and regular, while its source action is neither.

Theorem 30.8.2 (Complete C32 verification record). *A C32 permutation-quotient record is mathematically complete when it contains*

$$\begin{aligned} \mathcal{R}_{\text{perm}} := & (\text{source_id}, d, \text{side} = \text{left}, Z, (\rho(z))_{z \in Z}, \text{rel_cert}, \\ & P, L, \text{H_cert}, \text{K_cert}, \text{core_cert}, [G : H], [G : K], \\ & \text{transitive}, \text{source_faith_status}, \text{source_regular_status}, \\ & \text{source_faith_cert}, \text{source_regular_cert}, \text{image_faithful} = \text{true} \quad (32.40) \\ & \text{image_regular}, \text{phi8_status}, \text{descended_order}, \\ & \text{implementer_status}, r, \\ & \text{independentverification_hash}) \end{aligned}$$

where every Boolean and integer is recomputed from exact data. For each source classification, let $T = K$ and $\text{cert}_T = \text{source_faith_cert}$ for faithfulness, and let $T = H$ and $\text{cert}_T = \text{source_regular_cert}$ for regularity. Each pair has the exhaustive evidence rule

$$\begin{array}{l|l} \text{proved} & \text{cert}_T \text{ proves } T = 1_G, \\ \text{disproved} & \text{cert}_T \text{ contains a certified } t \in T \setminus \{1_G\}, \\ \text{undecided} & \text{no truth value is asserted and the certificate field is none.} \end{array} \quad (32.41)$$

A finite multiplication table never needs **undecided**, because the bit-vectors decide both tests. For a presented infinite source, **undecided** prevents an uncertified injectivity or trivial-stabilizer claim from being silently recorded as false.

The rotation fields have the following exhaustive presence rules:

phi8_status	descended_order	implementer_status	r
not_applicable	none	none	none
unstable	none	none	none
stable	1, 2, 4, or 8	absent	none
stable	1, 2, 4, or 8	present	least solution of (32.28).

(32.42)

Certificates and diagnostic hashes need not be unique. Define the certificate-free projection

$$\text{construction_key}(\mathcal{R}_{\text{perm}}) := (\text{source_id}, d, \text{side} = \text{left}, Z, (\rho(z))_{z \in Z}).$$

Two complete records encode the same labeled homomorphism relative to the declared source serialization exactly when their construction keys agree. Equality of full records is sufficient but not necessary. Minimizing the construction key over simultaneous point relabelings gives the corresponding unlabeled key.

Proof. The source bytes and ordered marking fix the domain. The relator certificate and generator permutations determine ρ , hence its image P and the image stabilizer $L = P_1$. The predicates (32.38) determine the typed preimages H and K , while Statement 30.4 proves the equality stored by **core_cert**. Statement 30.1–Statement 30.6 recompute transitivity, both indices, and all four scoped properties in (32.39). Rule (32.41) prevents an undecidable source-side assertion from masquerading as a Boolean result. Statement 30.7 recomputes stability, descended order, and the lexicographically least implementer when one exists. The rules (32.41)–(32.42) make every absent field unambiguous. Thus every field is independently verifiable.

Equality of construction keys gives bitwise equal source records, ordered markings, and generator images, hence identical labeled homomorphisms. The converse follows because a labeled homomorphism fixes those generator images. Proof witnesses may still differ, which is why they are absent from the key. Conjugating every generator image by one $u \in S_d$ is exactly a relabeling of the d points; minimizing over these conjugates therefore proves the unlabeled assertion. $\square$

The chapter therefore separates three quantities that must never be identified without proof:

$$\underbrace{d = [G : H]}_{\text{orbit degree}}, \quad \underbrace{|P| = [G : \text{core}_G(H)]}_{\text{quotient order}}, \quad \underbrace{|G|}_{\text{source order, when finite}}. \quad (32.43)$$

It also separates descent of ϕ_8 , implementation of that descent in S_d , and exact order eight. These are independent verification conditions for the systematic low-index construction that follows.

31 Systematic low-index construction

31.1 Finite enumeration of low-index actions

Fix $d \geq 1$, label the points by $\Omega_d = \{1, \dots, d\}$, and retain the ordered Bolza presentation

$$\Gamma_B = \langle a_1, b_1, a_2, b_2 \mid [a_1, b_1][a_2, b_2] = 1 \rangle. \quad (33.1)$$

For a tuple

$$T = (A_1, B_1, A_2, B_2) \in S_d^4, \quad P_T := \langle A_1, B_1, A_2, B_2 \rangle, \quad (33.2)$$

apply the two exact conditions

$$[A_1, B_1][A_2, B_2] = 1, \quad 1^{P_T} = \Omega_d. \quad (33.3)$$

Theorem 31.1.1 (Exhaustive tuple enumeration). *The finite scan of S_d^4 retaining exactly (33.3) produces every transitive degree- d homomorphism $\rho : \Gamma_B \rightarrow S_d$. Two retained tuples define isomorphic unlabeled Γ_B -sets exactly when*

$$T' = uTu^{-1} := (uA_1u^{-1}, uB_1u^{-1}, uA_2u^{-1}, uB_2u^{-1}) \quad \text{for some } u \in S_d. \quad (33.4)$$

Thus keeping the lexicographically least one-line vector in each simultaneous-conjugacy orbit is finite, exhaustive, and duplicate-free.

Proof. There are $(d!)^4$ tuples. The universal property of (33.1) says that the generator assignment extends to a homomorphism exactly when the first equation in (33.3) holds. Its image is P_T , and the action is transitive exactly when the orbit of one point, equivalently every point, is all of Ω_d . Hence the scan has exactly the claimed outputs.

An isomorphism between two actions on the same labeled carrier is a bijection $u \in S_d$ intertwining every source element. It suffices and is necessary to intertwine the four generators, which is precisely (33.4). Finite orbit minimization therefore selects one and only one representative of every unlabeled action. $\square$

The procedure first tests the relator, then computes the subgroup closure $P_T = \langle T \rangle$, and finally tests transitivity. The subgroup closure is computed before transitivity is tested. Every rejected tuple has an exact witness: the nonidentity relator permutation or the least point outside 1^{P_T} . At $d = 1$ there is one tuple, the trivial action; it passes transitivity but has trivial image.

The checkpoint after tuple index j contains d , the next mixed-radix index, the accepted canonical-prefix hash, and the software/version identity. Resume verifies the hash before continuing, so interruption cannot turn a prefix scan into a claim of exhaustive coverage.

31.2 Normal cores and marked finite quotients

For a retained tuple let

$$\rho_T : \Gamma_B \rightarrow S_d, \quad P_T = \text{im } \rho_T, \quad L_T := (P_T)_1, \quad (33.5)$$

and define the source subgroups

$$H_T := \rho_T^{-1}(L_T), \quad K_T := \ker \rho_T. \quad (33.6)$$

Theorem 31.2.1 (Core extraction). *The subgroup H_T has index d , and*

$$K_T = \text{core}_{\Gamma_B}(H_T) \trianglelefteq \Gamma_B, \quad \Gamma_B/K_T \cong P_T, \quad [\Gamma_B : K_T] = |P_T| \leq d!. \quad (33.7)$$

Proof. The transitive Γ_B -action has point stabilizer H_T , so Statement 30.1 gives $[\Gamma_B : H_T] = d$. Its kernel is the core of that stabilizer by Statement 30.4, while the same kernel is K_T by definition. Kernel normality, the first isomorphism theorem, and $P_T \leq S_d$ prove the remaining assertions. $\square$

The construction stores both exact membership predicates

$$\gamma \in H_T \iff \rho_T(\gamma)(1) = 1, \quad \gamma \in K_T \iff \rho_T(\gamma) = 1, \quad (33.8)$$

not the ill-typed equation $H_T = (P_T)_1$. The marked quotient record consists of the complete permutation table of P_T and the ordered image tuple T . This makes the epimorphism $\Gamma_B \twoheadrightarrow P_T$ independently verifiable from the presentation.

If H_T is normal, then $H_T = K_T$, $|P_T| = d$, and the image action is regular. If it is not normal, the quotient has order strictly greater than d . The natural degree-three action of S_3 is the finite boundary model for the latter phenomenon.

31.3 Conditional use of transitive-group catalogues

A transitive-group catalogue at degree d is admissible only with a theorem asserting that the embedded entries

$$\mathcal{T}_d = \{P_{d,1}, \dots, P_{d,m_d}\}, \quad P_{d,j} \leq S_d, \quad (33.9)$$

contain exactly one representative of every S_d -conjugacy class of transitive subgroups. The theorem, catalogue version, degree range, point convention, and exact generator data are input hypotheses, not conclusions of the search.

Theorem 31.3.1 (Catalogue completion criterion). *Assume (33.9) has certified coverage. For every entry P , scan all tuples in P^4 satisfying the surface relator and generating P , and quotient this tuple set by*

$$N_{S_d}(P) = \{u \in S_d : uPu^{-1} = P\}. \quad (33.10)$$

The resulting union over $P \in \mathcal{T}_d$ contains exactly one representative of every unlabeled transitive degree- d Γ_B -action.

Proof. Any transitive action has image $R \leq S_d$. Catalogue coverage supplies $u \in S_d$ and one entry P with $uRu^{-1} = P$. Conjugating the four generator images by u produces a relator-satisfying generating tuple in P^4 , so the scan reaches it. Within the fixed embedded group P , two tuples are simultaneously S_d -conjugate without leaving P exactly through an element of its normalizer. Distinct catalogue entries are not S_d -conjugate, which proves both coverage and uniqueness. $\square$

Listing each transitive subgroup once is not enough: different generating surface tuples in the same subgroup can have different kernels. Quotienting only by P -conjugacy can also leave duplicates created by outer automorphisms realized in $N_{S_d}(P)$. Conversely, quotienting by the abstract $\text{Aut}(P)$ is unsound for this fixed permutation embedding unless every used automorphism is certified to be realized by the normalizer.

If any catalogue hypothesis is missing, its entries are labeled **candidate_source**, not **exhaustive**. A direct S_d^4 scan from Statement 31.1 remains the unconditional finite fallback.

31.4 Order windows and the scope of negative results

Fix $1 \leq N_0 \leq N_1$. An abstract group order window is

$$\mathcal{W}[N_0, N_1] := \{Q / \cong : N_0 \leq |Q| \leq N_1\}. \quad (33.11)$$

It is certified complete only when every isomorphism class in this set is present with an exact group record.

Definition 31.4.1 (Complete marked order-window search). For each $Q \in \mathcal{W}[N_0, N_1]$, enumerate every surjection $\pi : \Gamma_B \twoheadrightarrow Q$ by the images of the four generators, impose the surface relator and generation, and deduplicate only by the kernel criterion of Statement 29.10. The resulting finite union is the complete marked order-window family.

Theorem 31.4.2 (What failure in a window proves). *Failure of every record in the complete marked order-window family at a stated verification condition proves nonexistence of an accepted quotient in that order window. The same conclusion is invalid for a selected list of simple groups, permutation groups, or catalogue entries whose coverage of (33.11) is unproved.*

Proof. Any quotient in the window has a target isomorphic to one listed group. Transporting its epimorphism across the isomorphism gives a scanned generating tuple, and kernel deduplication retains its cover class. Thus exhaustive failure contradicts existence. For an incomplete family, an omitted target or omitted marking is a logical possibility, so failure has only family-relative meaning. $\square$

Degree and order windows are related but not identical. A degree- d transitive action has image order at most $d!$, not at most d . Conversely, every finite quotient Q of order n has its regular representation of degree n . Hence the unconditional tuple scans for all $1 \leq d \leq N_1$ cover every quotient of order at most N_1 , after filtering by image order, although they also produce larger groups and are computationally expensive. A smaller degree cutoff needs an independent theorem that every target in the claimed order window has an admissible action within that cutoff.

The window record records inclusive endpoints, completeness source for the group list, excluded families, per-order marking counts, and whether a negative result is `absolute_in_window` or merely `relative_to_family`.

31.5 Parity-capable targets

Let Q be finite and let $(s_0, \dots, s_7)$ be a candidate shell generating Q .

Theorem 31.5.1 (Index-two parity criterion). *The following are equivalent:*

- (i) there is $\chi_Q : Q \rightarrow C_2$ with $\chi_Q(s_j) = 1$ for every j ,
 - (ii) there is $E \trianglelefteq Q$, $[Q : E] = 2$, with $s_j \notin E$ for every j .
- (33.12)

In this situation χ_Q is onto, $E = \ker \chi_Q$, and if $\alpha(s_j) = s_{j+1}$, then $\chi_Q \alpha = \chi_Q$.

Proof. Under (i), at least one generator has nonzero image, so χ_Q is onto and its kernel E is normal of index two. The eight value equations are exactly the exclusions $s_j \notin E$.

Conversely, every index-two subgroup is normal. The two cosets form the quotient group $Q/E \cong C_2$; its quotient map is zero on E and one on the other coset, proving (i). Finally, $\chi_Q \alpha(s_j) = \chi_Q(s_{j+1}) = 1 = \chi_Q(s_j)$. Because the shell generates Q , the two homomorphisms are equal. $\square$

Parity capability of the abstract group means merely that at least one epimorphism $Q \twoheadrightarrow C_2$ exists. Compatibility with a particular shell additionally requires all eight values in (33.12). The exact finite scan enumerates $\text{Hom}(Q, C_2)$ from generator images, checks all relations, retains surjections, and evaluates the shell table. Groups of odd order fail the group-level condition. A parity-capable group can still fail for a given shell if one s_j lies in every candidate kernel.

31.6 Exact-order-eight automorphism classes

For a canonically tabulated finite group Q , enumerate all maps on a fixed generating tuple, extend them by the multiplication table, and retain exactly the bijective homomorphisms. This constructs the finite group $\text{Aut}(Q)$ without assuming that a supplied subgroup is complete.

Define

$$\mathcal{A}_8(Q) := \{\alpha \in \text{Aut}(Q) : \alpha^8 = 1, \alpha^k \neq 1 \ (1 \leq k < 8)\}. \quad (33.13)$$

Theorem 31.6.1 (Automorphism-class reduction). *The finite power test (33.13) recognizes exact order eight. Conjugation by $\text{Aut}(Q)$ preserves $\mathcal{A}_8(Q)$, and one representative of each orbit is complete for seed construction up to target automorphism.*

Proof. The order of an element satisfying $\alpha^8 = 1$ divides eight; the seven nonidentity tests exclude every proper order. If $\alpha' = u\alpha u^{-1}$, then $(\alpha')^k = u\alpha^k u^{-1}$, so order is preserved. The map

$$(\alpha, s) \mapsto (u\alpha u^{-1}, u(s)) \quad (33.14)$$

is a bijection between the two seed domains and transports every group-word equation, generation condition, parity factor, and nonidentity test. It also postcomposes the marked quotient by u , leaving its kernel unchanged. Hence no kernel class is lost. $\square$

The certificate contains the full automorphism permutation vectors, composition table, inverse table, all eight powers of each retained representative, conjugacy-orbit membership, and the least vector chosen per orbit. Testing only $\alpha^8 = 1$ would incorrectly admit orders one, two, and four.

31.7 Centralizer-orbit seed parameterization

Fix $\alpha \in \mathcal{A}_8(Q)$ and put

$$\mathcal{S}_\alpha := \{s \in Q : \alpha^4(s) = s^{-1}\}, \quad C_\alpha := C_{\text{Aut}(Q)}(\alpha). \quad (33.15)$$

Theorem 31.7.1 (Centralizer action on seeds). *The rule $u \cdot s = u(s)$ defines an action $C_\alpha \curvearrowright \mathcal{S}_\alpha$. If*

$$\mathcal{O}_8(\alpha, s) = (s, \alpha(s), \dots, \alpha^7(s)), \quad (33.16)$$

then u transports this tuple componentwise to $\mathcal{O}_8(\alpha, u(s))$ and preserves all seed verification conditions. Thus one canonical representative per C_α -orbit is complete up to target automorphisms that fix α .

Proof. For $u\alpha = \alpha u$,

$$\alpha^4(u(s)) = u(\alpha^4(s)) = u(s^{-1}) = u(s)^{-1}, \quad (33.17)$$

so $\mathcal{S}_\alpha$ is invariant. The same commutation gives $\alpha^j(u(s)) = u(\alpha^j(s))$ for every j . Automorphisms preserve equality, inversion, group words, generated subgroups, and nonidentity. A parity map χ_Q transports to $\chi_Q u^{-1}$, and a quotient map postcomposes with u , so the parity and kernel conditions are also preserved. $\square$

Centralizing α only by inner automorphisms of Q can miss identifications induced by outer automorphisms. On the other hand, this reduction concerns abstract target/kernel equivalence. For a fixed labeled embedding $Q \leq S_d$, an automorphism u may be used as a point relabeling only after it is shown to lie in the image of the normalizer conjugation map from Statement 30.7.

Orbit enumeration stores the centralizer table, every seed orbit, its least carrier representative, the transport automorphism from each member to that representative, and a coverage hash.

31.8 Kernel-orbit reduction

Let $\pi_i : \Gamma_B \rightarrow Q_i$ be accepted surjections, each retaining the ordered images of (a_1, b_1, a_2, b_2) .

Theorem 31.8.1 (Exact kernel deduplication). *The following are equivalent:*

- (i) $\ker \pi_1 = \ker \pi_2$,
 - (ii) there is a unique isomorphism $\theta : Q_1 \rightarrow Q_2$ with $\theta\pi_1 = \pi_2$.
- (33.18)

Deleting every record except the canonical least member of each equivalence class preserves exactly one representative of every marked cover.

Proof. The equivalence is Statement 29.10. It can also be independently verified directly: under equal kernels define $\theta(\pi_1(\gamma)) = \pi_2(\gamma)$; equality makes the map well defined, and surjectivity makes it a unique isomorphism. Conversely an intertwining isomorphism carries identity exactly to identity, so the kernels agree. Canonical selection from each equivalence class plainly loses no class and leaves no duplicate. $\square$

Every property expressible solely through membership or invariance of the kernel is constant on such a class. In particular,

property	kernel formulation
C_8 -stability	$\phi_8(K) = K$
parity factorization	$K \subseteq \ker \chi$
dangerous-word separation	$K \cap \mathcal{D} = \emptyset$
quotient order	$[\Gamma_B : K]$.

(33.19)

Thus deduplication preserves these conditions. Abstractly isomorphic targets are not merged when their marked generator tuples fail the intertwining test; such maps can have different kernels.

The complete C33 record records

$$\begin{aligned} \mathcal{M}_{33} := (&\text{mode}, \text{scan_status}, \text{cursor}, \text{total_domain_size}, \\ &\text{degree_range}, \text{order_window}, \text{ordering_version}, \\ &\text{input_payload}, \text{input_hash}, \text{coverage_certificate}, \\ &\text{scan_chain_certificate}, \text{accepted_records_bytes}, \\ &\text{accepted_records_hash}, \text{kernel_classes_bytes}, \\ &\text{kernel_classes_hash}, \text{checkpoint_bytes}, \text{checkpoint_hash}, \\ &\text{failure_witness}, \text{derived_counts}). \end{aligned} \quad (33.20)$$

Every byte field is either inline or names a content-addressed object whose bytes must be retrievable; an unavailable object invalidates verification.

The tagged-union payload and range presence rules are

mode	input_payload	degree_range	order_window
DIRECT	D _{direct}	present	none
CATALOGUE	C _{catalogue}	present	none
ORDER_WINDOW	W _{groups}	none	present.

(33.21)

The direct payload serializes the mixed-radix tuple domain. The catalogue payload serializes every exact embedded group and its version. The order-window payload serializes every canonical group record and the inclusive endpoints. In every mode the coverage certificate is required to bind the mode, the present range, the literal payload bytes, and the ordering version.

For a compact presence table, write

$$\begin{aligned} I &:= (\text{input_payload}, \text{input_hash}, \text{coverage_certificate}), \\ S &:= \text{scan_chain_certificate}, \\ A &:= (\text{accepted_records_bytes}, \text{accepted_records_hash}), \\ K &:= (\text{kernel_classes_bytes}, \text{kernel_classes_hash}), \\ P &:= (\text{checkpoint_bytes}, \text{checkpoint_hash}), \\ F &:= \text{failure_witness}. \end{aligned} \quad (33.22)$$

Then all authoritative-field presence rules are

scan_status	cursor	required	none
COMPLETE	N	I, S, A, K	P, F
IN_PROGRESS	$0 \leq c < N$	I, S _{[0,c)} , A _{prefix} , P	K, F
INVALID	none	I, F	S, A, K, P.

(33.23)

Here $N = \text{total_domain_size}$. Diagnostic counters may remain present in every state, but never count as completeness evidence.

Theorem 31.8.2 (Deterministic terminal completeness condition). *A COMPLETE record is valid exactly when its presence row holds and deterministic verification establishes the following. Write $I_b, I_h, C_{\text{cov}}, A_b, A_h, K_b, K_h$ for the corresponding byte, hash, and coverage components in*

(33.22), and put $R_d = \text{degree_range}$, $R_o = \text{order_window}$, and $v = \text{ordering_version}$. Then

$$\begin{aligned}
& \text{Hash}(I_b) = I_h, \\
& \text{VerifyCoverage}(C_{\text{cov}}; \text{mode}, R_d, R_o, I_b, v) = \text{true}, \\
& \text{Replay}(I_b, 0, \dots, N-1; S) = A_b, \\
& \text{Hash}(A_b) = A_h, \\
& \text{KernelDedup}(A_b) = K_b, \\
& \text{Hash}(K_b) = K_h.
\end{aligned} \tag{33.24}$$

The coverage verification also checks the supplied coverage certificate against the literal arguments displayed above.

Proof. The first two equalities identify the exact finite input domain. The deterministic independent verification explicitly visits every index $0, \dots, N-1$ in the declared order, and the scan-chain certificate binds each successor to its predecessor; hence neither an omitted interior block nor a duplicated block can pass. The next hash binds the independent verification output bytes. Exact kernel deduplication of those literal bytes, followed by its own hash, binds the final class list. Therefore an interrupted prefix, unrelated group list, or merely matching pair of hash strings cannot satisfy (33.24). Conversely, these checks reconstruct the entire declared computation and prove completeness. $\square$

An order-window negative result is labeled **absolute_in_window** exactly when the mode is **ORDER_WINDOW**, (33.24) verifies in **COMPLETE** state, the coverage certificate proves the full window, and every retained kernel class fails the stated verification condition. A verified completed scan of a selected family is at most **relative_to_family**. **IN_PROGRESS** and **INVALID** scans have negative classification **UNDECIDED**.

For **IN_PROGRESS**, the prefix independent verification covers exactly $0, \dots, c-1$, and the checkpoint binds the next cursor c and rolling prefix hash; it must also satisfy $\text{Hash}(\text{checkpoint_bytes}) = \text{checkpoint_hash}$. **INVALID** requires a typed failure witness and makes all result fields nonauthoritative. Direct tuple enumeration, catalogue-assisted enumeration, order-window enumeration, automorphism conjugacy reduction, seed centralizer reduction, and kernel reduction are distinct stages and are never silently substituted for one another.

32 Minimizing representation degree

32.1 Minimum transitive degree

For a finite nontrivial group Q , define

$$\mu_t(Q) := \min_{L < Q} [Q : L]. \tag{34.1}$$

For the trivial group we set $\mu_t(1) = 1$, the degree of its unique nonempty transitive action.

Theorem 32.1.1 (Transitive-degree characterization). *The number $\mu_t(Q)$ is the least degree greater than one of a transitive Q -action. A subgroup $L < Q$ attains the minimum exactly when the coset action $Q \curvearrowright Q/L$ has degree $\mu_t(Q)$. Its kernel is $\text{core}_Q(L)$, which need not be trivial.*

Proof. Every nontrivial transitive action is Q -isomorphic to Q/L for a proper point stabilizer L , by Statement 30.2, and its degree is $[Q : L]$. Conversely every proper subgroup supplies such an action. The finite subgroup lattice is nonempty because $1 < Q$, so the minimum exists. Statement 30.4 identifies its kernel with the core, proving the final assertion. $\square$

For $Q = C_4$, the subgroup of order two gives $\mu_t(C_4) = 2$, but its core is itself, so this least action is not faithful. Likewise $\mu_t(S_3) = 2$, using A_3 , even though the natural faithful transitive action has degree three. These examples show that μ_t measures orbit compression, not faithful representability.

An exact computation enumerates subgroup conjugacy classes, since conjugate subgroups have equal index and isomorphic coset actions, and returns the least pair

$$([Q : L], \text{canonical_subgroup_record}(L)) \quad (34.2)$$

in lexicographic order. Completeness of the subgroup inventory is part of the lower-bound certificate.

32.2 Minimum faithful degree

For a finite group Q , retain the nonempty-action convention and define

$$\mu_f(Q) := \min \left\{ \sum_{i=1}^r [Q : L_i] : r \geq 1, \bigcap_{i=1}^r \text{core}_Q(L_i) = 1 \right\}. \quad (34.3)$$

Theorem 32.2.1 (Faithful-degree characterization). *Formula (34.3) is the least d for which Q embeds in S_d . It includes intransitive permutation representations, and*

$$\mu_f(1) = 1, \quad \mu_f(Q) \leq |Q| \quad (Q \neq 1). \quad (34.4)$$

Proof. Every finite Q -set is a disjoint union of orbits Q/L_i . Statement 30.3 gives its degree as the sum in (34.3) and its action kernel as the displayed core intersection. The action homomorphism $Q \rightarrow S_d$ is injective exactly when that kernel is trivial. Conversely every feasible family constructs such an embedding. The convention gives the first equality, and the regular action on $Q/1$ gives the upper bound. $\square$

In general $\mu_t(Q) \leq \mu_f(Q)$, but equality need not hold. For C_4 the values are two and four. Statement 30.3 gives the sharper intransitive phenomenon $\mu_f(C_2^3) = 6 < 8$, whereas every faithful transitive action of this abelian group is regular.

The output for μ_f contains both an attaining family and a lower-bound proof. A tuple of core bit-vectors with trivial intersection proves feasibility; it does not by itself exclude a smaller family.

32.3 Stabilizer structure and conjugacy reduction

For a Q -action and $y = gx$ in the same orbit,

$$Q_y = gQ_xg^{-1}, \quad \text{core}_Q(Q_y) = \text{core}_Q(Q_x). \quad (34.5)$$

Theorem 32.3.1 (Stabilizer-family criterion). *Choose one point x_i from each orbit of a finite Q -set X , and put $L_i = Q_{x_i}$. Then*

$$X \cong \bigsqcup_i Q/L_i, \quad \ker(Q \curvearrowright X) = \bigcap_i \text{core}_Q(L_i). \quad (34.6)$$

Thus faithfulness depends on the conjugacy classes of the orbit stabilizers, while the degree also counts their multiplicities.

Proof. For $y = gx$, an element q fixes y exactly when $g^{-1}qg$ fixes x , proving the first equation in (34.5). The core is normal and invariant under conjugating its defining subgroup, proving the second. Orbit-stabilizer gives each coset model in (34.6), and an element fixes all of X exactly when it belongs to every orbit kernel. $\square$

In a minimum faithful action of a nontrivial group, no orbit type is repeated: deleting one of two identical conjugacy-class components leaves the same kernel intersection and lowers the degree. A fixed-point orbit Q/Q is also absent, because its core is Q and it improves no intersection. These reductions are valid for minimum degree, but repeated or fixed orbits may be retained when the application imposes multiplicities.

The stabilizer certificate stores one canonical subgroup per orbit type, its conjugacy orbit, index, core bit-vector, and multiplicity. Recomputing (34.5)–(34.6) detects an accidental mix of based subgroups and unbased conjugacy classes.

32.4 Combining coset actions

For subgroups $L_1, \dots, L_r \leq Q$, define

$$X(L_1, \dots, L_r) := \bigsqcup_{i=1}^r Q/L_i \quad (34.7)$$

with left multiplication on each tagged component.

Theorem 32.4.1 (Disjoint-union construction). *The action (34.7) has*

$$|X| = \sum_{i=1}^r [Q : L_i], \quad \ker(Q \curvearrowright X) = \bigcap_{i=1}^r \text{core}_Q(L_i). \quad (34.8)$$

It is faithful exactly when the latter intersection is trivial.

Proof. The tags make the components disjoint, so their cardinalities add. An element acts trivially on the union exactly when it acts trivially on every component. Statement 30.4 identifies the component kernel with $\text{core}_Q(L_i)$, which proves (34.8) and the faithfulness statement. $\square$

The explicit embedding sends

$$q \mapsto \text{diag}(\rho_{L_1}(q), \dots, \rho_{L_r}(q)) \in S_{\sum_i [Q:L_i]}, \quad (34.9)$$

where diag means the block permutation on the tagged union. This is not a direct-product target declaration: the actual image is the diagonal subdirect copy of Q modulo the core intersection.

Canonical serialization orders blocks first by degree and then by canonical subgroup record, and renumbers points consecutively within blocks. It stores the full generator permutation vectors, not only the subgroup names.

32.5 Permutation degree versus computational cost

Let Z be an ordered generating set of size r . A faithful degree- d permutation model stores r one-line permutations, requiring rd point indices, or

$$O(rd \log d) \text{ bits} \quad (34.10)$$

under ordinary binary encoding. Multiplication and point action costs are therefore governed by d .

The quotient Hamiltonian constructed earlier, however, uses a position basis indexed by Q and a two-dimensional internal degree of freedom:

$$\mathcal{H}_Q = \mathbb{C}^2 \otimes \ell^2(Q) \cong \mathbb{C}^{2|Q|}. \quad (34.11)$$

Theorem 32.5.1 (No automatic physical compression). *A faithful embedding $Q \hookrightarrow S_d$ does not replace $\mathcal{H}_Q$ by $\mathbb{C}^{2d}$. Such a replacement preserves the complete Hamiltonian and observable algebra only after an explicit unitary intertwiner is proved; if $d \neq |Q|$, no unitary between the two full spaces exists.*

Proof. The dimensions are $2|Q|$ and $2d$, and finite-dimensional Hilbert spaces are unitarily isomorphic only when their dimensions agree. More structurally, the regular translations on $\ell^2(Q)$ and the degree- d permutation action have different matrix coefficients and generally different multiplicities. Faithfulness identifies abstract group elements, not these representations. A restriction to an invariant sector or an isometric encoding can define a different reduced problem, but preservation of every hopping term and observable then requires its own intertwining proof. $\square$

Thus d is a group-computation/storage cost, while $2|Q|$ is the default computational Hilbert dimension. Reports store both coordinates and never substitute one for the other.

32.6 Certified degree optimization

Let $Q \neq 1$, let $L_1, \dots, L_m$ be a complete list of subgroup conjugacy classes of Q , omit Q itself, and put

$$w_j = [Q : L_j], \quad C_j = \text{core}_Q(L_j). \quad (34.12)$$

The optimization problem is

$$\min_{x \in \{0,1\}^m} \left\{ \sum_{j=1}^m w_j x_j : \bigcap_{j:x_j=1} C_j = 1 \right\}. \quad (34.13)$$

Theorem 32.6.1 (Exact optimization certificate). *The optimum in (34.13) equals $\mu_f(Q)$. A claimed value D is certified exactly by*

$$\begin{aligned} \text{upper witness:} & \quad \text{a selected family of weight } D \text{ with trivial core intersection,} \\ \text{lower witness:} & \quad \text{a complete exclusion of every selection of weight less than } D. \end{aligned} \quad (34.14)$$

Proof. Statement 32.3 removes repeated conjugacy classes and fixed orbits from some optimal action, so every optimum appears among the binary vectors in (34.13). Statement 32.4 turns every feasible binary vector into a faithful action of the displayed weight. This proves equality of optima. The upper witness proves $\mu_f(Q) \leq D$, while the exhaustive lower witness proves $\mu_f(Q) \geq D$; together they are necessary and sufficient. $\square$

A independently verifiable lower witness may be a full list of lighter subsets, or a branch-and-bound tree. Every pruned node stores a valid lower bound on all of its descendants and the current core intersection. An incomplete subgroup inventory or an unsupported pruning inequality yields only an upper bound.

32.7 Optimizing quotient order

Fix the source, ϕ_8 , parity character χ , and a dangerous set $\mathcal{D} \subset \Gamma_B$. Let

$$\mathcal{K}_{\text{adm}} := \left\{ K \trianglelefteq \Gamma_B : \begin{array}{l} [\Gamma_B : K] < \infty, \quad \phi_8(K) = K, \\ K \subseteq \ker \chi, \quad K \cap \mathcal{D} = \emptyset, \\ \text{all remaining declared verification conditions hold} \end{array} \right\}. \quad (34.15)$$

Define

$$N_{\min} := \begin{cases} \min\{[\Gamma_B : K] : K \in \mathcal{K}_{\text{adm}}\}, & \mathcal{K}_{\text{adm}} \neq \emptyset, \\ +\infty, & \mathcal{K}_{\text{adm}} = \emptyset. \end{cases} \quad (34.16)$$

Theorem 32.7.1 (Candidate versus minimum). *An admissible quotient of order N proves $N_{\min} \leq N$. It proves equality only when every finite quotient of order less than N has been covered and rejected by a complete search with the same fixed predicates.*

Proof. For $Q = \Gamma_B/K$, its order is the index $[\Gamma_B : K]$, so a feasible kernel supplies the upper bound. If the admissible set is nonempty, well-ordering of the positive integers gives a minimum. To prove a candidate is that minimum, one must exclude every smaller possible index; complete rejection does so. Without such coverage an omitted smaller kernel remains possible, so feasibility alone cannot prove the reverse inequality. $\square$

The lower-order exclusion certificate uses the C33 terminal completeness condition for each covered window. Changing the dangerous set, parity character, symmetry, or any condition starts a new optimization problem and invalidates reuse of the old minimum claim.

32.8 Biobjective optimization

For an admissible kernel K , put

$$Q_K := \Gamma_B/K, \quad c(K) := (|Q_K|, \mu_f(Q_K)). \quad (34.17)$$

Say K_1 dominates K_2 when both coordinates of $c(K_1)$ are at most the corresponding coordinates of $c(K_2)$, with at least one strict inequality.

Theorem 32.8.1 (Pareto frontier). *Dominance is a strict partial order. The Pareto frontier*

$$\mathcal{P} := \{K \in \mathcal{K}_{\text{adm}} : \text{no admissible kernel dominates } K\} \quad (34.18)$$

contains exactly the candidates that cannot be improved in one cost without worsening the other. Applying a strictly increasing rescaling to either coordinate leaves $\mathcal{P}$ unchanged.

Proof. Irreflexivity follows from the required strict coordinate, and transitivity follows from coordinatewise transitivity with at least one strict inequality along the chain. The definition of $\mathcal{P}$ is exactly nondominance. Strictly increasing maps preserve and reflect both weak and strict inequalities, so they preserve domination and hence the frontier. $\square$

Neither coordinate substitutes for the other: Statement 32.5 separates permutation storage from physical Hilbert dimension. A scalar choice requires a declared rule such as

$$\lambda|Q| + (1 - \lambda)\mu_f(Q), \quad 0 \leq \lambda \leq 1, \quad \text{or a declared lexicographic order.} \quad (34.19)$$

Weighted sums can miss nondominated points in a discrete nonconvex cost set, so the unweighted frontier is the complete tradeoff report.

A frontier computed from an incomplete quotient family is marked **provisional**. It becomes certified in a bounded region only when C33-style coverage proves that every admissible candidate capable of dominating a reported point has been enumerated and when every displayed μ_f carries the Statement 32.6 lower-bound certificate.

Part XIII

Products, fiber products, and separators

33 Products of finite quotients

33.1 Finite direct products

Let I be finite and let Q_i be finite groups. On the Cartesian set define

$$Q^\times := \prod_{i \in I} Q_i, \quad (x_i)_{i \in I} (y_i)_{i \in I} := (x_i y_i)_{i \in I}. \quad (35.1)$$

Theorem 33.1.1 (Direct-product group). *The operation (35.1) makes $Q^\times$ a group with*

$$1_{Q^\times} = (1_{Q_i})_i, \quad (x_i)_i^{-1} = (x_i^{-1})_i, \quad |Q^\times| = \prod_{i \in I} |Q_i|. \quad (35.2)$$

Proof. Associativity, the identity law, and the inverse law hold in each coordinate and therefore in the tuple. Independent selection of one element in each finite coordinate gives the cardinality product. For $I = \emptyset$, the empty product is the trivial group and the empty numerical product is one. $\square$

The exact serialization orders I , uses the mixed-radix carrier index

$$\iota((x_i)_i) = \sum_{j=1}^{|I|} \iota_j(x_j) \prod_{k < j} |Q_k|, \quad (35.3)$$

and recomputes multiplication coordinatewise. Finiteness of both I and all factors is necessary for the finite-order formula; an infinite product remains a group but is not a finite quotient target.

33.2 Product homomorphisms and subdirect images

Given homomorphisms $\pi_i : G \rightarrow Q_i$, define

$$\Pi : G \longrightarrow Q^\times, \quad \Pi(\gamma) := (\pi_i(\gamma))_{i \in I}. \quad (35.4)$$

Theorem 33.2.1 (Actual target of a product map). *The map Π is a homomorphism. If every π_i is onto, then*

$$P := \text{im } \Pi \leq \prod_i Q_i, \quad \text{pr}_i(P) = Q_i \quad \text{for every } i, \quad (35.5)$$

so P is subdirect. Surjectivity of all coordinates does not imply $P = \prod_i Q_i$.

Proof. For $\gamma, \delta \in G$,

$$\Pi(\gamma\delta) = (\pi_i(\gamma)\pi_i(\delta))_i = \Pi(\gamma)\Pi(\delta). \quad (35.6)$$

The image is therefore a subgroup. If π_i is onto, then for each $q_i \in Q_i$ some γ has $\text{pr}_i(\Pi(\gamma)) = q_i$, proving (35.5). No choice of one preimage in each coordinate guarantees a single common γ , so full-product surjectivity does not follow. $\square$

For example, if $\pi_1 = \pi_2 : G \twoheadrightarrow Q$, then $\text{im}(\pi_1, \pi_2) = \{(q, q) : q \in Q\}$, the diagonal subgroup of order $|Q|$, not $Q \times Q$ of order $|Q|^2$. All computational and storage claims use the enumerated actual image P ; the ambient product is only a container.

33.3 Kernel intersections

Theorem 33.3.1 (Kernel of the product map). *For any finite family of homomorphisms,*

$$\ker \Pi = \bigcap_{i \in I} \ker \pi_i. \quad (35.7)$$

If every Q_i is finite, this intersection is normal of finite index and

$$G / \bigcap_i \ker \pi_i \cong \text{im } \Pi. \quad (35.8)$$

Proof. An element γ lies in $\ker \Pi$ exactly when $\Pi(\gamma) = (1_{Q_i})_i$, which is equivalent to $\pi_i(\gamma) = 1_{Q_i}$ for every i . This proves (35.7). It is the kernel of a homomorphism into a finite group, hence is normal and finite-index; the first isomorphism theorem gives (35.8). $\square$

For an empty family, (35.7) uses the convention $\bigcap_{\emptyset} \ker \pi_i = G$, matching the trivial product map. For a nonempty family of quotient maps, the certificate stores each $K_i = \ker \pi_i$, the product generator images, and the two membership predicates

$$\gamma \in \ker \Pi \iff \bigwedge_i \pi_i(\gamma) = 1 \iff \gamma \in \bigcap_i K_i. \quad (35.9)$$

33.4 Element-separation capacity

For a homomorphism π , let

$$\text{Sep}(\pi) := \{g \in G : \pi(g) \neq 1\}. \quad (35.10)$$

Theorem 33.4.1 (Union law for separation). *For every $g \in G$,*

$$\Pi(g) \neq 1 \iff \exists i \in I : \pi_i(g) \neq 1, \quad \text{Sep}(\Pi) = \bigcup_{i \in I} \text{Sep}(\pi_i). \quad (35.11)$$

Proof. The identity in a direct product is coordinatewise, so $\Pi(g) = 1$ exactly when all coordinates are identity. Negating this finite conjunction proves both assertions. $\square$

Thus a finite dangerous set $D \subset G \setminus \{1\}$ survives the product quotient exactly when every $d \in D$ is separated by at least one coordinate. Replacing the ambient product by the actual image changes neither $\Pi(g)$ nor its identity test. The separation certificate is the Boolean matrix

$$A_{di} := \mathbf{1}_{\{\pi_i(d) \neq 1\}}, \quad d \in D, \quad i \in I, \quad (35.12)$$

with exact nonidentity witnesses retained in every nonzero entry.

33.5 C_8 -equivariant products

Assume $I \neq \emptyset$, $G = \Gamma_B$, and, for every i ,

$$\alpha_i \pi_i = \pi_i \phi_8, \quad \alpha_i \in \text{Aut}(Q_i). \quad (35.13)$$

Define $\alpha^\times((q_i)_i) := (\alpha_i(q_i))_i$.

Theorem 33.5.1 (Equivariant product and exact order). *The map $\alpha^\times$ is an automorphism of $\prod_i Q_i$,*

$$\alpha^\times \Pi = \Pi \phi_8, \quad (35.14)$$

and the subdirect image $P = \text{im } \Pi$ is $\alpha^\times$ -stable. If all π_i are onto, then

$$\text{ord}(\alpha^\times|_P) = \text{lcm}_{i \in I} \text{ord}(\alpha_i). \quad (35.15)$$

Proof. The product of the coordinate automorphisms is a homomorphism with inverse $\prod_i \alpha_i^{-1}$. For every γ , its i th coordinate in $\alpha^\times \Pi(\gamma)$ is $\alpha_i \pi_i(\gamma) = \pi_i \phi_8(\gamma)$, proving (35.14) and image stability.

Let m be the order of the restriction to P . Each coordinate projection $P \twoheadrightarrow Q_i$ is onto. If the restriction raised to m is identity, projecting shows $\alpha_i^m = 1$ for every i , so the least common multiple divides m . Conversely that least common multiple kills every coordinate and hence P , proving equality. $\square$

If every coordinate order divides eight, the product order divides eight; it is exactly eight precisely when at least one coordinate order is eight. If a coordinate map is not onto, (35.15) must use the order of α_i on $\text{im } \pi_i$, not its order on all of Q_i . For $I = \emptyset$, the product image is trivial and the restricted automorphism has order one; no least-common-multiple formula is invoked.

33.6 Parity-compatible products

Let $K_i = \ker \pi_i$ and $K = \bigcap_i K_i$.

Theorem 33.6.1 (One parity factor suffices). *If $K_j \subseteq \ker \chi$ for at least one j , then*

$$K \subseteq \ker \chi, \quad (35.16)$$

so parity descends to the actual product image. Without such a factor, the augmented map

$$\hat{\Pi} := (\Pi, \chi) : G \rightarrow P \times C_2 \quad \text{has} \quad \ker \hat{\Pi} = K \cap \ker \chi \quad (35.17)$$

and

$$[G : \ker \hat{\Pi}] \in \{[G : K], 2[G : K]\}. \quad (35.18)$$

Proof. The intersection K is contained in every K_i , proving (35.16). The product-kernel formula gives (35.17). The induced map from K to C_2 has image of order one or two, and the index multiplication formula gives (35.18). $\square$

The absence of one individually parity-compatible factor does not imply $K \not\subseteq \ker \chi$: the intersection can satisfy this inclusion even when no K_i does. In that case (35.18) takes its unchanged-index branch and the added parity coordinate is redundant.

When every coordinate already carries a parity factor $\chi_i : Q_i \rightarrow C_2$ with $\chi_i \pi_i = \chi$, these coordinate values agree on P . Storing one induced map $\chi_P : P \rightarrow C_2$ is sufficient. Adjoining separate parity coordinates to every ambient factor inflates the container but does not further shrink the kernel.

33.7 Product-order growth

Assume now that $I \neq \emptyset$ and each $\pi_i : G \rightarrow Q_i$ is onto. Then

$$\max_i |Q_i| \leq |\text{im } \Pi| = [G : \bigcap_i K_i] \leq \prod_i |Q_i| = \prod_i [G : K_i]. \quad (35.19)$$

Theorem 33.7.1 (Equality criterion). *The upper inequalities in (35.19) are equalities exactly when Π is onto the full Cartesian product.*

Proof. Subdirectness makes every coordinate projection of the image onto, giving the lower bound. The image is a subgroup of the finite product, giving the upper bound. The first isomorphism theorem and (35.7) give the middle equality, while surjectivity of each coordinate gives $[G : K_i] = |Q_i|$. A subgroup of a finite group has the full group order exactly when it is the full group, which proves the criterion. $\square$

The ratio

$$\Delta := \frac{\prod_i |Q_i|}{|\text{im } \Pi|} \in \mathbb{N} \quad (35.20)$$

measures ambient redundancy. The diagonal example has $\Delta = |Q|$. A computational record records both the ambient order and actual image order, and allocates the quotient model using only the latter.

33.8 Minimum separating subsets

Let candidate quotient maps be indexed by $1, \dots, n$, give factor i a nonnegative rational cost w_i , and let D be a finite dangerous set. With the exact matrix from (35.12), solve

$$\begin{aligned} \min \quad & \sum_{i=1}^n w_i x_i \\ \text{subject to} \quad & \sum_{i=1}^n A_{di} x_i \geq 1 \quad (d \in D), \\ & x_i \in \{0, 1\}. \end{aligned} \tag{35.21}$$

Theorem 33.8.1 (Set-cover equivalence). *A binary vector is feasible in (35.21) exactly when the product of its selected quotient maps separates every element of D . A feasible vector of cost W is certified optimal only with a valid lower bound W for all feasible binary vectors.*

Proof. For fixed d , its constraint says that some selected factor has $\pi_i(d) \neq 1$. Statement 33.4 says this is equivalent to survival in the selected product. Applying this to every row proves feasibility equivalence. Feasibility gives an upper bound; a matching global lower bound gives the reverse inequality and hence optimality. $\square$

An exact certificate contains the selected indices, rational objective arithmetic, and either exhaustive enumeration or a branch-and-bound tree whose every pruned node has a verified lower bound for all feasible descendants, equivalently for its best feasible completion. An LP dual certifies the integer optimum only when its dual value matches the integer feasible cost. If $D = \emptyset$, the empty subset has cost zero unless parity, symmetry, or a nontrivial-target condition adds further declared constraints.

The final product record stores

$$\begin{aligned} \mathcal{R}_{\text{product}} := & (\text{selected_factors}, \text{factor_hashes}, A, \\ & \text{selected_cost}, \text{lower_bound_certificate}, \\ & \text{actual_image_hash}, \text{kernel_intersection_hash}). \end{aligned} \tag{35.22}$$

Hash fields protect serialized bytes; independent verification of the coordinate maps, image generation, and kernel/separation tests supplies the mathematical proof.

34 Subdirect and fiber products

34.1 Subdirect products

A subgroup $H \leq Q_1 \times Q_2$ is subdirect when both restricted coordinate projections are onto:

$$\text{pr}_1(H) = Q_1, \quad \text{pr}_2(H) = Q_2. \tag{36.1}$$

Theorem 34.1.1 (Product images are subdirect). *If $\pi_i : G \twoheadrightarrow Q_i$ are surjective, then*

$$H_\pi := \{(\pi_1(g), \pi_2(g)) : g \in G\} = \text{im}(\pi_1, \pi_2) \tag{36.2}$$

is a subdirect subgroup of $Q_1 \times Q_2$.

Proof. It is the image of a homomorphism, hence a subgroup. Given $q_i \in Q_i$, surjectivity of π_i supplies g whose i th coordinate is q_i ; therefore each restricted projection is onto. $\square$

Subdirect does not mean direct. For $\pi_1 = \pi_2 : G \twoheadrightarrow Q$, the image is the diagonal $\{(q, q) : q \in Q\}$. The serialized record stores the complete carrier of H_π , both projection tables, and surjectivity witnesses for every element of each factor. Merely storing the two factor names cannot certify the coupling.

34.2 Fiber-product construction

Let $f_i : Q_i \rightarrow C$ be homomorphisms. Define

$$Q_1 \times_C Q_2 := \{(x, y) \in Q_1 \times Q_2 : f_1(x) = f_2(y)\}. \quad (36.3)$$

Theorem 34.2.1 (Fiber subgroup). *The set in (36.3) is a subgroup of $Q_1 \times Q_2$. Surjectivity of the f_i is not required for closure.*

Proof. The identity pair satisfies the defining equality. If (x, y) and (x', y') satisfy it, then

$$\begin{aligned} f_1(xx'^{-1}) &= f_1(x)f_1(x')^{-1} \\ &= f_2(y)f_2(y')^{-1} = f_2(yy'^{-1}), \end{aligned} \quad (36.4)$$

so the one-step subgroup test applies. $\square$

If both f_i are onto, the fiber product is subdirect. For example, given $x \in Q_1$, choose $y \in Q_2$ with $f_2(y) = f_1(x)$; then (x, y) lies in the fiber. The reverse argument proves surjectivity onto Q_2 .

The finite construction sorts pairs lexicographically, tests the literal equality in the canonical table of C , and recomputes closure. It stores the maps f_i , because the same three abstract groups with different quotient maps can yield different embedded fiber subgroups.

34.3 Common quotients and the canonical coupling

Given surjections $\pi_i : G \twoheadrightarrow Q_i$, a common quotient *compatible with* (π_1, π_2) is the data

$$(C, f_1, f_2), \quad f_i : Q_i \twoheadrightarrow C, \quad (36.5)$$

such that $f_1\pi_1 = f_2\pi_2$. Thus it is not merely an abstract common quotient or an isomorphism class of groups. Postcomposing both maps with one automorphism of C leaves (36.3) unchanged; changing only one map generally changes it.

Write $K_i = \ker \pi_i$. Since the K_i are normal, $K_1K_2 \trianglelefteq G$, and define

$$C_\pi := G/(K_1K_2), \quad f_i(\pi_i(g)) := g(K_1K_2). \quad (36.6)$$

Theorem 34.3.1 (Canonical common-quotient description). *The maps in (36.6) are well-defined surjections and*

$$H_\pi = Q_1 \times_{C_\pi} Q_2. \quad (36.7)$$

Proof. Because $K_i \subseteq K_1K_2$, changing a representative of $\pi_i(g)$ does not change its coset modulo K_1K_2 ; surjectivity is immediate.

Every diagonal image pair has equal C_π -value. Conversely, suppose $f_1(\pi_1(g)) = f_2(\pi_2(h))$. Then $h^{-1}g \in K_1K_2 = K_2K_1$, so write $h^{-1}g = k_2k_1$. With $\gamma = gk_1^{-1} = hk_2$,

$$\pi_1(\gamma) = \pi_1(g), \quad \pi_2(\gamma) = \pi_2(h).$$

Thus the fiber pair is in H_π , proving equality. $\square$

The word “largest” has the following precise universal meaning. Let (C, f_1, f_2) be any compatible common quotient and put $\rho = f_1\pi_1 = f_2\pi_2$. Then $K_i \subseteq \ker \rho$, hence $K_1K_2 \subseteq \ker \rho$, and there is a unique surjection

$$u : C_\pi \twoheadrightarrow C, \quad u(g(K_1K_2)) = \rho(g), \quad f_i = u \circ f_i^\pi,$$

where f_i^π denotes the canonical map in (36.6). Existence, well-definedness, and surjectivity follow from the kernel inclusion and the surjectivity of ρ ; uniqueness follows because $G \twoheadrightarrow C_\pi$. Consequently C_π is the maximal compatible common quotient, or equivalently the initial object when arrows point from a common quotient to its factors. It is not a terminal object for that arrow convention and is not claimed maximal among incompatible abstract common quotients.

Thus C_π measures exactly the relation coupling the coordinates; $C_\pi = 1$ precisely when the product map is onto $Q_1 \times Q_2$.

34.4 Kernel coupling without a false kernel map

The fiber condition can be written safely as

$$Q_1 \times_C Q_2 = (f_1 \times f_2)^{-1}(\Delta_C), \quad \Delta_C := \{(c, c) : c \in C\} \leq C \times C. \quad (36.8)$$

This proves subgroup closure because $f_1 \times f_2$ is a homomorphism and Δ_C is a subgroup.

One is sometimes tempted to define

$$\delta(x, y) := f_1(x)f_2(y)^{-1}. \quad (36.9)$$

For nonabelian C , δ need not be a homomorphism, so the fiber product must not be called $\ker \delta$. Formula (36.8) is the correct preimage statement.

Theorem 34.4.1 (Projection kernels and extensions). *If both f_i are onto and $F = Q_1 \times_C Q_2$, then*

$$\begin{aligned} \ker(\text{pr}_1|_F) &= \{1\} \times \ker f_2, \\ \ker(\text{pr}_2|_F) &= \ker f_1 \times \{1\}, \end{aligned} \quad (36.10)$$

and there are short exact sequences

$$\begin{aligned} 1 \rightarrow \ker f_2 \rightarrow F \rightarrow Q_1 \rightarrow 1, \\ 1 \rightarrow \ker f_1 \rightarrow F \rightarrow Q_2 \rightarrow 1. \end{aligned} \quad (36.11)$$

Proof. A pair in the first kernel has first coordinate 1, and the fiber equality then says $f_2(y) = 1$; this proves the first line, and the second is symmetric. The restricted projections are onto by Statement 34.2, so the kernel descriptions give exactness. $\square$

34.5 Order compression

Theorem 34.5.1 (Fiber-product order). *For finite Q_i, C and surjective $f_i : Q_i \twoheadrightarrow C$,*

$$|Q_1 \times_C Q_2| = \frac{|Q_1||Q_2|}{|C|}. \quad (36.12)$$

Proof. Every fiber $f_i^{-1}(c)$ is a coset of $\ker f_i$, hence has size $|Q_i|/|C|$. Partitioning the fiber product by the common value gives

$$\sum_{c \in C} |f_1^{-1}(c)| |f_2^{-1}(c)| = |C| \frac{|Q_1|}{|C|} \frac{|Q_2|}{|C|} = \frac{|Q_1||Q_2|}{|C|}. \quad (36.13)$$

$\square$

Applied to (36.6)–(36.7), this yields the exact finite-index identity

$$[G : K_1 \cap K_2] = \frac{[G : K_1][G : K_2]}{[G : K_1 K_2]}. \quad (36.14)$$

The denominator is the coupling compression. It equals one exactly for the full direct product and equals $|Q|$ for the diagonal product of two identical quotient maps.

An order certificate counts the common-value blocks, verifies their uniform sizes, and independently enumerates the generated subdirect image. Agreement with (36.12) prevents the ambient Cartesian order from being reported as the actual quotient order.

34.6 C_8 -stable fiber products

Assume automorphisms $\alpha_i \in \text{Aut}(Q_i)$ and $\alpha_C \in \text{Aut}(C)$ satisfy

$$f_i \alpha_i = \alpha_C f_i \quad (i = 1, 2). \quad (36.15)$$

Theorem 34.6.1 (Equivariant fiber restriction). *The componentwise automorphism $\alpha_1 \times \alpha_2$ preserves $F = Q_1 \times_C Q_2$, and its restriction is an automorphism. If the projections $F \rightarrow Q_i$ are onto, then*

$$\text{ord}((\alpha_1 \times \alpha_2)|_F) = \text{lcm}(\text{ord } \alpha_1, \text{ord } \alpha_2). \quad (36.16)$$

Proof. For $(x, y) \in F$, compatibility gives

$$f_1(\alpha_1 x) = \alpha_C f_1(x) = \alpha_C f_2(y) = f_2(\alpha_2 y), \quad (36.17)$$

so the fiber is preserved. The componentwise inverse also preserves it, giving an automorphism. If a power is identity on F , surjectivity of each projection makes the corresponding power of each α_i identity; the converse holds coordinatewise. This proves the least-common-multiple formula. $\square$

For Bolza quotient maps satisfying $\alpha_i \pi_i = \pi_i \phi_8$, the canonical common quotient inherits an α_C precisely when $K_1 K_2$ is ϕ_8 -stable; this follows from $\phi_8(K_i) = K_i$. Hence (36.15) holds automatically for the induced maps. If both coordinate orders divide eight, the fiber order is exactly eight iff at least one coordinate order is eight.

34.7 Parity pullbacks

Let $\chi_i : Q_i \rightarrow C_2$ be surjective and define

$$F_\chi := Q_1 \times_{C_2} Q_2 = \{(x, y) : \chi_1(x) = \chi_2(y)\}. \quad (36.18)$$

Then

$$|F_\chi| = \frac{|Q_1||Q_2|}{2}. \quad (36.19)$$

Theorem 34.7.1 (Shared-parity image). *If quotient maps $\pi_i : G \rightarrow Q_i$ induce the same source character,*

$$\chi_1 \pi_1 = \chi = \chi_2 \pi_2, \quad (36.20)$$

then $\text{im}(\pi_1, \pi_2) \leq F_\chi$. On F_χ , the formula $\chi_F(x, y) = \chi_1(x) = \chi_2(y)$ defines a surjective parity map.

Proof. Equation (36.20) makes the two coordinates of every product image have the same parity, proving containment. The common-value formula is well defined on F_χ , multiplicative, and onto because the projections of the fiber product are onto and each χ_i is onto. $\square$

The containment can be strict: shared parity explains one coupling relation but need not explain all common quotients of the two factors. The actual image is enumerated and counted separately; (36.19) is an ambient fiber order, not automatically the quotient order.

34.8 Large-systole subdirect products

Let $\pi_i : \Gamma_B \rightarrow Q_i$ have kernels K_i , and put

$$K := K_1 \cap K_2, \quad P := \text{im}(\pi_1, \pi_2) \cong \Gamma_B / K. \quad (36.21)$$

Let $D_{\text{global}}(L)$ be the certified closed dangerous domain containing every nonidentity element of Γ_B with translation length at most L .

Theorem 34.8.1 (Intersection separator gives large systole). *If*

$$K \cap D_{\text{global}}(L) = \emptyset, \quad (36.22)$$

then

$$\text{sys}(\mathbb{H}^2/K) > L. \quad (36.23)$$

Proof. By definition of the dangerous domain, a nonidentity $k \in K$ with translation length at most L would lie in the intersection in (36.22), a contradiction. The subgroup K is finite-index in the cocompact surface group Γ_B , so it is again discrete and cocompact and its positive translation-length minimum is attained. Since no nonidentity length is at most L , that minimum is strictly greater than L . $\square$

Neither coordinate kernel must pass (36.22) separately. An element missed by one factor can be separated by the other, as in Statement 33.4. The certificate therefore evaluates every dangerous element in both coordinates, records at least one nonidentity witness, constructs the actual subdirect image, and applies the closed-domain completeness certificate before asserting the systole bound.

The final fiber record contains

$$\begin{aligned} \mathcal{R}_{\text{fiber}} := & (\text{factor_records}, \text{common_quotient_maps}, \text{fiber_carrier}, \\ & \text{actual_image}, \text{projection_kernels}, \text{order_certificate}, \\ & \text{symmetry_compatibility}, \text{parity_map}, \text{dangerous_set_certificate}). \end{aligned} \quad (36.24)$$

All group and map bytes are canonical; hashes protect transport, while table independent verification proves the subgroup, projection, order, equivariance, parity, and separation claims.

35 Separator construction

35.1 Typed finite dangerous sets

A separator problem begins with a finite, exact, deduplicated list

$$D = (d_1, \dots, d_m) \subset \Gamma_B \setminus \{1\}, \quad \text{kind}(D) \in \{\text{word}, \text{based}, \text{global}\}. \quad (37.1)$$

Each entry contains the triple

$$(\text{source_id}, \text{canonical_element_key}, \text{word_bytes}).$$

Equality uses the source identifier together with the independently validated canonical element key; the word spelling is a independent verification witness, not the equality key. An exact source-group independent verification must evaluate `word_bytes` and verify that the result equals `canonical_element_key`; a matching hash alone does not establish this semantic equality. A word dangerous set is closed by a declared word-length cutoff; a based set is closed by $d(o, \gamma o)$; a global set is closed by translation length after axis or conjugacy normalization. These are different predicates. The record may cite a proved conversion bound, but it never silently changes one kind into another.

Definition 35.1.1 (Closed dangerous-set record). A record for D is declared

$$\text{Closed}(D, L, \text{kind}) = \text{true} \quad (37.2)$$

only when it contains the enumeration rule, its exact terminal frontier, every accepted canonical element, and a independently verifiable proof that no unvisited state can satisfy the declared predicate at L .

Finiteness and closure are separate facts. A displayed finite list is not a certificate that the target domain is exhausted. Likewise, $1 \notin D$ is checked after source-group deduplication; deleting the identity by its word spelling alone is unsound.

35.2 Separation sets

Let $\pi_i : \Gamma_B \twoheadrightarrow Q_i$, $1 \leq i \leq n$, be exact finite quotients. Define

$$S_i := \{d \in D : \pi_i(d) \neq 1_{Q_i}\}, \quad F_i := D \setminus S_i = D \cap \ker \pi_i. \quad (37.3)$$

Every evaluation uses the stored source word, the certified images of the four presentation generators, and the exact multiplication law of Q_i . A nonidentity result is stored as its canonical carrier index; an identity result stores the complete word-evaluation trace.

Lemma 35.2.1 (Elementwise invariance). *Membership in S_i depends only on the element of Γ_B represented by a word.*

Proof. If words u, v represent the same source element, then $uv^{-1} = 1$ in Γ_B . Since π_i is a homomorphism,

$$\pi_i(u)\pi_i(v)^{-1} = 1 \implies \pi_i(u) = \pi_i(v). \quad (37.4)$$

Hence either both values are the identity or neither is. $\square$

Thus alternate spellings may be useful independent verification data, but they never create extra rows in the optimization problem.

35.3 The separator matrix

Fix the order of D and of the candidate quotients and put

$$A = (A_{ji}) \in \{0, 1\}^{m \times n}, \quad A_{ji} := \begin{cases} 1, & \pi_i(d_j) \neq 1, \\ 0, & \pi_i(d_j) = 1. \end{cases} \quad (37.5)$$

The matrix artifact contains more than bits: cell (j, i) links to the exact evaluation of d_j , and every one-cell records its canonical nonidentity value. Row and column permutations change the serialization hash, but only relabel the same covering instance.

Lemma 35.3.1 (Zero-row obstruction). *There exists a selection of the supplied factors separating all of D only if*

$$\sum_{i=1}^n A_{ji} \geq 1 \quad (1 \leq j \leq m). \quad (37.6)$$

Proof. If row j is zero, every candidate quotient kills d_j . Consequently every product of any subfamily also kills d_j . $\square$

The converse follows from selecting all columns, and will be included in the exact equivalence below.

35.4 Set cover is exactly separation

For $x = (x_1, \dots, x_n) \in \{0, 1\}^n$, let

$$I_x := \{i : x_i = 1\}, \quad \Pi_x(\gamma) := (\pi_i(\gamma))_{i \in I_x}. \quad (37.7)$$

Theorem 35.4.1 (Separator-cover equivalence). *The selected product map separates every element of D if and only if*

$$\sum_{i=1}^n A_{ji}x_i \geq 1 \quad (1 \leq j \leq m). \quad (37.8)$$

Proof. Suppose (37.8) holds. For every d_j there is a selected i with $A_{ji} = 1$, hence the i -th coordinate of $\Pi_x(d_j)$ is nonidentity; therefore $\Pi_x(d_j) \neq 1$.

Conversely, if $\Pi_x(d_j) \neq 1$, at least one selected coordinate is nonidentity. Its matrix entry is one, so the j -th inequality in (37.8) holds. Quantifying over all rows proves the equivalence.

$$D = \emptyset : x = 0 \text{ is feasible}; \quad D \neq \emptyset : x = 0 \text{ is infeasible.} \quad (37.9)$$

□

35.5 Minimum weighted separators

For certified rational weights $w_i \geq 0$, put $\mathcal{F} = \{x \in \{0, 1\}^n : Ax \geq \mathbf{1}\}$. The weighted separator value is

$$W_* := \begin{cases} \min_{x \in \mathcal{F}} \sum_i w_i x_i, & \mathcal{F} \neq \emptyset, \\ +\infty, & \mathcal{F} = \emptyset. \end{cases} \quad (37.10)$$

A zero row makes the classification **INFEASIBLE**; the certificate is the least row index j with $\sum_i A_{ji} = 0$. Conversely, if there is no zero row, the all-one vector is feasible, so this test is complete for the supplied finite factor family. A feasible vector and its row witnesses provide only an upper bound. One convenient independent lower certificate is a rational packing vector

$$y \in \mathbb{Q}_{\geq 0}^m, \quad A^\top y \leq w. \quad (37.11)$$

Theorem 35.5.1 (Checkable optimality). *Assume $\mathcal{F} \neq \emptyset$. For every feasible binary x and every y satisfying (37.11),*

$$\sum_j y_j \leq \sum_i w_i x_i. \quad (37.12)$$

If equality holds for a stored pair (x, y) , then x is exactly optimal for (37.10).

Proof. Because $Ax \geq \mathbf{1}$, $y \geq 0$, and $x \geq 0$,

$$\sum_j y_j \leq y^\top Ax = (A^\top y)^\top x \leq w^\top x.$$

Equality therefore matches a universal lower bound with the attained upper bound. □

The packing certificate need not close the integrality gap. In that case an exact branch-and-bound certificate records, at every node, a sound lower bound for all feasible descendants and either a branching partition, an infeasibility proof, or pruning against the incumbent. Floating solver summaries and rounded weights are not optimality proofs. The dual argument certifies finite feasible optima; it is not used to prove the value $+\infty$ in an infeasible instance.

35.6 The quotient determined by a cover

Put $K_i = \ker \pi_i$ and replace the ambient Cartesian target by

$$K_x := \bigcap_{i \in I_x} K_i, \quad Q_x := \text{im } \Pi_x \cong \Gamma_B / K_x. \quad (37.13)$$

Theorem 35.6.1 (Selected-product certificate). *If x satisfies (37.8), then*

$$D \cap K_x = \emptyset. \quad (37.14)$$

The exact quotient Q_x , rather than $\prod_{i \in I_x} Q_i$, has precisely the required kernel and is no larger than the ambient product.

Proof. The kernel-of-product identity gives $\ker \Pi_x = \bigcap_{i \in I_x} K_i = K_x$. For each d_j , (37.8) supplies a selected coordinate not killing d_j , so $d_j \notin K_x$. The first isomorphism theorem gives the displayed isomorphism, while

$$|Q_x| \leq \prod_{i \in I_x} |Q_i|. \quad (37.15)$$

□

The computational certificate enumerates Q_x , proves closure and generation by the four diagonal generator images, and recomputes its order. Merely multiplying factor orders reports an upper bound.

35.7 Actual-order optimization

Choosing

$$w_i = \log |Q_i| \implies \sum_i w_i x_i = \log \prod_{i \in I_x} |Q_i| \quad (37.16)$$

minimizes the Cartesian upper bound in (37.15), not necessarily $\log |Q_x|$. Common quotients and diagonal relations can make the subdirect image much smaller. Thus an exact actual-order claim uses

$$N_* := \begin{cases} \min_{x \in \mathcal{F}} |\operatorname{im} \Pi_x|, & \mathcal{F} \neq \emptyset, \\ +\infty, & \mathcal{F} = \emptyset. \end{cases} \quad (37.17)$$

Lemma 35.7.1 (Monotonicity under adding factors). *If $I \subseteq J$, then the coordinate-forgetting map $Q_J \twoheadrightarrow Q_I$ is onto and*

$$|Q_I| \mid |Q_J|, \quad |Q_I| \leq |Q_J|. \quad (37.18)$$

Proof. Both groups are diagonal images of Γ_B ; projection sends the larger tuple image onto the smaller one. The first isomorphism theorem for this projection gives divisibility and the inequality. □

Consequently exact enumeration may prune a branch only with a lower bound valid for every feasible descendant, such as the already attained partial-image order. A certificate for (37.17) stores every feasible leaf or a independently verifiable pruning reason and compares actual image orders. Factor deletion is accepted only after both the full row cover and the recomputed kernel/image certificate survive.

35.8 Nested separator towers

Let

$$D_1 \subseteq D_2 \subseteq \cdots, \quad \bigcup_{r \geq 1} D_r = \Gamma_B \setminus \{1\}, \quad (37.19)$$

where every D_r is finite and closed in its declared sense. The residual-finiteness theorem of Statement 16.2 and its constructive form Statement 16.3 give, for every $d \in D_r$, a finite quotient not killing d . Because D_r is finite, the diagonal product of these finitely many quotients separates all of D_r . Hence the required finite family exists at every stage. Moreover, either the direct constructive separator of Statement 16.3 or a deterministic complete and fair enumeration of all finite marked quotients, with exact verification of every candidate, eventually finds such a family. Without one of these complete mechanisms, the append-until instruction is only a conditional specification and is not a termination claim.

Starting from $K_0 = \Gamma_B$, at stage r retain every prior factor and append finitely many quotient maps until their product separates D_r . Let K_r be the intersection of all kernels retained through stage r .

Theorem 35.8.1 (Residual separator tower). *Every K_r is normal and finite-index,*

$$K_{r+1} \subseteq K_r, \quad D_r \cap K_r = \emptyset, \quad \bigcap_{r \geq 1} K_r = \{1\}. \quad (37.20)$$

Hence $Q_r = \Gamma_B/K_r$ form a residual inverse system with surjective bonding maps $Q_{r+1} \twoheadrightarrow Q_r$.

Proof. Finite intersections of finite-index normal kernels have the same properties. Retention of old factors gives nesting, and the stage cover gives disjointness. If $1 \neq \gamma \in \Gamma_B$, exhaustivity places γ in some D_r , whence $\gamma \notin K_r$ and, by nesting, in no later kernel. Therefore no nonidentity element lies in the total intersection. Kernel inclusion defines the bonding surjections. $\square$

If no new element of D_{r+1} survives in K_r , the stage may repeat $K_{r+1} = K_r$; strict descent is claimed only when an appended factor separates a witness still in K_r . When every retained kernel is C_8 -stable and lies in $\ker \chi$, these properties pass to each intersection and to every bonding map. The tower record records the typed closure certificate for D_r , the appended factors, the actual subdirect image, and the strict-or-repeat classification.

Final construction record. The mathematical certificate is the tuple

$$(D, \text{kind}, \text{classification}, A, x, \{\pi_i\}_{i \in I_x}, K_x, Q_x, \text{independent verification proofs}), \quad (37.21)$$

with an optional exact lower certificate for weighted or actual-order optimality. The operational record separately stores enumeration logs, carrier encodings, hashes, timing, memory, and solver versions. Hash agreement authenticates bytes; it does not replace any group, cover, closure, or optimality proof.

Part XIV

Exact finite-quotient search

36 A finite presentation as an exact constraint system

36.1 Finite carrier and variables

Fix a carrier $[N] = \{0, \dots, N-1\}$, an identity index e , and a multiplication table

$$m : [N] \times [N] \longrightarrow [N]. \quad (38.1)$$

The group-table condition verifies associativity, the two identity laws, and a two-sided inverse for every carrier element. Only after that condition may the table be denoted by Q . The marked search variables are

$$x_{a_1}, x_{b_1}, x_{a_2}, x_{b_2} \in Q, \quad \beta : Q \rightarrow Q, \quad \epsilon : Q \rightarrow C_2. \quad (38.2)$$

The first four entries propose images of the fixed presentation generators. The array β proposes the descended rotation and the bit array ϵ proposes descended parity.

Every candidate serializes the ordered carrier, table, identity, inverse array, marked tuple, and both maps. Renaming the carrier is allowed only through an identity-preserving table isomorphism carrying all marked data. A checksum authenticates this serialization but does not establish any of the equations below.

36.2 The surface relation

For $x, y \in Q$, set

$$[x, y] := xyx^{-1}y^{-1}, \quad r_Q := [x_{a_1}, x_{b_1}][x_{a_2}, x_{b_2}]. \quad (38.3)$$

Theorem 36.2.1 (Presentation criterion). *The marked tuple extends to a unique homomorphism $\pi : \Gamma_B \rightarrow Q$ precisely when*

$$r_Q = e. \quad (38.4)$$

Proof. The free group on a_1, b_1, a_2, b_2 admits a unique homomorphism to Q with the four prescribed values. It factors through

$$\Gamma_B = \langle a_1, b_1, a_2, b_2 \mid [a_1, b_1][a_2, b_2] = 1 \rangle \quad (38.5)$$

if and only if the normal closure of the relator lies in its kernel. That is equivalent to the relator itself evaluating to e , which is (38.4). The quotient universal property gives uniqueness. $\square$

The independent verification trace lists the carrier index after every multiplication and inversion in (38.3). Relation satisfaction does not imply that π is onto; generation is a separate constraint.

36.3 Automorphism and covariance equations

The proposed rotation must satisfy, for all $x, y \in Q$,

$$\beta(e) = e, \quad \beta(xy) = \beta(x)\beta(y), \quad \{\beta(x) : x \in Q\} = Q, \quad \beta^8 = \text{id}_Q. \quad (38.6)$$

The image-set equality is a finite bijectivity check. Let $W_s = \phi_8(s)$ be the fixed source words for $s \in \{a_1, b_1, a_2, b_2\}$. Covariance is the four-word condition

$$\beta(x_s) = W_s(x_{a_1}, x_{b_1}, x_{a_2}, x_{b_2}) \quad (s = a_1, b_1, a_2, b_2). \quad (38.7)$$

Theorem 36.3.1 (Finite covariance criterion). *Assume (38.4) and (38.6). Equation (38.7) is equivalent to*

$$\beta \circ \pi = \pi \circ \phi_8 \quad \text{on all of } \Gamma_B. \quad (38.8)$$

Proof. Both sides of (38.8) are homomorphisms $\Gamma_B \rightarrow Q$. Equation (38.7) says that they agree on the four presentation generators, so they agree everywhere. The converse follows by evaluating (38.8) on those generators. $\square$

The constraint $\beta^8 = \text{id}$ permits descended order dividing eight. Exact order eight additionally requires

$$\beta^d \neq \text{id}_Q \quad (d = 1, 2, 4), \quad (38.9)$$

verified by a stored carrier witness for each inequality. Since every proper divisor of eight is 1, 2, or 4, (38.6) and (38.9) are equivalent to exact order eight.

36.4 Parity equations

Write $C_2 = \{0, 1\}$ additively. The parity table must obey

$$\epsilon(e) = 0, \quad \epsilon(xy) = \epsilon(x) + \epsilon(y) \pmod{2} \quad (x, y \in Q), \quad (38.10)$$

and the marked generators satisfy

$$\epsilon(x_{a_1}) = \epsilon(x_{b_1}) = \epsilon(x_{a_2}) = \epsilon(x_{b_2}) = 1. \quad (38.11)$$

Theorem 36.4.1 (Parity descent). *Under (38.4), equations (38.10)–(38.11) imply*

$$\epsilon\pi = \chi. \quad (38.12)$$

In particular $\epsilon : Q \twoheadrightarrow C_2$ is surjective because a marked generator has value one. If the marked tuple generates Q , then $\pi : \Gamma_B \twoheadrightarrow Q$ is surjective.

Proof. Equation (38.10) makes ϵ a homomorphism. The two homomorphisms $\epsilon\pi, \chi : \Gamma_B \rightarrow C_2$ agree with value one on all four presentation generators, hence agree everywhere. Because value one occurs, ϵ is onto. Surjectivity of π follows separately from the generation constraint below. $\square$

If covariance is required simultaneously, the exact consistency check

$$\epsilon\beta = \epsilon \quad (38.13)$$

may be tested on all carrier elements. When π is onto it also follows from (38.8) and $\chi\phi_8 = \chi$, but recording the direct table check catches serialization errors.

36.5 Forbidden-word constraints

Let $D \subset \Gamma_B \setminus \{1\}$ be a finite, typed, closed dangerous-set record. For every stored word d , evaluate it in the marked tuple and impose

$$d(x_{a_1}, x_{b_1}, x_{a_2}, x_{b_2}) \neq e \quad (d \in D). \quad (38.14)$$

Lemma 36.5.1 (Exact separation semantics). *The finite conjunction (38.14) holds if and only if*

$$D \cap \ker \pi = \emptyset. \quad (38.15)$$

Proof. The evaluated carrier element in (38.14) is exactly $\pi(d)$. Thus each inequality is equivalent to $d \notin \ker \pi$, and the conjunction over D is precisely (38.15). $\square$

A certificate stores every final nonidentity carrier index and the full evaluation trace. If D is not accompanied by its independent closure proof, (38.15) certifies only the supplied list, not a ball or global systole target.

36.6 Generation and the actual quotient

Let X_0 contain e , the four marked elements, and their inverses, and iterate

$$X_{k+1} := X_k \cup \{xy : x, y \in X_k\}. \quad (38.16)$$

Since Q is finite, this increasing sequence stabilizes after at most N strict cardinality increases.

Theorem 36.6.1 (Generation criterion). *The map π is onto Q exactly when the stable set in (38.16) equals Q :*

$$X_\infty = Q \iff \langle x_{a_1}, x_{b_1}, x_{a_2}, x_{b_2} \rangle = Q. \quad (38.17)$$

Proof. Every X_k lies in the generated subgroup. At stabilization the set contains the identity and inverses of the generators and is closed under multiplication; finiteness then makes it a subgroup containing the marked tuple. Minimality of the generated subgroup gives equality. $\square$

If (38.17) fails, either reject the ambient candidate or replace it by the stable generated subgroup $H = X_\infty$. In the latter case H is a new candidate: after proving $\beta(H) = H$ and restricting β, ϵ , one must independently verify the group table, relation, automorphism, covariance, $\beta^8 = \text{id}_H$, all three exact-order witnesses in (38.9), parity, forbidden-word, generation, and order conditions on H . In particular, exact order eight on Q does not imply exact order eight for $\beta|_H$. Only a new candidate passing every requested condition is accepted. This replacement preserves the source kernel and prevents ambient group order from being reported as quotient order.

36.7 Order and representation certificates

For table models, an order bound is the finite disjunction

$$1 \leq N = |Q| \leq N_{\max}. \quad (38.18)$$

The exact order is the verified carrier cardinality after (38.17). For a permutation model $Q = \langle \sigma_1, \dots, \sigma_t \rangle \leq S_d$, the certificate supplies a base and strong generating set. If the successive base-point orbit sizes are $o_1, \dots, o_k$, then

$$|Q| = \prod_{\nu=1}^k o_\nu. \quad (38.19)$$

Lemma 36.7.1 (Stabilizer-chain order). *If each stored transversal is complete and each next group is exactly the point stabilizer of the preceding one, (38.19) is the exact order of the generated permutation group.*

Proof. At each step orbit–stabilizer gives $|G_{\nu-1}| = o_\nu |G_\nu|$. Multiplying the identities and ending at the certified trivial stabilizer yields (38.19). $\square$

The independent verification must also sift every original generator to the identity and reconstruct every strong generator from the originals; otherwise the chain may describe a different subgroup.

36.8 Lexicographic optimization

For every fully feasible, generated candidate define

$$\mathcal{C}(Q, \pi) := (|\pi(\Gamma_B)|, d_{\min}(Q), C_{\text{matvec}}(Q), C_{\text{cert}}(Q)), \quad (38.20)$$

ordered lexicographically. Here $d_{\min}(Q)$ is the least degree of any faithful permutation action, while the last two components use the fixed implementation and certificate cost models.

Theorem 36.8.1 (Meaning of a minimum certificate). *A claimed optimum $\mathcal{C}_*$ is certified only by both*

$$\text{a feasible candidate attaining } \mathcal{C}_*, \quad \text{independently verifiable exclusion of every admissible } \mathcal{C} <_{\text{lex}} \mathcal{C}_*. \quad (38.21)$$

Proof. The first object proves an upper bound in the lexicographic order. The exhaustive exclusions prove the matching lower statement that no smaller admissible tuple exists. Together they are exactly the definition of a minimum. Either component alone is insufficient. $\square$

The degree component requires a faithful action of the asserted degree and complete nonexistence certificates for smaller degrees; an available embedding is only an upper bound on $d_{\min}$. Order exclusions may be proof-producing SAT/SMT objects, complete canonical backtracking trees, or another independently verifiable finite exhaustion. Cost comparison begins only among candidates tied on all earlier components.

Final constraint record. The mathematical artifact is

$$(Q, m, e, x_{a_1}, x_{b_1}, x_{a_2}, x_{b_2}, \beta, \epsilon, D, \mathcal{C}, \text{proof objects}), \quad (38.22)$$

with separate classifications

$$\text{INVALID, FEASIBLE, OPTIMAL_ON_DECLARED_DOMAIN.} \quad (38.23)$$

The last classification names its order range, group-family universe, dangerous set, action-degree range, and fixed cost model. Runtime, memory, solver version, and hashes belong to an operational record:

$$\text{integrity evidence} \neq \text{relation, separation, or optimality evidence.} \quad (38.24)$$

37 Counterexample-guided quotient search

37.1 Initial finite constraints

Fix a finite target domain $D_{\text{target}} \subset \Gamma_{\text{B}} \setminus \{1\}$ with its typed closure certificate, and choose

$$D_0 \subseteq D_{\text{target}}. \quad (39.1)$$

The initial formula $\mathcal{F}_0$ contains the exact group-table, surface-relator, generation, C_8 -covariance, and parity constraints of C38, together with $\pi(d) \neq 1$ for $d \in D_0$. Its record freezes the admissible orders, group families, exact-order convention for the descended rotation, and canonicalization rule. Changing any of these data starts a different search.

37.2 Candidate synthesis

A solver model is decoded and independently verified to produce

$$(Q_k, \pi_k, \beta_k, \epsilon_k), \quad \pi_k : \Gamma_{\text{B}} \twoheadrightarrow Q_k. \quad (39.2)$$

If the decoded ambient group is not generated by the marked tuple, it is rejected or replaced by the generated image and every C38 condition is rerun there. Canonical labeled tables identify isomorphic marked models. Solver satisfaction without exact independent verification has classification **INDETERMINATE**, not **CANDIDATE**.

37.3 Kernel-witness extraction

Order D_{target} by the fixed geometric key

$$\kappa(d) = (\underline{\ell}(d), \text{tie}(d)), \quad (39.3)$$

where $\underline{\ell}$ is a certified lower-bound key and tie is a total canonical tie-breaker. Scan in this order, exactly evaluate $\pi_k(d)$, and return the first kernel hit

$$w_k \neq 1, \quad \pi_k(w_k) = 1, \quad \ell(w_k) \text{ certified}. \quad (39.4)$$

To call it the minimum geometric witness, the scan must also prove that every smaller key was visited and that the terminal geometric evaluation orders candidates correctly; lower-bound ordering alone does not certify the minimum exact length.

Lemma 37.3.1 (Verification dichotomy). *A complete scan returns either a independently verifiable witness (39.4) or the certificate*

$$D_{\text{target}} \cap \ker \pi_k = \emptyset. \quad (39.5)$$

Proof. Every target element is visited exactly once. An identity evaluation produces a witness; if none occurs, every element has nonidentity image, which is exactly (39.5). $\square$

37.4 Refinement by nontriviality

For a returned witness define

$$D_{k+1} := D_k \cup \{w_k\}, \quad \mathcal{F}_{k+1} := \mathcal{F}_k \wedge [\pi(w_k) \neq 1]. \quad (39.6)$$

Deduplication uses the validated source-element key. Because the current candidate satisfies $\pi_k(w_k) = 1$, it is not a model of $\mathcal{F}_{k+1}$. Every quotient valid on the full target remains a model, since it separates $w_k \in D_{\text{target}}$. Thus refinement is both excluding and sound.

37.5 The accumulated invariant

Theorem 37.5.1 (CEGIS invariant). *Every accepted model at iteration k satisfies*

$$D_k \cap \ker \pi_k = \emptyset, \quad D_0 \subseteq D_1 \subseteq \dots \subseteq D_{\text{target}}. \quad (39.7)$$

Proof. The inclusions follow from (39.6). Each constraint $\pi(d) \neq 1$ is retained in all later formulas. Exact model independent verification therefore proves nonidentity for every $d \in D_k$, giving the kernel statement. $\square$

The invariant is relative to the current witness set. Until (39.5) is obtained, it makes no assertion about unsampled members of the target domain.

37.6 Finite termination

Let $\mathcal{K}$ be the finite set of admissible kernel-equivalence classes after the order bound and all structural constraints are fixed.

Theorem 37.6.1 (Finite-domain termination). *If D_{target} and $\mathcal{K}$ are finite, synthesis is complete and fair, verification is complete, and each witness clause is retained, then the loop terminates after at most*

$$|\mathcal{K}| \quad \text{failed candidate classes} \quad (39.8)$$

with either a target-valid quotient or exhaustion.

Proof. A witness $w_k \in \ker \pi_k$ excludes every marked candidate with the same source kernel. A later candidate therefore belongs to a new class. There are only $|\mathcal{K}|$ classes. A complete scan either accepts one by (39.5), or all classes are eventually excluded and complete synthesis reports exhaustion. $\square$

The looser product bound $|\mathcal{K}||D_{\text{target}}|$ is valid but unnecessary. With an unbounded order or family universe, finiteness fails and termination needs an additional well-founded bound.

37.7 Replayable unsatisfiability

An exhaustion result at order bound N has the form

$$\mathcal{F}(D_k, \text{structure}, |Q| \leq N) \text{ is unsatisfiable.} \quad (39.9)$$

Acceptable evidence includes a checked DRAT/LRAT proof for the exact CNF, a proof-producing SMT object, a Gröbner basis reducing to 1 for the exact encoded ideal, or a complete canonical backtracking tree. The certificate binds the precise encoder, variable map, formula hash, and checker. A solver log, timeout, or statement that no model was found proves neither unsatisfiability nor completeness.

Lemma 37.7.1 (Independent independent verification rule). *The unsatisfiability classification is accepted only if the proof checker reconstructs the exact fixed formula and returns success:*

$$\text{CheckUnsat}(\mathcal{F}_k, \mathcal{P}) = \text{true.} \quad (39.10)$$

Proof. Soundness belongs to the specified proof calculus and checker. Binding $\mathcal{P}$ to the reconstructed formula prevents a valid proof for a different encoding from being reused. $\square$

37.8 Minimum-quotient certificates

Suppose a fully verified quotient of actual order N separates the closed target domain. It proves only

$$N_{\text{opt}} \leq N. \quad (39.11)$$

An exact minimum certificate additionally contains independently verifiable unsatisfiability proofs

$$\mathcal{F}(D_{\text{target}}, \text{structure}, |Q| = n) \text{ unsatisfiable} \quad (1 \leq n < N) \quad (39.12)$$

for every admissible smaller order, or one sound proof covering their finite disjunction.

Theorem 37.8.1 (Minimum-order condition). *The feasible certificate at order N together with (39.12) proves*

$$N_{\text{opt}} = N \quad (39.13)$$

on exactly the fixed family, symmetry, parity, and dangerous-set domain.

Proof. Feasibility gives the upper bound (39.11). The exclusions show that no admissible quotient has smaller order, giving the matching lower bound. $\square$

Final iteration record. Each row stores

$$(k, D_k, \mathcal{F}_k, \text{classification}, \text{candidate key}, w_k, \text{evidence}_k), \quad (39.14)$$

where field presence is typed as follows:

classification	candidate	w_k	evidence
REFINE	present	present	kernel-evaluation trace
VALID	present	NONE	complete successful target scan
EXHAUSTED	NONE	NONE	proof accepted by CheckUnsat
INCOMPLETE	optional	optional	independently verifiable checkpoint only
INVALID	optional	NONE	typed failure witness

Thus $w_k = \text{NONE}$ occurs for both successful validation and certified exhaustion, with different required evidence. The terminal classification is one of

$$\text{VALID}, \quad \text{EXHAUSTED}, \quad \text{INCOMPLETE}, \quad \text{INVALID}. \quad (39.15)$$

Hashes and timings are maintained separately:

$$\text{authenticated bytes} \neq \text{verified mathematics}. \quad (39.16)$$

38 CEGAR architecture

38.1 Finite abstraction

At stage k , let $D_k \subseteq D_{\text{target}}$ and define

$$\mathcal{A}_k := \mathcal{S} \wedge \bigwedge_{d \in D_k} [\pi(d) \neq 1], \quad (40.1)$$

where $\mathcal{S}$ is the fixed presentation, generation, order, covariance, and parity theory. Because only target clauses are omitted,

$$\text{Models}(\mathcal{F}_{\text{full}}) \subseteq \text{Models}(\mathcal{A}_k). \quad (40.2)$$

Thus the abstraction over-approximates fully valid quotients; it never certifies omitted geometry by itself.

38.2 Candidate synthesis

A model of $\mathcal{A}_k$ is decoded, replaced by its actual generated image if necessary, and fully independently verified as

$$c_k = (Q_k, \pi_k, \beta_k, \epsilon_k, \text{exact carrier data}). \quad (40.3)$$

Canonicalization includes the ordered marked tuple and all descended maps. The synthesizer is deterministic under its stored seed and tie-break order; completeness claims additionally require a finite, fair enumeration or a proof-producing decision procedure.

38.3 Independent exact verification

The verifier reconstructs rather than trusts the solver model and checks

$$\text{Group} \wedge \text{Relator} \wedge \text{Generated} \wedge \text{Order} \wedge \text{Covariant} \wedge \text{Parity} \wedge \text{Separates}(D_k). \quad (40.4)$$

All operations use integer permutations, table indices, or exact finite-field arithmetic. A candidate failing any conjunct is **INVALID**; a failure witness identifies the first canonical equation and both unequal exact values.

Lemma 38.3.1 (Verifier soundness). *If every conjunct in (40.4) replays successfully, c_k is a genuine finite quotient satisfying $\mathcal{A}_k$.*

Proof. The group and relator conditions construct π_k ; generation identifies its actual target. The remaining conditions are precisely the equations and nonidentity constraints in $\mathcal{A}_k$. Hence

$$\text{Verified}(c_k, \mathcal{A}_k) = \text{true}. \quad (40.5)$$

□

38.4 Deterministic counterexamples

If the complete target check fails, choose

$$w_k = \arg \min_{w \in D_{\text{target}} \cap \ker \pi_k} (\kappa_{\text{geom}}(w), \text{canonicalKey}(w)). \quad (40.6)$$

The first component must be an exact certified geometric key, or a lower-bound key accompanied by proof that its ordering agrees with the final exact comparison. The scan log proves closure below the returned key. This rule makes independent reruns select the same witness.

38.5 Sound refinement

Set

$$\mathcal{A}_{k+1} := \mathcal{A}_k \wedge [\pi(w_k) \neq 1]. \quad (40.7)$$

Theorem 38.5.1 (Refinement invariant). *The current candidate is excluded, while every model of the full target remains a model of $\mathcal{A}_{k+1}$:*

$$c_k \not\models \mathcal{A}_{k+1}, \quad \text{Models}(\mathcal{F}_{\text{full}}) \subseteq \text{Models}(\mathcal{A}_{k+1}). \quad (40.8)$$

Proof. The witness satisfies $\pi_k(w_k) = 1$, proving exclusion. Since $w_k \in D_{\text{target}}$, every fully valid quotient maps it to a nonidentity element, proving preservation. □

38.6 Symmetry-aware refinement

Assume D_{target} is ϕ_8 -stable and the abstraction enforces $\beta\pi = \pi\phi_8$. Replace one clause by the orbit family

$$\pi(\phi_8^j(w_k)) \neq 1 \quad (0 \leq j < 8), \quad (40.9)$$

after source-key deduplication.

Lemma 38.6.1 (Orbit-clause soundness). *Under covariance,*

$$\pi(\phi_8^j(w)) = \beta^j(\pi(w)). \quad (40.10)$$

Hence a quotient separates one orbit member if and only if it separates all of them.

Proof. Iterate the covariance equation. Since β is an automorphism, it maps the identity only to the identity and nonidentity elements to nonidentity elements. $\square$

Without target stability, adding an orbit element outside the target changes the specification; without verified covariance, (40.10) is unavailable.

38.7 Family-aware refinement

For an exact parameter space P , suppose witness evaluation has a proved symbolic form

$$\pi_p(w_k) = 1 \iff F_{w_k}(p) = 0 \quad (p \in P), \quad (40.11)$$

where F_{w_k} is a polynomial, ideal-membership, or congruence predicate over the fixed parameter ring. Then remove the entire failure locus

$$Z(F_{w_k}) = \{p \in P : F_{w_k}(p) = 0\}. \quad (40.12)$$

Soundness requires independent verification of both directions in (40.11), including denominator, field, and exceptional-parameter conditions. Sampled agreement supplies no certificate for unsampled parameters.

38.8 Global convergence

Theorem 38.8.1 (Finite CEGAR convergence). *If the canonical candidate set $\mathcal{C}$ is finite, synthesis is complete and fair, full verification decides validity, and every failed candidate is permanently removed, then after at most*

$$|\mathcal{C}| \text{ failed candidates} \quad (40.13)$$

CEGAR returns a fully valid quotient or a independently verifiable exhaustion proof.

Proof. No failed canonical candidate can recur. Each failure strictly reduces the finite remaining set, so either verification accepts a member or the set becomes empty. Completeness then certifies exhaustion. $\square$

For unbounded orders or parameter degrees, (40.13) does not apply. A valid extension needs a well-founded cost bound whose sublevel sets are finite and a declared terminal sublevel:

$$\mathcal{C}_{\leq B} := \{c : \text{cost}(c) \leq B\} \text{ finite.} \quad (40.14)$$

The iteration record records

$$(k, \mathcal{A}_k, \text{classification}, c_k, w_k, \text{evidence}_k), \quad (40.15)$$

as a tagged union with classifications

$$\text{REFINE, INVALID, VALID, EXHAUSTED_FINITE_DOMAIN, INCOMPLETE.} \quad (40.16)$$

Writing P, N, O for present, absent, and optional, respectively, field presence is

classification	c_k	w_k	required evidence
REFINE	P	P	kernel trace and sound clause
INVALID	O	N	typed verification failure
VALID	P	N	complete successful target scan
EXHAUSTED_FINITE_DOMAIN	N	N	coverage and checked exhaustion proof
INCOMPLETE	O	O	independently verifiable checkpoint

In particular VALID carries no refinement witness, exhaustion names and proves its finite domain, and INCOMPLETE makes no negative existence claim.

39 Exact solver formulations

39.1 SAT encoding

For carrier $[N]$, introduce one-hot variables m_{ij}^k meaning $i \cdot j = k$. Exactly-one clauses define a total table, while

$$(i \cdot j) \cdot r = i \cdot (j \cdot r), \quad e \cdot i = i \cdot e = i \quad (41.1)$$

are expanded into finite Boolean clauses. Inverse witnesses have both products equal to e . One-hot registers for the four marked generators feed word-evaluation circuits; a forbidden word is constrained away from the identity register. Bijection circuits alone do not encode an automorphism. The SAT instance also expands, for every carrier pair and every marked generator, the exact equations

$$\begin{aligned} \beta(e) = e, \quad \beta(xy) = \beta(x)\beta(y), \quad \beta^8 = \text{id}, \\ \beta(x_s) = \pi(\phi_8(s)), \quad \epsilon(xy) = \epsilon(x) + \epsilon(y), \quad \epsilon(x_s) = 1. \end{aligned}$$

If exact rotation order eight is requested, witness circuits impose $\beta^d \neq \text{id}$ for $d = 1, 2, 4$; compatibility may also impose $\epsilon\beta = \epsilon$. XOR or truth-table clauses encode ϵ , and every auxiliary condition has both implication directions. The DIMACS record maps each literal back to its mathematical predicate.

39.2 SMT encoding

Use a finite sort Q_N , multiplication m , identity e , marked constants, and exact arrays β, ϵ . Assert

$$m(m(x, y), z) = m(x, m(y, z)), \quad m(e, x) = m(x, e) = x, \quad \exists u [m(x, u) = m(u, x) = e]. \quad (41.2)$$

The SMT formula additionally asserts the finite expansions of the automorphism, bijectivity, covariance, β^8 , requested exact-order witnesses, parity-homomorphism, prescribed generator parities, compatibility, forbidden-word, and generation equations from C38. Omitting any requested family makes the formula only an over-approximation and forbids a completeness claim for the intended problem. Every quantifier is expanded over Q_N , or the solver must emit a proof object whose quantified reasoning is independently checked. Finite-sort distinctness and coverage prevent a smaller unintended carrier. A model is re-exported as a table and independently verified by C38; an SMT `sat` line is not itself a quotient certificate.

39.3 Finite-field matrix CSP

For a prescribed matrix group over $\mathbb{F}_q$, use entries of $X_{a_1}, X_{b_1}, X_{a_2}, X_{b_2}$ and inverse matrices as variables. Impose

$$X_s Y_s = Y_s X_s = I, \quad [X_{a_1}, X_{b_1}][X_{a_2}, X_{b_2}] = I. \quad (41.3)$$

Membership equations for determinant, symplectic, unitary, or projective constraints are family-specific and fixed. For each forbidden word w , impose that at least one entry of $X_w - I$ is nonzero; in a projective target, replace this by the exact condition that X_w is not an allowed scalar. The generated image, not the ambient classical group, is the quotient whose order is certified.

39.4 Gröbner-basis formulation

Translate all equality constraints into an ideal

$$I \subset \mathbb{F}_q[z_1, \dots, z_t]. \quad (41.4)$$

An inequality $f \neq 0$ is encoded by a new variable u and $uf - 1 = 0$, equivalently saturation by f . A disjunction $(f_1, \dots, f_r) \neq 0$ is encoded by

$$u_1 f_1 + \dots + u_r f_r - 1 = 0. \quad (41.5)$$

A checked Gröbner basis containing 1 proves infeasibility of the exact augmented ideal. Monomial order, coefficient field, generators, and reduction traces are part of the proof object; a basis computed for an unsaturated equality ideal does not prove a forbidden-word inequality.

39.5 Extension-field equations

Require and certify that p is prime, $f \geq 1$, and h is monic, irreducible, and of degree exactly f :

$$h(T) \in \mathbb{F}_p[T], \quad \deg h = f, \quad \mathbb{F}_q = \mathbb{F}_p[T]/(h). \quad (41.6)$$

Primality and irreducibility use independently verifiable integer and polynomial certificates; merely labeling p or h is insufficient. Every element uses its f -coordinate basis and multiplication is reduced modulo h . In an algebraic-closure encoding, add $x^q - x = 0$; in the fixed coordinate model this relation is verified rather than used to define an ambiguous field. Frobenius is the exact linear map $x \mapsto x^p$. A semilinear element (A, σ) uses

$$(A, \sigma)(B, \tau) = (A \sigma(B), \sigma \tau), \quad (41.7)$$

so ordinary matrix multiplication cannot replace the semilinear law.

39.6 Exact backtracking

Assign marked generator images in canonical order, propagate inverse and relator consequences, and maintain the generated subgroup and every stored witness evaluation. A branch may be pruned only by a proved monotone obstruction:

$$\text{relation contradiction,} \quad \text{forbidden word forced to 1,} \quad \text{generation upper closure too small.} \quad (41.8)$$

The final tree has a leaf for every canonical assignment or a recorded symmetry representative. Each pruned node stores the exact obstruction; heuristic ordering changes speed but not coverage.

39.7 Branch and bound

Let U be the incumbent feasible cost and let $L(\nu)$ be a lower bound for every feasible completion of node ν . The sound pruning rule is

$$L(\nu) \geq U. \quad (41.9)$$

Theorem 39.7.1 (Certified optimal search). *If every node is expanded into an exhaustive disjoint partition, certified infeasible, or pruned by (41.9), then the incumbent is optimal on the declared finite tree.*

Proof. Every feasible leaf belongs to one root-to-leaf branch. Infeasible branches contain none; a pruned branch has cost at least U ; expanded branches are covered by their children. Hence no feasible cost below U is omitted, which proves

$$U = \min\{\text{cost}(x) : x \text{ is feasible}\}. \quad (41.10)$$

□

39.8 Marked group-isomorphism reduction

Two table candidates (Q, x_s) , (Q', x'_s) are equivalent when an identity-preserving bijection $\theta : Q \rightarrow Q'$ satisfies

$$\theta(xy) = \theta(x)\theta(y), \quad \theta(x_s) = x'_s \quad \text{for all marked } s. \quad (41.11)$$

With descended data, also require

$$\theta\beta = \beta'\theta, \quad \epsilon'\theta = \epsilon. \quad (41.12)$$

Canonical labeled tables therefore select one representative without collapsing inequivalent marked homomorphisms.

39.9 Automorphism-centralizer reduction

After fixing β , an automorphism $\alpha \in C_{\text{Aut}(Q)}(\beta)$ sends a tuple (x_s) to $(\alpha(x_s))$. Then

$$\beta\alpha\pi = \alpha\beta\pi = \alpha\pi\phi_8, \quad \ker(\alpha\pi) = \ker \pi. \quad (41.13)$$

Thus the covariance equations and cover kernel are preserved. If ϵ is transported as part of the candidate, replace it by $\epsilon\alpha^{-1}$. If the search freezes the same parity array, reduce only by

$$C_{\text{Aut}(Q)}(\beta) \cap \text{Stab}_{\text{Aut}(Q)}(\epsilon). \quad (41.14)$$

Ignoring this distinction can merge differently marked parity records.

39.10 Kernel-equivalence

Theorem 39.10.1 (Kernel-equivalence criterion). *For surjections $\pi_i : \Gamma_B \twoheadrightarrow Q_i$,*

$$\ker \pi_1 = \ker \pi_2 \iff \exists! \theta : Q_1 \xrightarrow{\sim} Q_2 \text{ with } \theta\pi_1 = \pi_2. \quad (41.15)$$

Proof. For equal kernels K , both quotients identify canonically with Γ_B/K ; define $\theta(\pi_1(g)) = \pi_2(g)$. Kernel equality gives well-definedness and injectivity, while surjectivity of π_2 gives surjectivity. Conversely, an isomorphism is injective, so $\pi_2(g) = 1$ iff $\pi_1(g) = 1$. Uniqueness follows from surjectivity of π_1 . □

Kernel equivalence is the correct reduction for unmarked regular covers. When symmetry, parity, or a physical generator order is part of the object, the corresponding intertwining conditions (41.12) remain mandatory. The solver certificate closes with

$$(\text{encoding, canonical domain, proof object, independent independent verification}), \quad (41.16)$$

and never identifies

$$\text{solver termination} \quad \text{with} \quad \text{certified completeness}. \quad (41.17)$$

Part XV

Lower bounds and construction cost

40 Combinatorial order bounds

40.1 Cayley balls

For the fixed symmetric shell $S = S^{-1}$, $|S| = 8$, define

$$B_S(r) = \{\gamma \in \Gamma_B : |\gamma|_S \leq r\}. \quad (42.1)$$

The set is deduplicated in the source group. If π is injective on it, its distinct images lie in Q , hence

$$|Q| \geq |B_S(r)|. \quad (42.2)$$

Counting reduced words without source deduplication may overestimate $|B_S(r)|$ and is not a valid lower bound.

40.2 Exponential growth

Because the cocompact orbit map $\Gamma_B \rightarrow \mathbb{H}^2$ is a quasi-isometry and hyperbolic disk area grows exponentially, there are constants $c_-, c_+ > 0$, $\lambda_-, \lambda_+ > 1$, and r_0 such that

$$c_- \lambda_-^r \leq |B_S(r)| \leq c_+ \lambda_+^r \quad (r \geq r_0). \quad (42.3)$$

This is an asymptotic theorem. A numerical certificate must provide proved values of all four constants for this exact shell and range; fitting computed ball counts does not establish them.

40.3 The doubled-radius test

Theorem 40.3.1. *If*

$$\ker \pi \cap B_S(2r) = \{1\}, \quad (42.4)$$

then π is injective on $B_S(r)$ and $|Q| \geq |B_S(r)|$.

Proof. If $x, y \in B_S(r)$ and $\pi(x) = \pi(y)$, then $y^{-1}x \in \ker \pi$ and $|y^{-1}x|_S \leq |y|_S + |x|_S \leq 2r$. Equation (42.4) gives $x = y$; (42.2) finishes the proof. $\square$

40.4 Moore bounds

Assume the eight shell images are distinct, nonidentity, inverse-paired, and generate Q , so the undirected Cayley graph is simple eight-regular. If its girth is g , breadth-first expansion gives

$$|Q| \geq \begin{cases} 1 + 8 \sum_{j=0}^{(g-3)/2} 7^j, & g \text{ odd,} \\ 2 \sum_{j=0}^{g/2-1} 7^j, & g \text{ even.} \end{cases} \quad (42.5)$$

For odd girth expand from one vertex; for even girth expand from the two endpoints of an edge. Any collision before the stated depth joins two tree paths into a cycle shorter than g . The formula is not applied to a multigraph with loops or collapsed labels.

40.5 Bipartite correction

If $\bar{\chi} : Q \twoheadrightarrow C_2$ and every shell generator is odd, every edge changes parity. The Cayley graph is bipartite, so $g = 2k$ and

$$|Q| \geq 2(1 + 7 + \cdots + 7^{k-1}). \quad (42.6)$$

The two parts have equal size because either odd group element gives a bijection between them.

40.6 C_8 -orbit correction

Assume $\beta^8 = \text{id}$, so powers of β define a C_8 -action. For a β -stable counted set E , orbit-stabilizer gives

$$|\text{Orb}_\beta(x)| = \frac{8}{|\text{Stab}_{C_8}(x)|} \in \{1, 2, 4, 8\}. \quad (42.7)$$

An integer program may count complete orbits and their verified stabilizers. A blanket factor eight is sound only after proving every counted stabilizer trivial; fixed points and shorter orbits otherwise invalidate it.

40.7 Parity correction

Surjectivity of $\bar{\chi}$ yields

$$|Q| = 2|\ker \bar{\chi}|, \quad |\bar{\chi}^{-1}(0)| = |\bar{\chi}^{-1}(1)| = \frac{|Q|}{2}. \quad (42.8)$$

Consequently a certified lower bound M inside either one parity fiber implies $|Q| \geq 2M$. This doubling requires an injection or count confined to one fiber, not merely the fact that Q has even order.

40.8 Synthesis of lower bounds

Let $\mathcal{I}$ index only bounds whose hypotheses have been independently verified for the same construction domain. Then

$$N_{\min} \geq \begin{cases} \max_{i \in \mathcal{I}} M_i, & \mathcal{I} \neq \emptyset, \\ 1, & \mathcal{I} = \emptyset, \end{cases} \quad (42.9)$$

where the available M_i are selected from $|B_S(r)|$, M_{Moore} , M_{C_8} , M_{parity} , M_{geom} . The second branch is the fixed trivial lower bound for a nonempty finite quotient.

Proof. Every admissible quotient satisfies each applicable lower inequality, so it satisfies their maximum. Combining bounds by addition or multiplication would require a disjointness or independence theorem not supplied here. $\square$

The certificate records each term's exact counting domain, hypotheses, value, and independent verification object; an inapplicable term is marked NOT-APPLICABLE, never silently set to zero.

41 Geometric and arithmetic lower bounds

41.1 Area of a finite cover

Let $X = \mathbb{H}^2/\Gamma_B$ be the genus-two Bolza surface at curvature $-R^{-2}$, and let $K \triangleleft \Gamma_B$ have index N . The unbranched cover $X_N = \mathbb{H}^2/K \rightarrow X$ has degree N , hence

$$\text{area}(X_N) = N \text{area}(X) = 4\pi NR^2. \quad (43.1)$$

This uses multiplicativity of Riemannian area under a genuine covering; orbifold or branched quotients require correction terms.

41.2 Genus and order

Gauss–Bonnet gives

$$\text{area}(X_N) = 4\pi(g_N - 1)R^2. \quad (43.2)$$

Comparison with (43.1) yields the exact relation

$$g_N = N + 1. \quad (43.3)$$

Thus quotient order is the topological covering degree, not the order of a larger ambient group in which the actual image is embedded.

41.3 Conditional systolic upper inequalities

Assume a proved theorem applies to the present surface class and genus range with explicit $C > 0$:

$$\text{sys}(X_N) \leq 2R \log(Cg_N). \quad (43.4)$$

Then a target $\text{sys}(X_N) > L$ and (43.3) imply

$$N > C^{-1}e^{L/(2R)} - 1. \quad (43.5)$$

The strict sign follows from the strict target. Before numerical use, the certificate cites the theorem, normalization of curvature and systole, value of C , and applicable genera. A family-specific inequality is not silently promoted to all covers.

41.4 Arithmetic lower growth

For a certified congruence family one may have constants $c > 0, C_0$ and levels n such that

$$\text{sys}(X_n) \geq cR \log N_n - C_0R. \quad (43.6)$$

This proves efficient existence along that family: levels with $\log N_n > (L/R + C_0)/c$ pass the target. It is a lower bound on the systole of constructed members, not a lower bound on the order of every possible quotient, and sparse family orders need not realize each intermediate N .

41.5 Cayley girth versus geometry

Let $a, b > 0$ be validated comparison constants of Statement 12.8, schematically

$$a|\gamma|_S - b \leq d(o, \gamma o) \leq a^{-1}|\gamma|_S + b. \quad (43.7)$$

A word-ball kernel exclusion transfers to a based displacement bound through the appropriate direction of (43.7). Conversely, a short closed geodesic supplies a conjugacy class with short translation length; its representative may move the fixed basepoint far away. Therefore Cayley girth, based injectivity, and global systole are never identified without an explicit conversion theorem.

41.6 The global systole criterion

For the normal, torsion-free finite-index kernel $K = \ker \pi$,

$$\text{sys}(\mathbb{H}^2/K) = \min_{1 \neq k \in K} \ell(k), \quad (43.8)$$

where the minimum is attained by discreteness and cocompactness. With $D_{\text{global}}(L)$ the complete axis-normalized closed domain,

$$\text{sys}(\mathbb{H}^2/K) > L \iff K \cap D_{\text{global}}(L) = \emptyset. \quad (43.9)$$

The use of a closed cutoff is what certifies the strict conclusion.

41.7 Conditional asymptotic scale

Theorem 41.7.1 (Qualified exponential scale). *If (43.4) holds uniformly for all admissible covers, then any quotient with systole $> L$ has order $\exp(\Omega(L/R))$. If a certified residual family satisfies (43.6), let $N_{n(L)}$ be its first certified level with $\log N_{n(L)} > (L/R + C_0)/c$. Then it supplies a member of order*

$$N_{n(L)} = \exp((L/R + C_0)/c + \Delta(L)), \quad \Delta(L) := \log N_{n(L)} - (L/R + C_0)/c.$$

In particular the order is $\exp(O(L/R))$ only under the additional controlled-gap hypothesis $\Delta(L) = O(L/R)$, for example a uniformly bounded logarithmic level gap.

Proof. Equation (43.5) gives the necessary exponential lower scale. Rearranging (43.6) proves the selected level passes the target. The displayed identity isolates the possible overshoot; the final big- O conclusion follows exactly from the controlled-gap hypothesis. $\square$

41.8 A computable interval

For each fixed target L , report

$$N_{\text{lower}}(L) \leq N_{\text{opt}}(L) \leq N_{\text{achieved}}(L). \quad (43.10)$$

The left endpoint is the maximum of independently valid lower bounds; the right endpoint is the actual order of a fully verified quotient. If no quotient has been constructed the right endpoint is $+\infty$; if no positive lower certificate is available, use only the trivial admissibility constraints. Equality may be printed only after every intermediate admissible order is excluded by independently verifiable proofs.

42 Costs of enforcing structure

42.1 C_8 -equivariance cost

For $\pi : \Gamma_B \rightarrow Q$ with kernel K , the orbit core is

$$K_{C_8} = \bigcap_{j=0}^7 \phi_8^j(K), \quad Q_{C_8} = \Gamma_B / K_{C_8}. \quad (44.1)$$

It is ϕ_8 -stable and maps onto Q . The diagonal embedding into eight transported copies gives

$$|Q| \leq |Q_{C_8}| \leq |Q|^8. \quad (44.2)$$

For a minimal verified stable refinement within the declared kernel target, define

$$\Delta_{C_8} := \log |Q_{C_8}| - \log |Q| \in [0, 7 \log |Q|]. \quad (44.3)$$

This is a construction cost, not proof that the orbit core is globally minimal among all unrelated equivariant quotients.

42.2 Parity cost

The generated image of $\pi \times \chi$ has kernel $K \cap \ker \chi$, so

$$|Q| \leq |\text{im}(\pi, \chi)| \leq 2|Q|. \quad (44.4)$$

The factor is one exactly when χ already factors through Q , and otherwise two. Fiber products over an already shared parity map avoid duplicating the C_2 coordinate; ambient Cartesian order must not be charged when the actual subdirect image is smaller.

42.3 Normal-core cost

For $H \leq G$ of index n , the action on G/H has kernel $\text{core}_G(H)$. Hence

$$G/\text{core}_G(H) \hookrightarrow S_n, \quad [G : \text{core}_G(H)] \leq n!. \quad (44.5)$$

This exact worst-case bound follows from the faithful coset action. The actual image order is computed by a stabilizer chain and may be far below $n!$.

42.4 Representation degree

The permutation carrier cost is

$$\mu(Q) := \min\{d : Q \hookrightarrow S_d\}. \quad (44.6)$$

A displayed faithful action proves only an upper bound. Equality requires exhaustive exclusion of all smaller, possibly intransitive, faithful actions. For harmonic algorithms the relevant costs may instead be the verified irreducible block dimensions and multiplicities; neither is determined by $|Q|$ alone.

42.5 Joint certificate and operator cost

After fixing nonnegative weights and unit normalizations in advance, one possible objective is

$$C(Q) = w_1|Q| + w_2\mu(Q) + w_3|D_{\text{cert}}| + w_4 \text{nnz}(H_Q). \quad (44.7)$$

The definition records whether nnz counts directed entries, both layers, and interlayer orientations. Choosing weights after inspecting candidates changes the optimization problem and invalidates an earlier optimum certificate.

42.6 The computational window

Freeze memory, wall-time, and arithmetic budgets

$$(M_{\max}, T_{\max}, F_{\max}). \quad (44.8)$$

A candidate is computational admissible only after certified upper bounds

$$M(Q) \leq M_{\max}, \quad T(Q) \leq T_{\max}, \quad F(Q) \leq F_{\max} \quad (44.9)$$

include quotient storage, certificate independent verification, operator support, eigensolver workspaces, reorthogonalization, and the full parameter grid. Mathematical existence alone supplies none of these inequalities.

42.7 Optimal quotient scale

For a fixed support radius, observable set, error tolerance, and resource model, let $\mathcal{F}_{\text{prod}}$ be the set of quotients passing covariance, parity, global systole, and computational conditions. Define

$$C_* = \begin{cases} \min\{C(Q) : Q \in \mathcal{F}_{\text{prod}}\}, & \mathcal{F}_{\text{prod}} \neq \emptyset, \\ +\infty, & \mathcal{F}_{\text{prod}} = \emptyset. \end{cases} \quad (44.10)$$

The empty branch has classification **INFEASIBLE-ON-DECLARED-DOMAIN**. The feasible certificate gives an upper bound; exact optimality also requires lower-cost exhaustion. Changing the Hamiltonian support or observable can change both feasibility and the minimizing scale, so there is no scale-free universal optimum.

42.8 Construction accounting

Use disjoint ledger categories

$$C_{\text{total}} = C_{\text{search}} + C_{\text{exact verification}} + C_{\text{representation}} + C_{\text{physics}}. \quad (44.11)$$

Search includes failed candidates; verification includes all proof replays; representation includes classes, characters, and blocks; physics includes support assembly and numerical solves. Shared work is assigned once by a fixed rule to avoid double counting.

Theorem 42.8.1 (Existence versus computational). *A construction theorem proves a quotient exists with stated algebraic and geometric properties. A computational theorem additionally requires all inequalities (44.9) and a complete ledger (44.11).*

Proof. The first statement follows from the mathematical certificates. The second is exactly the conjunction of those certificates with the independent resource conditions; neither conjunct implies the other. $\square$

Part XVI

Certificate specifications

43 Finite-group certificate

43.1 Relation certificate

Store exact images A_1, B_1, A_2, B_2 and independent verification

$$[A_1, B_1][A_2, B_2] = 1_Q. \quad (45.1)$$

The certificate retains the four inverses, two commutators, and final product as canonical carrier indices. This proves the marked free-group map factors through Γ_B , but not generation.

43.2 Automorphism order

Store β as a carrier permutation and check

$$\beta(xy) = \beta(x)\beta(y), \quad \beta^8 = \text{id}, \quad \beta^d \neq \text{id} \ (d = 1, 2, 4). \quad (45.2)$$

Every inequality has an element moved by β^d . The last three tests are required only for a claim of exact order eight.

43.3 C_8 covariance

For each $s \in \{a_1, b_1, a_2, b_2\}$, verify

$$\beta(\pi(s)) = \pi(\phi_8(s)). \quad (45.3)$$

Both sides are homomorphisms in s ; agreement on the presentation generators proves $\beta\pi = \pi\phi_8$ on every source word.

43.4 Parity certificate

Provide $\bar{\chi} : Q \rightarrow C_2$, verify its homomorphism law on the complete table, and check

$$\bar{\chi}(\pi(a_i)) = \bar{\chi}(\pi(b_i)) = 1 \quad (i = 1, 2). \quad (45.4)$$

Then $\bar{\chi}\pi = \chi$. A generating-set shortcut is valid only when the proposed character is extended consistently and the extension is independently proved.

43.5 Generation certificate

A closure log or Schreier tree starts at 1_Q , follows the four images and inverses, and reaches exactly

$$|\langle A_1, B_1, A_2, B_2 \rangle| = |Q|. \quad (45.5)$$

Every tree edge records its parent, generator label, and recomputed endpoint. Coverage of every carrier element proves surjectivity.

43.6 Kernel certificate

The kernel is specified intensionally by the certified surjection:

$$K = \ker \pi, \quad \Gamma_B/K \cong Q. \quad (45.6)$$

It is not enumerated. Equality of two proposed kernels is proved by a quotient isomorphism intertwining the ordered source-generator images; testing finitely many sampled words is insufficient.

43.7 Order certificate

For a permutation group, store a base, strong generators, complete orbit transversals, and exact stabilizers. Repeated orbit–stabilizer gives

$$|Q| = \prod_j |\text{Orb}_{G_{j-1}}(b_j)|. \quad (45.7)$$

Sifting proves the original generators and certified chain generate the same group. For a table group, order is the validated carrier cardinality after the generation condition.

43.8 Isomorphism-class certificate

Accept one of: a canonical multiplication table; a library identifier plus explicit constructive isomorphism; or structural data plus maps θ, θ^{-1} satisfying

$$\theta(xy) = \theta(x)\theta(y), \quad \theta^{-1}\theta = \text{id}, \quad \theta\theta^{-1} = \text{id}. \quad (45.8)$$

Matching order, element-order statistics, or character degrees alone does not identify a group. If marked symmetry and parity are claimed, θ must also intertwine the generator tuple, β , and $\bar{\chi}$.

Certificate bundle. The finite-group condition accepts only the tuple

$$(\text{carrier, marked map, } \beta, \bar{\chi}, \text{generation, order, independent verification}), \quad (45.9)$$

with every requested component verified and version-bound.

44 Geometric certificate

44.1 Radius-three word-ball injection

For the corrected geometric shell S ,

$$\pi|_{B_S(3)} \text{ injective} \iff \ker \pi \cap B_S(6) = \{1\}. \quad (46.1)$$

The forward implication splits any word of length at most six as $y^{-1}x$ with $x, y \in B_S(3)$; a nontrivial kernel element would give equal images. The reverse implication is Statement 40.3. Enumeration uses the physical shell and source-group deduplication, not the presentation tuple regarded as an unordered generating set.

44.2 Based-dangerous scan

At the fixed basepoint o , enumerate

$$D_{\text{based}}(L) = \{\gamma \in \Gamma_B \setminus \{1\} : d(o, \gamma o) \leq L\}. \quad (46.2)$$

Proper discontinuity makes this set finite. A certificate contains exact element keys, certified distance comparisons, and an empty frontier proving completeness at the closed cutoff, followed by $\pi(\gamma) \neq 1$ for every listed element. This establishes based injectivity only.

44.3 Exact short-word scan

Enumerate freely reduced words through a proved cutoff, evaluate their matrices in the fixed exact number field and basis of Statement 5.2, and canonicalize projective signs. The analysis map

$$\text{word bytes} \mapsto \text{exact matrix} \mapsto \text{source element key} \quad (46.3)$$

is retained before deduplication. Floating matrices may guide search but do not decide equality or cutoff membership.

44.4 Axis-dangerous scan

For a hyperbolic element and a normalized determinant-one lift,

$$\cosh\left(\frac{\ell(g)}{2R}\right) = \frac{|\text{tr } g|}{2}. \quad (46.4)$$

Enumerate a certified finite outer set $E_{\text{axis}}(L)$ of canonical conjugacy representatives whose axes meet the chosen compact fundamental region. Its coarse closed frontier inequality may include elements longer than L , but it must be proved that every conjugacy class with $\ell \leq L$ has a representative in $E_{\text{axis}}(L)$. Completeness requires a geometric lower bound for every unexpanded frontier state.

44.5 Global-kernel scan

Let $E_{\text{axis}}(L)$ be the closed outer output of Statement 44.4. Evaluate the quotient exactly and report separately

$$\begin{aligned} N_{\text{all}} &= |E_{\text{axis}}(L)|, \\ N_{\text{ker}} &= |\{\gamma \in E_{\text{axis}}(L) : \pi(\gamma) = 1\}|, \\ N_{\leq L} &= |\{\gamma \in E_{\text{axis}}(L) : \pi(\gamma) = 1, \ell(\gamma) \leq L\}|. \end{aligned} \quad (46.5)$$

An empty frontier, source-key deduplication, exact image traces, and exact length comparisons are mandatory. N_{ker} may include enumerated elements beyond the final length condition and is not substituted for $N_{\leq L}$.

44.6 Exact systole lower bound

Theorem 44.6.1. *If the closed global scan is complete and $N_{\leq L} = 0$, then*

$$\text{sys}(\mathbb{H}^2 / \ker \pi) > L. \quad (46.6)$$

Proof. Every nontrivial kernel conjugacy class of translation length at most L has a representative in the complete axis domain. Such a representative would be counted by $N_{\leq L}$, contrary to zero. The cocompact kernel quotient has an attained positive minimum, and the closed cutoff excludes equality, proving the strict sign. $\square$

44.7 No-wraparound radius

For a closed hyperbolic surface,

$$r_{\text{inj}} = \frac{1}{2} \text{sys}. \quad (46.7)$$

Thus $\text{sys} > 6a_B$ implies

$$r_{\text{inj}} > 3a_B. \quad (46.8)$$

Every support ball of radius at most $3a_B$ is then strictly below the injectivity radius and embeds without deck identifications. If only a non-strict systole bound were known, equality at the boundary would require a separate check.

44.8 Geometry hierarchy

The final record has independent fields

$$(\text{word radius, based displacement, translation length, sys, } r_{\text{inj}}, \text{ operator support}). \quad (46.9)$$

Every implication cites its conversion theorem and strictness. Statuses are

$$\text{COMPLETE, INCOMPLETE, INVALID, NOT_APPLICABLE.} \quad (46.10)$$

No hash, based scan, or short-word scan is allowed to populate the global-systole field without the corresponding global proof.

45 Representation certificate

45.1 Conjugacy classes

Partition the exact carrier by $x \sim q x q^{-1}$. For each canonical representative x , enumerate its centralizer and verify

$$|\text{Cl}(x)| |C_Q(x)| = |Q|, \quad \sum_{[x]} |\text{Cl}(x)| = |Q|. \quad (47.1)$$

Every carrier element occurs in exactly one stored class; order statistics without this coverage proof are insufficient.

45.2 The character table

Store character values in a specified exact cyclotomic field. Row orthogonality is

$$\frac{1}{|Q|} \sum_{[x]} |\text{Cl}(x)| \chi_\lambda(x) \overline{\chi_\mu(x)} = \delta_{\lambda\mu}. \quad (47.2)$$

Column orthogonality, positive integral degrees $d_\lambda = \chi_\lambda(1)$, algebraic-integrality checks, and the unique trivial row are independently verified exactly. Decimal approximations are display aids only.

45.3 Realizing irreducibles

For every row construct generator matrices $\rho_\lambda(\pi(a_i)), \rho_\lambda(\pi(b_i))$. Extend them by certified straight-line words or store all carrier matrices and check

$$\rho_\lambda(xy) = \rho_\lambda(x)\rho_\lambda(y), \quad \text{tr } \rho_\lambda(x) = \chi_\lambda(x). \quad (47.3)$$

The matrices also independent verification the marked surface relation. Character rows alone do not provide operator blocks.

45.4 Dimension-sum completeness

Theorem 45.4.1. *Pairwise orthogonal irreducible rows form the complete dual precisely when*

$$\sum_{\lambda} d_\lambda^2 = |Q|. \quad (47.4)$$

Proof. The regular module contains an irreducible of degree d_λ with multiplicity d_λ . Orthogonality gives inequivalence, so the displayed sum is the dimension of the reconstructed regular submodule. Equality exhausts $\mathbb{C}[Q]$; a strict deficit leaves missing irreducibles. $\square$

45.5 Regular decomposition

With the complete list,

$$\mathbb{C}[Q] \cong \bigoplus_{\lambda \in \widehat{Q}} V_\lambda \otimes \mathbb{C}^{d_\lambda}. \quad (47.5)$$

The first factor carries ρ_λ , the second its multiplicity. The reconstructed regular character is checked against

$$\chi_{\text{reg}}(q) = \begin{cases} |Q|, & q = 1, \\ 0, & q \neq 1. \end{cases} \quad (47.6)$$

45.6 C_8 action on the dual

The descended automorphism acts by

$$\rho_\lambda \longmapsto \rho_\lambda \circ \beta^{-1}, \quad \chi_\lambda(x) \longmapsto \chi_\lambda(\beta^{-1}x). \quad (47.7)$$

Exact character comparison partitions $\widehat{Q}$ into orbits and computes stabilizers. For a fixed isomorphism class, an explicit intertwiner is separately required; extending it to a linear $Q \rtimes C_8$ representation may carry a cocycle obstruction and does not follow from character equality alone.

45.7 Parity pairing and vertex sectors

Tensoring by the one-dimensional character $(-1)^{\bar{\chi}}$ defines an involution

$$\lambda \longmapsto \lambda \otimes \bar{\chi} \quad (47.8)$$

of $\hat{Q}$. On $\ell^2(Q)$, define

$$J\delta_q = (-1)^{\bar{\chi}(q)}\delta_q, \quad P_{\pm} = \frac{1}{2}(I \pm J). \quad (47.9)$$

Then $P_{\pm}$ project onto even and odd vertex-support subspaces, each of dimension $|Q|/2$. Odd hopping interchanges these spaces; they are not asserted to be invariant irreducible Q -modules.

45.8 Completeness analysis

The representation condition checks, in order,

$$\begin{aligned} &\text{class coverage and centralizers; row and column orthogonality; } \sum d_{\lambda}^2 = |Q|; \\ &\text{realizing matrices; } C_8 \text{ orbit coverage; parity pairing; regular trace reconstruction.} \end{aligned} \quad (47.10)$$

Its classification is

$$\text{COMPLETE or INCOMPLETE(fields) or INVALID(witness)}. \quad (47.11)$$

The fixed artifact records

$$(Q, \text{classes, characters, matrices, } \beta, \bar{\chi}, \text{independent verification}). \quad (47.12)$$

No representation-resolved spectral claim passes while any listed field is incomplete.

46 Final construction claims

46.1 Construction domain

Every theorem begins with a domain tag chosen from

$$\text{ALL-FINITE, FAMILY, BOUNDED-CENSUS, FIXED-QUOTIENT}. \quad (48.1)$$

The tag names all structural equations, order/degree bounds, dangerous set, and equivalence relation. Evidence produced in one domain is not exported to a larger one.

46.2 Parameter domain

A family record states the rational primes, residue degrees, prime ideals or irreducible moduli, central-series levels, and every excluded value:

$$P_{\text{adm}} = P_{\text{declared}} \setminus (P_{\text{singular}} \cup P_{\text{nongenerating}} \cup P_{\text{exceptional}}). \quad (48.2)$$

Each exclusion has an exact predicate and proof; unexplored parameters are not called exceptional.

46.3 Kernel equivalence

Two surjections describe the same regular finite cover exactly when

$$\ker \pi_1 = \ker \pi_2 \iff \exists \theta : Q_1 \xrightarrow{\sim} Q_2, \quad \theta\pi_1 = \pi_2. \quad (48.3)$$

The four ordered generator images suffice to verify the intertwining once θ is a proved homomorphism. Marked C_8 , parity, or physical-shell equivalence requires those data to be intertwined too.

46.4 Minimum-order claims

A quotient of order N proves only

$$N_{\text{opt}} \leq N. \quad (48.4)$$

Equality requires the Statement 37.8 package excluding every smaller admissible order in the declared domain. A candidate lacking such an exhaustive package is not called minimum.

46.5 Minimum-degree claims

A faithful permutation action of degree d proves

$$\mu(Q) \leq d. \quad (48.5)$$

To prove equality, exclude every faithful action of smaller degree. Equivalently, exclude every finite family of subgroups $H_j \leq Q$ with

$$\sum_j [Q : H_j] < d, \quad \bigcap_j \text{core}_Q(H_j) = 1. \quad (48.6)$$

Restricting to one core-free subgroup covers only transitive actions and can miss a smaller intransitive faithful representation.

46.6 Uniqueness claims

Uniqueness means that the complete declared domain has exactly one kernel-equivalence class:

$$|\{\ker \pi : \pi \text{ admissible}\}| = 1. \quad (48.7)$$

It requires exhaustive enumeration plus pairwise equivalence certificates. A single construction, even at proved minimum order, does not establish uniqueness.

46.7 Family classification

A classification theorem supplies parameter coverage, exact exceptional loci, surjectivity and generation for each admitted parameter, and necessary-and-sufficient equivalence criteria. The finite search architecture developed here is instead a census:

$$\text{classification} \implies \text{exhaustive census}, \quad \text{finite census} \not\Rightarrow \text{classification}. \quad (48.8)$$

46.8 Artifact-bound Level A1 specification

An abstract Level A1 record is a tuple

$$\mathcal{A}_1 = (Q, \pi, \beta, \bar{\chi}, \mathcal{D}_{\text{ax}}(6a_B), \mathcal{C}_{\text{alg}}, \mathcal{C}_{\text{geo}}, \mathcal{C}_{\text{independentverification}}), \quad (48.9)$$

where $\pi : \Gamma_B \twoheadrightarrow Q$, the automorphism β and character $\bar{\chi}$ satisfy the exact C45 contracts, $\mathcal{D}_{\text{ax}}(6a_B)$ is the complete closed global domain of C46, and the three certificates are immutable and hash-bound. Define the Level A1 passage predicate by

$$\text{PASS}_{A1}(\mathcal{A}_1) \iff \begin{cases} \mathcal{C}_{\text{alg}} \text{ replays the marked surjection, exact } C_8, \text{ and parity,} \\ \mathcal{C}_{\text{geo}} \text{ proves complete closed-domain coverage and an empty frontier,} \\ \mathcal{D}_{\text{ax}}(6a_B) \cap \ker \pi = \emptyset, \\ \mathcal{C}_{\text{independentverification}} \text{ reproduces all component hashes and verifier outputs.} \end{cases} \quad (48.10)$$

Theorem 46.8.1 (Level A1 implication). *If $\text{Pass}_{A1}(\mathcal{A}_1)$, then*

$$\text{sys}(\mathbb{H}^2/\ker \pi) > 6a_B, \quad r_{\text{inj}} > 3a_B. \quad (48.11)$$

Proof. Under the hypotheses, the verified outer axis certificate and zero short-kernel count meet Statement 44.6, yielding the strict systole bound. Statement 44.7 then yields the injectivity-radius bound. $\square$

The verifier assigns the artifact-relative classification

$$\text{Classification}_{A1}(\mathcal{A}_1) = \begin{cases} \text{SATISFIED}, & \text{Pass}_{A1}(\mathcal{A}_1), \\ \text{VIOLATED}, & \text{a mandatory exact check has a independently verified counterexample}, \\ \text{INDETERMINATE}, & \text{otherwise.} \end{cases} \quad (48.12)$$

Thus a name, prose description, decimal estimate, or unattached scan count cannot discharge (48.10). A shortest element found inside a finite scan is only an upper bound for the true systole unless global coverage is proved. No minimum-order, minimum-degree, uniqueness, classification, tower extension, representation-completeness, or numerical-tractability conclusion is included.

Part XVII

Two quotient towers and their limits

47 Canonical characteristic tower

47.1 Nested characteristic kernels

For $M \geq 1$, define

$$K_M = \bigcap_{\substack{F \text{ finite} \\ |F| \leq M}} \bigcap_{\rho \in \text{Hom}(\Gamma_B, F)} \ker \rho. \quad (49.1)$$

There are finitely many group tables of size at most M , and a homomorphism from finitely generated Γ_B is determined by finitely many generator images. Thus (49.1) is a finite intersection of finite-index normal subgroups, so K_M is finite-index and normal. Precomposition by any automorphism permutes the homomorphisms, proving

$$K_{M+1} \subseteq K_M, \quad K_M \text{ characteristic.} \quad (49.2)$$

47.2 Bonding maps

Set $Q_M = \Gamma_B/K_M$. Inclusion of kernels induces the unique surjection

$$p_{M+1,M} : Q_{M+1} \twoheadrightarrow Q_M, \quad \gamma K_{M+1} \mapsto \gamma K_M. \quad (49.3)$$

Representative independence follows from the inclusion, and

$$p_{M+2,M} = p_{M+1,M} p_{M+2,M+1}. \quad (49.4)$$

47.3 C_8 -compatible tower

Characteristicity gives $\phi_8(K_M) = K_M$, hence

$$\beta_M(\gamma K_M) = \phi_8(\gamma) K_M \quad (49.5)$$

is a well-defined automorphism satisfying

$$\beta_M^8 = \text{id}, \quad p_{M+1,M} \beta_{M+1} = \beta_M p_{M+1,M}. \quad (49.6)$$

Its order divides eight; exact order eight at a particular level requires a separate moved-element witness.

47.4 Compatible parity

For $M \geq 2$, the map $\chi : \Gamma_B \rightarrow C_2$ occurs among the homomorphisms in (49.1), so $K_M \subseteq \ker \chi$. Therefore

$$\bar{\chi}_M(\gamma K_M) = \chi(\gamma) \quad (49.7)$$

is well defined, surjective, and satisfies $\bar{\chi}_M p_{M+1,M} = \bar{\chi}_{M+1}$.

47.5 Systole growth

Theorem 47.5.1.

$$\bigcap_{M \geq 1} K_M = \{1\}, \quad \text{sys}(\mathbb{H}^2/K_M) \longrightarrow \infty. \quad (49.8)$$

Proof. Statement 16.2 proves residual finiteness. For $1 \neq g$, choose a finite separator F ; then $g \notin K_M$ once $M \geq |F|$, proving the intersection. For fixed L , a compact axis normalization gives a finite closed set of all conjugacy classes of length at most L . Choose one finite separator for each representative and take the maximum target order. Normality makes kernel membership conjugacy invariant, so all short classes are absent from every later K_M . Statement 44.6 gives the strict systole bound. $\square$

47.6 Unbounded order

The bonding maps imply

$$|Q_M| \mid |Q_{M+1}|, \quad |Q_M| \leq |Q_{M+1}|. \quad (49.9)$$

If these orders were bounded, the integer sequence would stabilize. Equal successive indices together with $K_{M+1} \subseteq K_M$ forces equal subgroups, so the nested kernels would eventually be a fixed finite-index subgroup. Its intersection could not be trivial in the infinite group Γ_B , contradicting (49.8).

47.7 Representation pullback

Pullback along $p = p_{M+1,M}$ sends a representation ρ of Q_M to ρp of Q_{M+1} and preserves irreducibility because p is onto. On regular spaces,

$$(J_M f)(x) = |\ker p|^{-1/2} f(p(x)) \quad (49.10)$$

is an isometry. Direct substitution shows it intertwines left translation through p ; its range is the subspace constant on $\ker p$ -cosets.

47.8 The inverse limit

The compatible system defines

$$\widehat{\Gamma}_{\text{char}} := \varprojlim_M Q_M. \quad (49.11)$$

The diagonal map $\Gamma_B \rightarrow \widehat{\Gamma}_{\text{char}}$ is injective by (49.8). If $\rho : \Gamma_B \rightarrow F$ is any finite quotient, then for $M \geq |F|$, $K_M \subseteq \ker \rho$, so ρ factors through Q_M . Hence this system is cofinal among all finite quotients and

$$\widehat{\Gamma}_{\text{char}} \cong \widehat{\Gamma}_B, \quad (49.12)$$

the full profinite completion, not merely a selected tower completion.

48 A second, pro-2 tower and independence

48.1 Two notions of independence

Two towers are called algebraically independent here when infinitely many pairs of levels have incompatible finite composition factors. They are geometrically cofinal when both global injectivity radii tend to infinity:

$$\text{algebraic independence} \not\Longleftrightarrow \text{geometric cofinality}. \quad (50.1)$$

The first is finite-group data; the second is a kernel-separation property.

48.2 Distinct-prime congruence schema

Given a faithful integral realization $\rho : \Gamma_B \hookrightarrow \text{GL}_d(\mathcal{O}_K[S^{-1}])$, good prime ideals $\mathfrak{p}, \mathfrak{q} \notin S$ define reduction kernels

$$K_{\mathfrak{p}} = \ker \bar{\rho}_{\mathfrak{p}}, \quad K_{\mathfrak{q}} = \ker \bar{\rho}_{\mathfrak{q}}. \quad (50.2)$$

They need not be distinct merely because $\mathfrak{p} \neq \mathfrak{q}$; kernel inequality requires an explicit element lying in exactly one kernel. For each nonidentity element, separation follows only after choosing a prime avoiding a certified nonzero matrix entry. Distinct primes alone prove neither residuality nor independence of the resulting towers.

48.3 The lower exponent-2 central series

Define

$$P_1^{(2)} = \Gamma_B, \quad P_{n+1}^{(2)} = (P_n^{(2)})^2 [P_n^{(2)}, \Gamma_B], \quad R_n = \Gamma_B / P_n^{(2)}. \quad (50.3)$$

Endomorphisms preserve squares and commutators, so every term is characteristic. Inductively R_n is a finite 2-group and $P_n^{(2)}$ is finitely generated: the quotient $P_n^{(2)} / P_{n+1}^{(2)}$ is a finite elementary abelian 2-group.

The standard residual-2 theorem for closed orientable surface groups [5] says that every nonidentity element survives in some finite 2-group quotient. The lower exponent-2 central series of every finite 2-group terminates. Hence, if $\psi : \Gamma_B \rightarrow F$ is such a quotient, functoriality gives $\psi(P_n^{(2)}) \subseteq P_n^{(2)}(F) = 1$ for all sufficiently large n . An element surviving under ψ therefore does not belong to that $P_n^{(2)}$. Thus

$$|R_n| \text{ is a power of 2, } \quad \bigcap_n P_n^{(2)} = \{1\}. \quad (50.4)$$

The residual-2 theorem is an external mathematical input, not an inference from finite computations.

48.4 Distinct mechanisms

The characteristic tower (Q_M) is cofinal among all finite quotients. The tower (R_n) detects only finite 2-group quotients. A congruence tower from (50.2) is a third mechanism:

$$\text{same local limit} \not\Rightarrow \text{same kernels or inverse limit}. \quad (50.5)$$

Identification of any two levels requires an explicit kernel or intertwining-isomorphism proof.

48.5 Nonisomorphic levels

For $M \geq 3$, every homomorphism $\Gamma_B \rightarrow C_3$ occurs in (49.1). Choose a surjection, available from the abelianization $\Gamma_B^{\text{ab}} \cong \mathbb{Z}^4$. Then

$$K_M \subseteq \ker(\Gamma_B \twoheadrightarrow C_3), \quad Q_M \twoheadrightarrow C_3, \quad 3 \mid |Q_M|. \quad (50.6)$$

Since every R_n has 2-power order, no Q_M with $M \geq 3$ is isomorphic to any R_n .

48.6 Composition-factor separation

Every composition factor of the finite 2-group R_n is C_2 ; for $R_1 = 1$ the composition-factor multiset is empty. The quotient in (50.6) forces a C_3 composition factor of Q_M :

$$\begin{aligned} \text{Comp}(R_1) &= \emptyset, \\ \text{Comp}(R_n) &\subseteq \{C_2\} \quad (n \geq 2), \\ C_3 &\in \text{Comp}(Q_M) \quad (M \geq 3). \end{aligned} \tag{50.7}$$

Jordan–Hölder therefore gives a structural nonisomorphism certificate stronger than unequal order alone.

48.7 Common geometric limit

Theorem 48.7.1. *Both nested normal towers have global injectivity radius tending to infinity and pointed local limit $\mathbb{H}^2$:*

$$r_{\text{inj}}(\mathbb{H}^2/K_M) \rightarrow \infty, \quad r_{\text{inj}}(\mathbb{H}^2/P_n^{(2)}) \rightarrow \infty. \tag{50.8}$$

Proof. Each tower is residual. For a fixed closed finite global dangerous set, every element is excluded at some level; nesting and finiteness give one later level excluding them all. Statement 44.6–Statement 44.7 then give the claimed growth and local convergence. $\square$

48.8 Non-equivalence certificate

At selected levels store exact orders, certified normal/composition series, the explicit map $Q_M \twoheadrightarrow C_3$, and the C_2 factors of R_n . Replay verifies

$$Q_M \not\cong R_n \quad (M \geq 3). \tag{50.9}$$

This proves algebraic non-equivalence of the two named towers and, with (50.8), their common geometric limit. It does not classify a third arithmetic family or prove its independence without corresponding factor data.

49 Local, spectral, and differentiable convergence

49.1 Local labeled convergence

Let $K_n \triangleleft \Gamma_B$ be nested and residual. For fixed word radius r , once

$$K_n \cap B_S(2r) = \{1\}, \tag{51.1}$$

Statement 40.3 identifies the radius- r rooted labeled Cayley ball of Γ_B/K_n with that of Γ_B . Residuality plus finiteness of $B_S(2r)$ makes (51.1) hold eventually.

49.2 Benjamini–Schramm convergence

For the surface covers, a global lower bound $r_{\text{inj}}(X_n) > r$ holds at every point by definition of the global injectivity radius. Hence, if

$$r_{\text{inj}}(X_n) \longrightarrow \infty, \tag{51.2}$$

the probability that a uniformly chosen point has an embedded radius- r neighborhood tends to one (indeed is eventually one). Normality makes the covers regular but does not make the injectivity radius function pointwise constant; no such false homogeneity is used.

49.3 Strong finite-propagation convergence

Let H be bounded, equivariant, self-adjoint, and have propagation $\rho < \infty$. Identify a finitely supported vector u and its ρ -neighborhood with their exact copies in a sufficiently large quotient. Then

$$H_n J_n u = J_n H u \quad (51.3)$$

eventually. On the local inductive-limit representation, density of finitely supported vectors and $\sup_n \|H_n\| < \infty$ extend (51.3) to strong convergence. This is not norm convergence on a single canonically fixed global space.

49.4 Spectral-support convergence

Strong convergence alone does not exclude spectral pollution in a gap. Hausdorff convergence of the uniformly bounded spectral supports follows from two separately checked directions:

$$\begin{cases} \text{localized Weyl sequences give no loss,} \\ \text{uniform resolvent estimates give no pollution.} \end{cases} \quad (51.4)$$

The second direction must hold on every compact subset of $\mathbb{R} \setminus \sigma(H)$; local graph convergence alone does not supply it.

49.5 No loss

Theorem 49.5.1. *For every $\lambda \in \sigma(H)$,*

$$\text{dist}(\lambda, \sigma(H_n)) \longrightarrow 0. \quad (51.5)$$

Proof. Weyl's criterion gives unit vectors with residual tending to zero. Boundedness and density replace them by finitely supported unit approximants. Embed one sufficiently accurate support in all later quotients; (51.3) preserves its residual. For self-adjoint H_n , $\text{dist}(\lambda, \sigma(H_n)) \leq \|(H_n - \lambda)v\|$, proving the claim. $\square$

49.6 No pollution

Let $K \Subset \mathbb{R} \setminus \sigma(H)$. A sufficient independent hypothesis is that for some n_0, C_K ,

$$K \subset \rho(H_n), \quad \sup_{n \geq n_0} \sup_{\lambda \in K} \|(H_n - \lambda)^{-1}\| \leq C_K. \quad (51.6)$$

Then $K \cap \sigma(H_n) = \emptyset$ for every $n \geq n_0$. The resolvent-set inclusion is part of the hypothesis; writing an inverse norm where the inverse may not exist is not a proof. Establishing (51.6), for example by a uniform parametrix or coercive estimate, is a distinct analytic obligation.

49.7 Riesz projectors

After the local identifications, suppose a contour $\mathcal{C}$ lies in a common resolvent set and encloses an isolated cluster. Define

$$P_n = \frac{1}{2\pi i} \oint_{\mathcal{C}} (z - H_n)^{-1} dz, \quad P = \frac{1}{2\pi i} \oint_{\mathcal{C}} (z - H)^{-1} dz. \quad (51.7)$$

Uniformly bounded strong resolvent convergence on $\mathcal{C}$ gives

$$P_n \longrightarrow P \quad \text{strongly.} \quad (51.8)$$

Norm convergence of projectors requires norm-resolvent convergence; a closing contour gap invalidates both conclusions.

49.8 $C^0/C^1/C^2$ parameter convergence

Let θ range over one compact parameter domain and $0 \leq j \leq k \leq 2$. Assume, in a common identification,

$$\sup_{\theta} \|H_n^{(j)}(\theta) - H^{(j)}(\theta)\| \rightarrow 0 \quad (51.9)$$

in the topology required by the claimed conclusion, together with a uniform contour gap and uniform resolvent bounds. Differentiating

$$\partial_{\theta}(z - H)^{-1} = (z - H)^{-1} H' (z - H)^{-1} \quad (51.10)$$

and its second derivative gives uniform convergence of the Riesz calculus through order k . Thus the classification

$$\mathbf{C}^k\text{-CONVERGENCE} \quad (51.11)$$

requires derivative data through order k ; pointwise eigenvalue or spectral convergence does not imply it. The final analytic record is

$$(\rho, \sup \|H_n\|, \text{local maps, resolvent domain, gap, } \{H_n^{(j)}\}_{j \leq 2}). \quad (51.12)$$

Part XVIII

Interface with the bilayer Hamiltonian

50 Quotient regular representation

50.1 Left regular action

For $q \in Q$, define

$$(L_q f)(x) = f(q^{-1}x), \quad f \in \ell^2(Q). \quad (52.1)$$

Then $L_q L_r = L_{qr}$, $L_q^{-1} = L_{q^{-1}}$, and $L_q^{\dagger} = L_{q^{-1}}$; hence every L_q is unitary.

50.2 The physical shell

The ordered nearest-neighbor shell is

$$(g_0, \dots, g_7) = (a_1, b_1^{-1}, a_1^{-1} b_1^{-1} a_2, a_1^{-1} b_1^{-1} b_2^{-1}, \\ a_1^{-1}, b_1, a_2^{-1} b_1 a_1, b_2 b_1 a_1). \quad (52.2)$$

The eight permutations are $L_{\pi(g_j)}$. The order and word content are physical data; the symmetric presentation tuple is not a substitute.

50.3 Eight-label adjacency

Set

$$A_Q = \sum_{j=0}^7 L_{\pi(g_j)}. \quad (52.3)$$

Since $g_{j+4} = g_j^{-1}$, inverse-pair adjoints give

$$A_Q^{\dagger} = A_Q. \quad (52.4)$$

The labeled multigraph always has eight outgoing labels counting multiplicity. It is a simple eight-regular graph only when the shell images are nonidentity and pairwise distinct.

50.4 C_8 covariance

For the descended automorphism $\beta \in \text{Aut}(Q)$, let $(U_\beta f)(x) = f(\beta^{-1}(x))$. If $\beta(\pi(g_j)) = \pi(g_{j+1})$, direct substitution gives

$$U_\beta L_{\pi(g_j)} U_\beta^{-1} = L_{\pi(g_{j+1})}, \quad U_\beta A_Q U_\beta^{-1} = A_Q. \quad (52.5)$$

The second identity follows by cyclic reindexing, even if the descended β has order properly dividing eight.

50.5 Bipartite spectral symmetry

For descended parity define

$$(Jf)(x) = (-1)^{\bar{\chi}(x)} f(x). \quad (52.6)$$

Every shell element is odd, so

$$J A_Q J^{-1} = -A_Q. \quad (52.7)$$

If $A_Q v = \lambda v$, then $A_Q(Jv) = -\lambda Jv$. Thus eigenvalues occur as $\lambda, -\lambda$ with equal multiplicity, including the possible zero eigenspace.

50.6 Irreducible blocks

After the complete C47 analysis, the group Fourier transform gives

$$A_Q \cong \bigoplus_{\lambda \in \widehat{Q}} \left(\sum_{j=0}^7 \rho_\lambda(\pi(g_j)) \right) \otimes I_{d_\lambda}. \quad (52.8)$$

Each displayed operator block has size d_λ and multiplicity d_λ . Without a complete dual and realizing matrices, this formula is structural but not a computational decomposition.

50.7 Matrix-free action

Store the eight arrays implementing $x \mapsto \pi(g_j)^{-1}x$ in the convention (52.1). Then

$$(A_Q f)(x) = \sum_{j=0}^7 f(\pi(g_j)^{-1}x) \quad (52.9)$$

uses $8|Q|$ gathers and $O(8|Q|)$ additions, with $O(8|Q|)$ stored indices. A different forward-array convention is allowed only if its inverse relation is tested.

50.8 Infinite-cover limit

Along either residual tower, every fixed shell-labeled neighborhood eventually agrees with that of Γ_B . Consequently, on finitely supported vectors,

$$A_{Q_n} \longrightarrow A_\infty := \sum_{j=0}^7 L_{g_j} \quad \text{on } \ell^2(\Gamma_B), \quad (52.10)$$

and $\|A_{Q_n}\|, \|A_\infty\| \leq 8$. Density and this uniform bound give strong convergence in the local inductive-limit sense of Statement 49.3:

$$\text{propagation}(A) = 1, \quad \text{local equality} \implies \text{strong convergence}. \quad (52.11)$$

51 Monolayer and twisted bilayer operators

51.1 Monolayer Hamiltonian

Fix real parameters ω_0, t . On the Hilbert space $\ell^2(Q)$, with the left-regular convention of Statement 50.1, define

$$H_{\text{mono}}(Q) = \omega_0 I_{|Q|} - t A_Q. \quad (53.1)$$

This formula is also the matrix-free specification:

$$(H_{\text{mono}}u)(x) = \omega_0 u(x) - t \sum_{j=0}^7 u(\pi(g_j)^{-1}x). \quad (53.2)$$

Proposition 51.1.1. *$H_{\text{mono}}(Q)$ is self-adjoint, acts on exactly $|Q|$ complex degrees of freedom, and satisfies*

$$\|H_{\text{mono}}(Q)\| \leq |\omega_0| + 8|t|. \quad (53.3)$$

Proof. Statement 50.3 gives $A_Q^\dagger = A_Q$. Real scalar multiplication and addition of $\omega_0 I$ preserve self-adjointness. Each $L_{\pi(g_j)}$ is unitary, whence $\|A_Q\| \leq \sum_j \|L_{\pi(g_j)}\| = 8$; the triangle inequality gives (53.3). Equation (53.2) follows directly from (52.1), so it and an explicit sparse matrix must agree coordinate by coordinate. $\square$

As a boundary regression fixture, if $Q = \{1\}$, all eight labeled edges are loops and $H_{\text{mono}} = [\omega_0 - 8t]$. This fixture is deliberately not a simple-graph example: it detects any implementation that silently removes multiplicities.

51.2 Bilayer Hilbert space

The finite bilayer carrier and inner product are

$$\mathcal{H}_Q = \ell^2(Q)_{\text{top}} \oplus \ell^2(Q)_{\text{bottom}} \cong \ell^2(Q) \otimes \mathbb{C}^2, \quad \langle (u, v), (u', v') \rangle = \langle u, u' \rangle + \langle v, v' \rangle. \quad (53.4)$$

Thus $\dim_{\mathbb{C}} \mathcal{H}_Q = 2|Q|$. Layer exchange is

$$\tau(u, v) = (v, u), \quad \tau^\dagger = \tau, \quad \tau^2 = I. \quad (53.5)$$

The diagonal intralayer operator $\text{diag}(H_{\text{mono}}, H_{\text{mono}})$ commutes with τ . A full bilayer operator commutes with τ only when its two diagonal blocks agree and its off-diagonal block equals its own adjoint; layer exchange is therefore a testable symmetry, not an automatic consequence of using two copies.

51.3 Twist action and distance specification

Let z_x^t and z_y^b be the fixed in-plane representatives of top and bottom sites and let $\mathcal{R}_\theta$ rotate the top embedding about the fixed common center. With the declared in-plane metric $d_{\parallel}$, set

$$D_{xy}(\theta) = \sqrt{h^2 + d_{\parallel}(\mathcal{R}_\theta z_x^t, z_y^b)^2}. \quad (53.6)$$

The geometry serializer must record the angle convention, center, orientation, representatives, metric, and units. It must verify $\mathcal{R}_{\theta+\theta'} = \mathcal{R}_\theta \mathcal{R}_{\theta'}$, $\mathcal{R}_0 = I$, and the declared angular period. Equation (53.6) then places all twist dependence in $D_{xy}(\theta)$; A_Q is unchanged. No Q -equivariance of $D(\theta)$ follows merely from this definition. At a special angle the right-translation invariance needed below must instead be checked as

$$D_{xq,yq}(\theta) = D_{xy}(\theta) \quad (q, x, y \in Q). \quad (53.7)$$

Failure of (53.7) is the required negative regression test and disables the full Q -Fourier reduction below.

51.4 Full interlayer kernel

For $h > 0$, $\lambda_\perp > 0$, cutoff $D_c \geq h$, and amplitude $w \in \mathbb{C}$, define

$$[T_Q(\theta)]_{xy} = \begin{cases} w \exp(-(D_{xy}(\theta) - h)/\lambda_\perp), & D_{xy}(\theta) \leq D_c, \\ 0, & D_{xy}(\theta) > D_c. \end{cases} \quad (53.8)$$

This is the full radial kernel on the declared support, not an eight-shell replacement. Let Δ_r and Δ_c be certified maximum nonzero counts per row and column. Since $D_{xy} \geq h$, every nonzero entry has modulus at most $|w|$, and the Schur bound yields

$$\|T_Q(\theta)\|_2 \leq \sqrt{\|T_Q\|_1 \|T_Q\|_\infty} \leq |w| \sqrt{\Delta_r \Delta_c}. \quad (53.9)$$

The inequality $D = D_c$ belongs to the support. Tests must therefore include one pair strictly inside, one exactly on, and one strictly outside the cutoff; replacing $\leq$ by $<$ must fail that fixture.

51.5 Cutoff geometry and the no-wrap criterion

For arbitrary $0 < h \leq D_c$, the largest possible in-plane separation inside the three-dimensional cutoff is

$$r_{\parallel, \max} = \sqrt{D_c^2 - h^2}. \quad (53.10)$$

Proposition 51.5.1 (Sufficient geometric no-wrap criterion). *Suppose a independently verified global certificate establishes*

$$r_{\text{inj}}(\mathbb{H}^2 / \ker \pi) > r_{\parallel, \max}. \quad (53.11)$$

Then every supported in-plane displacement in (53.8) lies strictly below the injectivity radius and has a unique lift in the certified ball.

Proof. By (53.8), support membership gives $D_{xy}^2 = h^2 + d_\parallel^2 \leq D_c^2$, hence $d_\parallel \leq r_{\parallel, \max} < r_{\text{inj}}$. The covering map is injective on every ball of radius strictly below r_{inj} , so no supported pair can acquire a distinct quotient lift in that ball. $\square$

The implication is purely theorem-level. For any concrete quotient, the no-wrap field is **ESTABLISHED** only after a hash-bound global certificate replays (53.11); a missing premise yields **INDETERMINATE**, and a independently verified violation yields **VIOLATED**. Particular parameter values and artifact availability belong to a separate example or provenance record.

51.6 Finite-quotient Hamiltonian

On (53.4) define

$$H_Q(\theta) = \omega_0 I_{2|Q|} + \begin{pmatrix} -tA_Q & T_Q(\theta) \\ T_Q(\theta)^\dagger & -tA_Q \end{pmatrix}. \quad (53.12)$$

Proposition 51.6.1. *For real ω_0, t , (53.12) is self-adjoint for every θ , whether or not $T_Q(\theta)$ is self-adjoint, and*

$$\|H_Q(\theta)\| \leq |\omega_0| + 8|t| + \|T_Q(\theta)\|. \quad (53.13)$$

Proof. The diagonal blocks are self-adjoint by Statement 51.1, and the two off-diagonal blocks are mutual adjoints. The off-diagonal operator $\begin{pmatrix} 0 & T \\ T^\dagger & 0 \end{pmatrix}$ has norm $\|T\|$; combining this with $\|A_Q\| \leq 8$ proves (53.13). $\square$

For $Q = \{1\}$, the explicit regression matrix is

$$\begin{pmatrix} \omega_0 - 8t & T_{11}(\theta) \\ \overline{T_{11}(\theta)} & \omega_0 - 8t \end{pmatrix}, \quad (53.14)$$

which checks dimension, conjugation, and layer ordering.

51.7 Representation blocks: exact hypothesis

Define the right regular action by

$$(R_q f)(x) = f(xq), \quad \tilde{R}_q = \text{diag}(R_q, R_q). \quad (53.15)$$

Left and right translations commute: $L_s R_q = R_q L_s$. Hence A_Q , being a sum of left translations, commutes with every R_q . The full bilayer operator commutes with the right Q -action exactly when

$$R_q T_Q(\theta) = T_Q(\theta) R_q \quad (q \in Q). \quad (53.16)$$

For a kernel operator, (53.16) is equivalent to $T_{xq, yq} = T_{xy}$, which follows from the distance identity (53.7). It suffices to test (53.16) on a certified generating set of Q . When it holds and the C47 representation inventory is complete, the Fourier transform decomposes $H_Q(\theta)$ into $2d_\lambda$ by $2d_\lambda$ blocks, each repeated d_λ times. The dimension check is

$$\sum_{\lambda \in \widehat{Q}} d_\lambda(2d_\lambda) = 2 \sum_{\lambda} d_\lambda^2 = 2|Q|. \quad (53.17)$$

At a generic twist (53.16) can fail. Then only separately verified residual symmetries may be used; applying full Q -blocks is an invalid optimization and must be rejected.

51.8 Matrix-free computational conditions

The immutable input record consists of the ordered eight inverse permutations, canonical row-sorted interlayer neighbor lists with weights, $(\omega_0, t, w, h, \lambda_\perp, D_c, \theta)$, and hashes of the geometry and quotient enumerations. For $v = (u_t, u_b)$ the output is

$$\begin{aligned} y_t &= (\omega_0 I - t A_Q) u_t + T_Q(\theta) u_b, \\ y_b &= T_Q(\theta)^\dagger u_t + (\omega_0 I - t A_Q) u_b. \end{aligned} \quad (53.18)$$

Construction first validates array bijectivity and inverse pairing, then validates row offsets and cutoff membership, and only then exposes the map $v \mapsto y$. verification requires: coordinate equality with an explicitly assembled small matrix; the adjoint identity $\langle H v, z \rangle = \langle v, H z \rangle$; the norm bound (53.13); and every claimed symmetry commutator. Identical serialized inputs must produce identical neighbor order and hashes. Matrix-free storage does not, by itself, establish that a requested spectral procedure fits memory or time; that independent condition is C54.

The output record uses four independent typed fields:

field	allowed values	
<code>operator_status</code>	VALID, INVALID	(53.19)
<code>fourier_status</code>	AVAILABLE, DISABLED-NONEQUIVARIANT, INDETERMINATE	
<code>no_wrap_status</code>	ESTABLISHED, VIOLATED, INDETERMINATE	
<code>resource_status</code>	WITHIN-BUDGET, OUTSIDE-BUDGET, INDETERMINATE	

`operator_status=VALID` requires the base array, cutoff, adjoint, and small-matrix tests. Fourier fields are present only after both (53.16) and C47 completeness pass; failure of (53.16) disables only that optimization and does not invalidate the Hermitian operator. No-wrap classification requires a independently verified global hash-bound certificate. Resource classification is assigned only by the independent C54 budget condition.

52 Complexity and the computational boundary

52.1 Hilbert-space dimension and vector memory

Let Q be a certified finite quotient and put $q = |Q|$. The bilayer space has exact complex dimension

$$N := \dim_{\mathbb{C}} \mathcal{H}_Q = 2q. \quad (54.1)$$

If the scalar representation uses b_z bytes per complex entry, one uncompressed state vector occupies

$$M_{\text{vec}} = b_z N = 2b_z q \text{ bytes.} \quad (54.2)$$

Thus complex128, for which $b_z = 16$, uses $32q$ bytes per vector. An algorithm retaining m independent full vectors needs at least $2b_z m q$ bytes before indices, weights, basis metadata, alignment, and library workspaces. These identities are conditional only on the exact order q and the declared scalar format; they do not certify either input.

52.2 Intralayer sparsity

Each of the eight ordered shell labels supplies one permutation entry at every vertex. The two diagonal layers therefore have the exact labeled directed-entry count

$$C_{\text{intra}} = 2 \cdot 8q = 16q. \quad (54.3)$$

This count retains label and direction multiplicity. After coincident labels are summed, the number of distinct nonzero matrix positions can be smaller. Equality with the simple-graph nonzero count requires separate proofs that all shell images are nonidentity and pairwise distinct.

An explicit permutation-array implementation stores eight length- q arrays, shared by the two layer applications. If each index uses b_i bytes, the raw array payload is $8b_i q$ bytes. This describes that encoding; it is not a universal lower bound because an implementation may generate permutations on demand.

52.3 Interlayer support certificate

For the oriented top-to-bottom kernel define

$$\begin{aligned} \delta_x(\theta) &= \#\{y : D_{xy}(\theta) \leq D_c\}, & \delta^y(\theta) &= \#\{x : D_{xy}(\theta) \leq D_c\}, \\ \Delta_r &= \max_x \delta_x, & \Delta_c &= \max_y \delta^y, & S_T &= \sum_x \delta_x = \sum_y \delta^y. \end{aligned} \quad (54.4)$$

The last equality double-counts the same finite support set by rows and by columns. Here S_T is the geometric-support count. If $w \neq 0$, every exponential weight in (53.8) is nonzero and $\text{nnz}(T) = S_T$; if $w = 0$, then $\text{nnz}(T) = 0$ although S_T remains defined.

A claimed uniform support degree d is certified exactly by canonical row offsets and the exhaustive identities

$$\delta_x = d \quad (x \in Q), \quad S_T = dq. \quad (54.5)$$

A maximum-only certificate yields merely $S_T \leq dq$. The record also states whether a stored count describes T alone or both applications T and $T^\dagger$; the two matvec contributions are distinct even when one neighbor list encodes both.

52.4 Matrix-free operation and storage bounds

An implementation processing every stored interlayer support record uses $16q$ labeled intralayer gathers, S_T weighted gathers for T , and S_T for $T^\dagger$. Hence

$$C_{\text{mv}} = 16q + 2S_T. \quad (54.6)$$

If zero weights are omitted this is an upper bound for the compressed implementation. Under a certified uniform degree d ,

$$S_T = dq, \quad C_{\text{mv}} = (16 + 2d)q. \quad (54.7)$$

With b_c -byte column indices, b_o -byte row offsets, and complex128 weights, an explicit CSR representation of T has raw payload

$$M_{\text{CSR}} = (b_c + 16)S_T + b_o(q + 1). \quad (54.8)$$

The offset type must represent every integer from 0 through S_T . Thus a k -bit unsigned offset type is valid only if

$$S_T \leq 2^k - 1. \quad (54.9)$$

Equations (54.6)–(54.9) are encoding-specific exact arithmetic, not universal complexity lower bounds. A concrete substitution for (q, d, b_c, b_o) belongs in an example dossier whose inputs are independently certified.

52.5 Spectral-solver obligations

For an m -vector Krylov basis, vector storage alone is at least $2b_zmq$ bytes and m operator applications process at most mC_{mv} contributions in the full-support implementation. Full reorthogonalization adds order Nm^2 inner-product arithmetic; restarted methods must state the restart length and retained-vector count. A degree- k Chebyshev or kernel-polynomial computation needs k matvecs per probe after scaling by a certified spectral interval. The bounds (53.9) and (53.13) supply such an interval only after Δ_r and Δ_c are certified.

Every accepted Ritz pair carries the independently recomputed residual

$$r(v, \lambda) = \|H_Q v - \lambda v\|_2, \quad \|v\|_2 = 1. \quad (54.10)$$

The fixed procedure declares tolerance, precision, stopping rule, random seeds, and loss-of-orthogonality policy. A positive small fixture compares all eigenvalues with dense diagonalization. A negative fixture perturbs one canonical neighbor weight and must be rejected by an independent operator reconstruction or exact support-and-weight hash; a separately perturbed adjoint entry must fail the Hermiticity check.

52.6 Representation-resolved strategy

Block computation is permitted only when

$$\begin{cases} \sum_{\lambda \in \widehat{Q}} d_\lambda^2 = q & \text{with all C47 character and realization checks,} \\ [H_Q(\theta), \widetilde{R}_s] = 0 & \text{for a certified generating set } s \text{ of } Q. \end{cases} \quad (54.11)$$

The second line extends from generators to all of Q because commutation is preserved under products and inverses. When both lines hold, Statement 51.7 gives blocks of size $2d_\lambda$, repeated d_λ times, and $\sum_\lambda d_\lambda(2d_\lambda) = 2q$ checks complete multiplicity. The full Q -Fourier optimization applies when both lines hold; the matrix-free Hermitian operator remains valid independently.

52.7 Conditions for computational feasibility

Definition 52.7.1. A finite quotient is *producible for a fixed procedure* if and only if all four certificates pass:

certificate	required conclusion	
algebra	exact quotient, C_8 , and parity	
geometry	$r_{\text{inj}} > D_c$ from a complete global closed-domain scan	(54.12)
operator	canonical support, adjoint, and regression checks	
resources	memory, work, time, and residual bounds within the declared budget	

Each row is necessary by the procedure specification, and their conjunction is sufficient because it supplies every mandatory input and independent verification test. A based-radius result cannot fill the global geometry row, and matrix-free availability cannot fill the resource row. The classification is

$$\text{ProdStatus} = \begin{cases} \text{SATISFIED,} & \text{all four rows independent verification and pass,} \\ \text{VIOLATED,} & \text{a mandatory row has a independently verified violation,} \\ \text{INDETERMINATE,} & \text{otherwise.} \end{cases} \quad (54.13)$$

This trichotomy prevents absence of evidence from being recorded as either a pass or a mathematical counterexample.

52.8 Tower-to-physics correspondence

At level n , the immutable export contains generator and inverse permutations, exact group order and kernel identifier, C_8 and parity actions, a global injectivity-radius certificate, the ordered physical shell, interlayer row lists and cutoff data, operator-norm bounds, a representation inventory when requested, and resource estimates with the solver grid. Hashes bind every dependent field:

$$\mathcal{E}_n = \text{Hash}(\text{group, geometry, symmetry, support, operator, solver, budgets}). \quad (54.14)$$

Import first verifies component hashes, then replays algebra and geometry, reconstructs small operator fixtures, and finally applies the resource condition. Missing optional representation data disables only representation-resolved output. Missing any mandatory algebra, geometry, operator, or resource field prevents computational verification. A tower bonding map is exported separately and is never inferred from matching orders or filenames.

Part XIX

Explicit quotient construction and limiting theorems

53 Geometric and algebraic data

53.1 Geometry dataset

The geometry record is exact. The scale $R > 0$ is not stored as an untyped machine real: it is either a canonical positive rational or algebraic number, or the fixed symbolic scale token `R_UNIT`,

relative to which every length is represented. Together with $o = 0$, the record contains

$$\begin{aligned} \zeta &= e^{\pi i/4}, & \rho &= 2^{-1/4}, & v_j &= \rho e^{(2j+1)\pi i/8}, \\ A &= 1 + \sqrt{2}, & B &= \sqrt{2 + 2\sqrt{2}}, & G_j &= \begin{pmatrix} A & B\zeta^j \\ B\zeta^{-j} & A \end{pmatrix}, & g_j &= [G_j], \end{aligned} \quad (55.1)$$

the oriented sides $s_j = [v_{j-1}, v_j]$, the opposite-side pairing $g_j : s_{j+4} \rightarrow s_j$, and

$$a_B = 2R \operatorname{arcosh}(1 + \sqrt{2}). \quad (55.2)$$

Indices are reduced modulo eight, $G_{j+4} = G_j^{-1}$, and coordinates use the disk model with curvature $-R^{-2}$. A canonical serializer stores algebraic-number basis coefficients and the complete field tower rather than decimal approximations. For every algebraic generator it stores the defining polynomial, ordered basis, and a rational isolating interval or complex rectangle selecting the distinguished embedding. In particular the records select $\sqrt{2} > 0$, $\rho > 0$, $B > 0$, and the root ζ with positive real and imaginary parts. A hash-domain tag and schema version are also mandatory. The geometry hash is

$$h_{\text{geo}} = \text{SHA256}(\text{domain}, \text{version}, R, o, (v_j), (G_j), (s_j), a_B, \text{units}, \text{orientation}). \quad (55.3)$$

Replay checks $A^2 - B^2 = 1$, all endpoint identities of Statement 2.3, and the single eight-vertex cycle of Statement 2.4. Changing a side orientation, basepoint, unit, or algebraic coefficient must change the hash.

53.2 Presentation and Nielsen dictionary

Freeze the standard presentation

$$\Gamma_B = \langle a_1, b_1, a_2, b_2 \mid [a_1, b_1][a_2, b_2] = 1 \rangle \quad (55.4)$$

and the two inverse free-group substitutions

$$\begin{aligned} a_1 &= h_0, & b_1 &= h_1, & a_2 &= h_1 h_0 h_2, & b_2 &= h_3 h_0^{-1} h_1^{-1}, \\ h_0 &= a_1, & h_1 &= b_1, & h_2 &= a_1^{-1} b_1^{-1} a_2, & h_3 &= b_2 b_1 a_1. \end{aligned} \quad (55.5)$$

Proposition 53.2.1. *The record (55.5) is a genuine coordinate isomorphism, not merely a dictionary valid after quotienting.*

Proof. Substitute the second row into the first. The a_2 word reduces to $b_1 a_1 (a_1^{-1} b_1^{-1} a_2) = a_2$, and the b_2 word reduces to $(b_2 b_1 a_1) a_1^{-1} b_1^{-1} = b_2$; a_1, b_1 are immediate. Conversely, substitution returns h_2 and h_3 by the same free reductions. Thus both composites fix every free generator. $\square$

The serializer includes the source alphabet with every word. A test that interprets an h -word as an a, b -word without (55.5) must fail before group evaluation.

53.3 The exact C_8 automorphism

The four fixed standard-generator images are

$$\begin{aligned} \phi_8(a_1) &= b_1^{-1}, & \phi_8(b_1) &= a_2^{-1} b_1 a_1, \\ \phi_8(a_2) &= a_2^{-1} b_1 a_1 b_1^{-1} a_1^{-1} b_1^{-1} b_2^{-1}, & \phi_8(b_2) &= a_1 b_1 a_1^{-1} b_1^{-1} a_2. \end{aligned} \quad (55.6)$$

In geometric coordinates these are equivalent to

$$(h_0, h_1, h_2, h_3) \mapsto (h_1^{-1}, h_2^{-1}, h_3^{-1}, h_0). \quad (55.7)$$

Proposition 53.3.1. *The substitution (55.6) preserves the surface relator, sends g_j to g_{j+1} , and has exact order eight.*

Proof. C07 identifies the map with conjugation by the disk rotation $c_8(z) = \zeta z$. Exact covariance $CG_jC^{-1} = G_{j+1}$ gives the shell cycle and shows that the pairing group is normalized, so the relator is preserved. The projective rotation satisfies $c_8^8 = 1$; for $0 < k < 8$, its action on the eight distinct shell elements is a nontrivial k -step cyclic shift, hence $\phi_8^k \neq \text{id}$. Therefore the order is eight. $\square$

Replay performs the four word substitutions, relator reduction, all eight shell equations, and all four equations for ϕ_8^8 ; checking only one displayed generator image is insufficient.

53.4 Parity character

Define on presentation generators

$$\chi(a_1) = \chi(b_1) = \chi(a_2) = \chi(b_2) = 1 \quad \text{in } C_2. \quad (55.8)$$

Every commutator has total exponent sum zero modulo two, so the relator in (55.4) maps to zero and $\chi : \Gamma_B \rightarrow C_2$ is well defined. Substitution into the exact shell of Statement 53.5 gives

$$\chi(g_j) = 1 \quad (0 \leq j < 8). \quad (55.9)$$

Each word on the right of (55.6) also has odd total parity. Thus $\chi\phi_8 = \chi$ on the four generators and hence on all of Γ_B . A quotient map π receives a unique descended character precisely when

$$\ker \pi \subseteq \ker \chi. \quad (55.10)$$

Necessity follows because $\bar{\chi}(\pi(k)) = \bar{\chi}(1) = 0$; sufficiency and uniqueness follow by defining $\bar{\chi}(\pi(\gamma)) = \chi(\gamma)$. The identity and a deliberately even shell word provide the zero-parity boundary tests.

53.5 Ordered physical shell

The physical input is the ordered word tuple

$$\begin{aligned} S_{\text{geo}} = & (a_1, b_1^{-1}, a_1^{-1}b_1^{-1}a_2, a_1^{-1}b_1^{-1}b_2^{-1}, \\ & a_1^{-1}, b_1, a_2^{-1}b_1a_1, b_2b_1a_1). \end{aligned} \quad (55.11)$$

It obeys $g_{j+4} = g_j^{-1}$, $\phi_8(g_j) = g_{j+1}$, and $\chi(g_j) = 1$. The presentation alphabet

$$S_{\text{pres}} = (a_1, a_1^{-1}, b_1, b_1^{-1}, a_2, a_2^{-1}, b_2, b_2^{-1}) \quad (55.12)$$

is a separate object and is never substituted for (55.11) in the Hamiltonian. Canonical shell serialization stores the tuple position, freely reduced word bytes, source alphabet, inverse index, and exact matrix key. Reordering the same set changes the physical hash because the C_8 cycle and regression arrays are ordered data.

53.6 Strict injectivity-radius target

For a torsion-free normal subgroup $K \triangleleft \Gamma_B$, put

$$\text{sys}(\mathbb{H}^2/K) = \inf_{1 \neq k \in K} \ell(k), \quad r_{\text{inj}}(\mathbb{H}^2/K) = \frac{1}{2} \text{sys}(\mathbb{H}^2/K). \quad (55.13)$$

The fixed target is

$$r_{\text{inj}} > 3a_B \iff \text{sys} > 6a_B. \quad (55.14)$$

The equivalence is multiplication by the positive number two. The inequalities are strict: a kernel element with $\ell(k) = 6a_B$ fails the condition. Only the complete global axis-normalized closed domain $\ell(k) \leq 6a_B$ can certify (55.14); a based displacement bound at o cannot replace it.

53.7 Order target and optimization semantics

Before search, freeze either a cap $N_{\max}$ or an interval $[N_{\min}, N_{\max}]$, together with the objective and tie-breaking order. The typed branches are

order_mode	lower field	order predicate
CAP	$N_{\min} = \text{NONE}$	$ Q \leq N_{\max}$
INTERVAL	$1 \leq N_{\min} \leq N_{\max}$	$N_{\min} \leq Q \leq N_{\max}$.

(55.15)

A candidate is feasible only if its branch predicate and every non-order condition pass. A lower field in the CAP branch, or a missing lower field in the INTERVAL branch, is rejected rather than silently ignored. An achieved order is not an optimum unless every admissible family in the predeclared finite domain below it has been exhausted with independently verifiable rejection certificates. Feasibility and optimality are therefore separate typed fields: let $F_-(Q)$ mean that an exact independent verification violates the branch predicate, typing, or a non-order condition, and let $F_+(Q)$ mean that the exact order artifact replays and all those tests pass. These predicates are disjoint. Then

$$\text{FeasibilityStatus}(Q) = \begin{cases} \text{VIOLATED}, & F_-(Q), \\ \text{SATISFIED}, & \neg F_-(Q) \wedge F_+(Q), \\ \text{INDETERMINATE}, & \text{otherwise.} \end{cases} \quad (55.16a)$$

For optimality, let $O_-(Q)$ mean that $F_+(Q)$ holds and an exact independent verification either exhibits a smaller feasible case or invalidates a claimed exclusion. Let $O_+(Q)$ mean that $F_+(Q)$ holds and every smaller admissible case is excluded by a independently verifiable rejection certificate. Exact verification makes $O_-(Q)$ and $O_+(Q)$ disjoint, and

$$\text{OptimalityStatus}(Q) = \begin{cases} \text{VIOLATED}, & O_-(Q), \\ \text{SATISFIED}, & \neg O_-(Q) \wedge O_+(Q), \\ \text{INDETERMINATE}, & \text{otherwise.} \end{cases} \quad (55.16b)$$

The cases in each field are mutually exclusive. A independently verified counterexample takes priority over any incomplete claim, while missing data fall in INDETERMINATE. A numerical order quoted without a resolvable group artifact never promotes feasibility to SATISFIED; a feasible candidate with incomplete exclusions is not called optimal.

53.8 Numerical cost target

Freeze a budget vector before operator construction:

$$\mathcal{B} = (B_{\text{mem}}, B_{\text{mv}}, B_{\text{wall}}, \varepsilon_{\text{res}}, m_{\text{krylov}}, k_{\text{poly}}, \Theta_{\text{grid}}, n_{\text{probe}}, \text{precision}). \quad (55.17)$$

Units, hardware, concurrency, library versions, and whether bounds are measured or rigorous are part of the record. verification compares each certified cost component with its matching budget component; no weighted average may hide one exceeded component. Define

$$\text{ResourceStatus} = \begin{cases} \text{SATISFIED}, & \text{all components are certified and } \hat{\mathcal{C}} \preceq \mathcal{B}, \\ \text{VIOLATED}, & \text{some certified component exceeds its budget,} \\ \text{INDETERMINATE}, & \text{otherwise.} \end{cases} \quad (55.18)$$

The priority $\text{VIOLATED} > \text{INDETERMINATE} > \text{SATISFIED}$ prevents a missing field from hiding a known budget violation. A zero-support toy fixture tests the lower boundary, while a deliberately one-byte-too-large fixture tests strict rejection at the memory boundary. Concrete hardware measurements and candidate-specific budgets belong in separate example records, not in this universal specification.

54 Candidate quotient families

54.1 Arithmetic reduction family

Let K be a number field, S a finite set of primes, and suppose an injective representation is given by exact matrices

$$\rho : \Gamma_B \hookrightarrow \mathrm{GL}_d(\mathcal{O}_K[S^{-1}]). \quad (56.1)$$

For a prime ideal $\mathfrak{p} \notin S$ at which all matrix entries and determinants reduce, define

$$\bar{\rho}_{\mathfrak{p}} : \Gamma_B \longrightarrow \mathrm{GL}_d(\mathcal{O}_K/\mathfrak{p}), \quad Q_{\mathfrak{p}} := \langle \bar{\rho}_{\mathfrak{p}}(a_1), \bar{\rho}_{\mathfrak{p}}(b_1), \bar{\rho}_{\mathfrak{p}}(a_2), \bar{\rho}_{\mathfrak{p}}(b_2) \rangle. \quad (56.2)$$

Proposition 54.1.1. *$Q_{\mathfrak{p}}$ is a finite quotient of Γ_B , with kernel $\ker \bar{\rho}_{\mathfrak{p}}$. The ambient general linear group may be replaced by $Q_{\mathfrak{p}}$, but not conversely.*

Proof. The residue ring is a finite field, so its general linear group and the generated subgroup are finite. Evaluation of the surface relator before reduction is the identity and reduction is a ring homomorphism, so the four reduced matrices define a group homomorphism. Its image is exactly the subgroup in (56.2), and the first isomorphism theorem gives the quotient statement. $\square$

The record includes a membership/order certificate for the generated image. Distinct primes need not give distinct kernels. Descent of ϕ_8 also requires $\phi_8(\ker \bar{\rho}_{\mathfrak{p}}) = \ker \bar{\rho}_{\mathfrak{p}}$; it is not supplied by reduction alone.

54.2 Finite-field and p -quotient parameters

For a prime number p and an integer $f \geq 1$, an arithmetic residue field is fixed by

$$(p, f, q, F), \quad q = p^f, \quad \mathbb{F}_q = \mathbb{F}_p[X]/(F), \quad (56.3)$$

where F is monic irreducible of degree f , coefficients are in canonical order, and reduced generator matrices use the associated polynomial basis. If the field comes from a number-field prime, the ideal generators and the explicit residue map are also stored. The symbol q alone cannot reconstruct multiplication.

The independent lower exponent- p family is

$$P_1^{(p)} = \Gamma_B, \quad P_{n+1}^{(p)} = (P_n^{(p)})^p [P_n^{(p)}, \Gamma_B], \quad R_n^{(p)} = \Gamma_B / P_n^{(p)}. \quad (56.4)$$

Inductively, finite generation of $P_n^{(p)}$ makes $P_n^{(p)} / P_{n+1}^{(p)}$ a finite elementary abelian p -group, so each $R_n^{(p)}$ is finite. Residuality of this tower is a separate group theorem and must be cited or proved for the chosen p ; it cannot be inferred from (56.4). For $p = 2$, Statement 48.3 provides the required surface-group input.

54.3 Central-extension family and obstruction

Let

$$1 \longrightarrow Z \longrightarrow \tilde{Q} \xrightarrow{\nu} Q \longrightarrow 1 \quad (56.5)$$

be a finite central extension. Let $\pi : \Gamma_B \rightarrow Q$ be a certified homomorphism satisfying the surface relation, and let its generator-image tuple be

$$\mathbf{q} = (\alpha_1, \beta_1, \alpha_2, \beta_2) \in Q^4,$$

choose a lift tuple

$$\tilde{\mathbf{q}} = (\tilde{\alpha}_1, \tilde{\beta}_1, \tilde{\alpha}_2, \tilde{\beta}_2) \in \tilde{Q}^4$$

and compute

$$z_{\mathrm{obs}} := [\tilde{\alpha}_1, \tilde{\beta}_1][\tilde{\alpha}_2, \tilde{\beta}_2] \in Z. \quad (56.6)$$

Proposition 54.3.1. *The fixed four images lift to a homomorphism $\Gamma_B \rightarrow \tilde{Q}$ precisely when $z_{\text{obs}} = 1$. Moreover z_{obs} is independent of the chosen lifts.*

Proof. The presentation universal property says that the selected lifts define a homomorphism exactly when their relator is one. Its image under ν is the relator in Q , hence (56.6) lies in Z . Any other lift multiplies a selected lift by an element of central Z . Central factors cancel from every commutator, so (56.6) is unchanged. $\square$

Thus a nontrivial obstruction cannot be cancelled merely by changing lifts by kernel elements. One must change the underlying assignment in Q , change the extension, or reject the candidate. Here a *lift* always means a homomorphism $\tilde{\pi} : \Gamma_B \rightarrow \tilde{Q}$ satisfying $\nu\tilde{\pi} = \pi$.

54.4 Semilinear classical family

Represent a semilinear element by (A, σ) , where σ is a certified field automorphism. Multiplication and inverse are

$$\begin{aligned} (A, \sigma)(B, \tau) &= (A\sigma(B), \sigma\tau), \\ (A, \sigma)^{-1} &= (\sigma^{-1}(A^{-1}), \sigma^{-1}). \end{aligned} \tag{56.7}$$

Associativity follows from $\sigma(B\tau(C)) = \sigma(B)(\sigma\tau)(C)$; multiplying the two displayed expressions verifies the inverse. In projective groups, matrix equality is equality modulo the declared scalar center, and a canonical representative must be normalized after every product. The surface relator, generator inverses, ϕ_8 -covariance, and element order are all evaluated with (56.7), never with ordinary matrix multiplication alone. A negative fixture with nontrivial σ must distinguish $A\sigma(B)$ from AB .

54.5 Low-index permutation family

For a subgroup $H \leq \Gamma_B$ of finite index m , let Γ_B act on the left cosets by

$$\pi_H(g)(xH) = (gx)H, \quad \ker \pi_H = \bigcap_{x \in \Gamma_B} xHx^{-1} = \text{core}_{\Gamma_B}(H). \tag{56.8}$$

The kernel identity follows because g fixes every coset exactly when $x^{-1}gx \in H$ for every x . Hence the actual quotient is $\text{im } \pi_H \leq S_m$, not automatically S_m . A valid record contains a complete Todd–Coxeter or certified low-index enumeration, canonical coset representatives, four generator permutations, relator and transitivity checks, and the generated-image order. An incomplete enumeration or a coincidental subgroup of S_m is not a quotient certificate.

54.6 Products, subdirect images, and fiber products

Given quotient maps $\pi_i : \Gamma_B \twoheadrightarrow Q_i$, define

$$\Pi(\gamma) = (\pi_1(\gamma), \dots, \pi_r(\gamma)), \quad \ker \Pi = \bigcap_i \ker \pi_i. \tag{56.9}$$

The actual quotient is the generated diagonal image $\text{im } \Pi$, which is subdirect but need not be the full Cartesian product. If a witness survives in any component, it survives under Π , so products separate unions of witness sets.

If surjections $\epsilon_i : Q_i \twoheadrightarrow C_2$ satisfy $\epsilon_i \pi_i = \chi$, then

$$\text{im}(\pi_1, \pi_2) \leq Q_1 \times_{C_2} Q_2 := \{(x, y) : \epsilon_1(x) = \epsilon_2(y)\}. \tag{56.10}$$

The fiber product has order $|Q_1||Q_2|/2$, because the map $(x, y) \mapsto \epsilon_1(x)\epsilon_2(y)^{-1}$ is onto with that kernel. The diagonal image has this order only after surjectivity onto the entire fiber product is separately proved; otherwise its actual generated order must be used.

54.7 Characteristic-quotient family

For $M \geq 1$, use the canonical kernel

$$K_M = \bigcap_{\substack{|F| \leq M \\ \rho \in \text{Hom}(\Gamma_B, F)}} \ker \rho, \quad Q_M = \Gamma_B / K_M. \quad (56.11)$$

C49 proves finite index, nesting, characteristicity, trivial intersection, and compatible bonding maps. Every automorphism, including ϕ_8 , therefore descends. For $M \geq 2$, the homomorphism $\chi : \Gamma_B \rightarrow C_2$ occurs in the intersection, so $K_M \subseteq \ker \chi$ and parity descends. This is an unconditional existence family, but (56.11) may be computationally enormous; its canonical definition is not a tractability proof.

54.8 Hybrid closure family

Starting from any finite-index normal kernel K , form

$$K^{\text{hyb}} = \left(\bigcap_{j=0}^7 \phi_8^j(K) \right) \cap \ker \chi. \quad (56.12)$$

Proposition 54.8.1. *K^{hyb} is finite-index, normal, ϕ_8 -stable, contained in $\ker \chi$, and contained in K .*

Proof. Every automorphic image of a finite-index normal subgroup is again finite-index and normal; a finite intersection preserves both properties. Since $\phi_8^8 = \text{id}$, applying ϕ_8 cyclically permutes the eight factors. Statement 53.4 gives $\phi_8(\ker \chi) = \ker \chi$, proving stability. The $j = 0$ factor gives containment in K . $\square$

The quotient is constructed as the actual image of the product of the eight transported maps and χ . Because the new kernel is smaller, every separation already achieved by K is retained. Canonical factor order and actual-image minimization are mandatory; the ambient direct product order is only an upper bound.

55 Construction and refinement

55.1 Build the initial marked quotient

Input is a family identifier from C56, its complete typed parameters, and exact images

$$(A_1, B_1, A_2, B_2) \in G^4. \quad (57.1)$$

First evaluate $[A_1, B_1][A_2, B_2]$ in the declared group law and reject unless it is 1. Then compute

$$Q_0 = \langle A_1, B_1, A_2, B_2 \rangle, \quad \pi_0 : \Gamma_B \twoheadrightarrow Q_0. \quad (57.2)$$

The presentation universal property supplies the homomorphism and the definition of Q_0 supplies surjectivity. The ambient group G is not retained as the quotient unless equality $Q_0 = G$ is proved. The output stores a canonical enumeration, multiplication/inverse certificate, four marked generator keys, exact order, relator trace, and source-parameter hash. Re-evaluation must reproduce the same marked quotient key. An input satisfying the relator but generating a proper subgroup is the required boundary fixture.

55.2 Force descent and exact order of C_8

For $K_0 = \ker \pi_0$, either find an automorphism $\beta \in \text{Aut}(Q_0)$ satisfying

$$\beta\pi_0 = \pi_0\phi_8, \quad (57.3)$$

or replace the kernel by

$$K_0^{C_8} = \bigcap_{j=0}^7 \phi_8^j(K_0). \quad (57.4)$$

The latter kernel is finite-index and normal. Applying ϕ_8 cyclically permutes its factors because $\phi_8^8 = 1$, so it is stable and the rule

$$\beta(\gamma K_0^{C_8}) = \phi_8(\gamma) K_0^{C_8} \quad (57.5)$$

is well defined. It is realized computationally as the actual subdirect image of the eight transported quotient maps.

Stability proves only $\beta^8 = 1$, not $\text{ord}(\beta) = 8$. Exact order is checked on the marked generators. Since the order divides eight, it is eight exactly when $\beta^4 \neq 1$. If it is smaller, the candidate fails the exact C_8 condition; a refinement may separate a nonidentity word $\phi_8^4(s)s^{-1}$ for a generator s on which ϕ_8^4 acts nontrivially, followed again by (57.4).

55.3 Force parity descent

Given a current kernel K , parity descends if and only if $K \subseteq \ker \chi$. When this fails, replace

$$K^{\text{par}} = K \cap \ker \chi = \ker(\pi \times \chi), \quad (57.6)$$

and replace the ambient product by its actual generated image. The new kernel is finite-index, normal, contained in K , and permits the unique character $\bar{\chi}(\pi^{\text{par}}(\gamma)) = \chi(\gamma)$. Because $\chi\phi_8 = \chi$, applying parity closure to a ϕ_8 -stable kernel preserves ϕ_8 -stability. Replay checks all four generator parities and all eight odd shell images; merely finding some homomorphism from the finite group to C_2 does not identify the required descended character.

55.4 Force a local injectivity domain

For a word ball $B_S(r)$, injectivity of π on that ball is equivalent to

$$K \cap (B_S(r)^{-1}B_S(r)) = \{1\}. \quad (57.7)$$

Indeed $\pi(x) = \pi(y)$ exactly when $x^{-1}y \in K$. The difference set lies in $B_S(2r)$, so the stronger sufficient test $K \cap B_S(2r) = \{1\}$ is convenient but is not silently called equivalent unless the exact difference set was used.

For a chosen finite based-geometric or word domain D , enumerate it completely, deduplicate by canonical group keys, compute

$$W(K, D) = (K \cap D) \setminus \{1\}, \quad (57.8)$$

and, for every hit, intersect a finite separator kernel. Residual finiteness supplies a separator for each nonidentity hit; finiteness of D makes the construction finite. Apply the C8 and parity closures afterward. Because every operation decreases the kernel, previously separated elements cannot reappear.

55.5 Run the global witness scan

The global condition uses the exact axis-normalized closed dangerous domain

$$\mathcal{D}_{\text{glob}}(L) = \{w \neq 1 : w \text{ is the canonical representative of a conjugacy class with } \ell(w) \leq L\}, \quad (57.9)$$

with the finite covering domain, boundary convention, interval arithmetic, and empty-frontier proof specified in C46. It does not use the based set $d(o, wo) \leq L$.

The scan has exactly two valid result types. A SATISFIED record contains a complete-domain record and proves $K \cap \mathcal{D}_{\text{glob}}(L) = \emptyset$. A WITNESS record contains

$$(\text{word}(w), \text{groupkey}(w), [\ell_-(w), \ell_+(w)], \pi(w) = 1), \quad (57.10)$$

where $w \neq 1$ is proved exactly and the outward interval establishes $\ell_+(w) \leq L$. A partial enumeration with no hit has classification INCOMPLETE, never SATISFIED.

55.6 Strengthen against a certified witness

For a WITNESS w , choose a finite quotient $\rho_w : \Gamma_B \rightarrow F_w$ with $\rho_w(w) \neq 1$, and set

$$\pi' = \text{image}(\pi, \rho_w), \quad \ker \pi' = \ker \pi \cap \ker \rho_w. \quad (57.11)$$

Then $w \notin \ker \pi'$, while every element previously outside $\ker \pi$ remains outside the smaller kernel. Apply C8 orbit closure and parity closure; both further decrease the kernel and therefore preserve all separations. For a fixed complete finite dangerous set, repeating this operation terminates after at most the initial number of kernel hits, because each iteration permanently removes at least the selected hit. This is a finite-domain termination theorem, not a claim that arbitrary order-constrained search always succeeds.

55.7 Minimize without losing certificates

After feasibility, minimize a declared objective such as

$$\mathcal{J}(Q) = (|Q|, \text{faithful permutation degree}, B_{\text{mem}}, C_{\text{mv}}) \quad (57.12)$$

in the fixed lexicographic order. Candidate reductions include deleting product factors, replacing an ambient product by the actual subdirect image, quotienting by a certified redundant normal subgroup, and finding a lower-degree faithful action of the same marked group. Deleting a factor enlarges the kernel and may restore a dangerous element. Therefore every proposed reduction is accepted only after replaying the relator, generation, exact C_8 , parity, local, global, operator, and resource conditions. A faithful re-representation preserves the abstract marked kernel only when the marking is explicitly intertwined. Local optimality under these moves is not global minimality; the latter requires exhaustive coverage of the declared finite search domain.

55.8 Enumerate and canonicalize the candidate family

Each accepted candidate is keyed by its marked quotient, not by an unmarked abstract group:

$$\text{marked_key}(Q; A_1, B_1, A_2, B_2, \beta, \bar{\chi}). \quad (57.13)$$

The key is the lexicographically least serialization of the finite group table and displayed marked data under simultaneous relabeling. Two records are merged only after an explicit isomorphism carries every marked generator and typed action to its counterpart. Geometry, source-family, certificate, and cost hashes are attached as provenance records and do not decide marked-group isomorphism.

Let $\mathcal{P}$ be the predeclared finite, decidable parameter domain. Parameters are enumerated in canonical order, and each receives an accepted record or a machine-readable rejection with the first failed condition. The output classification obeys

$$\begin{array}{l|l} \text{COMPLETE} & \text{every } p \in \mathcal{P} \text{ has a recorded disposition} \\ \text{PARTIAL} & \text{some } p \in \mathcal{P} \text{ remains undisposed} \end{array} \quad (57.14)$$

Reaching a wall-time or resource limit while an item remains changes the classification to **PARTIAL**, even when all examined items have valid records. Only the **COMPLETE** branch yields

$$\text{a complete census of the declared finite domain,} \quad (57.15)$$

not an unrestricted classification of all finite quotients of Γ_B . Deterministic factor order, canonical word keys, and fixed tie-breaks make repeated runs byte-reproducible.

56 Verification criteria

56.1 Exact-relation condition

The algebra verifier consumes only canonical exact records. It checks: the finite-group multiplication, identity, and inverse tables; the four marked generator images; the surface relator; the two Nielsen substitutions; the eight ϕ_8 shell equations; the exact order of the descended β ; and the parity equations. Its SATISFIED predicate is

$$\begin{aligned} [A_1, B_1][A_2, B_2] &= 1, & Q &= \langle A_1, B_1, A_2, B_2 \rangle, \\ \beta\pi(s) &= \pi\phi_8(s) \quad (s = a_1, b_1, a_2, b_2), & \text{ord}(\beta) &= 8, \\ \beta(xy) &= \beta(x)\beta(y) \quad (x, y \in Q), & \beta : Q &\rightarrow Q \text{ bijective,} \\ \bar{\chi}(xy) &= \bar{\chi}(x) + \bar{\chi}(y) \quad (x, y \in Q), & \bar{\chi}(1) &= 0, \\ \bar{\chi}\pi(s) &= 1, & \bar{\chi}\beta &= \bar{\chi}. \end{aligned} \quad (58.1)$$

Generator checks extend to all words because both sides are homomorphisms. Generation is necessary because an ambient group may strictly exceed the actual image. Every input field, verifier version, and result is hashed. A single changed generator image must make at least the relator, covariance, parity, or input-hash test fail; otherwise the verifier is not independent.

56.2 Local-ball injectivity condition

Fix an ordered symmetric alphabet S and radius r . Breadth-first enumeration stores for every element of

$$B_S(2r) = \{\gamma : |\gamma|_S \leq 2r\} \quad (58.2)$$

its canonical key, least shortlex word, depth, parent key, and incoming label. A coverage record records layer counts and proves that expanding every vertex of depth below $2r$ yields only recorded vertices at the appropriate next depth. Deduplication is by exact group key, not text word.

Proposition 56.2.1. *If the complete record (58.2) has no nonidentity element in $\ker \pi$, then π is injective on $B_S(r)$.*

Proof. If $x, y \in B_S(r)$ and $\pi(x) = \pi(y)$, then $x^{-1}y \in \ker \pi$ and $|x^{-1}y|_S \leq |x|_S + |y|_S \leq 2r$. The condition forces $x^{-1}y = 1$, hence $x = y$. $\square$

Missing frontier coverage gives **INCOMPLETE**, not **SATISFIED**.

56.3 Based-geometry condition

For fixed basepoint o and closed cutoff L_{based} , define

$$D_o(L_{\text{based}}) = \{\gamma \neq 1 : d_R(o, \gamma o) \leq L_{\text{based}}\}. \quad (58.3)$$

The exact tile flood of C03 must certify completeness, canonical deduplication, and the closed boundary. Algebraic comparisons or outward intervals decide inclusion; an interval intersecting the boundary is refined until decided. SATISFIED means

$$D_o(L_{\text{based}}) \cap \ker \pi = \emptyset. \quad (58.4)$$

This proves injectivity for the declared based domain only. It does not prove a global systole bound, because an axis far from o can have short translation length while moving o farther than the cutoff. A verifier that labels (58.4) as global must be rejected.

56.4 Global systole condition

Let $\mathcal{D}_{\text{ax}}(L_{\text{global}})$ be the complete finite axis-normalized domain from C46, with exact canonical conjugacy keys, axis/fundamental-domain normalization, closed boundary $\ell(\gamma) \leq L_{\text{global}}$, and a independently verifiable empty-frontier certificate. The condition passes exactly when

$$\mathcal{D}_{\text{ax}}(L_{\text{global}}) \cap \ker \pi = \emptyset. \quad (58.5)$$

Theorem 56.4.1. *Under the completeness certificate for $\mathcal{D}_{\text{ax}}(L_{\text{global}})$, (58.5) implies*

$$\text{sys}(\mathbb{H}^2 / \ker \pi) > L_{\text{global}}. \quad (58.6)$$

Proof. If the systole were at most L_{global} , the kernel/closed-geodesic correspondence would give a nonidentity $k \in \ker \pi$ with $\ell(k) \leq L_{\text{global}}$. The global normalization theorem places a conjugate representative in $\mathcal{D}_{\text{ax}}$; normality of $\ker \pi$ keeps that conjugate in the kernel, contradicting (58.5). $\square$

Thus strictness in (58.6) comes from scanning the closed dangerous domain. Missing normalization or frontier data gives INDETERMINATE.

56.5 Representation-completeness condition

For each proposed irreducible ρ_λ , independent verification verifies the group law on marked generators, relations, matrix dimensions, and character values. Pairwise character inner products must be

$$\langle \chi_\lambda, \chi_\mu \rangle_Q = \frac{1}{|Q|} \sum_{q \in Q} \chi_\lambda(q) \overline{\chi_\mu(q)} = \delta_{\lambda\mu}, \quad (58.7)$$

and completeness requires

$$\sum_{\lambda} d_{\lambda}^2 = |Q|. \quad (58.8)$$

Orthogonality proves the listed representations are pairwise inequivalent; (58.8), together with their verified irreducibility, exhausts the regular representation. Realization matrices, basis ordering, multiplicities, and any C_8 intertwiner are hash-bound. Without this condition, representation-resolved spectra are not established, whereas the full-space Hermitian operator remains valid. The representation-resolved calculation is used only when these completeness conditions hold; otherwise the calculation is restricted to the full-space operator and produces no representation-resolved output.

56.6 Numerical-tractability condition

Let $\hat{\mathcal{C}}$ be the measured or rigorously bounded cost vector in the same typed coordinates as the fixed budget $\mathcal{B}$ of (55.17). The condition predicate is componentwise:

$$\hat{\mathcal{C}} \preceq \mathcal{B}, \quad r(v_i, \lambda_i) \leq \varepsilon_{\text{res}} \quad \text{for every accepted Ritz pair.} \quad (58.9)$$

Measured entries record hardware, concurrency, library, precision, peak-memory method, and repeated-run statistic. Rigorous work bounds record support counts, iteration/order counts, and rounding direction. Classification priority is deterministic: any certified exceeded coordinate or residual gives VIOLATED; otherwise any missing mandatory coordinate gives INDETERMINATE; only complete data satisfying every componentwise bound gives SATISFIED. An equality passes only when the corresponding budget uses a nonstrict bound. Matrix-free representation and mathematical admissibility do not waive (58.9).

56.7 Tower-extendability condition

From a certified level K_n , a next level must provide

$$K_{n+1} \subsetneq K_n, \quad p_{n+1,n}(\gamma K_{n+1}) = \gamma K_n, \quad L_{n+1} > L_n. \quad (58.10)$$

It must also prove $\phi_8(K_{n+1}) = K_{n+1}$, $K_{n+1} \subseteq \ker \chi$, compatibility of $\beta, \bar{\chi}$ with $p_{n+1,n}$, and the global condition through L_{n+1} . Containment makes the bonding map well defined and surjective; proper containment makes it nonisomorphic at the kernel level; the larger closed cutoff is a strictly stronger geometry certificate. A single finite quotient, a second quotient without a bonding map, or a larger order with the same kernel fails this condition.

56.8 Final data-consistency criterion

Let $G_1, \dots, G_7$ denote the preceding condition results. Define

$$\text{ModePass}_5(m) = \begin{cases} G_5 = \text{NOT-REQUESTED}, & m = \text{FULL-SPACE}, \\ G_5 = \text{SATISFIED}, & m = \text{REPRESENTATION-RESOLVED}. \end{cases} \quad (58.11)$$

Let HashOK mean that every mandatory component is present and hash-consistent, and let ReplayOK mean that an independent independent verification reproduces every mode-required result. The eighth condition is defined without circularity by

$$G_8 = \text{SATISFIED} \iff \left[\bigwedge_{i \in \{1,2,3,4,6,7\}} (G_i = \text{SATISFIED}) \right] \wedge \text{ModePass}_5(m) \wedge \text{HashOK} \wedge \text{ReplayOK}. \quad (58.12)$$

Computational freeze is permitted exactly in this SATISFIED branch. Operationally, the candidate record first binds immutable component hashes for geometry, marked quotient, algebra, local/global certificates, the representation inventory or canonical NONE, support, operator, solver grid, budgets, code, and external theorem versions. A fresh verifier then replays conditions Statement 56.1–Statement 56.7 from those components and emits the root hash

$$h_{\text{computational}} = \text{SHA256}(\text{domain}, \text{schema}, h_{\text{component},1}, \dots, h_{\text{component},m}, \text{independent verification results}). \quad (58.13)$$

Any changed or absent mandatory component invalidates the root. A violated or unsupported mandatory condition remains part of the record; the sole nonmandatory branch uses the explicit representation NONE sentinel above. A prose-only name, unattached scan count, decimal estimate, or unresolved certificate identifier supplies no mandatory component and therefore cannot pass Statement 56.4 or Statement 56.8.

57 Covering towers and physical limits

57.1 Level A1: artifact-bound specification

Let a proposed first level be the immutable tuple

$$\mathcal{A}_1 = (Q, \pi, \beta, \bar{\chi}, \mathcal{D}_{\text{ax}}(6a_B), \mathcal{S}_T, \mathcal{B}, \mathcal{C}_{\text{independent verification}}), \quad (59.1)$$

where the fields have the exact meanings fixed in C48 and C53–C58.

Theorem 57.1.1 (Artifact-bound Level A1 implication). *If immutable artifacts independent verification the exact marked quotient, the exact C_8 and parity actions, complete global closed-domain coverage with empty frontier, and*

$$\mathcal{D}_{\text{ax}}(6a_B) \cap \ker \pi = \emptyset,$$

then

$$\text{sys}(\mathbb{H}^2 / \ker \pi) > 6a_B, \quad r_{\text{inj}}(\mathbb{H}^2 / \ker \pi) > 3a_B. \quad (59.2)$$

If, in addition, the certified interlayer support obeys $r_{\parallel, \text{max}} < 3a_B$, then that support has no quotient wraparound.

Proof. The global scan and the kernel/closed-geodesic correspondence give $\text{sys} > 6a_B$ by Statement 56.4. Halving gives $r_{\text{inj}} > 3a_B$. The additional strict support inequality gives $r_{\parallel, \text{max}} < r_{\text{inj}}$, so Statement 51.5 supplies the unique-lift conclusion. $\square$

The tuple is SATISFIED only when all mode-mandatory C58 conditions independent verification, VIOLATED when an exact mandatory check produces a counterexample, and INDETERMINATE otherwise. In particular, a name or prose summary never supplies the artifacts required by the theorem.

57.2 Tower A: universal characteristic cutoffs

For $M \geq 1$, define

$$K_M = \bigcap_{\substack{|F| \leq M \\ \rho \in \text{Hom}(\Gamma_B, F)}} \ker \rho, \quad Q_M = \Gamma_B / K_M. \quad (59.3)$$

Theorem 57.2.1. *The K_M form a nested characteristic finite-index residual tower. Every ϕ_8 descends compatibly; parity descends for $M \geq 2$; the orders are unbounded; and*

$$\text{sys}(\mathbb{H}^2 / K_M) \longrightarrow \infty. \quad (59.4)$$

Proof. A finitely generated group has only finitely many homomorphisms to the finitely many group tables of order at most M . Thus (59.3) is a finite intersection of finite-index normal kernels and is finite-index normal. Precomposition by any automorphism permutes the homomorphisms, so K_M is characteristic. Enlarging M adds factors, hence $K_{M+1} \subseteq K_M$.

For $1 \neq \gamma \in \Gamma_B$, residual finiteness supplies a finite group F and $\rho(\gamma) \neq 1$; once $M \geq |F|$, $\gamma \notin K_M$. Hence $\bigcap_M K_M = \{1\}$. If the quotient orders were bounded, their nondecreasing integer indices would eventually be constant; nested subgroups of equal finite index are equal, giving a stabilized nontrivial finite-index intersection in the infinite group Γ_B , a contradiction. Characteristicity gives the compatible descended ϕ_8 . For $M \geq 2$, the map χ occurs in (59.3), so $K_M \subseteq \ker \chi$.

Finally, for each L the complete global dangerous set is finite. Residuality excludes each of its elements at some level, and nesting excludes all of them at one later level. Statement 56.4 then gives $\text{sys} > L$, proving (59.4). $\square$

The construction is canonical but is not claimed computationally efficient.

57.3 Tower B: lower exponent-2 central quotients

Put

$$P_1 = \Gamma_B, \quad P_{n+1} = P_n^2[P_n, \Gamma_B], \quad R_n = \Gamma_B/P_n. \quad (59.5)$$

Theorem 57.3.1. *Each P_n is characteristic, each R_n is a finite 2-group, and $\bigcap_n P_n = \{1\}$. After deleting repeated levels, the sequence is a strict nested tower of unbounded order. For $n \geq 2$, ϕ_8 and parity descend.*

Proof. Endomorphisms preserve squares and commutators, proving characteristicity. Starting from the finitely generated P_1 , the quotient P_n/P_{n+1} is a finite elementary abelian 2-group; induction makes R_n finite and P_n finite-index and therefore finitely generated. The residual-2 theorem for closed orientable surface groups [5] separates every $\gamma \neq 1$ in a finite 2-group F . The lower exponent-2 series of F terminates, and functoriality gives $\psi(P_n) \subseteq P_n(F) = 1$ for large n ; hence $\gamma \notin P_n$, proving trivial intersection.

Infinitely many strict drops follow from trivial intersection: eventual stabilization would leave a nontrivial finite-index subgroup. The same bounded-index argument as in Tower A gives unbounded orders. Finally,

$$P_n \subseteq P_2 = \Gamma_B^2[\Gamma_B, \Gamma_B] \subseteq \ker \chi \quad (n \geq 2), \quad (59.6)$$

so parity descends, while characteristicity descends ϕ_8 . $\square$

For $M \geq 3$, Tower A maps onto C_3 , whereas every R_n is a 2-group. Thus infinitely many cross-pairs are nonisomorphic, giving the stated algebraic non-equivalence without relying only on order size.

57.4 Common local limit

Let K_n denote either strict residual tower and $X_n = \mathbb{H}^2/K_n$. For every $L < \infty$, let $\mathcal{D}_{\text{ax}}(2L)$ be the finite complete axis-normalized representative domain of Statement 56.4. Residuality and nesting eventually give

$$K_n \cap \mathcal{D}_{\text{ax}}(2L) = \emptyset, \quad r_{\text{inj}}(X_n) > L. \quad (59.7)$$

Likewise, for every labeled radius r ,

$$K_n \cap B_{S_{\text{geo}}}(2r) = \{1\} \quad (59.8)$$

eventually holds. The proof is the same finite-set maximum argument: each nonidentity representative is absent at some level and nesting preserves absence; normality transfers the exclusion to its conjugacy class. Equations (59.7)–(59.8) show that fixed geometric balls and fixed labeled Cayley balls lift isometrically at all sufficiently large levels.

Now fix a bounded Γ_B -equivariant bulk operator H_∞ of propagation at most r_0 , together with its canonical finite coefficient record indexed by labeled displacements. Let H_n be the compatible quotient realization obtained by pushing that same coefficient record through $\Gamma_B \rightarrow \Gamma_B/K_n$; this compatibility is an operator hypothesis, not a consequence of local geometry alone. Let $J_{n,L}$ identify vectors supported in a fixed radius- L ball with their unique quotient copies. Once $r_{\text{inj}}(X_n) > L + r_0$,

$$H_n J_{n,L} u = J_{n,L+r_0} H_\infty u \quad (59.9)$$

for all u supported in the L -ball. Indeed, injectivity on the r_0 -neighborhood prevents two coefficient-record displacements from aliasing, while the quotient construction uses exactly the same coefficient for each surviving label. Thus every finite-support matrix coefficient is eventually equal, and uniform operator bounds extend this equality to strong convergence in the local inductive-limit sense. The compatible realizations on both towers have the same universal-cover limit because the fixed coefficient record on the right side of (59.9) is independent of the tower.

57.5 Spectral limit: loss versus pollution

Theorem 57.5.1 (No loss). *Let H_∞ and H_n be uniformly bounded self-adjoint finite-propagation operators satisfying (59.9). Then for every $\lambda \in \sigma(H_\infty)$,*

$$\text{dist}(\lambda, \sigma(H_n)) \longrightarrow 0. \quad (59.10)$$

Proof. By the Weyl criterion choose a unit vector u with $\|(H_\infty - \lambda)u\| < \varepsilon/2$. Finite-support vectors are dense, and boundedness of $H_\infty - \lambda$ permits replacing u by a normalized finite-support vector v with residual below ε . For large n , local equality transports v to a unit vector J_nv with $\|(H_n - \lambda)J_nv\| < \varepsilon$. The self-adjoint resolvent estimate $\text{dist}(\lambda, \sigma(H_n)) \leq \|(H_n - \lambda)J_nv\|$ proves (59.10). $\square$

The reverse inclusion is not a consequence of residuality. If for every compact $C \subset \mathbb{R} \setminus \sigma(H_\infty)$ there is $c_C > 0$ such that eventually

$$\|(H_n - \lambda)v\| \geq c_C \|v\| \quad (\lambda \in C, v \in \mathcal{H}_n), \quad (59.11)$$

then no spectral pollution occurs in C . Together with (59.10), uniform boundedness of the spectra, and a finite covering argument, (59.11) yields Hausdorff convergence of spectra. Without (59.11), the certified conclusion is no loss only.

57.6 Riesz-projector limit

Let $\mathcal{C}$ be a positively oriented contour enclosing an isolated spectral cluster of H_∞ . Assume $\mathcal{C}$ lies in the resolvent set of every sufficiently large H_n , that

$$\sup_{n, z \in \mathcal{C}} \|(z - H_n)^{-1}\| < \infty, \quad (59.12)$$

and that the locally identified resolvents converge strongly, uniformly for $z \in \mathcal{C}$. Define

$$P_n = \frac{1}{2\pi i} \oint_{\mathcal{C}} (z - H_n)^{-1} dz, \quad P = \frac{1}{2\pi i} \oint_{\mathcal{C}} (z - H_\infty)^{-1} dz. \quad (59.13)$$

Proposition 57.6.1. *Under these hypotheses $P_n \rightarrow P$ strongly after the stated identifications. Uniform norm-resolvent convergence on $\mathcal{C}$ implies norm convergence of the projectors.*

Proof. For each vector, strong uniform convergence of the integrand and the bound (59.12) permit passage through the contour integral, giving strong convergence. In the norm-resolvent case,

$$\|P_n - P\| \leq \frac{\text{len}(\mathcal{C})}{2\pi} \sup_{z \in \mathcal{C}} \|(z - H_n)^{-1} - (z - H_\infty)^{-1}\| \rightarrow 0.$$

$\square$

No conclusion is made when the contour gap closes or (59.12) fails.

57.7 Parameter-derivative limit

Let $H_n(\theta)$ and $H(\theta)$ be twice differentiable on a compact parameter interval I . Assume, after the common identifications,

$$\sup_{\theta \in I} \|H_n^{(j)}(\theta) - H^{(j)}(\theta)\| \longrightarrow 0 \quad (j = 0, 1, 2), \quad (59.14)$$

and assume one contour $\mathcal{C}$ has a uniform gap and resolvent bound for every $\theta \in I$ and all large n . With $R(z, \theta) = (z - H(\theta))^{-1}$, differentiation of $(z - H)R = I$ gives

$$\begin{aligned} \partial_\theta R &= RH'R, \\ \partial_\theta^2 R &= RH''R + 2RH'RH'R. \end{aligned} \quad (59.15)$$

The same identities hold for R_n . The resolvent identity and (59.14) give uniform convergence of R_n , then substitution in (59.15) gives uniform convergence of its first two derivatives. Integrating around $\mathcal{C}$ proves

$$\sup_{\theta \in I} \|P_n^{(j)}(\theta) - P^{(j)}(\theta)\| \rightarrow 0 \quad (j = 0, 1, 2). \quad (59.16)$$

Consequently bounded spectral observables built from this isolated Riesz cluster converge in C^2 . These hypotheses must be verified for the full cutoff kernel; smooth distances alone do not prevent support changes at $D_{xy}(\theta) = D_c$.

57.8 Bulk-limit hypotheses

Theorem 57.8.1 (Conditional full-bulk conclusion). *Let a strict tower of fixed levels pass the mode-dependent conditions Statement 56.1–Statement 56.8. Suppose every operator support radius is below the certified global injectivity radius and the uniform analytic hypotheses (59.11)–(59.16) hold. Then:*

- (i) *each finite-level Hamiltonian is locally identical to the bulk operator on its full support;*
- (ii) *the claimed C_8 , parity, and selected computational-mode symmetries are exact;*
- (iii) *spectral loss is absent, and spectral pollution is absent on the domains covered by (59.11);*
- (iv) *isolated Riesz projectors and their first two parameter derivatives converge as in (59.13)–(59.16).*

In REPRESENTATION-RESOLVED mode the complete C47 inventory also makes the block spectrum exhaustive. In FULL-SPACE mode no representation-resolved claim is made.

Proof. The global radius and support inequality give unique lifting of every matrix entry. Statement 56.1 and the operator regression condition give exact algebraic and operator symmetries; Statement 56.5 supplies exhaustive blocks only in its requested mode. Statement 57.5 proves no loss and uses (59.11) for no pollution. Statement 57.6–Statement 57.7 prove the projector and derivative claims. Statement 56.8 binds these inputs and independent verification results to each computational level. $\square$

For any proposed computational tower, define the conclusion classification by

$$\text{FullBulkStatus} = \begin{cases} \text{SATISFIED,} & \text{all hypotheses of the theorem independent verification,} \\ \text{VIOLATED,} & \text{a mandatory exact condition has a independently verified violation,} \\ \text{INDETERMINATE,} & \text{otherwise.} \end{cases} \quad (59.17)$$

Thus residuality, a single finite quotient, or local convergence alone cannot yield a FULL-BULK passage classification.

58 Unified construction theorem

Theorem 58.0.1 (Unconditional existence and local limit). *The Bolza surface group admits two algebraically non-equivalent nested characteristic residual towers of finite-index normal subgroups: the universal characteristic cutoff tower (K_M) and, after deleting repetitions, the lower exponent-2 tower (P_n). Both carry the octagonal automorphism, parity from the stated levels onward, divergent systole and injectivity radius, and the same labeled and geometric local limit. Every fixed bounded Γ_B -equivariant finite-propagation bulk operator and its compatible quotient realizations converge strongly locally and satisfy the no-loss conclusion (59.10).*

Proof. Tower A is proved in Statement 57.2 and Tower B in Statement 57.3. Their algebraic non-equivalence follows from the C_3 quotient of Q_M for $M \geq 3$ versus the exclusively 2-primary groups R_n . Statement 57.4 gives both local limits and Statement 57.5 gives no loss. None of these arguments depends on a separately chosen finite example. $\square$

Corollary 58.0.2 (What remains conditional). *For either tower, full Hausdorff spectral convergence, Riesz-projector convergence, C^2 parameter convergence, representation-exhaustive block spectra, and a computational FULL-BULK claim hold only under their respective hypotheses in C58 and Statement 57.5–Statement 57.8. In particular,*

$$\text{residuality} \not\Rightarrow \text{no pollution or numerical tractability.} \quad (59.18)$$

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